UESCA Running Coach Course
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Welcome to the UESCA Running Course

The ***United Endurance Sports Coaching Academy's* (UESCA)** running certification is a comprehensive certification that will provide you with everything you need to know to be an effective running coach. This is the second edition of the UESCA running coach certification, with many updates from the first version. As a certified running coach, your athletes will expect you to know up-to-date information on everything from carbon-soled shoes to how to create and implement training programs from 5K through marathon distances. This certification discusses all these things and more.

After obtaining this certification, you will be a comprehensive resource for an athlete who wants to train for a running race event … even if you have no previous knowledge of the sport. As such, this certification will prepare you to answer almost any question or concern an athlete may have.

This certification aims to create a well-rounded running coach who understands the fundamentals of exercise science and can apply this information. In addition, this certification goes beyond the physiological and biomechanical aspects of training. We believe that if an athlete has a question regarding any aspect of the sport, the coach should be knowledgeable in that area, even if that means knowing when to refer an athlete to a specialist. **This certification was developed in conjunction with a team of**

expert advisors and contributors across many disciplines to ensure that the quality of the certification is second to none.

Just like a successful athlete, a successful coach must be confident in their knowledge base and the results they can deliver to an athlete. Unless you have all the tools and resources to do this, your coaching will not be optimal, which will be evident to an athlete.

This certification will give you the tools necessary to become a knowledgeable, confident, and all-encompassing resource.

Certification Structure

The UESCA running coach certification is not designed to function like a Paint-By- Numbers picture, nor is it designed to offer various coaching templates from which to choose. This certification aims to educate and inform an individual on all aspects of what is required to be a successful running coach. Athletes will come to coaches with a wide array of goals and issues. Therefore, the focus of this certification is not to teach exactly how to coach XYZ athlete or what program to assign to ABC athlete because every athlete's program is different due to the many factors involved in program creation.

An analogy to illustrate this would be an ice cream shop with 20 different flavors of ice cream and 30 different toppings. There would be hundreds of other ice cream and topping combinations. This is analogous to how many potential running-training programs would be out there if one were to factor in all the various options available. The only difference is that instead of hundreds of different ice cream possibilities, there would be thousands of training programs. At UESCA, we certainly do not have the time to create this many programs, and we are also quite sure that you would not have the time to read through them all to select the best one for your athlete! However, this certification provides all the requisite theory, information, and tools for you to meet your athletes' individual needs.

Staying current with what is happening within the sport of running and sports science is critical to being an effective coach. Due to the rapid evolution of the sport, many training practices and concepts may not exist in established research to determine their validity. However, by researching the theory behind the training practice, you will likely be able to deduce whether the training practice is valid and applicable to your athletes.

UESCA rarely takes a definitive standpoint on training practices, with a few exceptions. Why is this? We believe that what works well for one athlete might not work so well for another. In other words, **people respond differently to the same training stimuli.** Therefore, it is short-sighted to expect that one training methodology adheres to the "one size fits all" theory. This certification provides a coach with the science and training theories to advise athletes. Therefore, it is the coach's responsibility, not UESCA, to decide what training practice is in the best interest of an athlete.

The certification is comprised of text, images, and videos. At the end of each of the modules are practice quizzes.

No Tiers

UESCA does not produce tiered coaching certifications, which means this certification is not a beginner-level certification where there are still one or two more advanced running certification levels above it. As stated earlier, if you are a *certified* running coach, your athlete will expect you to know everything from how to taper for a marathon to what the *Rectus Femoris* is. At UESCA, we believe a certification program should provide you with all the information to ensure you are a well-educated and up-to-date running coach.

UESCA Certification Principles

- One must understand training theory before it can be applied
- Training must be grounded in science
- Coaching must take place within your scope of practice
- Random training = random results
- Challenge the status quo: do not accept long-standing training practices at face value.
- You are regressing if you are not constantly learning and applying this knowledge.
- Educated athletes are engaged athletes.
- Coaching success is predicated mainly on thorough training feedback from athletes.
- The coaching process is dynamic and not static.
- All coaching decisions must be based on the individual.

Certification Content

As noted previously, this is the second edition of the UESCA running coach certification. One of the significant differences between the two editions is that we sought out even more specialist advisors to contribute to this certification. The main contributor is Ben Rosario, former head coach and current executive director of the Hoka Northern Arizona Elite Running Team, one of the top professional running teams in the US.

Some of our other contributors include:

- Bob Seebohar – Nutrition
- Dr. Justin Ross – Psychology
- Corrine Malcolm – Environmental Physiology
- Alexandra Coates – Physiology
- Dr. Leah Roberts – Female Physiology
- Dr. Nick Studholme – Running Mechanics

Many well-established training principles and “rules” exist solely in folklore or through casual and coincidental relationships.

This is not to say that all current, well-established training practices are incorrect or ineffective, as this is not the case. That said, the development of this certification program did not take any training theory or practice at face value. Each training methodology described in this certification has been thoroughly vetted to ensure its basis in science and application. We have examined the validity of everything from the 10% increase volume ‘rule’ to the proposed benefits of carbon fiber soled shoes.

If nothing else, the goal of this certification is to challenge those obtaining it never to accept established norms just because they have been around for a long time. In the words of Nobel Laureate Sir Harold Kroto, ***“We need everyone to know how to think rather than accept unquestioningly what we are told. It is amazing to me what people will believe without any proof”*** (303).

It is essential to recognize that accurate training information is a moving target. As new research results become available, current training methodologies are validated, debunked, or modified to reflect the new findings. It is your professional responsibility as well as UESCA to stay abreast of the latest training information.

Module 1: Run Coaching 101

To be a successful running coach, you must be versatile, knowledgeable, and dependable. The areas noted in this module represent the different functions that a successful running coach performs. As a coach, involvement in your athletes' training and racing can range from relatively basic to highly complex and involved.

In the following video, Ben Rosario discusses what he views as his main goal as a coach.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

So, what exactly do coaches do, and what are their responsibilities? To some degree, a coach’s role depends on what an athlete is asking for. However, a coach must be well-rounded, as it is an all-encompassing role.

Upon completion of this module, you should have an understanding of the following areas:

- Roles of a coach
  - Physiologist
  - Psychologist
  - Leader
  - Motivator
  - Disciplinarian
  - Operations manager
  - Instructor and educator
- What constitutes being a professional coach
- How to communicate effectively
- Value of being a comprehensive resource
- Fallacies of being a coach

Traits of a Coach

A running coach must be part exercise physiologist, psychologist, leader, motivator, disciplinarian, operations manager, and, last but not least, an instructor and educator. All of these things are essential to being a successful coach, as your athletes will expect you to be a resource in all areas. While you will not serve as a psychologist or exercise physiologist after completing this certification, you will have enough education to develop high-level programs for your athletes and also have the knowledge base to identify when to refer them to someone more specialized, if needed.

How athletes respond to you will likely vary significantly from one to another. **As a coach, you will need to figure out how each athlete learns best and how to motivate each one.** For example, some people tend to be more visual than auditory learners. If this is the case and you are discussing cornering, drawing a diagram, and explaining the concept might prove best.

Concerning motivation, you might have athletes who respond best to being yelled at (i.e., boot camp), while other athletes might feel as though they are being scolded. Therefore, the way you motivate an athlete is extremely important. As a successful coach, you must discern the best way to motivate each athlete. It is important to modify your tone, body language, and verbiage to suit each athlete.

Leader

Above everything else, a successful leader garners respect from those the person leads. An athlete must be confident that you are the total package. So, what is the “total package?”

- Organized and structured
- Educated
- Experienced
- Puts safety first
- Attuned to an athlete’s psychological needs
- A resource
- Flexible
- Dependable and accountable
- Consistent in the quality delivered
- Honest and trustworthy

Perhaps the most critical aspect of being a leader is understanding what you are and are not. First and foremost, you are their coach. This is not to say you cannot be friends with your athletes. However, this should not be the intent or focus of your relationship.

Motivator

In order to get the most out of an athlete, you will need to be a source of motivation. It is understood that some athletes will require more motivation than others, but all will require encouragement to some degree. When taking into account all the other things that a runner must balance with training (i.e., family, job, social life, etc.), it is often difficult to maintain an athlete's motivation at the necessary level. **The methods used to motivate an athlete will vary by the individual as not all athletes will respond the same way to a particular method of motivation.**

It is equally important to know when to back off. This is where the listening component comes into play. For example, if an athlete experiences a death in the family it is advisable to give them all the space and time needed prior to resuming training. Other things in people's lives are far more important than training for running events, and this big-picture approach always is best kept in mind when working with athletes.

Disciplinarian

What comes to mind when you hear the word *disciplinarian*? Most likely it is associated with something negative. **Pertaining to coaching, discipline should not be viewed as a negative thing, but rather something positive.** Being a disciplinarian ties in closely with being a motivator. While it is important to be flexible with athletes about their training programs, it is equally important to make sure they stay on track to elicit the results they are looking for.

The disciplinarian role of a coach is less a function of being a relentless taskmaster than it is of holding athletes accountable. The training programs you develop for your athletes are useful only if they perform them the way they were intended. **Therefore, your expectations must be clearly stated to your athletes.**

Some athletes will need to be held more accountable than others in regard to sticking to their program. However, oversight of all athletes' programs is necessary to ensure the proper training progression is accomplished.

Operations Manager

Depending on your level of involvement, you may find yourself thrust into the role of an operations and logistics manager. Some of the aspects of this role might include:

- Travel plans
- Equipment ordering/management
- Event planning
- Contingency planning
- Route scouting

It goes without saying that successful coaching requires strong organizational skills, and the more involved you are as a coach, the more things you will need to juggle. Let's say that you run a successful online coaching business with 50 athletes whose goals and training plans vary. First and foremost, you need to remember when you need to contact your athletes via either email or phone. Each athlete should feel as though he or she is your only athlete instead of one out of 50.

Keeping track of your athletes' communication and progress requires organization and structure. While this might not sound like a big deal … it is. When an athlete is paying a fee for your services and you promise to email a weekly program on Sunday and follow up with a call each Tuesday at 6 p.m., and you forget to do both, you can assume that you will not have this athlete for much longer.

It is also your responsibility to have full awareness of the rules of the sport and, more specifically, the rules of any event your athletes are participating in.

Instructor And Educator

As an instructor and educator, it is your job to ensure your athletes understand precisely what they should do and the reasoning behind it. It is your responsibility to convey information to your athletes in an effective manner.

Part of an effective message is continuity. **This means sending no mixed signals to athletes relative to their training program.** For example, if you state that Monday is always a rest day no matter what, and then a few weeks into the program, you have the athlete do a hard workout on Monday, this sends a mixed message and erodes trust and confidence over time between you and your athlete.

Not all athletes will want instruction the same way; however, some basic principles should underlie how you instruct your athletes. Some of these are:

- Encourage your athlete
- Provide explanations of your program design
- Be as clear and concise as possible
- Provide feedback and honest assessments
- Perform goal-setting exercises
- Be positive
- Be realistic in your expectations of the athlete
- Continuity

It is helpful to teach your athletes the rationale behind their program. While some athletes will want to know more than others, they must understand the program's basic reasoning. The more you educate your athletes, the better they will understand the reason behind their program and become active participants in its design and development, leading to a more effective program.

An analogy would be the relationship between a race car driver and their pit crew. Unless the driver understands how the car handles based on its inner workings, the driver will not be able to communicate effectively with the pit crew on how to adjust the car to make it faster. Sharing your training strategy with your athletes is critical to ensuring that you are both on the same page.

The coach/athlete relationship is a two-way street. The athlete is learning from you, and you are continually learning from the athlete.

Application of Information

It is not enough to understand training theory and the physiology of the body. To be an effective coach, you must understand how to apply this information on an individual basis to get the best possible results for your athletes.

Professional

It is essential to convey to your athlete what you know and do not know. **The guiding principle always should be your athletes' best interests.** For example, just because you know why an athlete's knee is hurting unless you are a doctor or physical therapist (or similar specialist), you are not qualified to diagnose this. The correct decision is always to refer the person to a specialist. Likewise, if an athlete asks you something you do not know, it is correct to tell them that you will get back to them with the answer or refer them to someone who will have the answer. **By trying to be everything to your athlete, you are doing a disservice to them, and in some cases, your actions could be illegal and potentially injurious.**

Honesty

A key component of professionalism is honesty. While this seems simple, it is often the most difficult rule to follow. For example, you start coaching an athlete who, despite having results in the bottom half of most races, the runner's goal is to win a prestigious half marathon. While goals are necessary and stretch goals are a good idea, an athlete's goal must be within the realm of possibility. Therefore, in this situation, it is your professional responsibility to inform the athlete that the person's expectations are most likely too high given their current level of conditioning and the competition at the event. This is not always an easy conversation, but to progress the athlete properly, it is a necessary one. This is an example of treating an athlete honestly, with integrity, and with respect. There are ways to make this conversation as painless as possible, but at the end of the day, it is of crucial importance that you convey an accurate message to the athlete.

Part of being honest also involves knowing when to say you do not understand something. Coaches often feel they must know everything because an athlete has hired them and therefore must be an all-knowing resource. First, you will never know everything there is to know about the sport of running. Second, and as noted previously, it is okay to say, "I don't know," to your athlete. It is much better to say, "I'm not sure, let me get back to you," than to guess, lie, or bluff your way through an answer. Not only is this disingenuous, but it could lead athletes off track with their training or, worse, to injury.

Verbal and Physical Actions

Being professional also relates to your verbal and physical actions. If you are working with an athlete one-on-one, you should aim to arrive at the meeting location five minutes ahead of the designated time. **During your session, your athlete is the *only* person that matters and must have 100 percent of your focus.** For example, if your session takes place at a gym, you should not be watching TV, checking your email (turn your phone off or vibrate), talking to other staff or members, or taking the focus off your athlete. To that same end, your body language and attire say a lot about you as a professional, especially with new athletes who do not know you well. During your session, you should always stand or kneel if you need to watch their form from a lower level. You should never sit, lie down, or lean against a wall. When this occurs, the perception from an athlete's point of view is often that of laziness and a lack of professionalism. Regarding attire, your attire should not be sloppy, nor should it have potentially offensive verbiage or images.

Lastly, the verbiage that you use with your athlete reflects on you. If you swear, tell off-color jokes, and use poor grammar or slang, this paints a picture of someone who is unprofessional and does not take the occupation seriously.

Stay Current

A professional in any field must remain current on all the latest developments. This is the only way to stay relevant. **For a coach, this means staying up to date on training and coaching methodologies and equipment.** A coach's credibility is not limited to areas that involve the actual training process but the sport as a whole.

Effective Communicator

This is perhaps the most critical aspect of developing an effective training program. **Without effective communication between you and the athlete, it is almost certain that the resulting program will be ineffective.** As noted, communication between you and your athlete is a two-way street. Initially, athletes need to communicate their goals, their health and athletic background, their bandwidth regarding training capacity, and perceived strengths and weaknesses. Once training begins, athletes must communicate their feelings and perceptions of the program’s effectiveness. You need to share your coaching philosophy along with your expectations. Throughout the training process, you need to listen to the athlete’s feedback and communicate effectively regarding modifications to the training program and the reasons behind them.

Most coaching program failures are the result of poor communication. To a large degree, the effectiveness of communication is based on honesty. If an athlete was supposed to run 10 miles but only ran 4 and reports doing 10, this will become an issue because you will progress the athlete based on 10 miles. This may result in overtraining, injury, or both. If an athlete has never run more than 4 miles at a time but signed up for a marathon taking place in two months and wants you to help in preparation for it, it is your responsibility to advise them that this is not possible, and the person should revise the goal. **As a coach, you need to learn how to manage your athletes' expectations, which sometimes means having hard conversations about what is or is not possible.**

Typically, losing an athlete is considered a bad thing, although sometimes it is for the best. Like all relationships, there will be difficulties, and sometimes relationships do not work out. This is not good or bad, but simply a reality. Relationships between coaches and athletes do not work out for various reasons. Some of the more common ones are:

- Personality conflicts
- Coaching philosophy disagreements
- Lack of perceived results on the part of the athlete
- Inflexibility of time by either party
- Lack of professionalism by either party

While it is the hope that some or all these issues could be mitigated early in the relationship through proper and honest communication, it is a reality of coaching that not all athletes will work out for one reason or another. A professional coach does not take these things personally and knows when to recommend terminating a coaching relationship with an athlete.

Professionalism needs to be evident in all areas of the coaching/athlete relationship. Below are some of the areas:

Email

Your emails should be proper and professional. What does this mean? It means no emoticons, slang, texting-style abbreviations, spelling errors, all CAPS, unnecessary exclamation points, or improperly constructed sentences.

In short, your emails should be constructed like a thank-you letter for an interview. They should be to the point and with a clear and concise message. While this sounds trivial, the ability to compose a professional email is of utmost importance. It tells the athlete that you are professional and you care. Be sure to reread all emails before sending them.

Using a different email account for your coaching business is strongly advised. Make sure that your email address is also professional.

Phone

Do not use slang words and speak clearly and professionally at all times. Email and phone are similar in that you do not have the advantage of seeing the other person’s facial expressions or body language. Therefore, your words and tone can easily be misconstrued. Because of this, you want to ensure that you always present yourself professionally to minimize the chance of misinterpretation.

Text

It is important to note that texts are a different beast than email. As texts have largely replaced email and phone calls, it should be noted that the way an individual uses text is different from email. For example, the use of abbreviations is standard in text versus email (ex: ‘u’ instead of ‘you’). Additionally, while a text is easy to fire off, unless a coach lets an athlete know that they are accessible after work hours, a coach should not be expected to reply to texts outside of the designated timeframe. Also, the text exchanges should be reasonable – meaning, if a text exchange is longer than two or three texts, it is best to move the conversation to a phone call.

Timeliness

As noted previously, if you tell an athlete you will call or send an email at a particular time, you must do so. If you agree to meet your athlete at a designated location and set time, you should arrive at least five minutes before the specified time. It is immaterial whether your athlete gets there late or not. Your athletes are paying you to provide them with a service and deserve a coach who respects their time.

Assuming your athlete’s training session has a pre-determined duration, you must structure the session to accomplish what you need to within the established time. Your athlete might not be wearing a watch and might need to leave right at the session’s end. Also, if you exceed the time limit, you risk cutting into your next athlete’s appointment time. You are responsible for running the session effectively and paying attention to the session duration.

Sensitivity

Being sensitive to your athlete’s emotions and current issues is important. Many of these areas you will learn about only during the training process. For example, perhaps you have athletes who are very sensitive about their weight. If this is the case, you will need to devise a communication strategy regarding communicating effectively with them while not offending them.

As noted previously, you should avoid discussing areas of their personal life such as finances, religion, human rights, politics, and any topic of a sexual nature. **You are hired to be a running coach – no more, no less.** Of course, you will venture into areas of small talk with your athletes but try to keep the conversation light. If your athlete asks for your opinion on any of the issues above, respond by saying that you would rather not discuss it.

Boundaries

This is an important issue for all coaches. Many coaches become “friends” with their athletes, especially if they have worked with them for a long time. To some degree, this is normal and acceptable. What you do with your athlete determines what is acceptable or not.

Some things not to do are:

- Meet for activities other than coaching or running-related events
- Emails and phone calls about anything other than coaching
- Discussing your personal lives beyond what may be perceived as comfortable (e.g., talking about your athlete’s kids’ soccer game is okay, but talking about their messy divorce is not okay)
- Date your athlete

- Assume that based on the length of your coaching relationship, you are free to ask your athlete questions about their personal life

If you move outside professional boundaries, you are diminishing your professional relationship and risk losing athletes altogether. In summation, it is your responsibility as a professional coach to use sound judgment regarding what is acceptable to discuss or do with your athletes.

Areas of Study a Coach Must Understand

While there are many areas that a coach must be knowledgeable in, the areas of physiology and psychology are paramount to a coach being able to understand their athletes.

Physiology

The goal of this certification is to provide you with enough information so you will have a thorough knowledge of basic human anatomy and physiology as it applies to exercise and, more specifically, the sport of running. Having this knowledge base is critical to understanding how to develop and implement programs. While it is not necessary to pass this information along to athletes, it is important that you know it. If you were to put this in the context of a visit with your doctor, while the physician most likely communicates with you in layman's terms instead of using medical terminology, you fully expect that the medical professional has the medical training and knowledge to understand and diagnose your illness.

One way to think of the science of exercise is as the foundation of a pyramid. **Without this broad knowledge base there can be no application and, therefore, no solid programming for an athlete.** This is not to say that you cannot develop a program without an understanding of anatomy and physiology. However, a program developed without this knowledge would most likely be based solely on the coach's personal experiences and/or through reviewing other coaching plans. While this method of coaching may or may not develop a successful plan, the bottom line is that you will not have the ability to explain why the program is laid out a certain way or know how to adjust it if it does not work for an athlete.

Psychology

It is of paramount importance to listen and relate to your athletes. If you are unable to listen to them and take what they say into consideration, you will most likely not have much success as a coach. What your athletes tell you might not always directly relate to running or training. For example, they might be having difficulties with work, personal relationships, or finances. They may or may not verbally express what else is going on in their life, but, as their coach, it is important to notice changes in their motivation, energy, and attitude. The longer you work with an athlete, the easier it will be for you to identify changes in their behavior. Generally, the longer an athlete works with a coach, the more open the individual becomes about other areas affecting him or her.

While it is not your place as a running coach to advise athletes on matters outside of the sport (i.e., finances, relationships, etc.), you do need to be aware that for some athletes, having somebody to speak to about things is an important aspect of the coach/athlete relationship. It is strongly advised that you do not offer advice on areas outside of running. If an athlete asks you for your advice in these matters, it is in your best interest to say that it is not your area of expertise and the person would be better served speaking to someone in that subject or industry.

It is not enough only to notice changes in your athletes. You also must know how to modify their programs accordingly. For example, if an individual is going through a divorce that is demanding a lot of time and energy, you might need to adjust their program to something that is more time and energy efficient while still keeping the ultimate goal in mind.

Coaching Fallacies

While there are likely quite a few coaching fallacies, this certification will discuss three that are unfortunately quite commonplace in the coaching industry.

You Must Be Fast!

Question: Do you think Patrick Sang, coach of marathon world record holder Eliud Kipchoge, could beat Eliud in a race? Not a chance!

It is not uncommon for running coaches to promote their business by using their personal records or the number of races run. While personal running experience is helpful to the coaching process, the ability to run "fast," have a set number of races under one's belt, or be a faster runner than one's athletes is not a prerequisite to being a quality coach. Conversely, one could make the argument that being a talented and fast runner may result in coaches being less effective, as they cannot relate to slower runners. **The bottom line is that there is no correlation between being a fast runner and being a great coach.** It is much more important for a coach to be inquisitive, constantly learning, and humble. Also, note that what constitutes "being fast" is very much subjective.

This does not mean that potential athletes won't be impressed by fast personal records and seek out like coaches. Therefore, it is important to understand that if this topic comes up when speaking to prospective athletes, you need to educate them on what constitutes real value when hiring a running coach.

Many "fast" coaches rely on their personal race résumé as a marketing tool and also derive their coaching practices based on their own training. This is misguided and akin to a research study with one participant. While one could argue that this is evidence-based, it is hardly statistically significant! Remember, a training methodology that works great for one person may not work great for another. **This is the genesis of this certification – put your trust in science and statistically significant evidence-based findings and you will become a professional and effective coach.**

Holding Out Information

A theory among some sports and fitness professionals holds that if they give athletes too much information, they will fire them as they now know to coach themselves. First, let's address this issue by remembering why you want to become a coach. **As a professional running coach, it is your responsibility to act in the best interest of your athletes to help them reach their goals.** This typically involves explaining the training process and the science behind it – at least in an abbreviated fashion – when applicable and warranted. To hold back training information from athletes for fear that they will leave you is unprofessional at best and unethical at worst. If this is your desired future coaching direction, stop reading and contact UESCA, and we will refund your money, as this is not the type of coach we want to be affiliated with.

It speaks volumes of your overall coaching practice if you feel that acquiring information in the short term is the only reason an athlete hires you. As noted previously in this module, the role of a professional running coach is multifaceted. You are there to provide moral support, tweak a program based on near-limitless variables, devise a realistic and manageable race strategy, and recommend and/or help facilitate third-party interventions such as physical therapists and physicians when needed.

Your ability and willingness to explain the science behind your training programs should legitimize you and your coaching practice and increase the value of your services, not the other way around.

An Extensive Race Resume'

Sure, it's advantageous to have running experience and ideally to have run some races but it's not a prerequisite to being a good or for that matter, a great coach. That said, there is a myth that in order to "qualify" to be a running coach, you must have a lot of races under your belt. While 'a lot' is subjective, many runners that think that unless they have an extensive personal race resume', they aren't qualified to coach others.

To be clear, having run a lot of races is not a prerequisite to be a coach. As noted above, it's always a good thing to have race experience so that you understand the experiences and nuances that your athletes will be dealing with on race day but there is no magic number of races that makes you a credible coach.

The most important thing is to be a student of the sport and each and every training run or race that you do, try to take something away from it that you can learn from.

Everyone Responds the Same Way to a Training Stimuli

As alluded to previously, many coaches coach their athletes the way they coach themselves. While there are likely many reasons for this, one of the more likely reasons is that the coach assumes that if a particular training method works for them, it will work for everyone. This is incorrect.

For example, if this theory were correct, everyone would respond the same way to diets and exercise; thus, weight loss would be easy to figure out and implement. Of course. we know that this is not true.

Therefore, if a coach prescribes the same workouts to their athletes that they personally do and/or prescribes the same activity (s) to all of their athletes – it is safe to say that they are a bad coach that does not understand how the body works.

Factors of Athlete Success

As a coach, it is important to understand key factors that will make your athletes successful. In the following video, Ben Rosario discusses the three factors that he believes to be the most important for athlete success.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Most Common Running Mistakes

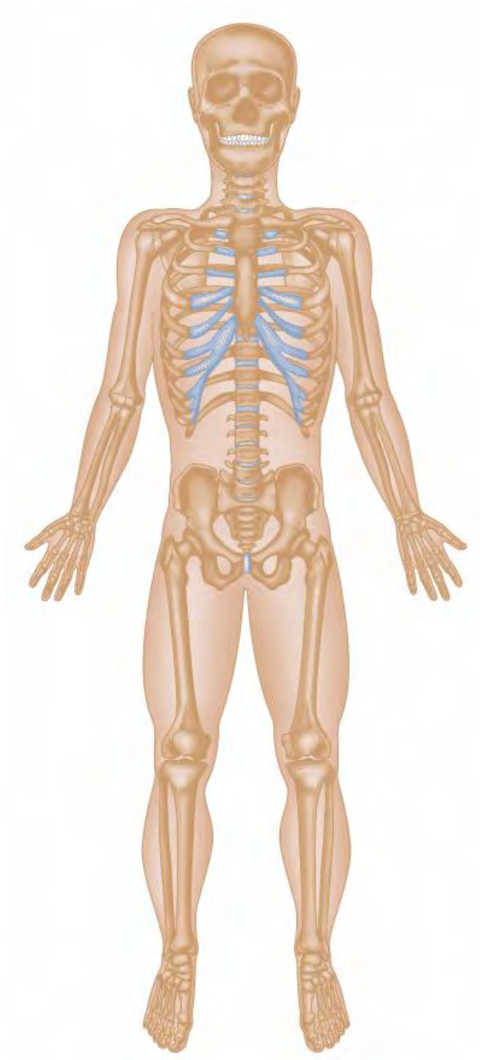
Like factors of success, it is also important as a coach to be aware of common running mistakes so that you can be aware of them and identify them when they occur... or better yet, avoid them in the first place with your athletes. In the following video, Ben Rosario discusses his top three mistakes the runners make.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Summary

- To be a successful running coach, you must be versatile, knowledgeable, and dependable.
- A running coach must be part exercise physiologist, psychologist, leader, motivator, disciplinarian, operations manager, instructor, and educator.
- As a coach, you will need to figure out how each athlete learns best and how to motivate each one.
- Effective communication is critical to the success of a running coach
- A running coach must understand physiology and the psychology of their athletes to be truly effective.
- A running coach does not need to be ‘fast’ or have an extensive personal race resume’.
- It is important to understand that everyone will respond differently to the same training stimuli.

Module 2: Skeletal System

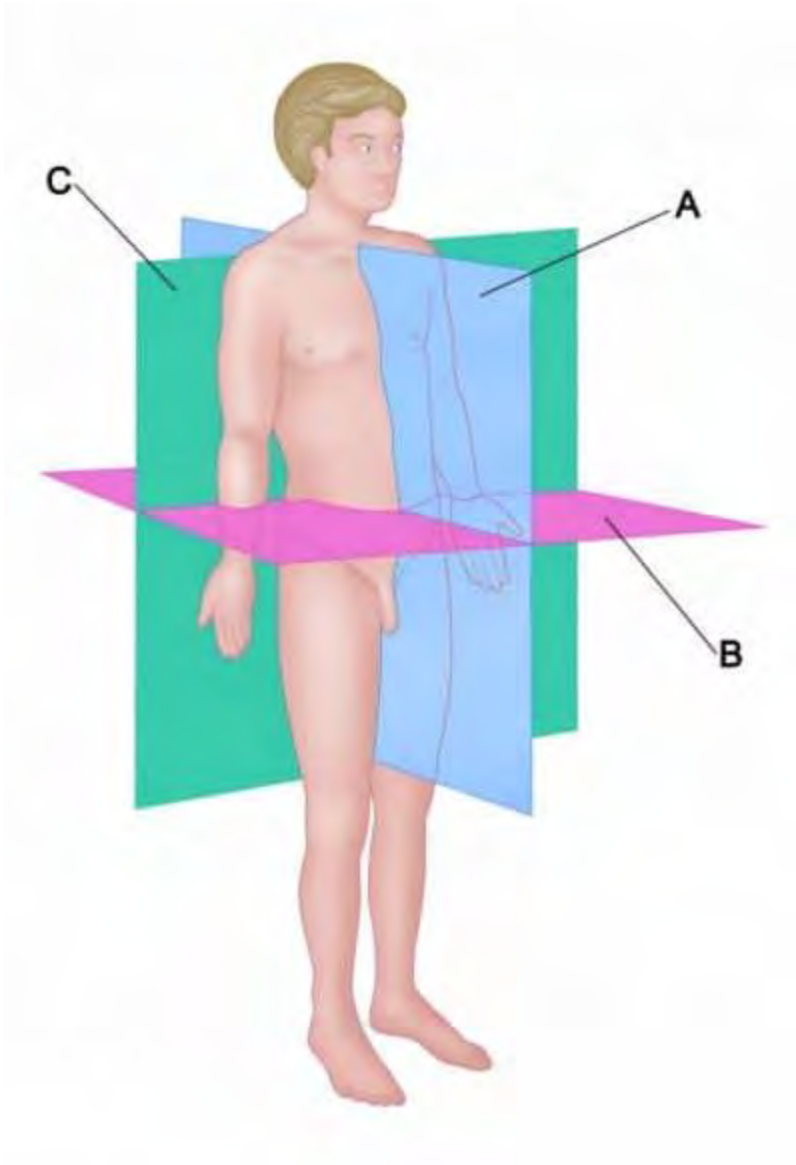


This section discusses the skeletal system as it relates to directional references, identification of bones, spinal injury, and the foot/ankle complex.

- Directional and landmark terminology
- Skeletal system
- Herniated disk
- Regions of the spine
- UESCA definition of *neutral spine*
- Spinal injuries
- Landmarks of pelvis
- Types of joints
- Joints of the foot/ankle and their functions

Directional Terminology

Within this certification program will be references to the planes of the body and directional terminology. There are three primary anatomical reference planes of the body:



A: Sagittal

Divides the body into right and left halves (e.g., the legs move in the sagittal plane when pedaling)

B: Transverse

Divides the body into lower and top halves (e.g., rotating the upper body from left to right)

C: Frontal (Coronal)

Divides the body into front and back halves (e.g., side-to-side or lateral motions such as side steps)

When describing the relationship between body parts, movement of body parts, or the location of an external object to the body, the use of directional terms is necessary. Directional terms are paired, which represent opposing actions or locations. Some of the most common directional terms are:

Anterior: toward the front of the body (e.g., toes are anterior to the heel)

Posterior: Toward the back of the body (e.g., the spine is posterior to the collarbone – or clavicle)

Medial: toward the midline of the body (e.g., the arm swing of a baseball player hitting a ball moves medially through the first half of the range of motion)

Lateral: away from the midline of the body (e.g., a backhand tennis return moves the arm laterally away from the body)

Superficial: toward the surface of the body (e.g., skin is superficial to subcutaneous fat)

Deep: inside the body, away from the surface (e.g., the intestines are deep relative to the skin)

Adduction (movement): toward the midline of the body (e.g., lowering arms to the legs from a starting position of the arms being extended out to the side)

Abduction (movement): away from the midline of the body (e.g., raising arms to horizontal from a starting position by the side of the legs)

Flexion: a movement that decreases the angle between body parts (e.g., bending forward at the waist flexes the abdominal muscles)

Extension: a movement that increases the angle between body parts (e.g., going from a seated position to a standing position extends the knees)

Plantar Flexion: increase angle between lower leg and foot (e.g., women who wear high-heeled shoes are in a constant state of plantar flexion)

Dorsi Flexion: decrease the angle between lower leg and foot (e.g., walking on the heels of the feet results in the feet being dorsiflexed)

Elevation: movement in the superior direction (e.g., when shrugging your shoulders, the shoulder blades – or scapulas – elevate)

Depression: movement in the inferior direction (e.g., when lowering your shoulders, the scapulas depress)

Medial Rotation (internal rotation): rotation toward the midline of the body (e.g., someone who is “knock-kneed” has medial rotation of the upper leg – or femur)

Lateral Rotation (external rotation): rotation away from the midline of the body (e.g., the leg is externally rotated if the feet are turned outward from the body’s midline)

Unilateral: movement occurring on one side of the body (e.g., lifting just the right arm)

Bilateral: movement occurring on both sides of the body (e.g., raising both arms)

Ipsilateral: on the same side as another structure (e.g., when lifting the right arm, it is ipsilateral to the right leg)

Contralateral: on the opposite side of another structure (e.g., when lifting the right arm, it is contralateral to the left leg)

Varus: outward angle of bone or joint (e.g., bowlegged)

Valgus: inward angle of bone or joint (e.g., knock-kneed)

– Varus and valgus often refer to substantial biomechanical abnormalities. However, these terms can also describe normal biomechanical angles, such as the forefoot.

Protraction (anteriorly): moves forward (e.g., shoulders that are rounded to the front of the body are in a state of protraction)

Retraction (posteriorly): moves backward (e.g., squeezing the shoulder blades together results in retraction of the shoulder blades)

Proximal: point of attachment, closer to the center of the body (e.g., the femur is proximal to the tibia)

Distal: away from the point of attachment, farther from the center of the body (e.g., the tibia is distal to the femur)

Prone: facing downward (e.g., when people lie on their stomach, they are in the prone body position)

Supine: facing upward (e.g., when people lie on their back, they are in a supine position)

Inversion: turning inward (e.g., when the sole of the foot is facing medially, the foot is inverted)

Eversion: turning outward (e.g., when the sole of the foot is facing laterally, the foot is everted)

Rotation: circular motion around a fixed point (e.g., turning – or rotating – the torso in the transverse plane while in an upright, standing position)

Circumduction: the combination of abduction, adduction, flexion, and extension (e.g., moving a limb in a circular motion)

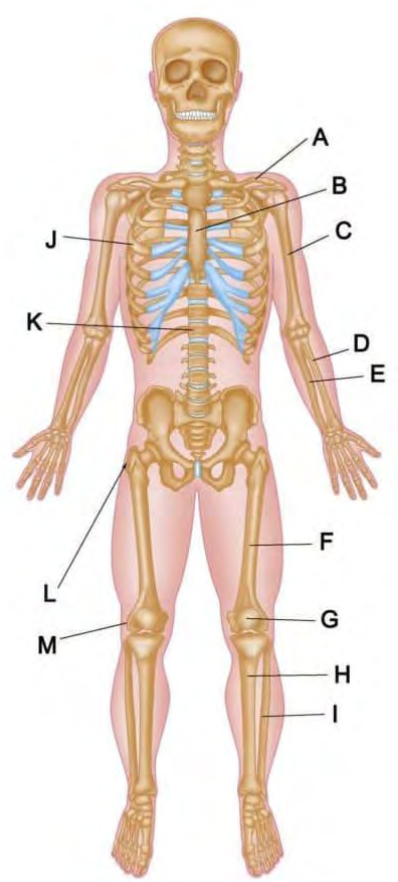
Superior: closer to the head (e.g., the torso is superior to the legs)

Inferior: farther away from the head (e.g., the feet are inferior to the knees)

ROM: Range **O**f **M**otion

Skeletal System

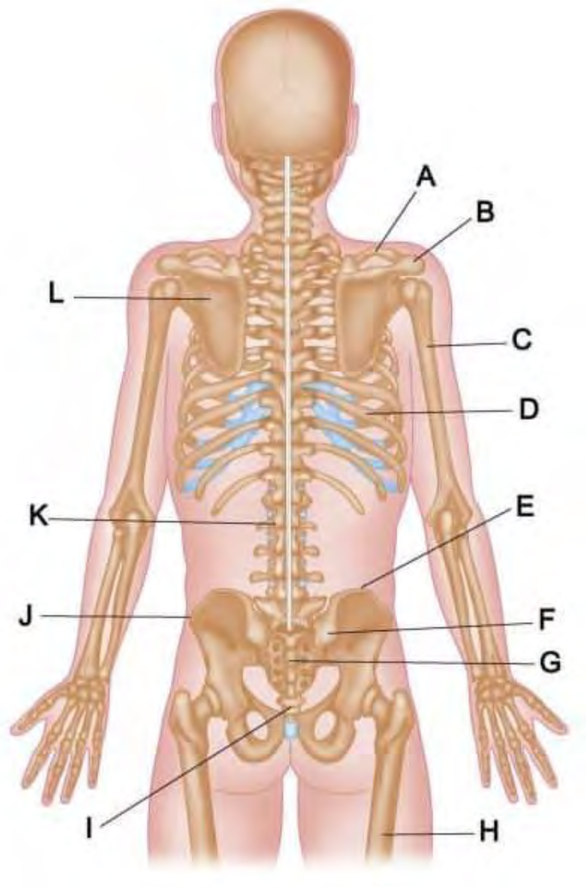
The adult human body comprises 206 bones that constitute the skeletal system. At birth, the body has more than 206 bones. However, some bones fuse to create the adult skeletal system (212).



Anterior View

A. Clavicle
B. Sternum
C. Humerous
D. Radius
E. Ulna
F. Femur
G. Patella
H. Tibia
I. Fibula
J. Rib

K. Vertebrae
L. Greater Trochanter (top of the femur)
M. Lateral Femoral Epicondyle



Posterior View

A. Clavicle
B. Acromion Process
C. Humerous
D. Rib
E. Pelvic Crest
F. Posterior Superior Iliac Spine (PSIS)
G. Sacrum
H. Femur
I. Coccyx
J. Anterior Superior Iliac Spine (ASIS)
K. Vertebrae
L. Scapula

The skeletal system serves as the structural framework and levers of the body. The skeletal system can be broken into two distinct regions: *axial* and *appendicular*.

Axial Skeleton

It forms the central axis that is responsible for providing support to the appendicular skeleton. Comprises the skull, spine, ribs, and sternum (breastbone).

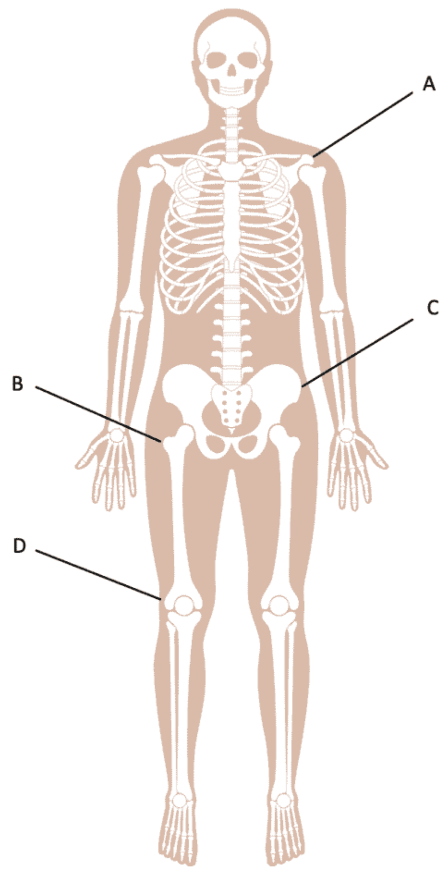
Appendicular Skeleton

It forms the central axis that is responsible for providing support to the appendicular skeleton. Comprises the skull, spine, ribs, and sternum (breastbone).

Represented by the body’s limbs and the bones that attach the limbs to the axial skeleton. Comprises the bones of the legs, arms, scapulas, clavicles, hands, feet, and pelvis.

Landmark Terminology

This section defines skeletal points of the body that are often used for landmarks.



A: Acromion Process: Bony aspect of the shoulder blade (scapula) that extends above the shoulder joint. It is joined to the collarbone (clavicle) via the acromioclavicular joint (AC joint).

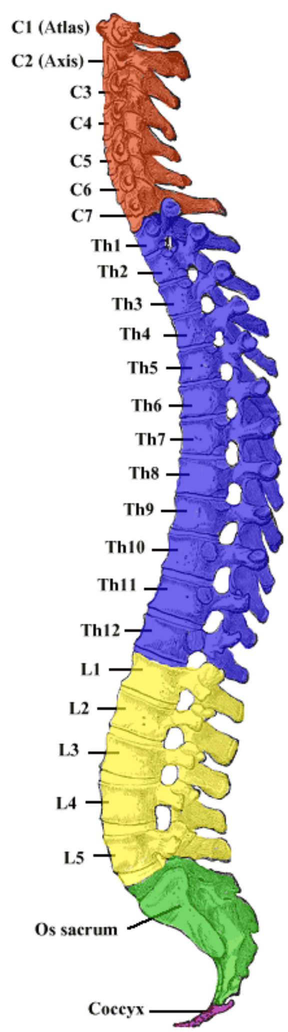
B: Greater Trochanter: Lateral protrusion at the top of the femur

C: Anterior Superior Iliac Spine (ASIS): Noted later in the section regarding the pelvis, the ASIS is the anterior or forward-most part of the pelvis. The ASIS is located by running the fingers down the side of the body until the top of the pelvic bone is felt. Then, trace the pelvic bone inferiorly until the farthest anterior point of the ilium (pelvis) is felt.

D: Lateral Femoral Epicondyle: A small, lateral, protruding aspect at the inferior aspect of the femur. Often used as a landmark, it is located on the lateral side of the knee. This landmark is often referenced as a source of pain from iliotibial band syndrome (IT band syndrome).

Spine

The spine is the critical element to the movement and stabilization of the body. The spine comprises 33 sections – or vertebrae – and between each vertebra is an intervertebral disk made of fibrocartilage. Intervertebral disks are what allow the spine to move and absorb compression. There are five distinct spine regions, and each spinal region assigns a number to individual vertebrae. The letter(s) in front of the number corresponds to the section of the spine (ex: T=thoracic). The numbers correspond to specific vertebrae, with the lowest number (i.e., 1) being the superior vertebrae in the section.



Cervical: (C1–C7) – Red
Thoracic: (T1–T12) – Blue

Lumbar: (L1–L5) – Yellow
Sacral: (S1–S5) – Green
Coccyx: (Co1–Co4) – Purple

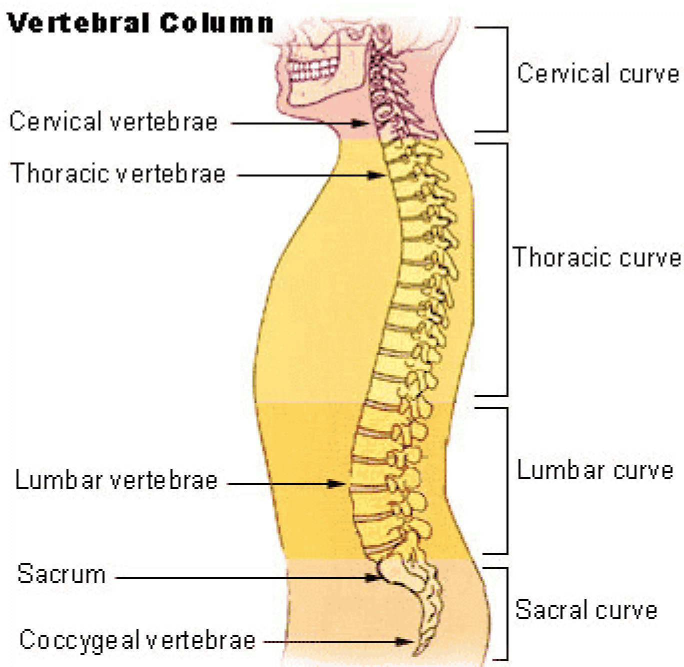
There are two primary types of curves in the spine:

1. **Kyphotic**: refers to an anterior curve of the spine
2. **Lordotic**: refers to a posterior curve of the spine

These terms are often used to denote extreme spinal curvatures. More specifically, these conditions are termed:

1. **Kyphosis**: an extreme anterior curve in the spine (i.e., hunchback)
2. **Lordosis**: an extreme posterior curve in the spine (i.e., sway back)

Below are the locations of the four spinal curvatures along with their respective curve type:



Cervical Curvature (Lordotic)

Thoracic Curvature (Kyphotic)

Lumbar Curvature (Lordotic)

Sacral Curvature (Kyphotic)

Neutral Spine Position

Neutral spine is a concept that has existed for a long time, and like other areas of exercise science, it has acquired multiple definitions. While many people refer to neutral spine as the perfect alignment of the spine, the reality is that neutral spine is different for everyone and is mainly genetic and activity-based. For example, someone who slouches at a desk all day will have a very different neutral spine position than a professional runner.

For this certification, the definition of neutral spine will be based on a neuromuscular approach:

Neutral spine is the position of the spine in which minimal neuromuscular activity is required to maintain a standing, relaxed posture.

The phrase “find your neutral spine” generally refers to the action taken to establish the ideal spinal position relative to the spine’s natural curves. This typically involves a predetermined alignment based on a lateral perspective from the head to the toes. Your athlete may or may not be able to establish or maintain this position.

Regarding neutral spine, the goal of a coach is to work with athletes to help them establish and maintain a spinal position that improves posture, stability, and movement patterns, as well as maintains proper muscle length-tension relationships. As noted in the definition, the goal is for the athlete to be able to maintain these positions with minimal neuromuscular activity.

Lumbar Spine

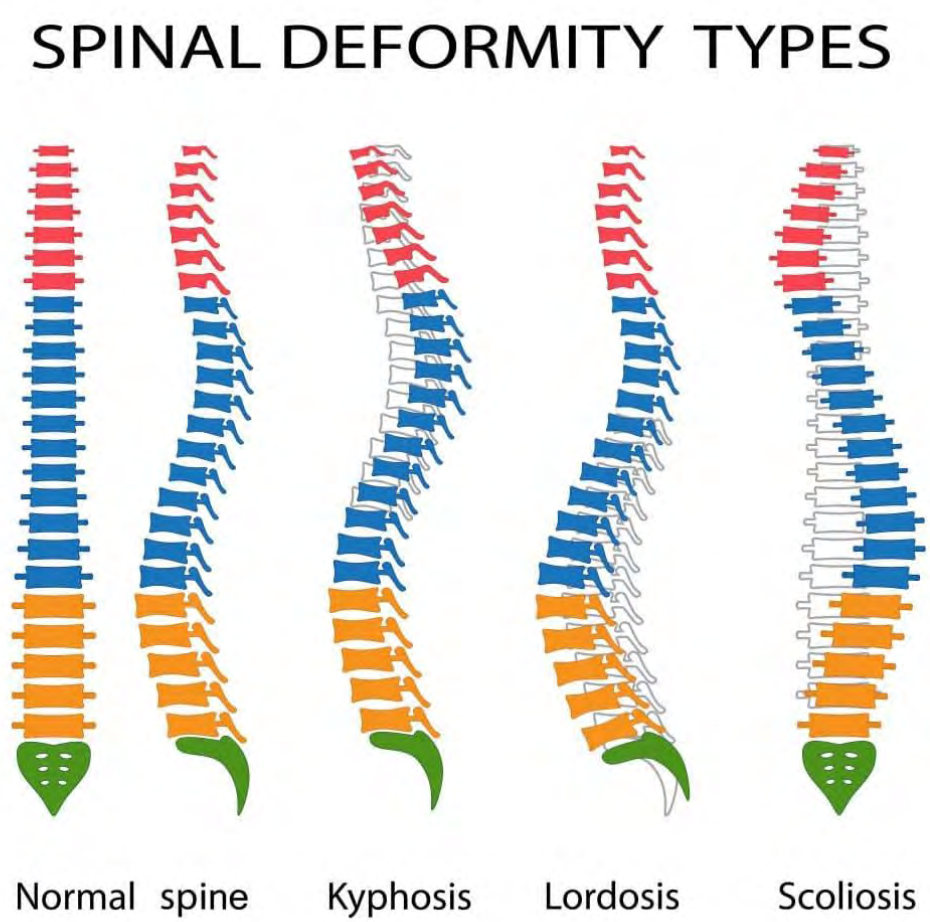
Often referred to as the *low back*, this section of the spine is emphasized because of the important role it plays in stabilization and is the region of the spine where pain and injury are most often localized. While the cervical spine supports the weight of the head and the thoracic spine gains rigidity from the rib cage, the lumbar spine largely depends on muscular support to provide stabilization and mobility both superiorly and inferiorly. The lumbar spine is located between two stable structures, the ribs and the pelvis.

Inefficient muscular stabilization of the lumbar spine is associated with lower back pain.

According to the *National Academy of Sports Medicine,* **over $85 billion is spent annually to alleviate back pain** (predominately lower back pain) in the US. Over 80 percent of Americans will experience back pain within their lifetime (213).

Spinal Abnormalities

While it is natural to have spinal curvatures as described above, they are considered abnormal if the curves are excessive. Any factors such as osteoporosis, disease, accidents, obesity, pregnancy, or poor posture for extended periods can cause abnormal spinal curvature.



There are three classifications of spinal abnormalities:

1. **Lordosis**: excessive lumbar spinal curvature (hyperextension)
2. **Kyphosis**: excessive thoracic spinal curvature (hyperflexion)
3. **Scoliosis**: spinal curvature in the frontal plane (lateral curvature)

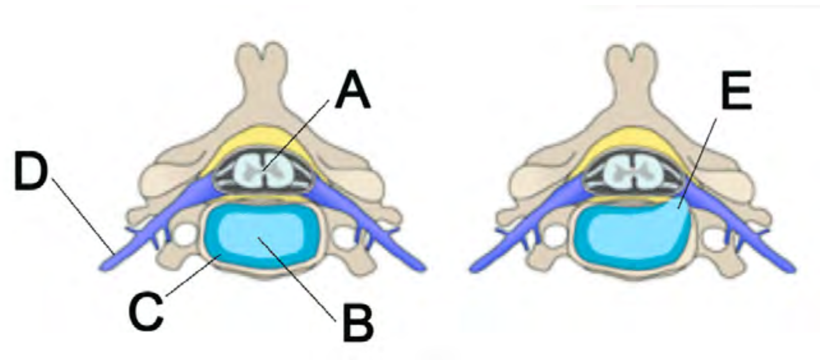
The two classifications of scoliosis are structural and functional. Functional scoliosis originates elsewhere in the body, such as a leg length discrepancy, whereas structural scoliosis originates because of a growth abnormality in the spine. Structural scoliosis often presents with rotation or twisting of the spine and lateral curvature. Visible signs of scoliosis are uneven hips and shoulders.

If you suspect your athlete has any of these three spinal abnormalities, you should refer the person to a physical therapist or physician for an evaluation.

Injury

A common spinal injury is a **herniated disk,** also known as a ruptured disk, which occurs when an intervertebral disk is injured. Each disk is made up of two main components. The cartilage aspect of the disk is circular and is called the annulus fibrosus. The disk's center is composed of a gelatinous substance called the nucleus pulposus. A herniated disk is caused when excessive stress placed on a disk causes the annulus fibrosis to crack, and some of the nucleus pulposus leaks out (119). The leaked nucleus pulposus can put pressure on the spinal cord or spinal nerves. This pressure causes pain. Physical therapy and/or surgery are often prescribed to individuals with a herniated disk.

In the image below, the disk on the left is normal, and the disk on the right exhibits a herniation as it shows the nucleus pulposus leaking out and putting pressure on the spinal nerve.



A. Spinal Canal
B. Nucleus Pulposus
C. Annulus Fibrosis
D. Spinal Nerve
E. Herniation of Disk

A herniated disk is commonly referred to as a “slipped disk.” Technically, this is incorrect, as intervertebral disks cannot slip because they are attached to vertebrae (4).

Following are other causes of pain that involve the spine:

Spondylolisthesis

A slipped vertebra in relation to the one adjacent to it. Most commonly, the vertebra moves anteriorly. However, it can move laterally and posteriorly (355).

Spinal Stenosis

Narrowing of the spinal canal because of degeneration, instability, or a lesion. The *spinal canal* is a vertical hole within the vertebrae that allows the spinal cord to pass through it (354).

Radiculopathy

Pain in the extremities (arms, hand, feet) originates from spinal nerve compression.

- **Sciatica:** A type of radicular pain. Presents with pain down the leg’s posterior and/or lateral aspect. This is more of a general term than a specific condition. Sciatica can manifest because of compression (i.e., herniated disk), spondylolisthesis, or muscle contraction.

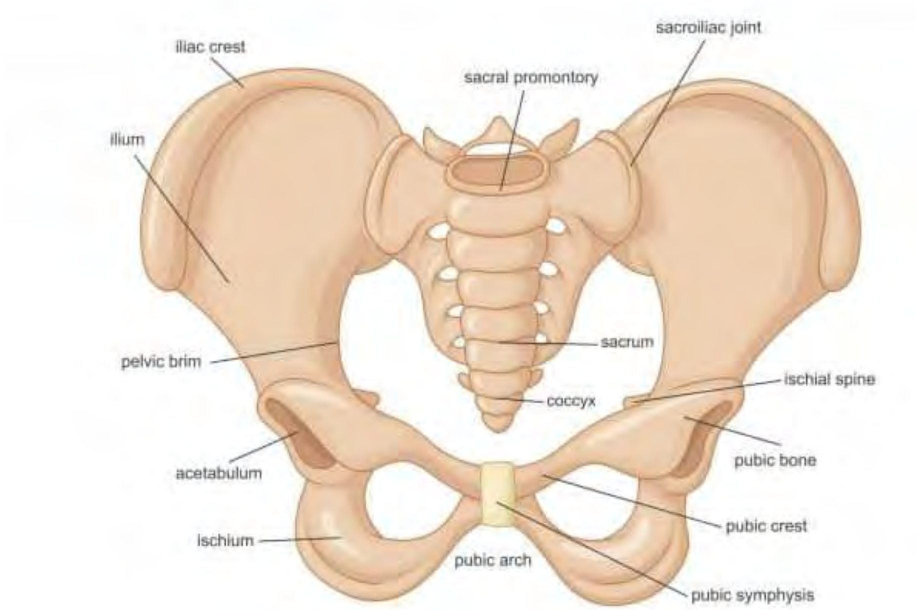
If an athlete experience pain in either the back or extremities that is not simply resolved by a change in body position or rest, they must be referred to a physician or physical therapist.

Pelvis

The pelvis is a fundamental structure in the human body. The pelvis is commonly referred to as the hips and separates the lower body from the torso. Because of its location, movement of the pelvis affects both the upper and lower body. From a skeletal perspective, the pelvis represents the spine's base and the location where the femurs insert.

Mechanically speaking, the primary purpose of the pelvis is to support the upper body and transfer the energy from the legs to the spine and upper body. The pelvis protects the reproductive and digestive organs of the lower torso.

Areas of the Pelvis



Iliac Crest

The most superior aspect of the ilium is called the *iliac crest,* which is commonly used as an anatomical landmark.

Ilium

Located on either side of the sacrum, the pelvis's most prominent bone(s) is the ilium. The primary purpose of the ilium is to protect internal organs.

Ischium

Often referred to as the *sits bones*, the ischium bones are located where one's body weight is distributed in a seated position.

Coccyx

Commonly referred to as the *tailbone*, the coccyx is the lowest part of the vertebral column.

Pubic Symphysis

Connects the two sides of the lower pelvis. This joint is a semi-movable joint and is connected by ligaments and cartilage. The pubic symphysis expands as the distance between the legs expands. Pain in the pubic symphysis can occur when the joint does not return to the normal position (i.e., it is misaligned).

Acetabulum

The location of the pelvis that the head of the femur sits in.

Anterior Superior Iliac Spine (ASIS)

The ASIS is the forward-most part of the pelvis. The ASIS is located by running the fingers down the side of the body until the top of the pelvic bone is felt. Then, trace the pelvic bone inferiorly until the farthest anterior point of the ilium is felt.

Sacroiliac Joint

Often termed the *SI joint.* The SI joint is the connection between the sacrum and the ilium, located on both sides of the spine.

Strong ligaments stabilize the SI joint. The purpose of the joint is to absorb shock. The joint “locks” during walking and running to provide a solid base of support during the foot push-off aspect of the gait cycle (245). Inflammation of the SI joint can be caused by too much or too little motion and typically presents with pain in the SI joint and low back region (244).

Sacrum

It comprises the base of the spine and is made up of five fused vertebrae and is located between the lowest lumbar vertebrae (L5) and the coccyx.

Joints

There are three primary classifications of joints within the body:

1. **Fibrous** (e.g., joints of the skull)
2. **Cartilaginous** (e.g., intervertebral disks)
3. **Synovial** (e.g., shoulder joint)

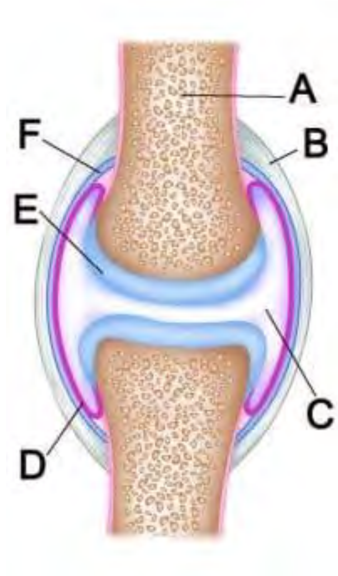
Joints can be further classified by the amount of movement. The three functional classifications of joint movement are:

1. **Immoveable:** fibrous joints
2. **Slightly Moveable:** cartilaginous joints
3. **Freely Moveable:** synovial joints

Synovial joints are by far the most prevalent in the body of these three classifications of joints.

Synovial joints are made up of the following components:

1. **Synovial fluid**: a collagen-like substance that surrounds a joint
2. **Synovial membrane** or **synovial cavity**: the inner layer of a joint capsule that secretes synovial fluid – a lubricating liquid
3. **Cartilage**: pads the ends of the bones
4. **Joint capsule**: fibrous sac that contains the synovial membrane/fluid



A. Bone
B. Ligament
C. Synovial Fluid
D. Synovial Membrane
E. Cartilage
F. Joint Capsule

‘Cracking’ Joints

Most associated with the knuckles, joints can make a cracking or popping noise when moved into particular positions. This issue is noted because athletes often ask if cracking joints mean they are injured or if there is something wrong with them. This noise does not signify that an injury has occurred. As noted previously, a synovial joint consists of a fluid-filled cavity. When the bones of a joint are pulled away from each other (e.g., knuckles cracking), the synovial cavity space increases in size.

Old Theory

It was previously thought that the cracking was caused by decreased synovial fluid pressure within the cavity. This decrease in fluid pressure was thought to cause bubbles to form within the cavity through a process called cavitation. Therefore, if the bones of the joint continued to be pulled away from each other, the pressure in the cavity would drop low enough that the bubbles pop, thus causing the audible cracking noise (246).

Current Theory

Research by Kawchuk et al. found that the cracking noise emitted by joints was not due to a bubble popping within the joint but rather the rapid creation of a temporary cavity within the joint (termed tribonucleation), thus debunking the prior theory (770). This new theory was discovered by viewing a real-time Cine MRI of the joint separating and cracking. Regardless of the actual cause, there seems to be no danger in cracking joints.

Foot and Ankle Complex

The foot and ankle complex is a very busy area. This complex has 26 bones, 33 joints, and over 100 muscles, tendons, and ligaments.

Skeletal Structure

– **Leg Bones:** tibia, fibula
– **Ankle Bone:** talus
– **Heel Bone:** calcaneus
– **Midfoot Bones (foot arch):** cuboid, navicular, cuneiform (3)
– **Forefoot Bones:** metatarsals, phalanges (toes)

The foot and ankle complex plays a major role in running. In spite of this, it is often an overlooked and misunderstood part of the body, specifically in regard to its mechanics.

While not noted in the image above, there are typically two small, pea-sized bones (sesamoid) located underneath the first metatarsal bone (behind the big toe). These bones are embedded into a tendon and functionally are used to provide leverage when walking and running. Because of the stress placed on these bones when running, they can break and/or the surrounding tendon can become irritated, called **sesamoiditis**. Sesamoiditis is discussed in greater detail in the *Injury and Illness* module.

Foot / Ankle Joints

Two primary joints function within the ankle.

Talocrural Joint

Connects the tibia and fibula with the talus bone in the foot. This is a hinge joint and provides dorsi flexion and plantar flexion.

Subtalar Joint

It connects the talus bone and calcaneus bone. This is a gliding joint and allows inversion and eversion of the ankle to occur.
The Achilles tendon covers the subtalar joint when viewing the leg from a posterior aspect.

Two primary joints in the foot affect gait and foot stability.

Midtarsal (Transverse Tarsal and Talocalcaneonavicular)

This joint spans the arch of the foot and primarily controls inversion and eversion of the foot (see below image). This joint ranges from flexible to rigid based on what aspect of the support and drive phase the foot is in. As the foot moves under the body's center of gravity and prepares to push off, the joint locks up and becomes rigid to provide a stable base of support to push off from (214).

Tarsometatarsal

Like the midtarsal joint, this joint also spans across the width of the foot arch. However, it is located posterior to the metatarsals, effectively at the junction between the midfoot and forefoot. This joint has more stability and less range of mobility than the midtarsal joint. The below image illustrates the bottom of the foot concerning the midtarsal and tarsometatarsal joints.

The foot and ankle joints act together to allow and restrict motion so that proper mechanics can occur.

Genetic Factors

It is common knowledge that weight-bearing activities such as running and weightlifting are beneficial for increasing bone density. The more stress put on a bone(s), the stronger it will become. However, bone strength has also been found to have a genetic component. A 2010 study by Libby Cowgill found that differences in bone strength are evident as early as one year of age (491). **So, while bone can become denser through training, there is also a large genetic factor.**

In the book *The Sports Gene*, the author cites research by Francis Holway that compares the human skeleton to that of a bookcase (466). The stronger and heavier a bookcase is, the more books it can carry. However, Holway's research found that there is a limit to how much muscle weight a skeleton can carry. Specifically, 2.2 pounds of bone can support a maximum of 11 pounds of muscle, or a five-to-one ratio of muscle to bone. Once this ratio is reached, any additional weight gain is typically fat versus muscle (466).

Summary

- The body is divided into three planes: *sagittal, frontal,* and *transverse.*
- The spine is made up of five sections: *cervical, thoracic, lumbar, sacral,* and *coccyx.*
- The skeletal system can be broken down into two distinct regions: *axial* and *appendicular*.
- The spine is made up of 33 vertebrae.
- "Cracking joints" is not indicative of injury.
- Kyphosis refers to an anterior curve of the spine.
- Lordosis refers to a posterior curve of the spine.
- *Neutral spine* is the position of the spine in which minimal neuromuscular activity is required to maintain a standing, relaxed posture.
- The lower back (i.e., lumbar spine) is the area of the spine most commonly associated with pain.
- Synovial joints are the most common type of joint in the body.
- A herniated disk causes the nucleus pulposus to leak out and results in pressure on the spinal nerve.
- Sciatica typically causes pain to radiate down the leg.
- The primary purpose of the pelvis is to support the upper body and to transfer the energy from the legs to the spine and upper body.
- Weight-bearing activities increase bone density.

- There is a genetic component to bone strength.
- The three primary classifications of joints are:
    - Fibrous
    - Cartilaginous
    - Synovial (most prevalent joint type)
- Four major joints in the foot/ankle affect gait and foot stability:
    - Talocrural (ankle)
    - Subtalar (ankle)
    - Midtarsal (foot)
    - Tarsometatarsal (foot)

Module 3: Muscular System

This module discusses muscle physiology. As anatomy and physiology is a very broad, all-encompassing subject, it is sometimes difficult to know where to start, especially when discussing running-specific training methodologies.

Without a basic understanding of muscle physiology, there is no foundation on which to build a solid training program. To coach athletes of any sport, the ability to apply sports science to form an appropriate training regimen is an absolute requirement. It is not enough to rely on one's past sport-specific experience to coach athletes.

Experience is a great asset. However, integration with a working knowledge of sports science is required to be a genuinely effective and resourceful coach.

Upon completion of this module, you should understand the following areas:

• Skeletal muscle structure, function, and terminology
• Contractile properties of a muscle

• Muscle origin and insertion
• Muscle fiber types
• Muscle fiber recruitment
• Muscle hypertrophy
• Nerve innervation of muscles
• Types of connective tissue
• Role of fascia
• Primary muscles and their actions (as noted in body images)
• Inner and outer core musculature
• Kinetic linking
• "Warming up" muscles
• Delayed Onset Muscle Soreness (DOMS)/Muscle Burn

Muscle Physiology

Terminology

Concentric muscle contraction: shortening of a muscle
– Flexing the elbow concentrically contracts the bicep muscle.

Eccentric muscle contraction: lengthening of a muscle
– Extending the elbow eccentrically contracts the bicep muscle.

Isometric muscle contraction: A muscle contraction occurs, but no change in muscle length results.
– Holding a weight in your hand with your elbow at 90 degrees with no motion is an isometric contraction.

Isotonic muscle contraction: When an isotonic muscle contraction occurs, the muscle tension remains the same, but the length of the muscle changes. An isotonic contraction can be either concentric or eccentric.
– A push-up results from isotonic chest muscle (pectoralis major) contractions.

Hypertrophy (hi-per-tro-fee): increase in muscle size.
– A bodybuilder's muscles are the result of hypertrophy.

Atrophy: decrease in muscle size.
– Individuals on bed rest for long periods often experience muscle atrophy.

Agonist: muscle or muscles primarily responsible for movement around a joint; also called a *prime mover.*
– The quadriceps muscles are the agonist muscles during leg extension.

Antagonist: muscle or muscles that act in opposition to the agonist's muscles.
– The triceps are the antagonist to the biceps.

The three types of muscle tissue in the body are:

1. **Cardiac muscle:** located only in the heart and is solely responsible for contracting/relaxing the heart to pump blood to the lungs and the rest of the body.
2. **Smooth muscle:** involuntary muscles that surround organs to protect against outside forces.
3. **Skeletal muscle:** any muscle that is neither cardiac nor smooth. These muscles provide movement of the body and account for 35–45 percent of a human's body weight (5). There are approximately 640 skeletal muscles in the body, most of which are grouped in pairs.

There are two types of muscle contractions: ***voluntary*** and ***involuntary***.

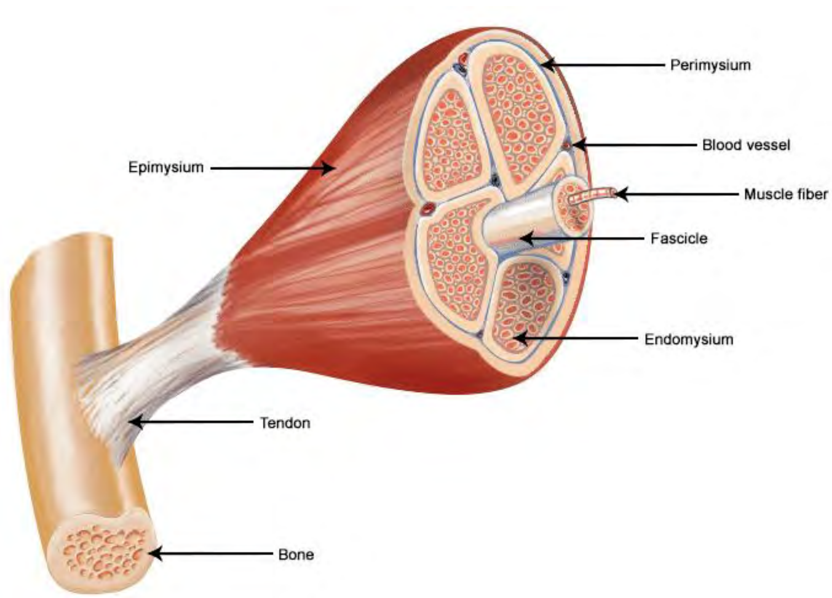
1. **Voluntary Muscle Contractions:** These are muscle contractions controlled by the central nervous system (CNS), and the origin of the contraction is a conscious thought by the brain. An example of this is a bicep curl.
2. **Involuntary Muscle Contraction:** This muscle contraction is controlled by the autonomic nervous system (ANS) and occurs without conscious thought. An example of this is the contraction of the heart.

Skeletal Muscle Structure

The skeletal muscle structure can be considered a thick cable on a suspension bridge. If you examined the cable that suspends a bridge, you would find that the large cable is made up of many smaller cables that, in turn, are made up of even smaller cables. At the smallest level, each wire strand is woven to make up a single cable that might be less than a millimeter thick. However, after bundling these wires together to create a cable, bundling those cables together to create a larger cable, and repeating this process many times, the resulting cable can support the road and the traffic.

Likewise, skeletal muscle is not a singular “cable” but a series of smaller cables that make up the whole muscle.

Below is an image that denotes the layers that make up a muscle, starting at the muscle fiber layer (17).



Contractile Properties

To understand how a muscle contracts, it is helpful to break down the composition of a *muscle fiber* (i.e., muscle cell) via a hierarchy.

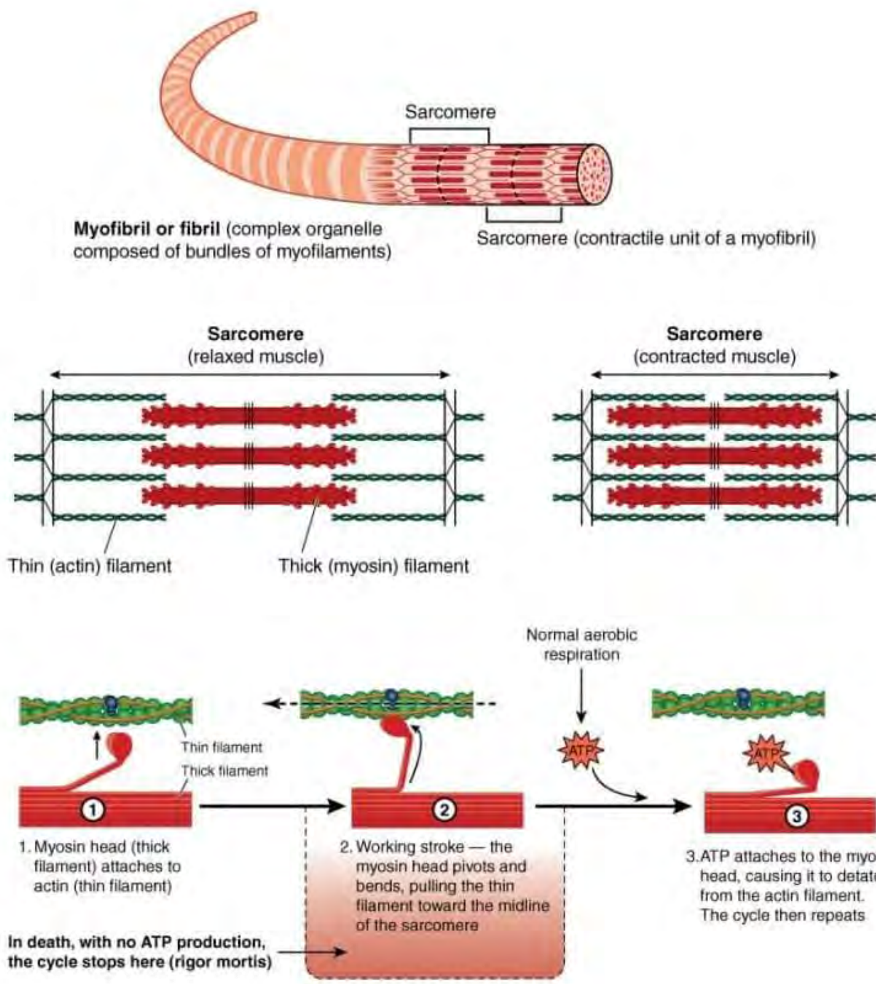
- A muscle fiber is made up of ***myofibrils***
- Myofibrils are made up of ***sarcomeres***
- Sarcomeres are made up of ***actin*** and ***myosin***

Myofibril: Contractile fibers that run along the length of a muscle fiber. The ability of a myofibril to contract is due to the sarcomere. When myofibrils contract, the muscle contracts.

Research in muscle fatigue suggests that the capacity of a myofibril to produce force may be affected by fatigue (556, 557).

Sarcomere: A sarcomere is a unit within the myofibril responsible for muscle contraction. In essence, it is the functional unit of the muscle fiber. A sarcomere contains three protein types that have different roles. These roles and associated proteins are:

1. **Contractile:** *Actin* (thin filament) and *myosin* (thick filament) produce a contraction and shorten the sarcomere.
2. **Regulatory:** Made up of *tropomyosin* and *troponin* that allow or disallow the contractile process to take place.
3. **Structural:** Made up of *titin* and *myomesin* that keep the contractile filaments (*actin* and *myosin*) aligned (18).



The shortening of the sarcomere, and therefore the muscle contraction, is caused by the **sliding filament model** (14). This model states that within the sarcomere, actin and myosin are alternately layered on top of one another. Myosin ratchets along the actin and causes the actin to shorten.

This compression causes the sarcomere to shorten. Each sarcomere is connected to the next sarcomere by a *Z line*, or *Z-disk*. When a muscle fiber contracts, all of the sarcomeres contract at the same time. This simultaneous contraction produces force at each end of the muscle fiber, causing a muscle contraction.

The speed and force of a muscle contraction relate to how much power is expended by a muscle. **The greater the rate of a muscle contraction and/or the more muscle contractions, the greater the amount of ATP is used.** This correlates to an increase in power being produced (705).

If the actin and myosin are too far apart or too close together, the contractile properties of the muscle will be significantly diminished. This is termed the ***muscle length/tension relationship*** and is described later in this module.

Myonuclei And Satellite Cells

The nucleus of a muscle fiber is called the myonuclei (481). There are multiple myonuclei per muscle fiber. It is theorized that the gain and loss of myonuclei within a muscle fiber are directly related to muscle fiber hypertrophy and atrophy, respectively (482). This directly relates to the theory that myonuclei are responsible for regulating protein synthesis in the surrounding area (482, 483).

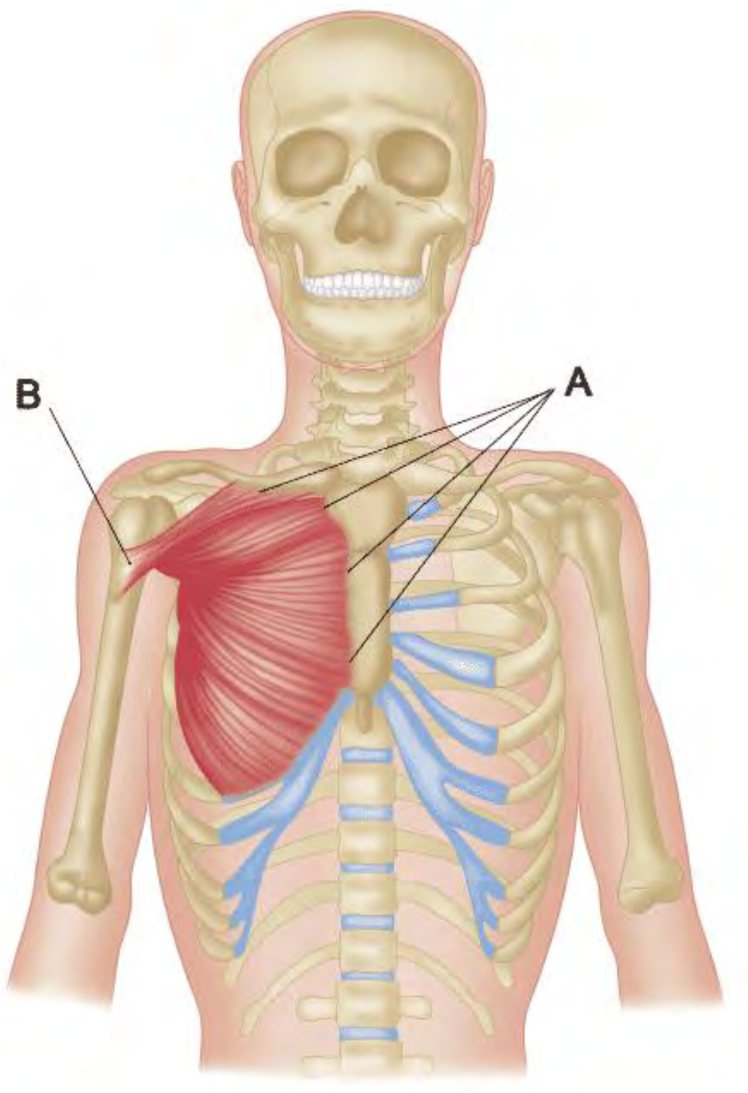
Satellite cells, or myosatellite cells, are found in muscle. These cells typically lie dormant in the muscle until needed. **The primary role of satellite cells is to assist in muscle hypertrophy, recovery from an injured muscle, or maintenance of muscle mass** (484, 485). When activated, these cells proliferate and attach to existing muscle fibers or even fuse to create new muscle fibers (486).

Simply put, exercise that overloads a muscle (even stretching) causes muscle damage. As a result, satellite cells are activated to assist muscle regeneration (481).

Muscle: Origin and Insertion

Muscles are anchored to bones in two places by *tendons*. These attachments to bone have two distinct connections. The ***origin*** is where the tendon anchors to a non-moving bone. The other attachment point is called the ***insertion***, where the tendon attaches to the bone that moves.

Let us take, for example, the *pectoralis major* (below image). The origin is the *clavicle* (collarbone) and *sternum* (breastbone), and the insertion is the superior aspect of the *humerus* (top of the arm bone closest to the shoulder).



A = Origin
B = Insertion

So, what does this tell us? When looking at the direction of the muscle fibers and the location of the insertion and origin, it tells us that the muscle acts to adduct the humerus toward the midline of the body. Therefore, an exercise such as bringing the arms together in front of the chest (chest fly) works the pectoralis major in the range of motion specific to the direction of the muscle fibers and origin/insertion.

By understanding the insertion and origin of muscles, you will understand what specific action the muscles perform and what exercises would be most effective for targeting a particular muscle or muscles.

Muscle Fiber Types

Not all muscle fibers are created equal, or it might be more accurate to say that not all muscle fibers are created the same. One does not have to be overly observant to realize that the body type and, therefore, the muscle size of a competitive bodybuilder does not look like that of a runner. Why is this the case? Obviously, the training methodology of a bodybuilder versus that of a runner is quite different, but is that enough to make such a profound difference in body type? Many reasons could play into the difference in physical appearance between these two athletes. Some of these reasons are:

- Diet
- Strength training program
- Cardiovascular training program

While the athletic performance and muscle size of an individual are certainly affected by the previously cited variables, the absolute performance and muscle size of an individual are primarily dictated by genetics.

There are two primary classifications of muscle fibers in the body, fast-twitch (Type II-A and II-B) and slow-twitch (Type I) (118).

Type I (Oxidative)

Type I fibers are characterized by their ability to fire repeatedly with minimal fatigue. They fire more slowly and with less force than Type II fibers and, as a result, are better geared toward endurance events, such as running races.

Type II-A (Fast Oxidative Glycolytic)

Type II-A is an intermediate fast-twitch fiber. This fiber type is a hybrid between Type I and Type II, meaning its contractile properties are faster than Type I but not quite as fast as Type II-B. Because of this, Type II-A fibers have midrange endurance capabilities. As such, this muscle fiber type could also be termed ***Intermediate Twitch***.

Type II-B (Fast Glycolytic)

Type II-B fibers are the classic fast-twitch fibers, which means their contractile properties are the fastest of all fiber types. Still, they cannot fire repeatedly without fatiguing. As a result, Type II-B fibers are best suited for short, explosive efforts.

The chart below demonstrates the characteristics of the muscle fiber types across a wide range of criteria (118).

PROPERTIES	SLOW TWITCH (Type I)	FAST/INTERMEDIATE TWITCH (Type II-A)	FAST TWITCH (Type II-B)
Size (Diameter)	Small	Medium	Large
ATP Synthesis (Energy Source)	Aerobic (Triglycerides)	Aerobic/Anaerobic (Creatine Phospate/Glycogen)	Anaerobic (Glycogen)
Mitochondrial Density	High	Medium	Low
Capillary Density	High	Medium	Low
Endurance	High	Medium	Low
Force of Contraction	Low	High	High
Speed of Contraction	Low	High	High

While muscle size can be increased with strength training, the degree of hypertrophy is primarily determined by the fiber type.

Some muscles within the body are composed of the same fiber type, regardless of genetics. This is mainly dependent on the function of the muscle or muscles. For example, the soleus (calf muscle) is predominately made up of slow-twitch fibers (19). The reason for this is that the soleus is responsible for the ankle's stability; therefore, it is constantly active and must have exceptional endurance properties.

Fast-twitch muscles produce more lactate than slow-twitch muscles. Therefore, the intensity of exercise and the fiber type ratio of an individual influence their lactate threshold (583)

Myostatin

While the type of muscle fiber plays a significant role in determining the size of the fiber, a protein in the body called **myostatin** regulates how large a muscle fiber can become. This occurs by the myostatin regulating the number of resources the muscles consume to grow in size (417). Individuals or animals with mutations to the myostatin protein have abnormally large muscles due to the lack of muscle size regulation (417).

Muscle Fiber Recruitment

To understand how muscles respond to their demands, it is essential to know how muscle fibers are recruited – or engaged. Muscle fiber recruitment results from a neuromuscular reaction in which the central nervous system sends an electrochemical impulse to a particular muscle or muscles, at which point a contraction occurs. As noted previously, there are three types of muscle fiber types: **Type I** (oxidative), **Type II-A** (fast oxidative glycolytic), and **Type II-B** (fast glycolytic).

Type I ➡ Type IIA ➡ Type IIB

Muscle Recruitment Hierarchy

When the intensity of an activity progresses from low or moderate to high, there is a hierarchy as to which types of muscle fibers are recruited. This is referred to as **Henneman's Size Principle** (356). First, Type I fibers are recruited, and as the intensity level rises, Type II-A and then Type II-B are recruited. Suppose the intensity is increased over a very short period and to a very high level (i.e., explosive-type movements). In that case, the central nervous system will recruit all three muscle types at the same time.

A muscle fiber cannot partially contract, or more specifically, a muscle fiber cannot contract with varying force depending on the effort required. **A muscle fiber contracts either 100 percent or not at all**. Therefore, the degree of muscle contraction is based on the number of muscle fibers recruited to perform an action, not the degree of contraction per muscle fiber.

Can Muscles Change Their Type?

Whether or not muscle fibers can change their type has long been debated, and to this day, the debate has not been settled. In rare cases, muscles have demonstrated the ability to convert from Type I to Type II when the muscle was subjected to substantial deconditioning, in the case of injury, for example (21). **There has not been much evidence to support the theory that muscles can change from Type I to Type II through changes in training routines** (289). However, the lack of verification might be due to insufficient research in this area.

While not attributable to a change in muscle fiber type, some research has shown that through specific types of training, **the contraction speed of Type I fibers can increase** (290, 291).

Regardless of a muscle fiber's ability to change type, people can continually improve their performance even if it is at odds with their genetic muscle fiber type composition. For example, let us say an athlete has fast-twitch muscle fibers predominately and wants to compete in an endurance event such as a 100-mile (161 km). Through the development and application of the proper training regimen, they can become very proficient despite their genetic makeup to the extent that the individual can perform quite well in distance running events.

Muscle Hyperplasia vs. Muscular Hypertrophy

Hypertrophy refers to the increase in the size of a muscle fiber, whereas hyperplasia refers to the increase in the number of muscle fibers. While muscle hypertrophy has long been established as the method by which muscles increase in size, skeletal muscle hyperplasia is a much-debated topic. Skeletal muscle hyperplasia has been shown to exist in mammals and human cardiac muscle (415).

However, evidence of skeletal muscle hyperplasia in humans is still largely inconclusive. A theorized reason for this is that the existence of skeletal muscle hyperplasia is impossible to detect via a standard muscle biopsy (416). Therefore, more research is required to determine whether human skeletal muscle hyperplasia exists.

No Pain, No Gain?

It is widely accepted that muscles are damaged during intense exercise bouts and are rebuilt (i.e., remodeled, restructured) during the recovery phase, thus allowing for strength gains. However, detectable damage (i.e., plasma levels, pain, muscle soreness) to muscles is not required for strength gains. In other words, **just because an exercise does not cause pain does not mean that positive results aren't realized regarding strength/performance.**

A 2010 study by Flann et al. put two groups (trained/untrained) through training programs that elicited muscle damage in one group (untrained) but not in the other (trained) (683). It was found that both groups experienced similar increases in muscle rebuilding, strength gains, and muscle hypertrophy. Therefore, **muscle damage is not required for muscle rebuilding and strength increases.**

Relationship Between Muscle Fiber Type and Fat Metabolism

Metabolism, or metabolic rate, is typically associated with how efficient one's metabolism is. While many variables likely determine one's metabolism, several studies have demonstrated a link between muscle fiber type and fat metabolism (489, 490). Not surprisingly, the studies found that individuals with a high percentage of slow-twitch muscle fibers were more efficient at burning fat than those with high percentages of fast-twitch fibers.

As fat is the primary fuel for slow-twitch fibers, the greater the ratio of slow-twitch to fast-twitch fibers, the greater the capacity to burn fat is (489, 490). Therefore, **individuals with a high percentage of fast-twitch fibers may find losing fat more difficult than** someone with a high percentage of slow-twitch fibers.

Skeletal Muscle Function and Characteristics

Muscle has four primary characteristics (16):

- **Excitability:** responds to stimuli (e.g., nervous impulses)
- **Contractibility:** shortens in length
- **Extensibility:** stretches when pulled
- **Elasticity:** returns to original shape and length after contraction

The three primary functions of muscle are:

1. **Heat Production** (up to 70 percent of body heat is generated via muscle tissue) (8)

2. **Maintain Posture**
3. **Movement**

Understanding that **muscle tissue generates up to 70 percent of a human's body heat** and how it relates to an athlete's performance and recovery is important, especially regarding training and competing in cold and hot weather.

The contraction and relaxation of muscles profoundly affect physiological changes in the body. When exercising at an intensity of 4 liters of oxygen/minute, a muscle's oxygen consumption increases approximately 70 times over the resting level (9)! Increased blood flow is channeled to the working muscles to meet this demand. When the working muscles contract, the blood flow to those muscles decreases, and when the muscles are relaxing, the blood flow increases. This pumping action helps to facilitate blood flow through the muscles and back to the heart. Capillaries, dormant under normal conditions, are utilized so that up to 4,000 deliver blood to each square millimeter of muscle (cross-section) during high-intensity exertion (10).

When a muscle is contracted to approximately 60 percent of its maximum contractile capacity, blood flow is occluded because of intramuscular pressure (10, 11).

Muscle Memory

Muscle memory is often used to denote an individual remembering how to perform a task or the body's ability to rapidly regain muscular strength.

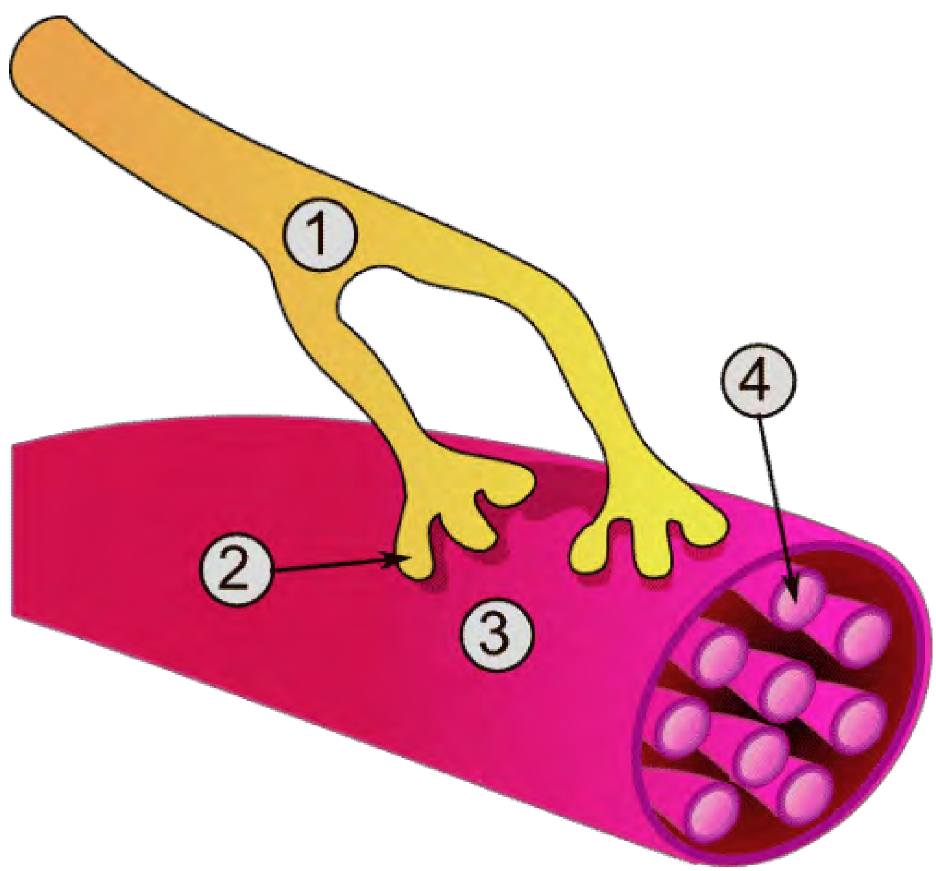
To be clear, muscles do not have a brain and thus do not have a memory. The ability of an individual to perform physical tasks by memory is the result of performing enough repetitions of a particular task that the motor skills required are memorized (277).

It is believed that learning a new skill (e.g., motor learning) occurs in the motor cortex of the frontal lobe of the brain. Once the skill has been memorized, the brain primarily utilizes the cerebellum to execute the skill (277).

Muscle memory is not a muscle's function but the brain's.

Nerve Innervation

Nerves are the pathways that deliver electrical impulses from the brain to the muscles, thereby eliciting a muscle contraction.



1 – Motor Neuron (nerve)
2– **Synapse** (connection from nerve to muscle)
Stimulate the muscle to contract. The connection point of muscle fiber and synapse is the **neuromuscular junction.**
3 – Muscle Fiber
4 – Myofibril

Of the three types of muscle tissues, only skeletal muscles have voluntary contractions. This means the brain sends signals through the central nervous system (CNS) to the motor neurons that innervate muscle fibers and produce muscle contractions (6). This process is termed neural signaling. As individuals become better trained, their neural signaling can be improved (520).

A **motor unit** is a motor neuron and all the muscle fibers it innervates (above image).

Cardiac and smooth muscles rely upon the **autonomic nervous system (ANS)** to provide contractions. This means that cardiac and smooth muscle have involuntary contractions and therefore do not require the CNS to elicit a muscle contraction. More to the point, the contractions occur without conscious thought. The ANS is part of the peripheral nervous system and has two primary areas of function:

1. **Sensory (afferent):** an example of this would be physical pain experienced when you touch a hot surface
2. **Motor (efferent):** an example of this is heart contractions

The ANS is broken into the **sympathetic (SNS)** and **parasympathetic nervous systems (PNS).** The actions of the SNS and PNS typically oppose each other. For example, the SNS increases cardiac output (the amount of blood pumped out of the heart in one minute), whereas the PNS decreases cardiac output (7). Therefore, the ANS aims to keep the body in a state of equilibrium (homeostasis) by self-regulating itself via constant physiological adjustments.

As the nervous system is the catalyst for muscle contractions, an injury to a part of the nervous system may affect the ability of the related muscles to function properly. A common nerve issue is sciatica. Sciatica occurs when compression on the sciatic nerve causes pain and/or weakness in the leg to which the compressed sciatic nerve connects. This link demonstrates the relationship between the nerves and muscles.

Connective Tissue

Connective tissue is responsible for providing support and protection to the body and supporting the musculoskeletal system's movement.

Tendons: connect skeletal muscle to bone (e.g., the Achilles tendon connects the calf muscles to the heel – or calcaneus)

Ligaments: connect bone to bone (e.g., the patellar ligament connects the knee – or patella – to the lower leg – or tibia)

Cartilage: There are three types of cartilage: hyaline (articular), fibrocartilage, and elastic. Relative to movement, hyaline cartilage is the most important type to be aware of. Hyaline cartilage is found on the surfaces of most joints and protects the joints while acting as a barrier between bones. Cartilage primarily comprises collagen and water.

Labrum: Found around the edge of the shoulder and hip joints. Its primary purpose is to keep the ball aspect of the humerus and femur within the shoulder and hip socket, respectively. It comprises fibrocartilage. Injury to the labrum is called a labral tear.

Fascia

When most people learn anatomy and how it applies to sports, the focus is typically on bones, muscles, and tendons/ligaments. This learning approach leaves out an important part of the picture – fascia. Fascia influences posture and facilitates body movement. If tension on the fascia is too high, it can cause pain and postural abnormalities because of restricted movement (270). Thomas Myers, author of Anatomy Trains, states, “Fascia is the missing element in the movement/stability equation” (330).

Fascia is a soft-tissue component of connective tissue (collagen fibers) that covers, connects, and separates all muscles, organs, bones, and nerves (331). Interestingly, while the fascia surrounds each of the areas above, it is one large, continuous tissue (330). Fascia comprises multiple layers, from superficial (just below the skin’s surface) to deep (covers internal organs). While thin, it is very strong and fibrous. If you have ever pulled the skin off raw chicken, you have encountered fascia. The fascia is the thin, near-white translucent membrane above the muscle and below the skin of the chicken. Perhaps the best way to think of **fascia is a weblike structure that permeates the whole body.**

Fascia provides support to the musculoskeletal system. According to the International Fascia Research Congress (IFRC), while the basics of fascia are understood, most of the information about fascia is broadly not researched (187). The IFRC theorizes that one of the reasons for the lack of understanding of the fascia is that, because of its immense scope, researchers are often unable to divide it up into distinct areas on which to classify and focus. Additionally, the general population and even those who have taken anatomy classes disregard the fascia because most anatomical diagrams/images show the body with the fascia removed (187).

Under normal conditions, fascia is pliable and flexible. However, when fascia is under stress because of trauma, dehydration, and biomechanical/neuromuscular issues, it becomes inflamed, which causes it to constrict and become less pliable and slippery. **This leads to tightness, pain, and a reduced range of motion.** If the tightness is substantial enough, it can alter proper body mechanics. Overly tight fascia can produce tension up to 2,000 pounds per square inch (269)!

Like the body, fascia works better when it is hydrated. The fascia is wet and slippery in properly hydrated individuals (330). This allows the fascial surfaces to move unrestricted. However, when an individual is dehydrated, the fascia can become tight and “sticky,” thus reducing and altering normal movement patterns and ranges of motion.

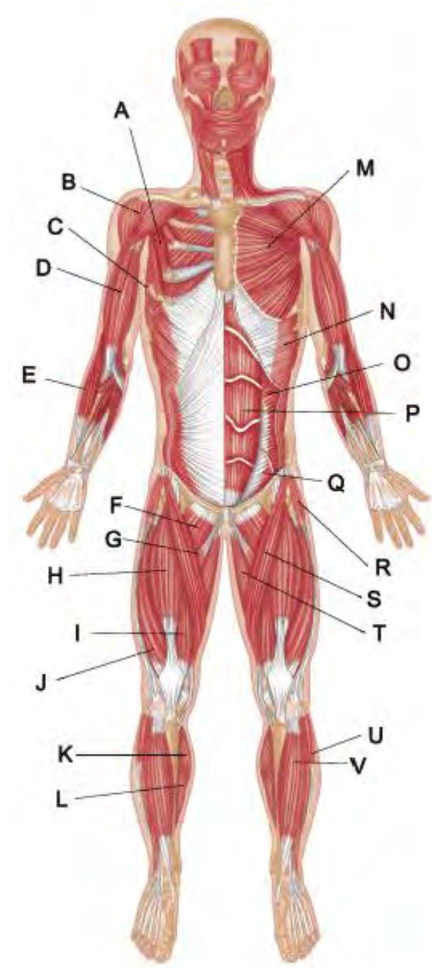
Diagnostic tests such as MRIs, CT scans, and X-rays do not show restricted fascia (332). As a result, individuals whose symptoms are caused primarily by restricted fascia often go undiagnosed or are incorrectly diagnosed (269). Fascia can contract with or independently of muscles (330).

The IFRC claims that fascia plays a significant role in joint stability, biomechanical issues, chronic stress, respiratory dysfunction, and lower back pain (187). This is further corroborated by a 2012 study by Willard et al. in which the fascia of the lumbar and thoracic areas of the spine is discussed in relation to the stabilization of the spine and pelvis (188).

For all of the reasons just listed, it is clear that fascia is not a passive wrapping but rather a dynamic and vital component in the body’s support and movement structure.

Muscles of the Body

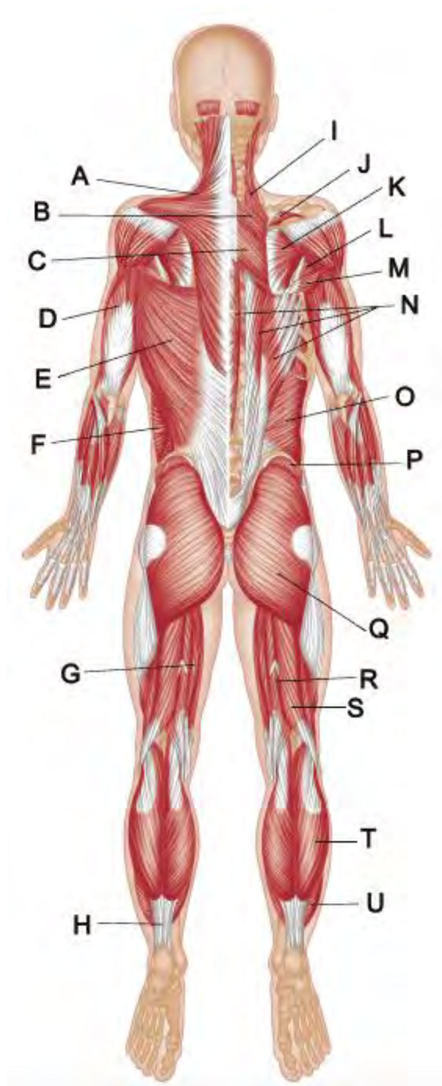
Muscle View – Anterior



A – Pectoralis Minor
B – Deltoid
C – Serratus Anterior
D – Biceps Brachii
E – Brachioradialis
F – Pectineus
G – Adductor Longus
H – Rectus Femoris
I – Vastus Medialis
J – Vastus Lateralis
K – Gastrocnemius
L – Soleus
M – Pectoralis Major

N – External Oblique
O – Internal Oblique
P – Rectus Abdominis
Q – Transverse Abdominis
R – Tensor Fasciae Latae
S – Sartorius
T – Gracilis
U – Peroneals
V – Tibialis Anterior

Muscle View – Posterior



A – Trapezius
B – Rhomboid Minor
C – Rhomboid Major

D – Triceps
E – Latissimus Dorsi
F – External Oblique
G – Semimembranosus
H – Achilles Tendon
I – Levator Scapulae
J – Supraspinatus
K – Infraspinatus
L – Teres Major
M – Teres Minor
N – Erector Spinae
O – Internal Oblique
P – Gluteus Medius
Q – Gluteus Maximus
R – Semitendinosus
S – Biceps Femoris
T – Gastrocnemius
U – Soleus

Muscle Classification

Some of the muscles listed next could be classified into multiple categories, such as the hamstrings (core and lower body). However, for our purposes, the category in which a muscle is listed here is based on its significance within that area. For example, while the hamstrings could be listed as "lower-body" muscles, they are listed as "core" muscles because they stabilize the lumbar spine/pelvic/hip region. Additionally, not all muscles of the body are noted in the certification.

Core Musculature

The core is perhaps the most over-hyped, over-marketed, and least understood body area. When most people refer to the core, they think of six-pack abs and that the more "ripped" one's abs are, the stronger the core is. This could not be further from the truth.

Like other areas of exercise science, one of the primary issues in defining the core is that there are multiple definitions, many of which are driven by product marketing and a lack of knowledge.

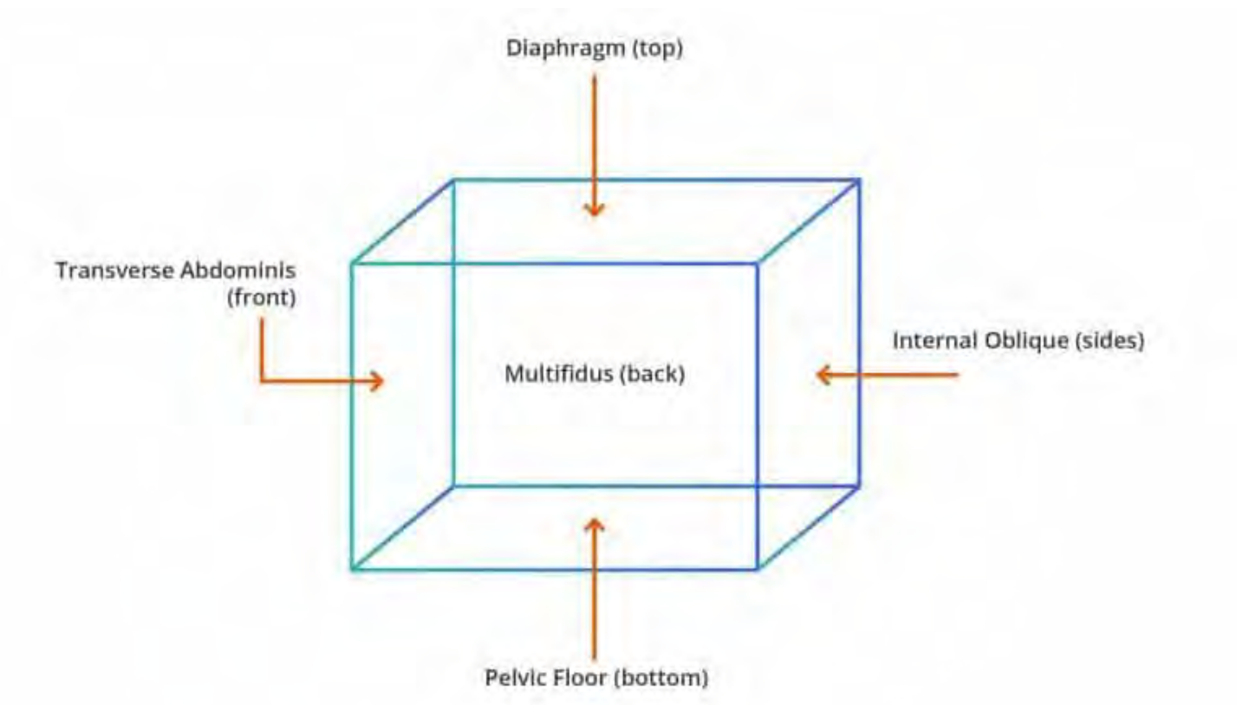
Within this certification, when the term "**inner core**" is used in relation to abdominal, oblique, and back muscles, it refers to deep muscles, meaning muscles that are at least one layer below the surface muscles that most people refer to as the core. In other words, if you can see a muscle (i.e., superficial), it is not a primary core muscle.

The core area is also called the **lumbo-pelvic-hip complex (LPHC).** This term is more appropriate than simply “core,” as it specifically addresses the area that stabilizes the upper and lower body. Therefore, muscles that act to stabilize this complex can be viewed as core musculature. The LPHC encompasses the inner (primary) and outer unit core muscles.

The primary core muscles are:

- **Multifidus (lumbar)**
- **Pelvic Floor Muscles**
- **Transverse Abdominis (TVA)**
- **Internal Oblique**
- **Diaphragm**

A good visual cue for the core (primary) muscles is to imagine a box (147). All sides must provide stability for a box to be strong.



As a point of reference, the muscle that most people consider the core is the **rectus abdominis.** The rectus abdominis is the "six-pack" muscle. This is a superficial muscle, and while it does provide some stability to the LPHC, its primary job is anterior spinal flexion.

The primary purpose of the core musculature is to provide stability and rigidity to the area of the spine and pelvic region. The force production of the core muscles is not great, as their primary function is to create stability, not gross movement. The synergy between the core musculature and ligaments allows for substantial stability.

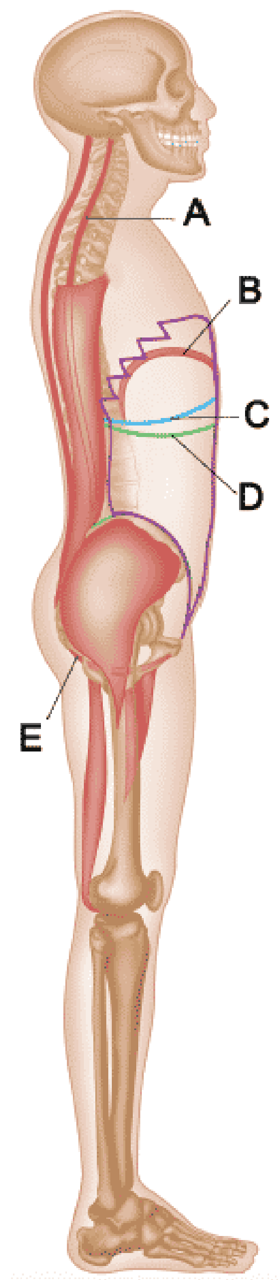
Given the confusion over what muscle or muscles constitute the core, it is advised to separate the core into two areas: the inner and ***outer unit***.

The inner unit comprises the muscles just described as **primary core muscles**. The outer unit is made up of the following muscles:

- **Rectus Abdominis**
- **External Obliques**
- **Erector Spinae**
- **Quadratus Lumborum**
- **Iliopsoas (made up of Ilicus and Psoas)**
- **Rectus Femoris**
- **Hip Adductors**
- **Glute Maximus**
- **Hamstrings**

While the outer unit muscles provide stabilization to the LPHC, they are not as specialized for this task as the inner unit. As noted earlier, the primary function of the outer unit muscles is to provide movement, not stabilization. Therefore, when the outer unit muscles become overactive, they shut off the inner unit musculature. This dramatically reduces the amount of LPHC stabilization. The inner unit is the primary LPHC stabilizer, and the outer unit is the secondary stabilizer.

Inner Unit Core Musculature – Lateral View

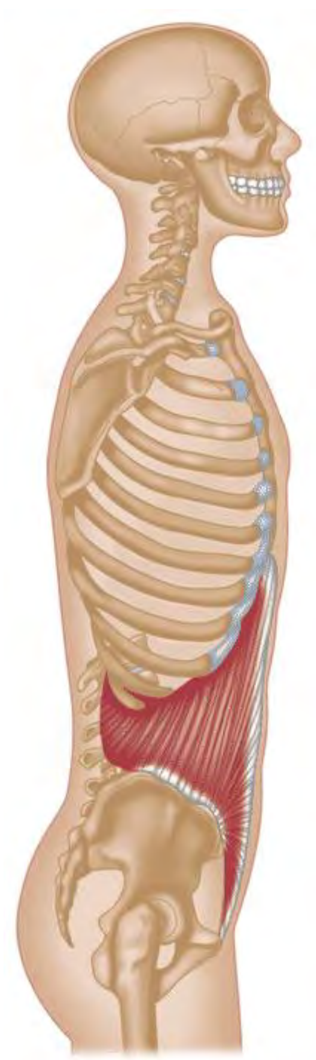


A – Multifidus
– Extends from the sacrum to the cervical spine
B – Diaphragm
C – Internal Oblique
D – Transverse Abdominis
E – Pelvic Floor Muscles

As most people view the outer unit as the core, it is common for people to overwork their outer unit muscles, which, as just noted, can shut down the inner unit core musculature. Individuals who perform countless abdominal crunches and leg raises believe they are strengthening their core muscles and thus increasing their lumbar support when, in fact, the opposite is often the case.

The inner unit, specifically the TVA, stabilizes the lumbar spine to allow both upper and lower body motion by contracting 90 milliseconds before upper- body motion and 120 milliseconds before lower-body motion (146). This illustrates the importance of the TVA in providing stability to the lumbar spine.

Inner Unit Core Musculature



Internal Oblique

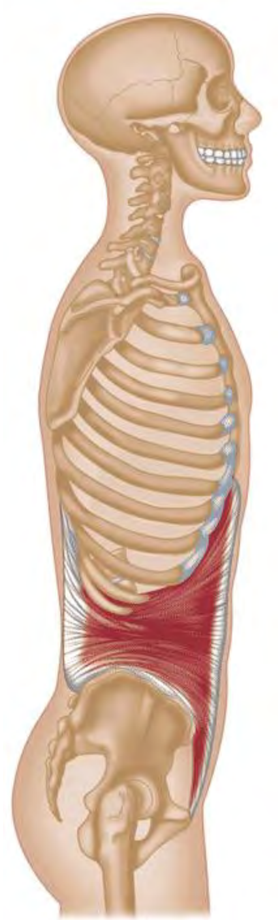
This muscle can be considered the intermediate muscle of the core because of its placement, as it lies between the transverse abdominis and the external oblique. In addition to providing spinal stability, it has two additional functions.

Function One:

The contraction and relaxation of the internal oblique results in movement of the diaphragm, which is responsible for inhalation and exhalation.

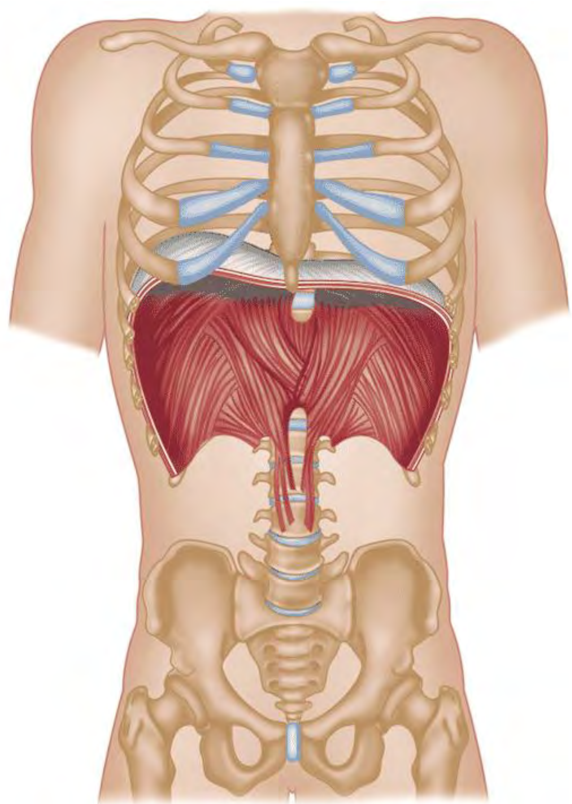
Function Two:

The internal oblique is responsible for the upper body's rotation and lateral flexion (side-to-side bending).



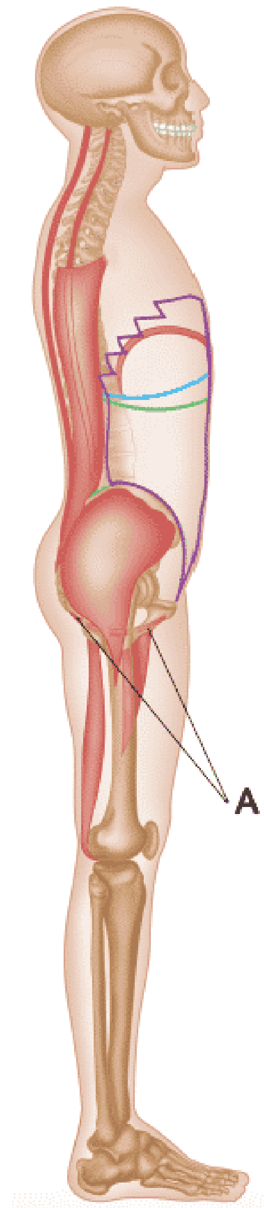
Transverse Abdominis (TVA)

This is the deepest of all the abdominal muscles, and its primary action is to provide lumbar stability. The TVA acts as the body's natural weight belt by providing hoop tension around an individual's midsection.



Diaphragm

The diaphragm is a 'dome-like' sheet of muscle between the thoracic and abdominal cavities. It is responsible for increasing and decreasing lung size when inhaling (contraction of the diaphragm) and exhaling (relaxation of the diaphragm), respectively. The diaphragm also provides lumbar stability by working with the other inner unit core muscles.

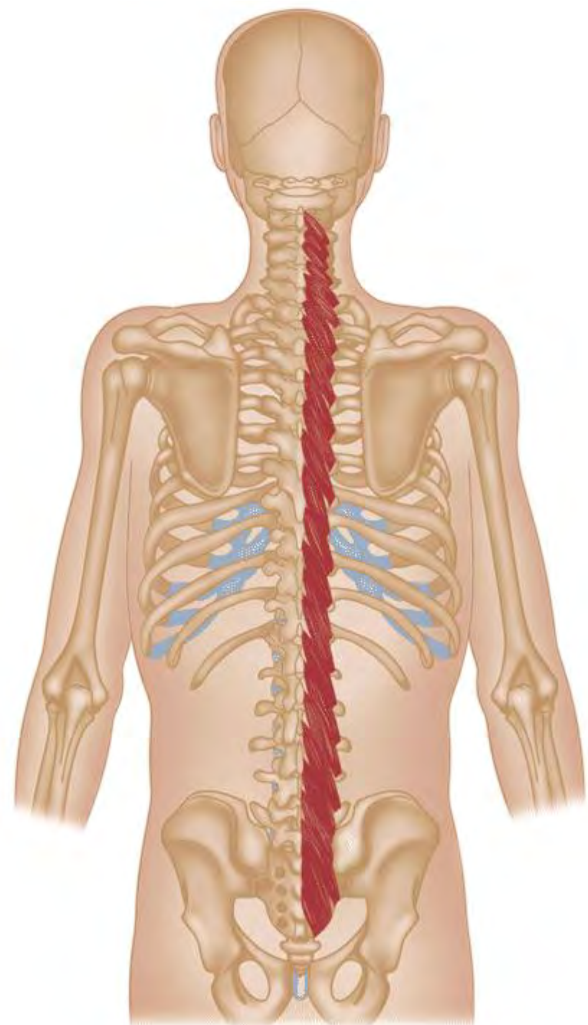


Pelvic Floor Muscles (PF)

Comprising the **levator ani**, ***coccygeus***, and connective tissue, the pelvic floor muscles essentially form a "hammock" that supports pelvic organs (e.g., bladder).

The PF muscles also assist in bladder and bowel function. Specifically, in relation to core musculature, the PF muscles assist in the stabilization of the spine and pelvis.

A: Pelvic Floor Muscles



Multifidus (Lumbar)

The multifidus (mull-tiff-a-duss) is a series of small muscles that attach laterally to both sides of the spinal column and is deep to the erector spinae. The multifidus is broken down into segments with the insertion point several vertebrae higher than the origin. This segmental construction means that each multifidus segment is responsible for controlling a small section of the spine.

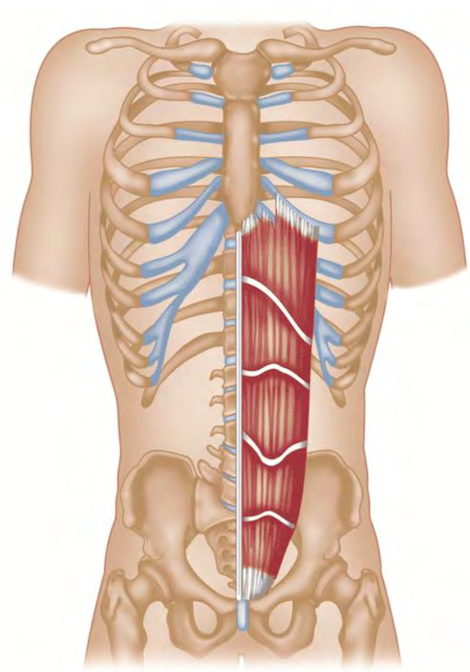
The multifidus can be broken down into superficial and deep muscles. The superficial multifidus is responsible for lumbar back extension, rotation, and lateral flexion, while the deep multifidus is responsible for spinal stabilization.

The correlation between the multifidus's strength (spinal stability) and lower back pain was documented in a 2002 study by Danneels et al. published in the European Spine Journal. In this study, a group of individuals with healthy backs was compared with a group suffering from lower back pain. It was demonstrated via EMG testing of the multifidus that the group with lower back pain showed much lower multifidus activation than the healthy group. Additionally, the lower back pain group had a reduced ability to voluntarily contract their multifidus over the healthy group (145).

The multifidus co-contracts with the TVA to provide spinal stability. Like the TVA, the multifidus contracts before most body movements as a protective mechanism for the spine.

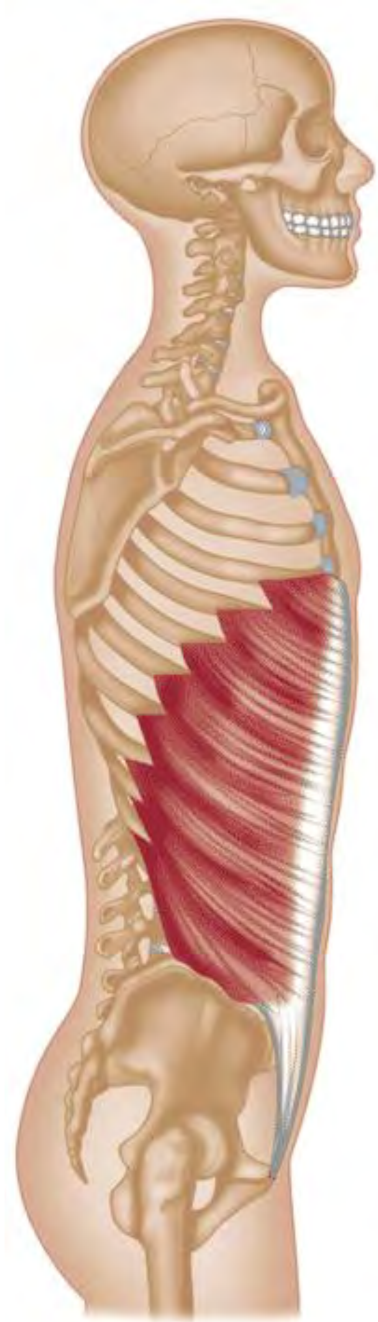
Outer Core Musculature

The primary purpose of the outer core musculature is to move and, secondarily, to assist in providing stabilization to the LPHC.



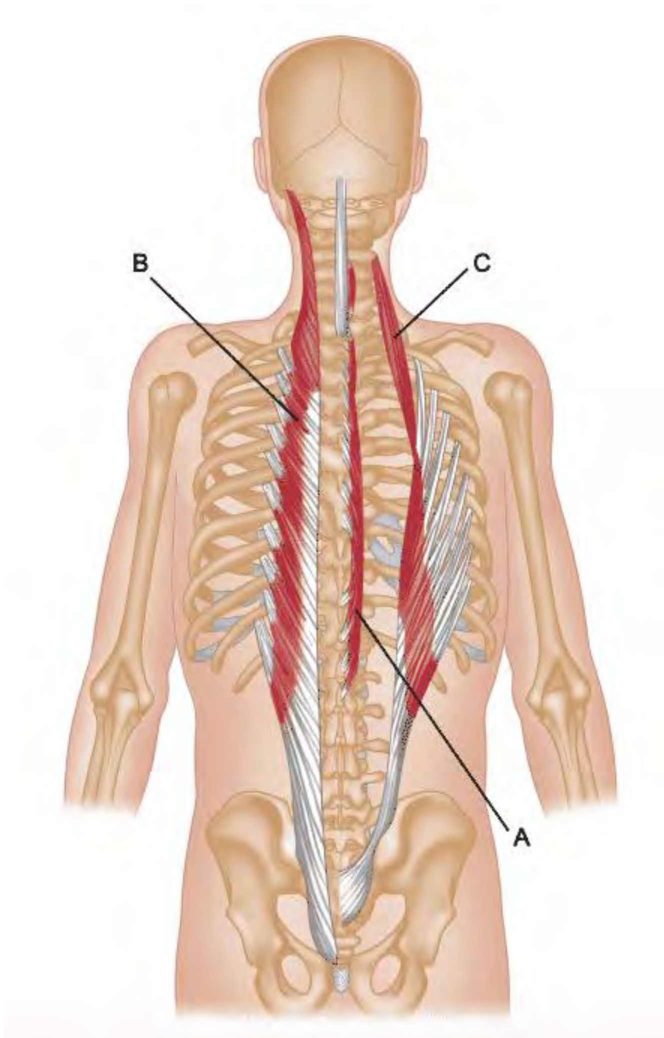
Rectus Abdominis

This muscle group comprises two muscles that run parallel to each other and are separated by connective tissue (linea alba). The rectus abdominis is primarily responsible for flexion of the lumbar spine. Secondarily, the rectus abdominis assists in stabilizing the lumbar spine.



External Oblique

This muscle is located on either side of the rectus abdominis. It is responsible for the spine's flexion and rotational and lateral flexion (bending side to side). Like the rectus abdominis, it secondarily assists in stabilizing the lumbar spine.

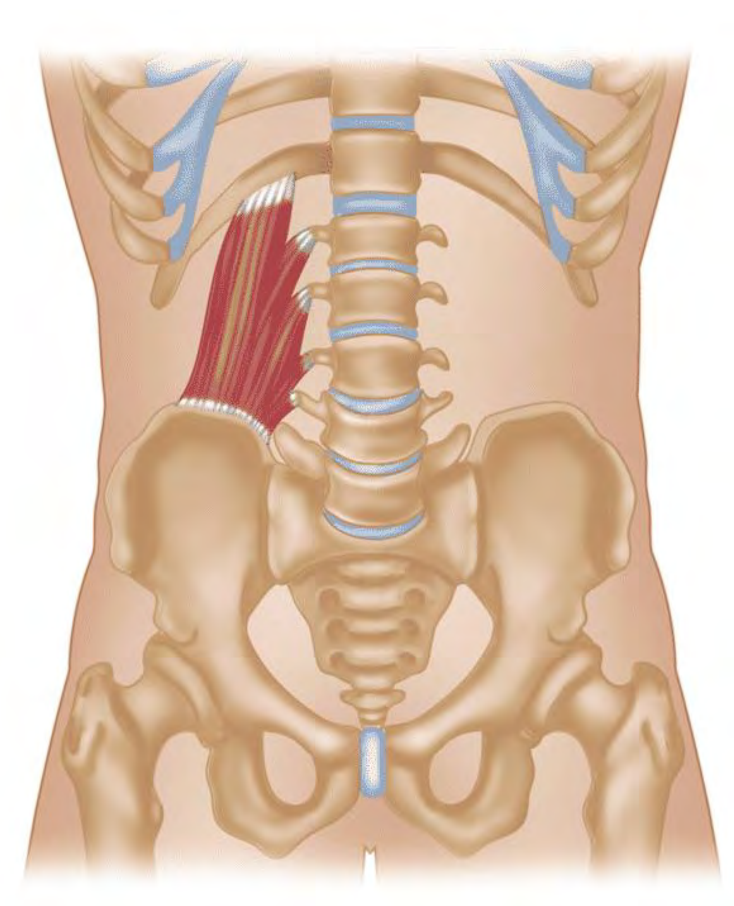


Erector Spinae

Also called **extensor spinae**, this group of muscles runs parallel to the vertebral column from its base (lumbar) through the top (cervical) section. The three groups of muscles are:

A – Spinalis (Medial)
B – Longissimus (Center)
C – Iliocostalis (Lateral)

The primary function is extension and lateral flexion of the spine. Secondarily, it provides stabilization to the spine.



Quadratus Lumborum (QL)

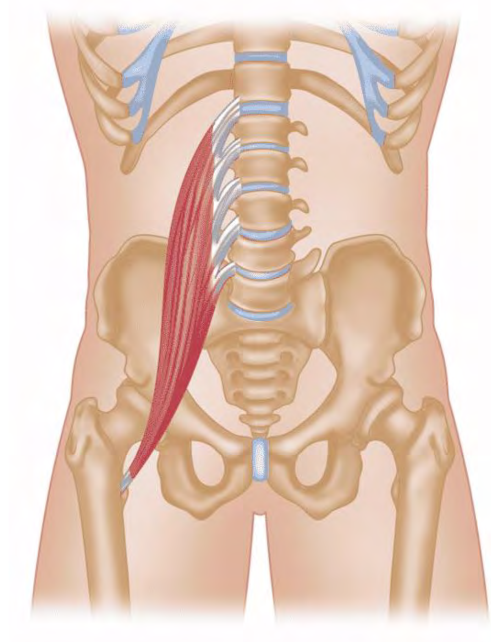
Originates at the iliac crest and inserts into the lowest rib and side of the lumbar vertebrae. The QL acts to stabilize the pelvis and laterally flex the lumbar spine. If the glute medius is underactive, the QL can initiate hip abduction, often leading to the QL becoming overactive.

Rotation of the pelvis also can result in the QL becoming overactive.

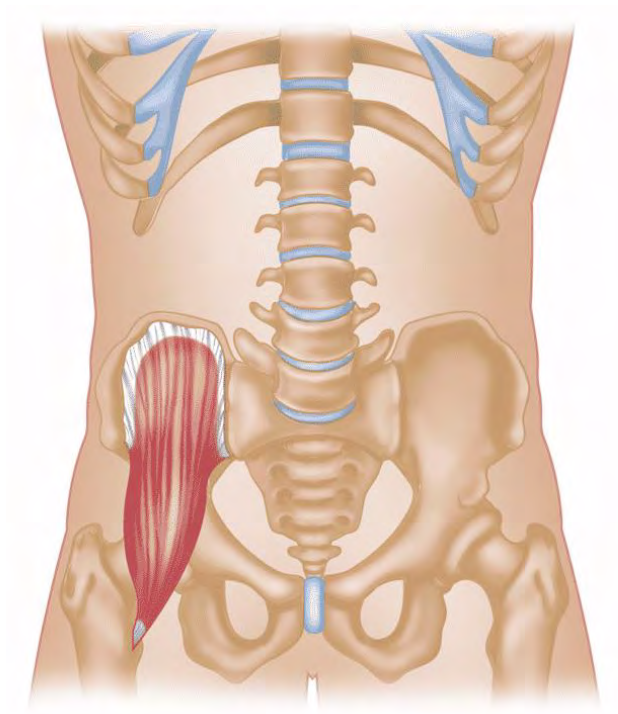
Hip Flexors

The psoas (so-as) and iliacus (ill-e-ack-us) comprise a group of muscles commonly termed hip flexors due to their primary role of hip flexion.

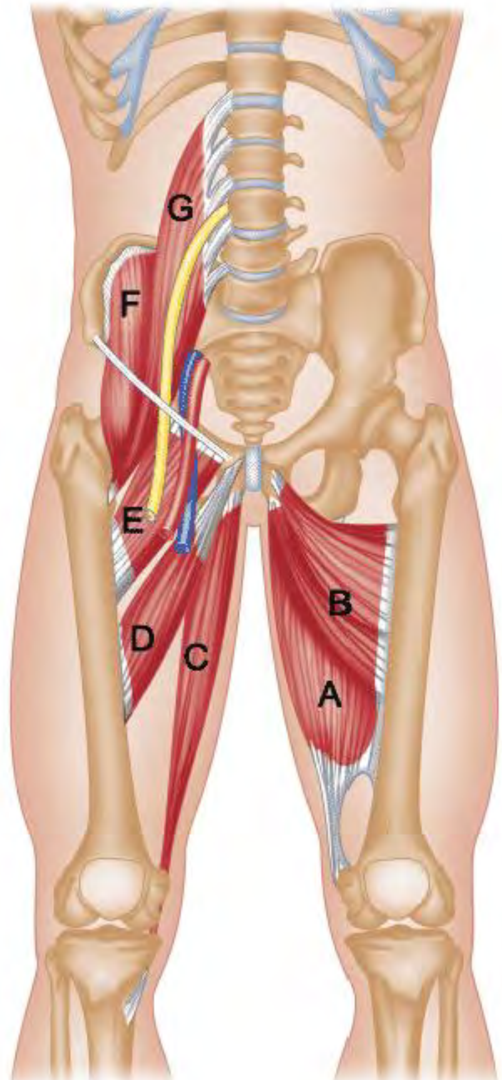
1. **Psoas**
2. **Iliacus**



Psoas



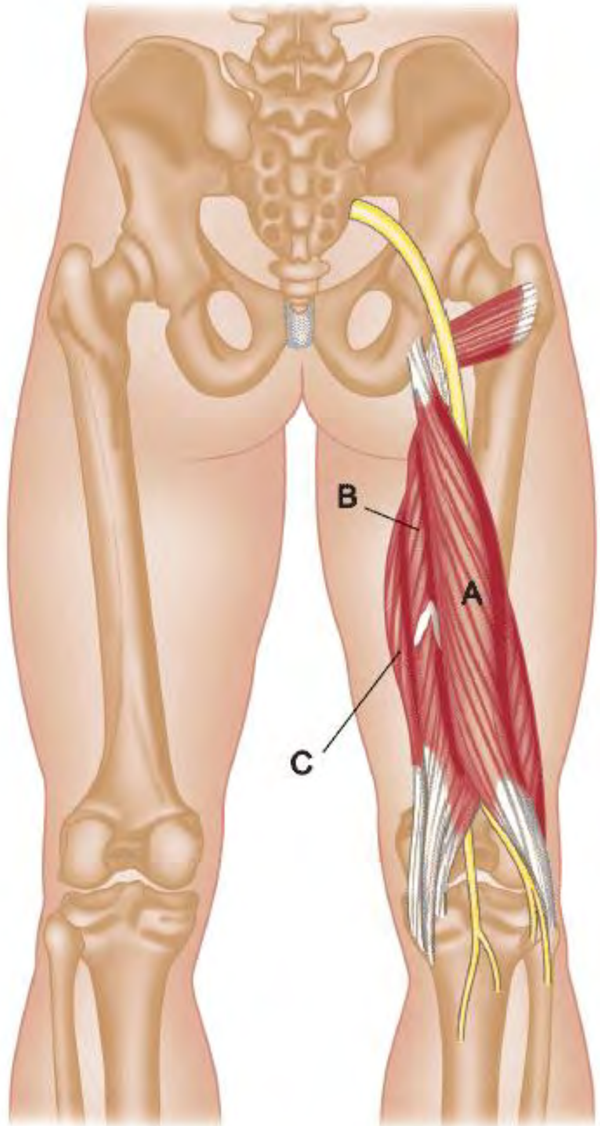
Iliacus



Hip Adductors

Below are the muscles responsible for hip adduction:

A – Adductor Magnus
B – Adductor Brevis
C – Gracilis
D – Adductor Longus
E – Pectineus
F – Iliacus
G – Psoas Major

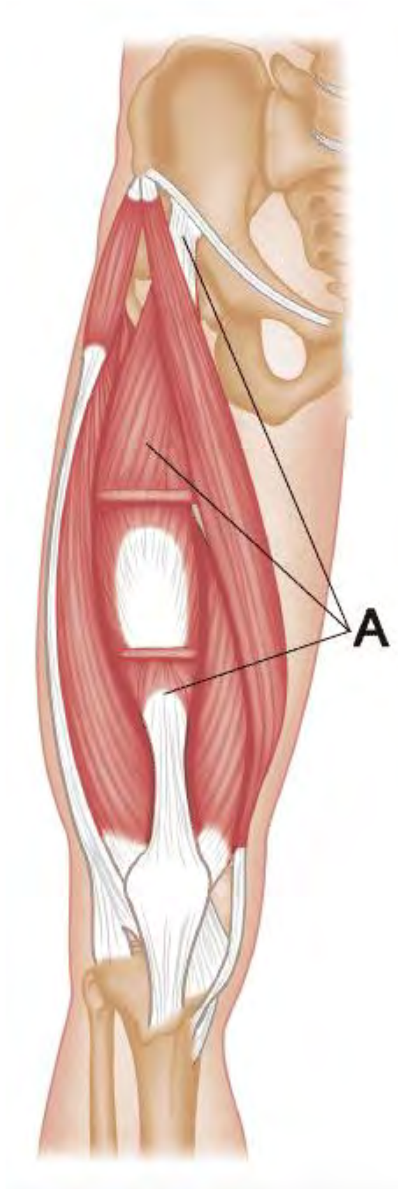


Hamstrings

The hamstrings comprise three muscles (posterior view):

A – Biceps Femoris
B – Semitendinosus
C – Semimembranosus

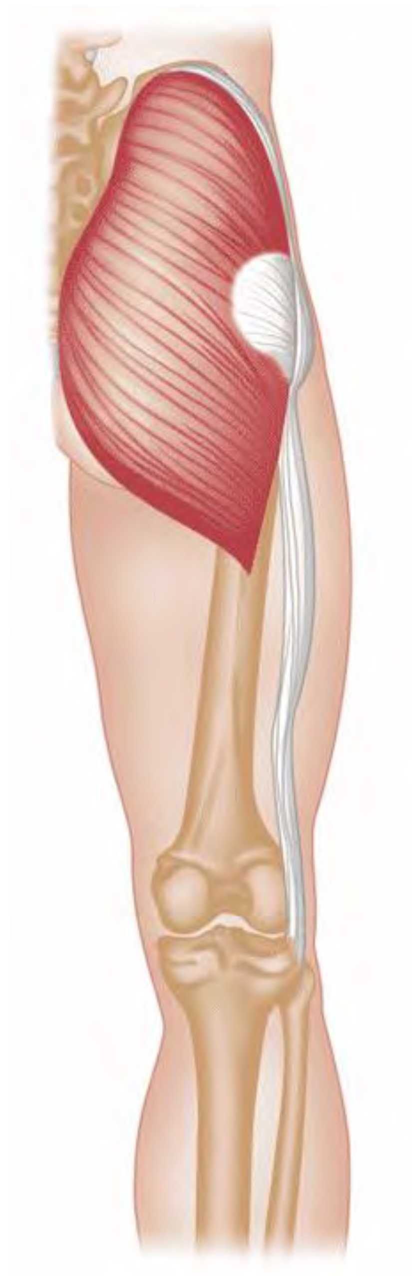
These muscles stabilize the LPHC, flex the lower leg, and extend (posterior) the femur.



Rectus Femoris

The rectus femoris is one of four muscles that make up the quadriceps and is visually the center muscle of the four when viewing the muscle in the frontal plane. The primary role of the rectus femoris is knee extension and hip flexion. However, it also plays a role in stabilizing the LPHC and the knee.

A – Rectus Femoris (cut)

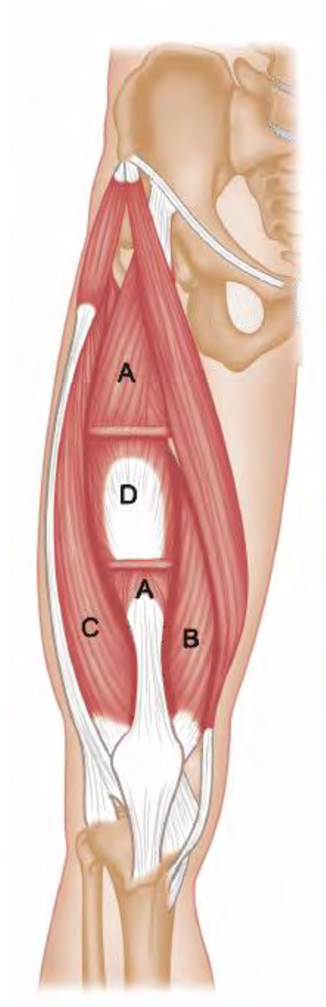


Gluteus Maximus

The gluteus maximus (GMax) is the most superficial of the three gluteal muscles (gluteus medius, gluteus minimus). Its primary function is support of the LPHC and extension/external rotation of the hip.

Due to its role in hip extension, the GMax is one of the most important muscles for runners.

Lower Body Musculature



Quadriceps

The “quads,” as this muscle group is often called, consist of four different muscles on the upper leg’s anterior side.

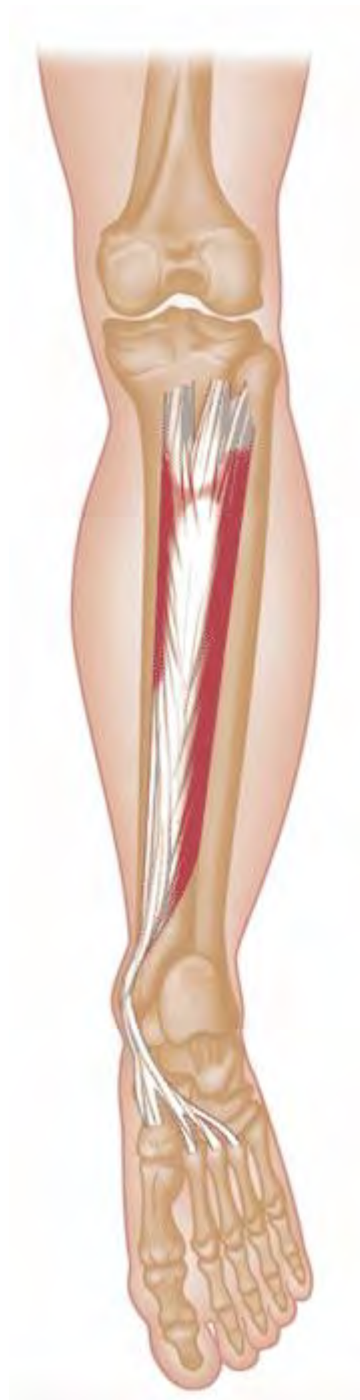
A – Rectus Femoris (cut)
B – Vastus Medialis
C – Vastus Lateralis
D – Vastus Intermedius

These muscles are used to extend the lower leg and stabilize the knee. The rectus femoris also acts to flex the hip, as noted previously in the Outer Core Musculature section.

Foot Extensors and Flexors

Tibialis Anterior

Location: Front of the tibia
Action: Dorsiflexion and foot inversion

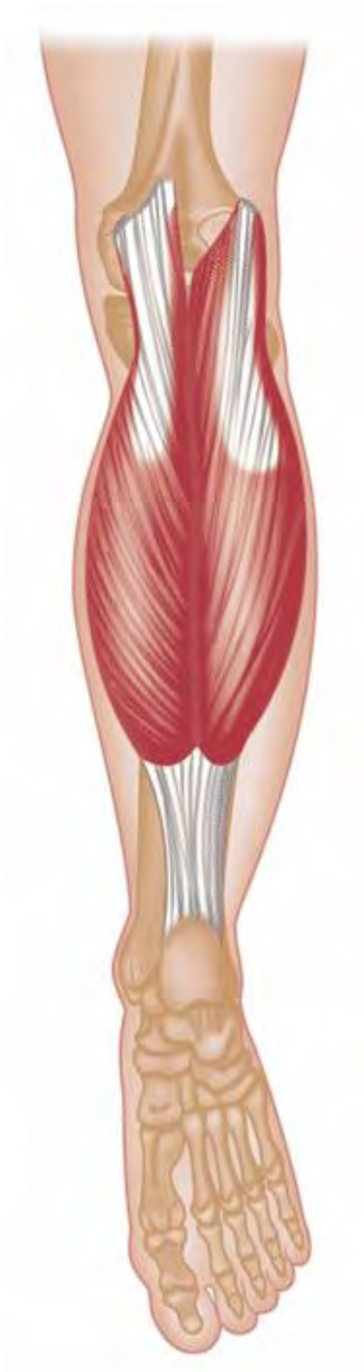


Tibialis Posterior

Location: Deep muscle located behind the tibia.
Action: Plantar flexion, foot inversion, and stabilization

Soleus

Location: Deep muscle located behind the tibia and underneath the gastrocnemius.
Action: Plantar flexion and stabilization



Gastrocnemius

Location: Superficial, located on top of soleus.
Action: Plantar flexion and stabilization

The term **triceps surae** (three heads) is often used to refer to the gastrocnemius and the soleus.

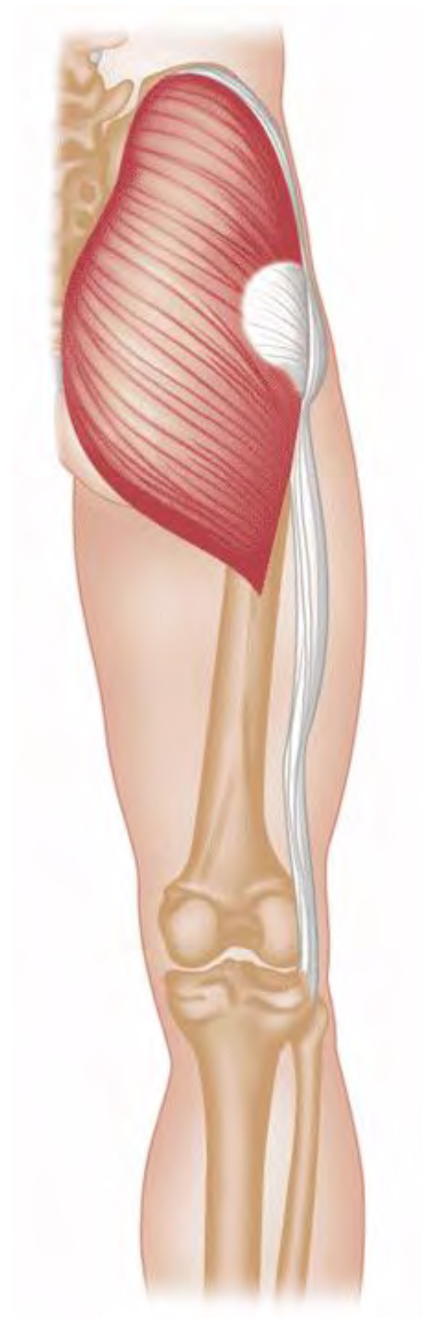
Peroneus Longus

Location: Lateral aspect of the lower leg
Action: Plantar flexion and foot eversion

Peroneus Brevis

Location: Lateral aspect of the lower leg
Action: Plantar flexion and foot eversion

Gluteals



Gluteus Maximus

Location: The most superficial of the three gluteal muscles (gluteus medius, gluteus minimus).
Action: Its primary function is support of the LPHC and extension/external rotation of the hip.

Gluteus Medius

Responsible for abduction and extension, as well as internal rotation (when the hip is extended) and external rotation (when the hip is flexed) of the hip.

These muscles also stabilize the pelvis.

Gluteus Minimus

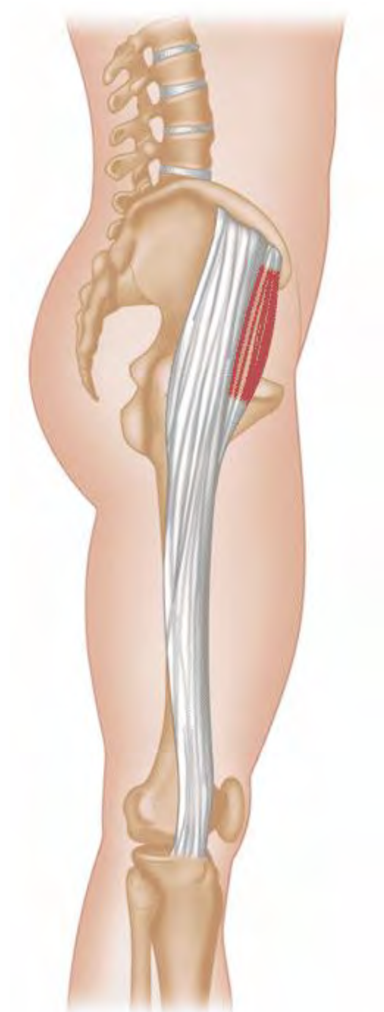
Responsible for abduction and extension, as well as internal rotation (when the hip is extended) and external rotation (when the hip is flexed) of the hip.

These muscles also stabilize the pelvis.

Glute Medius Weakness

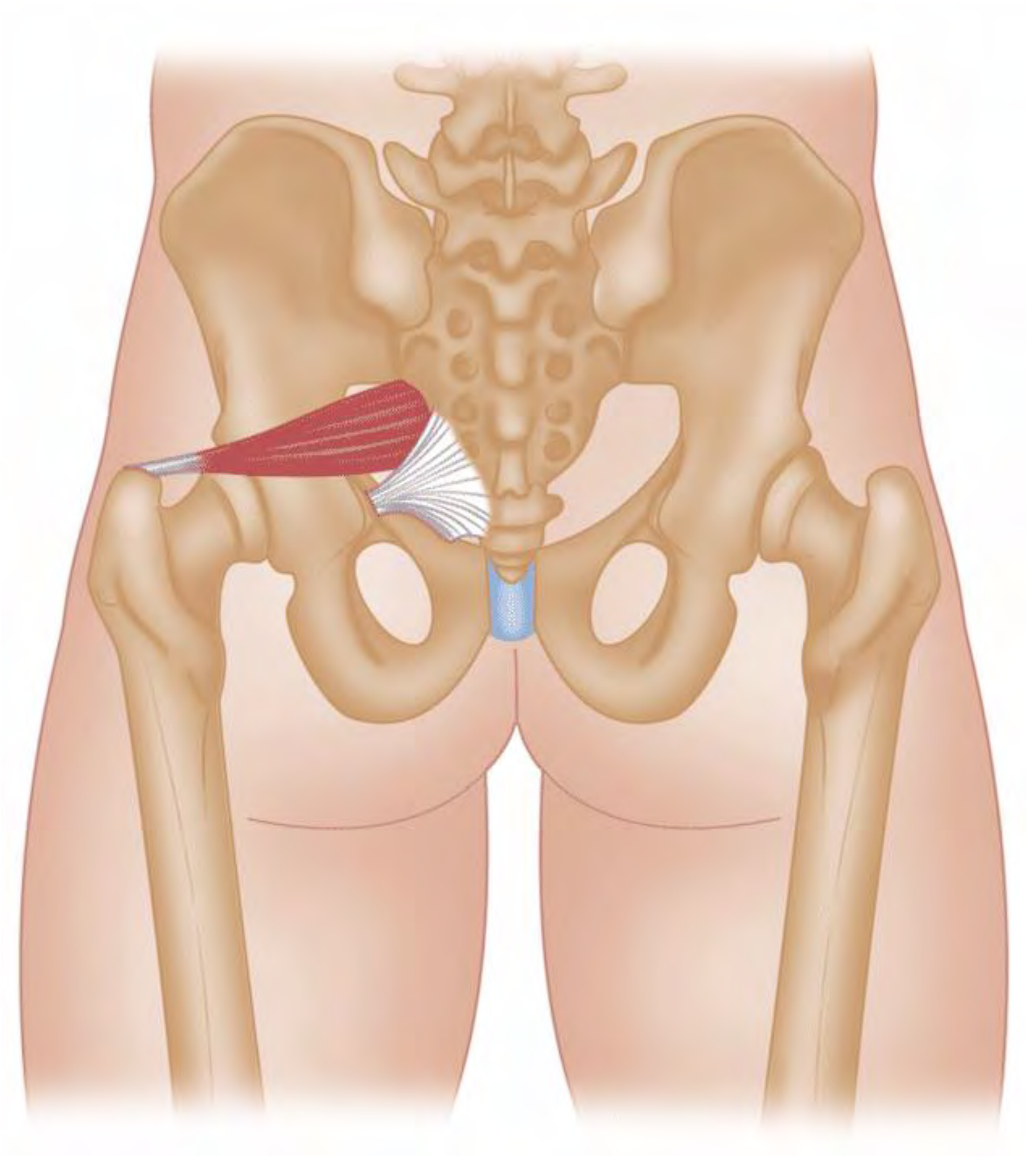
When the glute medius is weak, the quadratus lumborum on the opposing side often becomes overactive to stabilize the pelvis since the glute medius isn't assisting in pelvic stabilization. This often causes low back pain.

The glute medius is responsible for external hip rotation when the hip is flexed, as occurs upon footstrike. Therefore, if there is internal rotation of the hip and femur upon footstrike, this may indicate an underactive glute medius.



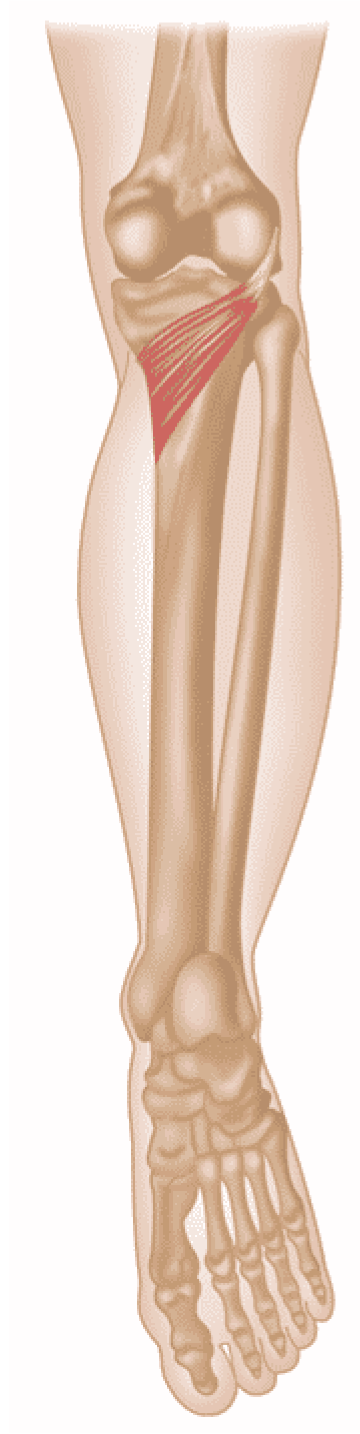
Tensor Fasciae Latae

In addition to the gluteals, the tensor fasciae latae (commonly referred to as the TFL) also acts to abduct the hip. The TFL also extends and externally rotates the hips. The TFL is located at the side of the hip.



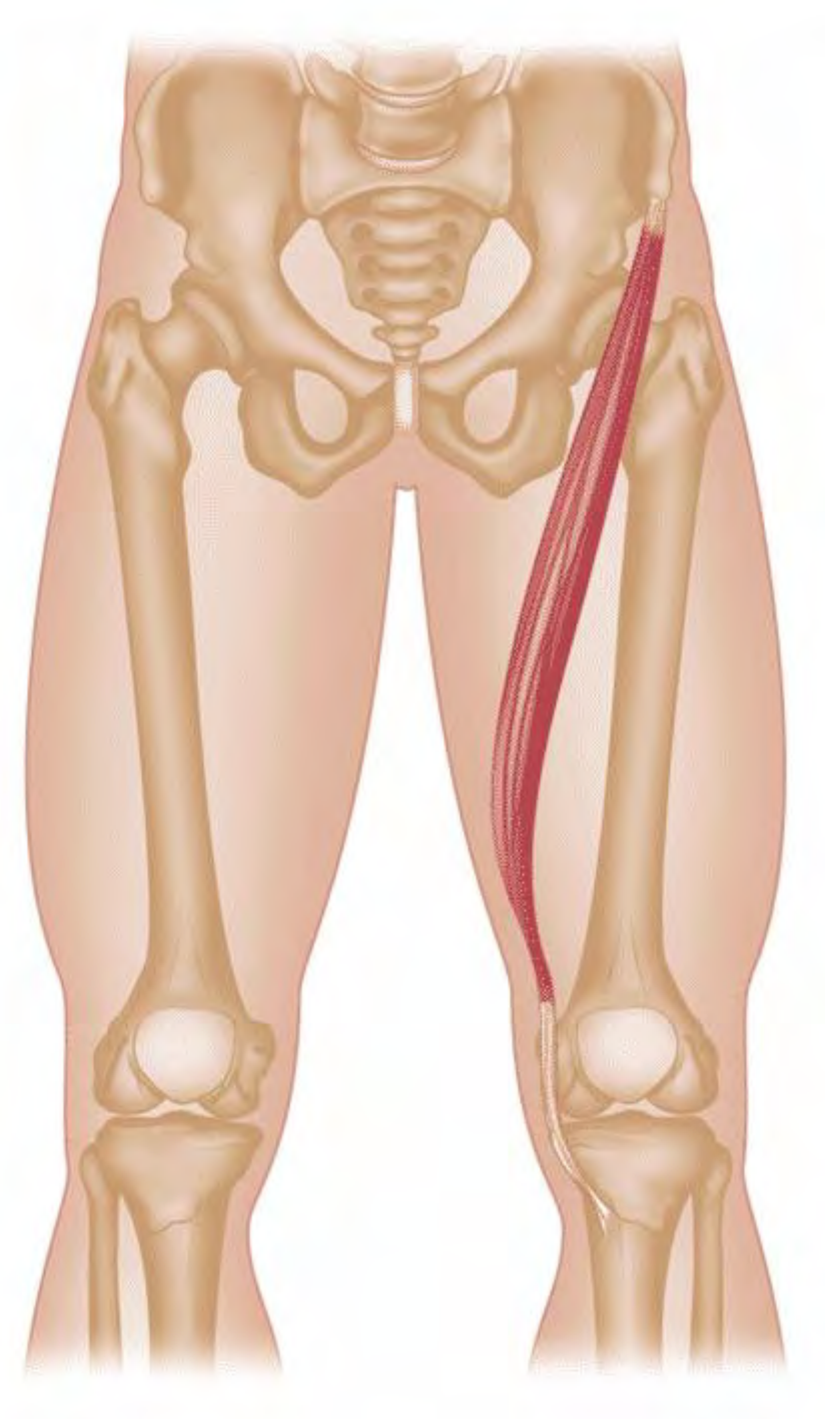
Piriformis

Abducts the femur when the hip is flexed. For many individuals, the sciatic nerve passes through the piriformis. When the piriformis tightens, the sciatic nerve may become compressed and induce sciatica (pain that originates at the gluteals and often travels down the leg via the sciatic nerve). This is also called **piriformis syndrome**.



Popliteus

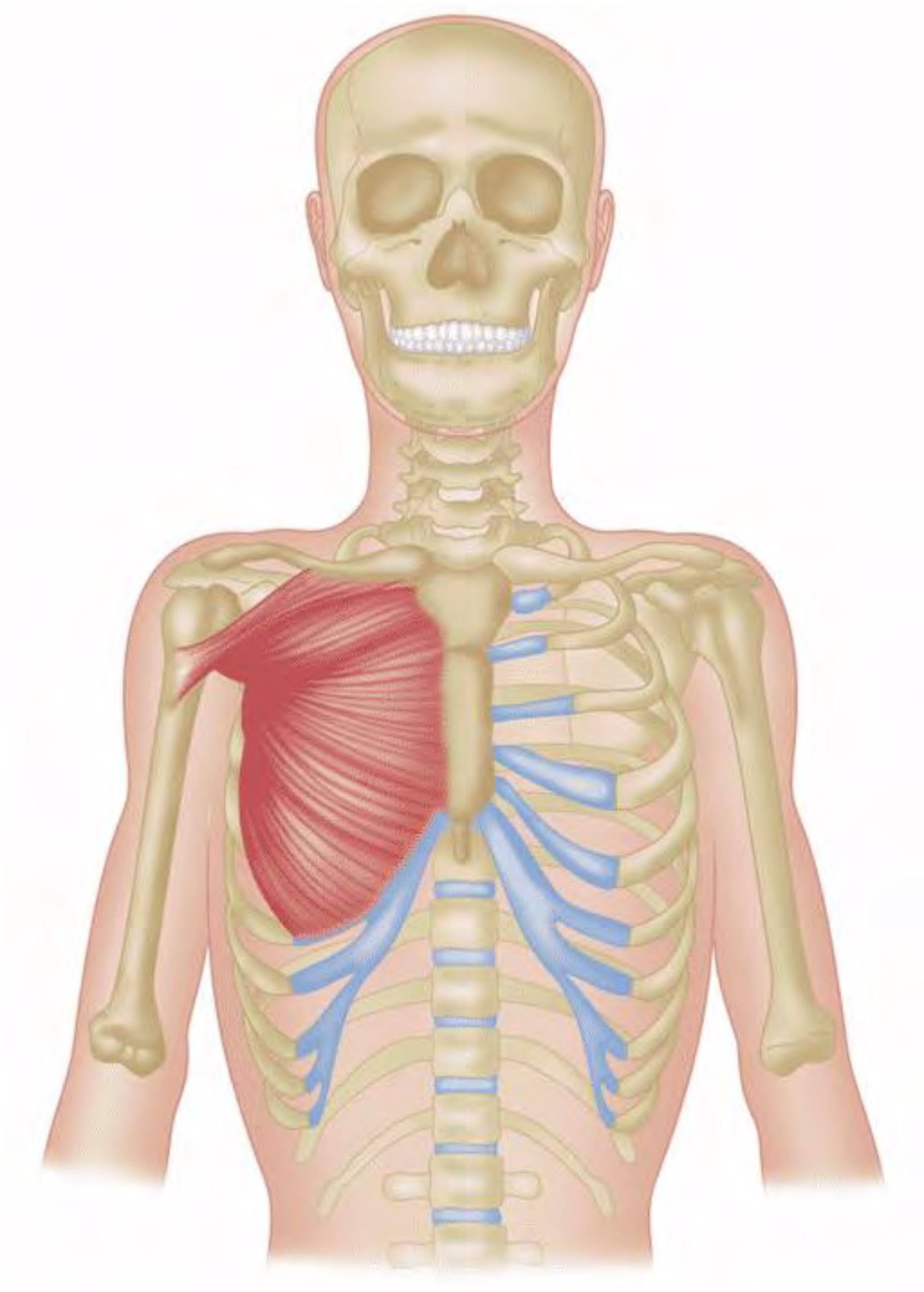
This is a relatively small muscle located behind the knee. The primary action is knee flexion; more specifically, it is responsible for ‘unlocking’ the knee from a straightened position. It also assists with internal rotation of the tibia.



Sartorius

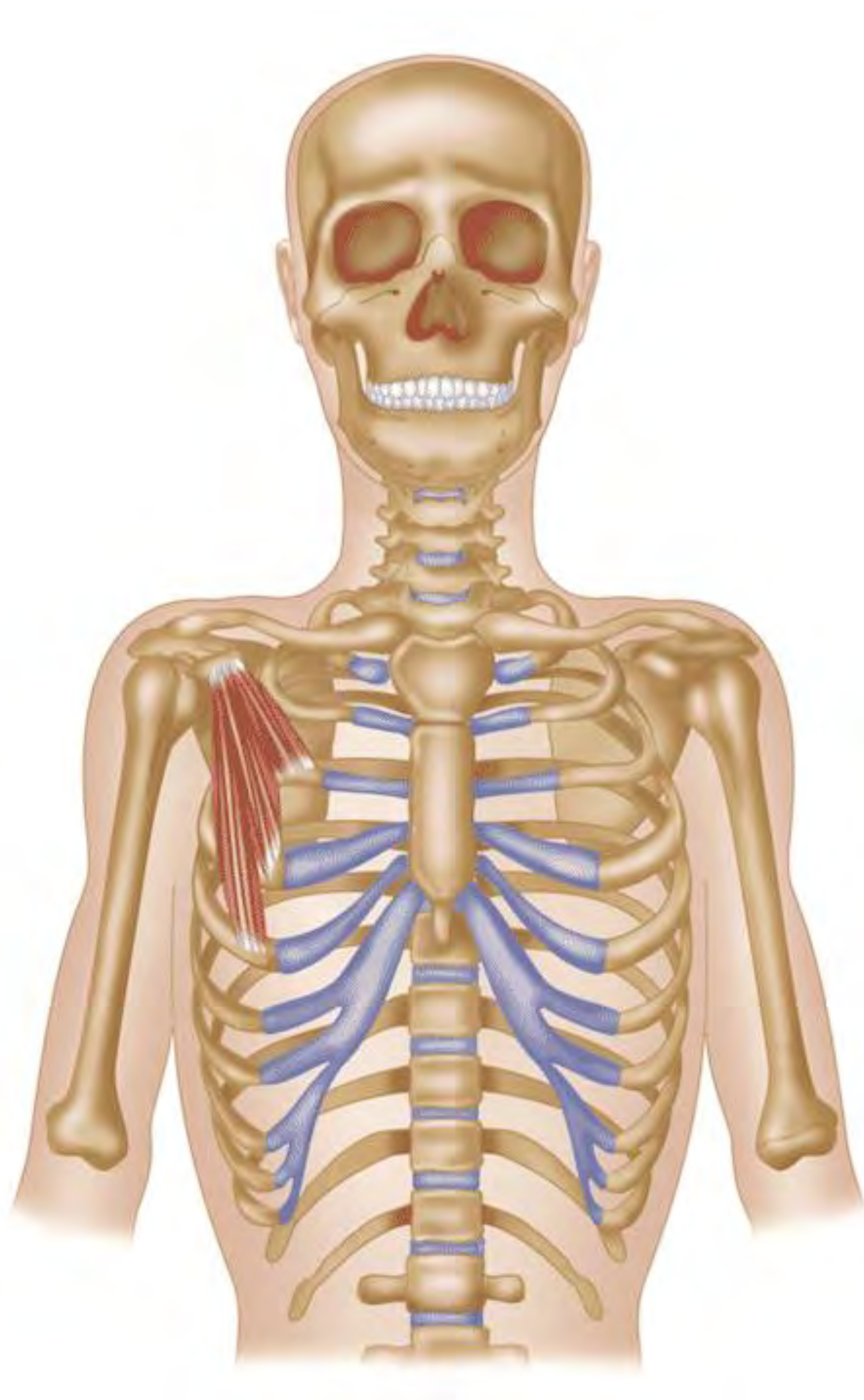
This is the longest muscle in the body and begins (originates) below the ASIS and ends (insertion) on the superior and medial aspect of the tibia. This muscle acts to flex, abduct, and externally rotate the hip. It also aids in knee flexion.

Upper Body Musculature



Pectoralis Major

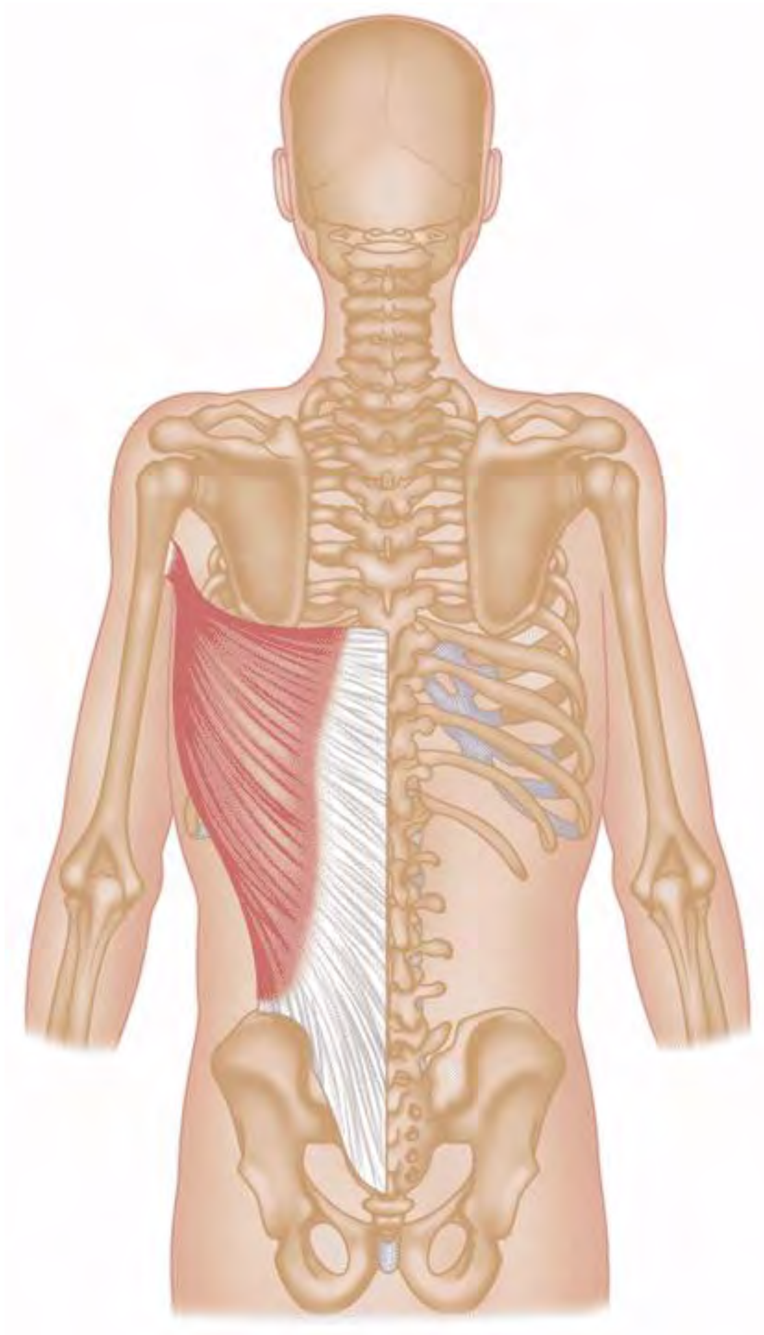
Chest muscle responsible for adduction and internal rotation of the humerus.



Pectoralis Minor

Chest muscle that moves the scapula forward and downward.

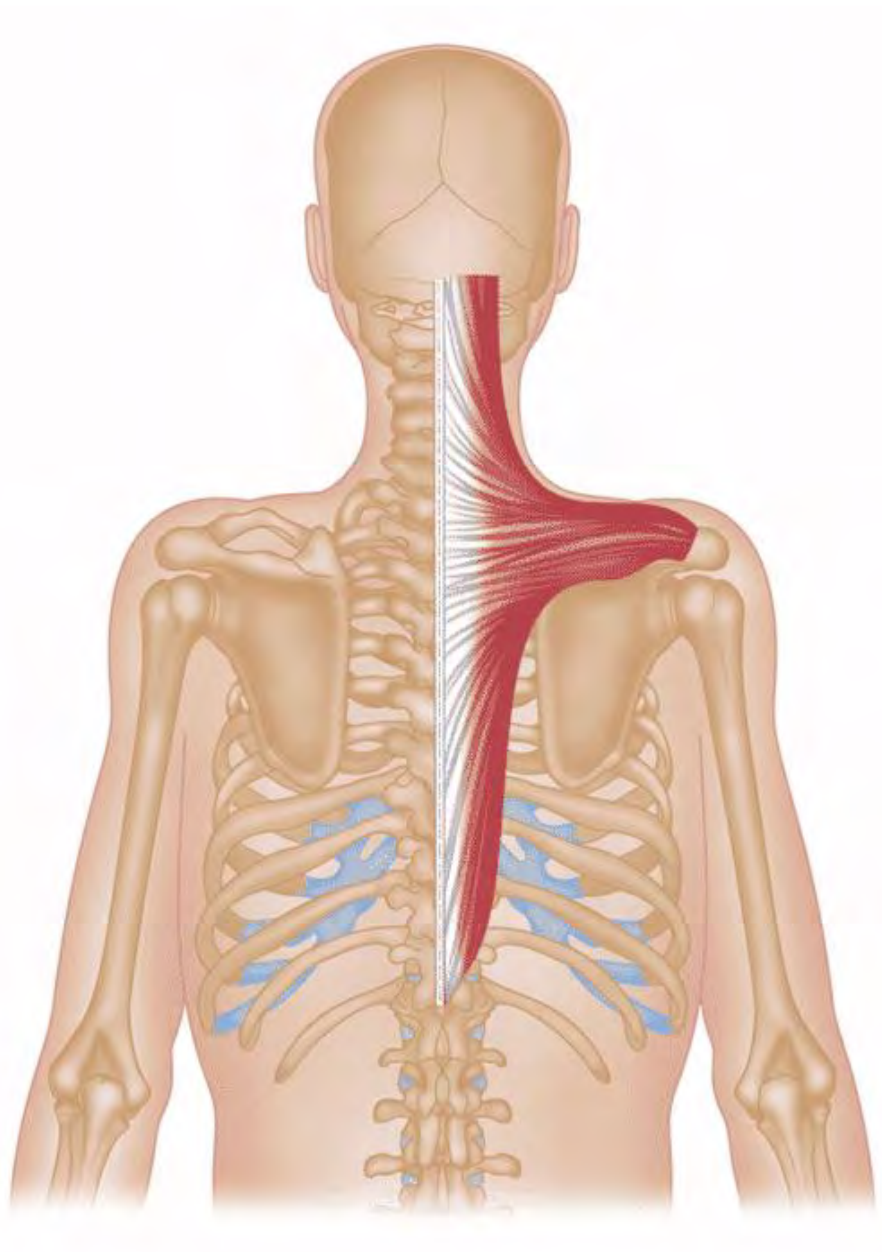
It is located underneath the pectoralis major.



Latissimus Dorsi

The largest muscle in the back and primarily responsible for adduction, extension, horizontal abduction, and internal rotation of the shoulder joint.

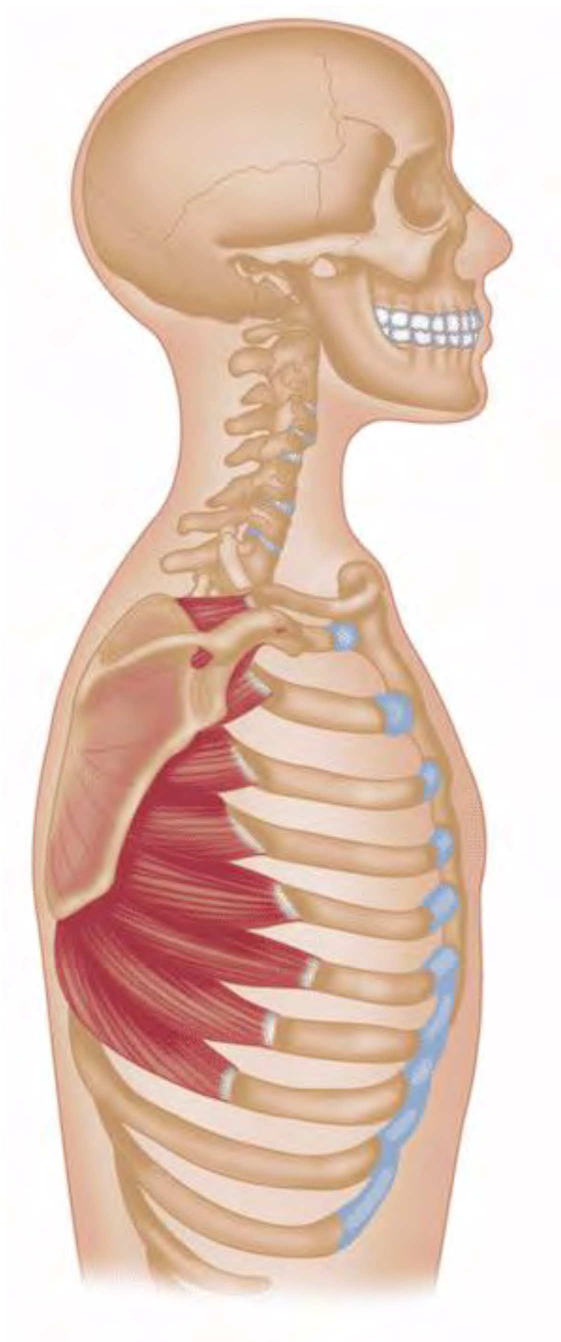
It also assists in stabilizing the LPHC.



Trapezius

The large, diamond-shaped muscle is on either side of the mid-upper spine. The primary action of the trapezius is to move the scapula and provide support to the upper arm.

The trapezius stabilizes, elevates, rotates, depresses, and retracts the scapula. The upper trapezius produces cervical extension, lateral flexion, and rotation.

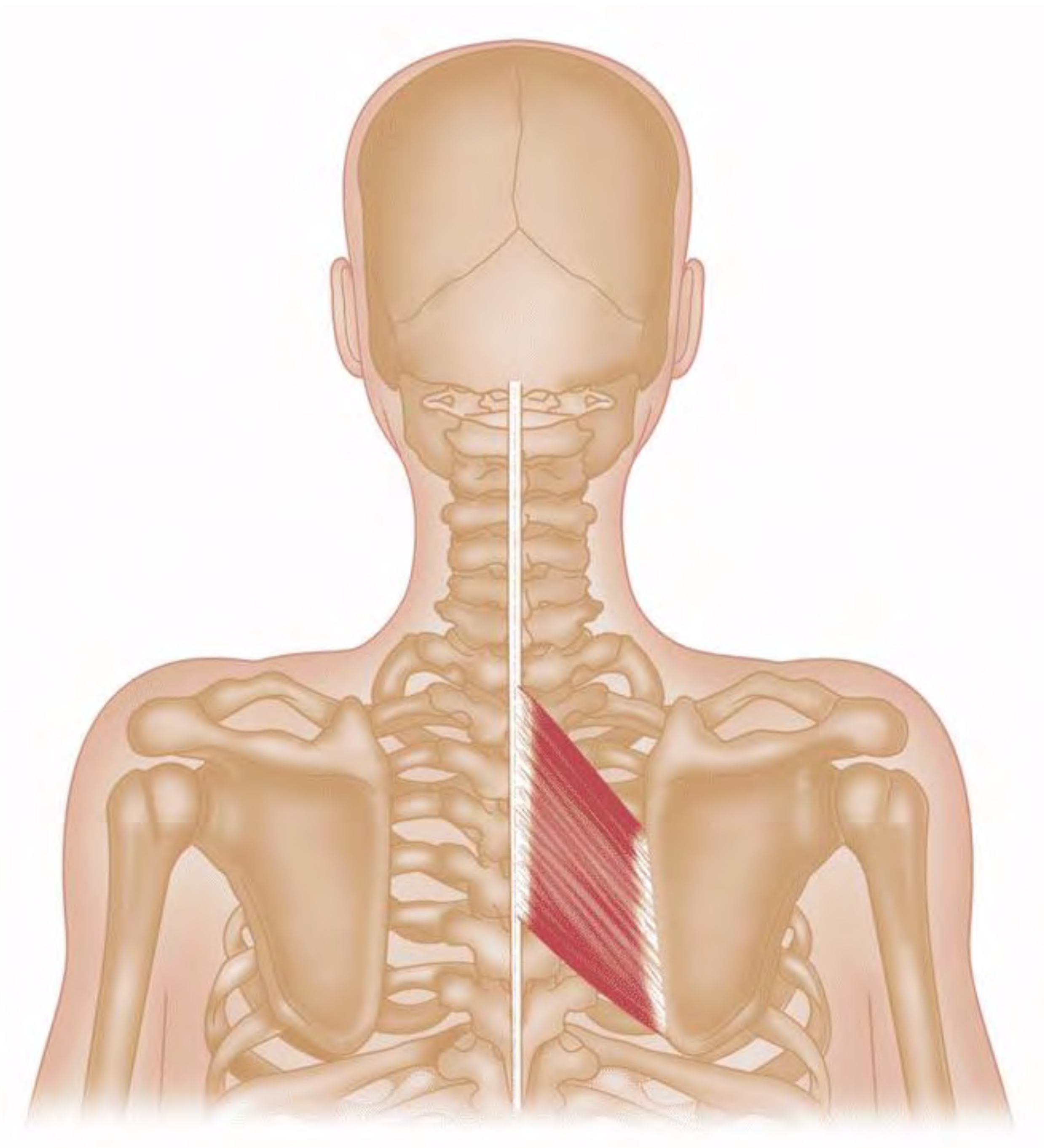


Serratus Anterior

It originates at the ribs and attaches to the medial side of the scapula. Its function is to protract, stabilize, and provide upward rotation of the scapula.

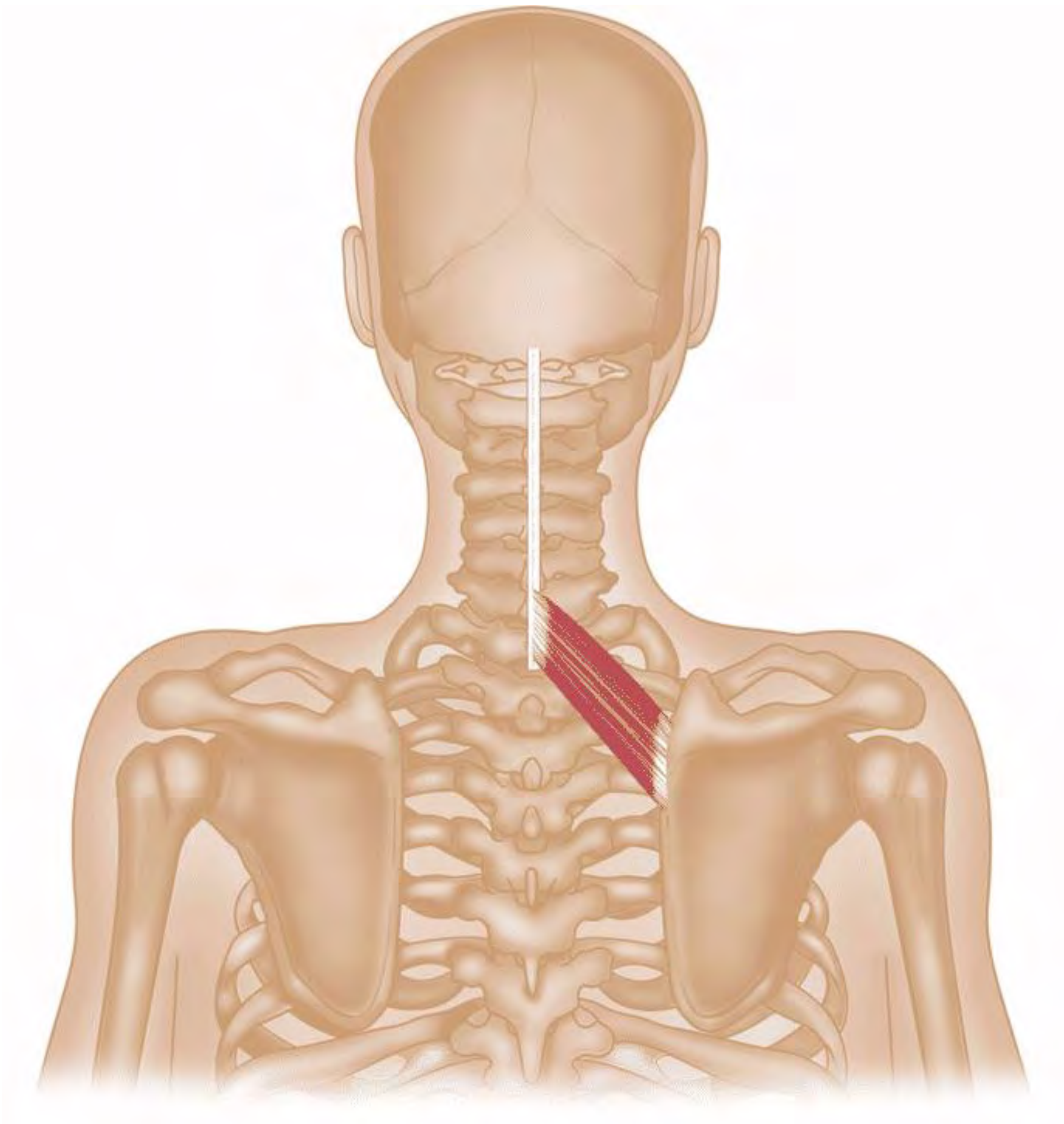
This muscle is often termed the boxer's muscle as it pulls the scapula forward, which occurs when throwing a punch.

Rhomboids



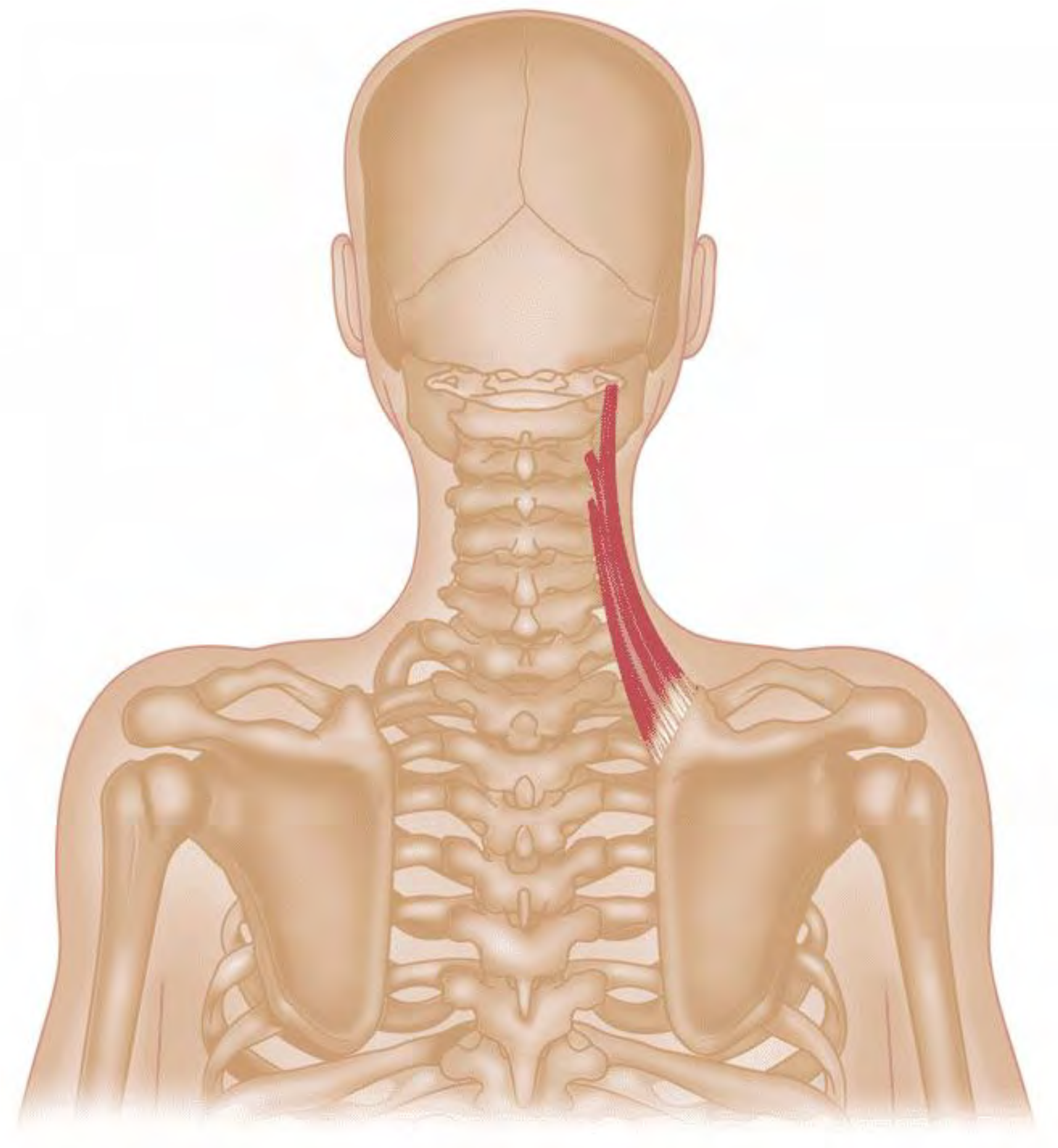
Rhomboid Major

It originates on the spine and attaches to the scapula. Stabilizes, retracts, and rotates the scapula.



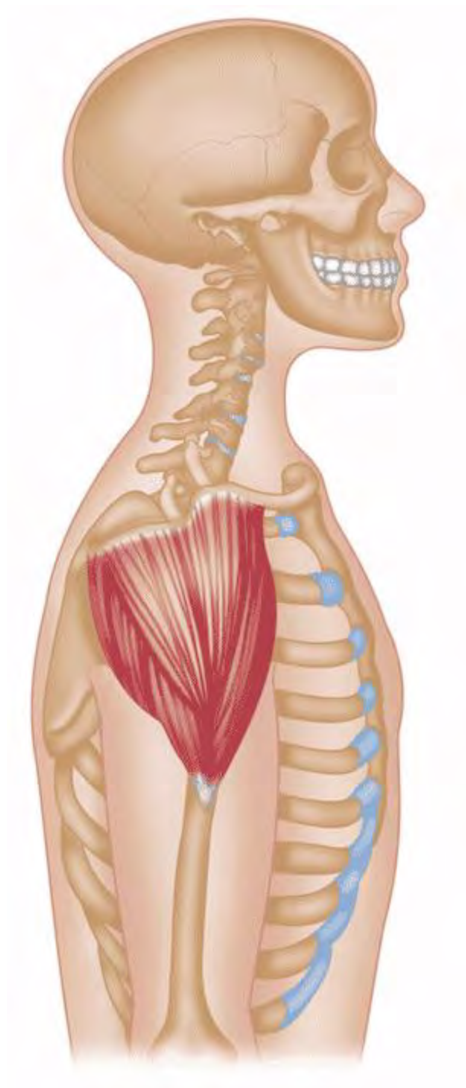
Rhomboid Minor

It originates on the spine and attaches to the scapula. Stabilizes, retracts, and rotates the scapula.



Levator Scapulae

Elevates the scapula—assists in scapular retraction, lateral flexion, and neck extension.



Deltoids (Shoulder)

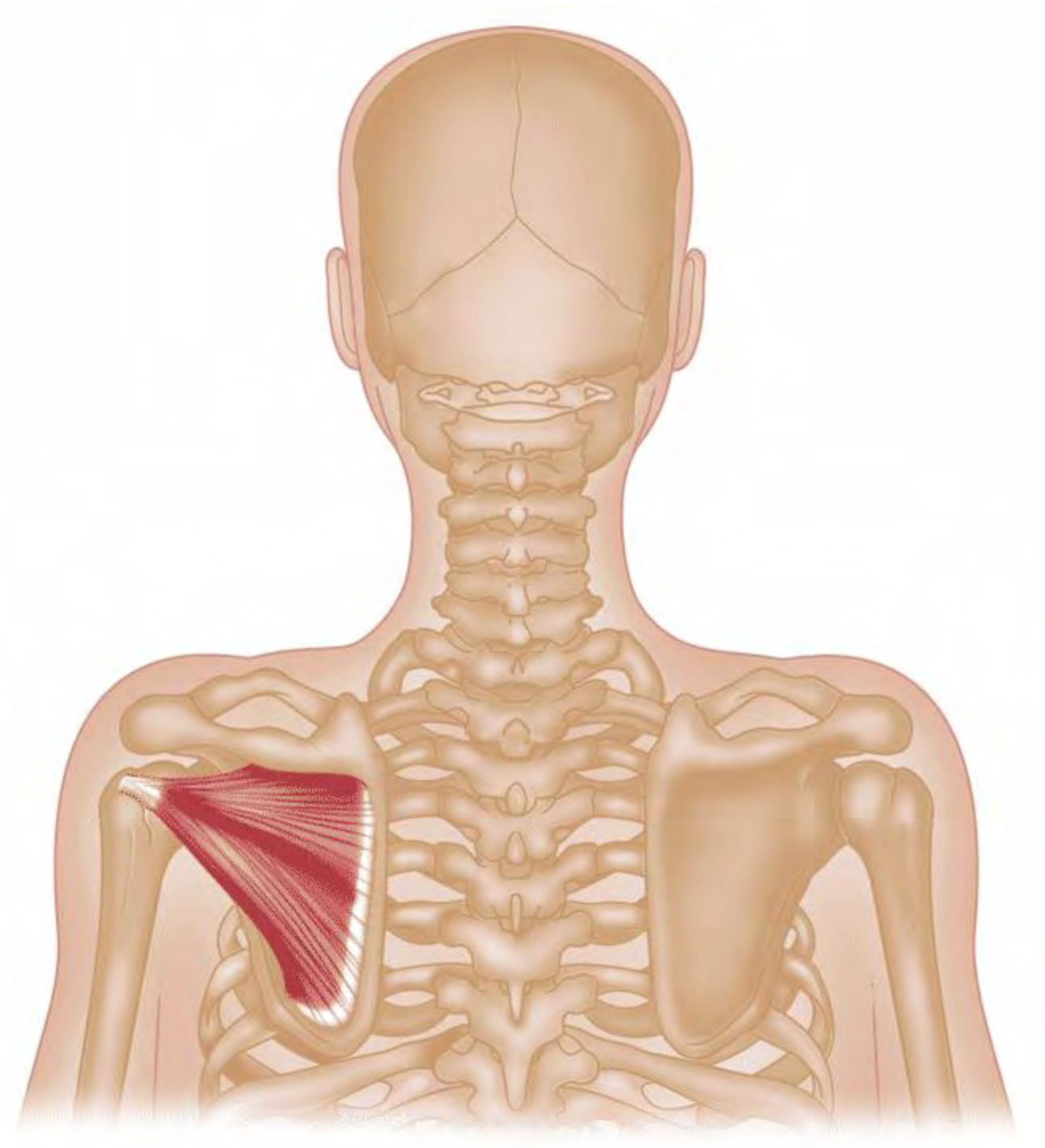
Many muscles act on the shoulder (glenohumeral) joint. For this certification, we will focus on the deltoid muscles (anterior, posterior, and lateral) and the **rotator cuff** muscles, which comprise the supraspinatus, infraspinatus, teres minor, and subscapularis.

The deltoid muscle consists of three main heads:
Anterior (shoulder flexion, internally rotate humerus), **Lateral** (arm abduction), and **Posterior** (shoulder extension).

The rotator cuff muscles provide static and dynamic support to the glenohumeral joint (159). The actions of the four rotator cuff muscles are noted in the following image.

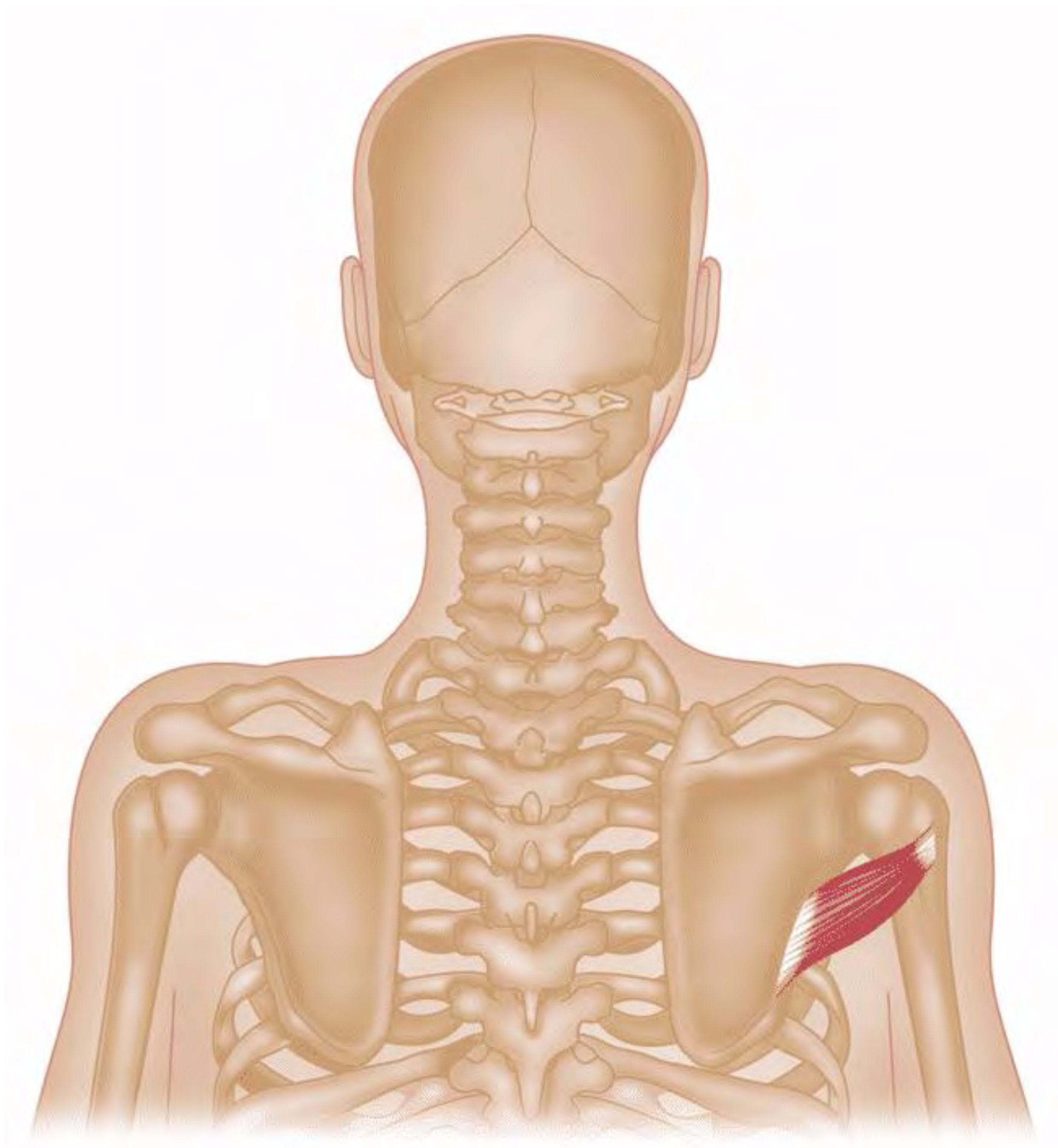
Rotator Cuff

The rotator cuff muscles provide mobility and stability to the glenohumeral joint. The rhomboids and trapezius provide mobility and stability to the scapula.



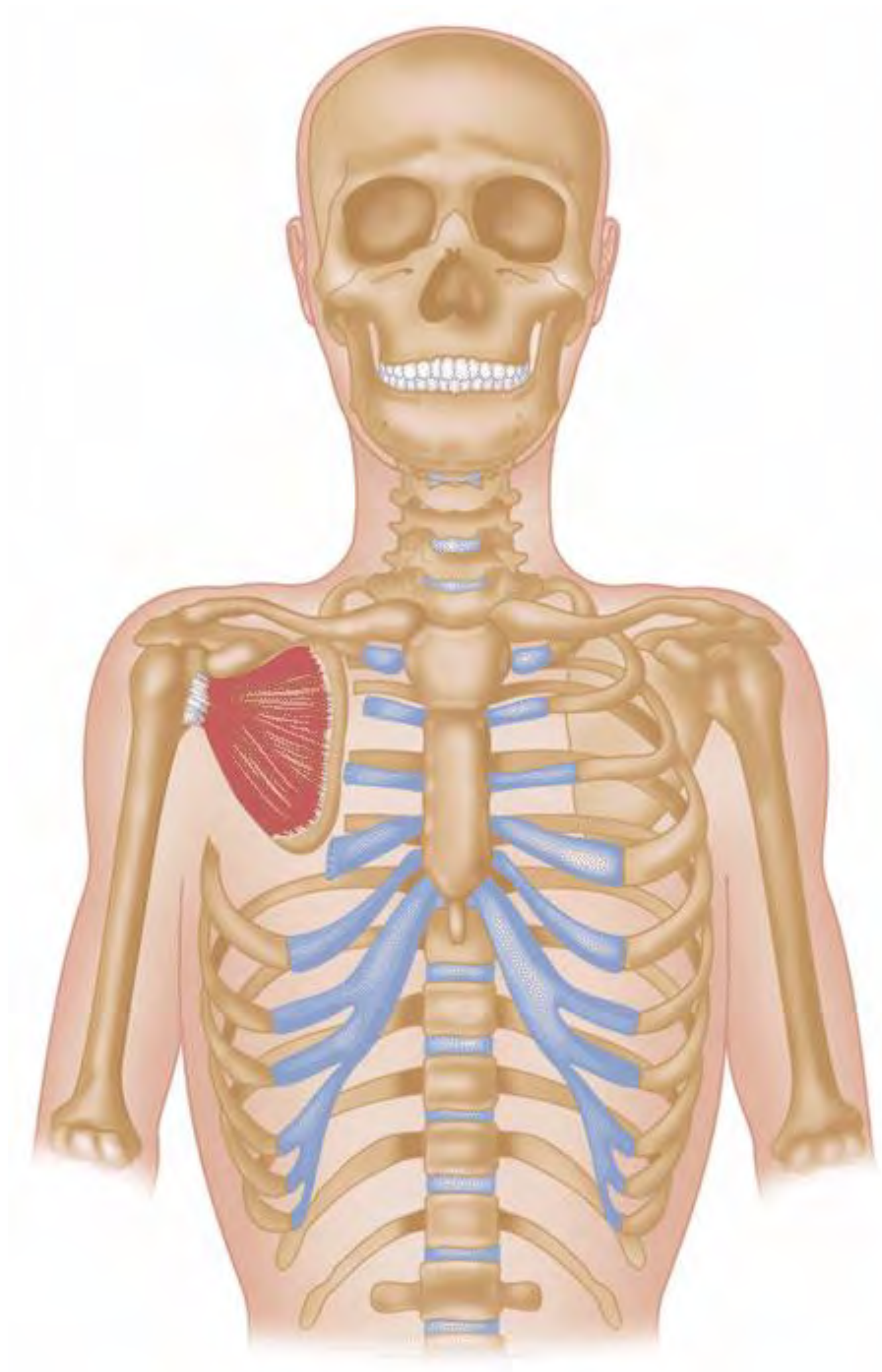
Infraspinatus

Action: External shoulder rotation



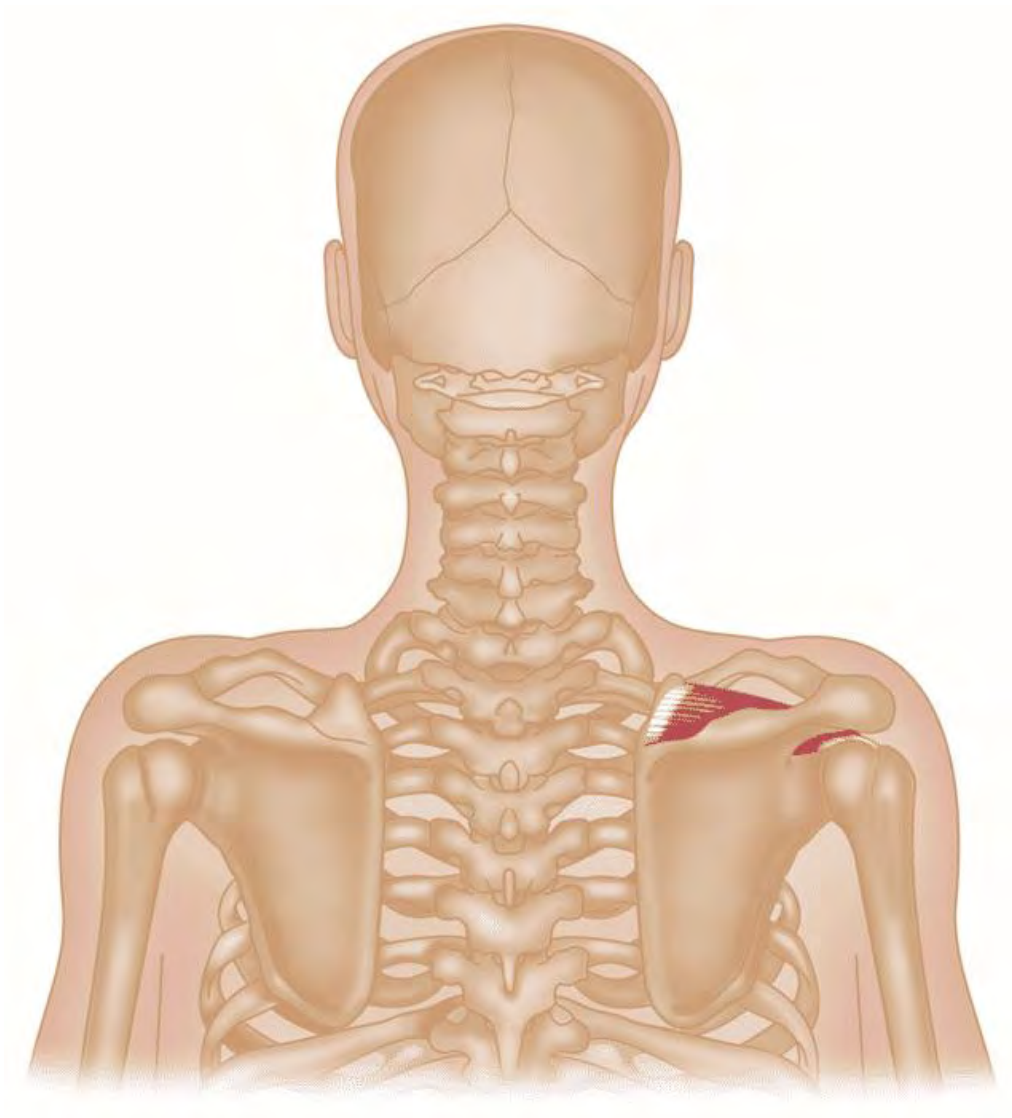
Teres Minor

Action: External shoulder rotation



Subscapularis

Action: Internal shoulder rotation



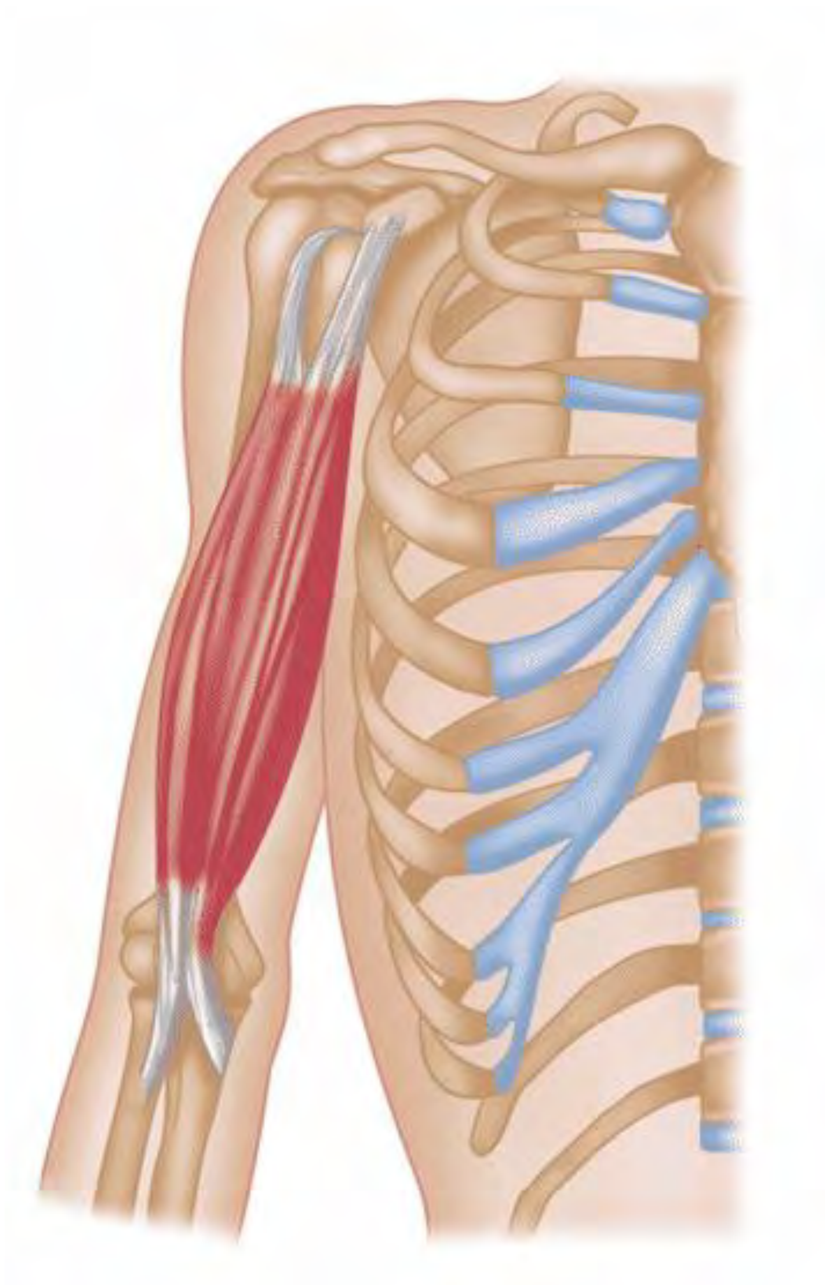
Supraspinatus

Action: Shoulder abduction

Arms

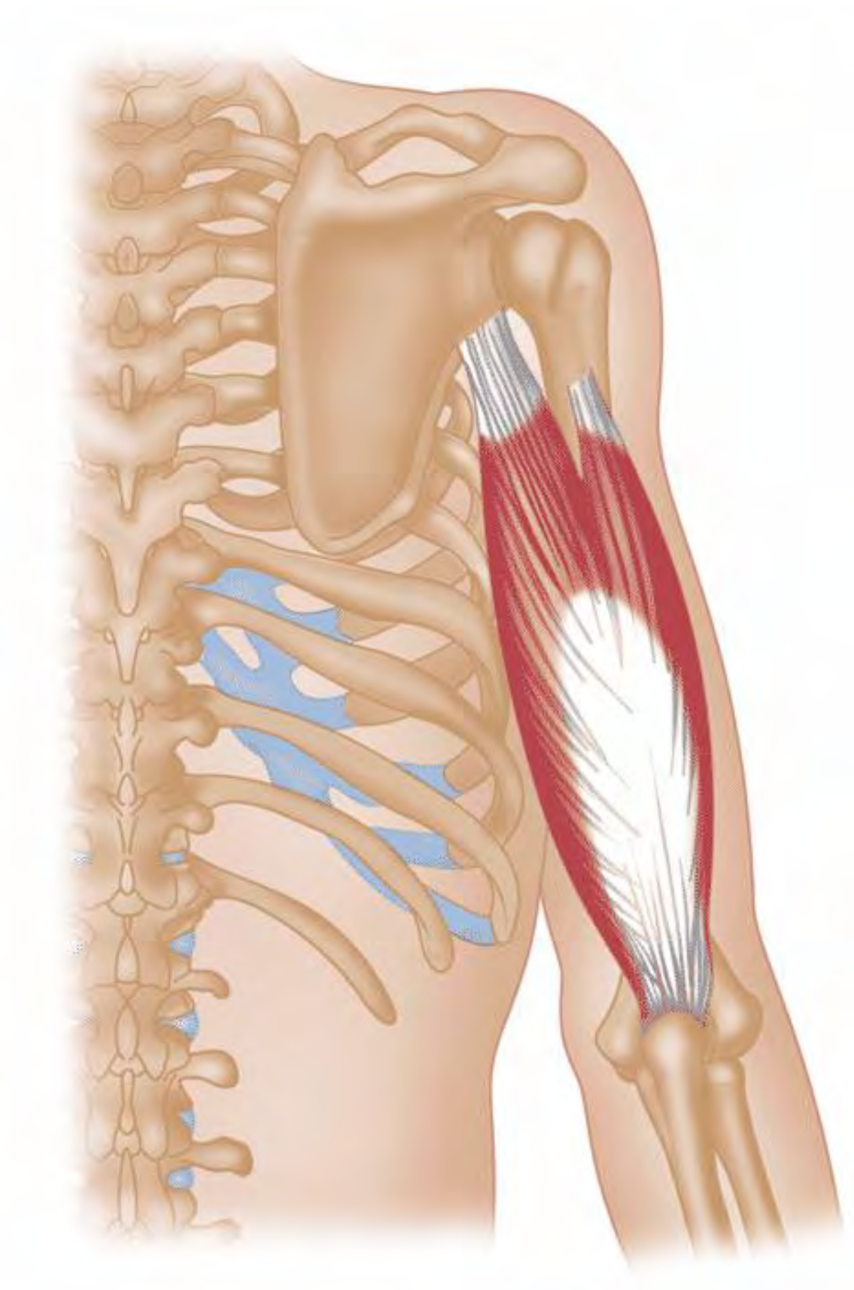
The arm muscles mentioned in the certification are the **Biceps Brachii**, **Triceps Brachii**, and the **Brachioradialis**.

In addition to these are over 20 muscles that make up the forearm that act to move the wrist, forearm, and digits (fingers). While not all the forearm muscles will be mentioned in this certification, it is important to know that they are responsible for forearm rotation and wrist and digit extension/flexion.



Biceps Brachii

Action: Elbow flexion



Triceps Brachii

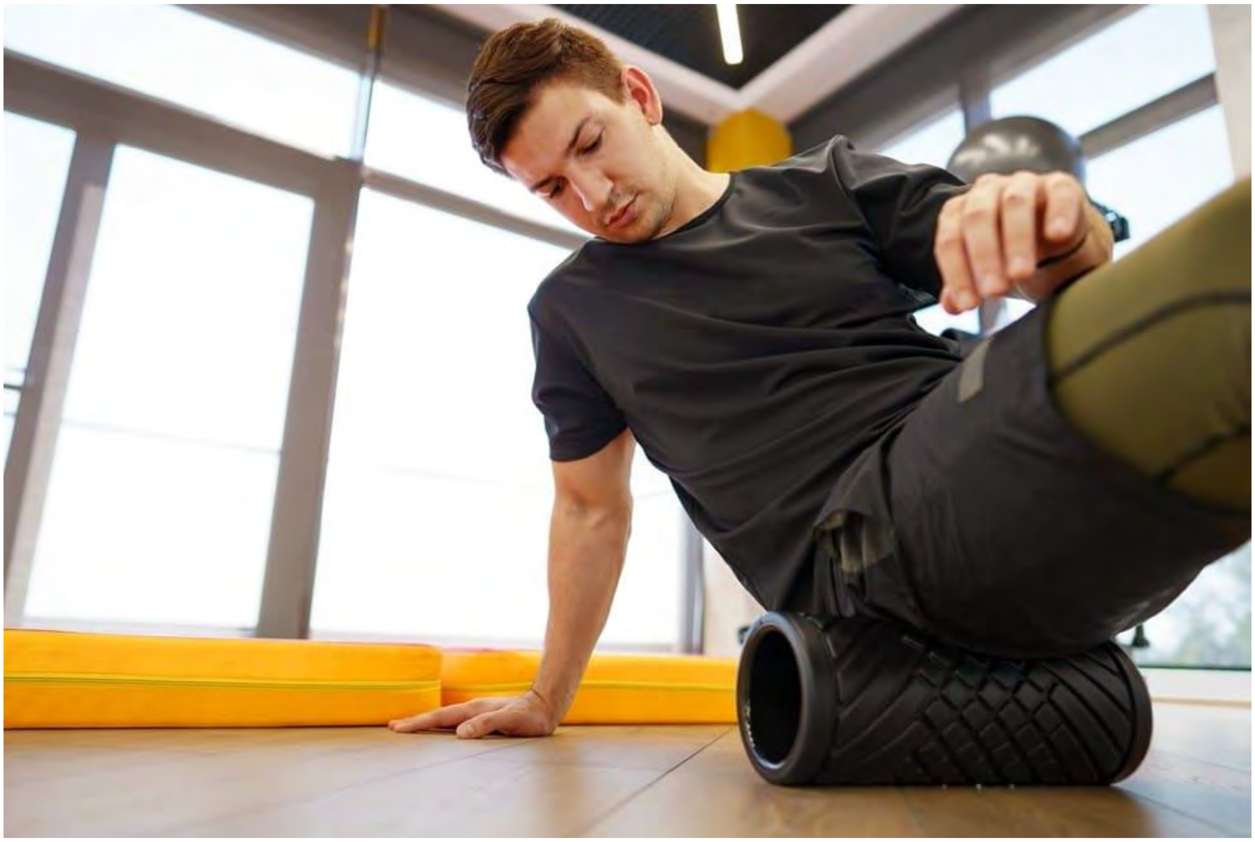
Action: Elbow extension

Brachioradialis

Action: Wrist extension

Intramuscular Performance Factors

This section discusses warm-ups/cool-downs, delayed onset muscle soreness (DOMS), ‘muscle burn,’ muscle synergy, and muscle imbalances.



Warm-Up

For shorter distance events such criteriums, a proper warm-up is a necessity because of the short distance and intensity of the race. The primary purpose of a warm-up is to increase one’s core temperature (625). Increased core body temperature may also decrease the chance of injury such as muscle strains (625, 626).

Below are several areas that a proper warm-up enhances:

- Increased force capacity of a muscle (627)
- Decreased chance for injury (626)
- Increased range of motion (628)

Physiology of the Warm-Up

Competitive and recreational athletes typically perform a warm-up to prepare for moderate to strenuous exercise. Doing so has been found to enhance performance and prevent injury. The primary mechanism of the warm-up is to increase the body temperature, which has a corresponding physiological response that has shown to do the following according to FG Shellock, WE Prentice Sport Med Journal.

- Increase in muscle blood flow
- Reduction in muscle viscosity
- Activation of metabolic chemical reactions
- Increase in speed of nervous impulses

There are three main warm-up techniques that when used together, can best prepare the body for intense physical activity.

1. Passive warm-up: increasing body temperature by an external source.
2. Generalized warm-up: non-sport-specific body movement to increase body temperature
3. Specific warm-up: increases temperature and circulation to sport-specific body parts associated with the most activated systems for a given activity.

Note: Of the three strategies, the specific warm-up has shown to significantly impact performance as it activates direct muscle groups used during a given activity. A warm-up strategy is particular to each athlete, their physical preparation, and environmental conditions that may require adaptations to warm-up strategies.

While little literature exists around the ideal warm-up protocol for endurance athletes, research has found that static stretching, followed by 5 minutes post-warm-up, peak power, time to peak power, and improved performance was present when compared to a control group that had just performed a moderate 5 minute on the bike cycling warm-up. The results of this study by DM O'Connor, MJ Crowe, WL Spinks, Journal of Sports Medicine, and Physical Fitness are shown below.

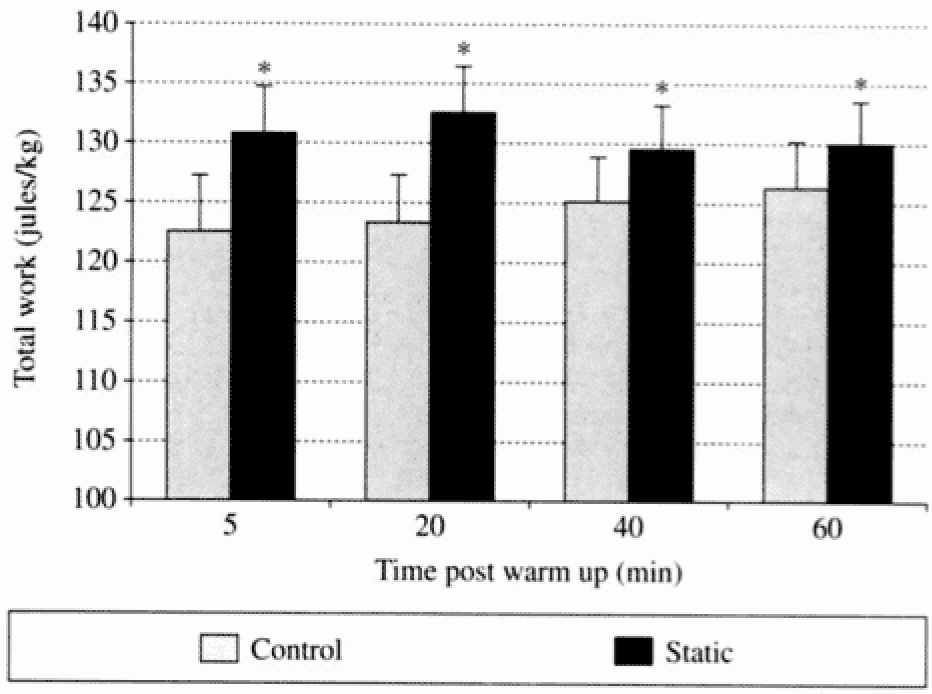


Figure 3.—Mean (+SE) relative total work (J/kg) generated during the 10 s leg power tests at 5, 20, 40 and 60 min following the control and static warm-up. * Significantly different from the respective control.

Stretching is typically incorporated into a warm-up, but research has shown contradictory evidence of its effect on performance. Research has shown it decreases injury, but there are differing conclusions regarding performance gains. Maintaining good flexibility aids in preventing injury to the musculoskeletal system by increasing the range of motion and muscle elasticity, meaning higher tension can be achieved on a muscle before damage occurs.

Stretching vs. Warm-Up: Are They the Same?

It is typical for stretching and warming up to be lumped into the same category. This is often the case as athletes are commonly instructed to warm up their muscle(s) before stretching to reduce the chance of injury while stretching. In this scenario, both can be considered part of the warm-up. However, as the name suggests, a warm-up primarily relates to warming the body, including increasing muscle temperature.

When researched in isolation, a warm-up reduced the incidence of lower-limb injury, whereas stretching alone did not (629). The primary reason for stretching is to increase the range of motion around a joint. This is pertinent as research has shown that stretching a muscle before a bout of exercise can reduce the contractile force of a muscle (608).

In other words, stretching can be part of a warm-up routine, but it is not advised in isolation.

Active vs. Passive Warm-Up

An active warm-up occurs when the changes in body temperature are due to muscle activity (i.e., running). In contrast, a passive warm-up is due to an external source (i.e., a heating pad). It is advised to perform an active warm-up rather than a passive warm-up.

A **Related Warm-Up** is a subset of an active warm-up and denotes warming up (via active warm-up) the same muscles used in the activity being performed and, ideally, in the same range of motion. This is the advised type of warm-up.

Warm-Up Guidelines

A proper active warm-up aims to find the sweet spot between the muscles becoming sufficiently warm but not physically taxing the individual. As a result, a proper active warm-up can be difficult to prescribe and perform.

Generally speaking, the more conditioned an individual is from a cardiovascular standpoint, the longer the warm-up will be. Research by Stewart and Sleivert found that **an active warm-up of 15 minutes and at an intensity of approximately 65 percent of one's VO2 max was most beneficial regarding anaerobic activities** (631). Another study by Tomares et al. found that regarding sprint track cycling, shorter and lower-intensity warm-ups elicited better performance than longer warm-ups (>20 min of cycling at varying intensities) (632). While these studies primarily examined a warm-up's effect on anaerobic activity, it seems to demonstrate that a long warm-up (>20 minutes) may have detrimental effects on an individual's energy and power output.

Try out different warm-up protocols in training to determine which protocol elicits the best result for your athlete(s).

Perhaps the most critical finding regarding warming up is not the exact intensity or duration of the warm-up but rather the time duration between the cessation of the warm-up and the start of the competition. **The benefits gained via increased muscle temperature are lost approximately 15 minutes after the warm-up period ends.** Therefore, the time between the end of the warm-up and the start of the competition should be no longer than 15 minutes (633). While not discussed, it stands to reason that the environmental temperature would impact the desired window between warm-up and event start (i.e., cold weather = shorter period than 15 minutes). This information is especially pertinent to races that will require a hard effort right from the beginning, such as a time trial.

Regardless of the warm-up duration and activity, muscles work better when they are warm versus cold. Muscles work optimally when the body core temperature is approximately 100.4°F and a muscle temperature of 101.3°F. However, like most aspects of human performance, the temperature at which muscles work optimally exists on a bell curve. For example, the contractile force of a muscle decreases once the muscle temperature goes above 103.3°F (760).

More discussion and review of running studies regarding warm-ups are noted in lesson 21 – Race Preparation and Execution.

Delayed Onset Muscle Soreness (DOMS)

While there may be muscle soreness and weakness directly after a workout or a bit the next morning, the most severe pain, or DOMS, usually occurs within 24– 48 hours following strenuous activity.

For many years DOMS was a mystery and not fully understood. Many studies have been published on this subject, yet most do not fully explain what DOMS is. In 1981, Jan Frieden published his findings based on taking muscle biopsies from subjects who repeatedly walked down flights of stairs. This was the first objective evidence of what DOMS is and what it is not (23).

Jan Frieden discovered that **DOMS primarily results from eccentric contractions of muscles** (23). An eccentric contraction is when a muscle contracts to decelerate a limb with a load. Essentially, an eccentrically contracting muscle acts as a brake.

For example, after performing a bicep curl, the bicep muscle must contract to decelerate the weight on the way back down to the starting position (arm extended downward). During an eccentric contraction, the muscle fibers get pulled lengthwise to the point where damage occurs at a subcellular level. This damage induces an inflammatory response in a muscle (260). The amazing thing about the body is that, given the proper amount of rest, the damaged muscles rebuild stronger than before.

Additionally, eccentric contractions involve fewer motor units than concentric contractions to elicit the same force production. Therefore, during an eccentric contraction, there is more significant stress per muscle fiber than during a concentric muscle contraction. This added stress per muscle fiber may contribute to DOMS because of disruption of the myofibrillar structures of the muscle (634).

While many people view DOMS as bad, it is a good thing in some respects. The pain that people feel with **DOMS is, in fact, muscle rebuilding rather than breaking down**. One way to look at DOMS is to relate it to the flu. While most people view the side effects of the flu as negative because they feel horrible because of high body temperature, chills, and aches, the side effects represent the recovery phase. The high temperature is the body's protective response to infection and injury. Just as one would not be active when having a temperature, one should also not stress the same muscles affected by DOMS while sore. As the soreness is representative of the healing phase, stressing the muscles affected by DOMS will inhibit healing and further break down the muscle fibers. Many athletes and coaches subscribe to the "no pain, no gain"' philosophy. However, when it comes to DOMS, **if you try to push through DOMS by exercising the affected muscles, you will end up doing more harm than good.**

DOMS is not limited to strength training. Any time a muscle is stressed to a level to which it is not adapted, damage can occur. DOMS is often the resulting aftereffect. Runners who increase their mileage too much on a particular day will often feel the effects of DOMS.

Potential Role of Estrogen on DOMS

From a biochemical standpoint, a marker for muscle damage (i.e., DOMS) is the enzyme creatine kinase (CK) in the blood. Research has shown that estrogen may affect muscle enzymes, lowering blood CK levels. Therefore, it can be theorized that women may have less severe bouts of DOMS than men.

Cause of Muscle Soreness and Weakness

The prevailing thought is that DOMS-related muscle weakness is due to sub-cellular muscle damage. However, research has discovered that undamaged muscle fibers can exhibit muscle weakness (635). Therefore, muscle weakness is likely due to a muscle's inability to activate contractile structures due to the over-stretching of sarcomeres (636).

As the time course of inflammation is typically the same as for muscle soreness, the soreness that is felt during DOMS is likely attributable to the inflammation response (637). As noted above, the over-stretching of the sarcomeres likely creates a loss of cellular calcium homeostasis, which is the reason for the muscle soreness (638).

Is a Cool Down Necessary?

A cool down is characterized by a significant reduction in intensity after an exercise session is over. The popular purported benefit of cooling down is bringing down the heart rate steadily while allowing the muscles to return to normal tension levels gradually. Is all this necessary?

It is theorized that warming up before exercise can slightly reduce DOMS, while a cool down does not reduce DOMS (335). According to Dr. Hirofumi Tanaka, an exercise physiology professor at the University of Texas, **the cool down is largely a myth** that like most myths in the fitness and exercise realms, keeps getting passed along without much thought given to its validity (336).

The one subject all physiologists seem to agree on is that after abruptly ceasing intense exercise such as running, the blood vessels in the legs are dilated, and therefore, blood can pool quickly in the legs and feet. This can make an individual feel dizzy and potentially pass out because of a lack of blood flow to the brain (336).

One could deduce that lying down with the legs elevated to reduce or eliminate blood pooling would be just as, if not more, effective than a short cool down after intense exercise.

Muscle Burn

For runners, one of the most feared byproducts of exertion is the dreaded burn they feel in their legs as the intensity and/or distance increases. So, what exactly causes this burn?

For many years (and still to this day for some people), the thought was that an increase in blood lactate within the body was the cause of the burning sensation. The primary reason for this belief was a study done in the 1920s by Dr. Otto Meyerhof. Using electric stimulation to a deceased frog's leg muscles, Dr. Meyerhof observed that the leg muscles of the frog twitched initially when subjected to the electric stimulation.

However, after a while, the twitching ceased. The cessation of muscle activation correlated with high levels of blood lactate. Thus, Dr. Meyerhof concluded that an increase in lactate causes muscular fatigue (24). This theory was challenged and ultimately put to bed by the research of Dr. George Brooks, who while studying for his Ph.D. dissertation, found the opposite of what Dr. Meyerhof had proposed was true.

Dr. Brooks discovered that blood **lactate is an important muscle fuel, not a waste byproduct that causes the burn**, as earlier thought (25). Pyruvate can be oxidized and used as fuel for the Krebs cycle, or it can be converted to lactate, which, in turn, is converted into glycogen in the liver via the Cori cycle. This increase in glycogen means that the glycogen stores are larger and last longer.

When a person begins to exercise, the oxygen consumption of the individual increases; however, at a certain point, an oxygen consumption plateau is reached. This plateau is often referred to as steady-state or steady rate. This means that there is a balance between the energy used by the muscles doing the work and the energy being created aerobically (more on this topic in the next module). During this steady-state condition, pyruvate accumulation is minimal. This is because any pyruvate (an acidic byproduct of glycolysis) being produced is either oxidized or converted to lactate, which is converted to glycogen, as previously mentioned (27).

Intramuscular lactate levels can elevate rapidly if the intensity of exercise is extremely high. Gaitanos et al. found that intramuscular lactate levels can increase by seven mM (a substantial increase) in just six seconds when performing high-intensity cycle efforts (559). Blood lactate levels increase to approximately 40 times their resting level from rest to maximal exercise intensity (566).

So, what causes the burn? **The burn results from a buildup of acidity in the muscle cells.** More specifically, this acidity is caused by the release of hydrogen ions during the fast turnover of ATP (26). This release of hydrogen ions within the muscle cells causes an interference with the ability of the muscle fibers to contract properly. Unless the intensity of the exercise decreases, the high level of burn (acidosis) will eventually cause cessation of the exercise (28). The role of lactate is to help buffer and neutralize the hydrogen ions, thereby delaying the burn (26).

While most athletes view the burning sensation as bad, it is the body's way of self-regulation so that it does not push itself too far and cause damage. The burning sensation is akin to a governor in a car motor that limits how fast the car can go. The theory of a governor is discussed in the Pacing module.

- Lactate threshold is discussed in a future module

Training to Delay Onset of Blood Lactate Accumulation

In endurance sports, a key marker for athletic performance is lactate threshold. Lactate threshold is the limit to which the body starts to produce more lactate than it can remove. This is where the "muscle burn" sets in. As blood lactate accumulates, overall muscle work capacity decreases until an effort falls below the lactate threshold or the activity is stopped. At rest, the body can eliminate or buffer lactate via release into urine, used as fuel by other organs like the brain, heart, and other muscles, or processed in the liver back to glucose.

The athlete that can delay this Onset of Blood Lactate Accumulation (OBLA) the longest usually corresponds with a better athletic performance. Delaying OBLA is achieved primarily through training, specifically training lactate threshold. The idea is to train just below the lactate threshold for extended intervals and complete rest between intervals. By intentionally focusing training sessions on lactate threshold development, an athlete can sustain a more strenuous effort for a longer duration before OBLA occurs.

Below is a sample progression of lactate threshold efforts to gradually increase power output at threshold.

Lactate Threshold Training Examples

- 10-minute warm-up with a total of 20-30 minutes at an endurance pace
- 3 X 10 minutes at 95% of threshold heart rate or power output with 8 minutes of rest

- 3 X 15 minutes at 95% of LT heart rate or power output with 10 minutes of rest
- 2 X 20 minutes at 95% of LT heart rate or power with 15 minutes of rest

Exercise-Associated Muscle Cramps

Previously thought to be the result of a lack of hydration and/or electrolytes, **the most recent research seems to indicate that exercise-associated muscle cramps (EAMC) are primarily the result of a muscle or muscles being overworked (i.e., fatigue).** If a muscle has a substantially greater load applied to it that requires a faster or stronger contraction than it is used to, or if a muscle is activated for longer durations of time without rest than it is used to, cramping is often the result.

For example, if you never do squats as an exercise and you proceed to do so with a heavy weight, cramps may likely result. Were you dehydrated? Likely no. Were your electrolyte levels low? Probably not.

According to Dr. Martin Schwellnus of the University of Cape Town, EAMC results from an abnormality of neuromuscular control in response to muscle fatigue – not a depletion of electrolytes (758).

In a 2011 study by Schwellnus et al., 210 triathletes were recruited to examine risk factors for EAMC (759). Of the 210 triathletes, 43 reported suffering from EAMC while 166 did not experience EAMC. Those who suffered from EAMC predicted faster race times and had faster times than those who did not suffer from EAMC. This is important because the two groups (EAMC and non-EAMC) had similar pre-race training preparation and performance histories. Lastly, the pre and post-race blood serum electrolyte concentrations and body weight were the same in both groups. **These findings suggest that dehydration and electrolyte balance are not a cause of EAMC, whereas exercise intensity is a major contributing factor.**

Muscle Synergy

Even the simplest muscle movement requires synergy between muscles for movement to occur. From a macro point of view, the human body is made up of many opposing muscles. Let us use the elbow as an example. The bicep muscle is responsible for flexing the elbow, whereas the triceps muscle is responsible for extending the elbow. When one muscle shortens, the opposing muscle lengthens, and vice versa. While this example illustrates only one movement of one joint, when an action requires a multi-joint, multi-planar movement, the muscle synergy is substantially more complex.

For a runner to be as efficient as possible, the relationships between working muscles need to be as synergistic as possible. **When this synergy breaks down for any reason, performance is diminished, and the potential for injury dramatically increases.**

Kinetic Linking

The term kinetic linking (also referred to as the kinetic chain) corresponds to the order in which muscles activate and work together to allow the body to move. From a performance-based standpoint, it also pertains to how muscles work to give an individual a biomechanical advantage.

Kinetic linking is all about optimizing performance. **The goal is to have the body move in the correct pattern to offer a biomechanical advantage.** This requires the muscles to fire in the proper sequence so movement can occur seamlessly and in the correct position(s).

Muscles and body parts do not work in isolation in functional body movement. Consequently, incorrect movement patterns in one area of the body may be caused by dysfunction in another area of the body.

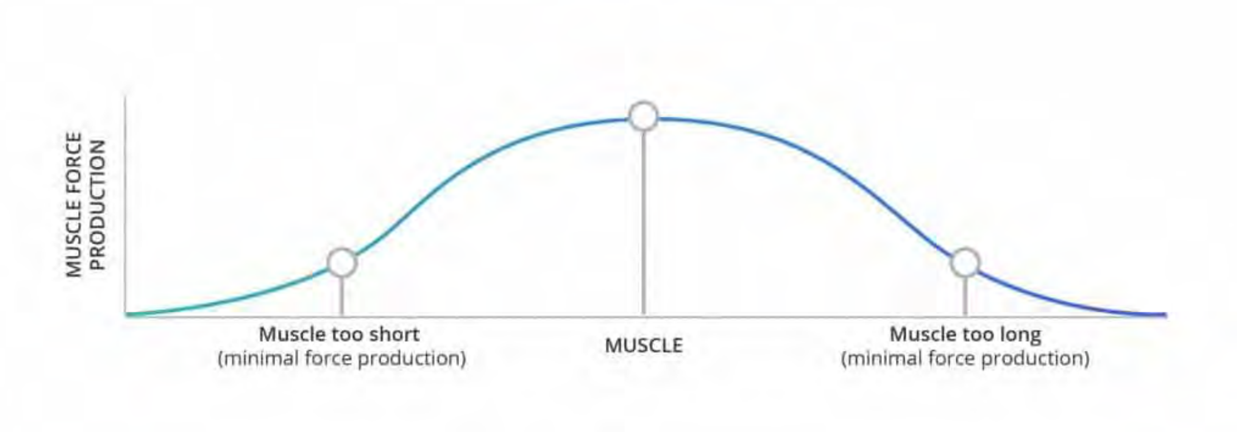
The term **shut down** is often used to denote muscles that contract minimally or not at all. Using the downstroke of the pedal stroke as an example, the glutes and hamstrings should fire. For athletes whose glutes are shut down, the glute muscles are not recruited to the degree they should be. Therefore, the hamstrings are forced to take over the role of the glutes. In essence, the hamstrings are performing two jobs at once. This can lead to muscle strains and tendonitis of the hamstrings. If your athlete cannot activate muscles through training or often incurs hamstring injuries, it is advised to refer the person to a physical therapist.

Length / Tension Relationship

A significant determinant of a muscle’s force production is its length. When the word length is used in this section, it has to do with the amount a muscle can stretch, not the actual length of a muscle.

- **Short Muscle:** Muscle is not elongated enough – or is too tight – to provide adequate force production
- **Long Muscle:** Muscle is too elongated – or stretched too long – to provide adequate force production. This is often termed an “inhibited” muscle.

If a muscle is shortened or lengthened too much, force production of the muscle will diminish to the point where minimal or no force production is possible (22). The relationship between a muscle’s length and the ability to produce force resembles a bell curve.



This is an important relationship as it works hand in hand with biomechanics. **No matter how great a training program may be, if a runner is not biomechanically efficient enough to maximize force production, the person will not perform optimally.**

Reduced Muscle Activation

When a muscle is too short or too long, impulses from the nervous system to the muscle are decreased. This decrease also affects the ability of the opposing (antagonist) muscle to contract properly. As you can see, this relates directly back to the muscle length/tension relationship (22).

For example, let us say the gastrocnemius muscle of a runner is exceptionally tight (i.e., short). This tightness will also inhibit the optimal stretch of the soleus. This might mean that the runner cannot extend their hip to the degree that it needs to due to limited dorsiflexion of the ankle.

Fatigue is also a factor in reduced muscle activation. **Muscle unit recruitment and firing rate decrease with fatigue** (562, 563).

Additionally, muscle co-activation decreases the muscular force of agonist and antagonist muscles surrounding a joint (564). It is theorized that a benefit of muscle co-activation is to stabilize a joint, as the opposing muscle forces of the antagonist and agonists effectively cancel each other out (565).

“Shut Down” Muscles

The term **shut down** is often used to denote muscles that contract minimally or not at all. Using the downstroke of the pedal stroke as an example, the glutes and hamstrings should fire. For athletes whose glutes are shut down, the glute muscles are not recruited to the degree they should be. Therefore, the hamstrings are forced to take over the role of the glutes. In essence, the hamstrings are performing two jobs at once. This can lead to muscle strains and tendonitis of the hamstrings. If your athlete cannot activate muscles through training or often incurs hamstring injuries, it is advised to refer the person to a physical therapist.

Muscular Compensation

This goes hand in hand with reduced muscle activation. Muscular compensation is typically caused by reduced muscle activation and results from a muscle (e.g., prime mover muscle) unable to produce the required force. Therefore, a synergist or support muscle takes over the primary muscle’s responsibility.

As noted earlier, muscle activation of the gluteal muscles is reduced for many endurance athletes, and the hamstrings assume force production. This leads to premature fatigue of the hamstrings and potential overuse injuries of the hamstrings as they are supposed to be synergist muscles to the glutes, not the primary movers. Nobody likes to do two jobs while getting paid for only one – muscles are no different!

Workplace Postural Issues

The type of job your athlete has is often reflected in what areas of the body are imbalanced. Many of your athletes likely have desk jobs. This means that for eight or more hours a day, they are in a seated position. If you look at the typical seated position, you will often see the following postural issues:

– **Hamstrings:** shortened
– **Hip Flexors:** shortened
– **Spine:** excessive spinal flexion in the lumbar and thoracic regions, which often results in excessive cervical spinal extension
– **Shoulders:** internally rotated and elevated
– **Core:** weak inner unit core musculature

As you can imagine, no matter what you do during your one-on-one sessions with your athlete, unless you have them stretch/strength train on their own and change their workplace posture, these issues will not improve. Therefore, your athlete needs to have an ergonomically correct workstation.

Ergonomics is the science of adapting postural and equipment changes at a workstation. For companies, the result of employees with poor workstation configurations is often decreased productivity due to lower back, neck, and wrist/forearm pain. For these reasons, many companies employ certified ergonomic specialists who ensure the workforce is set up at workstations correctly.

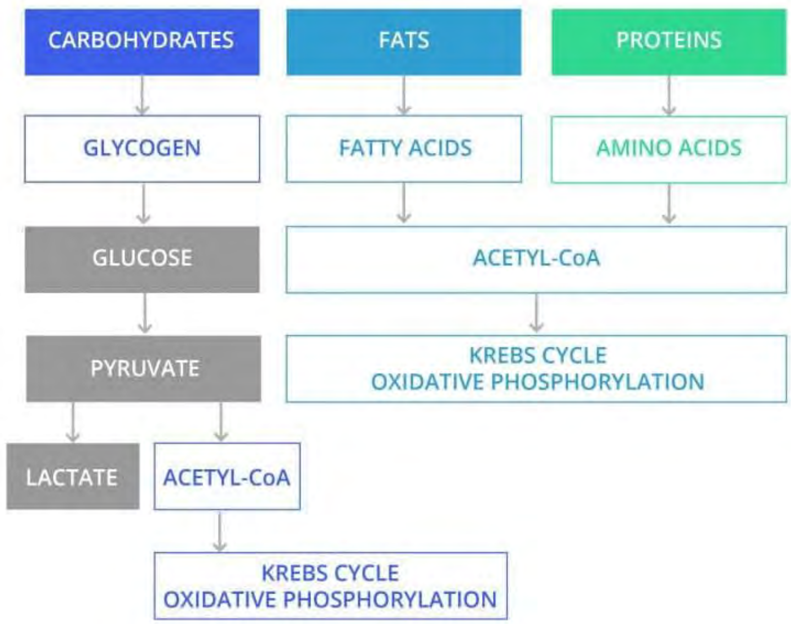
Summary

- Muscles do not have their own “memory.”
- A concentric muscle contraction relates to the shortening of a muscle
- An eccentric muscle contraction relates to the lengthening of a muscle
- Hypertrophy is an increase in muscle size
- Atrophy is a decrease in muscle size
- There are two types of muscle contractions: *voluntary* and *involuntary*
- Up to 70 percent of body heat is generated by the muscles
- The structure of skeletal muscle can be viewed as a thick cable that is made up of many smaller cables
- *Actin* and *myosin* allow the sarcomere to contract
- The *sliding filament model* demonstrates how sarcomeres and thus how a muscle contracts

- Understanding a muscle's origin and insertion is important in determining the muscle's effect on the body
- Type I muscle fibers are oxidative and are the muscle fiber primarily used by runners
- Hypertrophy refers to the increase in size of a muscle fiber whereas hyperplasia refers to the increase in the number of muscle fibers
- Nerve innervation (electric stimulation) of muscles is what makes a muscle contract.
- A muscle fiber cannot partially contract
- Tendons connect muscle to bone
- Ligaments connect bone to bone
- Fascia permeates the entire body, and based on the tension of the fascia, it can cause pain and postural abnormalities
- Core musculature is divided into an *inner unit* and *outer unit*.
- *Muscle burn* is the result of a buildup of acidity in the muscle cells. More specifically, this acidity is caused by the release of hydrogen ions during the fast turnover of ATP
- DOMS largely results from eccentric muscle contractions
- There should be no more than 15 minutes from the end of a warm-up period to the beginning of a running session
- As a coach, you are not allowed to advise or prescribe any medications to your athlete – including NSAID's
- Muscles work together in a set pattern to elicit body movement in what is called the *kinetic chain.*
- Exercise-induced muscle cramps are the result of muscle fatigue, not dehydration or low electrolyte levels
- Poor workplace posture often creates and magnifies muscle imbalances
- If a muscle is too long or too short, the muscle's force production will be greatly diminished.
    - If a muscle's force production is greatly diminished, this often requires another muscle(s) to assist in force production. This is termed *muscular*

Module 4: Energy and Cardiopulmonary Systems

This module discusses the circulatory and cardiovascular systems, energy systems, measurements of intensity, and the physiological effects on the body due to intense exercise.



Upon completion of this module, you should have an understanding of the following areas:

- Energy system terminology
- Circulatory system
- Function of lungs
- Role of cells in energy production
- Cori Cycle
- Three energy systems
- Blood glucose regulation

• Physiological response to aerobic training

• Effect of hypoxic training

• Physiological effects of intense exercise

• Glycogen depletion

Terminology

Anaerobic: This term means *without oxygen*. This is a slight misnomer, as being in an anaerobic state typically has more to do with a lack of utilization of oxygen rather than an absence of oxygen.

Adenosine Triphosphate (ATP): A molecule responsible for storing and releasing energy in the body. Biologists have coined ATP the *"currency of life."*

ATP is generated in the mitochondria of cells and can be produced both aerobically and anaerobically.

The body stores only a small amount of ATP at any given time. Therefore, ATP must be continually regenerated on a cellular level.

When ATP is broken down for energy, one of the phosphates breaks away from the molecule. Since there are only two phosphates, the molecule is now adenosine diphosphate (ADP). Through a process called *phosphorylation*, a phosphate is added to the ADP molecule to create ATP. Phosphorylation can occur aerobically or anaerobically.

Mitochondria: An organelle that is the power plant of cells; converts potential energy from food molecules into ATP

Glucose: A simple sugar in the blood

Glycogen: The main form of carbohydrate storage in the body (muscles/liver) and can be converted to glucose

Glycolysis: Metabolism of glucose.

Pyruvate: An acid that is the result of glycolysis.

Krebs Cycle: A cycle of complex chemical reactions that cells go through in the presence of oxygen to create energy (cellular respiration). They are also referred to as the *citric acid cycle*.

Lactate: A substance found in body tissue and a byproduct of pyruvate. During high-intensity exertion, when oxygen supply is limited, pyruvate produces lactate, which allows for the breakdown of glucose for energy. At lower intensities, lactate is used as fuel by the mitochondria. Lactate also helps to buffer cellular acidosis *(refer to "Muscle Burn" in the previous module for more information on buffering cellular acidosis).* Lactate is produced at all times.

Lactate Threshold: Representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.

Maximum Lactate Steady State (MLSS): The highest intensity level at which blood lactate concentrations are maintained at a steady-state (equilibrium) level during exercise bouts of approximately 60 minutes.

Onset of Blood Lactate Accumulation (OBLA): Represented when blood lactate levels reach four millimoles per liter (mmol/l) during exercise bouts (125). Since this can be ascertained only by using a blood lactate analyzer, this assessment will not be used in this certification.

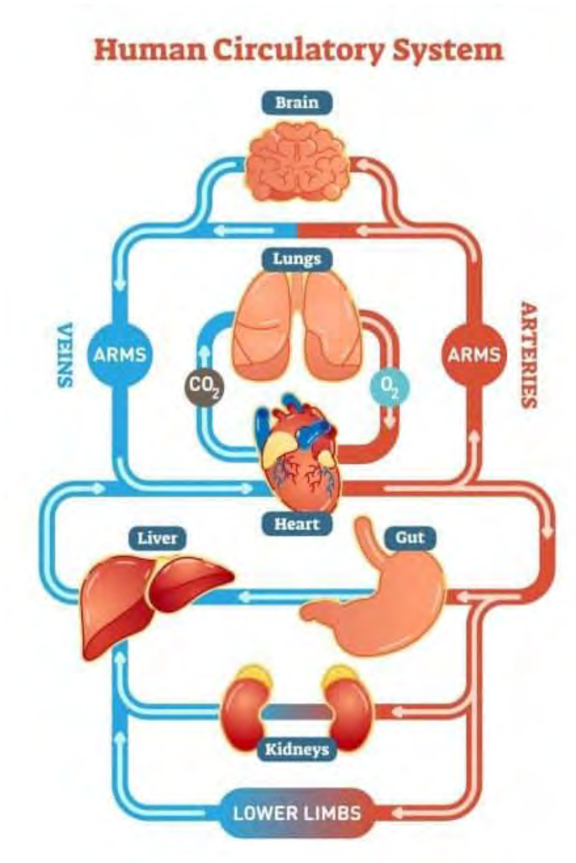
Acetyl-CoA: A molecule that results from the oxidation of fatty acid, amino acids, and pyruvate. Acetyl-CoA is broken down and used for energy production in the Krebs cycle.

Creatine Phosphate (CP): A high-energy phosphate compound that aids in the resynthesis or regeneration of ATP.

Ventilatory Threshold (VT): The point at which the ventilation (breathing) rate increases faster than the workload. Until the VT is reached, the workload and respiration rate increase linearly.

Note – You will not be tested on formulas or specific energy system processes (ex: NADH + Pyruvate). However, they are noted to enhance your understanding of energy systems.

Circulatory System



There are two main circulatory systems within the body

1. **Pulmonary:** This system involves the heart and the lungs. Deoxygenated blood leaves the heart and goes to the lungs, where it is reoxygenated and returned to the heart.
2. **Systemic:** This system relates to circulation throughout the body except for the lungs. Oxygenated blood leaves the heart via arteries to supply the body, and the deoxygenated blood comes back to the heart via the veins.

Muscles rely on a steady oxygen supply to perform aerobic activity. Oxygen is transported to and from muscles via blood vessels. There are three primary classifications of blood vessels in the body:

1. **Arteries:** Carry oxygenated blood away from the heart to the rest of the body.

2. **Veins:** Carry deoxygenated (without oxygen) blood to the heart.
3. **Capillaries:** The smallest blood vessels in the body. They increase in density based on how much blood supply is needed in a particular area. They are found in the human body's tissues and transport blood from the arteries to the veins. Because of the thinness of the walls of capillaries, O_2, CO_2, waste products, and nutrients can pass through the walls. The process through which this occurs is called *diffusion* (301).

In addition to providing core muscle stability, the *diaphragm* plays a significant role in the circulatory system. This sheet-like muscle contracts and relaxes to facilitate breathing. When the diaphragm contracts, oxygen is pulled into the lungs, and when the diaphragm is relaxing, carbon dioxide (CO_2) is expelled from the lungs.

The cardiovascular system is a "closed" system, meaning blood does not leave the vessels at any point. As noted in the preceding description of capillaries, diffusion allows oxygen and nutrients to pass through the vessel walls and into cells. Conversely, waste products and CO_2 can pass from the cells through the vessel walls. The space between blood vessels and cells is occupied by a fluid substance called *interstitial fluid* (i.e., *tissue fluid)* (302).

There are three types of blood cells: *red, white,* and *platelets*. Below are their primary functions:

- **Red:** carries oxygen to the body
- **White:** helps the body fight off infection
- **Platelets:** clots blood

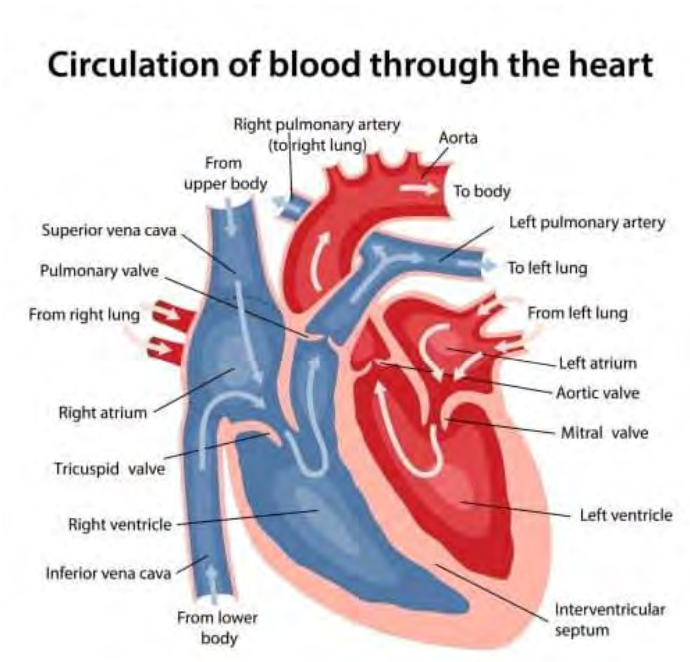
Hemoglobin is a protein that is found in red blood cells. Its function is to transfer oxygen from the blood to the muscles. ***Myoglobin*** is a protein found in muscle that receives oxygen from the blood (red blood cells) and transports it to the mitochondria of the muscle cells (726).

Blood (i.e., red blood cells) goes where it is most needed. As a result, the legs are the primary recipients of blood volume when running. At rest, 15–20 percent of one's total blood volume goes to muscles, whereas during exercise, 85–90 percent of blood volume goes to muscles (727). The process by which this occurs is called ***blood shunting***. This process dilates and constricts the arteries to determine what percentage of blood is received by working muscles.

Heart

This organ is the pump that supplies the body with blood and, therefore, oxygen. The heart is divided into four chambers.

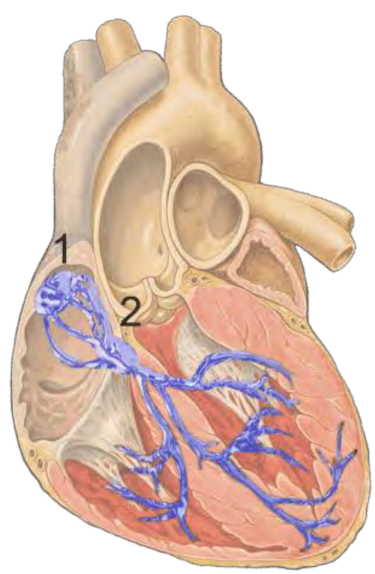
- Right Atrium
- Right Ventricle
- Left Atrium
- Left Ventricle



The four chambers of the heart are illustrated in the above image (right and left atrium/ventricle). The blue areas denote deoxygenated blood (veins) and the red regions denote oxygenated blood (arteries). The circulatory system process:

1. The right atrium receives deoxygenated blood via the superior and inferior vena cava (veins).
2. Blood passes from the right atrium through the tricuspid valve into the right ventricle, which is pumped to the lungs to be reoxygenated.
3. Reoxygenated blood returns from the lungs to the heart (left atrium).
4. Blood passes from the left atrium, through the mitral valve, and into the left ventricle.
5. Oxygenated blood in the left ventricle is pumped out of the aortic valve and supplies the body with oxygen.

Heart contractions are caused by electrical impulses at the heart's **sinoatrial node (SA node).** This is the heart's natural pacemaker. The SA node sends an electric signal across the atria (i.e., left and right atriums) that causes a contraction. This electric signal also influences the ***Purkinje fibers***, which stimulates the ventricles to contract. The Purkinje fibers are represented in the figure below by the blue branches in the middle to lower parts of the heart (358).



1 – SA node
2 – AV node

If the SA node fails, the **atrioventricular node (AV node)** takes over the pacemaking responsibility. Lastly, the Purkinje fibers can act as a weak pacemaker if the AV node fails. The AV node primarily acts to regulate the signal from the SA node (357). The heart's rate of contraction is a function of the *autonomic nervous system* (ANS). This demonstrates the heart's built-in contingency system in case the SA or AV nodes fail.

Blood flow allocation within the body is based on need. For example, a runner would have greater blood flow to the legs than the arms because of the greater demand for oxygen in that region. This is most notable in individuals exposed to extreme cold for an extended period. To keep the body functioning, blood flow is concentrated to life-sustaining organs and diverted away from the extremities.

When exercising, heart rate is influenced by one's aerobic conditioning level and movement intensity. **The greater one's aerobic capacity, the more blood is ejected with each heart contraction, which translates to a lower heart rate than someone not as aerobically conditioned.** An increase in heart rate correlates with increased physical activity because of the body's increased demand for oxygen.

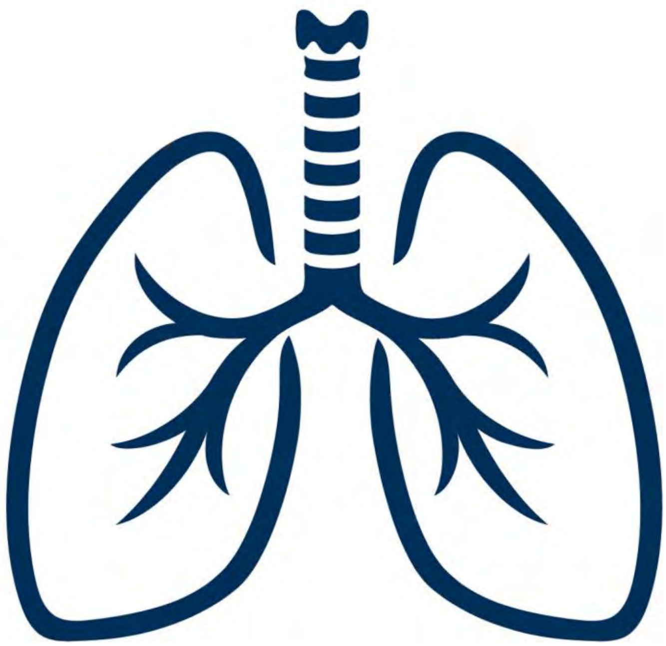
Blood Pressure

A standard measure of health, blood pressure represents the amount of pressure against the blood vessel walls. A blood pressure reading has two numbers, a top and bottom number. Normal blood pressure is 120/80. The top number is called the *systolic pressure* and represents the amount of pressure against vessel walls when the heart is contracting. The bottom number is called the *diastolic pressure* and represents the pressure against the vessel walls when the heart is relaxing.

The numbers represent millimeters of mercury (mmHg), the unit used to measure blood pressure. When taking one's blood pressure, the systolic number is the first beat heard, and the diastolic is the last one heard.

Under resting conditions, high blood pressure is considered 140/90 or higher. However, during exercise, it is normal for blood pressure to increase substantially. Those with high blood pressure are often termed ***hypertensive***. If your athlete is hypertensive, the person should be under a physician’s care and guidance. Hypertension is typically managed by diet, exercise, and medication.

Lungs



The lungs are located on either side of the heart. Deoxygenated blood is pumped from the heart to the lungs via the pulmonary artery. In the lungs, oxygen is diffused into the blood, and carbon dioxide is transported out of the blood, where it is exhaled. This exchange of oxygen and carbon dioxide occurs in the alveoli, tiny air sacs in the lungs. Air (oxygen) is inhaled through the trachea (windpipe) and ends up in the alveoli, where it is used to reoxygenate the blood. The reoxygenated blood is returned to the heart via the pulmonary veins, where it is then pumped out of the heart and into the systemic circulatory system.

The diaphragm’s contraction and relaxation is the primary way the lungs inflate and deflate. As noted previously, when the diaphragm contracts, the diaphragm moves downward, allowing the lungs to fill with air. Conversely, relaxation of the diaphragm causes the air to flow out of the lungs (*think:* balloon deflating and losing its air).

Diaphragmatic Breathing

This type of breathing also referred to as ***belly*** or ***abdominal breathing***, is the most efficient way to breathe and relies on taking a full breath to make the diaphragm move downward and upward through its full range. Diaphragmatic breathing is characterized by the abdominal section of the body rising as the breath is inhaled. This is in contrast to *chest breathing* in which the lungs expand and make the chest rise.

The primary reason why diaphragmatic breathing is more beneficial over chest breathing (especially to athletes) is because more oxygen is inhaled, and more carbon dioxide is exhaled with each breath. This equates to more oxygen for the working muscles (440).

Introduction to Energy Systems

The energy system classification used by the body during exercise is determined by exercise intensity, duration, and fuel availability. The body's energy systems largely function as a result of chemical reactions. Knowing all the specific chemical reactions that occur is not necessary for this certification. However, it is important to know and understand the results of the chemical reactions on a macro level.

Understanding how the energy systems work in the body is paramount to being a successful coach. By understanding these systems, you will be better able to design effective programs related to intensity.

While endurance sports focus primarily on the aerobic (oxidative) aspect of training, anaerobic and neuromuscular aspects also contribute to an endurance athlete's performance (589–591).

The body utilizes three classifications of energy systems during exercise:

- **Anaerobic Alactic: ATP-PC (phosphocreatine) system**– Provides ATP for ~10sec of maximal effort (913)

- **Anaerobic Lactic: Glycolysis** – Provides ATP for ~60-90sec of intense effort (914)
- **Aerobic: Oxidative phosphorylation** – The Krebs cycle and electron transport change – provides ATP for prolonged sub-maximal efforts (914)

While all energy systems start simultaneously, only one system typically provides most of the energy at any given time. Therefore, when this certification denotes the first, second, and third energy systems, it refers to the energy system performing the majority of work.

The terms ‘aerobic’ and ‘anaerobic’ are often thrown around concerning when one stops (aerobic) and another starts (anaerobic). There is no switch in the body that does this. As noted above, while one energy system for a particular intensity level of exercise provides the majority of the energy, that does not mean that the other energy systems are ‘shut off.’ Therefore, **focusing on an intensity to target a particular energy system will improve others, just to a lesser degree.**

At the most basic level, energy systems represent the body’s ability to convert chemical energy stored in food into mechanical work. **When discussing energy systems, we are talking about how and in what manner ATP is produced.** This is because for a muscle contraction to occur, ATP is required. The greater the amount of ATP supplied to working muscles, the longer the muscles can work. Conversely, the less ATP, the shorter the exercise session will be.

This relates to the two primary ways ATP is produced. **If oxygen is not utilized, then ATP will be produced anaerobically, and if oxygen is utilized, then ATP will be produced aerobically.** When ATP is produced anaerobically, it does not elicit a high number of ATP molecules for the working muscles to use as energy. However, it can produce a lot of energy for a short period. When ATP is produced aerobically, many more ATP molecules are produced compared with the amount produced anaerobically. This allows exercise to continue for much longer before fatigue occurs. Endurance sports such as running are primarily aerobic activities.

While ATP levels fluctuate based on the intensity of exercise, the levels of ATP never get to levels that would be considered dangerously low (561). While it is theorized that muscle can match ATP production and consumption, ATP levels are not stable during exercise, especially during high-intensity bouts (560).

Cells

The body's cells take in raw material (nutrients from food) and convert them into energy. The macro view of how this process occurs is as follows (509):

1. Nutrients are broken down within cells
2. ATP is formed
3. When needed, ATP is broken down. This breakdown, or catabolism of ATP, creates usable energy for the cell.

Bioenergetics

Adenosine triphosphate (ATP) is created by combining adenosine diphosphate (ADP) and inorganic **phosphate** (Pi). This process **consumes** energy. The process of adding a phosphate group is called **"phosphorylation."**

$$\text{ADP} + \text{Pi} + \text{Energy} \longrightarrow \text{ATP}$$

The chemical bond between ADP and Pi stores energy, and when the enzyme ATPase breaks this bond, energy is **released** for muscular contraction.

$$\text{ATP} + H_2O \underset{\text{ATPase}}{\longrightarrow} \text{ADP} + \text{Pi} + H^+ + \text{Energy}$$

In the muscle cells, only enough ATP is stored for about 15sec of moderate exercise or under 2sec of maximal exercise (913). Furthermore, ATP is highly protected – which means that our body aims to maintain ATP concentrations, even during strenuous exercise. As such, the continuous creation of ATP during exercise is critical (914).

Anaerobic Lactic: ATP-PC System

The fastest way to create ATP is through the Phosphocreatine system. This is because it’s a one-step process, where the phosphate of phosphocreatine (PCr) is donated to ADP to create ATP and creatine.

$$ADP + PCr + H^+ \longrightarrow ATP + Cr$$
$$\text{Creatine kinase}$$

The phosphorylation of ADP with phosphocreatine (PCr) through creatine kinase provides rapid ATP upon **initiation of exercise** and up to 10 seconds of maximal exercise intensity (913). This system is commonly referred to as the Phosphagen system.

Anaerobic Lactic: Glycolysis

The second energy system, glycolysis, is also an anaerobic process. While it’s not as quick as the breakdown of phosphocreatine, it is still a reasonably quick process compared to aerobic metabolism. As such, glycolysis will be the second system that will kick in when exercise is initiated and will provide enough energy for 60-90sec of very intense exercise before exhaustion (914). Glycolysis involves either the breakdown of glucose from the blood or glycogen from the muscles (termed glycogenolysis), and it ends with the formation of pyruvate and 2-3 ATP. Pyruvate will enter the mitochondria and contribute to oxidative phosphorylation if oxygen is present or will be converted to lactate in the absence of oxygen. This system is commonly referred to as the Glycolytic system.

Glycolysis involves 8 reactions and glycogenolysis involves 9 reactions, but the simplified formulas are:

$$\text{Glycogen} + 3ADP + 2NAD^+ \longrightarrow 2\ \text{pyruvate} + 3ATP + 2NADH$$

$$\text{Glucose} + 2\ ADP + 2NAD^+ \longrightarrow 2\ \text{pyruvate} + 2ATP + 2NADH$$

The conversion of pyruvate to lactate is a very simple reaction catalyzed by lactate dehydrogenase that looks like this:

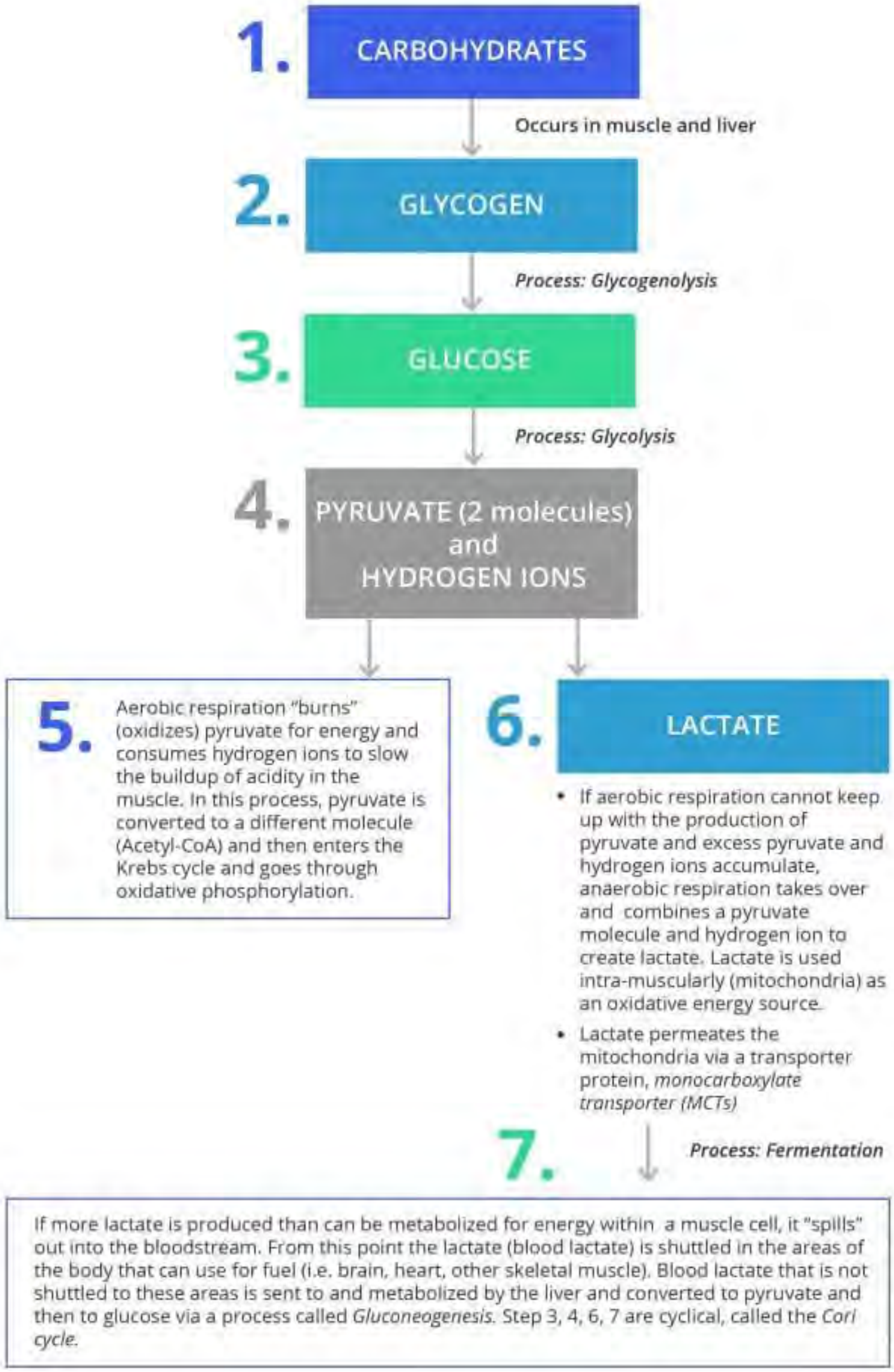
$$\text{Pyruvate} + \text{NADH} \xrightarrow{\text{Lactate dehydrogenase}} \text{lactate} + \text{NAD}^+ + \text{H}^+$$

The formation of lactate is extremely important as it frees up the electron acceptor NAD^+, which is required for further glycolysis and allows ATP to continue to be formed despite the lack of oxygen. Lactate can be converted to pyruvate and then transformed back into glucose in the liver. This recycling of lactate is called the Cori cycle.

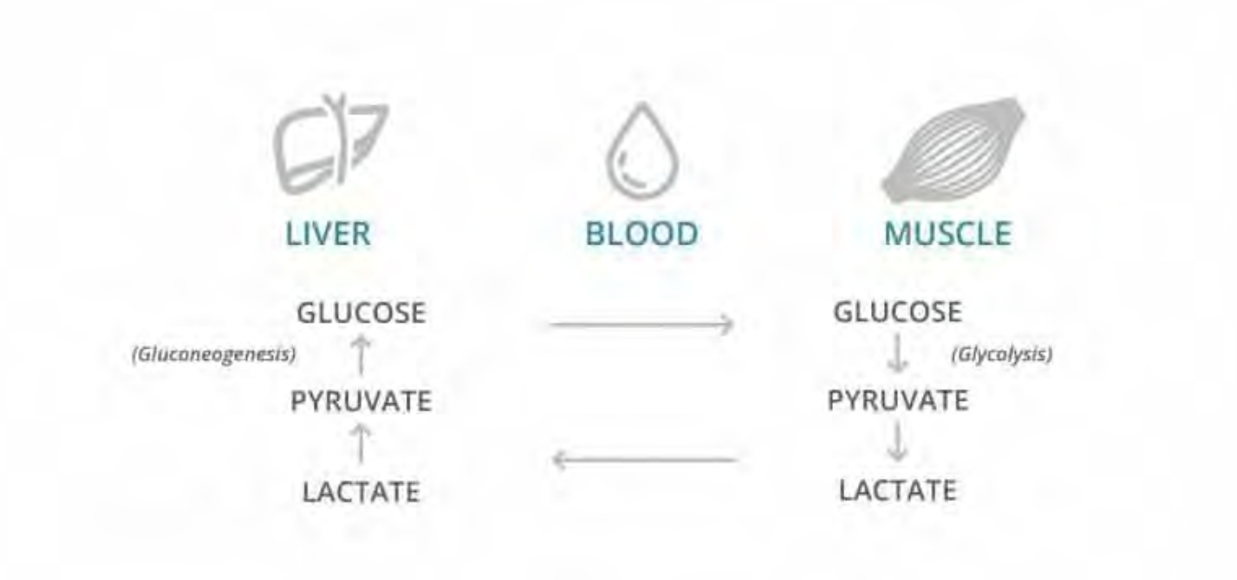
The formation of lactate (step 6 – below diagram) does not necessarily occur because of an absence of oxygen but rather without a substantial utilization of it. This is an important distinction because the lack of oxygen is not required to form lactate (139).

1. Glycogen is converted into glucose via a process called *glycogenolysis* (33).
2. Through the process of glycolysis, glucose is converted into pyruvate.
3. Lactate is formed from pyruvate and hydrogen ions if aerobic respiration cannot keep up with pyruvate production.
4. Lactate is converted to pyruvate and then to glucose in the liver. The glucose is then shuttled back to the muscles via the blood to be used for energy.
5. The glucose in the muscles is converted back to pyruvate, then lactate.

Steps 3, 4, and 5 in the process noted below are cyclical and called the **Cori Cycle.**


1.
CARBOHYDRATES
Occurs in muscle and liver
2.
GLYCOGEN
Process: Glycogenolysis
3.
GLUCOSE
Process: Glycolysis
4.
PYRUVATE (2 molecules) and HYDROGEN IONS
5.
Aerobic respiration "burns" (oxidizes) pyruvate for energy and consumes hydrogen ions to slow the buildup of acidity in the muscle. In this process, pyruvate is converted to a different molecule (Acetyl-CoA) and then enters the Krebs cycle and goes through oxidative phosphorylation.
6.
LACTATE
• If aerobic respiration cannot keep up with the production of pyruvate and excess pyruvate and hydrogen ions accumulate, anaerobic respiration takes over and combines a pyruvate molecule and hydrogen ion to create lactate. Lactate is used intra-muscularly (mitochondria) as an oxidative energy source.
• Lactate permeates the mitochondria via a transporter protein, *monocarboxylate transporter (MCTs)*
7.
Process: Fermentation
If more lactate is produced than can be metabolized for energy within a muscle cell, it "spills" out into the bloodstream. From this point the lactate (blood lactate) is shuttled in the areas of the body that can use for fuel (i.e. brain, heart, other skeletal muscle). Blood lactate that is not shuttled to these areas is sent to and metabolized by the liver and converted to pyruvate and then to glucose via a process called *Gluconeogenesis.* Step 3, 4, 6, 7 are cyclical, called the *Cori cycle.*

The below diagram represents the Cori Cycle and is representative of steps 3, 4, and 5 in the diagram above.

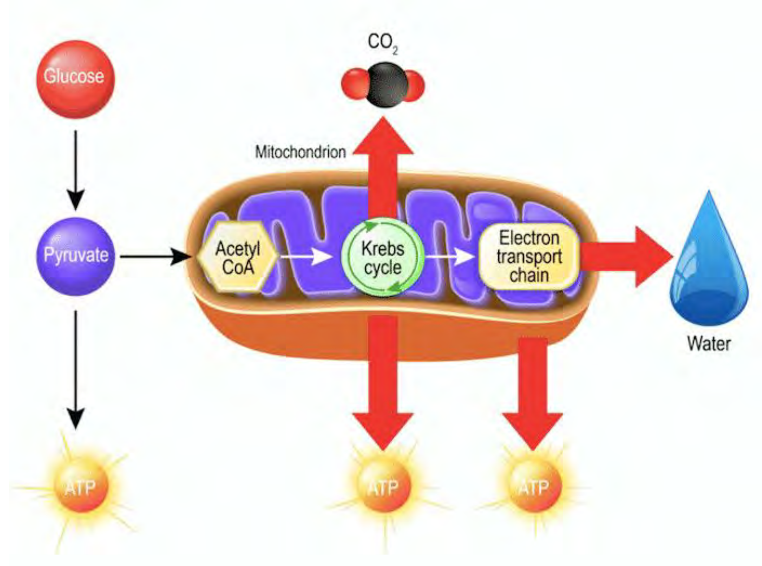


The video below by physiologist Alexandra Coates discusses energy systems.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Aerobic: Oxidative Phosphorylation

Aerobic production of ATP occurs inside the mitochondria. It involves the Krebs cycle and the electron transport chain to produce ~32 ATP molecules per one molecule of glucose or ~130 ATP molecules per one molecule of the fatty acid palmitate. As you can see, aerobic metabolism creates a lot of ATP. Still, as many reactions are involved, it takes some time for the aerobic energy system to take over when exercise is initiated. This is why it takes about 1-3 minutes before oxygen consumption will reach a steady state when exercise is initiated at a given load. In the meantime, the anaerobic systems mentioned previously will create the ATP required to perform that exercise. This system is commonly referred to as the aerobic, aerobic pathway, or oxidative system.



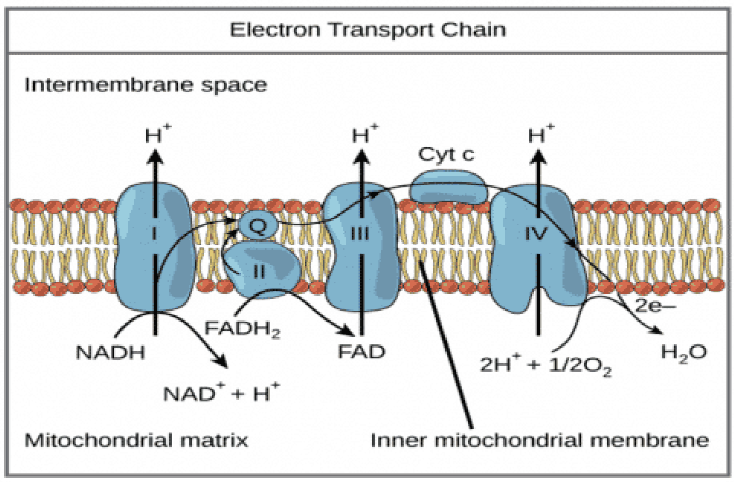
To begin, the Krebs cycle (also termed citric acid cycle or tricarboxylic acid cycle) is initiated by a molecule of acetyl-CoA, which will come from either a) pyruvate which breaks down to form acetyl-CoA and CO_2 b) beta-oxidation which transforms fatty acids from fat cells or muscle fat stores to acetyl-CoA or c) the transformation of protein (amino acids) into acetyl-CoA (only 2-15% of energy during exercise comes from protein) (3).

The Krebs Cycle

Once acetyl-CoA is available, the Krebs cycle will begin. The primary mechanism of energy production in the Krebs cycle is through the release of hydrogen ions which are accepted by the electron acceptors NAD^+ and FAD to create NADH and FADH. As seen below, with each turn of the Krebs cycle, three molecules of NADH and two molecules of FADH will be formed. These electron acceptor molecules will then move to the electron transport chain to release their hydrogen ions and power further ATP production. For every pair of electrons passed through the electron transport chain from NADH, enough energy is available to form 2.5 ATP molecules. For every electron passed through the electron transport chain from FADH, 1.5 ATP molecules are formed. In addition to the production of NADH and FADH, the Krebs cycle also directly produces one GTP molecule, which is used to create ATP, resulting in one ATP molecule per cycle. One glucose molecule will be broken into two pyruvate molecules, hence two acetyl-CoA molecules. Therefore, one glucose molecule will always turn the Krebs cycle twice (915).

The Electron Transport Chain: Oxidative Phosphorylation

This final step in aerobic metabolism is most rewarding. Essentially, the electron carriers NADH and FADH produced from glycolysis, and the Krebs cycle will release their electrons to a series of electron carriers called cytochromes. As the electrons are passed down the cytochrome chain, energy is released to phosphorylate ADP to ATP. At the end of the electron transport chain, oxygen accepts the electrons and combines with hydrogen to form water. As such, if oxygen wasn't there to accept the electrons, the electron transport chain would stop working – which occurs during high-intensity exercise when the demand for ATP exceeds the speed at which oxygen can be delivered and accept the electrons (915).



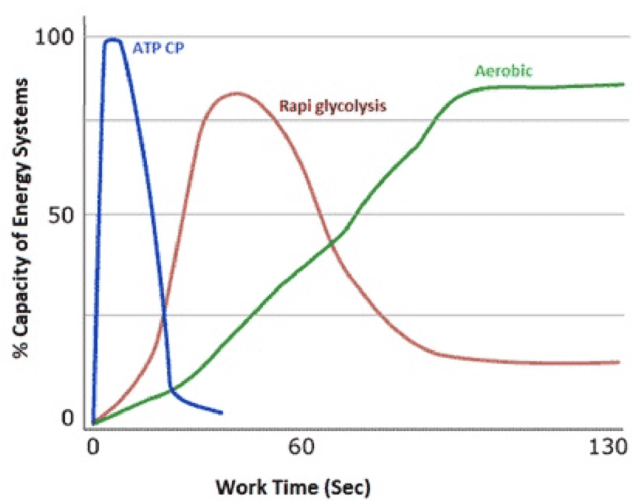
As mentioned above, ~32 ATP molecules are formed per one molecule of glucose and ~130 ATP molecules per one molecule of the fatty acid palmitate (915). As such, it would seem as though fatty acids are the ideal fuel source for creating a lot of ATP! This is undoubtedly true. However, the metabolism of fatty acids also requires more oxygen (23 O_2 molecules for one palmitate) than the breakdown of glucose (6 O_2 molecules for one glucose). This explains why we cannot continue to burn fat at high intensities of exercise, as we cannot deliver that much oxygen at those intensities (915).

Duration of Energy Systems

The following graph depicts the three energy systems and their contribution of total energy-related to the duration of exertion. As noted previously, while all three systems start simultaneously, typically, only one predominant system provides energy at any given time (39). For example, regardless of how intense an exercise level is, some amounts of ATP are always generated aerobically (oxidative phosphorylation) (538).

The Wingate test (WAT) is a specific example of how the different energy systems contribute to an exercise bout, is the Wingate test (WAT). The WAT is a 30-second maximal effort cycling assessment designed to assess an individual's anaerobic fitness capacity. At least 20-30 percent of the energy utilized during the WAT comes from aerobic sources (558). This shows that a seemingly purely anaerobic activity such as the WAT has an aerobic component.

Below is a chart that denotes the contribution from the various energy systems from a work time perspective.



An analogy that describes how energy systems behave sequentially is a car with an automatic transmission. For this analogy, the car has three gears.

- **First Gear (Phosphagen):** quick acceleration but tops out quickly
- **Second Gear (Glycolytic):** medium-range gear; can be driven at high speeds but can only go so fast before it burns out

- **Third Gear (Oxidative):** long-range gear where the car is most efficient; slow to accelerate but is built to last for long hauls.

While anaerobic energy sources are used quickly, they can power far greater maximal efforts than the aerobic energy system. In a VO2max test, an individual might top out at 350W – the maximum power they can achieve aerobically. However, that same individual (with the same muscles) can hit a peak power of over 1000W in a 30s Wingate. This implies that the anaerobic energy systems are highly effective for a short period.

Fuel Sources

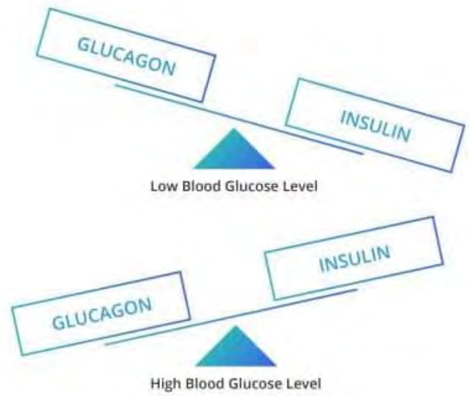
Glucose in the bloodstream (plasma glucose) can be used for immediate energy or stored in the liver and muscles as glycogen. When the glycogen stores are called upon to provide energy, the glycogen is converted to glucose (29). Plasma (blood) glucose typically stores 20 kcal of glucose or less. Compare this with liver glycogen stores, which range from 350–650* kcal, and leg muscle glycogen stores, which vary from 1,250–2,270* kcal (based on 70kg body weight), and it is easy to see that plasma glucose, while technically a carbohydrate reservoir, is largely negligible (705).

Glucose

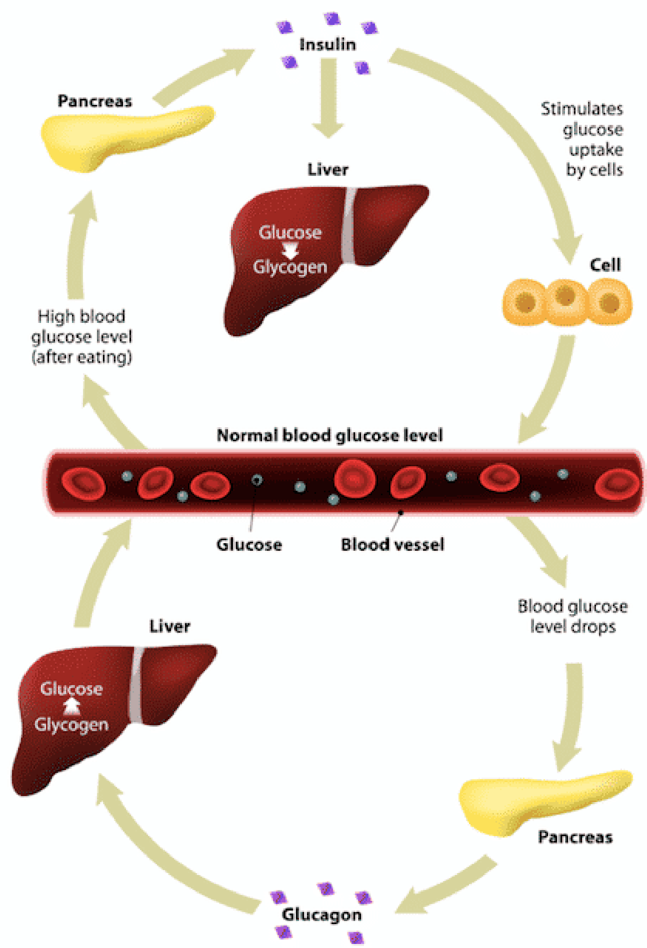
Like most aspects of physiology, regulating blood glucose (additionally termed: plasma glucose, blood sugar) is a series of checks and balances that looks to maintain homeostasis.

The primary players involved in maintaining blood glucose homeostasis are ***insulin*** and ***glucagon***. Insulin is a hormone that is produced in the pancreas. When blood glucose levels get too high, insulin reduces blood glucose concentration by facilitating the uptake of blood glucose into skeletal muscle and fat. As a result, fat is not used as an energy source.

Conversely, when blood glucose levels fall too low, glucagon, a hormone also produced in the pancreas, facilitates the increase in blood glucose level by converting glycogen in the liver to glucose, which is then released into the bloodstream. Blood glucose levels are also increased by ingesting carbohydrates.



Below is a more detailed illustration of the relationship between insulin and glucagon.



Implications For Runners

Maintaining some degree of blood glucose/glycogen is required to not run out of energy (i.e., hit the wall). While blood glucose is not viewed as a substantial energy reservoir, ingesting carbohydrates while running can increase endurance by increasing the percentage of fuel contributed by blood glucose as opposed to stored glycogen (707, 708). This of course is critical to runners. Due to the duration of many running races, ingesting carbohydrates consistently while running is a necessity if running over 90 minutes.

Carbohydrates must be ingested well before one's suspected point of fatigue due to glycogen depletion (approximately 30 minutes). Interestingly, so long as the carbohydrates are consumed at least 30 minutes prior to fatigue setting in, the exact time they are ingested mid-race has little influence on their effectiveness (709).

Being in "good" aerobic condition correlates with the body's ability to utilize fat as an energy source more efficiently. This reduces an individual's reliance on carbohydrates as a fuel source, thus preserving more carbohydrates to use as fuel at a later time and/or at greater exercise intensities.

Muscle Glycogen Storage

As noted previously, glycogen is primarily stored in the liver and the muscles. Between the two, more **glycogen is stored in the muscles versus the liver.**

While the liver size of individuals is typically the same (approximately 2.5 percent of total body mass) (714), the leg muscle mass of individuals varies substantially (18–22.5 percent in women, 14–27.5 percent in men) (711–713). As a result, muscle glycogen storage capacity between individuals will vary significantly based on their leg muscle mass (705, 706).

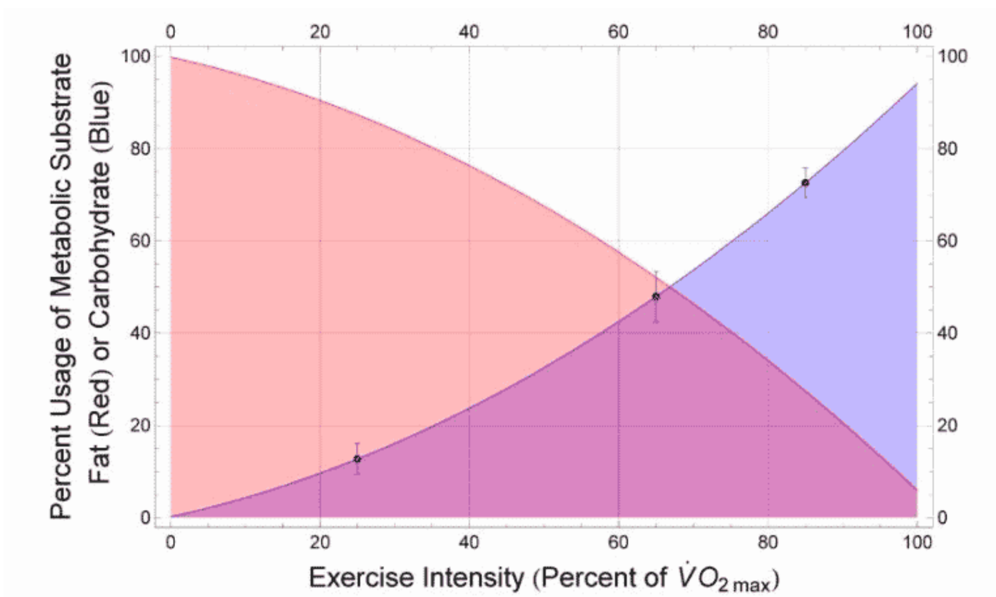
Glycolytic and Oxidative Relationship

While the aforementioned automatic transmission analogy identifies glycolytic and oxidative energy systems as two separately occurring phases, they co-occur, as noted in the following chart. While most of the energy source is based on the intensity of the exercise, even at low intensities, glycogen is still an energy source.

As much more ATP is generated from fat than from carbohydrates, it would be great if fat could be the sole energy source and save the carbohydrate (glycogen) stores for when they are needed. However, varying amounts of glycogen are always utilized for energy (705).

Intensity-Based Fuel Sources

The following chart demonstrates the fuel source (fat, carbohydrate) based on the intensity of exercise in relation to one's VO2 max. This illustrates that at all aerobic exercise intensity levels, both fat and carbohydrates are energy sources in varying percentages based on the intensity level (705).



Fatty Acids

An important aspect of the body's energy production is the use of fat for energy. The primary location for fat storage is **adipose tissue** (i.e., body fat) within the body. Most adipose tissue used for energy is called *subcutaneous fat* (below the skin). This is not to be confused with visceral fat, which is fat located in the intra-abdominal cavity, also known as abdominal fat.

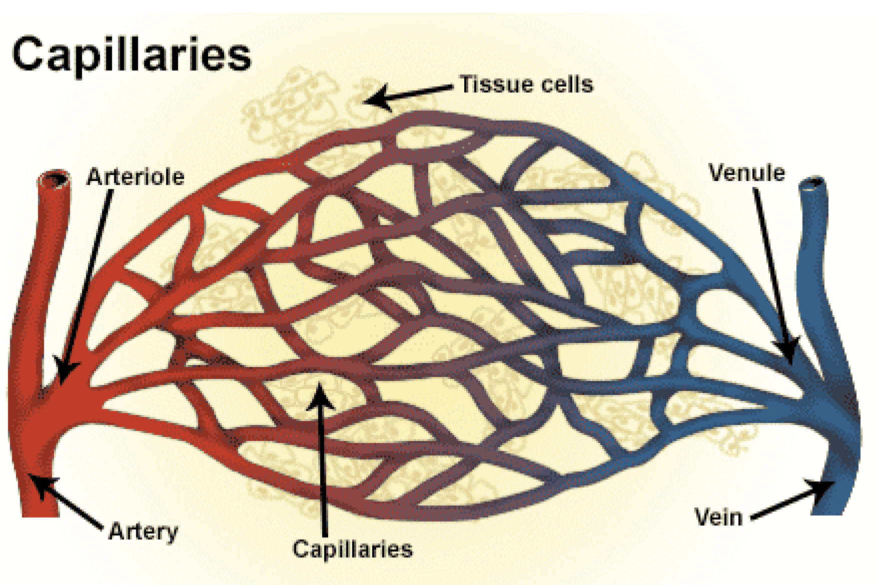
Adipose tissue is composed of **adipocytes** or fat cells. Adipocytes are where *triglycerides* are stored and synthesized. Triglycerides are compounds containing glycerol and free fatty acids. Glycerol is a chemical compound that is the primary chemical support structure of triglycerides (37). Glycerol and free fatty acids are the elements *catabolized* (broken down) by the body for energy. Glycerol is broken down anaerobically through glycolysis, whereas fatty acids are catabolized through **beta-oxidation**. During beta-oxidation, fatty acid is converted into *acetyl-CoA* and then enters the Krebs cycle to be oxidized.

Interestingly, skeletal muscle also stores fat (called intramuscular triglycerides), and highly-trained athletes have more of these fat stores than untrained individuals. Intramuscular fat stores provide faster energy than releasing fat from adipose tissue.

Physiological Response to Aerobic Training

As a running coach, it is important to understand the physiological effects of aerobic training on an individual. Below are physiological adaptations that occur as a result of aerobic exercise (36):

- Decrease in resting heart rate
- Increase in stroke volume
- Greater capacity to use fat as fuel
- Less reliance upon blood glucose for fuel
- Increase in capillary density
- Increase in mitochondrial density



The first two adaptations are cyclical, as an increase in stroke volume decreases resting heart rate. An increase in stroke volume means the heart is pumping more blood per stroke, thereby increasing its efficiency. Because more blood is pumped per stroke, the absolute number of strokes is reduced, lowering the resting heart rate.

Capillaries can increase in number and size in areas where additional blood supply (i.e., oxygen) is needed. Because of the oxygen requirements of endurance athletes, well-trained athletes have a much greater capillary density than non-active individuals. Capillaries grouped together are called a **capillary bed** (38).

The greater the capillaries, the larger and denser the capillary bed is. **As the capillary bed becomes larger, the oxygen supply to muscles increases because of the increased blood flow.** As noted previously, a runner would have a much greater capillary density in the legs versus the arms. For most runners, the arms would have a similar capillary density to a non-endurance athlete.

Another adaptation of aerobic training takes place on a cellular level and is represented by an increase in ***mitochondrial density***. **Mitochondria** are the structures within cells that produce **adenosine triphosphate (**ATP). Because of increased mitochondria, well-trained skeletal muscle (such as runners) has an increased capacity to generate ATP aerobically relative to non-trained skeletal muscle.

Signs of Being Aerobically Deconditioned

Deconditioned individuals typically present with some or all of the following physiological effects, as compared with those who are aerobically conditioned:

- Higher heart rate at rest and while exercising at low to moderate intensities
    - Until VT/LT is reached, one's RPE is often low in relation to the heart rate. Once VT/LT is reached, the RPE typically elevates rapidly.
    - Heart rate at moderate intensity is at a similar level to someone who is aerobically conditioned at a high intensity.
- Slow heart rate recovery

Effects of Intense Exercise

Runners who strenuously exert themselves for long periods risk encountering one or all three of the following issues: *glycogen depletion*, *oxygen debt, and* becoming *“out of breath.”* The common theme among these three things is that they are all the result of intense exercise. Additionally, all of these things relate to being fatigued. *Muscle fatigue* is a general term that regarding running equates to a reduction in the force that a runner generates.

Glycogen Depletion

Glycogen depletion is a technical term for *hitting the wall*, *bonking*, or *blowing up* … get the picture? If you are an endurance athlete, you have probably experienced glycogen depletion at some point, and if so, you also know why it is termed ***hitting the wall***. This is because glycogen depletion feels like you ran into a wall and are stopped dead in your tracks.

Glycogen depletion occurs when the muscles and liver are empty of all glycogen, and the body must rely solely on fat for energy. While individuals can continue, they must do so at a much lower intensity rate. The reason for this is that converting fat to use as an energy source requires a substantial amount of oxygen and thus occurs at a relatively low-intensity level.

Rapoport's 2010 study (running-specific) noted that many exercise physiology resources infer that because of insufficient glycogen stores, hitting the wall during a marathon is inevitable unless a runner's glycogen stores are full because of carbohydrate loading (705). However, what Rapoport found was that many factors play into one's ability to complete a marathon regarding glycogen reserves, such as:

1. Muscle mass distribution
2. Liver and muscle glycogen density
3. Running speed

While even the leanest runners have enough non-essential fat to run near-endless distances, **the primary performance limiter is carbohydrates (glycogen) and, more specifically, their relatively small reserves compared with fat (705).** This is an issue as runners utilize substantial carbohydrate reserves.

Generally speaking, **the more aerobically conditioned runners are, the more efficient they are at utilizing fat instead of glycogen for energy.** Conversely, less aerobically conditioned athletes are more reliant on glycogen. The average male has about 90,000–110,000 kcal of energy from fat reserves (1 gram of fat has 9 kcal of energy). In contrast, the carbohydrate energy (glycogen) reserve is less than 2,000 kcal in a non-carbohydrate-loaded state (302). This demonstrates that untrained individuals who rely on glycogen as an energy source will find that they cannot exercise for very long at a medium to high intensity before running out of energy. As an analogy, think of burning a piece of paper versus a wood log. The piece of paper catches fire immediately but burns out quickly. The log, however, burns for quite a while. In this analogy, the piece of paper is glycogen, and the wood is fat.

As mentioned earlier, the liver and muscles are the two primary locations where glycogen is stored. While the glycogen concentration in the liver is higher than in muscles, **the absolute amount of glycogen stored in the muscles is 75 percent versus 25 percent in the liver** (41). If glycogen were utilized as the sole energy source, an athlete would be able to race for approximately two hours before total glycogen depletion occurs, depending on the intensity level. However, if a runner started with low glycogen levels, the time to depletion would be less.

Using fat as the primary energy source instead of glycogen is important because an endurance athlete needs an energy reserve. This reserve is glycogen, and if it is used up due to an athlete's inefficiency in using fat as fuel, the athlete's performance will decrease substantially.

As glycogen depletion requires a runner to slow down or stop, this is often viewed as catastrophic from a performance standpoint. As a result, proper pacing during a long event, such as a 20-mile run, is critical to minimizing the chance of *hitting the wall*.

Many factors are involved in human performance concerning not ending up in a glycogen-depleted state. These factors are nutrition (carbohydrate loading and pre-race fueling), pacing, and other related risk factors. These areas are discussed in the *Sports Nutrition* and *Pacing* modules, respectively.

Approximately two in five runners report hitting the wall during a marathon and 1-2 percent drop out before finishing (705).

Glycogen Depletion Training (GDT)

Low glycogen is typically viewed as a bad thing in endurance sports, as it represents running low on energy and moving closer to the dreaded *bonk*. However, a training practice called glycogen depletion training (GDT) has individuals perform physical training sessions in a glycogen-depleted state (no glycogen before or during activity).

The purported benefit of GDT is to reduce the body's reliance on glycogen by making it more efficient at utilizing fat as a fuel source.

While there is not much research in this area, a study by Van Proeyen et al. found that individuals who performed GDT had increased fat utilization over a group that ingested a carbohydrate-rich breakfast. Additionally, the GDT group had increased enzymes associated with fat metabolism, while the non-GDT group did not (643). It should be noted that the non-GDT group did not show increased performance over the GDT group.

Training in a glycogen-depleted state stresses the body. As a result, GDT should not be undertaken regularly as it is likely that the repeated stress will decrease overall energy and performance. GDT should be used sparingly and solely during training, not in competition.

Lastly, as the purported benefit of GDT is to increase the body's efficiency of utilizing oxygen as an energy source, GDT is more applicable to long-distance races. For shorter events such as criteriums, it can be theorized that GDT would not have a substantial benefit, as glycogen plays a more significant role in providing energy for these races (644).

GDT is most often implemented by athletes with a history of bonking late in a race.

It is advised that when performing GDT, runners train on a relatively short course so that if and when they bonk, they are not too far from home or close to a source (i.e., a restaurant) where they can refuel. Carrying emergency glycogen such as a gel or carbohydrate drink is strongly recommended.

The Science / Reasoning Behind GDT

The stress of GDT increases the activity and amount of a protein (PGC-1á) responsible for increasing fat oxidation and production of mitochondria (645).

Oxygen Debt

Anyone who has exercised at a high degree of intensity has experienced oxygen debt. Oxygen debt is most often characterized by heavy breathing after an intense exercise bout ends.

The definition of oxygen debt is the difference between oxygen consumption at rest and an elevated oxygen consumption rate following an exercise bout. When an exercise session ends, the level of oxygen consumption does not immediately return to a resting baseline. There is a slow return to this baseline level. Oxygen debt is indicative of this return to a baseline.

Perhaps the best way to think of oxygen debt is in terms of a credit card. If you use your credit card to purchase an item, you have accrued debt and owe this amount to the credit card company. You have a zero balance once you have paid the credit card in full. This is analogous to oxygen debt because one must repay the oxygen used until the body returns to a zero baseline.

The repayment of oxygen is termed *excess post-exercise oxygen consumption*, commonly referred to as EPOC. EPOC represents the intake of oxygen after exercise ceases. EPOC aims to reduce and eliminate the oxygen debt, thus returning the body to normal (non-exercising) oxygen levels. EPOC occurs with both aerobic and anaerobic exercise. The higher the intensity level of exercise is, the greater the oxygen debt and EPOC (42).

The reason why high-intensity training that is primarily anaerobic also trains the aerobic energy system is due to EPOC. The oxidative energy system (aerobic) works very hard to bring an individual back to baseline after intense anaerobic work. This is primarily why high-intensity, interval-based training increases both anaerobic and aerobic fitness levels.

Studies have shown that while the oxygen level consumed at rest (i.e., the baseline) is similar between trained and untrained individuals, **aerobically trained individuals return to baseline faster than untrained individuals.** This is because untrained individuals rely more on anaerobic energy systems during strenuous exercise than trained individuals, and therefore the resulting oxygen debt is higher.

Three areas that oxygen debt addresses when "repaying" debt is (120):

1. Oxidizing lactate
2. Supplying the demands of an increased metabolic rate
3. Replenishing high-energy phosphate stores
    1. Refer to *First Energy System: Phosphagen*

'Out Of Breath'

From a postural standpoint, positions can decrease or increase the lungs' ability to get a full breath. However, it is interesting that studies have shown little difference between sedentary and trained individuals' pulmonary function (43). Two areas of pulmonary function are ***total lung capacity*** **(TLC)** and ***forced vital capacity*** **(FVC).** TLC represents the largest volume of air the lungs can hold, and the FVC represents the largest amount of air that can be expelled after taking a deep breath. So, if pulmonary function is not substantially increased with training, why do people who are deconditioned become “out of breath” faster than those who are aerobically well trained?

The major limiting factor in exercise is cardiac output (44). Cardiac output pertains to the volume of blood pumped out of the heart in one minute. There are two ways that cardiac output can be increased:

1. Increase in heart rate
2. Increase in stroke volume (the amount of blood pumped out of the heart with each stroke).

Aerobically conditioned individuals have a higher cardiac output than aerobically deconditioned individuals and are therefore able to supply more oxygen to their working muscles than those who are aerobically deconditioned. **Therefore, the primary reason a deconditioned individual gets out of breath faster than someone who is aerobically conditioned is that their body does not utilize oxygen for energy efficiently (43).**

Being out of breath usually coincides with increased and relatively shallow breathing. Untrained individuals who exercise often do not have optimal oxygen delivery to the muscles because of inadequate cardiac output. The body naturally looks to compensate for this lack of oxygen by increasing the number of breaths taken to improve the oxygen saturation of the blood. This increase in respiration rate is what most people relate to being *out of breath*. **The point where a person becomes out of breath is also called the *ventilatory threshold* (VT).** It is important to note that anyone who exercises at high intensity can experience being out of breath, regardless of their conditioning level. However, aerobically conditioned individuals will reach this level much later than those who are aerobically deconditioned.

Summary

- The body produces substantially more ATP aerobically than anaerobically.
- The *Cori cycle* is the process by which the body produces ATP anaerobically.
- The three energy systems are *phosphagen*, *glycolytic,* and *oxidative*.

- The body stores only a small amount of ATP at any given time. Therefore, ATP must be continually regenerated on a cellular level.
- Aerobic training results in both increased mitochondrial and capillary density.
- Fat yields more ATP generation than glucose.
- *Lactate threshold* (LT) is representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.
- Therefore, other measures (i.e., lactate threshold) are often used to infer AnT.
- The two main circulatory systems in the body are:
    - Pulmonary
    - Systemic
- Red blood cells carry oxygen to the body
- The *sinoatrial node* (SA node) is the body's natural pacemaker
- Diaphragmatic breathing is beneficial over chest breathing
- Insulin and glucagon work together to maintain homeostasis
- Glycogen depletion occurs when the muscles and liver are empty of all glycogen.
- The definition of oxygen debt is the difference between oxygen consumption at rest and an elevated rate of oxygen consumption after an exercise bout
- *Hypoxic training* is done to increase one's red blood cells. This results in the body being able to deliver more oxygen to working muscles
- An aerobically trained individual will reduce oxygen debt faster than someone who is aerobically deconditioned
- *Cardiac output* is the primary factor in an individual's being "out of breath," not pulmonary function
- Aerobically deconditioned individuals often have elevated heart rates and slow heart rate recovery times
- An anaerobic state does not mean a lack of oxygen but rather the lack of oxygen utilization

Module 5: Environmental Physiology

Many of the athletes that you will work with train and race in a variety of environmental conditions such as extreme heat, cold and high altitude. Understanding how these conditions affect individuals will help you prepare your athletes as best as possible in both training and racing scenarios.

Altitude

History of Altitude Acclimation

Altitude training became a hot topic after the 1968 Olympics in Mexico City, Mexico. Most endurance events had subpar performances, and most sprint and other anaerobic events had record-breaking times (50, 52).

Mexico City sits at 7,350 feet above sea level (450). High-altitude training is defined as training at altitudes of 5,000 feet or above. When researching the 1968 Games was that because of the decrease in partial pressure of oxygen, it's a lot harder to get the oxygen both into the lungs and from the lungs into the circulatory system – which limited performance in endurance events. While participants of anaerobic events (e.g., sprints) were subjected to the same decrease in partial pressure of oxygen, it did not impact them as much since their reliance on oxygen was minimal given the anaerobic nature of their event(s).

It was determined that anaerobic athletes broke so many records because the air was less dense at a higher altitude than at sea level, and therefore, there was less air resistance, leading to faster times. The 1968 Games were the catalyst in determining what physiologically occurs in the body at altitude and how it can be exploited to enhance human performance (50).

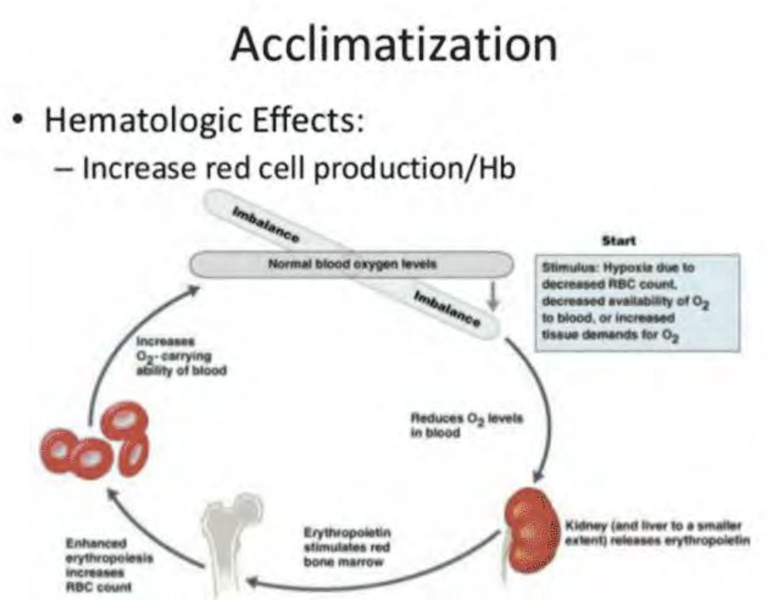
Physiology of Altitude Acclimation

If you haven't said it, you've probably heard it, *"There's no air up here!"* But is that really the case? It turns out that even as you go up in elevation, say from sea level to the top of Pikes Peak (14,115 feet), the **gases that make up the air around you remain the same**, 20.93% oxygen (O_2), 0.03% carbon dioxide (CO_2), and 79.04% nitrogen). So, why do you feel so terrible if there isn't "less oxygen"? Well, it turns out that as you go up in elevation, there is a decrease in partial pressure of oxygen, and your lungs rely on that partial pressure gradient for inhalation and exhalation. Additionally, that pressure gradient also influences gas exchange between your alveoli and capillary beds (873). **So, while there is not less oxygen, it's a lot harder to get the oxygen both into your lungs and from your lungs into your circulatory system (and to your working muscles, brain, etc.).**

So how does your body adapt to not only survive but thrive at different elevations? When you arrive at higher elevations, your body must undergo a series of adaptations incredibly quickly to counter your sudden decreased ability to transport oxygen efficiently (remember the pressure gradient from above?). Some of these adaptations take place very quickly, within the first 24 hours at altitude, while others may take as long as several weeks to fully develop. So what are these changes?

Immediately your body tries to artificially boost the density of your red blood cells through a process known as **haemoconcentration**. This temporarily boosts your hematocrit to transport oxygen more effectively until your body can produce more red blood cells to help (874, 875). This process, however, decreases your blood plasma volume by 10-25%. **This is why your heart rate is generally elevated when you initially travel to altitude.** Think about it, cardiac output (Q) = heart rate x stroke volume – and if your blood volume decreases (stroke volume), your heart rate has to increase to keep cardiac output stable. **This is generally why athletes commonly feel worst between 24 to 72 hours after arriving at altitude.**

From the respiratory side, something occurs called a *Hypoxic Ventilator Response*. At altitude, because there is a decrease in O_2 coming into your bloodstream (again, remember the issue with the pressure gradient above?), your ventilation rate and depth of breathing increase. This ultimately shifts your pH balance, and if we know anything about the human body, it's that it likes to operate in pretty finite windows. This shift in blood pH causes your kidneys to get involved, primarily via fluid balance from increased urination, to bring you back into equilibrium. This hypoxia-induced pH shift does one more important thing; it triggers the release of more erythropoietin (EPO) (876). You might be thinking, EPO like the illicit drug? Well, yes, however, EPO is also naturally produced to help stimulate the production of red blood cells. This process, called *erythropoiesis*, takes at least one week but as many as two to three weeks to allow red blood cells to mature and become fully functional. **These long-term changes ultimately lead to an expanded blood plasma volume and increased hematocrit and hemoglobin.** The ability to carry more oxygen is the primary physiological goal of altitude exposure.



Source: D. (2017, July 17). Acute Mountain Sickness Explained – South Africa Adventures. Retrieved from http://www.southafricaadventures.com/acute-mountain-sickness-ams/

Altitude Acclimation Considerations

There's a distinction to be made when thinking about altitude; **are you trying to use altitude to improve general endurance performance, or are you specifically trying to perform in an event at altitude?** For most athletes, outside of elites with means for extended training camps (21-28 days in length), our primary concern is performing at altitude (i.e., your athlete from Wisconsin competing at Leadville). From there, you need to consider several factors. These include: at what elevation does the athlete live, at what elevation does the athlete train, at what elevation is their event, and does the athlete have the means to arrive at the event early?

Based on what we know, the most basic recommendation for racing at altitude is to arrive at least one week before race day. This is because the likelihood is that you:

1) Are past the 'worst of it' in the first 24-72 hours at altitude
2) Have already started to positively adapt to the higher elevation

This is also why it's often recommended **that if you cannot arrive at altitude early (at least one week), you are better off arriving the night before or on the morning of the event.** Once again, you are missing the 'worst of it' window before your body starts to work through incredibly taxing changes.

There are two primary types of altitude acclimation: hypobaric hypoxia exposure (normal altitude exposure) and **normobaric hypoxia exposure** (simulated altitude exposure, think altitude tent). Both have been shown to work, but the key difference is a question of dosage and potentially practicality. For hypobaric hypoxia, i.e., going to altitude, it appears the critical value for adaptation is having chronic exposure to elevations between 5,905 feet (1800 meters) to 9,842 feet (3000 meters) for ideally at least two weeks and up to four weeks to reap maximum benefits (red blood cell production/maturation) (877). It should be noted that going above 9,842 feet is not inherently negative. However, there appears to be no added benefit from an adaptation standpoint.

But what if your athlete does not have the luxury to get away to altitude for that long? Enter normobaric hypoxia exposure, or rather, the utilization of an altitude tent. It appears that the time course for adaptation remains about the same 3-4 weeks of accumulative time… but there's the catch, accumulative time. When utilizing an altitude tent or room, you are not spending 3-4 weeks in there. At most, you are only getting 8-10 hours a day. Spending longer than that in bed veers dangerously towards bed rest, which can negatively impact blood plasma volume. These things combined make the utilization of an altitude tent less straightforward. Given the sheer volume of time that may be necessary in an altitude tent or room to produce adaptations, you need to weigh all the factors. These include the following:

- Can the athlete afford the tent?
- Does it decrease the quality of their training by impacting their sleep quality?
- What elevation do they currently live at (there are more positive indications for altitude tents for those living closer to the minimum required altitude)?
- Is it possible the athlete can make it out to their race more than 72 hours prior to their event?

Use of Heat to Acclimate to Altitude

As having access to an altitude tent or spending time at high altitude weeks in advance of a race are likely impractical due to cost and cost/time perspectives, respectively, it makes sense that other methods (cross-acclimation) to acclimate to altitude would be investigated and utilized.

Training in the heat may also simulate training at altitude. Because of the body’s need to cool itself, blood is shunted away from the working muscles and toward the skin. This reduction in blood flow equates to less oxygen to the active muscles during exercise – similar to the effects of training at altitude. This is why exercising in the heat is more difficult than at lower temperatures. The physiological adaptation from training in the heat increases blood plasma volume (580, 581). Increased blood plasma may increase stroke volume and thus lower exercise heart rates (582).

This practice is getting attention due to some of the purported benefits. In other words, utilizing heat training stress to induce adaptations that are present at altitude. Of the main purported benefits, heat acclimation may reduce physiological strain at rest and during moderate-intensity exercise at altitude – primarily via adaptations associated with improved oxygen delivery, increases in plasma volume, and reductions in body temperature (888).

However, as this is an area of emerging research and due to the potential for stress on the kidneys, it is not advised at present. However, this cross-acclimation area should be paid attention to as more research and findings emerge.

Airflow Restriction Devices (ARD)

Airflow Restriction Devices (ARD) are precisely what they sound like. They are masks that restrict/reduce the amount of air an individual can breathe in – think: breathing in through a straw. This certification will not spend much time on this topic other than to state that ARDs do not help in altitude acclimation, as none of the physiological responses present with altitude acclimation occur with an ARD. This has been noted by several studies and sources (896, 897, 898).

Altitude Sickness

Altitude sickness has various symptoms depending on the elevation traveling from and to. The greater the spread between elevations, the greater the severity of symptoms. Symptoms usually begin 12 – 48 hours of exposure and range from mild to severe.

Mild Symptoms

- Dizziness
- Fatigue and loss of energy
- Shortness of breath
- Loss of appetite
- Sleep disturbance

Moderate Symptoms

- Worsening fatigue, weakness, and shortness of breath
- Difficulty with activity
- Severe headache, nausea, and vomiting
- Chest tightness

Severe Symptoms

- Shortness of breath at rest
- Inability to walk
- Confusion
- Fluid buildup in the lungs and/or brain

The cause of altitude sickness is primarily due to the change in air pressure, which reduces the oxygen exchange between the lungs and blood. Fitness level does not seem to change an athlete's susceptibility to altitude sickness. It is not recommended that an athlete traveling to altitude from a lower elevation do strenuous exercise or training for a few days to allow the body to adapt without additional stress optimally. As noted previously, because symptoms usually manifest within 12 – 48 hours, it is sometimes recommended that an athlete arrive at an event either as close to the race as possible (24 hours or less) or with enough days to acclimate prior to competing. This method can sometimes require arriving 4-6 days prior. This should be addressed on a case-by-case basis depending on how an athlete typically responds to altitude exposure.

Intervention

There are a variety of interventions to minimize altitude sickness, depending on the severity. For mild sickness, some over-the-counter medication can reduce the severity of symptoms, but time is the most effective medicine. It must be done under the guidance of a runner's physician. Within 3-4 days, the body naturally creates more red blood cells, allowing for a greater oxygen-carrying capacity in the blood. When this adaptation takes place, symptoms tend to subside correspondingly. Moderate symptoms can also be improved with medication or, if possible, going to a place that may be 1,000 to 2,000 ft lower elevation than where symptoms began. **In severe cases**, **immediately traveling below 4,000ft is recommended, along with a healthcare provider consultation.**

The most effective way to prevent altitude sickness is through acclimatization or a slow progressive exposure to increasing altitudes for increasing amounts of time. Sometimes this is not possible geographically or financially. Athletes primarily sea level-based and, if economically feasible, can purchase altitude tents that allow for exposure to a hypoxic room in their home. **This option is expensive and creates an unnatural sleeping environment that should be addressed on a case-by-case basis with an athlete**. In all instances of treatment or prevention, medication is available through a healthcare provider. However, for US-based athletes that are in an anti-doping testing pool or at an event where anti-doping testing may occur, any prescribed medication should be cross-referenced with the US Anti-doping Agency's in and out of competition drug reference database. For runners outside of the US, referencing their national anti-doping agency’s drug reference database or that of the World Anti-Doping Association (WADA) is recommended.

Hypoxic Training

The oxygen saturation levels at which your athlete’s train has profound physiological effects on their bodies and their performance capabilities.

- **Hypoxic Training:** Training occurs in an environment substantially depleted of oxygen (e.g., training at a high altitude). It can be done in either natural (altitude training) or simulated environments (altitude tent).

When an individual is at altitude, the primary physiological effect is a decrease in oxygen hemoglobin saturation (49). ***Hemoglobin transports oxygen in the blood from the lungs to the body’s muscles***. **The body’s response to this decrease in hemoglobin is to increase the production of erythropoietin (EPO), a hormone secreted by the kidneys that stimulate red blood cell production (51).** This is the primary reason why athletes train at altitude. The thought process is that by training at altitude, athletes will increase their red blood cell count and therefore can deliver more oxygen to working muscles. This increase in oxygen to muscles enables the athlete to perform at a higher level.

The effects of altitude training have been shown to last for up to 15 days after athletes return to sea level (48). Therefore, many athletes schedule their altitude training to have the benefits of altitude training when competing at lower altitudes. It is important to note that EPO secretion can be increased with altitude training, but only by a finite amount. If too much EPO were to be secreted by the kidneys, too many red blood cells would be produced, which could have harmful side effects. This is an example of how the body regulates itself.

Hematocrit refers to the volume (percent) of red blood cells in the blood. Through hypoxic training, the goal is to raise one’s hematocrit level, thereby increasing the amount of oxygen delivered to working muscles. A male’s hematocrit level is approximately 45 percent, whereas a female’s is typically 40 percent (381).

There is no refuting that altitude training stimulates red blood cell production, but an individual cannot train as hard in that environment because of decreased oxygen levels. Therefore, the question is if increased red blood cell production benefits are canceled out while training at altitude. Because of this, a theory emerged. This theory is termed ***Sleep High, Train Low*** (53). Some of you might be thinking that this seems a bit odd and wonder, *how can you sleep at altitude, then train at sea level?* You are correct that this is highly impractical.

The logic behind this concept is that by sleeping at altitude, you will gain all the benefits of high-altitude training (increase in red blood cells) but will be able to train at sea level, where you can train at normal intensity. This type of training is possible only by using altitude tents, which decrease the oxygen level that you breathe in. This practice has proved to provide the best of both worlds. Some of these tents can replicate altitudes as high as 12K feet (275).

Altitude tents are also helpful if an individual is heading to a high altitude and must acclimate before the trip. If an athlete is planning on utilizing an altitude tent or going to altitude to train, it is advised for them to consult with their doctor before performing this type of training.

It is important to note that individuals will adapt to altitude at different rates.

Heat

Contributor: Corrine Malcolm

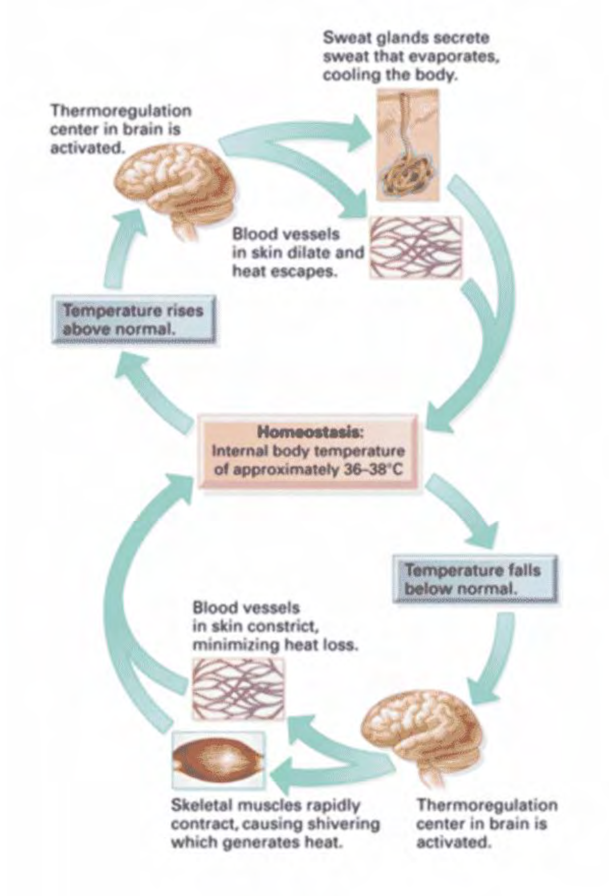
Physiology of Heat Acclimation

Although humans are uniquely suited for many complicated tasks, it turns out we are also inherently inefficient. For example, when it comes to ‘doing work’ like propelling yourself up a hill (and all the metabolic processes that make that happen), most of that work simply generates heat without having anywhere to go. To manage this, **the body has its own thermostat, the hypothalamus**, the thermoregulation center of the brain. It utilizes a negative feedback loop; when your skin and core temperature drops, you experience vasoconstriction and shivering to keep warm blood near your core and produce heat. Conversely, when you heat up, you vasodilate – sending blood out towards the skin surface, allowing heat to dissipate into the environment while activating sweat glands to utilize evaporative cooling.

As you run, two significant factors contribute to you heating up: the metabolic heat you are producing internally and the heat the environment applies to you externally. When it comes to environmental factors, you will be impacted differently depending on where you are riding. For example, ambient air temperature is greatly influenced by the surface you are on, direct vs. indirect sunlight, and even the direction of the wind.

These impact the four categories of heat transfer; conduction (direct transfer of energy), convection (air or water moving over your skin), radiation (direct or indirect transfer of sunlight), and evaporation (conversion of liquid to gas) (878).

All of these things work together to either heat you up or cool you off, but their effect largely depends on the temperature gradient between you and the surrounding environment because even heat moves along a gradient (high to low concentration). A temperature gradient describes what direction and at what rate temperature changes for a particular location.



Heat Acclimation Considerations

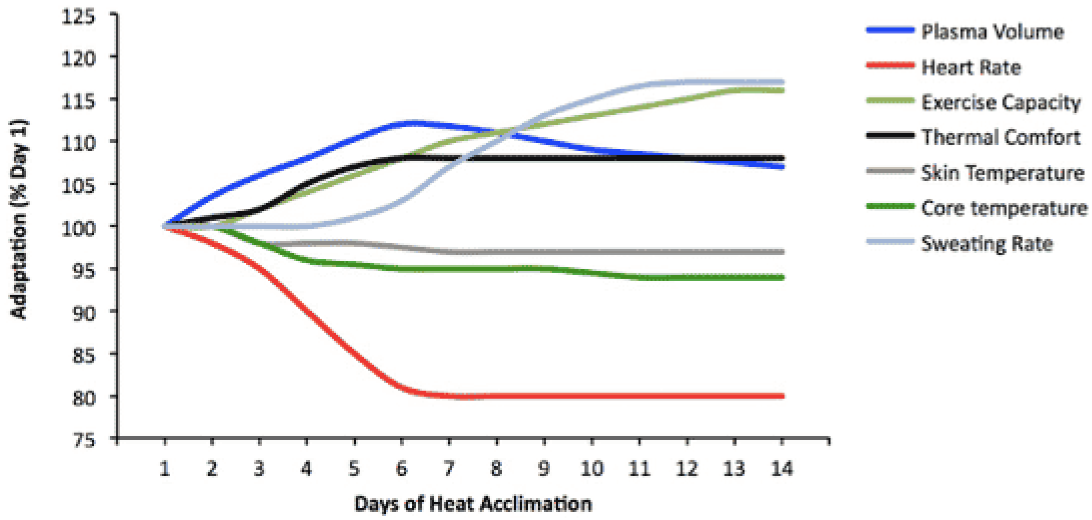
The human body is incredibly smart, and with chronic, daily heat exposure, your body can make changes that allow you to thrive and survive in hot environments (and even benefit us in cooler environments). These adaptations include:

- Increase in sweat rate and a corresponding decrease in sweat concentration
- Lower skin and core temperature that comes with blood plasma volume expansion
- Lower heart rate during exercise (back to normal heart rate)
- Increased fluid and cardiovascular stability
- Decrease in the metabolic cost of exercise (879, 880).

When you add up all these factors, you have the most important adaptation: the ability to tolerate and feel more comfortable in the heat on race day. What's really quite striking, compared to altitude acclimation, **heat acclimation happens fairly rapidly, with the**

bulk of adaptation occurring in 6-7 days and is complete by ~14 days of heat exposure (880).

The body has approximately 2 million to 4 million sweat glands (595). Sweat secreted by the glands is cooled via evaporation. The purpose of cooling the skin is to cool the blood shunted to the skin from deeper aspects of the body (organs, muscles) (595). However, if exercising in humid conditions, the degree of evaporation is substantially lessened. Thus, **exercising in hot and humid conditions can prove dangerous because of a lack of evaporative cooling.** As such, wiping away sweat before it has had a chance to evaporate greatly reduces the cooling benefit of sweating.



Source: Périard, J. D., Racinais, S., & Sawka, M. N. (2015). Adaptations and mechanisms of human heat acclimation: Applications for competitive athletes and sports. *Scandinavian Journal of Medicine & Science in Sports,25*, 20-38. doi:10.1111/sms.12408

Pre-Race

When managing heat, it is advantageous to break it up into two main periods, **pre-race** and **during-race cooling**. The pre-race period is the most effective way an athlete can prepare for heat exposure. This happens naturally as an athlete is exposed to warmer temps during spring and summer, but that's not always convenient (largely due to one's training potentially not happening in warm enough temps based on location and time of year).

There are many physiological benefits of heat acclimation. Some of these benefits are as simple as jump-starting your preparation for warmer temps, while others come from the positive effects of things like blood plasma volume expansion etc. **So how does one heat acclimate?**

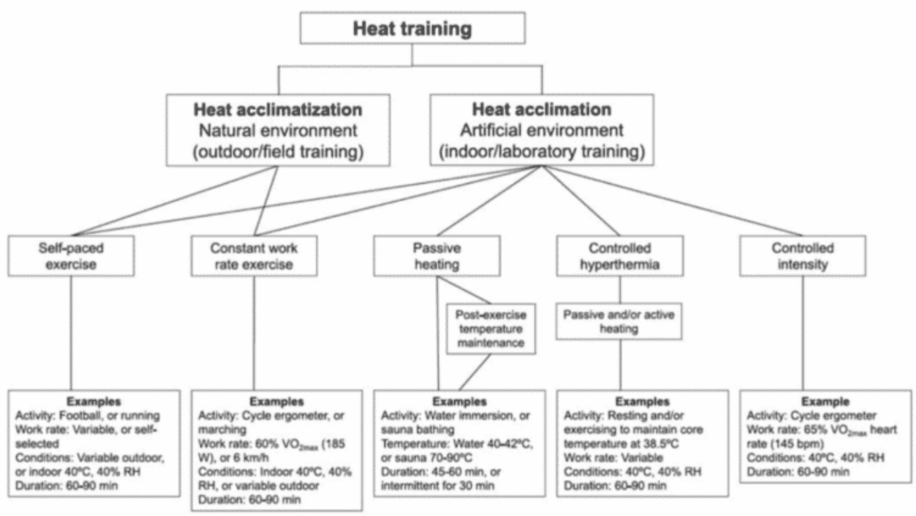
It's important right off the bat to get on board with one thing, and that is, as a coach, you need to ask yourself, ***"how do I elicit the adaptation I want with the lowest possible dose?"*** This is of utmost importance because anything you add to training is stress, and there will always be possible tradeoffs whenever you add additional stress (i.e., what do we have to remove to make room for the additional stress we are adding). From the literature, it is clear that **daily increases in core body temperature through exercise in warm temps, environmental stress, passive heat interventions (sauna or hot water immersion), or a combination of the above will induce a meaningful physiological change that will ultimately increase thermal tolerance** (881). However, when it comes to what is best for your athlete, it boils down to what you have at your disposal and what your athlete can tolerate.

The methods that athletes most commonly use include:

- Natural heat acclimatization (exercising in a naturally warm environment)
- Controlled hyperthermia (overdressing)
- Controlled intensity (via exercise in a climatic chamber)
- Passive heating (post-exercise heating via sauna or hot water immersion) (882).

As mentioned above, **all these methods work, but they all require different doses to elicit change.** This is because the physiological tipping point requires your core body temperature to reach ~101F (38.5C) for approximately 90 minutes. It should be noted that more is not better; more time in a session is not better, twice a day is not better, and weeks on end is not better. *So where does that leave us?*

Of the methods mentioned, you have to ask yourself **which one maintains your athletes' ability to maintain the highest quality training and add the least amount of additional workload?** Methods like overdressing or training in a climatic chamber often mean sacrificing the quality of the individual workouts by cutting down the volume or intensity. Traditional time in a dry sauna, steam room, or hot tub requires a lot of additional time added to your day, 60-90 minutes on top of normal training, which might not be manageable for your athlete. So in considering your athlete's needs, not impairing their training, and the dose needed, **passive heating, which combines heat exposure immediately post-exercise, rises to the top of this pile**. Why is that? Because you allow the athlete to maintain the quality of their workouts by training in their natural environment, and also spend less time in the actual heat because you go into the heat exposure 'pre-warmed' from exercising. This means suddenly an athlete only has to spend 20-30 minutes in the sauna instead of 60-90 minutes, which is both easier on the body and also easier for athletes to time manage.



Source: Daanen, H. A., Racinais, S., & Périard, J. D. (2017). Heat Acclimation Decay and Re-Induction: A Systematic Review and Meta-Analysis. *Sports Medicine,48*(2), 409-430. doi:10.1007/s40279-017-0808-x

Cooling During a Race

We've discussed prepping pre-race from an acclimation standpoint, but what about the other pieces of this puzzle? You've done the work, and your athlete is fit and as ready as they can be, so what other strategies can they employ to mitigate heat-related issues on race day?

In shorter races, there are some pre-race cooling strategies you can utilize – these generally include preemptively lowering core body temperature to allow for extra heat storage capacity. However, ultras are generally considered too long for something like this to be effective. This leads us to during-race cooling strategies. These strategies aim to help rapidly remove excess heat from the body. **Unlike pre-cooling, the purpose of during-race cooling is not so much about actually lowering core body temperature but to keep skin temperature low enough to allow a temperature gradient to be present, which helps to move heat out into the environment by promoting vasodilation.**

Suggested strategies to employ are:

- Packing ice on your torso, icing your neck and head (via ice bandanas, ice collars, and special ice hats)
- Icing your peripheral arteries (putting ice down arm sleeves)
- Wearing loose and light-colored clothing
- Limit conductive heat (avoid dark colors)
- Increasing evaporative cooling (getting wet)
- Drinking cool fluids (883, 884).

Other Heat Acclimation Information and Considerations

1. Individuals will differ greatly in their ability to deal with heat and humidity.
2. The more body fat an individual has, the less efficient they will be at dissipating body heat.
3. Generally, the larger the runner, the more energy is required to reduce body heat. This is because the smaller the individual, the larger the skin surface compared with body volume. This enables small individuals to dissipate heat better than large individuals (487).
4. Wind speed, humidity, and shade all affect the runner. Sweat must evaporate to cool the body. Therefore, the chance for

overheating increases in humid conditions where sweat does not evaporate as well as in non-humid conditions.

5. After ten days of heat acclimatization, the capacity for sweating nearly doubles (596).
6. Novice endurance athletes and deconditioned individuals are often the most negatively affected by hot weather.
7. The physiological benefits of heat acclimatization are lost between two and three weeks post-acclimation training(594).
8. Individuals who are heat acclimatized can sweat up to 4 liters/hour, while those who are not acclimatized will sweat approximately 1.5 liters/hour (732). This demonstrates the increased efficiency regarding thermoregulation due to heat acclimatization training.

Humidity

But what about humidity? As anyone who has experienced humidity can tell you, it makes everything feel harder, and there is some truth to that. This is because it hampers your primary mode of getting rid of heat via evaporative cooling. Remember heat and moisture, among other things, want to move along a gradient from areas of high concentration to areas of lower concentration. Just as this is true for heat (it’s easier to cool off when the temperature is lower than core and skin temperature), it is also true for sweat (or any moisture). For evaporative cooling to work, there needs to be ‘room’ for that to happen; ‘room’ for that moisture to move from your skin and into the air, taking heat from your body. When the humidity increases, your ability to utilize evaporative cooling from sweat or other water (such as melted ice or water you’ve drank) decreases.

So, humidity doesn’t only make everything feel harder; it makes everything harder. This has led some to believe that it’s important to acclimate to the type of heat you might be exposed to on event day (humid vs. dry heat). However, there is concern that adaptation to humid heat (via steam rooms, etc.) might actually blunt your sweat response as it is not as useful in those environments. Because of this, we would suggest sticking with a traditional dry heat acclimation with the option to add in some supplementary sessions in a humid environment primarily to build a tolerance to it.

There is presently a lot of research being done on this topic, and there will likely be updates to this field.

Heat Illness Overview

Despite how much preparation an athlete undertakes or how much they know about hydration or cooling, heat illness can sneak up on even the most experienced athletes in training or competition. There are two categories of heat illness:

1. **Classic**
2. **Exertional**

Classic heat illness is a direct result of the environment; high temperature and humidity, strong direct sun exposure, and still air. Exertional heat illness, however, results from an athlete's own heat production, which can happen in all types of weather.

Heat-related illnesses can range from mildly uncomfortable to life-threatening and include: heat edema (swelling), heat rash, heat syncope, heat cramps, heat exhaustion, and heat stroke (885). Below is a chart that notes the various illnesses, associated symptoms, and signs.

Criteria for Diagnosis of Heat Illness[a]

Condition	Core Temperature °F (°C)	Associated Symptoms	Associated Signs
Heat edema	Normal	None	Mild edema in dependent areas (ankles, feet, hands)
Heat rash	Normal	Pruritic rash	Papulovesicular skin eruption over clothed areas
Heat syncope	Normal	Dizziness, generalized weakness	Loss of postural control, rapid mental status recovery once supine
Heat cramps	Normal or elevated but <104°F (40°C)	Painful muscle contractions (calf, quadriceps, abdominal)	Affected muscles may feel firm to palpation.
Heat exhaustion	98.6°F-104°F (37°C-40°C)	Dizziness, malaise, fatigue, nausea, vomiting, headache	Flushed, profuse sweating, cold clammy skin, normal mental status
Heat stroke	>104°F (40°C)	Possible history of heat exhaustion symptoms before mental status change	Hot skin with or without sweating, CNS disturbance (confusion, ataxia, irritability, coma)

[a]CNS, central nervous system.

There are many risk factors for developing more serious heat-related illnesses of both the environmental and physiological nature. These factors include how hard an athlete is working, clothing choices, lack of shade/direct sunlight, dehydration, high air temperature and humidity, age (people under the age of 15 or over the age of 65 are more susceptible to heat illness), recent alcohol consumption, certain medication or supplement interactions, recent illness, recent head injury, sunburn, a history of heat-related illness, or insufficient heat acclimation (885).

So how can an athlete treat heat illness? Like many other medical emergencies, time is perhaps the most important factor in properly treating heat-related illnesses. The primary goal is to rapidly decrease core body temperature (below 100.4F or 38C); this protects the athlete's brain and vital organs. To do this, get the athlete into a shaded, cooler environment and use whatever you have available to cool the athlete off – dousing with a hose, shower, wrapping in cold towels, applying ice packs, ingesting cold fluids, or best yet, immersion in an ice bath.

There are three main categories of heat-related illnesses that relate to elevated core temperature:

1. Heat Cramps
2. Heat Exhaustion
3. Heat Stroke

The body has approximately 2 million to 4 million sweat glands (595). Sweat secreted by the glands is cooled via evaporation. The purpose of cooling the skin is to cool the blood shunted to the skin from deeper aspects of the body (organs, muscles) (595). However, if exercising in humid conditions, the degree of evaporation is substantially lessened. Thus, **exercising in hot and humid conditions can prove dangerous because of a lack of evaporative cooling.** As such, wiping away sweat before it has had a chance to evaporate greatly reduces the cooling benefit of sweating.

Heat Cramps

Characterized by muscle cramps and sweating, heat cramps (not exercise-based) are often treated with cessation of activity, moving to a cool, shaded area, and drinking an electrolyte-based beverage. Stretching and massaging the cramped area can also help to alleviate acute pain.

Heat Exhaustion

This occurs when the body's natural temperature regulation system begins to break down. Because of a lack of hydration and minerals, the capillaries of the body reduce in size, thereby reducing the body's effectiveness in cooling itself. Common symptoms include (297):

- Dizziness
- Heavy sweating
- Confusion
- Nausea
- Weak, rapid pulse
- Low blood pressure
- Excessive thirst
- Hyperventilation
- Loss of appetite
- Anxiety

The athlete must stop activity and move to a shady, cool location. The individual should be given an electrolyte drink and cooled with a fan and/or wet cloths. Additionally, the person should be positioned in the supine position with legs elevated. Medical personnel should be called to check and monitor the athlete.

Heat Stroke

This is the most serious of the three categories of heat sickness and can be fatal. While an individual may have experienced heat cramps and exhaustion prior to heat stroke, it is not always the case. Heat stroke is caused by the body's lack of water and electrolytes. Effectively, the body is shutting down. Symptoms include (297).

- Rapid heart rate
- No sweating
- High body temperature (>103)
- Red, dry skin
- Confusion
- Vomiting
- Nausea
- Erratic behavior
- Difficulty breathing
- Constricted pupils

The most important thing to do is lower the individual's body temperature immediately and get emergency personnel on-site. Call 911.

Prevention

By staying properly hydrated, acclimating to heat gradually, taking breaks, wearing proper clothing (e.g., light colors, wicking), and being aware of the body's response to the heat, the chance of developing a serious heat-related injury can be greatly mitigated. Pre-hydrating with cold fluids before training or racing in the heat has been shown to reduce one's core temperature (592).

Cold Weather

Athletes are always interested in cold, particularly after thinking about how humans adapt to both heat and altitude, but the truth is it isn't as glamorous. This isn't because humans cannot adapt to the cold, but rather the physiological changes that occur from cold exposure do not produce the same degree of performance benefit as the adaptations to heat or altitude. The key takeaway from cold adaptation is that with time, from chronic exposure (this can be seasonal), you make minor changes that allow you to shiver less and vasoconstrict more precisely. You become, primarily, habituated to the cold. But are there any special considerations for runners braving the cold?

With any sort of long-term exposure to extreme cold, there is a risk of sustaining a cold injury. These are most likely to happen during long winter runs. These include frostnip (freezing the top layers of skin tissue), frostbite (deeper freezing of tissues), and trench foot (prolonged cold, wet exposure). Additionally, you can suffer from eye injuries such as frozen corneas and snow blindness (sunburn of the eyes). The best way to protect yourself or your athletes? Practice logic. Dress appropriately for the weather conditions while maximizing loose (loft/allows for warmed air pockets), heat-insulating layers and avoiding constriction of body parts (can promote vasoconstriction), opt for mittens over gloves if you get cold easily, stay dry by utilizing wicking materials like wool as opposed to cotton to move moisture away from the skin, and maintain adequate hydration and nutrition to maintain normal fluid and electrolyte balances.

Hypothermia

Hypothermia is the opposite of heat-related illnesses. Hypothermia is the body's inability to maintain normal metabolic functioning and is representative of a body temperature of 95 degrees F or less (normal body temperature = 98.6 degrees F).

If an athlete is training or racing in cold temperatures without proper clothing, hypothermia can also set in.

Physiologically, the body looks to preserve heat in the most vital areas. Therefore, the body moves blood away from the extremities to more vital, life-sustaining organs. Muscle contractions can increase heat production two to four times (299). This is why the body shivers – to produce heat.

Water increases heat loss. Therefore, wearing wicking clothing is very important when running in cold weather. When clothing becomes wet with sweat, body heat loss is accelerated.

Like heat-related illnesses, there are several stages of hypothermia (451):

Mild Hypothermia

The most common visible symptom is shivering and cold or numb extremities due to the constriction of blood vessels (vasoconstriction). Moving to a warm area, taking off wet clothes, and putting on dry and adequate amounts of clothing are the most common solutions to eliminate mild hypothermia.

Moderate Hypothermia

The most common visible symptoms are excessive shivering, a loss of muscular coordination, and extremities – including the lips and ears – turning blue because of a lack of blood flow to these areas. An individual must be immediately moved to a warm area and changed into dry clothing. A warm hat is also suggested. Medical attention is required.

Severe Hypothermia

At this point, the body begins to shut down and is represented by the heart rate, blood pressure, and respiration rate, all declining substantially. Individuals will also likely have difficulty walking and talking, and their cognitive ability is typically compromised. Heart and respiration rates continue to fall in relation to moderate hypothermia (300). Unless an individual is moved to a warm area and medical attention is provided, the organs of the body will shut down, and death will result. Medical attention must be provided.

Prevention

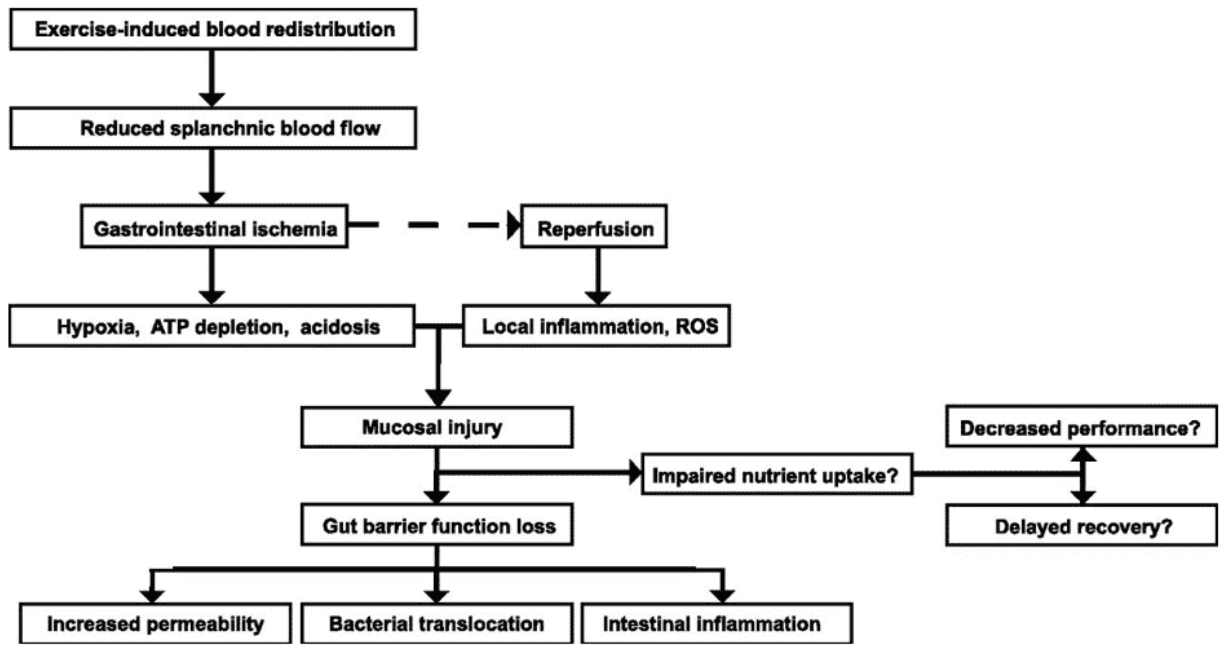
First and foremost, athletes must be aware of their environment and the weather forecast for their training and racing area. Proper clothing selection is critical to avoiding hypothermia. They should wear sweat-wicking fabrics and dress in layers. Additionally, extremities and normally exposed body parts (e.g., ears, nose) must be covered in extremely cold conditions. **When training in cold conditions, athletes must always have a partner to train with in case of an emergency.** It is a good idea to ensure they are training in an area near civilization in case of an emergency, and they must always tell someone of their planned route. If at all possible, it is advised to train in a concentrated area near their residence instead of covering a large geographic area. If athletes get too cold, they must know they are not far from home. It is also a good idea to carry a cell phone in case of an emergency.

Regardless of the planned training for the day, athletes must be attentive to their body's response to the cold, and if they feel they are getting too cold, they must head for home or seek shelter in a warm environment.

Environmental Impact on the Gut

One of the most challenging things about performing at high elevations or in the heat is the toll it takes on your gut, specifically on the gut's ability to function as it should. This is due to splanchnic hypoperfusion, or the decreased blood supply to the tissues of the organs of the GI tract. During exercise, blood flow to your GI tract normally reduces by up to 80% compared to when at rest (930). This is because that blood is shunted towards active muscles and the skin instead of the gut. This is logical, **your muscles need more oxygen to be delivered during exercise, and your body benefits greatly if you can dissipate heat by sending warm blood to the surface of the skin.**

As such, you can imagine that this process becomes more extreme in hot weather and at altitude, pulling more and more blood away from the tissues of the GI tract. This can, in some circumstances, lead to a 'gut injury.' These gut injuries can cause changes to the permeability and function of the small intestine; the most common symptoms can include nausea, vomiting, abdominal pain, diarrhea, and loss of iron through microscopic GI bleeding (930). These symptoms can be short-lived, generally as blood supply is restored during rest. However, sometimes these can include prolonged symptoms like inflammation which can cause longer term gut permeability issues.



Hyponatremia

This is a metabolic condition that occurs when there is not enough sodium (salt) in the body fluids outside the cells. Low sodium levels in the blood can cause cerebral edema (brain swelling). Hyponatremia is primarily caused by excessive drinking behavior, four or more hours of consistent exercise, low body weight, being of the female sex, slow pace, race inexperience, high availability of drinking fluids, and extremely hot environmental conditions. Hyponatremia can be life-threatening.

Drinking in excess of hourly sweat losses may result in hyponatremia.

Symptoms: bloating, nausea and vomiting, headache, confusion, disorientation, agitation, seizures, respiratory distress, and unresponsiveness.

In the event of hyponatremia, an athlete should be transferred immediately to a medical facility.

Prevention of hyponatremia can be achieved by determining an athlete's hourly sweat rate and the person drinking only enough liquid to match the fluid loss from sweating. Athletes can add extra sodium to their diet in the days leading up to a race and avoid over-drinking days or hours before a race to prevent lowering blood sodium levels.

Sports drinks containing sodium can help to maintain plasma sodium levels and may reduce the risk of hyponatremia. However, if an athlete drinks excessively, sodium-rich sports drinks will not prevent hyponatremia. This is because typically the main cause of hyponatremia is over-hydration, not underconsumption of sodium (395).

Dehydration

Dehydration occurs when the body loses excessive amounts of fluids. The body then cannot carry out its normal functions and becomes overheated. Severe dehydration can lead to overheating and heat stroke. **Consuming fluids during exercise decreases the risk of heat-related illness, maintains physiological function, and improves athletic performance.** The symptoms of dehydration may not be obvious until severe dehydration occurs (411). Symptoms of dehydration include dark urine, a small volume of urine, elevated heart rate, and headache.

There is debate regarding what percent dehydration causes a decrease in performance and, more importantly, presents a health risk to an individual. Traditionally, it has been thought that a 2 percent dehydration level negatively affected performance, as the chart below denotes. This determination is based on a paper by the *American College of Sports Medicine* (Sawka et al.) (771). Based on this guideline, pre-hydrating before exercising and drinking at set intervals was advised.

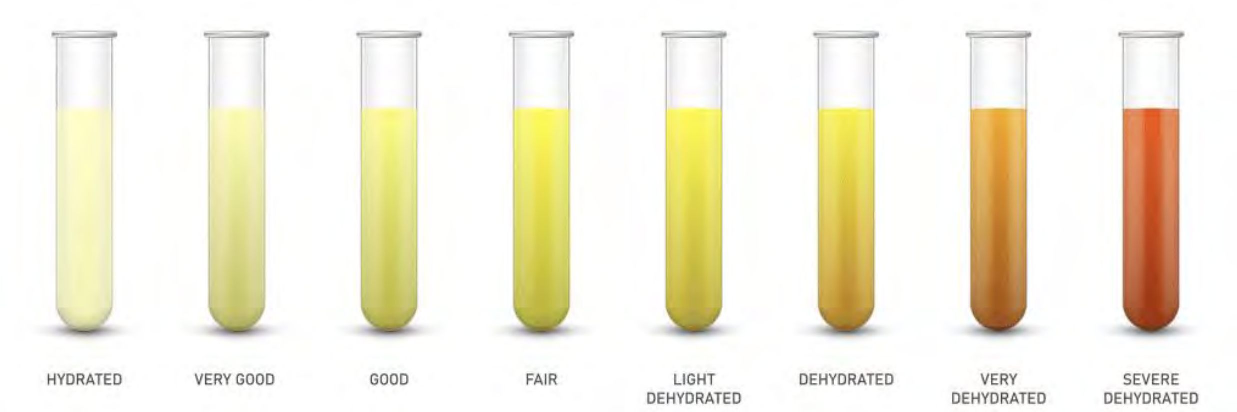
Research by Dr. Noakes found that a dehydration level of 2 percent does not necessarily decrease performance. His theory is that a safe level of dehydration is individually based. He notes that for some individuals, this may be 1 percent, whereas, for others, it may be 12 percent (397). Regardless of the exact percentage, Dr. Noakes's dehydration theory is that one's thirst response will determine what is appropriate and safe regarding a hydration strategy.

PERCENT DEHYDRATION	PHYSICAL EFFECTS
1%	Increased Body Temperature
2%	Impaired Performance
5%	G.I. Issues, Heat Exhaustion
7%	Hallucinations
10%	Circulatory Collapse

Urine Color Chart

Prevention of dehydration requires an individualized hydration plan and consuming fluid to minimize loss of body weight without overhydrating. Remember to avoid drinking too much or too little fluid, as both can be dangerous. Weigh immediately before and after activity to determine the average sweat rate. Each pound lost (through sweat) after a workout equals 16 ounces of water lost. Knowing an athlete's sweat rate can help with creating a hydration strategy.

A runner will likely sweat more than the individual can replace via fluid intake. This is especially true for long-distance events such as marathons and ultramarathons.



Determining Sweat Rate

The calculation for sweat rate is as follows:

Pre-weight – post-weight + fluid intake during activity = athlete's individual sweat rate

Following is an example of this calculation:

- **Pre-workout weight:** 140 lbs
- **Post-workout weight:** 139 lbs
- **Fluid intake:** 16 oz
- **Exercise duration:** 1 hour

140–139 lbs = 1 lb lost (16 oz) + 16 oz of fluid consumed = 32 oz (2 lbs) of sweat loss/hour.

Based on two years' worth of data collected by Dr. Noakes, the average sweat range of runners and rowers was 23.7 to 44 fl oz/hr (397). Assuming the average intake of fluids per hour is 16 fluid ounces during a running race, an individual will likely become dehydrated, which equates to a decrease in body weight. The important issue is what level of dehydration will negatively affect performance.

Clothing selection also affects sweat rate. If an individual wears too many layers, clothes that do not "breathe," or dark colors, sweat rate will be higher than it should be.

Environmental Conditions and Lung Health

Frequent seasonal questions have to do with when it is or isn't a good idea to run outside. This could be in the winter months as temperatures plummet or during fire season, reducing air quality for days to weeks. These conditions affect everyone differently, and most of us will have our own personal threshold for temperature or air quality. Still, the likelihood is we've all gone for a run at some point in conditions we shouldn't have. Be it cold or dry air, or air high in particulate matter (from diesel exhaust or wildfire smoke), both are taxing on your lungs and can lead to a breathing condition like exercise-induced bronchospasm (EIB), formerly called both exercise-induced asthma and hyper-reactive airway.

EIB is not an acute condition in that it's generally not a one-off event but one that is caused by chronic exposure to something. In many respects, it could be referred to as an overuse injury. The cells that make up the lining of your airways are incredible in that they can repair themselves quickly. However, with repeated injury and repair, they can also become maladapted, becoming more reactive over time. While at rest, your ventilation rate is only 12-20 breaths per minute, but while running, you may be breathing at a rate of 50 breaths per minute or more. This increase in ventilation combined with an irritant (i.e., cold, dry, or polluted air) exposes athletes to more risk. So, while you are breathing, and breathing hard, you cause water to evaporate from the airway liquid surface that lines your bronchioles at a rate you cannot keep up with. This drying then causes a shift in the osmotic gradient (again a gradient) in the cells that make up the airway, stimulating the release of inflammatory mediators. These inflammatory mediators, such as histamine, act on the smooth muscles of the bronchioles, causing constriction, mucus buildup, and edema (swelling) of the bronchioles.

As mentioned above, this is not an acute problem but a slow and progressive one that worsens over time. One day of training in extreme cold or smoky skies will not guarantee you a lifetime of breathing issues. However, continued training in them does heighten your risk. So, what can you do to protect yourself and your lungs?

- You don't lose fitness overnight. Fitness decay takes two weeks of zero exercise to set in. So, take a day off, or two, or three, and support your athlete in that. Their long-term health is always more important.

- You can alter the plan. Move the intervals inside or cut the intensity when air quality or temperature drops. Remember, as ventilation increases, exposure also increases.
- Shorten the total duration. Not only is it how hard you are working but how long you are working. If you stay outside, consider cutting the workout duration to limit exposure.
- Be flexible. There are generally times of day better than others for air quality or temperature. Air quality tends to be worse midday and better early in the morning and later in the evening. Air temperature tends to warm as the day goes on. Try to get intervals in during warmer afternoon periods during cold months.
- Know your threshold. There is not one threshold for everyone, but with health in mind, err on the side of caution. For cold temperatures, you can use the international ski federation cut-off of -4F (-20C). To be fair, this cut-off is more about exposure (cold injury risk) and less about the lungs, and I would caution most athletes to move intervals inside when temperatures fall below 10F. If you are coughing after a hard workout, that is generally a side effect of drying of that airway surface. For air quality, I'd encourage you to avoid intervals if AQI is above 75 (unless you have a history of asthma, then the threshold is lower), cut the duration of workout if air quality is between 100-150, and avoid outside exercise if air quality is above 150.

Daily AQI Color	Levels of Concern	Values of Index	Description of Air Quality
Green	Good	0 to 50	Air quality is satisfactory, and air pollution poses little or no risk.
Yellow	Moderate	51 to 100	Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution.
Orange	Unhealthy for Sensitive Groups	101 to 150	Members of sensitive groups may experience health effects. The general public is less likely to be affected.
Red	Unhealthy	151 to 200	Some members of the general public may experience health effects; members of sensitive groups may experience more serious health effects.
Purple	Very Unhealthy	201 to 300	Health alert: The risk of health effects is increased for everyone.
Maroon	Hazardous	301 and higher	Health warning of emergency conditions: everyone is more likely to be affected.

Summary

- There is not less oxygen at high altitudes, but a change in the pressure gradient
- An athlete is advised to arrive at altitude either at least one week in advance, or the night before/day of the race.
- It is not advised to use heat to cross acclimate to the altitude
- The hypothalamus is the body's thermoregulator
- Heat acclimation occurs faster than altitude acclimation
- The use of a dry sauna is the most practical way to acclimate to heat while still maintaining one's training load.
- Humidity increases one's effort level via reduced evaporative cooling.
- Environmental conditions can affect both the GI tract and the lungs

Module 6: Performance Assessments and Metrics

Various methods are used to benchmark a runner's physical condition and performance. These methods are often correlated to assess an athlete's level of conditioning accurately.

The areas discussed in this module are:

- **Anaerobic Threshold**
- **Lactate Threshold**
- **Heart Rate**
- **Rate of Perceived Exertion (RPE)**
- **VO2 Max**

Rate of Perceived Exertion

An athlete's rate of perceived exertion is the measure of how physically and mentally difficult an individual feels an exercise or activity is.

The **rate of perceived exertion (RPE) scale** is a subjective method to assess one's physical exertion or pain level. This scale was created in the 1960s by Dr. Gunnar Borg, a Swedish psychologist. Dr. Borg created two RPE scales, the 6– 20 scale and the CR10 scale (79). The 6–20 scale is based on a scale from 6 to 20, with 6 representing the lowest exertion/pain level and 20 being the highest level. The CR10 scale is similar to the 6–20 scale.

However, it ranges from 0–10, with 0 being no exertion and 10 being the highest. The RPE scale is used in both sports and medical settings.

RPE Scale	Rate of Perceived Exertion
10	**Max Effort Activity** Feels almost impossible to keep going. Completely out of breath, unable to talk. Cannot maintain for more than a very short time.
9	**Very Hard Activity** Very difficult to maintain exercise intensity. Can barely breath and speak only a few words
7-8	**Vigorous Activity** Borderline uncomfortable. Short of breath, can speak a sentence.
4-6	**Moderate Activity** Breathing heavily, can hold short conversation. Still somewhat comfortable, but becoming noticeably more challenging.
2-3	**Light Activity** Feels like you can maintain for hours. Easy to breathe and carry a conversation
1	**Very Light Activity** Hardly any exertion, but more than sleeping, watching TV, etc

UESCA advises its coaches to use the Borg CR10 scale, as the 0–10 scale is typically perceived as more intuitive than the 6–20 scale. The figure above shows a modified Borg CR10 RPE Scale.

Regardless of the RPE scale used, the scale is based on the athlete's perception of exertion or pain and therefore is highly subjective. For example, someone with a high pain tolerance might rate running at 95 percent of their lactate threshold as a 6 on the RPE scale (Borg CR10), whereas someone with a lower pain tolerance might rate that effort level as a 9.

One of the key advantages of using RPE versus other metrics such as heart rate is that RPE is not affected by any other variable such as environmental factors. Additionally, it gives an athlete a real-time assessment of their exertion level at any point.

Heart Rate

Heart rate training has long been the standard when analyzing and assessing cardiovascular effort. A heart rate monitor (HRM) is a relatively inexpensive and beneficial way to assess effort levels.

Most heart rate monitors consist of two pieces, a watch, and a chest strap. Some activity trackers pick up heart rate at the wrist. The benefits of using an HRM are:

1 – Athletes can tell when they are not recovered. Without an HRM, this is hard to determine and forces athletes to rely on how they feel, which is not always the best measure. Below are three metrics that typically correlate to not being recovered and rested:

- Heart rate is slow to return to baseline after an effort
- Abnormally high/low resting heart rate. An abnormally low heart rate is the most common

- Heart rate does not reach normal peak levels when putting out a corresponding effort

2 – It is especially useful during preseason training when athletes work on their base fitness level. Using an HRM, athletes can monitor their heart rate throughout the exercise session to ensure they do not push too hard. This is very important because, as a general rule of thumb, the more de-conditioned athletes are, the higher their heart rate will be and the faster their heart rate will elevate with increased effort. This elevation in heart rate does not always correspond to the athlete feeling tired or exerting excessively. If you do not use an HRM with athletes to measure heart rate during base fitness training, it is very easy for them to exercise too hard and not get the most out of the training process.

3 – Training with an HRM allows athletes to set heart rate training zones. By setting these zones, an athlete can accurately focus on varying intensities.

4 – Determine running pace

Estimated Maximum Heart Rate

A long-established standard for determining an individual's maximum heart rate has been the formula:

220 – Age = Maximum Heart Rate

Based on the result of this formula, heart rate zones could be set to determine varying intensity workloads. There is just one problem with this formula ... it's a formula! **A formula is not practical in determining an individual's maximum heart rate**, as there is too much of a range as to what one's maximum heart rate could be. This needs to be assessed individually, not based on a set formula. Regardless, there is no scientific validity to the formula.

The *220–Age* formula was developed in the late 1960s by cardiologist Sam Fox and physiologist William Haskell (350). The formula was derived to estimate maximum heart rate for individuals undergoing stress testing. The formula was developed using a small sample of individuals. Interestingly, even Haskell noted that the formula was not intended to determine athletes' maximum heart rates (349).

Conclusion

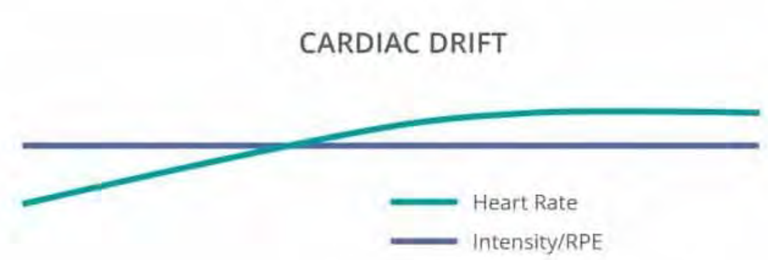
Since this formula was developed, there have been attempts to identify a more accurate and scientific way to predict and estimate maximum heart rate. However, UESCA recommends using lactate threshold as the starting place via the Functional Threshold Heart Rate assessment (FTHR) noted later in this module. After this is established, training zones can be determined based on LT-based heart rate.

Cardiac Drift (CD)

This is a phenomenon that relates to a natural increase or upward “drift” in heart rate while the overall intensity (e.g., respiration rate, effort level, caloric burn) remains the same (359). There are two primary reasons for *cardiac drift* (359):

1. Increase in body temperature raises heart rate
2. Decrease in muscular efficiency due to fatigue

CD has implications when designing a heart rate-based training program. **If an athlete experiences cardiac drift during training sessions based on heart rate, there is a strong likelihood that the individual will be underperforming during the training sessions.**



For example, let us say an athlete is supposed to train between a heart rate of 155 and 165 beats per minute (BPM). For the first half-hour, the athlete can run at this pace easily while maintaining a 7:30 min/mile pace. However, approximately 35 minutes into the run, cardiac drift brings the heart rate up to 168. At this point, the athlete slows down to stay in the appropriate heart rate training zone. They are reducing their workload by decreasing the effort level to accommodate for CD. In contrast, if they remained at the same effort level (even though the heart rate would rise past the prescribed heart rate zone), they would perform at the appropriate level.

Therefore, **it is important to inform the athlete about CD and to judge effort not just on heart rate but also on perceived exertion.** If you know that your athlete experiences CD and how many heartbeats the individual drifts upward, you may consider assigning them a higher heart rate zone to factor in CD.

In a 2000 study of 10 highly conditioned cyclists performing an MLSS test, CD was noted from the beginning to the end of the test (140).

Environmental Factors

While heart rate can be an effective way to assess intensity/effort, one major drawback is the effect that environmental factors have on it. For example, heat and altitude typically increase one's resting and sub-maximal heart rates (901). Additionally, a 2016 study by Che et al. found that relative humidity increases one's heart rate (902). Exercising in the cold will also likely raise one's heart rate as the cold weather can cause the blood vessels and arteries to narrow, which forces the heart to work harder to pump blood throughout the body.

Resting Heart Rate

This assessment is performed by taking one's heart rate upon waking and before getting out of bed. This gives a good base measurement in terms of your athlete becoming more aerobically fit and when the person is recovered. **Generally speaking, the lower the resting heart rate, the more aerobically fit someone is.** Individuals naturally have varying resting heart rate levels, so the value of this test is not the initial resting heart rate but rather the reassessments. Resting heart rate is affected by many things, so to obtain the most accurate baseline measurement, your athlete will want to take their resting heart rate on three or four consecutive days and average the results. Your athlete should see the average resting heart rate decline throughout the training process, especially throughout the base training phase.

A decrease in resting heart rate corresponds to an increase in stroke volume (amount of blood ejected with each heart stroke). When stroke volume increases, the heart beats less to supply the same amount of blood to the body as it would at a lower stroke volume. This physiological adaptation indicates the cardiovascular system is becoming more efficient, and the heart does not have to work as hard to supply the body with blood. Typically, the greater one's age, the lower the person's resting heart rate.

A significantly low or high resting heart rate can be indicative of fatigue.

Recovery Heart Rate

In addition to the resting heart rate test, the recovery heart rate assessment measures cardiovascular fitness and, more important, improvement. An athlete's recovery heart rate is assessed by having the individual get their heart rate up to a predetermined level, then ceasing activity. **The recovery heart rate is represented by the number of beats the heart rate drops in one minute after cessation of exercise.**

The greater the heart rate drop, the more aerobically conditioned an athlete is. The speed at which the heart rate drops can also be attributed to the fatigue level of an athlete. If an athlete is not recovered from a previous workout or is sick, the heart rate will not drop as fast as it would otherwise. Therefore, the same test should be done on different days, and the average calculated to accurately represent the correct heart rate drop. As your athlete becomes more aerobically fit, the person's average heart rate recovery time should decrease.

Heart Rate Variability

Heart Rate Variability (HRV), while previously applied to predict sudden cardiac death and diabetic abnormalities, has been found to have a profound application in its ability to help guide coaches and athletes in training, recovery, and predictive performance capabilities.

In basic terms, HRV measures variation in time between each heartbeat. The autonomic nervous system (the primitive part of the nervous system) works behind the scenes to regulate heart rate, blood pressure, breathing, and digestion. Originally measured via electrocardiogram, companies are now launching wearable technology linked with smartphone or tablet apps to measure HRV during physical activity, sleep, and rest. HRV measured via wearable technology is a non-invasive way to identify imbalances in the autonomic nervous system. HRV is affected by the different experiences we have in a day, such as physical stress, diet, emotional stress, sleep, and the environment.

According to a Harvard Health report, there are still questions about the accuracy and reliability of tracking HRV and its use with athlete training and physical activity. Tracking HRV has proven beneficial but should be considered as one piece of health data along with other markers indicating an athlete's rest, recovery, and physical preparedness.

VO2 Max

The term ***maximum effort heart rate*** or ***VO2 max*** is used a lot among endurance athletes and relates to maximal oxygen consumption or, more specifically, an individual's maximal capacity to transport and use oxygen during exercise. The *V* stands for *volume*, *O2* stands for *oxygen*, and max refers to *maximum*.

VO2 max also commonly refers to the test to determine a subject's maximal oxygen consumption. So what does VO2 really represent? A VO2 max test can be done on any apparatus (e.g., bike, treadmill, rower, etc.); however, the general testing method is the same. A subject's oxygen consumption is measured while the intensity of the effort increases.

As the intensity of the effort increases, oxygen consumption increases, initially as a linear relationship. VO2 max is represented when a subject's oxygen consumption plateaus, even if the subject can continue increasing effort. It is important to note that when performing a VO2 max test, some individuals cannot determine their VO2 max because they fatigue before their oxygen consumption hits a plateau.

Additionally, while it is normal to increase one's VO2 max through exercise, **the absolute limit of one's potential VO2 max is largely determined by genetics**. As noted below, some individuals do not reach a plateau.

According to the *American College of Sports Medicine* (ACSM), an **individual's VO2 max declines 5-15 percent per decade past the age of 30** (45). The good news is that older adults respond to aerobic exercise equally as well as younger adults in terms of increasing VO2 max. This means that older adults can improve their VO2 max by 10-30 percent through aerobic exercise (45).

Regarding increasing one's VO2 max, studies have found that increases in VO2 are mostly limited to recreational athletes versus well-trained (elite) athletes (597, 598). Therefore, **performance increases in well-trained athletes are likely due to increases in running economy versus increases in VO2 max.**

An individual's VO2 max is typically expressed as milliliters of oxygen per kilogram of body weight per minute (ml/kg/min). For individuals who either should not or cannot do the VO2 max test, there are estimated tests that do not take the individual to their absolute maximum. These tests are called *submaximal tests* and use equations to estimate VO2 max. While the VO2 test determines aerobic fitness, **this test in isolation has often been found to be a poor determining factor of success in endurance athletes as many other factors influence athletic performance** (46). Therefore, determining one's VO2 max is a great way to benchmark fitness and use for reassessment purposes, but regarding performance in the real world, it is only one potential factor.

Exercise Economy

The primary reason VO2 max is often viewed as a poor performance indicator is because of *exercise economy*. **Exercise economy relates to how efficient an athlete is.** Many factors, such as aerobic fitness, biomechanics, and muscular endurance, influence one's exercise economy. Let's look at two hypothetical athletes to demonstrate how exercise economy fits into overall performance. Both athletes (athlete A and athlete B) have a VO2 max of 70. However, athlete B is much more biomechanically efficient than athlete A. If one were to look at just VO2 max, they would be identical regarding potential. However, looking at the big picture (exercise economy), it becomes clear that athlete B will outperform athlete A. In other words, athlete B has greater exercise economy than athlete A.

When assessing athletes regarding their potential as well as their current level, you must take exercise economy into account.

To drive home this point, take Paula Radcliffe, holder of the women's marathon record (2:15:25). When tested (VO2 max) initially as an 18-year-old, and 11 years later, her VO2 max did not change despite her vast improvement. Therefore it is easy to see in this case that her improvement was largely due to an increase in running economy, not an increase in VO2 max (494).

Sport – Specific Fitness?

How can an individual run a 2:30 marathon but barely make it one lap in a pool without hyperventilating and experiencing a skyrocketing heart rate? Is the person out of shape? Not likely if the individual can run a 2:30 marathon. However, they are inefficient swimming, and the wasted energy equates to rapid fatigue and elevated heart rate. This individual's performance in the pool does not correlate whatsoever to aerobic fitness level. It is just a question of what sport the person is most efficient at.

While cardiovascular fitness is not sport-specific, an individual's overall efficiency is, which directly impacts the demand on the cardiovascular system.

Running Economy

While many variables influence running economy, leg girth and, more specifically, the girth of the lower leg can significantly affect a runner's efficiency. The heavier the leg, the more energy it takes to swing it. The closer to the foot the weight is, the truer this becomes. For example, a runner would have greater exercise economy if they had large quadriceps and small calves versus large calves and small quadriceps.

A study by Myers et al. looked to replicate this scenario using weights applied to different body parts while running (492). When 8 pounds were placed at waist level, it required 4 percent more energy than running without any extra weight. However, a 4-pound weight attached to each ankle required the runner to exert 24 percent more energy to run at the same pace.

Another study compared two groups of boys from different countries. The average volume/thickness of one group's calves was 15–17 percent less than the other group. When running, this equates to energy savings of 8 percent per kilometer (493)!

Of course, leg girth is not the only factor of efficiency, which has a significant genetic component. The area of running economy most often discussed is that of running form and muscle/connective tissue stiffness. While these areas are discussed in the *Running Mechanics* module, the areas focused on are stride rate, stride length, spine/hip rotation, vertical oscillation, and overall running form.

Relating VO2 Max to Burning Fat / Calories

If you were to ask ten people what the correct workout intensity is to lose weight, burn calories, and "get in shape," you would most likely get ten different answers. One reason for this is that you are asking three different questions.

Many people associate a great workout with high intensity. However, given the physiological response to aerobic training, this is not necessarily the case and largely depends on a person's fitness goals. Most cardiovascular machines (e.g., treadmills) have a chart depicting heart rate zones optimal for burning fat or calories and getting into aerobic shape. It is true that the lower the intensity at which an individual trains, the greater the percentage of energy that comes from fat. However, one must look at it from the perspective of the total energy expended. This was demonstrated by Johannes Romijn et al. in a 1993 research paper titled *Regulation of endogenous fat and carbohydrate metabolism in relation to exercise intensity and duration* (47).

Romijn demonstrated that while at 25 percent of an individual's VO2 max, the percent energy from fat was 85 percent. At 65 percent of an individual's VO2 max, the percent energy from fat decreased to 60 percent. However, due to the increased energy expenditure at 65 percent of VO2 max, the 60 percent of energy from fat accounted for more (about 40 percent more!) than the energy from fat at 25 percent VO2 max. An individual at 85 percent of their VO2 max uses only 30 percent of energy from fat. Even though this percentage is low, it still represents more total energy from fat than someone at 25 percent VO2 max, but less than someone at 65 percent VO2 max (47). The following chart graphically illustrates this (CHO = Carbohydrates).

Therefore, when discussing which intensity is more optimal for burning fat, there are two answers: at 25 percent VO2 max, you will have the greatest percentage of energy coming from fat. However, **the intensity where you will burn the most fat is at approximately 65 percent VO2 max**. To address the calorie issue, this one is pretty easy. The greater the intensity, the greater the caloric expenditure will be.

The above chart is also a graphic representation of how exercise intensity correlates with exercise sustainability. The greater the intensity, the greater the reliance upon carbohydrates; therefore, the less sustainable the exercise is for long durations of time.

Should You Assess VO2 Max?

The VO2 max test is considered the gold standard in determining aerobic capacity. This test requires specialized equipment and training to administer. Submaximal tests are used to estimate the VO2 max for individuals for whom a true VO2 test would be unsafe or for those who do not have access to a VO2 max test or simply do not want the full test. As a side note, a VO2 max test typically costs $100–$300 and is often done with a blood lactate threshold test.

As noted previously, **VO2 max is often a poor performance indicator, as it does not consider exercise economy**. This is where correlating blood lactate is valuable to know. For example, two athletes could have the exact same VO2 max but different blood lactate levels. VO2 max relates to aerobic capacity, but lactate threshold (LT) determines the percentage of VO2 they can perform and for how long. This was demonstrated in a 1982 study by Bertil Sjodin (499). In this study, runners performed training sessions at lactate threshold for 14 weeks. At the end of the 14 weeks, the subjects decreased their average mile time from 5:43/mile to 5:29/mile. Interestingly, the VO2 max of the runners did not increase at all!

A 2008 study by Tim Noakes found that **during a VO2 max test, only 47 percent of subjects reached a plateau** (728). While this finding suggests that the limit of a VO2 max test may not solely depend on oxygen, it is clear that the defining point of a VO2 max test (plateau) is likely unreliable.

Several assessments/tests are used to estimate (submaximal) VO2 max. The *Bruce protocol test* (70) and the *YMCA cycle ergometer test* are two of the more widely used assessments. Concerning a true VO2 max assessment, estimated VO2 max assessments are relatively easy to administer and do not require extensive equipment or training.

While it might be interesting to know one’s VO2 max, from a training perspective, you should be more focused on the heart rate – specifically the final heart rate when the athlete can no longer continue. This is because VO2 max expressed as ml/kg/min cannot be correlated directly to other aerobic fitness benchmarks, whereas heart rate can.

When compared to a true VO2 max test, many of the estimated VO2 max tests underestimated one’s VO2 max (454, 455). Specifically, the *Bruce protocol test* and the *YMCA cycle ergometer test* underestimated VO2 max by an average of 14 percent (455). This certification will not include an estimated VO2 max assessment for the preceding reasons.

Anaerobic Threshold

The term **anaerobic threshold (AnT)** is used extensively in the world of endurance sports and is most commonly used to describe the working upper limit of one's aerobic capacity. **AnT is difficult to define as it is more of a concept than an established metric.** It should be noted that some individuals do not believe that AnT exists (131). Because of this, there are many different interpretations regarding what AnT is. In addition to confusion about the definition of AnT, it makes sense that there is equal confusion over how to assess and estimate it (441). Some of the most common benchmarks used to interpret AnT are:

Lactate Threshold (LT)

Representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.

Maximum Lactate Steady State (MLSS)

The highest intensity level at which blood lactate concentrations are maintained at a steady-state (equilibrium) level during exercise bouts of at least 60 minutes (441). This may or may not be higher than OBLA.

Ventilatory Threshold (VT)

The point at which the ventilation rate increases faster than the workload. Until the VT is reached, the workload and respiration rate increase linearly. It is often correlated with LT.

Onset of Blood Lactate Accumulation (OBLA)

OBLA is often used synonymously with LT. However, OBLA occurs at a slightly higher level than LT (71, 125). Technically speaking, OBLA is reached when the blood lactate accumulates to a certain level in the blood (4mmol) (382).

A primary reason for the increase in popularity of quantifying AnT in sports is the poor correlation between VO2 max and athletic performance (137, 138). As a result, finding a submaximal threshold that can be maintained over long periods has gained traction within the endurance sports community (138).

While the AnT interpretations just noted have value to runners when gauging intensity, **they should not be used to infer AnT (127) directly**. This is primarily because of a lack of correlation between assessment methods. For example, a 1993 study out of the University of British Columbia found that many of the correlations appeared to be coincidental when using VT and LT to estimate AnT (123).

The theory that a coincidental correlation exists between AnT, VT, and LT is further supported by a 1982 study that looked at hyperventilation rates in people with *glucose storage disease-type 5* (commonly called *McArdle syndrome*) (141, 142). Individuals with McArdle syndrome cannot break down glycogen in the muscles. As **blood lactate results from glycogen breakdown (catabolism)**, these individuals do not produce lactate above resting levels with exercise. In this study, the participants' VT increases at the point at which their proposed AnT would be. This demonstrates that VT is not caused by an increase in LT.

Therefore, the assessments noted above and others should be used as stand-alone assessments but not for inferring AnT.

A proposed AnT implies that once individuals exceed it, they no longer use oxygen for energy but instead rely on glycogen exclusively for energy. This is erroneous as there is no evidence that the body "switches" from using oxygen to not using oxygen at a particular intensity level. **Oxygen is present at all levels of intensity, albeit in varying amounts.** At one's proposed AnT, the anaerobic energy systems do not take over for the aerobic energy system but supplement it to meet the increased energy demand (126).

Conconi Test

A popular estimated anaerobic threshold (AnT) test is the ***Conconi Test***. The Conconi test was developed in 1982 by Francesco Conconi, an Italian sports doctor (259). This test can be administered on either a treadmill or bike. This test determines AnT by plotting a subject's heart rate and looking for a plateau. This plateau is known as the ***deflection point*** and represents the AnT. The theory behind the Conconi test is that one's heart rate increases linearly with the increase in workload. When the linear relationship ends because of the deflection point, it represents the AnT.

Individuals can continue exercising past their AnT, and their heart rate will continue to rise. However, the individual will probably not be able to continue for much longer. Several studies have challenged the validity of the Conconi test, stating that not all subjects have shown deflection points on the test, meaning the results are not consistently replicated (73). Also, studies show a lack of correlation between the Conconi test and true LT tests. The lack of correlation demonstrates that **the Conconi test likely overestimates LT** (74). While UESCA will not use the Conconi test for the reasons above, it is important to be aware of this test regarding its intended use and the challenges to its validity.

Origin Of Anaerobic Threshold

Wasserman and McIlroy coined the term anaerobic threshold in 1964 to identify an intensity that was challenging and safe for cardiac patients (136). The goal of anaerobic threshold was to identify a reliable, submaximal effort level so that cardiac patients were not subjected to maximal cardiovascular tests. The premise of their anaerobic threshold theory was that blood lactate accumulation is the result of poor oxygen levels (127).

Lactate Threshold

Lactate threshold is the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.

While lactate is always produced at rest or at low to moderate exercise intensities, the lactate is "cleared" within the muscle. So, what exactly does "cleared" mean, and how does this influence LT? Regarding LT, when you see the term "cleared," think "used for energy." **Lactate is cleared intra-muscularly** by oxidizing the lactate and converting it back to pyruvate (oxidative fuel) within the muscle in which lactate was formed (319).

Lactate is used for fuel in the mitochondria. Lactate permeates the mitochondrial wall via a special transporter protein called ***monocarboxylate transporters*** **(MCTs)** (512). MCTs can be increased via training, thus facilitating increased lactate clearance (76).

This clearing method prevents the intra-muscular lactate levels from increasing too fast to the point where lactate levels reach the intra-muscular threshold and spill out into the bloodstream. Therefore, intra-muscular lactate levels can increase substantially (up to five times resting levels) without any increase in blood lactate levels (513). However, when the intra-muscular lactate threshold is breached, some lactate exits the muscle and enters the circulatory system, thus increasing blood lactate levels. This is representative of the lactate threshold.

Once blood lactate enters the circulatory system, it is shuttled to areas of the body that can use it for fuel (i.e., brain, heart, and other skeletal muscle). Blood lactate that is not utilized is sent to the liver, where it is converted to pyruvate and then to glucose. The glucose is sent (via blood) to the muscles to be used as fuel. This is representative of the ***Cori Cycle*** (515).

When exercising at MLSS, most lactate is oxidized for fuel by converting lactate to pyruvate and back to lactate. When the intensity increases much beyond MLSS, the lactate enters the circulatory system.

The process by which lactate moves intra-muscularly and throughout the circulatory system is termed the ***lactate shuttle*** (319, 511).

As lactate can be added and removed from the blood, the amount of blood lactate at any given point is representative of blood lactate accumulation, not production.

An analogy to LT would be to equate a muscle fiber to a drinking glass, the water within the glass to lactate, and any area outside the glass to the circulatory system. Levels of water (i.e., lactate) can increase and decrease, but so long as the levels do not breach the top of the glass (i.e., lactate threshold), blood lactate levels will not increase.

However, once the water spills over the top of the glass, the blood lactate levels increase as there is an increase in lactate within the circulatory system.

Historical Reference

In addition to Dr. Meyerhof's frog study (referenced in *Module 3*), in which he concluded that high lactate levels caused muscular fatigue (24), two other studies have profoundly impacted the erroneous perception of lactate.

In 1857, biochemist Louis Pasteur deduced that lactic acid formation was possible only in the absence of oxygen (anaerobic). This is called the *Pasteur effect* (509).

Secondly, in 1923, research by physiologist A.V. Hill theorized a correlation between high work levels and high blood lactate levels. More specifically, he theorized that the aerobic system was inadequate to provide fuel to the body at high intensities. Therefore the anaerobic/lactic acid system would “switch on” and take over from the oxidative system (510).

In the years since then, the following has been found to be true:

- Lactic acid does not exist within the human body
- Lactate is a fuel
- Lactate is produced at all times
- The proposed anaerobic system does not “switch on” after the oxidative system is exhausted
- Lactate is not responsible for muscle burn

Despite these findings, there is still a large consensus that believes that lactate is a waste product and responsible for a reduction in human athletic performance.

Is Lactic Acid and Lactate the Same?

Lactic acid does not really exist in the body – this is because the blood’s pH is too neutral, and acids require a very low pH. Therefore, immediately after lactic acid is produced in the body, it splits into lactate and hydrogen (908). So while some might think of it as semantics, when people use the term ‘lactic acid,’ they are really referring to lactate.

Benchmarking Lactate Threshold

LT has become a common benchmark for gauging intensity during training and racing in endurance sports. Like anaerobic threshold, the method by which LT is assessed and quantified is confusing. LT benchmarks such as OBLA, MLSS, and LT itself are often viewed as synonymous. As noted previously, LT, OBLA, and MLSS occur at slightly different intensity levels.

LT has become a common benchmark for gauging intensity during training and racing in endurance sports. Like anaerobic threshold, the method by which LT is assessed and quantified is confusing.

For this certification, we will provide a few ways to determine LT and corresponding heart rate zones to provide intensity zones for your athletes effectively.

Lactate threshold typically corresponds to approximately 85 percent of one's maximum heart rate (737). UESCA recommends using the Functional Threshold Heart Rate Assessment (FTHR) to estimate one's LT. This assessment is discussed below.

Blood Lactate Clearance

Lactate clearance was discussed earlier in terms of being cleared intra-muscularly. However, once lactate threshold is reached and lactate enters the circulatory system, lactate becomes blood lactate and is cleared in one of two ways (29):

1. Lactate is shuttled to areas of the body that can use it for fuel (i.e., brain, heart, other skeletal muscle).
2. Blood lactate that is not utilized is sent to the liver, where it is first converted to pyruvate, then to glucose. This is representative of the *Cori Cycle.*

When blood lactate levels increase past what can be cleared, blood lactate accumulates.

In a 1989 study by McMaster et al., swimmers were assessed to determine the best intensity level to clear lactate after a maximal effort. McMaster et al. found that when swimmers swam at 65 percent of their maximum effort, they cleared the most lactate (320). This study was performed using swimmers because during meets, swimmers often have to compete in multiple events, thus necessitating proper recovery from one event to the next.

Another study found that exercising near one's LT cleared the most lactate. This study also found that **lactate was cleared faster via active rest than at rest** (506).

While the first study noted is specific to swimming, the science likely holds true for that of running. It is applicable from both a perspective of recovering during competition and training when performing interval-type sessions.

The positive effects of active recovery regarding clearing lactate seem to be limited to 20 minutes or less (507).

Training Implication

While physical training has no impact on the body regarding lactate production, it does influence the ability of an individual to clear lactate. **Interval-based training is the most efficient way to increase one's ability to clear blood lactate.**

Why Not Directly Test for Lactate Threshold?

LT can be assessed directly by taking blood samples for analysis and running them through a lactate blood analyzer. However, unless you are trained on how to administer and analyze an LT test, it is recommended that you outsource this test to a qualified professional. If you are qualified to administer the test, you must use all proper blood handling/disposal procedures and must check with local laws to ensure you are complying with current health regulations. For most coaches this is not an option; therefore, an estimated assessment of LT (MLSS) is the best course of action (72). An estimated LT assessment such as MLSS is an accurate, safe, and legal way to assess an individual's LT.

Functional Threshold Heart Rate Assessment (FTHR)

FTHR represents the maximum effort that can be sustained at a steady rate for one hour (77). FTHR typically correlates to LT (77). Using FTHR as a benchmark, you can establish training zones for your athlete. FTHR can be estimated by a 20-30 minute assessment rather than a full hour.

As FTHR pertains to the average heart rate an individual can likely sustain for 1 hour, races shorter or longer than 1 hour will result in different heart rate targets.

FTHR Assessment

The FTHR assessment is a great way to determine an athlete's specific training needs and race preparation. Heart rate response over target durations can help gain insight into areas of an athlete's relative strengths and weaknesses.

Adapted from the Functional Threshold Power (FTP) assessment originally developed by Allen & Coggan, it is a popular system upon which many cyclists and coaches base their intensity control. It is the highest power that a rider can maintain in a quasi-steady state for approximately an hour.

Concerning the 20-minute FTHR test, it is critical to perform an effective warm-up. While the exact warm-up structure will vary between athletes, the main component of a solid warm-up is easy running with several short (10-20 second) hard efforts about two-thirds to three-quarters of the way through the warm-up.

While an FTHR assessment aims to maintain a maximum effort for 20-minutes, it is important to not under or overshoot this effort. Therefore, it is critical not to start too hard and not be able to finish the test or drastically reduce one's intensity.

20-Minute Test

This test can be done on a treadmill or outdoors. The most important thing is that the runner's pace is uninterrupted (i.e., stoplights). For this reason, many individuals find a treadmill or running track to be the most optimal environments. The intensity should be at the maximum level they can sustain for 20 minutes. Therefore, running substantially faster or slower than this desired pace will result in an inaccurate test result. Since heart rate is being assessed, having a heart rate monitor that can average heart rate over a set workout is necessary.

The duration of the FTHR assessment is a total of 30 minutes, with the first 10 minutes being the warm-up. At the 10-minute point, the runner increases their effort to what they believe they can sustain for the next 20 minutes. At this point, the athlete starts their heart rate monitor to record their heart rate so they can take the average of the 20 minutes at the end of the assessment.

It is suggested that this test be used for training intensity prescription and regular monitoring, as the 20-minute time-trial performance is reliable and is sensitive to training adaptations and running ability. Repeating testing is beneficial to get appropriate pacing strategy and effort benchmarks. One's FTHR is 95% of their average heart rate for the 20-minute test.

Training Zone Integration Of FTHR

Once an athlete has their FTHR, they can establish their training zones. Below are the zones and associated intensities, as well as a hypothetical athlete's zones. It is important to note and reiterate that the body's energy systems do not function as zones – one does not 'turn off,' and the next zone 'turns on' as intensity increases. The zones are noted as general intensities and guidelines to train at to stimulate and improve particular physiological adaptations to improve running performance. Therefore, while a particular adaption might be noted in the 'Primary Training Application' section, that adaption is likely being developed in another zone(s) as well, albeit to a lesser degree.

TRAINING ZONE	PRIMARY TRAINING APPLICATION(S)	AVERAGE HEART RATE (% OF FTHR)
ZONE 1: RECOVERY (Active)	Recovery	< 68%
ZONE 2: ENDURANCE	Improve fat metabolism, increase endurance, Increase cardiovascular/base fitness	69 - 83%
ZONE 3: STEADY STATE	Increase aerobic fitness	84 - 94%
ZONE 4: LACTATE THRESHOLD	Increase lactate threshold, improve sustainable pace	95 - 105%
ZONE 5: VO2 MAX	Increase ceiling for aerobic fitness	> 106%
ZONE 6: ANAEROBIC POWER	Anything above VO2 Max (ex: sprinting)	N/A

After performing a 20-minute FTHR test, our hypothetical subject, Brian, calculated his FTHR to be 165. Based on this information, below are Brian's heart rate zones as it relates to the above zone percentages.

TRAINING ZONE	PRIMARY TRAINING APPLICATION(S)	% OF FTHR	BRIAN'S FTHR ZONES
ZONE 1: RECOVERY (Active)	Recovery	< 68%	< 112 bpm
ZONE 2: ENDURANCE	Improve fat metabolism, increase endurance, Increase cardiovascular fitness	69 - 83%	113 - 137 bpm
ZONE 3: STEADY STATE	Increase aerobic fitness	84 - 94%	139 - 155 bpm
ZONE 4: LACTATE THRESHOLD	Increase lactate threshold	95 - 105%	157 - 173 bpm
ZONE 5: VO2 MAX	Increase ceiling for aerobic fitness	> 106%	> 175 bpm
ZONE 6: ANAEROBIC POWER	Sprinting	N/A	

Training Zones

Zone 1: Recovery

This intensity zone is representative of active recovery. Regarding intensity, it should feel extremely easy.

Zone 2: Endurance

Base training and any type of long workout to build or maintain endurance fall into this category.

- **Training Applications:** Long run

Zone 3: Steady State

This is a slightly more intense version of Zone 2. During this phase, lactate production increases but can still be cleared effectively.

- **Training Application:** Runs at a slight increase in intensity from an endurance run.

Zone 4: Lactate Threshold

During this phase, lactate accumulation continues to increase to the point where it is around one’s lactate threshold.

- **Training Application:** tempo

Zone 5: VO2 Max

At this point, lactate accumulates faster than it can be cleared and represents the high end of one’s aerobic capacity.

- **Training Applications:** intervals, repeats

Zone 6: Anaerobic Power

During this zone, an individual works above their VO2 Max, which is not sustainable for long periods.

- **Training Applications:** All-out sprints.

Programming and Pacing

In addition to stating exact heart rate ranges, you can direct athletes using the zones.

By using training zones with respect to the primary energy system they target, you can utilize this information for pacing guidelines for various race/training distances and types. Other metrics such as rate of perceived exertion (RPE) and pace per mile can also be integrated into pacing guidelines.

If using heart rate to base pacing off of, environmental factors such as heat and cardiac drift should be considered as they may influence an athlete’s heart rate.

Specificity of Sport Disciplines

The greater an athlete’s proficiency in a particular sports discipline, the lower the heart rate response. This is a function of one’s efficiency.

Preferred Analysis Tools

In the video below, Ben Rosario discusses two primary analysis tools he favors for athletes.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Emerging Technology

Like any field or industry, sports science is always evolving, and with it, the means to assess and track athlete data. While there are a lot of areas of emerging technology in the realm of endurance sports, two of the most significant areas currently are blood biomarkers – specifically, the assessment of one's blood for the purposes of health and performance and the other being constant glucose monitors for performance.

Blood Analysis / Markers

Blood analysis provides a unique window into an athlete's health and performance. Certain biomarkers like vitamin D and ferritin can directly impact athletic performance, while others like cortisol and creatine kinase can provide insights into overtraining and injury prevention. Athletes can now utilize blood testing to pinpoint areas of focus and implement effective changes to improve performance.

Of course, it is outside the scope of practice and knowledge of an athlete or coach to perform blood draws. And even if an athlete or coach could, they likely would not be able to analyze the data and make meaningful and actionable changes to a training program based on the information. Companies such as InsideTracker are focused on analyzing blood biomarkers specific to endurance athletes, not just overall health, as with physician-referral blood tests.

Constant Glucose Monitoring (CGM)

Glucose, or blood sugar, is a critical element to an endurance athlete's performance and too little results in a decrease in athletic performance – often referred to as the dreaded 'bonk.' CGM technology is not new. Those with diabetes have used it to monitor their glucose levels. However, this technology is now being used to track one's glucose levels in real-time to reduce the chance that an athlete will not run out of glucose/glycogen. In other words, it is a means to inform an athlete when they need to ingest fuel.

The technology works via a patch/sensor with a tiny needle, and in the case of Supersapiens (a CGM company), the sensor is paired to an app that reads the sensor's data. According to Supersapiens, the CGM has benefits both while exercising and not exercising. The advantage while exercising is that an athlete can fuel in the proper amount (i.e., not over or under fuel). While not exercising, the CGM can inform an athlete of the right amount of fuel to ingest to optimize muscle glycogen replenishment (1015).

Other

Some other areas to be aware of are sweat sensor monitors and core body temperature sensors.

Summary

- A heart rate monitor is an important tool to assess intensity as well as recovery.
    - Using a heart rate monitor to assess one's resting and recovery heart rate helps to assess recovery as well as increases in aerobic fitness level
- *Rate of perceived exertion* (RPE) scale is a subjective test that measures physical exertion or pain
- Heart rate variability (HRV) measures variation in time between each heartbeat
- VO2 max in isolation is often a poor determining factor of performance
- It is not advised to use one's estimated maximum heart rate to base exercise intensities off of
- The greatest amount of fat "burn" occurs at approximately 65 percent of VO2 Max

- Functional Threshold Heart Rate (FTHR) correlates to lactate threshold
- The *Conconi test* has been shown to often overestimate lactate threshold
- The *talk test* can be used in isolation or to establish a benchmark for training.
- *Anaerobic threshold* (AnT) is difficult to define as it is more of a concept than an established metric
- Of all the AnT and LT benchmarks, FTHR has the highest correlation to training for and competing in distance running events and endurance sports as a whole
- It is advised to create training intensity charts based off the FTHR.
- *Cardiac drift* relates to a natural increase or upward “drift” in heart rate while the overall intensity remains the same
- Constant glucose monitoring and blood biometric markers are considered emerging technology that provides an athlete with more data points to help guide their training and racing.

Module 7: Running Mechanics and Drills

Running is an impact sport and therefore tends to have a relatively high injury rate compared to other endurance sports. Because of this, a runner's form and, more specifically, the running gait is often analyzed to improve efficiency and prevent injury.

It is important to reiterate that unless you are a physical therapist, chiropractor, physician, nurse, physician assistant, etc., you cannot diagnose or treat a suspected injury. You must always practice within your scope of knowledge and training. Dealing with an injury is not within your scope of practice, and aside from telling an athlete to ice a sore area, the advice should always be to seek a qualified medical opinion.

Biomechanical assessments are the first place to start when evaluating your athlete. This will tell you a lot about areas that are potentially susceptible to injury or, at the very least, inefficient. The next phase is to assess athletes while they are running.

The mechanics, and therefore the runner's form, is often termed ***running economy***. Many factors influence running economy, such as calf muscle girth, bone length, running form, and stride rate. Some factors are genetic, while others are not. **Running economy relates to how efficient a runner is.** A study found that the variation in efficiency between well-trained runners with good running economy and untrained runners with poor running economy is between 5 and 7 percent (715). It is likely that on an individual basis, the difference is even more significant (705).

Just because an individual runs doesn't mean the person is efficient at it. Proper running form does not just happen. Running is a learned skill, just like other sports (731).

Powerpoint videos and assessment videos in this lesson are provided by UESCA contributor and running biomechanist Dr. Nick Studholme DC.

Terminology

Pronation: Inward (medial) rotation of the foot. It is normal during a proper gait pattern for the foot to pronate. Foot pronation helps the body absorb stress from the impact of running and walking

– **Overpronation:** Excessive pronation

Supination (under pronation): Occurs when the foot "rolls" laterally. This places the majority of the weight on the outside aspect of the foot. Essentially, supination is a lack of pronation. During supination, the subtalar joint does not flex.

Heel-to-Toe Drop: Difference in height between the heel and forefoot of a shoe

Ground Reaction Force (GRF): Based on Newton's third law, GRF is the force exerted by the ground on a body in contact with it (665).

Spinal Engine: Developed by Dr. Serge Gracovetsky, this model states that counter-rotation of the spine results in transverse hip/pelvic rotation and is a major influence on the gait cycle. More specifically, Dr. Gracovetsky theorized that upon foot strike while running, energy was not absorbed by the ground but rather utilized up the body's kinetic chain to assist with locomotion.

Coupled Motion: This relates to a sequence of movements by different body parts to create an efficient and powerful movement.

Dorsiflexion: Pointing of the foot upward (flexion of ankle joint)

Plantar Flexion: Pointing of the foot downward (extension of ankle joint)

Passive Energy: This relates to energy being stored (potential energy) via the stretch of a muscle and connective tissue. By utilizing this free energy, the active muscle requirement is reduced while still producing the required amount of force.

Stretch-Shortening Cycle (SSC): Representative of an eccentric contraction of a muscle followed by a rapid concentric contraction of the same muscle. Concerning running, the quadriceps, obliques, and calves represent the muscles most influenced by the SSC.

Passive Movement: Representative of body movement that is not the result of active muscle contractions. Specific to running, lower leg (tibia/fibula) swing and hip flexion are several body movements influenced exclusively by passive movement.

Pedestrian Gait Model: The legs are the primary aspect of the body responsible for locomotion, and the upper body is largely passive.

Windlass Mechanism: Before the foot landing when running, the toes are dorsiflexed, causing the plantar fascia to tighten, thus creating a stiff foot arch.

Vertical Oscillation: Vertical motion of a runner. While some vertical movement is normal and necessary during the running gait cycle, excessive vertical oscillation reduces a runner's efficiency.

Torsion: Twisting of an object, especially from one end of an object relative to the other end.

Aponeurosis: Tendon that is typically flat and broad (i.e., sheet-like). The plantar fascia is an example of aponeurosis.

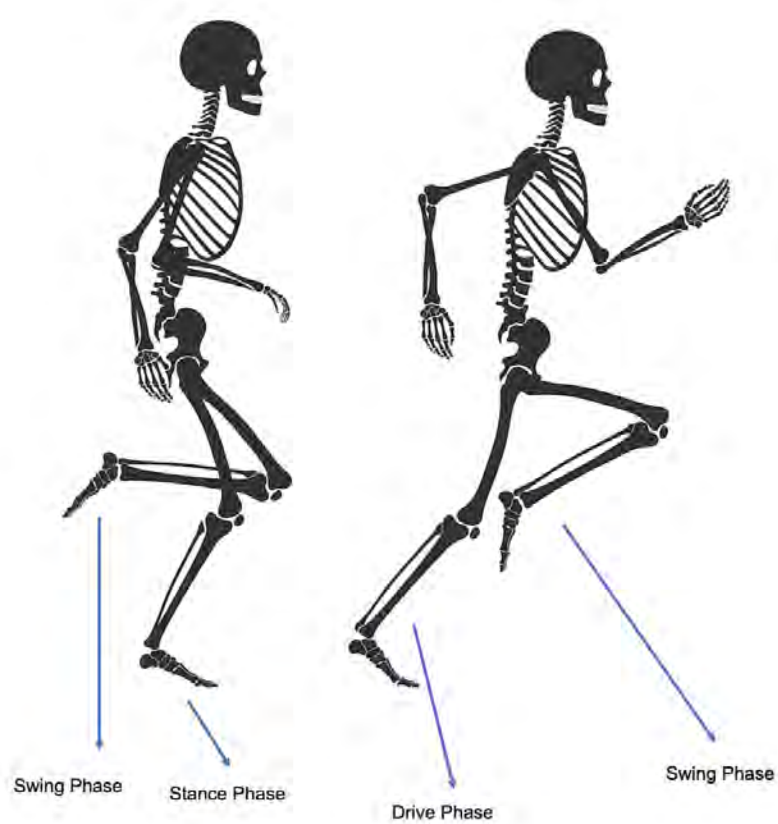
In this module, there will be several videos by our biomechanics contributor, Dr. Nick Studholme. In some of these videos, Dr. Studholme will reference slides that will be sequentially below the video. Therefore, as a video is playing, please scroll down to reference the slide Dr. Studholme is referring to.

Gait Phases

Running gait (cycle) has three phases:

1. **Recovery (Swing):** Any time a foot is in the air
2. **Drive:** The drive phase starts when the leg is slightly behind the runner’s hips and continues until right before the foot leaves the ground
3. **Support/Stance:** The support (*stance phase*) starts when the foot hits the ground until the leg is directly underneath the hips. With a correct midfoot strike, the support phase is eliminated or substantially minimized as the foot hits the ground under or just in front of the hips, thus beginning the drive phase.

These phases are illustrated below.



Ground Reaction Force

Ground reaction force *(GRF)* is important as it relates to a runner's efficiency and the potential for injury. GRF relates to Newton's third law:

For every action, there is an equal and opposite reaction.

Regarding running, this means that an equal force acted upon the body in relation to the force of the foot strike on the ground. On average, each foot strike exerts a force 2.5 times one's bodyweight. This force is transmitted through your foot and up the leg. A runner's form in relation to the person's biomechanics will influence the distribution of force on the musculoskeletal system (668).

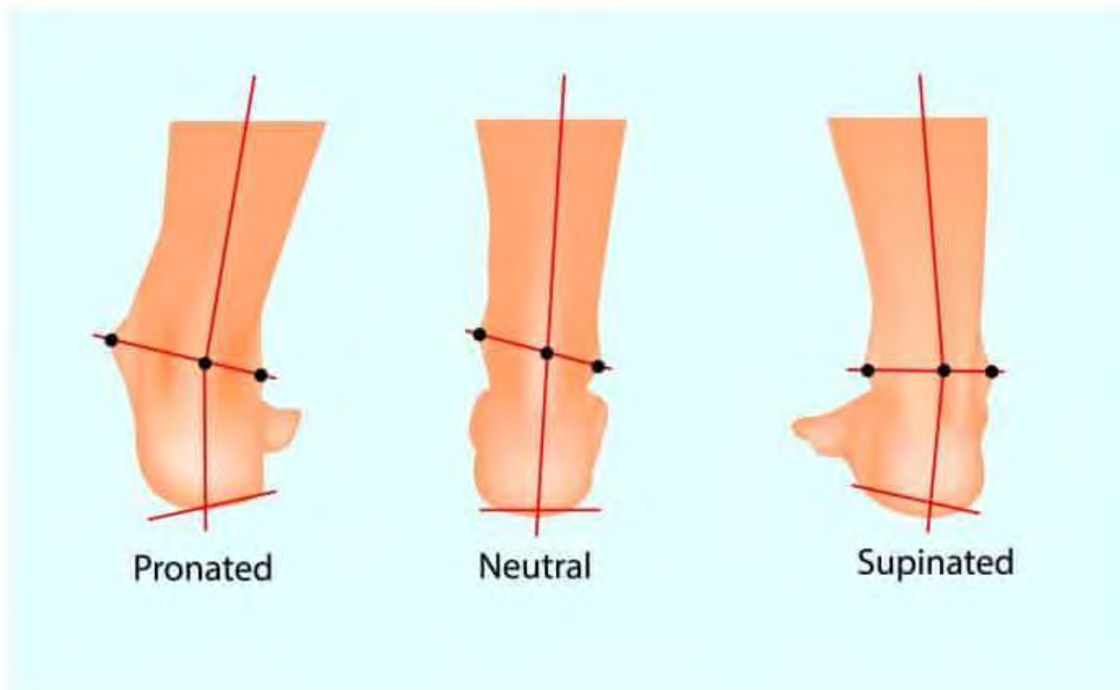
Ground reaction forces change based on the gait of an individual. In a study by Keller et al., vertical GRF was evaluated during walking, slow jogging, and running (666). The study found that vertical GRF was 50 percent higher during slow jogging than walking or running. This increase in GRF was associated with greater vertical oscillation.

The higher the GRF, the greater the chance for injury (667). More specifically, an individual's specific biomechanics will likely influence the location and type of injury sustained (668).

Pronation and Supination

There are three common designations of foot movement throughout the gait cycle.

1. **Pronation (mild)**
2. **Overpronation**
3. **Supination (i.e., under pronation)**



These foot patterns are commonly evaluated statically in a standing position and dynamically during running. Since there is no contest in running events for how long one can stand statically, this is not an important benchmark. More specifically, it has been shown that foot and ankle mechanics vary greatly between standing and running (421, 422, 423).

A common assessment of pronation in a static position is the *wet test*. This assessment is performed by having an individual wet the bottom of the feet and then stand on something that absorbs water (i.e., a brown paper bag). The tester can then determine if the foot or feet are neutral, pronate slightly, overpronate, or supinate by assessing the footprint left by the watermarks. As noted previously, the issue with this assessment is that foot and ankle mechanics are not the same when standing and running. Additionally, **studies by the U.S. military demonstrated that assigning footwear based on one's footprint did little to reduce injury** (432, 433).

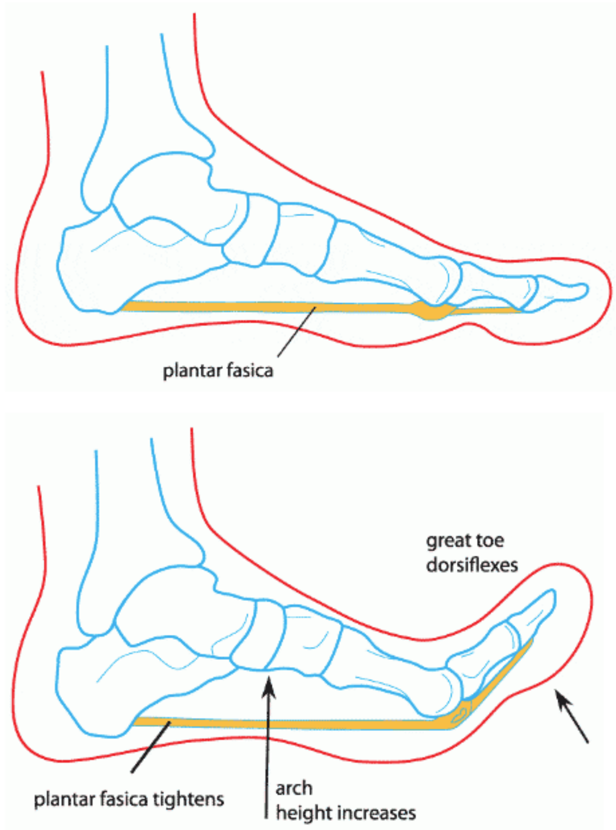
Role of the Big Toe

Within the world of running, there is a lot of information about biomechanics – primarily with respect to the knees, hips, and ankles. The toes and, more specifically, the big toe is often left out. The body is one big kinetic chain, meaning that no body part functions in isolation when running. Regarding the big toe, its function will affect the areas above, such as the hips, knees, and ankles.

During running, the big toe (first metatarsophalangeal joint) plays a key role in the following:

- Stabilize the foot
- Regulate the degree of foot pronation
- Forward propulsion

The big toe, in relation to the other toes, is responsible for a much more significant percentage of foot and body stabilization and forward propulsion.



Credit: https://www.docpods.com/foot-pain-info/the-windlass-mechanism-in-the-foot-and-foot-pain/

The ***windlass mechanism*** of the foot pertains to the lifting (dorsiflexion) of the toes. As is seen in the illustration above, the origin of the plantar fascia is the heel (calcaneus) and inserts at the head of the metatarsals.

Prior to the foot landing when running, the toes are dorsiflexed, causing the plantar fascia to tighten, thus creating a stiff foot arch. This is representative of the *windlass mechanism* (611).

Upon landing, the toes and plantar flex, and the arch flattens to absorb shock.

The toes begin to dorsiflex when the leg moves into the drive phase. This action turns the arch into a rigid lever. This allows for a solid base of support to push off against.

As noted in the image to the left, the sesamoid bones are two small (pea-size) bones embedded into a tendon (flexor hallucis brevis). The bones sit under the ball of the foot, at the big-toe joint. The sesamoid bones act as a fulcrum to provide the foot leverage when pushing off the ground (612).

Big Toe Facts

- Carries 12 times more weight than the small toe
- Only toe made up of two bones as opposed to three
- It has a separate set of control muscles and tendon insertions than the rest of the toes

Shoe Implications

Running shoes that are too narrow crowd the toes in the toe box. **This negatively affects the aforementioned roles of the big toe, thus reducing the efficiency and stability of a runner**. A reduction in an ultrarunner's efficiency due to a cramped toe box likely increases their metabolic cost and decreases their stability. Therefore, running in shoes with a wide enough toe box is advised to allow the forefoot and toes to exist in their normal state (e.g., not cramped). Chronically wearing footwear with a crowded toe box can cause a bunion (biomechanical foot issue).

Shoe fit for runners is not just a function of efficiency but also minimizing the chance for hotspot and blister formation. Therefore, the interface of the socks and shoes, as well as the overall fit, is important for preventing blisters and running performance.

One shoe company, *Altra*, has addressed this issue by making shoes with a wide toe box. While other shoemakers offer shoes in varying widths, *Altra* incorporates this aspect in all of their shoes.

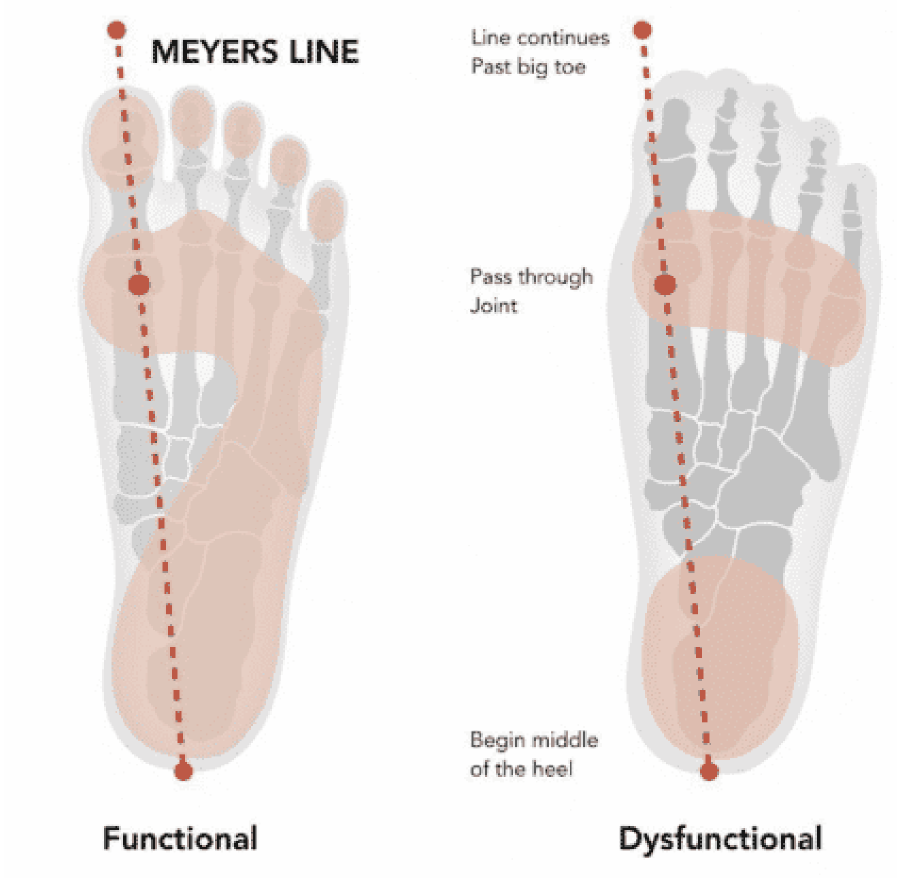
Dysfunction

As noted above, cramming the foot in the toe box of running shoes reduces the big toe's role and the overall size of the foot pushing off the ground, thus minimizing the power generated.

In terms of running gait and power, there are a lot of areas that one could discuss, including but not limited to foot arch/Achilles tendon stiffness. However, as this section is focused on the big toe, there is one main area of focus – the base of support.

Whether due to a cramped toe box, or a structural foot issue, any reduction in the ‘size’ of the forefoot due to malalignment of the toes will negatively affect a runner.

The *Meyer’s Line* is a straight axis line that, on a functional foot, goes from the center of the heel, through the big toe joint, and through the center of the top of the big toe (as noted in the below image). However, when the big toe is angled toward the second toe, in terms of running, its role in providing support, controlling pronation, and facilitating forward propulsion is greatly reduced.



Often caused by footwear that restricts/crams the toes together, a bunion (pictured below) is a deformity of the big toe joint that presents with the big toe angling toward the second toe and a medial bony protrusion. The greater the angle, the greater the dysfunction. For some, a bunion can be painful and, if severe enough, may require surgery.

Get Stronger and More Flexible

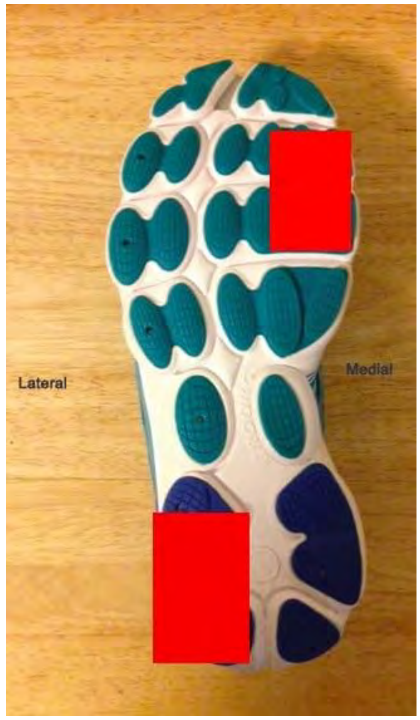
Any exercise or movement challenging the big toe to stabilize and strengthen is good. Exercises such as balancing on one leg, plantar flexing the toes, and spreading the toes are good for strengthening the big toe. Additionally, it is also important that the big toe have enough range of motion. So stretching the big toe (dorsiflexion), either manually or via a standing stretch are good mobility drills.

Shoe Wear Patterns

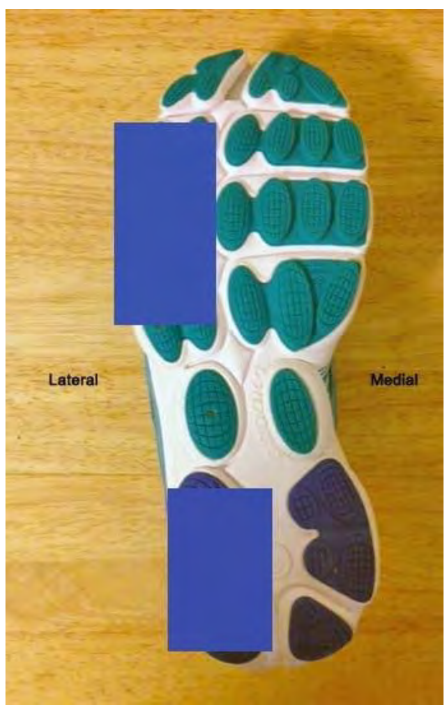
If your athlete has an old pair of running shoes, **looking at the wear pattern on the bottom of the soles will tell you a lot about the person's foot strike**. For example, if there is very little or no wear on the heel, the athlete likely runs with a midfoot or forefoot strike.

If you look at the worn sole of a pronator, typically the lateral aspect of the heel is worn because this represents the first place of impact, as the foot is supinated. As the foot rolls forward and moves into pronation, the middle to the inside (medial) of the top third part of the sole tends to be the most worn. Normal pronation is also associated with compression of the foot arch.

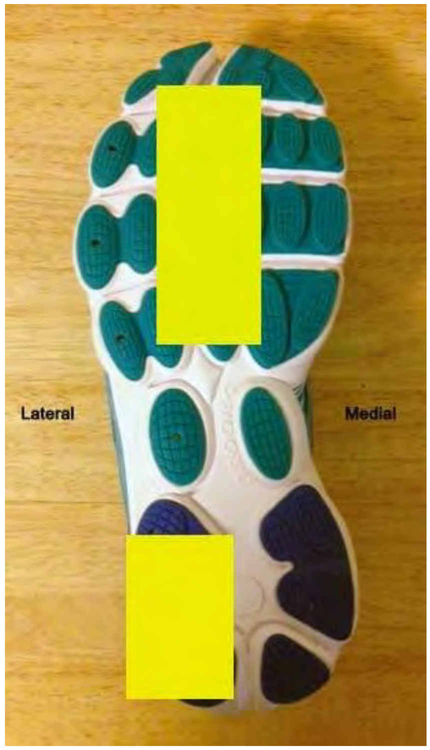
The images below denote three common wear patterns and associated foot/ankle mechanics.



PRONATION



SUPINATION



NEUTRAL

If an athlete walks and runs in running shoes, the wear pattern will likely resemble that of a heel striker, even if the person runs with a midfoot strike. This is because when walking, the foot initially lands on the heel.

Pronation

Foot pronation refers to the inward (medial) rotation of the foot. **It is normal during a proper gait pattern for the foot to pronate. Foot pronation helps the body absorb stress from the impact of running and walking.** Many people assume that foot pronation is bad, but as noted above, it is a natural and necessary movement.

The origin of pronation comes primarily from the ankle (subtalar and midtarsal joints). Below are the four primary phases of a normal gait cycle regarding foot pronation:

1. **Foot strikes the ground:** slightly supinated (land on the outside heel)
2. **Foot travels through the support phase:** Foot transitions from supinated to a pronated position

3. **Foot begins the drive phase:** Foot is in a pronated position
4. **Foot prepares to exit the drive phase and enter the recovery phase:** Foot begins to supinate. The big toe plays a large role in the stability of the foot, regulating the degree of foot pronation and forward propulsion

Mild Pronation (Rear View)

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Mild Pronation (Front View)

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Overpronation

There is not a clear consensus on exactly how many degrees equal *overpronation*. However, many clinicians believe that a subtalar **joint angle greater than 8–10 degrees equates to overpronation.**

Overpronators typically have the same lateral heel wear pattern as someone who pronates, but because of the extreme medial roll of the foot, the wear on the top part of the shoe is almost exclusively on the medial side.

The more a foot pronates, the more the femur and tibia internally rotate. If your athlete presents with excessive pronation accompanied by pain while running, it is advisable to have the individual meet with a specialist such as a podiatrist or physical therapist. Custom orthotics are often prescribed to help alleviate stress and pain caused by overpronation. Custom orthotics are made by taking a mold of an individual's feet. There are also over-the-counter orthotics that are not very costly. However, if you think your athlete would benefit from orthotics, referring the person to a professional is best. Motion-control running shoes are often recommended for runners who overpronate.

Overpronation is associated with increased ground reactive forces and foot-contact time (664).

A 2008 study by Cheung et al. found that the degree of foot pronation tended to increase with an increase in the distance run (68). Weak hip abductor muscles (i.e., gluteus medius) can also contribute to overpronation.

Flat Feet

Flat feet are often called *fallen* or *collapsed* arches. The medical term for flat feet is *pes planus*. Individuals with flat feet have little to no arch in the feet in the standing position. Those who overpronate often have flat feet, but this is not always the case.

A study by Menz (1998) found that static assessments to determine arch height classifications, such as low or high arches, were unreliable (610).

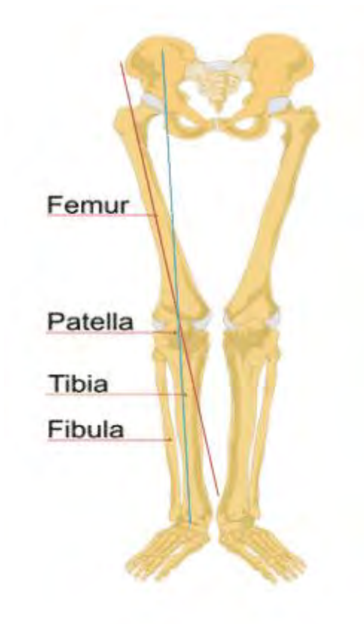
Orthotics and arch-strengthening exercises are often prescribed for those with flat feet.

What Is a Q-Angle?

Regarding overpronation, the term ***Q angle*** is often a possible culprit. The Q angle is assessed by measuring the angle between two lines (57):

- **Red Line:** ASIS to the middle of the patella
- **Blue Line:** middle of the patella to the tibial tubercle (also known as *tibial tuberosity*, it is the bump at the top/middle of the tibia, immediately below the patella)

In the following diagram, the Q angle is represented by the angle between the red and blue lines.



Women typically have a greater Q angle than men because of a wider pelvis. The normal Q angle range for adults is 15−22 degrees, with women typically trending toward the larger angle (58, 59).

A significant Q angle is often noted as a potential cause of patellofemoral syndrome (PFS). However, recent research has shown no strong correlation between an individual's Q angle and the incidence of PFS (646). Therefore, **a large Q angle should not be used to infer PFS.**

Supination

Supination (also called ***under pronation***) is essentially a lack of pronation. During pronation, the ankle joint (subtalar joint) flexes to allow medial rotation of the foot. During supination, the subtalar joint does not flex. As noted previously, the foot is in a slightly supinated position right before striking the ground. The foot stays in this position throughout the running gait cycle for individuals who supinate. This results in individuals "running" on the lateral aspect of their feet and pushing off the ground directly behind the small "pinkie" toe. This is evident if you look at the wear pattern on a shoe of someone who supinates. You will find that the outer part of the sole is worn, but not much else.

Those who supinate typically have a high and rigid foot arch. Based on the Cheung et al. study, this may or may not put the runner at increased risk for injury (194, 195, 196).

Assessing Pronation and Supination

Like overpronation, **there is no clear consensus on how many degrees constitute normal or mild pronation.** Depending on the source or study, mild pronation ranges from 4−8 degrees of the subtalar joint angle (67).

Because of this lack of consensus, it is advantageous and practical to look for signs of excessive pronation/supination and note of pain and injuries when assessing pronation and supination.

Visual Signs of Overpronation

- Knock-knees
- During the recovery phase, the lower leg swings excessively outward in relation to the upper leg. The foot also tends to turn outward (toe pointed laterally) at the rearmost part of the recovery phase. This often appears as if the foot briefly “flicks” outward
- Substantial inward angle of the ankle

Visual Signs of Supination

- Bowlegged
- During the recovery phase, the lower leg swings excessively medially in relation to the upper leg
- Outward angle of the ankle

Injury Considerations

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Pronation is Not Indicative or Predictive of Injury

It is a popular belief that pronation is bad and will likely lead to injury. As noted earlier, **pronation is a natural occurrence and acts to absorb stress, transfer the load on the feet, provide elastic energy, and facilitate a stable platform for the feet to push off of.**

Within the running shoe world, there is often a drive to correct any foot/ankle articulation that is not considered a *neutral* or *normal* foot position − meaning a foot/ankle position that neither pronates nor supinates. Since most individuals' feet pronate to some degree, it is odd that a neutral foot position is viewed as "normal" (431). One could hypothesize that the driving force behind this are shoe manufacturers that stand to profit by selling shoes that "correct" foot/ankle movements such as pronation.

It is advantageous to not view pronation as being indicative of pain or injury, as study after study has proven this (424, 425, 426). In fact, several studies theorize that running injuries stem more from training errors (429) and hip dysfunction (430).

Knee Valgus

This refers to the knee moving medially, often termed ***excessive medial knee displacement (EMKD)***. During EMKD, both the femur and tibia medially rotate. This medial rotation is what causes the knee to move inward. Potential causes of EMKD are:

- Large Q angle

- Foot overpronation
- Overactive hip adductors
- Underactive hip abductors – primarily the gluteus medius
- Lack of ankle dorsiflexion

Of the aforementioned potential causes of EMKD, overpronation and underactive (i.e., weak) gluteus medius are the most cited.

Ankle Dorsiflexion

In a study by Fong et al., individuals with poor ankle dorsiflexion during foot impact (i.e., running) had greater ground reaction forces. Higher ground reaction forces are associated with EMKD (780). This is because ankle dorsiflexion is essentially a shock absorber and acts to absorb energy. **A reduction in ankle dorsiflexion corresponds to a lack of energy absorption.** This energy has to go somewhere, resulting in increased ground reactive force and likely EMKD.

Running Form Components

Ask a non-figure skater to perform a triple axel on an ice rink, and you'll likely get a look of bewilderment followed by, "*No freaking way!"* However, ask a non-runner to go for a run, and there is a strong likelihood that you'll get a, "*Sure, why not? Let's do it!"* While this is an extreme contrast, the point is that running is often viewed as a skill set that everyone knows how to do. While in its most basic element, this is likely correct. However, as a coach, you must understand that there is a massive difference between running and running correctly. Correct running form and proper running mechanics should not be viewed as a skill set everyone naturally has but rather a learned skill that requires training to improve upon.

Energy Allocation

To appreciate the most important aspects of running form, it is helpful to understand what elements of running form use the most energy.

The majority of energy goes to supporting body weight, followed by energy required for forward propulsion (drive phase of gait cycle) and energy required to swing the recovery leg forward. (585). Arm swing and balance have a minimal metabolic cost (586).

The following are 11 running-form principles based on biomechanics and running economy.

1 – Look Straight Ahead, Not Down or Up

When looking down, runners are not able to see obstacles or hazards that are in their path. The other primary reason for looking ahead is that the eyes cue the head position, which, in turn, cues the upper-body position. **Runners who keep their eyes focused down tend to tilt their head downward**, which causes the shoulders to roll anteriorly and rotate internally. Further down the kinetic chain, this can lead to excessive spinal flexion.

Conversely, looking upward or tilting the head backward cues the lower back to extend, which stresses the erector spinae muscles. This arched back position (hyperextension) results in the feet striking the ground substantially in front of the body, thus lessening the runner’s efficiency.

2 – Upper Body Erect, Slight Forward Lean

An excessive anterior or posterior lean angle of the upper body results in excessive stress on the LPHC musculature. **A correct slight forward lean does not come from the pelvis but rather from the ankles.** When running, the body should have a slight forward lean from the heel to the shoulders. A good way to teach this is to have athletes stand with their feet in the frontal plane and lean forward from the ankles. They will automatically move one leg forward to support themselves as their body moves forward.

It is common to see runners leaning back with substantial lumbar extension. This is especially true when a runner becomes fatigued.

Center of Mass (COM)

While running, one's center of mass is maintained over the base of support (i.e., legs/feet). By leaning slightly forward, one's COM is maintained over the base of support versus behind it. Balance is what enables a runner to control the COM.

As the body is in contact motion (vertical oscillation, planes: sagittal, frontal, transverse) when running, the COM is constantly changing.

A 2010 study by Hamner et al. found that the quadriceps and foot plantar-flexors (gastrocnemius, soleus) are the primary contributors to accelerating the body's center of mass during running (695). However, this study did not consider the spinal engine model – noted later in this module.

Most people believe that the majority of the energy used while running is for forward propulsion. A study by Chang et al. found that **73 percent of a runner's total energy expenditure is used to support body weight, and only approximately 20 percent is used for forward propulsion** (730). This demonstrates the importance of proper form regarding COM.

3 – Avoid a High or Long Stride

As with any exercise, efficiency is of utmost importance. A stride that pushes runners upward wastes energy that should be used to propel them forward. One way to cue this is to wear a hat with a brim and use the brim as a reference point. The brim of the hat should not rise or fall substantially. To further illustrate this concept, think about throwing a football. If you were asked to throw the ball 10 feet, you could throw it in a straight line and hit your target. However, if you were asked to throw it 75 feet, you would have to aim higher in the air to put an arc on it.

This is the same with a long stride versus a short stride. Longer strides require excessive vertical movement (inefficient), whereas short strides require very little vertical movement (efficient).

Vertical Oscillation

This relates to the degree of vertical motion of the body while running. As previously noted, the correct running gait will always have some degree of vertical oscillation. However, excessive vertical oscillation is inefficient. The more vertical oscillation one has, the more energy the person must expend to counteract gravity. This concept is discussed in greater detail later in this module.

4 – Foot Strikes the Ground Slightly in Front of or Behind the Center of Mass

Whether your athlete runs with a heel or midfoot strike, the correct form is for the foot to strike the ground directly under or slightly ahead of the hips. While more natural with a midfoot strike, it can also be accomplished with a heel strike. This relates directly to one's center of mass. There are two primary reasons why the foot should strike the ground under the body:

1. **Hamstring Stress**: When the leg is excessively elongated in front of the body (termed *overstriding*), it puts substantial eccentric stress on the hamstrings.
2. **Direct Power Transmission:** When the foot strikes the ground under the body, it immediately pushes down and rearward, thus propelling the runner forward.

Refer to the video in the section, *Midfoot Strike vs. Heel Strike*, for a dynamic visual representation of foot-strike location. The video contrasts heel- and midfoot-based strides and is representative of the foot striking the ground ahead of and under the body, respectively.

5 – Transverse Pelvic Rotation (Hips)

Most individuals run with too little or too much pelvic rotation in the transverse plane. It is natural for the pelvis to rotate when walking/running, and a longer stride length typically equates to increased pelvic rotation because of increased hip extension.

By eliminating pelvic rotation, you effectively leave out an important aspect of the body’s natural kinetic chain.

Pelvic rotation helps to minimize the impact of running. Running with no pelvic rotation results in the feet hitting the ground and then moving rearward. This form substantially impacts the legs and body (e.g., ground reaction force). When running with the correct amount of pelvic rotation, by the time the foot strikes the ground, that side of the pelvis is already rotating posteriorly, which reduces the impact on the body (224). This is because the impact of the foot strike is dissipated slightly through the rotation of the pelvis.

While the pelvis needs to rotate, it should not over-rotate as this puts excessive pressure on the LPHC musculature and decreases the runner’s efficiency.

There are no hard numbers to determine the correct amount of pelvic rotation. Therefore, **visual observation is typically the most practical and efficient way to assess pelvic rotation and, more specifically, over-rotation.** Running with excessive pelvic rotation requires the upper body to rotate substantially to counterbalance the pelvis and legs. If your athlete’s torso rotates 45 degrees or more (noted at the shoulders) in the frontal plane, there is likely too much pelvic rotation. Fifteen to 30 degrees of upper-body rotation typically equates to the appropriate amount of pelvic rotation in the transverse plane.

As noted earlier, the degree of pelvic rotation is primarily influenced by stride length and, more specifically, the amount of hip/leg extension.

When running with the correct amount of pelvic rotation, the hips and glutes drive the motion and the legs “*follow through*.” This reduces the reliance on the hamstrings as the glutes and hip rotation initiate the drive phase. It could be theorized that those who run with static hips would be more susceptible to hamstring injury due to the minimized role of the glutes.

6 – Sweep, Don’t Pound

This correlates with rotating the pelvis in the transverse plane while running. When the foot strikes the ground, it should already be moving in a sweeping motion toward the rear of the body. **A good way to visualize this is a cat "pawing" at the ground.** As noted earlier, when the foot strikes the ground without traveling rearward simultaneously, it creates a pounding effect on the ground that decreases efficiency and likely increases the chance for injury. A sweeping foot strike creates a fluid and powerful stride, whereas a pounding foot strike creates a broken and non-fluid stride. Typically, **runners with a pounding foot strike have more vertical oscillation than those who sweep their feet.**

Rotation of the pelvis in the transverse plane initiates and facilitates a sweeping foot strike.

7 – Keep Hands, Shoulders, and Face Relaxed

The goal of a runner is to have as little wasted energy as possible. When a runner's face is tightened, hands clenched, and shoulders shrugged (scapular elevation), the individual diverts energy from the legs and core muscles.

8 – Arm Movement Should Primary Come from Shoulders, Not Elbows

Arm movement should come primarily from the shoulders, not the elbows. **Active flexion and extension of the elbows waste energy.** Arm movement should not be forced, as arm motion does not drive the body forward. Arm motion is used to counterbalance the momentum of the legs (695), and as such, excessive arm movement should not occur. If an athlete looks like they are throwing uppercuts at Mike Tyson while running, they have too much arm movement!

Arms do not contribute substantially to propulsion when running (695). As just noted, **the primary role of the arms is to counterbalance the momentum of the legs.**

Another study found that the arms act to reduce torso and head rotation (696). The study also found that upper-body rotation is primarily a result of lower-body movement, not arm movement. A study by Kram et al. found that swinging the arms (not excessively) saved a runner approximately 4 percent in energy cost versus not swinging the arms. This is likely due to reduced muscular demand due to improved counterbalance and facilitating spinal counter-rotation via stretching the latissimus dorsi (586).

Arm Crossover

There is a long-standing myth that when running, the arms should not cross the vertical midline of the body (imaginary vertical line from the middle of the nose to the center of the belly button). The most commonly cited reason for the arms not crossing the body's midline is that it facilitates torso rotation. This is interesting because most runners lack the appropriate degree of transverse torso rotation. As a result of this flawed myth, the common advice is to run with the arms in a forward motion with no inward movement. Aside from looking very odd, this form puts excessive strain on the external rotator cuff muscles of the shoulders. Some degree of medial arm movement is suggested.

9 – Short Front Stride with Fast Leg Turnover

Typically, runners who land on the heel have a slower turnover than those who land midfoot. Regardless of the type of foot strike, **the proper stride should focus on a relatively fast turnover with ample hip/leg extension.** Shortening one's stride pertains to eliminating or significantly reducing the foot strike ahead of the body, thus placing the foot strike closer to the center of mass. This is not intuitive for most runners and takes practice to learn. Conversely, leg/hip extension (leg movement behind the runner) typically should not be shortened unless a runner has too much extension. Stride length and hip rotation are discussed in greater detail later in this module.

While a short stride is typically favored over a long stride, **too short of a stride can also be inefficient.** This is largely because the runner is not taking full advantage of the body's stretch-shortening cycle (SSC). The ideal stride rate and length function on a bell curve (573).

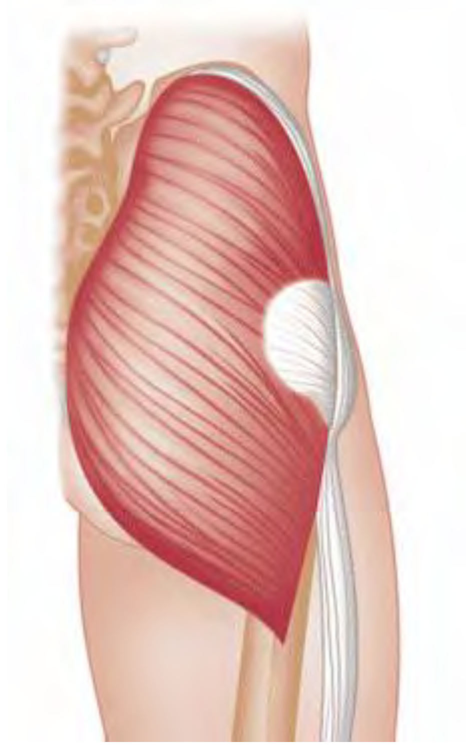
10 – Hip Extension

Hip extension (e.g., leg behind the body) is essential to proper running form. Individuals who lack sufficient hip extension because of tight hip flexors (iliopsoas) or a lack of transverse hip rotation often hyperextend their lumbar spine as a compensatory measure. **Hip extension should come from the hip, not the lumbar spine.** Proper hip extension is also a function of transverse pelvic rotation.

As noted later in the section discussing *passive movement*, proper hip extension allows for greater passive movement during the swing aspect of the gait cycle. This increases the efficiency of a runner.

Role of the Glutes

Most runners will tell you that the glutes are some of the most important muscles for runners to develop – if not the most important – as this is where a runner's power comes from.



While this statement is largely correct, there are a few issues here.

- When most people refer to the "glutes," they are referring to the gluteus maximus (GMax) – not the gluteus minimus or medius.
- While the GMax is critical for proper running form, other muscles are also very important to a runner's form and performance.
- A runner's form has to be correct to utilize the GMax correctly and adequately.

While all runners utilize the GMax when running, most runners do not have the proper degree of hip extension either because of an extremely short posterior stride and little to no transverse hip rotation. As a result, **the GMax is likely underactive as the primary role of the GMax is hip extension.**

Concerning running, the GMax is used concurrently to move the runner vertically and forward (upward diagonal motion). Too much vertical movement (i.e., vertical oscillation) results in wasted energy, increased impact on the body, decreased running economy, and a slower running speed. This form requires excessive GMax activation to move the runner vertically. Conversely, too little vertical oscillation results in a runner shuffling along.

If runners lack flexibility in their hip flexors, they will likely lack adequate hip extension. This reduces the involvement of the GMax. Runners with poor hip flexor range of motion often compensate via hyperextension of the lumbar spine.

In summary, the GMax is an important muscle for running, but it is utilized correctly only if a runner has the appropriate level of vertical oscillation and posterior hip extension.

11 – Neutral Spine

It is common to see runners with excessive posterior or anterior pelvic tilts. Of the two, running with an anterior pelvic tilt is more common. This often results in runners with a substantial arch (lordosis) in their lower back. As the hamstrings attach to the pelvis, if the pelvis is tilted too anteriorly, it “pulls” on the hamstrings and creates excessive tension. As a result, during the swing phase of the gait cycle, when the leg is in front of the body, the hamstrings are at a greater risk for injury because of increased tension under eccentric stress.

Hyperextension

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Excessive Flexion

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Putting It All Together – Correct Form

So now that we've identified multiple elements that make up 'good form,' what does good form actually look like?

The below video denotes proper form.

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Assessing Form: Treadmill vs. Outdoors

A common assessment for evaluating a runner's form is to watch the person run on a treadmill. This is a valuable tool as it provides a controlled environment to assess a runner. However, one must be careful about what they are assessing. A treadmill is a good tool to assess foot/ankle position (i.e., pronation) and foot strike type. However, it has its weaknesses when assessing overall form.

The foot pushes downward and rearward when running outdoors to propel a runner forward. However, when running on a treadmill, when the foot impacts the treadmill belt, it is swept rearward. In other words, when running outdoors, a runner must push off the ground, whereas when running on a treadmill, a runner must keep up with the treadmill belt. **From a neuromuscular standpoint, these two running environments are different.**

Many runners do not rotate their pelvis enough when running on solid ground. Conversely, when running on a treadmill, runners often have increased pelvic rotation due to the treadmill belt sweeping the legs rearward. Therefore running on a treadmill can artificially induce pelvic rotation that is not present when running on solid ground.

It is strongly advised to assess pelvic rotation and a runner's form in its entirety while watching the person run outdoors at varying speeds.

Fatigue has been shown to alter the biomechanics of the lower extremities, specifically the knee and hips, in the sagittal and transverse planes (337). This means that the starting form of an athlete at the beginning of a run is likely not the same form they will have at the end. This may increase the chance of injury. It doesn't take a rocket scientist to appreciate that the form of runners at the start of a marathon is likely quite different from their form at the end!

Here is a summary of the benefits and drawbacks of treadmill-based assessments:

Benefits of Treadmill Assessments

- Easy to assess form because of the runner being stationary
- Can manipulate speed and incline
- Most gyms have them

Drawbacks of Treadmill Assessments

- A treadmill may induce transverse hip rotation that is not present when running outdoors

- When running on a treadmill, the belt often facilitates the drive foot moving rearward versus the foot pushing rearward because of hip rotation/extension.

Despite the noted possible drawbacks of treadmill assessments, a treadmill is the most accurate and easy way to assess form.

Self-Selection Of Running Form

Some sources state that runners naturally self-select the most efficient running form/gait. This directly relates to a runner's economy and efficiency. So, do runners self-select the most optimal running cadence, foot strike, overall form, etc.? The answer is… it depends.

For new runners, the answer is no. Don't believe this? Watch a marathon. Observe the form of elite/professionals as they run by. While they might not all have what is considered "textbook" form, they are very efficient as every body movement is purpose-driven to propel them forward with minimal wasted energy. Assuming that the form of elite runners is as perfect as one can get, compare this form with those of recreational runners in the rest of the race. If the theory that all runners self-select the most efficient stride/form is correct, why would there be a difference in form between elites and recreational runners?

The most reasonable answer is that most elite runners have naturally good biomechanics, whereas most age-group runners do not have the same level of naturally good biomechanics.

A 2014 study by De Ruiter et al. evaluated running efficiency based on stride rate (SR) and found that when self-selecting running SR, no novice runners selected the most efficient rate (779). Experienced runners in the study selected a more efficient SR than novice runners. While both novice and experienced runners didn't self-select the most optimal SR, the trend demonstrates that the more experienced runners are, the closer they run to their ideal SR.

The good news is that **simply by running more, runners can improve their running economy** (778). However, it is important to note that while a runner can improve their form by running more, the improvements will not likely be to the extent that they will see significant biomechanical changes to the point where they have the form of a completely different runner.

The likely ‘answer’ regarding the self-selection of running is that it is a balance. Meaning, that a runner will improve their form to a degree by running more, and they will also improve their form to a degree by focusing on improving their form. Therefore, combining both ‘strategies’ is likely the correct route.

Passive Energy

Within the body exist structures and movement patterns that allow runners to significantly reduce muscle activation while maintaining the same, if not greater, running performance. The two areas that this certification will focus on regarding passive energy are:

1. **Energy Return**
2. **Passive Movement**

Passive movement and energy return are interrelated. The genesis of passive energy is that when the feet impact the ground while running, energy is absorbed and stored. The goal of a runner is to optimize running form (mechanics) to convert as much of this stored energy as possible into forces that assist in forward movement.

Regarding improving efficiency and performance, the five areas that this certification will focus on in relation to passive energy are hip rotation in the transverse plane, hip flexion, foot plantar flexion, knee flexion/extension, and foot-arch compression.

Maximizing runners' passive energy contribution can improve their running economy. By utilizing passive energy, active muscle requirement is reduced while still producing the required amount of force (620).

Stretch Shortening Cycle

"Energy cannot be created or destroyed; it can only be changed from one form to another." – Albert Einstein

This quote pertains to a law of physics, the *conservation of energy*. Regarding the SSC, it is applicable as potential elastic energy is converted to kinetic energy.

By definition, the SSC is representative of an eccentric contraction of a muscle followed by a rapid concentric contraction of the same muscle. Concerning running, the quadriceps, obliques, and calves represent the muscles most influenced by the SSC.

When the feet hit the ground, the quadriceps and calves eccentrically contract (lengthen) and then rapidly concentrically contract (shorten) during the drive phase to provide forward propulsion (616). The stretch aspect of the muscle and tendon during the SSC is quite small, 6 to 7 percent (615).

To illustrate how the **stretch-shortening cycle** works, think of shooting a rubber band. The more you stretch the band, the farther it will travel in the air once released. However, if you stretch the band too much, it could break. Additionally, the more tension the band has (harder to pull back), the farther it will go. Concerning the human body, muscles and tendons are the rubber band. To recap, tendons connect bone to muscle, and therefore muscles and tendons are often considered two parts of a working whole, the **muscle-tendon unit** (524). The variables that affect the degree of elastic return of the SSC are (525):

1. Length of the stretch
2. Speed of the stretch (loading)
3. Stiffness of the muscle and tendon
4. Time between the stretch and the contraction

From a running perspective, the legs act as springs. **The springs compress during the first half of the support phase and rebound during the drive phase.**

The stiffer a muscle is, the greater the amount of energy that can be stored and released. However, to not increase the chance of injury, a muscle must have full mobility (608, 523).

Tendons

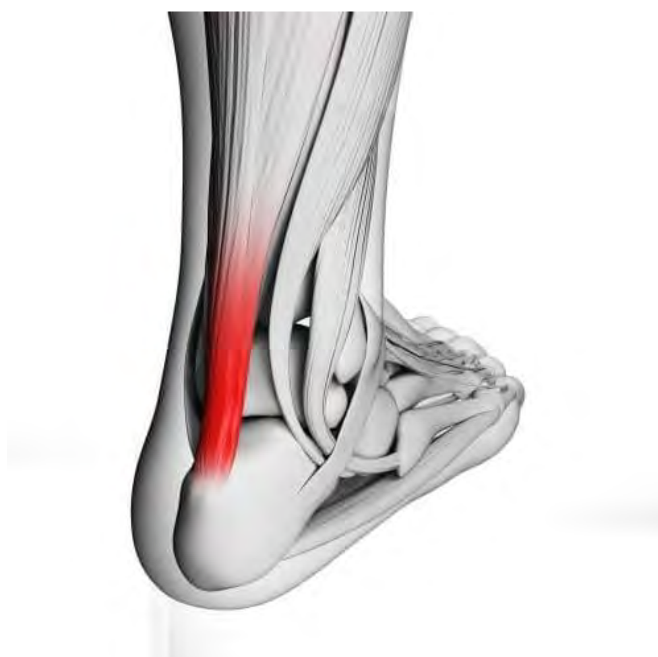
It is important to note that depending on the location of a tendon in the body, it will vary in thickness, shape, and length. These variables affect the stiffness of the tendon and thus the capacity for force production. Tendons have elastic properties that allow them to stretch. The stretch of a tendon (or muscle) stores energy, and when the stretch is unloaded, the stored energy is released. **Utilizing this stored energy properly can greatly minimize the metabolic cost of movement** (521).

The optimal stretch of a tendon should be viewed as a modified bell curve – meaning that too little or too much of a stretch is not optimal. If stretched beyond the end point of its range of motion, a tendon could tear completely. However, before this point is reached, a tendon could still be overstretched. When this occurs, structural changes occur to a tendon that effectively alters tendon length and thus reduces the stretch reflex (522). The degree of stretch to a tendon that elicits the ideal stretch reflex is called the **elastic region**. Once the stretch extends past this region, it is called the **plastic region**. At this point, the tendon's structure changes and therefore changes the tendon length (522).

The below image illustrates the elastic and plastic regions on a curve-based model (load/deformation curve).

Achilles Tendon

Regarding running, the Achilles tendon greatly impacts performance with respect to free energy. A study that examined the link between resistance training and Achilles tendon stiffness found that a 16 percent increase in tricep surae tendon (gastrocnemius, soleus, Achilles tendon) stiffness via resistance training decreased the rate of oxygen consumption during running by 4 percent, thereby increasing running economy (530).



Another study confirmed these findings by noting that differences in Achilles tendon mechanical properties were primarily influenced by muscle strength (531).

While the amount of energy that can be stored in the Achilles tendon and the resulting increase in running economy vary, **any increase in muscle strength of the tricep surae will increase Achilles tendon stiffness, resulting in improved running economy.**

There appears to be no difference between men and women regarding the effect of Achilles stiffness on running economy (529).

Mechanics

As rapid dorsiflexion (loading) and plantar flexion (unloading) of the ankle are responsible for the spring action of the Achilles tendon, if the ankle is not dorsiflexed enough, the energy stored in the Achilles tendon will not be as high as it could be. The implication from a mechanical perspective is that the heel of the foot during the drive phase should touch the ground while the hip is extended. **Potential energy and running economy are reduced if the heel does not touch the ground during this phase.**

Shoes with a high heel in relation to the forefoot likely (significant heel-to-top drop) reduce the amount of stored energy. This is because of reduced ankle dorsiflexion, which places the Achilles tendon and calf muscles in a shortened position. Insufficient range of motion (flexibility) of the Achilles tendon also limits the amount of potential energy able to be stored.

Outside Influences

Running economy is not just based on storing energy but also on utilizing it. The interface of the foot and the ground influence the amount of energy that can be utilized. The more solid the interface is, the more efficient a runner will be. If the running surface is very soft (e.g., sand) and a shoe has a lot of cushioning, this will reduce the amount of energy the runner can use as a large portion of the energy is absorbed by the ground and shoe respectively.

Some absorption, however, is likely a positive thing in relation to running economy and decreasing the chance of injury. Studies have shown **a higher metabolic cost to running barefoot than**

with minimal shoes because of increased muscular demand (532).

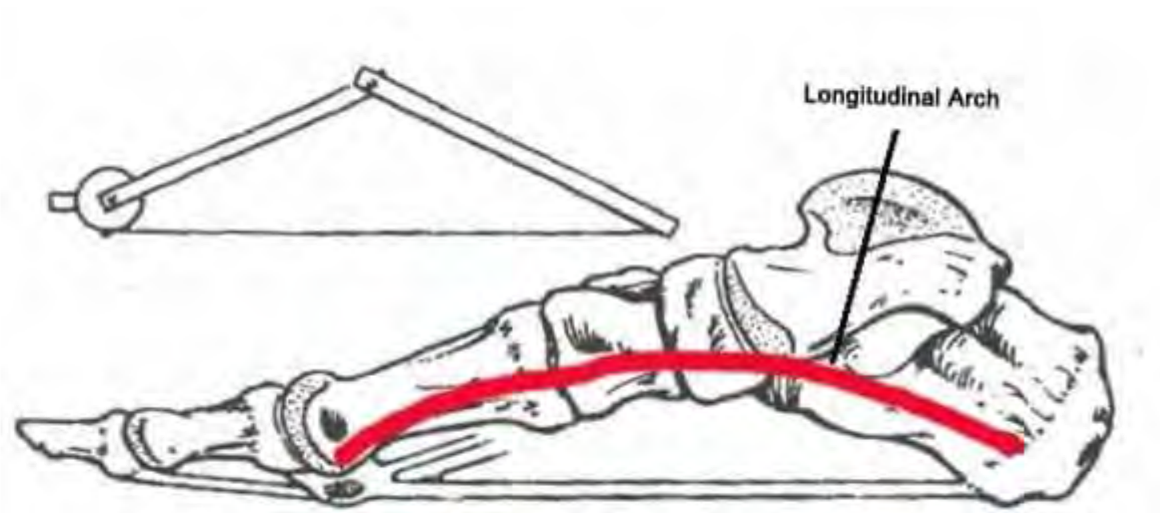
Trainability

Many distance runners do not perform strength and sprint training because of the perceived lack of specificity. While there are many benefits to distance runners, if they did sprint training only to stiffen the Achilles tendon to more efficiently utilize elastic energy return, it would be well worth the effort.

Foot Arch

The primary functions of the foot regarding running are shock absorption and propulsion. The foot plays a large part with respect to free energy, specifically, the longitudinal arch (below image). The muscles and connective tissue of the foot affect the longitudinal arch, which acts as a spring to provide passive energy while running (601). The red arch in the image below denotes how the foot acts as a spring in relation to the plantar foot muscles and connective tissue.

A common argument for running in bare feet or minimal footwear is that since the arch of the foot is supported in conventional running shoes, the elastic energy of the longitudinal arch is negated – thus greatly reducing an energy source used for forward propulsion.



Muscles

Like tendons, **the stiffer a muscle is, the greater the elastic return will be.** Pre-activation of leg muscles when running prepares the body and leg musculature for foot impact. Pre-activation of leg muscles is theorized to decrease stress to leg muscles and increase the cushioning upon landing (520). The body would collapse upon foot strike if the leg muscles did not contract before and during landing.

Leg muscle “stiffness” is controlled consciously and unconsciously (526). The degree of leg stiffness directly affects the amount of knee flexion. Regarding conscious control of leg muscle stiffness, an individual can control stride rate and length (527). Leg stiffness is also influenced by the geometry of the leg at impact. This is because depending on the angle of the leg at foot impact, there will be varying loads on the leg that the muscles must counteract (526).

The type of surface that the foot lands on also correlates to the degree of muscle stiffness. When running, the body looks to maintain the same degree of total vertical stiffness (surface stiffness + leg stiffness) at all times. Therefore when running on different surfaces such as pavement and sand, the degree of leg stiffness will change to maintain the same degree of total vertical stiffness. For example, if running on sand, the legs must become stiffer to compensate for the decrease in surface stiffness (528).

Passive Movement

Go for a walk. Did you consciously contract muscles to swing your arms? This is representative of **passive movement**. Passive movement is representative of body movement that is not the result of active muscle contractions (519, 520). Specific to running, several body movements are influenced exclusively by passive movement:

- Lower leg (tibia/fibula) swing
- Hip flexion
- Knee flexion

The degree of passive movement largely depends on actions done before the passive movement occurs (520). For example, the lower leg swing (the aspect of the gait when the foot is in the air) results from how extended the hip is at the end of the drive phase. The greater the hip extension is, the larger the lower leg swing will likely be (think: pendulum). When examining the movement of the leg swing, the knee flexes after the foot leaves the ground behind the runner and extends prior to the foot hitting the ground. These leg movements are synonymous with hamstring and quadriceps activation, respectively. However, concerning the leg-swing aspect of the running gait, these muscles are not activated because of passive movement.

Concerning hip flexion, during the middle of the leg-swing phase, the hip flexes to allow the leg to move forward so that the foot can strike the ground and propel the body forward. In isolation, the hip flexors are largely responsible for flexing the hip. =One possible exception would be running uphill, in which case the hip flexors may be consciously activated.

While a primary action of the hamstrings is to flex the knee during the gait cycle, this action is largely passive versus active. Therefore, the hamstrings are most utilized while running to decelerate the lower leg during the swing phase and to assist in hip extension.

Training Implications

The sport of running traditionally incorporates a lot of drills. Two of the more common drills are ***butt kicks*** and ***high knees***. Butt kicks are exactly what they sound like – runners stand in the same spot and try to “kick their butt” with their heels. This rapid knee flexion/extension trains a runner to develop a fast kick and leg turnover. Muscularly speaking, this drill primarily focuses on the hamstrings. However, as noted previously, this aspect of the running gait is largely passive. Therefore, by performing this drill, the runner is taking a passive movement and turning it into an active one (e.g., muscle activation required) (579). However, these drills do have value, especially from a neuromuscular standpoint and that of ballistically strengthening the ankle and foot complex.

High Knees

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Butt Kicks

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The below 'drills' are great for strengthening the muscle/tendon unit(s) to increase one's running efficiency.

While exaggerated skipping or bounding may look similar to the high knees drill, it is important to note that the skipping drill focuses on the body's whole kinetic chain rather than primarily isolating the hip flexors.

Strides have a multitude of benefits and applications, however, in respect to increasing one's efficiency, strides are a functional way to train the spring systems of the whole body.

Exaggerated Skipping

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Strides (Slow Motion and Full Speed)

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Allocation of Energy

Watch an individual run. A lot is going on – legs and arms swinging, torso rotating, ankles flexing and extending. This leads to the question, how much energy is required to make each of these things occur, and how does it relate to an individual's total energy expenditure? Fortunately, Dr. Rodger Kram from the University of Colorado also wondered about this and performed several studies to determine the answer (584–588).

Dr. Kram et al. discovered that the following areas were responsible for consuming the most energy while running (585):

- **69 percent** – Supporting body weight
- **29 percent** – Forward propulsion

Taking muscle synergy into account, Kram et al. deduced that supporting one's body weight and forward propulsion equates to 98 percent of the energy used by a runner (587). While these percentages can likely be varied to some degree based on the efficiency of a runner, the primary thing to take note of is the distribution of percentages as it relates to areas of focus. For example, reducing body weight would be most advantageous from a performance standpoint.

As noted previously, Dr. Kram et al. found that swinging the arms reduced a runner's energy cost by approximately 4 percent (586). This is likely due to the counterbalancing effect.

Lastly, Dr. Kram et al. found that the balance required by a runner equated to 2 percent of their energy cost. The remaining aspects that likely influence the energy cost of a runner are the running surface, shoes, and step width (587, 588).

Rotational Driven Running

While hip (pelvic) rotation in the transverse plane is an important aspect of the running gait, it is also one of the least-discussed aspects of running form. By leaving out pelvic rotation, a major aspect of force production is eliminated.

Dr. Robert Lovett introduced the rotational concept of locomotion in the early 1900s (621), developed upon by Dr. Serge Gracovetsky (617) and modified further by Dr. Erik Dalton (619). Before the introduction of this concept of locomotion, the traditional locomotion model was called the ***pedestrian model of gait***. This model states that the legs are the primary aspect of the body responsible for locomotion and the upper body is largely passive (622).

Rotational-driven running reduces compression on the spine by adding a rotary element. This rotation stores and releases energy via torsion (775).

The Spinal Engine

Dr. Gracovetsky developed this model (spinal engine). The catalyst for the spinal engine originated from observing an individual with no legs walking upright by rotating the pelvis, thereby “walking” on the ischial tuberosities (landmarks on the base of the pelvis) (617). This model consists of two primary aspects (617):

1. **Spinal Rotation**
2. **Coupled Motion**

A key aspect of the *spinal engine* model is that, unlike the *pedestrian model of gait*, Dr. Gracovetsky theorized that upon foot strike while running, energy was not absorbed by the ground but rather utilized up the body’s kinetic chain to assist with locomotion (619).

As referred to earlier, this relates to *ground reaction force*, which is based on Newton’s third law:

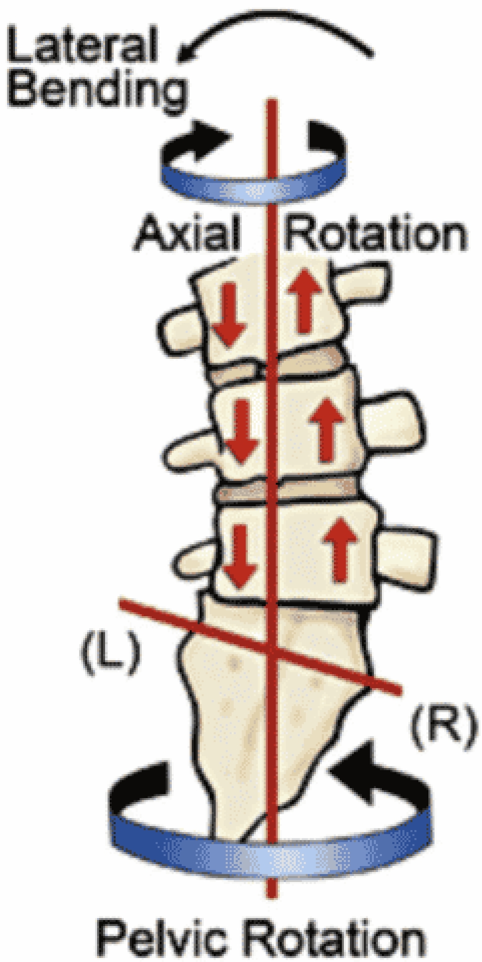
For every action, there is an equal and opposite reaction.

Here is how the *pedestrian model of gait* and the *spinal engine* interpret ground reaction force (Newton’s third law).

- **Pedestrian Model of Gait:** Energy is absorbed by the body
- **Spinal Energy:** Energy is utilized via counter-rotation of the spine to facilitate locomotion

Spinal Rotation

When running, the spine rotates (i.e., twists) in opposite directions. This is representative of counter-rotation of the hips and torso, as noted in the images below. This counter-rotation involves the ***stretch-shortening cycle*** **(SSC)** to utilize potential energy to stabilize the body and assist in locomotion via transverse rotation of the hips (774).



The above left image illustrates how the spine counter-rotates and laterally flexes while running. Spinal counter-rotation is represented by a runner in the above right image. In this image, you can see that the runner's shoulders (axial rotation) are rotated to the right while the pelvis is rotated to the left as the left hip is extended. The image above right is a real-life example of the image to its left (note the slight lateral spinal flexion to the left).

It is normal for individuals to differ regarding how much spinal counter-rotation they have. A rough assessment for this is to have an athlete lie on the floor with their knees and hips at 90-degrees and have them rotate their lower body down to one side without raising their shoulders off the ground. Some individuals can rotate their legs all the way down to the ground, whereas others may be able to get their legs only halfway down before the contralateral shoulder lifts off the ground.

Regarding movement in the transverse plane, the shoulders rotate much more than the hips (pelvis). As a result, **it can be difficult to decipher visually if an individual is running with transverse hip rotation.**

The videos below denote proper torso rotation.

Spinal Counter Rotation (Posterior View)

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Torso Rotation

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Spiral Spring System

According to Dr. Gracovetsky, intervertebral disks of the spine are not for shock absorption but rather to store and unload energy via torsion of the spine (775).

The best way to think of Dr. Gracovetsky's spinal engine is to think of the spine as a long spring. Visualize a long metal coil spring – using both hands, grab the coil spring at each end. Twist the spring in opposing directions and then let it go. The coil will allow some rotation to occur, and when you let it go, it springs back to the starting position. **The spring twisting in opposing directions represents spinal rotation and thus transverse rotation of the hips and torso.**

While the spinal engine model states that most of the energy stored via the spiral spring system is located in the collagen of the vertebral disks (annulus) and the fascia of the low back, other aspects of the body, such as the core musculature (inner-/outer-unit core muscles), play a role in the storage and release of energy (775).

The muscles involved in the spiral spring system are noted in the next section, ***Coupled Motion*****.**

Regarding running, the spinal engine (SE) is most associated with a heel-strike gait and, more specifically, the stretch of the biceps femoris during a heel strike. As a result, running with a midfoot or forefoot strike may reduce the activation of the biceps femoris regarding being a major contributor to spinal rotation. This is described in greater detail in the PSSS section below.

Coupled Motion

With proper running form, power is initiated through rotation of the hips and the resulting follow-through of the legs. This principle is the same as the boxer described earlier regarding ***kinetic linking***. In that example, for the boxer to effectively throw a punch, the following things occurred sequentially:

1 – Hips rotate
2 – Rotation of the torso
3 – Arm extension

This allows the boxer to throw a powerful punch with the origin of the punch coming from transverse hip rotation.

Dr. Dalton modified Dr. Gracovetsky’s spinal engine model (myoskeletal engine model) to include a modified version of the body’s natural spring system (774). In doing so, Dr. Dalton identified what he termed the **anterior spiraling spring system (ASSS)** and the **posterior spiraling spring system (PSSS)** (776). These spring systems comprise muscle and fascia (myofascial).

These spring systems, in combination with spinal rotation, represent the "engine" that powers proper running locomotion. In other words, these spring systems work together to facilitate proper running form.

Dr. Dalton identified two other spring systems that aid in locomotion, the **lateral spring system** and the **stirrup spring system**. However, this certification is focused on the ASSS and PSSS systems, as they represent the most relevant systems regarding spinal rotation.

For example, while the spine bends slightly laterally during the gait cycle, this lateral bending (lateral spring system) is more of an automatic process, whereas running with proper spinal rotation is not.

– In addition to Dr. Gracovetsky, Dr. Dalton's PSSS and ASSS systems were influenced by Tom Myers (Anatomy Trains) and Dr. Andry Vleeming, who developed his own muscular spring models: **anterior oblique system** and **posterior oblique system** (777).

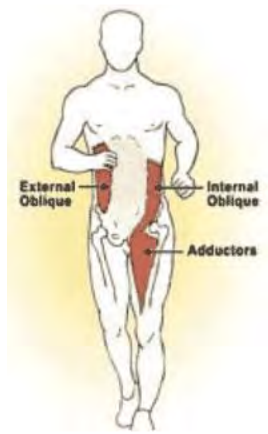
How Do the Spring Systems Work?

The hips (pelvis) and shoulders rotate in opposite directions.

- When the hips and shoulders are counter-rotated the most, this represents the point at which the muscles, tendons, and fascia of the spring systems are stretched the most. This relates to the point in the gait cycle where the most potential energy is stored. This also corresponds to the legs being the farthest apart (fore/aft) in the sagittal plane.
- As the hips and shoulders begin to rotate back toward each other, this is representative of the potential energy stored in the spine, muscles, tendons, and fascia converting to kinetic energy to rotate the spine and thus the hips and shoulders.
- When the hips and shoulders are in the same frontal plane (the legs pass by each other under the torso at this point), this represents the least tension (stretch) on the spring systems and also represents the point at which the stretch of the spring systems alternates to the opposing side of the body.

Anterior and Posterior Spring Systems

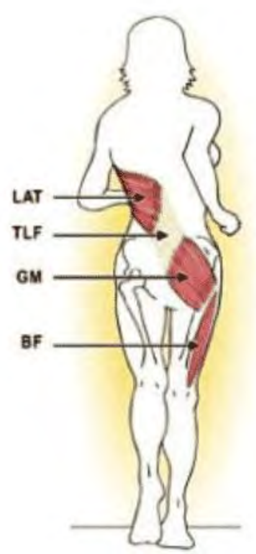
The **anterior spiraling spring system** comprises the following muscle groups (776):



Anterior Spiral Spring System (ASSS)

- External Oblique
- Abdominal fascia and external oblique aponeurosis (tendon sheath)
- Contralateral Internal Oblique
- Contralateral Adductors

The **posterior spiraling spring system** comprises the following muscle groups (776):

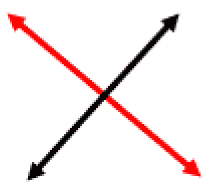


Posterior Spiral Spring System (PSSS)

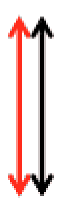
- Latissimus Dorsi
- Fascia of the thoracic and lumbar spine region
- Contralateral Glute Max
- Contralateral Biceps Femoris

These spiraling spring systems (PSSS and ASSS) gain and release energy via the stretch-shortening cycle. As you can see from the preceding PSSS and ASSS images, the spring systems form diagonal lines. These diagonal lines stretch and contract in a kinetic chain to facilitate spinal rotation and, therefore transverse pelvic rotation. Stretching and contracting of the PSSS and ASSS represent an elastic response.

The PSSS and ASSS oppose each other, meaning the ASSS crosses the body diagonally in the front, whereas the PSSS crosses the body diagonally in the back and at an opposing angle (see below image).

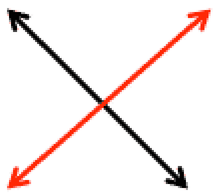


A



B

Red Line = ASSS
Black Line = PSSS



C

Diagrams A, B, and C are denoted below with respect to a runner's form:

A: Right leg forward, left leg rearward, pelvis rotated left, shoulders rotated right

B: Both legs directly underneath pelvis, pelvis and shoulders aligned in the frontal plane (no stretch of either ASSS or PSSS system)

C: Left leg forward, right leg rearward, pelvis rotated right, shoulders rotated left

Arm Motion

As the latissimus dorsi is primarily stretched by forward movement of the arm (humerus), running with no arm movement takes away an element of the ASSS. It, therefore, reduces the total force of the spiral spring system. Conversely, too much arm motion wastes energy. Like most elements of human performance, the optimal amount of arm swing/movement exists on a bell curve.

Too Much Arm Movement

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Too Little Arm Movement

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Did you ever wonder why track sprinters have very muscular upper bodies and a lot of arm movement compared with distance runners?

Did you ever notice that your arm movement becomes more pronounced when you go from running at a slow pace to a sprint?

This is because the body needs to generate more energy when more speed is required. This is accomplished by increasing spinal rotation, and therefore a larger and more rapid stretch/contraction of the PSSS and ASSS is elicited. By increasing arm movement, a greater stretch is placed on the latissimus dorsi, thus increasing the potential and kinetic energy of the PSSS.

Do 'The Arms Drive the Legs?'

This is a common explanation for why arm movement while running is important. **In the simplest of terms, arm motion does not drive the legs.** As discussed previously, arm movement is primarily used to assist in counterbalancing the body and stretching the latissimus dorsi (via PSSS) to create potential energy to assist in spinal rotation.

Therefore, arm movement can assist in driving the legs (via spinal rotation). However, this is likely not what most people mean when they refer to this statement.

Range of Motion

Too much or too little transverse spinal rotation will likely decrease the effectiveness of the spinal engine. Too little rotation will not create enough stored (potential) energy, whereas too much rotation will decrease running economy because of excessive motion. The ideal amount of rotation exists somewhere in between the two extremes.

Spinal Engine Summary

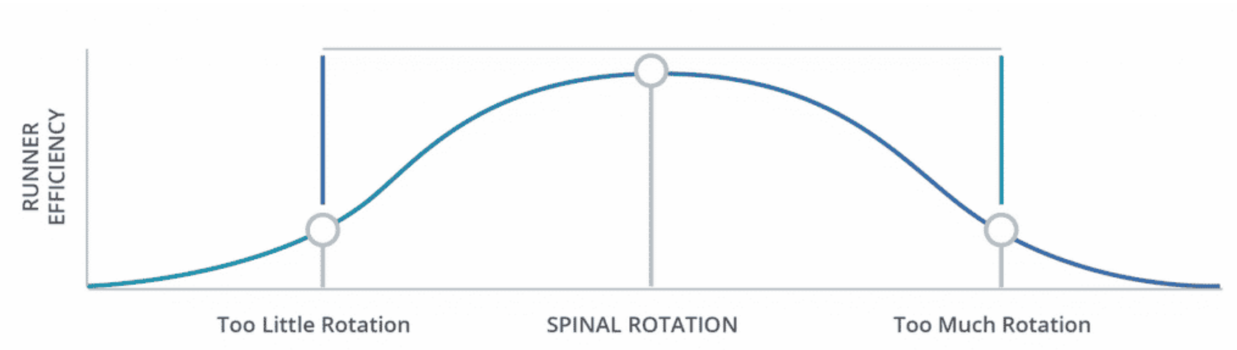
Specifics of the SSC and spinal engine can get quite complex, especially regarding the exact muscles and aspects involved with spinal rotation. However, the following sums up spinal rotation regarding its actions and benefits:

Counter-rotation of the spine facilitates hip extension via hip rotation in the transverse plane. This allows for increased glute and core (primarily obliques) involvement and reduced ground reaction force (GRF). This results in less reliance on the hamstrings, reduced compression of the spine, and increased force production.

Net effect = A faster and more efficient/economical runner due to a reduced metabolic cost of gait.

Relationship Between Spinal Rotation and Efficiency

The speed of the runner affects the optimal degree of spinal rotation. Concerning the total spectrum of hip rotation, it is likely that a slow running speed would relate to a small amount of spinal rotation being most efficient, whereas a fast running speed would likely translate to a greater amount of spinal rotation being optimal.



In the image above, the rectangle denotes the optimal degree of spinal rotation, with the vertical blue line indicating ideal spinal rotation when running slowly and the green vertical line denoting ideal spinal rotation when running fast. Therefore, the degree of spinal rotation outside the rectangle denotes too little and too much spinal rotation. Be aware that *slow* and *fast* are subjective based on the individual. While transverse hip rotation should occur to some degree at all running speeds, it often does not.

Elastic Response

As noted previously, the elastic stretch response is part of the stretch-shortening cycle (SSC). As the hips rotate in the transverse plane in one direction, the torso rotates in the opposing direction. This stretch involves the intervertebral disks, and muscles/tendons/fascia noted in the preceding PSSS and ASSS illustrations (774).

Increased muscle tension and stiffness increases elastic response. Therefore, the more engaged the core musculature is, the greater the elastic return of the hips will be.

Hip rotation (toward the side of the drive leg) begins before the foot hits the ground. This illustrates how hip rotation initiates the running drive phase and how the core musculature plays a key role in force production while running.

Cueing Hip Rotation

For runners who do not rotate or over-rotate the hips when running, learning the correct amount of transverse hip rotation can be challenging. The degree of hip rotation is directly correlated to the degree of rotation of the torso and, more specifically, the shoulders. Don’t believe it? Try running without rotating the upper body while rotating the hips ... awkward! Conversely, try running while substantially rotating the torso but not the hips. As you can see, this does not work very well. The mechanical reason for this is counterbalance. As defined by *Merriam- Webster*, counterbalance is (533):

A force or influence that offsets or checks an opposing force

Be aware that when teaching athletes to run with the proper mechanics and, more specifically, the correct degree of hip rotation, you should not expect to see them “get it” immediately. Like other areas of motor development, it is a process, and individuals will acquire proper form at different rates.

The most important aspect when teaching proper form is that athletes fundamentally understand why the correct technique is more efficient and economical than what they are currently doing and understand the theory and mechanics involved. Even once an athlete can run with proper form, it takes a lot of practice for it to become repeatable to the point where it becomes the normal, default form.

Cue Shoulders

Because of spinal counter-rotation, if your athlete under or over-rotates the hips, a good starting place is the shoulders. By instructing your athlete to increase or decrease rotation of the shoulders, it is likely you can directly affect the rotation of the hips. To demonstrate this, clasp your hands together with your arms pointed straight ahead. Now begin running and make sure there is no lateral movement of the arms. This prevents the torso (shoulders) from rotating, which in turn greatly reduces hip rotation because of a lack of spinal counter-rotation.

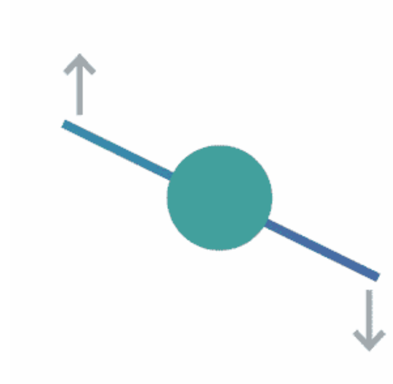
Run From the Shoulders, Not The Arms

While countless resources discuss “correct” arm position and movement, there are few, if any, resources that discuss proper torso rotation. As a result, most runners focus on arm movement, not torso rotation. As the following image denotes, an effective way to cue runners who under-rotate their torso is to have them focus on moving their shoulders forward while running. While the focus is on the shoulders moving forward, the action is greater rotation in the transverse plane.

Drill

This “cue” in the image assists runners in attaining a greater degree of torso rotation. The purpose of this drill is not to have athletes achieve the perfect degree of torso rotation but rather to have them understand how shoulder movement affects torso rotation and to have them “feel” how increased torso rotation affects hip rotation and gait mechanics.

Once an athlete has greater awareness of these areas, they can work to adjust their mechanics to achieve the optimal degree of shoulder/torso rotation.



The **Grapevine**, or **Carioca Drill**, is great for working on transverse hip mobility and range of motion. It is also a great drill as it trains the body to move in the frontal plane (see below video).

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Cueing Leg Extension

Another way to cue hip rotation is to instruct runners to focus on getting full leg/hip extension behind their body. **Cueing athletes to push down and back with a foot until the leg is at full extension will cause the hips to rotate to the side of the extended leg.** While this may over-exaggerate hip rotation initially, it can be a valuable learning tool for runners to understand the link between leg extension and hip rotation.

Cueing Arms

While the arms in isolation do not aid in locomotion, fore/aft arm (humerus) movement contributes to stretching the latissimus dorsi, which is part of the kinetic chain involved with the posterior spiraling spring system (PSSS).

Relationship Between Ankle Flexibility and Hip Extension

Hip extension is typically related to transverse hip rotation, hip flexor flexibility, and glute max/hamstring activation. This is true from a static position. However, regarding running and, more specifically, proper running form, when the leg is in the drive phase (posterior to the body), **the ankle must have adequate dorsiflexion or else the drive phase of the stride will be cut short** – thus reducing hip extension.

When people discuss *shortening one's stride*, this is (or at least should be) referring to **reducing stride length in front of the body, not behind it**. While reducing the stride length in front of the body (termed ***overstriding***) is typically a good thing, the posterior stride length should not be shortened. In fact, for most runners, the posterior aspect of the drive phase is too short.

Therefore, if your athlete has poor ankle flexibility (i.e., poor ankle dorsiflexion), you should work on stretching the calf muscles (gastrocnemius and soleus) and utilizing a foam roller. When stretching the calf, ideally it will be in a position specific to running – hip extended with the leg straight and the foot as flat to the floor as possible and without pain.

If you have athletes who run with a very short posterior stride and/or prematurely lift their heel during their posterior stride (or always run on their forefoot), a lack of adequate ankle dorsiflexion may be the culprit.

Midfoot Strike

Many runners have attempted to convert from a rearfoot strike to a midfoot strike. The catalyst for the interest in midfoot striking coincided with the minimal shoe (or barefoot) movement that occurred around 2009-2010 in conjunction with the popularity of the book, *Born to Run* by Christopher McDougall (69). This book profiles the Tarahumara tribe in Mexico, where some members run long distances wearing little more than thin rubber soles. The book discusses not only footwear but also the midfoot-strike running style.

As noted previously, an important factor in an efficient running gait is the foot striking the ground below or slightly in front of the body.

Many runners find that this form is best facilitated via a midfoot strike.

For many runners, **the topic of foot strike is quite polarizing**. This section examines the differences between foot-strike types and the research surrounding this topic. To be clear, the topic of heel versus midfoot strike is not a right and wrong type of argument as there is no one ‘correct’ foot strike type.

Is A Midfoot Strike Right for Everybody?

Probably not. As noted in the certification, a heel strike puts more stress on the skeletal system, whereas a midfoot strike puts more stress on the muscular system – especially the calves. **Regardless of the adaptation length, some athletes will have muscular/connective tissue issues when running with a midfoot strike.**

This section is included to discuss the pros and cons of a midfoot strike, not to necessarily encourage you to convert your athletes to this foot-strike type. This must be ascertained on an individual basis. Like most other areas of training, some athletes will find more success with one type of training (or foot-strike type) than others.

Heel vs. Midfoot Strike

Heel Strike

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Midfoot Strike

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When comparing these two running styles, biomechanically, a midfoot strike produces less impact force on the body than running with a heel strike (321). This is because **a heel strike is more reliant on the skeletal system for support and stability than a midfoot strike, which relies more on the musculature system.**

If your athlete heel strikes, they should not necessarily convert to a midfoot strike.

As a midfoot strike tends to reduce the impact on the knees, hips, and back and lessen stress on the hamstrings, if an athlete is not experiencing any issues in these areas, the *if it ain't broke don't fix it* theory may be applied. However, if you feel that because of biomechanical or flexibility reasons they might benefit from a midfoot strike, it would be advantageous to speak with the person about slowly converting. A midfoot strike tends to stress the soleus and Achilles tendon, especially for those converting to this foot-strike type.

The video below denotes a runner with a forefoot strike – notice that the heel does not hit the ground.

While a midfoot strike reduces the magnitude of impact forces on the feet relative to a heel strike, it does not mean that it will feel better or be more comfortable, especially to those in the process of converting (321). You must be patient with your athletes while they are making the switch from heel to midfoot running. Remember, they have been walking and running with a heel strike their whole lives, so the change will likely not come quickly, easily, or without some degree of frustration.

As noted previously, a heel strike that occurs substantially in front of the body (overstriding) puts the hamstrings under substantial eccentric stress. If the stride is really long, hyperextension of the knee is also possible. When running with a midfoot strike, the aforementioned issues are largely corrected. This results in the ability to run with less chance of injury to the hamstrings.

Forefoot Strike

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Foot Strike Misnomer

Only The Balls of The Feet Contact the Ground During a Midfoot Strike

Many people who adopt a midfoot strike impact the ground only with the balls of their feet, leaving the heel of the foot in the air. This stresses the Achilles tendon and calf muscles, primarily the soleus. This type of foot strike is commonly referred to as a forefoot strike.

While the emphasis of foot contact with a midfoot strike is the ball of the foot, it should not be the only point of contact. Once the ball of the foot impacts the ground, immediately after that, the rest of the foot should lightly impact the ground. While the ball of the foot is the area that pushes off the ground, contact (ground) between the middle of the foot and heel is used for support.

This support decreases the stress on the Achilles tendon and calf muscles. When watching a runner with a proper midfoot strike, it should almost look as though the entire length of the foot hits the ground at the same time. The only time the ball of the foot should hit the ground exclusively is when running up a steep hill or sprinting. This is called a *forefoot strike.*

Drills: Midfoot Form

To understand what a midfoot strike is supposed to look like and, more important, how it is supposed to feel, there are two very good drills:

Uphill Running

Have an athlete run up a gradual (2–5 percent grade) hill. The postural traits of a midfoot strike are almost identical to that of running uphill.

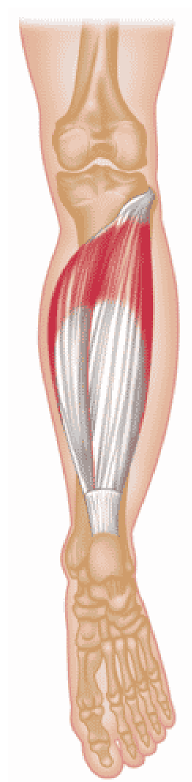
- Slight forward upper-body lean (from the ankles, not the hips)
- Foot lands on midfoot versus heel
- Feet land under the body

Resistance Running

Another way to teach proper midfoot strike form is to wrap a long theraband or rope around your athlete's waist while you hold the ends behind the person (see the video to the right). Cue them to lean forward at the ankles, not the waist. You should be at least three feet behind them. Then instruct them to run forward against the resistance. During this time, you need to be braking to slow them. Your form should be similar to walking down a steep hill – low center of gravity, hips shifted down and posteriorly, knees bent, and upright torso. This drill is effective because the runner must lean forward and strike with the midfoot. This drill can also be accomplished by having the athlete pull a weight-loaded sled attached to the waist by a harness/rope or running with a running chute.

Converting to Midfoot Form

Many runners who convert to a midfoot strike from a heel strike will notice increased calf tightness and, occasionally, injury to the soleus and Achilles tendon. While soleus tightness is expected to some degree, injury should not be a part of this stride transition. To reduce the chance of injury and to facilitate an effective transition to running with a midfoot strike, the following are several important things to address:



Soleus

1. Allow the midfoot and heel to impact the ground almost simultaneously (the ball of the foot impacts the ground slightly before the heel) to reduce the stress on the calf muscles. However, this action should not be cued by asking the athlete to land on a certain part of their foot; rather, they should be

cued to shorten their forward stride and increase their stride rate.
2. Perform exercises to strengthen the soleus and Achilles tendon (e.g., jumping rope, walking on balls of feet)
3. Right before the foot hits the ground, it should be moving rearward, and the hips should rotate toward the side of the body of the foot strike.
4. Do not bounce excessively when running. Bouncing causes the calf muscles and Achilles tendon to work excessively to accelerate and decelerate the body vertically. Aside from being potentially more self-injurious, it is also a slower way to run as energy should be used for forward, not vertical propulsion.
5. Gradually introduce midfoot striking into runs and increase the duration of time running with a midfoot strike so long as no injuries or excessive calf tightness occur.

Athletes who convert to barefoot or minimalist running with no progression have an increased chance of injury. Individuals converting from a heel to a midfoot strike often do not have the structural arch support or muscle and connective tissue strength to run in an unsupported shoe. Muscles and connective tissue adapt to what they have been doing for long periods. Adapting to a new running shoe type or foot strike requires proper progression.

When Not to Make Changes

If you acquire an athlete who is currently heel striking and has a race in the near future (anything less than five months), it is advisable to not convert the individual to a midfoot strike for the event. Running efficiently with a heel strike is a learned skill, as is running with a midfoot strike. Unless the athlete becomes injured because of heel striking, trying to unlearn heel striking and learn to run with a midfoot strike when a race is not too far away is not a smart idea and will most likely lead to slower times, frustration, and possible injury.

This is not to say that you cannot introduce small doses of midfoot striking into the training program, but you should not expect an athlete to successfully run the event with the new foot strike. **All biomechanically based changes must be introduced in very small doses and over extended periods.**

Go Slower to Go Faster

When converting to a midfoot strike, a runner's speed will initially likely decrease. This is normal and must be conveyed to a runner prior to and during the conversion to midfoot striking.

Should I Convert My Athlete to A Midfoot Strike?

Depends. Do you think an athlete could be more efficient? Is the individual experiencing issues that may be rectified with a midfoot strike? If the answer to one or both of these questions is yes, then conversion to a midfoot strike may be the right course of action.

Contrary to many running articles, striking with the heel versus the midfoot is not wrong or incorrect, just different. Therefore, if your athlete is heel striking and doing fine with that, there is no reason to change. In the same respect, if your athlete is currently running with a midfoot strike and is having issues (i.e., chronically tight soleus), the right course of action may be to work with the athlete to convert to a heel strike.

More important than the type of foot strike is the angle of the tibia upon foot strike. This is discussed in the next section.

It's All About the Tibia

Running gait (stride) efficiency correlates directly to the angle of the tibia in relation to the ground at the time of foot strike. Upon foot strike, the angle of the tibia in relation to flat ground should be at or close to 90 degrees.

In the above image, the blue line represents 90 degrees, and the distance between the yellow and green lines represents the acceptable tibia angle range during the foot strike. Therefore, the foot-strike angle (red line) is outside what should be considered an acceptable range.

A foot-strike (tibia) angle at or near 90 degrees in respect to the ground allows forward propulsion to occur almost immediately as the foot is directly under, or close to being under, the body's center of gravity. If the foot strikes substantially in front of the body, the ground reaction force creates an upward and rearward moment that does not facilitate forward motion (215). In other words, when the foot is too far in front of the body at the time of foot strike, a "dead spot" in the stride is created because of a lack of forward propulsion occurring with the drive leg. Research has demonstrated that landing on the heel significantly ahead of the body increases the loading rate versus a midfoot strike with the leg under or just in front of the body. **It is likely that the closer a heel strike is to the body, the lower the loading rate is in relation to a heel strike farther in front of the body.**

If you have an athlete running with a heel strike and overstriding, the goal should be to shorten their stride in front of their body so the feet land under, or close to being under, the body's center of gravity. This leads to a reduction in stride length and an increase in their stride rate (leg turnover). Additionally, reducing stride length reduces the angle that the foot is pointed diagonally upward when the heel impacts the ground. When the foot is pointed upward excessively, it causes the foot to “slap” down on the ground, increasing stress on the legs and body. This slap also causes a slight delay in time before the foot can act to push down and rearward to propel the body forward, though the negative effect of this on performance is negligible.

Drill

To demonstrate why the tibia needs to be at or close to a vertical position upon foot strike, have an athlete stand on one leg with the knee behind the ankle (the athlete will need to hold on to something for stabilization). Then instruct them to push off the foot to move the body forward. They will find that forward propulsion is impossible until the knee is slightly ahead of the ankle. This directly correlates with the body being over the foot to generate forward propulsion.

Natural Foot Gait

Generally speaking, the faster or more uphill an individual runs, the farther forward on the foot they tend to run. Conversely, the slower a person runs (or walks), the more natural heel striking often feels. If you look at the sole of a track sprinter’s shoe, you will see that the spikes are under only the balls of the feet and the toes. This is because when people sprint, they are forced to emphasize forward propulsion at all times. Conversely, ask your athlete (or try it yourself) to walk with a midfoot strike. This is extremely awkward and feels very odd. The faster a person runs, the more natural a midfoot strike will likely feel. This is substantiated in the study discussed next.

Bare Feet = Midfoot Strike?

There is an assumption by many that the body automatically converts to a midfoot strike when running in bare feet or minimalist shoes. The thought process behind this is that barefoot runners naturally run with a midfoot strike to avoid the high stress (impact forces) associated with running with a heel strike. A 2013 study out of George Washington University (GWU) found this not to be the case (321). The idea that running on bare feet naturally converts a runner to a midfoot strike resulted from studying just one population of barefoot runners. This research was then marketed in popular running journals.

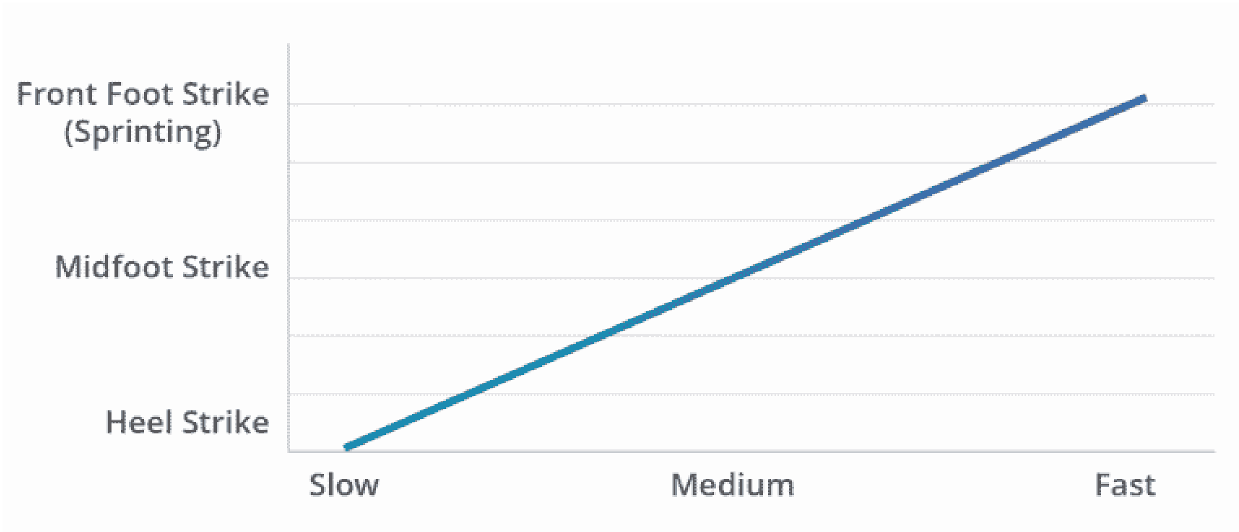
The GWU study researched 38 runners from a habitually barefoot population in northwest Kenya called the Daasanach. The study had the runners run around a track at varying speeds. The study found that while running with a midfoot strike lowered impact forces, most runners ran with a heel strike. While some runners switched to a midfoot strike while running at faster speeds, a midfoot strike was not a constant theme when running at faster speeds. **This research contradicts the theory that running in bare feet automatically changes one's running gait from a heel to a midfoot or forefoot strike** (321).

Foot Contact Time

Foot-contact time relates to how long a foot is in contact with the ground during the running-gait cycle. Running with a heel strike results in a longer contact time than a midfoot strike because the feet typically hit the ground in front of the body, thus elongating the stride length. A 2007 study by Hasegawa et al. used video cameras at the 15K point in a half marathon to capture the foot-strike type and foot-contact time in which beginners to elite competitors competed. The results showed the following (322):

- A higher percentage of faster runners ran with a midfoot strike than slower runners
- Faster runners had less foot contact time than slower runners

While not all "fast" runners use a midfoot strike, the tendency is that the faster one runs, contact occurs farther forward on the foot. The following chart illustrates natural running gait transitions based on the running speed. The chart below illustrates the correlation between running speed and the type of foot strike.



Another tendency when converting to a midfoot strike or when getting tired during a run is that the foot impacts the ground while the leg is still moving forward and not rearward. This causes the stride to look like a shuffle and increases foot-contact time. This results in one's toes sliding forward in the shoes and jamming in the front part of them (toe box). With a midfoot strike, any time the foot is on the ground, it should push down and back to propel the body forward.

The Achilles tendon also plays a role in foot-contact time, and therefore one's running efficiency. From a biomechanical perspective, the Achilles tendon is a spring that loads and unloads energy. The stiffer and longer the tendon, the greater the rebounding effect of the "spring" (496). While the length of the tendon is predetermined at birth, **the stiffness of the tendon can be improved through strength training, running, and any other activity that stresses the Achilles tendon.**

Lastly, a study that observed the foot strike of 286 marathon runners found that when comparing the foot-strike type at 10K (6.2 miles) with 32K (20 miles), a large number of runners who ran with a midfoot strike at the 10K point converted to a heel strike at 32K (324). While not noted in the study, it could be hypothesized that fatigue and a slower pace were potential reasons for the change in foot-strike type.

Form Analysis

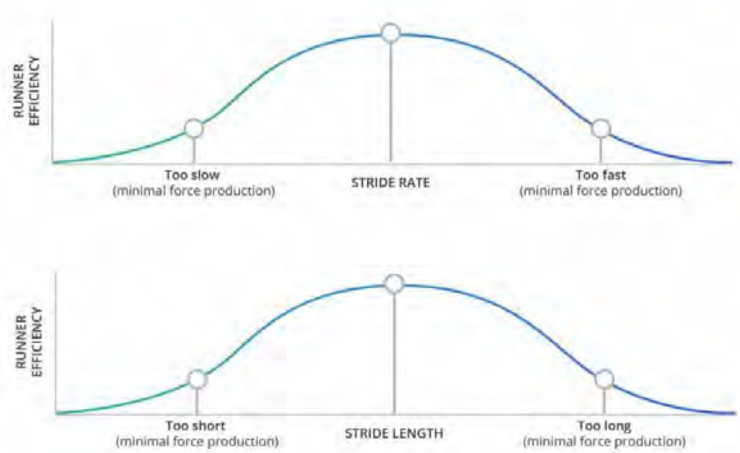
Recording an athlete on video while running on a treadmill for immediate feedback and reassessment is of substantial value. Video of running on a treadmill is especially valuable for assessing foot strike, foot movement (pronation, supination, etc.), and overall body position.

While you can assess transverse hip rotation on the treadmill, it is often more accurately assessed outdoors, as the treadmill may facilitate hip rotation. The easiest way to assess runners outdoors is to either run alongside them or have them run on a track or back and forth in front of you. If running with an athlete, make sure that the road ahead is clear before looking at your athlete to the side. Video is a good medium for this as you can record the athlete running past you. You can also record them running while you ride a bike beside the person. If filming while riding a bike, you must ensure that you are looking forward and in control of the bike at all times.

Stride Length and Rate

Stride length and stride rate go hand in hand because of their effect on each other. The only way to increase running speed is to increase stride rate and/or stride length (568). Therefore, a commonly asked question is, how long and fast should one's stride be?

This is the million-dollar question with unfortunately no clear answer. Before we delve too deep into this question, the most important thing to appreciate is that, like most aspects of exercise science, stride rate and length function on a bell curve. Too slow or too fast of a stride rate and too long or too short of a stride length will negatively impact a runner (573). Therefore, a runner is most efficient somewhere in between the two extremes.



Argument Against Changing Stride Length / Type

Two studies examined whether or not changing the foot-strike type and stride length would increase performance or, more to the point, running economy (724, 725). Both studies found that changes in the stride did not improve running economy but rather had the opposite effect. So is changing one's stride a bad idea?

Not necessarily. The studies noted here occurred over a relatively short period. This is in contrast with the time likely required for neuromuscular adaptations to occur that would make a runner more efficient. If an individual has been running with a set stride for a long time, it almost guarantees that when initially switching strides, the person would have a decrease in running economy. Changes in biomechanical form require a long neuromuscular adaptation period.

If a runner is injury-free and appears to have good running economy, there is likely no reason to change form. However, if an athlete is injured or has a form that is "less than ideal" and you believe the issue is biomechanical, working to change the stride might be the correct way to proceed.

Correlation Between Stride Rate and Stride Length

Initial increases in speed rely more on the increase in stride length versus stride rate. However, at higher running speeds, speed increases depend more on increasing stride rate and slightly decreasing stride length (569).

Interestingly, leg length, height, and mass (girth) are not primary factors in determining one's preferred stride rate and length (570).

It is more energy efficient to lengthen the stride within an acceptable range than to increase the stride rate (571). As heel strikers typically have a longer stride length than midfoot strikers, this may explain why toward the end of races, there is often a conversion from individuals striking with the midfoot to the heel (572).

Stride Length

An individual's stride length is typically self-selected based on running speed. Some studies suggest that individuals self-select their running style (stride rate/length) based on what is most efficient, and when forced to change their running gait, they become less efficient (574). This is likely due to an individual adapting to a particular running gait versus it being mechanically and physiologically most efficient. **Typically, when an individual changes established form, there will be an initial decrease in performance and efficiency until the body adapts to the new form.**

To illustrate this point, watch a running race, and you will see all sorts of stride lengths and rates – many of which are inefficient and not economical. If the theory that people self-select the ideal, and therefore most efficient, running gait was correct, all runners would be efficient regarding their gait characteristics (foot strike, stride rate/length). This is not true.

Short Stride

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Overstriding

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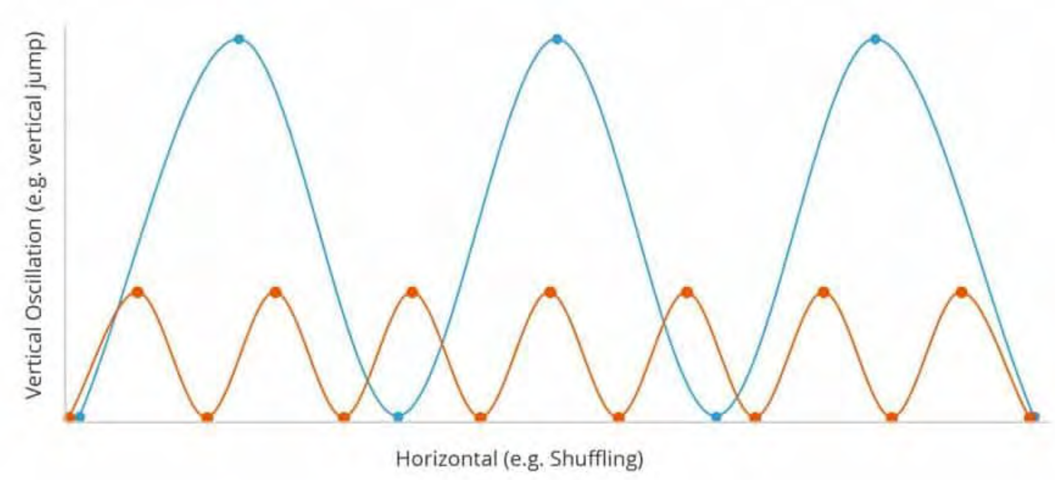
Vertical Oscillation

An aspect of stride length often not discussed is that of vertical oscillation. This pertains to the vertical movement of the body while running. From an efficiency standpoint, moving the body vertically takes much more energy than it does horizontally.
Therefore, **running with as little vertical oscillation as possible is suggested.** Too much vertical oscillation results in a runner "bounding" or "bouncing." Conversely, a runner with minimal vertical oscillation typically shuffles. As such, the ideal amount of vertical oscillation exists on a bell curve. Additionally, greater vertical oscillation equates to higher ground forces (i.e., impact on the body).



Orange = Low Vertical Oscillation
Blue = Large Vertical Oscillation

Regarding stride length, the longer the stride, the more significant the vertical oscillation. An analogy to this is throwing a ball. To throw a ball far (e.g., long stride), the ball must have a substantial arc (e.g., a lot of vertical oscillation).

The below video denotes excessive vertical oscillation.

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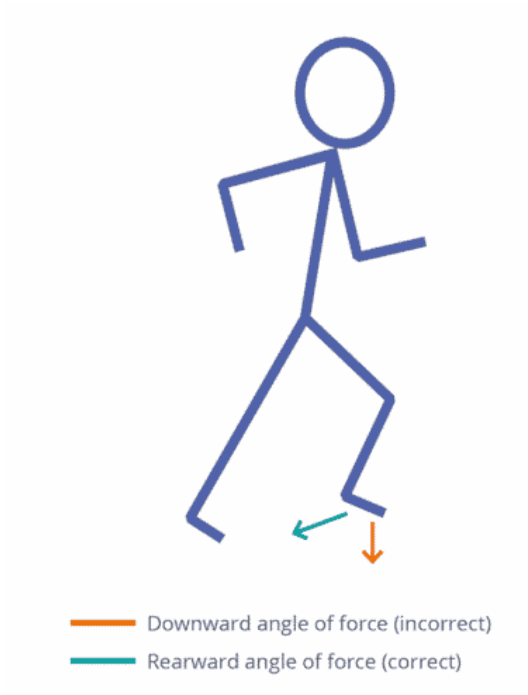
In addition to a shorter stride length, transverse rotation of the pelvis (hip rotation) assists in reducing vertical oscillation due to the energy being used for forward versus vertical propulsion.



The primary determinant regarding the amount of vertical oscillation pertains to how the drive leg pushes against the ground. The correct angle of force is downward and rearward (diagonally rearward). When a runner pushes directly down into the ground, the body moves primarily upward, not forward.

Stride Rate

Stride rate (SR) directly affects stride length (SL) and vice versa. The topic of stride rate is often discussed from the perspective that a fast stride rate is better than a slow stride rate. Two things to consider:

1. Running with a fast stride rate does not necessarily equate to having proper form or efficiency.
2. What constitutes a “fast” SR is subjective and primarily based on the runner’s speed.

Running with a high SR is typically indicative of running at a relatively high rate of speed. Therefore, increasing a runner’s SR without increasing speed will likely reduce efficiency. **Increasing one’s SL is primarily a function of increased force production, whereas increasing SR is likely due to increased neuromuscular and cardiovascular capacity** (578). Therefore, it is probable that a runner currently running with a low SR cannot increase SR dramatically without a decrease in form because of inadequate neural connections and cardiovascular fitness.

At the 1984 Olympics, renowned running coach Jack Daniels observed the stride rates of runners and found that all but one runner had a stride rate of 180 or more per minute (576). **This data point has found its way into running folklore representing the low end of what constitutes a “fast” or “high” stride rate.** Therefore, an SR of 180 has become a benchmark for many runners looking to increase their SR and overall running performance.

As a runner’s speed largely influences SR, it is not surprising that Olympic- caliber runners have high SRs, even for long-distance events (576). However, as most runners do not have the leg turnover of Usain Bolt, it is expected that the SR of a good majority of runners will be below 180. Additionally, a runner who attains an SR of 180 on a fast training workout may likely be below 180 on a long run.

As noted by running coach Steve Magness, if an individual runs at an 11- minute/mile pace with an SR of 180, this equates to a stride length of only 32 inches – hardly an efficient SL (577)!

Conclusions

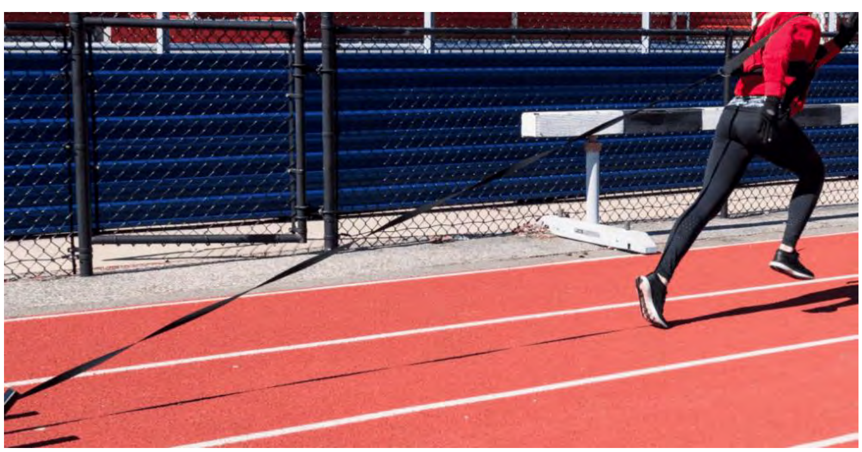
Decreasing SL and increasing SR are common guidelines for improving running performance. However, **any SL and/or SR changes must be done with a focus on proper form.** While these guidelines are not incorrect per se, they must be appreciated in the context of a runner as a whole. As noted previously, any change in form typically equates to an initial decrease in performance due to the adaptation process.

It is natural for SR and SL to change during a race or workout (577, 578). This is especially the case when fatigue sets in (577).

An interesting point of view regarding SL and SR comes from Steve Magness, who views these two data points as feedback. In other words, focus on the other elements of form (foot strike, hip rotation, upper-body position, etc.), and the SL and SR will change accordingly (577).

Lastly, **as a coach, do not become fixated on a set number regarding the optimal SR and SL.** These are influenced individually; therefore, there are no right or wrong SR and SL numbers.

Drills



This section consists of running-form drills to integrate into a training program. This is not a complete list but rather a comprehensive overview of popular drills and workout types. You will inevitably create new drills that will best serve your athlete. Some of the following drills are also noted in the biomechanical sport-specific modules.

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Run Uphill

The form used to run up a slight hill (5–8 percent grade) matches that used when learning how to run with a midfoot strike.

- The body leans slightly forward.
- The lead leg does not lift up excessively but simply moves forward to support the upper body.
- The lead foot hits the ground with the midfoot, not the toes or heel.
- The lead foot lands when the body is directly over the foot, not in front of the body.

Run Downhill

When running downhill, the tendency is to elongate the running stride and lean back into the hill. This is primarily done to control one's speed (i.e., braking). The stride length should stay about the same throughout a run, regardless of whether the grade is flat, uphill, or downhill. Therefore, the form when running downhill should not vary much from the form at any other time.

Run with a Slight Forward Lean

There should be a slight forward lean from the ankles to the top of the head when running.

This is primarily for the purpose of the body being "over top" of the legs at the beginning of the drive phase. This allows for immediate forward propulsion. When the body is behind the legs at the beginning of the drive phase, the body must catch up to the drive foot for forward propulsion. The other reason for a slight forward lean is to reduce the chance of the knees hyperextending via excessive lengthening of the hamstrings.

Partner Pull Back Drill

Place a harness around your client's waist (can use a long theraband or padded rope) and stand behind the person holding the ends of the rope. Then have your client start running at a moderate pace. You will run behind (braking), trying to reduce the client's speed. This restriction forces the client to have a slight forward lean and drive off the midfoot versus the heel. This drill can also be replicated using a running/sprint chute or weighted sled (see image above).

Strides

Strides are essentially sprints (not 100 percent effort) that are done before and/ or after runs/races. The purpose of performing them before a race is to warm up the legs to increase blood flow and get the mind adapted to what race pace will feel like. Strides also work to enhance the neuromuscular efficiency of a runner.

When done at the end of long or easy runs, strides train the neuromuscular system to increase speed rapidly. This enhances not only one's form but also one's speed – especially at the finish line.

Strides are typically done independently of a run. When speed increases are integrated into a run, they normally fall into the category of *speedwork*, which strides are not. While strides increase one's heart rate, the primary physiological adaptation is neuromuscular, not cardiovascular.

If strides are done before an event, a runner must be sufficiently warmed up before performing them. To perform strides, a runner starts running several steps at a jog and then increases the pace quickly to be between 80 and 90 percent of their maximum speed within 15 to 25 foot strikes. Once this speed has been attained, the goal is to hold this pace for another 10 to 20 foot strikes before slowing down. A critical aspect of strides, including the deceleration, is maintaining as close to perfect form as possible. As strides train the body from a neuromuscular standpoint, performing strides with incorrect form trains the body to have incorrect mechanical patterns.

An athlete should do no more than five to 10 strides. If your athlete begins to feel tired from the strides, the individual has done too many.

Study

A 2015 study by Barnes et al. found that runners who performed strides wearing weight vests (20 percent of BW) elicited the following results ten minutes after performing the strides (672):

– 2.9 percent increase in peak running speed

– 20.4 percent increase in leg stiffness

– 6 percent increase in running economy

– Slight decrease in cardiorespiratory demand

These results can all be attributed to an increase in leg stiffness, specifically regarding the muscles of the lower leg and Achilles tendon.

Based on these results, the question regarding conventional strides (not wearing a weight vest) is whether they would elicit a meaningful percentage of the aforementioned results.

Discussion

Instead of a weight vest before the start of a race/run, unilateral plyometric leg jumps focused primarily on ankle plantar flexion would be an appropriate substitute.

Conclusion

When warming up for a short- to middle-distance event (10K), a combination of unilateral plyometrics and standard strides is recommended approximately 10 minutes before the start of the event. Post-run or race, standard strides are recommended for neuromuscular training. However, as noted above, more research in this area is needed.

Biomechanical Running Assessments

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The following videos by UESCA contributor and biomechanist, Dr. Nick Studholme are noted for the purpose of identifying potential biomechanical issues as it relates to running economy and a possible increased risk for injury. However, remember that as a running coach, you cannot diagnose or treat an injury.

Tibial Angle of Inclination

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Pelvic Drop

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Peak Knee Flexion

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Knee Window

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Hip Extension

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Hip and Knee Flexion Angles

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Hip Adduction

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Foot Bisection

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Ankle Dorsiflexion

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Center of Mass to Ankle

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Odds and Ends

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Summary

- The three phases of running gait are:
  - Drive
  - Support (Stance)
  - Recovery (Swing)
- Mild pronation is normal. Supination and overpronation are considered abnormal foot/gait patterns.
- Ground reaction force relates to the force acting upon the body in relation to the force of the foot strike on the ground.
- Pronation is not predictive of pain or injury.

- The *Meyer's Line* is a straight axis line that, on a functional foot, goes from the center of the heel, through the big toe joint, and through the center of the top of the big toe.
- Q angle relates to the angle of the femur.
- Utilization of passive energy reduces the active muscle requirement while still producing the required amount of force.
- The foot should strike the ground under the runner with the tibia at or close to 90 degrees in relation to the ground during the foot strike.
- Most of the energy when running goes to supporting the body.
- The hips should rotate slightly in the transverse plane when running. This is a source of energy termed *the spinal engine.*
- The PSSS and ASSS systems utilize the stretch-shortening cycle to store potential energy and use kinetic energy.
- Excessive vertical body movement equates to wasted energy.
- Generally, the faster one runs, the foot strike occurs farther forward on the foot.
- A heel strike places more stress on the skeletal system whereas a midfoot strike places more stress on the musculature system.
- Running with a short stride and fast turnover is often preferable to running with a long stride and a slow turnover.

Module 8: Injury and Illness

Injuries and illnesses happen, and as a coach, you need to know how to deal with these situations as they arise and see how the training process may be impacted.

Injury Terminology

Sprain: Injury to a ligament (ligaments connect bone to bone). Sprains are categorized as Level 1–3, 3 being the most severe.

Strain: Injury to muscle or tendon (tendon connects bone to muscle).

Muscle Cramp (exercise-based): Involuntary spasm of a muscle or muscles. Cramps can be caused by many factors, including overuse of a muscle in duration and/or intensity and nerve-related issues. Mild dehydration and low electrolyte levels, previously associated with cramping, have largely been determined not to be a valid catalyst for cramps (763–765).

Fracture: A break in a bone; varying degrees of severity.

Tendinosis: (often incorrectly called "tendonitis") Damage to a tendon. Often caused by overuse.

Bursitis: Inflammation of a bursa sac. A bursa sac is a synovial fluid-filled sac located between bone, muscle, tendons, etc. The purpose of a bursa sac is to provide cushioning and reduce friction.

Nerve Impingement: A nerve that has pressure applied to it. Often referred to as a *pinched nerve* or *compressed nerve*.

RICE: Acronym for ***R**est–**I**ce–**C**ompression–**E**levation*.

Chafing: Location on the body where the top layer of skin is irritated or worn away either due to skin-on-skin rubbing (e.g., thighs) or, more commonly, from skin rubbing against clothing (e.g., bra straps). While not an injury exactly, it can be very painful, and, like an open wound, it can get infected if not treated. The most common places for chafing are the nipples (men), inner thighs, areas where the bra rubs (women), and under the arms.

To reduce the chance of chafing, here are some preventative measures:

– Try new clothes on short runs before longer runs
-Men: wear nipple covers
-Women: make sure the bra fits properly and is tight
– Apply a skin lubricant such as *Body Glide*® in areas that are likely to get chafed
– Clothing with hidden seams is preferable to garments with regular seams

Scope of Practice

Successfully passing this certification will qualify you to advise and educate a runner in all areas pertaining to training and racing. It is important to remember there is a big difference between advising/educating and prescribing/diagnosing. As a running coach, you cannot do the following:

- Diagnose or treat an injury or illness

- Prescribe a nutritional or supplementation program in most states unless you are a registered dietician or certified nutritionist
    - Check with local or state laws
- Hands-on muscular (i.e., massage) or skeletal manipulation
- While not illegal, any substantial biomechanical deviation should be referred to a specialist.

If you do not feel qualified to advise or educate on a specific area and decide to proceed anyway, you are practicing outside your scope of knowledge.

This is an important distinction. **You can work within your scope of practice but outside your scope of knowledge.** Conversely, you can work within your scope of knowledge but outside your scope of practice. An example of someone working within their scope of practice but outside their scope of knowledge would be a new running coach whose athlete wants them to advise them on the difference between two different shoes. While this question is within a coach's scope of practice, if they do not know the answer and proceed to fabricate one, they are practicing outside their scope of knowledge.

An example of someone practicing within their scope of knowledge but outside their scope of practice would be a seasoned running coach that is 99.9% sure their athlete has patellofemoral syndrome and manually massages their athlete's leg to relieve the symptoms. While the coach may be correct in their 'diagnosis,' they cannot legally diagnose a condition and cannot massage their athlete's leg to relieve the symptoms – even if they are sure this will work.

The distinction between what is legal and illegal when working with athletes can be unclear. It is always best to err on the side of caution when working with athletes and refer them to specialists.

Team Approach

A coach's legal and professional obligation to refer their athletes to specialists may also present learning opportunities for a coach. Often, a specialist will advise a coach on what protocol to follow or why an athlete is experiencing a particular issue. Knowledge gained in this fashion is invaluable and will help support further the development and education of a coach.

It is suggested that a coach coordinate a 'team' of specialists to whom they feel comfortable referring athletes. This team approach generally elicits the best and fastest results for athletes and releases a coach from potential liability by overstepping their professional boundaries.

A coach can't have all the resources required by an athlete. If a coach is not implementing a team approach, they are not providing their athletes with the best possible training. Unless a coach is all of the following: doctor, physical therapist, registered dietician, podiatrist, sports psychologist, etc., they need other members on their team. Just because a coach has knowledge or experience in a particular area does not necessarily mean that there is not someone better equipped to assist in that area. For example, a full-time biomechanist is likely better suited to perform an accurate gait analysis than a running coach. This is not a negative reflection on the running coach but rather demonstrates the specialized expertise of a biomechanist. Referring athletes to others is not a sign of weakness but indicates a coach's intelligence and dedication to their athletes.

Injury – need to add content

Running is a sport with a relatively high injury rate, and as such, understanding methodologies to possibly prevent injuries is of critical importance.

Your athletes will experience discomfort and pain from time to time. You and your athlete must assess the situation, use the best judgment, and determine whether medical intervention is necessary or if the issue is something more benign. **If you or your athlete is unsure if medical intervention is necessary, seek medical intervention!**

The following is a list of possible scenarios and the advised courses of action.

Injury Scenarios

Discomfort that appears to be due to a biomechanical issue

Work to correct the biomechanical issue to see if the discomfort diminishes or disappears. If not, refer to a specialist. If a biomechanical issue is substantial or you do not feel comfortable addressing the issue, refer immediately to a specialist.

Acute pain or trauma that is severe

Refer to a specialist or emergency care personnel

Pain that appears to be the result of overuse

Reduce duration and intensity. If the pain does not diminish or disappear, refer to a specialist.

Random pain

Refer to a specialist

Discomfort or pain due to illness

Cease physical activity, and depending on the severity of the illness, the client should contact a doctor.

Chronic Pain

Chronic pain is characterized as pain that has been present for six months or more. It is advised to have the client see a medical professional.

You must work within your scope of knowledge and practice at all times. In other words, know and respect your limits. If you feel your athlete's situation is at the threshold of your knowledge or practice, you must refer the individual to a specialist. As such, this certification will not cover how to diagnose or treat injuries, as this constitutes an area outside the scope of practice of a running coach. Additionally, you cannot recommend or prescribe medications. This includes over-the-counter medications such as ibuprofen and other anti-inflammatories.

While diagnosis and treatment of injuries are outside the scope of a running coach, it does not mean that you should be unfamiliar with the most common types of injuries and their potential symptoms.

Emergency Situations

Emergency personnel must be alerted immediately if any of the following situations arise.

Note: Contacting emergency personnel is not limited to the following situations. Do not leave the individual and/or scene until emergency personnel arrives.

- Broken bone

- Loss of consciousness
- Shock
- Choking
- Cardiac arrest
- Drowning
- Respiratory event (e.g., asthma attack)
- Allergic reaction
- Unresponsiveness
- Uncontrolled bleeding

While coaching athletes in person, you should always have a cell phone available to call emergency personnel (#911).

Inhaler Use

If your athlete is asthmatic and, in particular, suffers from exercise-induced asthma attacks, it is advised the person carries a fast-acting inhaler while exercising.

Contents of an inhaler are a prescription medicine. Therefore, you are not legally allowed to administer the medicine to athletes – even if they suffer an asthma attack. Athletes **MUST** administer the drug/inhaler themselves.

Injury Prevalence and Location

A study by Van Mechelen found that running injuries among recreation runners ranged from 37–56 percent annually (192). Other pertinent findings from this study:

- Most running injuries involve the lower extremities
- **50–75 percent of injuries are due to overuse (repetitive stress)**
- Recurrences of running injuries range from 20–70 percent

The Van Mechelen study also found that many of the commonly associated risk factors for running injury are unclear, in large part due to contradictory or scarce research findings. Some of these factors include:

- Warm-up/Stretching
- Muscular imbalances
- Lack of range of motion
- Orthotics

Therefore, more research is required to assess the validity of these areas regarding their correlation to running injury.

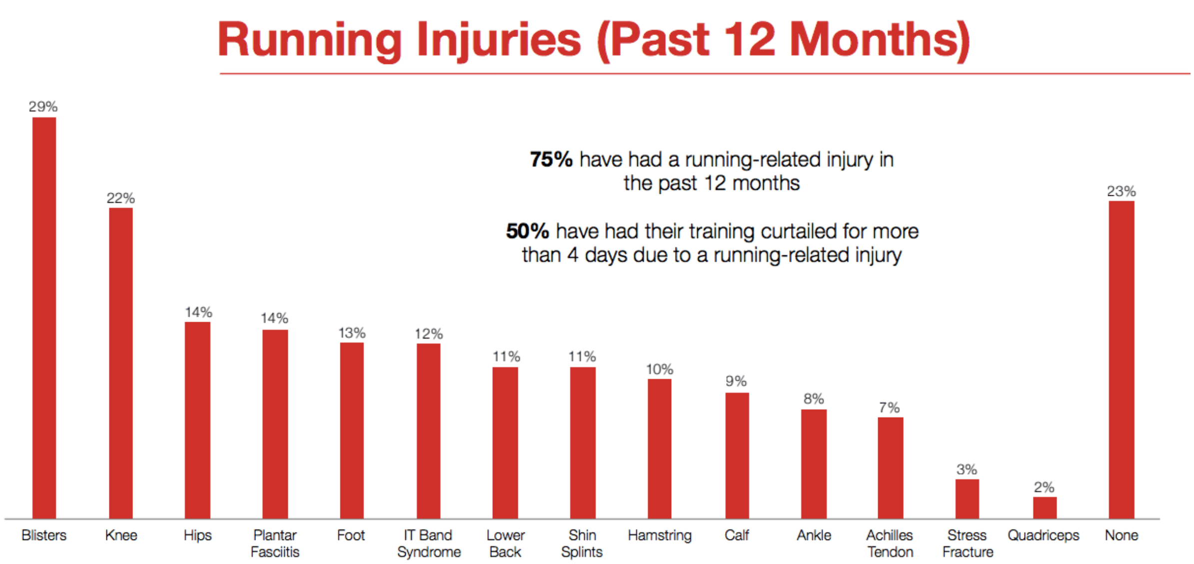
Many factors influence injury prevalence and location. A 2012 study by Daoud et al. examined repetitive stress injuries among collegiate runners and found that **74 percent of runners**

experienced a moderate to severe injury annually (648). Some of the study's specific findings:

- The most common injuries were:
  - Shin splints
  - IT band syndrome
  - Patellofemoral syndrome
  - Achilles tendon pain

Runners in this study ran 40–45 miles/week, performed high-intensity training sessions, and ran between 12 and 18 races per year. As such, runners in this study are classified as competitive versus recreational. Therefore, it is probable that factors such as high mileage, fast running speeds, and frequent racing contribute to an increased injury rate (648).

The below infographic from the *USA Running's 2017 National Runner Survey* found the following results regarding the most prevalent injuries:



Influence of Foot Strike Type

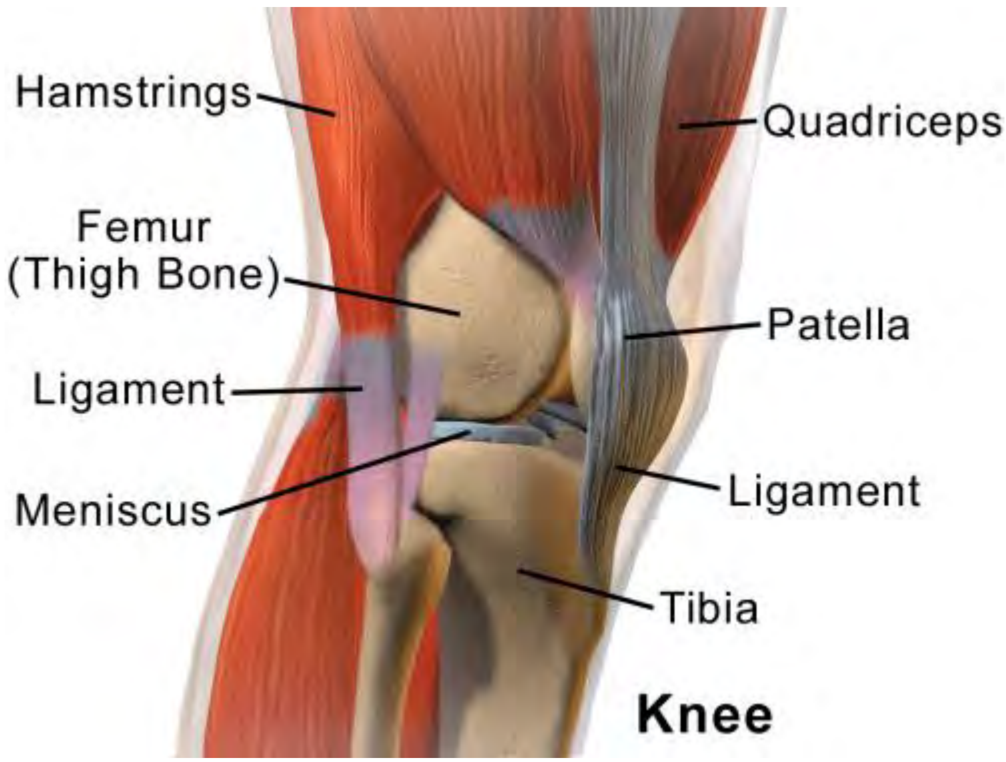
The aforementioned study (Daoud et al.) found that regarding repetitive stress injuries, runners who ran with a rear foot strike were two times more likely to sustain a repetitive stress injury than those who ran with a midfoot strike (648).

Anatomical Factors

A 2010 study examining factors involved in preventing injuries found that two anatomical factors correlate highly with running injuries (649):

1. High-arched feet
2. Leg-length inequality (functional or structural)

Is Running 'Hard on the Knees?'



Running is a sport that is often associated with being *hard on the knees*. The common association is that compressive forces on the knees due to the impact nature of running cause knee injuries.

A 2013 study by Hinterwimmer et al. examined the effect of marathon training on cartilage changes (volume and thickness) of the knees of beginner marathoners (690). It was found that while there were some cartilage changes, they were not clinically significant. Therefore long-distance running is well tolerated by the cartilage in the knees of beginner marathoners. More research is required regarding changes/loss of cartilage in seasoned marathon runners.

Dr. Michael Fredericson of the Stanford Running Clinic notes (772),

"At faster speeds, runners tend to have better hip mechanics, leading to reduced knee loads."

This primarily has to do with hip rotation in the transverse plane.

A recent study by Petersen et al. also found that running at slower versus faster speeds increased the load on the knees. This, however, was due to a decrease in the overall number of strides to cover the same distance (773).

Lastly, Dr. Fredericson states that running with a foot strike closer to the front of the foot reduces the cumulative load on the knees.

Injury Prevention

While injuries occur for various reasons, you can take measures to mitigate this risk for your athletes.

Volume Increases

Traditionally, a 10 percent maximum increase in weekly training volume and a 10 percent maximum increase of the longest training day (week-to-week) are recommended. The thought process behind the *10 percent rule* is that volume increases greater than 10 percent will increase the chance for injury.

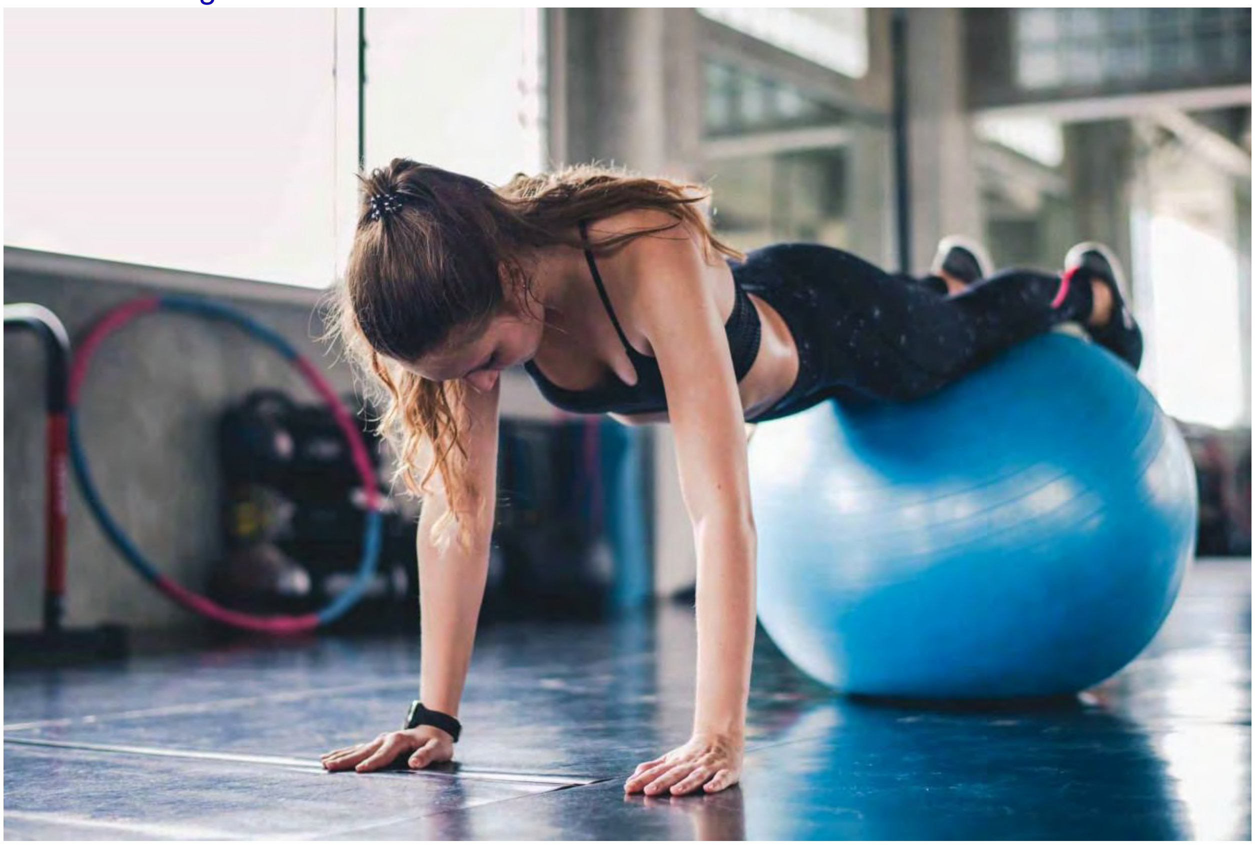
The viewpoint of this certification program is that **while 10 percent is ideal for some athletes, it may be too high or too low for others.** Therefore, while the 10 percent rule is a general guideline, changes in training volume must be based on feedback from an athlete. If an athlete is starting at the absolute lowest training volume, initial volume increases can typically afford to be larger than 10 percent.

A study by Fields et al. found that regarding preventative injury strategies, only reducing weekly running mileage correlated strongly with decreased running injuries (649).

Intensity

The level of intensity and the number of intensity-based workouts should increase incrementally. The more intense a workout is, the greater the musculoskeletal system demands. Without proper progression, muscles and connective tissue will not have the proper amount of time to adapt to the increased workload, thereby increasing the risk of injury. When intensity is initially added to a program (e.g., hill repeats, fartleks, repeats, etc.), only one day should be allocated to this purpose. As the body adapts, additional intensity days can be added.

Cross-Training



By definition, cross-training is when individuals participate in a sport or activity other than the sport they compete in for performance enhancement. Common cross-training activities are:

- Resistance training
- Cycling
- Swimming/water jogging
- Yoga and Pilates
- Other cardiovascular (e.g., rowing, elliptical, boxing, soccer, basketball, cross-country skiing)

While these are some of the more popular activities, any activity other than running is considered cross-training. As most of the movements involved with running occur in the sagittal and transverse planes, **cross-training in the frontal plane is recommended.**

A 2013 study by Malisoux et al. found that cross-training is a viable strategy to reduce the chance of running injury due to varying the load applied to the musculoskeletal system (687).

Cross-training has many benefits, including increasing cardiovascular fitness, injury prevention, increasing strength, staying mentally fresh, and, perhaps most important— having fun!

Fatigue

When individuals become fatigued, their form can break down and may lead to injury. An athlete must always be conscious of fatigue during training and racing. If their fatigue gets to the point where form breaks down significantly, or the person cannot maintain focus and clarity, the activity level needs to be reduced or ceased.

In addition, an athlete likely needs to refuel when the energy level drops substantially. Exercising must cease if the fatigue level is so high that an individual’s safety is at risk.

Pre-Stretching?

It is commonplace to see runners stretching before training runs and races. There are likely two reasons for this.

1. Decrease the chance of injury
2. Warm up the muscles

Interestingly, in both cases, stretching is not the panacea that many runners think it is. Concerning decreasing the chance for injury, a large meta-analysis of 12 studies comprising over 8806 runners found no evidence that stretching before running decreased the injury rate (2022). The most substantial evidence of preventing injury is reducing volume.

Kinesio Tape

Kinesiology tape (often called *Kinesio Tape*) is a stretchable adhesive tape primarily used for injury prevention/intervention or to increase performance. Dr. Kenzo Kase developed this tape in 1973 (214). Specifically, kinesiology tape is used for muscle and joint support, enhancement of biomechanical efficiency, and aid muscle recovery and endurance.

Studies researching the benefits of kinesiology tape have resulted in largely inconsistent results (207, 209). More research is needed in this area to determine the tape's effectiveness regarding its purported benefits. It can be assumed that using kinesiology tape will not diminish performance or reduce joint stability or functioning, possibly enhancing these areas (208). Therefore, if athletes ask about kinesiology taping or want to try it, it is best to refer them to a specialist with experience in this area.

Running Shoes

While no evidence-based research demonstrates that running shoe type reduces injury, research suggests that swapping out different types of running shoes during training reduces the risk for injury (687).

It is theorized that using different shoe types throughout the training process reduces the potential for injury because of the variation of load applied to the musculoskeletal system.

Running Injuries

Running is synonymous with injury because of one primary factor – impact. According to Van Gent et al., between 37 and 56 percent of runners sustain at least one injury annually, and between 20 and 70 percent of those seek medical attention (191). Another study found that between 30 and 90 percent of running injuries caused runners to decrease or stop their training (192).

Two studies, Marti et al. (1987) and Taunton et al. (2002), examined runners to identify risk factors that lead to running injuries. The most common injuries involved the knees and feet. Interestingly, both studies came across four factors that placed runners at greater risk for injury (205, 206):

1. Running high mileage (25–37 miles/week)
    - Mileage greater than this did not increase the chance for injury.
2. Competitive runners with a dedicated focus on training and racing
3. Younger than 34
    - Runners over 34 were at greater risk for Achilles and calf injuries.
4. Newly active (active less than 8.5 years)

Other Factors (205):

- **Weight:** This follows a bell-shaped curve, meaning unless runners were substantially under or overweight, they were not at an increased risk for injury.
- **Shoes:** Shoe type did not affect injury rate.
- **Gender:** Women are at a higher risk of injury than men.

Highlighted Gender-Specific Injury Areas:

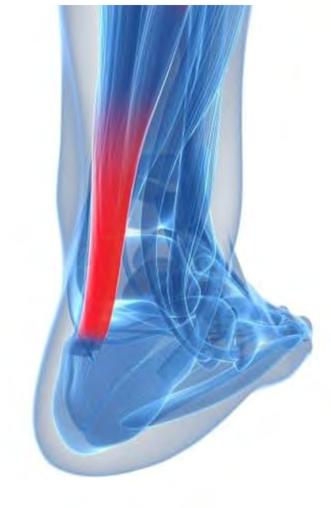
- **Women:** knee, hip, shin
- **Men:** patellar and Achilles tendons

These risk factors correlate to a 2014 study by Kluitenberg et al. that examined risk factors for novice runners. This study had 1,696 participants, ranging in age from 18–65. Of the novice runners, 10.9 percent sustained a running-related injury, and the identified risk factors based on those injured were as follows (647):

- Older runners
- Higher *body mass index (BMI)*
- Previous musculoskeletal complaints not related to running
- No prior running experience

What Is Tendonitis?

The term and diagnosis of tendonitis are quite common among endurance athletes, especially runners. To understand what this term means, attention must be paid to the suffix *-itis.* This is a pathological/medical term that denotes inflammation. Therefore, *tendonitis* means inflammation of a tendon.



The issue with the term *tendonitis* is that recent research has shown that in conditions diagnosed as tendonitis, there is little to no inflammatory cells present (606). Therefore, the term **tendinosis** is more applicable. **The primary finding regarding the cause of tendinosis is a disruption of collagen fibers of the tendon, not inflammation.** Because of the disruption of collagen fibers, the tendon weakens and cannot support high loads (603, 604). Collagen is a structural protein in connective tissue, such as tendons.

As there is likely no inflammation present with tendinosis, common treatment methods such as anti-inflammatory medications and corticosteroid injections have increased recovery time and accelerated the degenerative process (603-606). **Eccentric strengthening has been shown to assist in improving the collagen and strength of a tendon** (603, 604).

If an athlete believes they have tendinosis, it is advised to refer the person to a specialist.

Common Causes of Running Injuries

Common factors can cause or exacerbate an injury due to stress. In addition to and including some of the aforementioned risk factors, the following are the five potential risk factors regarding running injury.

Poor Running Mechanics

While some mechanical issues can be easily identified, such as foot-strike type and excessive pronation or supination, it does not mean they are easy to change. This is because poor running mechanics are often the result of muscle imbalances and kinetic chain issues. Retraining the body to run "properly" is not an overnight solution but rather a long process.

Worn Shoes

This is primarily an issue for those who heel strike, as a midfoot strike uses primarily knee flexion for shock absorption rather than heel padding. As noted previously, while the body adapts to a reduction in shoe cushioning, excessive wear is likely a contributing factor to injury – especially when an individual utilizes shoes for biomechanical foot and ankle support.

Lack of Range of Motion

Lack of range of motion, whether hamstrings, lower back, quadriceps, or any other muscle group, will negatively affect an individual. Tight hamstrings are often the primary cause of injury regarding a lack of flexibility. This is especially true with those who run with a heel strike, as this stresses the hamstrings eccentrically.

Mileage and Intensity

Starting a training program with mileage greater than what a client is ready for or increasing mileage too fast is a common cause of injury. Additionally, integrating speed work into a training program before a client is ready for it can also cause injury. Lastly, doing too many days of speed work or high mileage during a week can result in excessive stress on the musculoskeletal system. This often results in a runner being unable to recover sufficiently and places the individual at higher risk for injury.

Metabolic fitness precedes structural readiness (190). This essentially means that one's aerobic capacity/fitness develops at a faster rate than muscular and connective tissue strength. Therefore, runners tend to increase before their bodies can handle it.

Like incremental increases in distance, intensity must also progress at the correct rate. Intensity should be introduced into a training program at a relatively low starting point.

Muscle Weakness

This goes hand in hand with *poor running mechanics*. Minor muscle weaknesses are normal, as hardly any synergistic muscle groups are equal in strength and flexibility. However, substantial muscle weakness is not only inefficient but is also potentially injurious to a runner.

Running injuries can range in severity from a minor muscle strain to a major issue, like a fracture. The key to running injury-free is to respect the impact on the body and therefore perform due diligence regarding the training process. Following are areas to take note of when prescribing a running program:

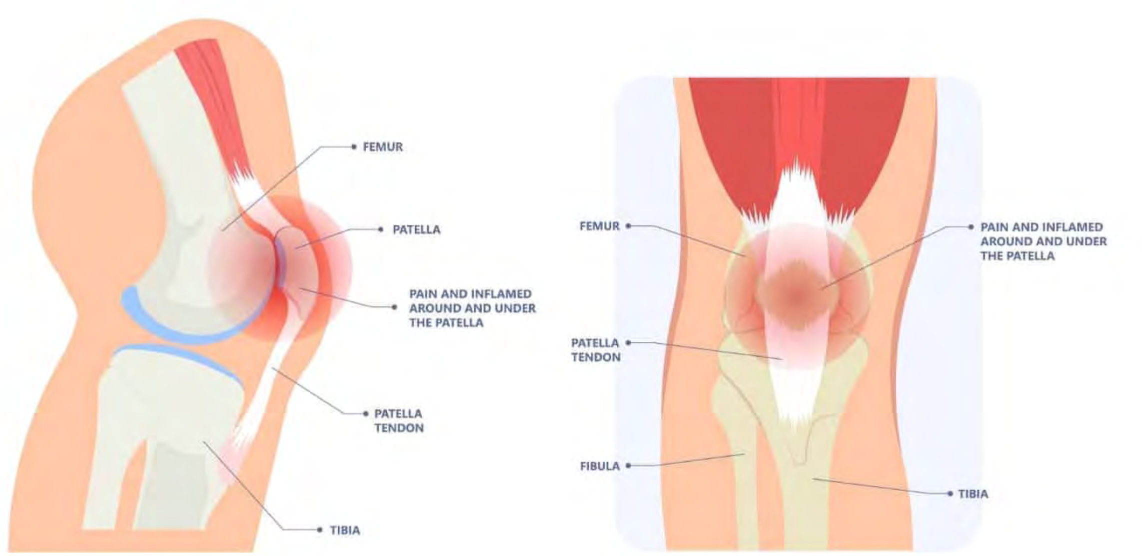
- Always warm up properly.
- Be aware of high-volume increases.
- If performing speed work, start with only one day per week and no more than two days per week, maximum.
- Work to have your athlete improve their running gait.
- Do not start speed work until a solid base mileage has been established.
- If your athlete experiences muscular or skeletal discomfort on a run that is not a normal byproduct of the intensity of the workout, have the person stop. If the pain continues, refer the individual to a specialist.
- Allow for enough rest in the training program.
- If your athlete is injured and has sought a doctor's advice, ensure the person follows the advice, even if it seems that the individual is "healed" sooner than the doctor thought.
- Cross-train
- Train your athlete in frontal and transverse planes, not just the sagittal plane.
- Listen to the body (e.g., if athletes sense they are still very sore from a previous workout, it is advisable to rest or cross-train versus run – even if it is not scheduled in the program).

Even if athletes follow all the "rules" of safe running, there is no guarantee they will not get injured. However, training intelligently mitigates the risk of getting injured; at the end of the day, this is the best one can hope for.

Running-Related Injuries

If you suspect an athlete has an injury, including those noted below, you must advise them to seek a specialist such as a doctor or physical therapist.

Patellofemoral Syndrome (PFS)



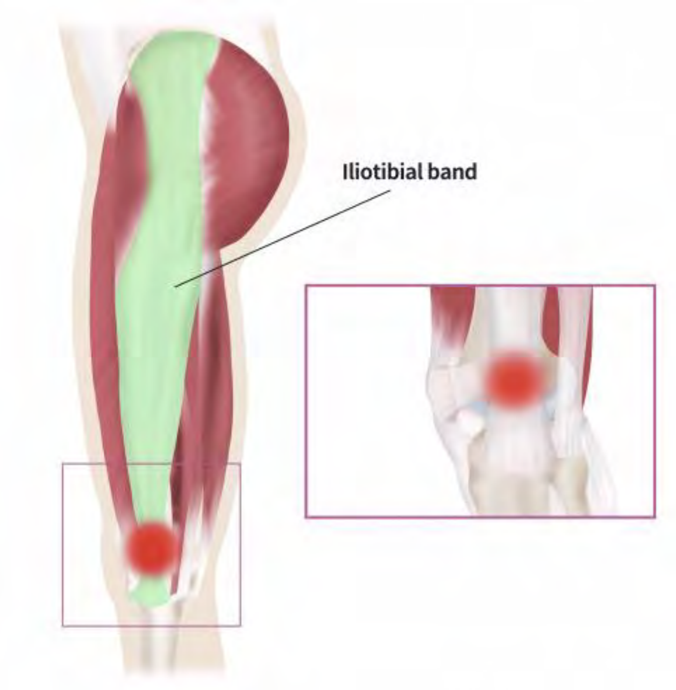
Frequently termed *"Runner's Knee,"* this is a generic term used to describe pain around and under the kneecap. There are many theories concerning what causes PFS. The reality is that the definitive cause is largely unknown. Below are some of the possible and hypothesized causes:

- Tight connective tissue as well as patellar malalignment. Patellar malalignment is often found in athletes with a large Q-angle (89).
- Compressive and shearing forces and hip rotation asymmetry have been shown to potentially cause PFS (343).
- ‘Tight’ quadriceps and a lack of gluteal strength.
- Muscle imbalance between medial and lateral heads of the quadricep muscles.
- Degeneration of cartilage under the kneecap

Another issue regarding an accurate diagnosis for PFS is that the knee region is a very busy area. Often athletes and clinicians alike will label any pain in the knee region as PFS. Below are three things that may be diagnosed as PFS but are unrelated. As you can see, some of these are also noted above as possible symptoms of PFS.

1. **Patella Tendonitis** – Inflammation of the patellar tendon. The tendon originates at the bottom of the patella (kneecap) and attaches to the top of the tibia.
2. **Chondromalacia** – Softening and degeneration of the cartilage under the knee
3. **IT Band Syndrome** – Noted below

Iliotibial Band Syndrome (IT Band Syndrome)



Before we go any further, let’s define the Iliotibial Band (ITB). The IT band (or tract) is a thick band of connective tissue (fascia) that originates at the top of the pelvic crest and inserts on the lateral head (lateral condyle) of the tibia, right below the knee. It runs down the lateral side of the thigh. Concerning running, the primary role of the ITB is to stabilize the knee.

Friction Syndrome?

Due to where the ITB inserts on the tibia, pain at the side of the knee while running has traditionally been thought to be caused by the ITB rubbing against the protrusion on the lateral side of the femur where it meets the knee, essentially a shearing-type movement. This bony protrusion is called the lateral femoral epicondyle. This is why ITBS is also commonly termed **‘IT Band Friction Syndrome (ITBFS).’**

The ITBFS theory largely assumes that there is a bursa sac between the ITB and the lateral femoral epicondyle. A bursa sac is a fluid-filled sac that aims to reduce friction between surfaces. ITBFS is therefore theorized to be the result of inflammation of the bursa sac (bursitis).

However, research by Fairclough et al. found no presence of a bursa sac between the ITB and lateral femoral epicondyle. What was found was a layer of high innervated fat (i.e., lots of nerves) below the ITB… more on this later.

Movement Of the IT Band

ITBFS is thought to be caused by the ITB passing or rolling over the lateral femoral condyle in a front-to-back motion when the knee bends… therefore causing friction.

The same study by Fairclough et al. also found that the ITB is firmly anchored to the femur and is also part of the tensor fascia lata (TFL); thus, the theory that it ‘passes over’ the lateral femoral epicondyle is incorrect.

Compression

In the Fairclough study, two runners that presented with ITBS showed no changes in the ITB based on magnetic resonance scans in relation to asymptomatic runners, but changes in the fatty layer under the ITB were identified. It was also discovered in the study that the ITB compressed against the lateral femoral epicondyle at 30 degrees of knee flexion. This was largely due to internal rotation of the tibia. Therefore, **ITB pain is most likely due to compression of the fat layer beneath the ITB versus friction caused by the ITB 'rolling over' the lateral femoral epicondyle.** In other words, the ITB or bursa sac (which is not present) is not what is irritated; it's the fat layer underneath it.

Training Implications

The common advice and treatment options for ITBS typically revolve around stretching and foam rolling.

Let's first address stretching. The ITB is a very tough structure. In a 2010 study by Falvey et al., a strain gauge was used to test various stretches on the ITB, and the result was that the ITB was unaffected.

Therefore, stretching is highly unlikely to affect the 'tightness' of the ITB.

Now let's discuss foam rolling. This is pretty simple. **Stop doing it!** Anyone that has ever performed foam rolling on their ITB will tell you that it is insanely painful. **The erroneous and commonly held belief is that if it's painful, it must be tight; therefore, more foam rolling is the answer.** However, let's refer back to what is likely the cause of ITB pain – compression. Assuming this is true, or at least a very strong contributing factor, common sense would dictate that adding compression via foam rolling will only exacerbate the issue. Therefore, whether to prevent ITBS or intervene once painful, foam rolling directly on the ITB should likely be avoided.

So, what should one do if they have pain in the side of the knee? As ITBS is largely an overuse injury, the first logical step would be to rest or, at the very least, reduce training volume. Like many other pain-related issues, the localized area of ITBS pain is likely the result of dysfunction in one or more areas. In the case of ITBS, according to Fairclough et al., it is likely that hip dysfunction may be a leading cause. Therefore, seeking a trained clinician is the correct course of action.

Shin Splints

Shin splint is also a relatively generic term used to describe pain in the region of the tibia. Shin splints can be categorized into three main areas: muscular, tibial fascia, and skeletal. If an athlete experiences shin pain, it is advised to have them seek out medical advice, as the type of treatment will likely differ based on the “type” of shin splint.

- **Muscular:** This is often referred to as ***exertional compartment syndrome.*** In the case of shin splints, the tibialis anterior is the muscle affected. During exercise, increased blood flow increases the size of the tibialis anterior. However, like all muscles, the tibialis anterior is encased in fascia and can increase in size only so much. This increase in pressure on the fascia is what causes pain. Individuals who have tight fascia surrounding the tibialis anterior are the ones who most often experience exertional compartment syndrome (385). Typically once one stops running, the pain gradually subsides.

As the tibialis anterior dorsiflexes the foot/ankle, when an individual experiences exertional compartment syndrome, the ankle likely has less ability to plantar flex. This lack of plantar flexion results in decreased shock absorption. It could be presumed that this decrease in foot plantar flexion would put more stress on the tibia, thereby increasing the chance of injury to the tibia.

- **Skeletal:** This relates to injury or stress to the bone (tibia). This is commonly referred to as ***medial tibial stress syndrome*** (MTSS). However, this term is also used to denote connective tissue/fascia related to shin pain. This typically presents with pain on the inside of the tibia.

If the pain is on the front of the tibia (tibial spine), the individual should seek a physician, as this type of injury is typically classified as more serious (420). Continuing to run with a shin splint can lead to a stress fracture (small crack in a bone).

- **Tibial Fascia (Tibial Fasciitis):** Inflammation of the deep fascia surrounding the tibia can result in shin pain (442). This is often confused with bone pain.

Hamstring Injuries

Hamstring injuries are often caused by hyperextension of the knee. The hamstrings act to eccentrically decelerate the lower leg prior to the support phase of the gait and to extend the hip during the drive phase.

Runners who heel strike and overstride versus midfoot strike are at a higher risk of having hamstring issues. This is because with a heel strike, the runner's stride is typically longer; therefore, the lower leg moves through a larger range of motion than a midfoot strike. This large range of motion requires a substantial eccentric contraction on the part of the hamstrings to decelerate the lower leg.

As the hamstrings contract eccentrically during the swing phase of the running gait, if the pelvis is tilted anteriorly when running, it will put more tension on the hamstrings. Therefore, **running with a substantial anterior pelvic tilt will likely increase the chance of hamstring injury because of the increased tension on the hamstrings during the swing phase of the gait cycle.**

Running too fast or with too much force puts substantial concentric strain on the hamstrings and can result in injury (hamstring strain).

Ankle Injuries

Runners who "twist" their ankle are often susceptible to twisting it again. A twisted ankle is also called a *rolled* or *sprained ankle.* This involves spraining one or more of the ligaments in the ankle. Research done by Gehring et al. demonstrated that the cause of lateral (inversion) ankle sprains is not just isolated to the ankle (310). Compromised movement patterns of the pelvis and femur and neuromuscular reflexes contribute to lateral ankle sprains (310).

Achilles tendinosis refers to damage to the Achilles

Popliteus Injury

The ***popliteus*** muscle is a small and relatively overlooked muscle behind the knee that profoundly affects running.

The popliteus has three primary roles:

- To “unlock” the knee (flex) from a straightened position
- Prevent knee hyperextension
- Laterally rotate the femur when the foot is planted

Injury to the popliteus can occur due to overpronation of the foot/ankle complex and knee hyperextension. When the popliteus is injured, it shortens, so full knee extension is often not possible or only with pain.

Plantar Fasciitis

Plantar fascia is a thick band of fascia that begins at the back of the heel and extends under the foot to the heads of the metatarsals.

Pain associated with plantar fasciitis is most commonly found at the beginning of the arch of the foot, closest to the heel. When running, pain most commonly occurs when the foot pushes off the ground. **Tight calf muscles often contribute to plantar fasciitis.** Additionally, weak intrinsic foot muscles may increase the load on the plantar fascia (600).

Side Stitch

While not an injury, a side stitch can be quite painful and result in the cessation of exercise. Also known as ***exercise-related transient abdominal pain*** **(ETAP)** (437), ETAP is most commonly associated with running (437).

ETAP is characterized by cramping, or sharp pain toward the bottom of the rib cage and most commonly occurs on the right side of the body.

While there are many theories as to why ETAP occurs, no one cause has been identified (437). Some of these theories are a lack of electrolytes (438), stretching of ligaments associated with the diaphragm (437), and shallow breathing (438). Another study noted that individuals who demonstrated back kyphosis (excessive flexion) when running were more apt to experience ETAP versus those without kyphosis (439). **The most common treatments are stopping the exercise, stretching, and deep breathing.**

Preventing side stitches often involves strengthening the inner-core musculature, slowly easing into increasing running mileage and intensity, increasing one's fitness level, and getting a proper warm-up.

It should be noted that some individuals experience pain in the shoulder (most often the right shoulder) when running, which is thought to be referred pain from the diaphragm via the phrenic nerve (437).

Metatarsalgia (Hot Spots)

A relatively common issue for runners is 'hot feet' or 'hot spots' on their feet – also referred to as ***Metatarsalgia***. Metatarsalgia is a generic term that refers to pain and inflammation on the ball of the foot. Despite the prevalence of this issue, there is no singular ‘fix’ as the cause is likely multifactorial. The pain from metatarsalgia can range from slightly annoying to unbearably painful.

Two issues may induce metatarsalgia:

Morton’s Neuroma

Commonly associated with pain in the middle of the ball of the foot, Morton’s neuroma is caused by the thickening of the tissue around a nerve in the ball of the foot (652). The pain is often in the form of a burning sensation.

Morton’s neuroma has been associated with wearing high-heeled shoes. As a result, wearing a shoe with a lower heel and wearing a shoe with a wider toe box may help (652).

Sesamoiditis

Under the big-toe joint exists two pea-sized bones (sesamoids). The sesamoids are embedded in a tendon to assist with the mechanical efficiency of the foot when running or walking (653).

The pain associated with *sesamoiditis* is due to irritation of the tendon in which the sesamoid bones are embedded. Rest and placing a pad on the fore/midfoot with a cutout for the big-toe joint (termed dancer’s pad) can assist in taking pressure off the sesamoids and thus enhancing recovery (653).

A bunion (Hallux Valgus)

A bunion is an abnormal bone growth that causes the big toe to slant inward toward the other toes. This slant of the toe causes the outside of the big toe joint (first metatarsophalangeal joint) to jut outward (medially). A bunion causes pain and, in severe cases, requires surgery (651).

Some evidence shows that wearing tight shoes (in the toe box area) may contribute to bunions. As a result, it is advised to run in shoes that have ample room in the toe box (651).

Blisters

Blisters are a common running injury. A blister is characterized by a bubble of skin with fluid on the inside. There are two common causes of blisters:

1. Friction between the skin and sock or the skin and the shoe
2. Wet feet due to water or excessive sweating

Blisters may or may not be painful. If your athlete is running in a race and is experiencing a painful blister, have the person seek out attention at a medical tent.

Ways to prevent blisters (292):

- Make sure feet are dry
- Wear socks that wick sweat
- Do not shave off calluses, as these act to protect the feet
- Run only in properly fitted shoes
- Do not lace up shoes too tight
- Use a small amount of skin lubricant such as *Vaseline*® or *Body Glide*® in areas that are prone to blistering

Response To Injury

Runners respond to injury in various ways. Runners likely take stock of how serious they feel an injury is and, based on that, determine the course of action. Additionally, if an injury or pain does not improve, runners will likely seek external help, such as a physical therapist or doctor.

The infographic below is from USA Running's 2017 National Runner Survey regarding how people address injury.

Running Injury Summary

While more research needs to be done on the causes and prevention of running injuries, it is important to be aware of factors that may contribute to them. It is important to keep in mind that athletes must be assessed on an individual basis. For example, your athlete may present with many of the proposed running injury risk factors but never get injured, whereas another athlete may seemingly be perfect regarding mechanics and has no significant risk for running injury – but experiences a high injury rate.

While many studies contradict each other regarding potential causes for injury, most studies agree on two significant contributing factors for running injury: high mileage and increasing mileage too fast.

Illness

Dealing with illness is unfortunately part of the training process. This section will focus on guidelines and common environmental factors that can cause illness.

Terminology

Hyponatremia: Electrolyte imbalance (i.e., sodium) where sodium levels are lower than normal. This can be a result of overhydration.

Dehydration: Occurs when the body takes in less fluid than it uses.

Hypothermia: Occurs when the body's core temperature drops below what is necessary for body function and metabolism to take place at a normal rate.

Heat Illnesses: A range of heat-related illnesses that can range from fairly benign (e.g., cramps) to heat stroke, which can be fatal.

Colds And Flu

This is a common ailment for many distance runners. Endurance exercise has been shown to decrease one's immune system. Therefore, unless athletes get enough rest/recovery, eat healthy foods, and take care of their bodies, they are likely at a greater risk of getting sick than those who do not train for endurance sports (295).

When an individual gets sick from a cold or flu, the body undergoes a physiological response to protect and help it heal. Unfortunately, the mechanisms that help to heal the body also decrease its ability to perform at peak levels. Below are several physiological responses that occur with illness (296):

1. The body breaks down muscle (catabolism)
    - Studies done on individuals with fevers showed microscopic damage to muscles.
    - Muscle strength decreases, and it can take up to two weeks to regain strength (from a three-day illness associated with a fever).
2. Fat metabolism is decreased
    - **Body is primarily utilizing muscle protein for fuel, not fat.**
3. Aerobic metabolism is diminished
    - VO_2 max and lactate threshold will decrease.

Exercise should not be done when fighting off a cold or something more serious like the flu. The focus must be on rest and recovery. Even if an athlete were to exercise when sick, there would be little to no improvement in fitness, and likely the opposite would occur.

Rest is often hard for athletes to adhere to as they view lost training days as exactly that – lost. It is your responsibility to inform them that they will not see any physiological gains when training while sick and that by doing so, they will only extend their time being “out of commission.”

Guidelines

- If an athlete is symptomatic (e.g., runny nose, elevated temperature, congestion, body ache, chills, etc.), they should not exercise.
- If an athlete thinks they might have a simple cold but is asymptomatic, it is best to either not exercise or exercise at an extremely low level for a short duration.

- If a head cold has “moved” into the chest, it is imperative not to exercise.
- Rest and recovery are the best bet until the athlete feels better and shows no signs of illness.

Summary

Injury

- In emergency situations, emergency personnel must be called (911).
- As a certified UESCA coach, we recommend that you be CPR/AED certified if working with athletes in-person.
- To mitigate the chance for injury, be attentive to the following areas:
    - Volume increases
    - Intensity-level increases
    - Monitoring fatigue level
    - Integrating cross-training
- When working with athletes with an injury, it is best to adopt a team approach with other professionals and clinicians
- Running has a high injury rate, primarily because of the impact nature of the activity.
- There is no definitive cause of side stitches.
- There are many potential origins of shin splints. If your athlete has developed a shin splint, aside from not running, the person should seek out a sports physician or physical therapist.
- If your athlete suffers a sports-related injury, the individual should seek out a specialist such as a physical therapist or a sports medicine physician.
- As a coach, when dealing with injury, you must always work within your scope of knowledge and practice.
- You cannot diagnose an athlete’s injury.
- ITBS is likely the result of compression versus friction
- Ice has been shown to delay the healing process

Illness

- If your athlete has a cold and is symptomatic, the person should not exercise.

- When sick, the body becomes catabolic and fat metabolism decreases.
- As a coach, when dealing with illness, you must always work within your scope of knowledge and practice.

Module 9: Resistance Training

For many endurance athletes, strength training is not done at all or only during the off-season. Strength training is highly beneficial from a performance standpoint, and those who do not actively incorporate it into their training program are at a significant disadvantage compared to those who do. **Runners who perform strength training only in the off-season will lose the benefits gained as the season progresses.** In this case, the saying *use it or lose it* is very applicable. People would not expect that if they did cardiovascular training for only four months, the results would last for the other eight months of the year. Despite this, many endurance athletes take this approach to strength training.

Strength training can be interpreted in many different ways. For example, in theory, going for a run constitutes strength training, as there is tension on the muscles. Others consider strength training a workout only if they lift weights in a gym. **For this certification, resistance training relates to exercises that may or may not utilize weights or strength-training equipment.** Moreover, resistance training equates to any type of resistance on muscles for performance enhancement via strength and/or power gains.

The terms ***resistance training*** and ***strength training*** are considered synonymous and thus are used interchangeably.

This module focuses primarily on resistance training theory, while the following focuses on resistance training exercises.

Why Resistance Training?

Just as it is important to understand why runners resistance train, it is equally important to understand why many runners do not do it. In this section, both of these areas will be discussed.

Our Standpoint on Resistance Training

- Resistance training is a critical aspect of a running program, specifically regarding improving running economy.
- The majority of endurance athletes have pronounced muscle imbalances, weak core (inner unit), and upper-body musculature. Therefore, resistance training is important for all runners.
- Resistance training is just as important as cardiovascular training.
- Resistance training can help to maintain lean muscle mass.

Improve Running Economy

A 2008 study of distance runners by Storen et al. found that heavy weight training over eight weeks improved the economy and time to exhaustion of the runners (no change in VO2 max or body weight) (654). This study correlates with the results of a study by Paavolainen et al. (282) that found **explosive strength training improved 5K running times by improving running economy.**

A study by Mikkola et al. prescribed three groups of recreational runners varied strength-training programs, all while concurrently running (treadmill-based). The three strength-training groups were:

1. Heavy resistance
2. Explosive resistance
3. Light to medium resistance

All of the runners improved their running performance. However, only the heavy and explosive resistance-training groups improved their neuromuscular fitness levels (655).

Do It Right or Not At All

While correct form is important during all aspects of training, it is absolutely critical for resistance training.

As a coach, you must know what constitutes correct and incorrect form and accept nothing less than proper form. Incorrect form greatly increases the chance of injury. Correct versus incorrect form is often measured in millimeters or degrees. For example, picking up a weight with a flat lumbar spine and stable core musculature is a safe movement. Add a few degrees of flexion of the lumbar spine, and it can become extremely dangerous because of shearing forces on the vertebrae. For exercises involving multiple joints and/or planes of movement, ensuring correct form becomes significantly more difficult than single-joint movements.

If you notice your athlete using incorrect form, stop them immediately and correct it. **There is no such thing as "slightly incorrect form."** It is either correct or incorrect. Allowing your athletes to proceed with bad form not only risks injury and diminishes the exercise's effectiveness, but it tells them that the incorrect form is correct since they were not told otherwise. Do not allow and reinforce incorrect movement patterns.

Time Constraints

Many runners do not strength train because of a lack of time. While training for distance running can be time-consuming, most runners look at strength training incorrectly from a time perspective. First, strength training should not be viewed as an *if I have a little extra time, I'll do it* sort of thing. Second, strength training can be done anywhere at any time. **A runner does not need a gym to do strength training.** For example, athletes could do a wall sit exercise while waiting for their eggs to cook.

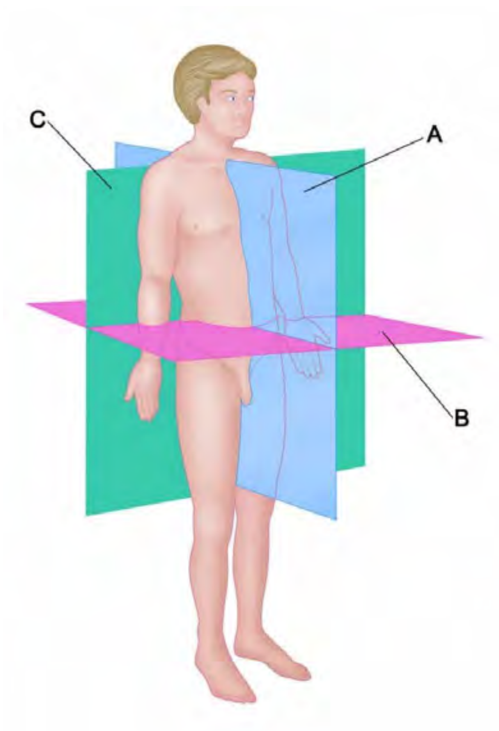
Strength training can be done throughout the day; therefore, it is unnecessary to lump it into one set time frame. The key to an effective program is being flexible and appreciating the "whole picture" when designing a program. If this means spreading ten strength-training exercises throughout the day, so be it − as long as it does not negatively affect other aspects of the training program. Integrating strength training into a running program is discussed later in this section under ***Concurrent Training***.

Movement Planes

Body movements of runners primarily occur in the sagittal plane (A) and transverse plane – hips, torso (B). If runners have not trained for movements outside the sagittal plane, when they are forced to move in other planes, the strain on the muscles, ligaments, and tendons often results in soreness at best and injury at worst.

While the body does not move much in the frontal plane (C) while running, there is a need to stress the muscles in this plane as many stabilizer muscles are strengthened this way. These stabilizer muscles help to support the primary mover muscles during sagittal plane movements. Therefore, while frontal plane movements may not mimic sport-specific movements, strengthening stabilizer muscles will help enhance overall running performance.

When running, the body expends energy to move forward, stay upright and balanced, and absorb ground reaction forces. Regarding the percent of one's body weight, 250 percent is absorbed vertically upon each foot strike, 50 percent is used to stabilize in the sagittal plane (resist falling forward/backward), and approximately 20 percent is used to stabilize in the frontal plane (resist falling over sideways) (669). This requires runners to control their bodies in multiple planes of movement.



Cardiovascular Fitness Precedes Musculoskeletal Readiness

This concept is mentioned in other sections of this certification and means that one's **cardiovascular system adapts to training**

faster than the musculoskeletal system. This is likely a big reason for the high injury rate among runners. For example, just because someone has the cardiovascular fitness to run 20 miles does not necessarily mean the person’s muscles, bones, and connective tissue are ready for this sustained impact.

By performing specific resistance-training exercises, a runner can speed up the musculoskeletal adaptation to reduce the chance of injury. While it is unlikely that even with strength training, a runner can adapt at the same speed in both cardiovascular and musculoskeletal areas, it will likely reduce the adaptation gap between the two.

Myths Surrounding Resistance Training

For this certification, it would be advantageous not to think of Arnold Schwarzenegger in the movie *Pumping Iron* when considering resistance training. While putting on muscle might be the goal for some runners, becoming a competitive bodybuilder or powerlifter is certainly not the goal. The purpose of resistance training is to go faster and get stronger, not necessarily bigger, or at least not substantially so.

One pervasive theme within the endurance sports community is that resistance training will add muscle bulk and therefore should be avoided at all costs. This common misconception is often rationalized by using two erroneous validations:

1. **Muscle bulk will slow me down**
2. **Power-to-weight ratio will decrease**

The focus is often so high on the cardiovascular facet that all other aspects of fitness go out the window. The question to these runners is that while they may be aerobic monsters, how much faster could they be if they had a stronger core or fewer muscle imbalances? By not doing strength training, any gains in cardiovascular fitness are marginalized because the athlete is missing out on a major fitness component.

Regarding misconception #1 (strength training will add muscle bulk), a 2015 study by Beattie et al. found that over 40 weeks, distance **runners who participated in a strength-training program could improve their leg strength while not adding any “unwanted” muscle mass.** It is proposed that this adaptation is primarily due to improved neuromuscular function (692).

Because strength training is either negated completely or performed in minute quantities by most runners, it will become the secret weapon to your athlete’s success. Resistance training with the right selection of exercises and proper form and progression can significantly improve a runner’s performance.

Anyone who has tried to put on a lot of muscle while maintaining an intense cardiovascular endurance-based program will tell you it is next to impossible. The myth is that if resistance training is performed, all of a sudden you will look like a bodybuilder. This could not be further from the truth. To put on substantial muscle size (*hypertrophy*), you would have to eat a considerable sum of calories each day and do specific weight-training exercises in set cycles that would not resemble an endurance-based strength-training program. Additionally, you would have to do a minimal amount of cardiovascular exercise, as this inhibits muscle hypertrophy through the burning of calories that could be going to building muscle. As you can see, this is hardly a runner’s program.

The *power-to-weight* argument is synonymous with putting on muscle bulk. By **performing strength training specific to endurance sports training, one’s power-to-weight ratio will increase.** Again, the assumption by those who feel strength training will decrease their power-to-weight ratio is they will put on substantial amounts of muscle.

Genetics do come into play to some degree when discussing muscle hypertrophy. For example, if an athlete has a large proportion of fast-twitch muscle fibers, the person will likely increase muscle mass to a greater extent than someone with predominantly slow-twitch fibers. However, assuming the individual is eating a normal amount of calories, strength training for muscular endurance, and performing a substantial amount of cardiovascular exercise, the actual hypertrophy would be minimal.

Physiology of Resistance Training

There are two primary areas that this certification discusses regarding resistance training physiology:

- **Overload Principle**
- **Neuroendocrine Response**

Overload Principle

Benefits reaped from strength training are the result of the ***overload principle***. The overload principle states that greater-than-normal stress on the body must be present for a training adaptation (108). The overload principle was developed by Thomas Delorme, M.D., who used heavy-resistance training exercises on soldiers who had just returned from WWII. Rapid rehabilitation of soldiers was necessary as there was a shortage of hospital beds then. In 1951, Dr. Delorme and Arthur Watkins published their findings in a book called *Progressive Resistance Exercises* (113).

While the overload principle originated from a resistance-training perspective, **all training adaptations are based on this principle.** Regarding strength training, the overload principle can relate to different physiological adaptations, such as muscular endurance or power.

Neuroendocrine Response

An area rarely discussed regarding strength training is the neuroendocrine response. This response refers to the interactions between the nervous and endocrine systems that occur when performing specific strength-training movements. The greatest neuroendocrine response occurs when performing exercises that recruit large muscle masses and are done at a relatively high volume with moderate to high resistance (109, 189). **This response increases testosterone and hormones (i.e., growth hormone) that stimulate muscle growth/strength.**

Testosterone is a hormone within the body responsible for muscle growth (20). Testosterone is a steroid hormone from the androgen group responsible for protein synthesis (12). Therefore, testosterone is responsible for an increase in muscle size/strength, hair growth, and bone density. In males, testosterone is secreted by the testes, and it is secreted by the ovaries in females. Additionally, the adrenal glands release a very small amount of testosterone. Adult males produce approximately seven to eight times more testosterone than females (13).

While the strength program of a runner won't contain high neuroendocrine response exercises exclusively, it is crucial to include them in a program. Exercises with a high degree of neuroendocrine response incorporate multiple large-muscle groups such as the squat, deadlift, and other Olympic-style lift exercises (20).

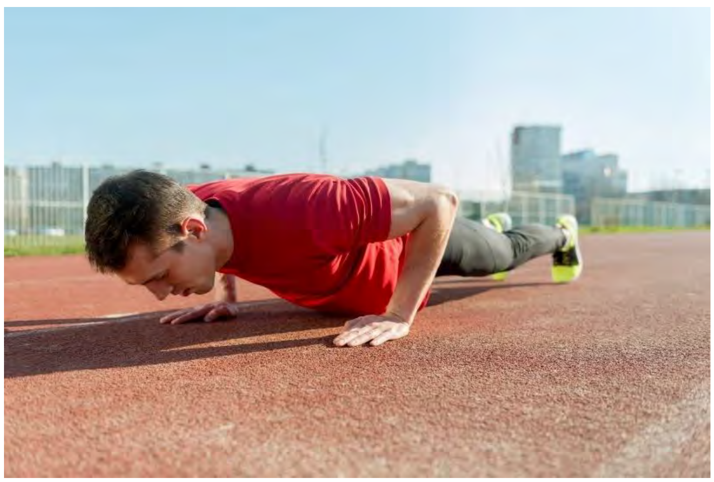
The neuroendocrine response also influences the order in which exercises should be completed. Exercises that have a high neuroendocrine response increase, for a short time, the amount of testosterone circulating in the blood. Therefore, exercises that do not elicit a high neuroendocrine response (e.g., shoulder raise) should typically be done after exercises with a large response. These smaller muscles (low neuroendocrine response) will be able to reap some of the benefits of the response because of the increase in testosterone in the bloodstream.

Muscle Hierarchy

Muscles are often categorized as *primary* and *secondary* movers:

- **Primary Mover:** is a muscle or muscle group that is primarily responsible for producing a movement
- **Secondary Mover:** (also termed “stabilizer”) is a muscle or muscle group that contributes indirectly or secondarily to produce movement. A secondary mover supports/assists the primary mover. For example, during a push-up, the pectoralis major is the primary mover, and the triceps are the secondary movers

Resistance Training Protocol



Resistance training is integrated into a training program for many different reasons. Some of these include:

1. **Corrective:** Target specific muscle(s) to correct postural issues and muscle imbalances
2. **Sport Replication:** Exercises that mimic running movements
3. **Increase Power:** Exercises are typically called *plyometric exercises*, meaning explosive-type movements
4. **Increase Strength:** While all exercises increase muscle strength, not all strength exercises fall into the just-noted categories. For example, you might prescribe a *wall sit* to an athlete to increase quadriceps strength but not necessarily to

correct a muscle imbalance, replicate a sport movement, or increase power.

Most resistance-training programs for sports and general fitness typically assign a range of 10 to 15 repetitions with a two- or three-set range. Why?

Strength-Endurance Continuum

The strength-endurance continuum is the genesis of the 10- to 15-rep, two- to three-set exercise prescription. This continuum states that **strength and endurance are inversely related** (179). Regarding resistance training, strength increases are associated with high loads and low repetitions, whereas endurance increases are associated with low loads and high repetitions. A repetition range of 10−15 repetitions seeks to capitalize on the benefits of both strength and endurance, as the repetition range falls somewhere on the spectrum between what is typically considered primarily strength and primarily endurance.

In a 2002 study by Campos et al. (180), 32 untrained men participated in an eight-week resistance training study to establish the validity of the strength-endurance continuum. The subjects were divided into four groups: High Repetitions (20-28 reps), Intermediate Repetitions (9-11 reps), Low Repetitions (3-5 reps), and the control group (no exercise).

Subjects were assessed pre/post in the areas of strength (1-RM), muscle endurance (max reps performed at 60% of 1-RM), and cardiovascular (VO2 max, time to exhaustion, pulmonary ventilation, maximum aerobic power).

Not surprisingly, the low-rep group saw the most significant strength gains, and the high-rep group realized the highest gains in muscle endurance and significant gains in max aerobic power and time to exhaustion. Interestingly, the low- and intermediate-rep groups did not realize any gain in the latter two areas.

Based on this information, repetition range is not something that should be assigned randomly or doled out the same to all athletes. **The repetition range assigned to an exercise must be purpose-driven and with a specific goal in mind.**

Low to medium repetitions (three to 11- rep range) are advised if you are looking for large strength gains. A higher-repetition range (20– 28 reps) is likely the correct route for muscular endurance.

PURPOSE	WEIGHT	REPETITIONS
Large Strength Gains	High	3 to 10
General Fitness	Low to Medium	10 to 15
Muscular Endurance	Low to Medium	20 to 28

Given the rep ranges assigned to the different groups in the study, it could be theorized that **the 10- to 15-rep range so commonly assigned to athletes falls in between true strength and endurance benefits, and thus the benefits are minimized because of a lack of specificity.** Further research is needed to determine if this hypothesis is correct.

However, for athletes new to strength training, the repetition range should be midrange with a relatively low weight. Specificity of repetition range should come into play only once the muscle and connective tissue adapt to the strength training.

- **Repetition (Rep):** One full exercise movement. Performing one repetition at maximum intensity is referred to as *one repetition maximum*, or *1-RM*.
- **Set:** A group of predefined repetitions.

Alternate Resistance Training Duration Prescription

Rate of Perceived Exertion

Another way to guide your athlete regarding intensity level is the ***rate of perceived exertion (RPE) scale***. Very easy intensity would rank a “1,” and extremely difficult (almost to muscular failure) would rank a “10.” For example, an RPE of “8” would represent a difficult effort but not to the point of muscular failure. The RPE scale is very useful in determining rep ranges, so long as you know that your athlete’s interpretation of their intensity level regarding the RPE scale is the same as yours.

For example, if you tell your athlete to exercise at a level 10, but because the individual does not have a high tolerance for intense training (i.e., pain tolerance), they stop the exercise at what appears to be a level 5 or 6, you would be better off using a set rep range.

Time

Using *time* versus *repetitions* to prescribe strength-training set durations is useful for any exercise, but especially for exercises that take a relatively long time, where counting reps would become monotonous. Using time is also helpful for isometric exercises such as a wall sit where there are no repetitions.

Areas Of Focus

While most all areas of the body could benefit from strength training regarding running performance, several areas should be focused on:

Calves

As noted in the *Running Mechanics* module, the stiffness of the calf muscles (especially the gastrocnemius) and Achilles tendon increases the performance of the ankle spring mechanism that assists with propulsion. Therefore, exercises that focus on strengthening this area will help improve an individual's running economy. Assuming there are no contraindications, ballistic exercises such as plyometrics would be beneficial. The soleus muscle is the only leg muscle that is active throughout all phases of the gait cycle, including the stance phase. Therefore, strengthening the soleus is of critical importance.

Feet

Like the ankle, the arch of the foot also acts as a spring mechanism to assist in the propulsion of a runner. The arch of the foot compresses upon foot landing and then acts as a spring to reform the arch. While one can do exercises to focus on the arch of the foot, walking around barefoot (indoors) is a great way to help strengthen it. Additionally, performing unilateral barefoot plyometrics can greatly strengthen the foot arch.

Core (LPHC)

The inner core (LPHC) provides stability to a runner. Performing exercises such as planks, front shoulder lifts, and single-arm cable rows functionally train the inner core musculature.

Gluteals

The gluteals, specifically the gluteus maximus and medius, are important for runners. The gluteus maximus is a very large and powerful muscle that generates a lot of force and therefore is largely responsible for the forward propulsion of a runner, assuming correct form is used. The gluteus medius acts to prevent knee valgus (inward movement of the knee) and stabilizes the pelvis when standing on one leg during the gait cycle.

Plyometrics

The most common type of power training is called ***plyometrics*** and refers to explosive-type movements. Almost any movement can be turned into a plyometric exercise. For example, doing push-ups where the individual pushes off as hard as possible so that the hands leave the ground would be considered a plyometric exercise. While these types of movements are fairly basic looking, from a physiological point of view, they are quite complex and therefore are considered advanced exercises.

The characteristics of a plyometric movement are a fast stretching phase (often termed the *loading phase*) of a muscle followed by a rapid contraction of the same muscle (110). **There is a direct relationship between the speed/distance of the loading phase and the resulting muscular contraction.** An example of this would be jumping off a box and then jumping back up onto another box. Landing after jumping off the box is the loading phase and jumping back up onto the next box is the explosive phase.

The primary purpose of performing plyometric movements is to train the body to recruit as many muscle fibers simultaneously within the muscles doing the work. **The recruitment of a large number of muscle fibers simultaneously increases force production over a short period compared with standard strength training.** The goal of conventional strength training is to increase the amount of work that can be done without regard to time. Running requires both long-term strength and explosive power. Therefore, both conventional strength training and plyometric training apply to running. In the most direct application, running is a series of many plyometric movements.

A 2013 study by Hudgins et al. found that standing long jumps (plyometric exercise) benefited both sprinters and middle- and long-distance runners (693). Therefore, increased power is important to long-distance runners, so plyometric training should be a part of any distance-running program.

Another study by Skovgaard et al. examined whether repeated 30-second sprints and heavy resistance training performed in succession would increase performance in distance runners. The study found that this training protocol improved distance-running performance (1,500m), running economy, and dynamic muscle strength (694).

Because of the explosive nature of plyometrics, they **should be done only by those with a solid strength-training base and who can perform a movement with proper form throughout the prescribed range of motion.** When starting a plyometric program, you should limit your athlete to just one or two plyometric exercises per session and limit the range of motion, acceleration, and duration of the exercise. The range of motion, quantity, and intensity level of plyometric exercises per session can be increased as they progress.

Make sure that the athlete warms up sufficiently and moves through the range of motion that will be done during the plyometric exercises.

Like any exercise, plyometrics can be self-injurious if done incorrectly. Common causes of injury include:

– **Not sufficiently warmed up**
– **Improper form/poor biomechanics**
– **Undiagnosed or overlooked preexisting musculoskeletal issues**
– **Progressed too quickly**

Awareness of these reasons will significantly reduce the chance of injury. Perhaps the most important aspect of avoiding injury during jumping plyometric exercises is to avoid lumbar and thoracic spinal flexion, especially during acceleration and deceleration. This will ensure that the lumbar and thoracic aspects of the spine are supported throughout the exercise.

Correlation to the Stretch-Shortening Cycle (SSC)

As noted previously, running is a plyometric exercise – meaning, running is a series of explosive movements. While excessive vertical movement during running (i.e., bounding) is inefficient, shuffling is not efficient either. Therefore, an element of explosiveness is required for a running-gait pattern to be efficient.

This relates directly to the *stretch-shortening cycle (SSC)*. To recap, the SSC is representative of an eccentric contraction of a muscle followed by a rapid concentric contraction of the same muscle. Regarding running, the SSC is most representative of the calves and Achilles tendon unit. Assuming the ankle joint has full mobility, the stiffer the calves and Achilles tendon unit is, the more efficient a runner will be because of a greater energy return.

Programming

The programming aspect of strength training is often where many coaches run into problems. The two main areas that coaches tend to have issues with are progression and integration.

Progression

Like cardiovascular training, strength training does not adhere to the 10 percent increase rule. For example, let us say your athlete started off being able to bench press 130 pounds. Using the 10-percent-per-week rule, eight weeks into the program, they would be able to bench press 253 pounds. As you can see, this is not feasible, correct, or sport-specific. Strength training typically progresses much slower than cardiovascular training and does not typically increase every week or every training session.

Muscles adapt much faster to strength training than connective tissue does. Therefore, the **progression of strength training must be very gradual** regarding time, weight, speed, and repetitions to avoid connective tissue injury.

If you are working with an athlete who has never done strength training or has not done it for a while (over one month), you must be very careful to start slowly and with a low weight. You also might introduce just a few exercises for several repetitions for the first strength-training session. Remember, this is not a personal training session where it has to take an hour and consist of all weight lifting.

You should increase the weight and repetitions (or time) in very small amounts and only after an athlete has shown the ability to perform at the current level without notable muscle soreness in the following days.

Structure

As with the cardiovascular aspect of training, an individual will benefit from strength training to a greater extent with structure than with random strength training.

The following chart illustrates how the phases of a cardiovascular-training program can resemble that of a strength-training program regarding progression.

CARDIOVASCULAR	STRENGTH
Base	General Conditioning
Development	Strength and Corrective
Peak	Power and Sport-Specific

While the progression from one phase to the next is not set in stone, the overriding principle is that for an athlete to gain strength and power safely, there must be some form of progression.

While you can emphasize a particular type of resistance training, there should not be exclusively one type performed at any given time.

As muscles and connective tissue do not rapidly increase in strength from week to week, there will likely be several weeks where the same weight and repetitions are repeated. This is both normal and acceptable. Going this route is much better than progressing an athlete too rapidly and risking injury.

From a safety standpoint, the progression of strength training is as follows:

1. **General Conditioning:** midrange repetition range (10–15 reps), two or three sets, light to medium weight
2. **Strength:** low repetition-high weight
3. **Power:** Add speed aspect to programming

The strength-training recovery phases (weeks) should coincide with the cardiovascular program's recovery phases.

While this is the general progression, your athlete may or may not progress to all these areas. For example, your athlete may stay in the general conditioning phase throughout the training program, or the individual might not progress to the power phase because of an injury. The degree of progression is completely based on the individual.

Strength Training Taper

There is not much research regarding strength-training tapering for runners. As such, the following are rough guidelines:

- For 5K, 10K, and half marathon running events, lower-body strength training should be eliminated
- For marathons, lower-body resistance training can be done during the first week of the taper but in limited amounts (based on a three- to four-week taper). After the first week of the taper, lower-body resistance training should be ceased.

The aforementioned guidelines are noted specifically regarding lower-body exercises. Core and upper-body exercises can still be done during the taper. However, it should be done in limited quantities and intensity. Individuals should not begin a strength-training program or introduce high intensity (HIIT) during a race taper period.

Muscle Specificity

It is common to associate fitness in one area with fitness in another. For example, even if your athlete is an elite-level runner who can run a four-minute mile, if the person has never done strength training, they will be quite sore the day after an initial strength-training session. Conversely, if an avid weightlifter who doesn't do cardiovascular exercise ran six miles at a high level of intensity, they would be in a world of hurt the following days.

This also holds true for exercise selection. If your athlete has been doing squats for the legs and you change to leg extensions on a machine, the person's quadriceps will likely be sore the next day.

The specific use of muscles, weight, speed of contraction, and repetitions all factor into the specificity of muscle use.

Concurrent Training

Your ability to successfully integrate resistance training into a training program is necessary to have a well-rounded program and thus a well-rounded athlete. A critical factor when integrating strength training into a cardiovascular program is not to diminish the athlete's ability to perform at the desired levels in other areas of the program. You must design a resistance-training program so that an athlete's muscles are not too sore to perform optimally when running.

How your athlete responds to strength training is individually based. These are general recommendations that should be modified on an individual basis.

- Because of muscle specificity, a sore muscle in one range of motion might not be affected in another range of motion. For example, an individual's hamstrings might be tight from performing squats the day before, but when running, the hamstrings do not bother them.
- Strength training does not need to take place in a single session but can be spread out to accommodate an athlete's personal, work, and training schedule.

As noted in the study cited at the beginning of this module (Ronnestad et al.) (287), resistance training just one day a week was shown to maintain lean muscle mass and increase power output.

Argument For and Against the Integration of Strength Training

Some research challenges the effectiveness and validity of integrating cardiovascular and resistance training. The main point of contention is a lack of muscle strength development when done in tandem with a program designed primarily around the cardiovascular aspect.

Effects of Strength Training and Running Order

A study by Doma and Deakin looked to determine if either of the following scenarios was preferable (656):

1. Strength-training session followed by running (six hours between training sessions)
2. Running followed by strength training (six hours between training sessions)

Doma and Deakin found that concerning running performance the day after the aforementioned training scenarios, the group that performed strength training before **running showed greater running impairment** (i.e., decreased running economy) versus the group that performed running followed by strength training.

Against Integration

This argument was first brought to light in 1980 by Robert Hickson and was termed ***concurrent training****.* Hickson's research on concurrent strength and endurance training found that there was an inhibition of strength development when both strength and endurance training was done concurrently (181).

Hickson's 10-week study placed subjects into three groups:

1. Endurance
2. Strength
3. Combination of the two (Strength and Endurance)

An important aspect of this study was that the strength-training component was done with as much weight as possible (80 percent of 1-RM), and the repetition range was low (five repetitions) for all exercises (squat, leg press, leg extension, leg curl) except calf raises, which was 20 repetitions.

The strength and endurance group realized significant increases in strength until week six-seven, when strength peaked, then decreased during the last two weeks.

A 1985 study by Dudley and Djamil (182) also found that an inverse relationship between strength training and endurance might exist. This study integrated high-intensity cycling intervals and isokinetic strength testing. The primary finding was that a reduction of strength development occurred during high-velocity strength training but not at low velocities.

In both the Dudley and Hickson studies, it was determined that a lack of strength gain occurred only in muscles that were used for both endurance and strength training modalities. Therefore, the researchers determined that **concurrent training affects muscles on an acute and local level rather than systemically** (181, 182).

For Integration

A 1999 study by Paavolainen et al. found that when explosive strength training was integrated with endurance training, performance in a 5K race improved compared with those who did not integrate explosive strength training (282). Paavolainen theorizes that this increase in performance was due to increased neuromuscular characteristics, which equated to improved running economy, as the subject's VO2 max levels did not improve.

In terms of running, a decrease in leg muscle activation often equates to longer foot contact with the ground. In this study, it is additionally theorized that the **neuromuscular adaptations from explosive strength training decreased ground contact time because of consistently high muscle activation** (i.e., force production). The result was faster 5K running times.

These findings were further corroborated by Mikkola et al., who found that concurrent endurance and explosive training increased anaerobic capacity without decreasing aerobic capacity (283).

Timing

Perhaps the most relevant study regarding integration of strength training was by Sporer and Wenger (2003). They looked at the effect of the duration of time between aerobic exercise and strength training (185). Sixteen male subjects were divided into two groups. One group performed high-intensity cycling intervals while the other performed a steady-state submaximal cycle test. The two groups performed an incline leg press (75 percent of 1-RM) at four, eight, and 24 hours after the cycle tests. Each strength-training assessment was done at least 72 hours after the previous assessment to ensure fatigue from the previous test was not a factor. Below are the results:

	Control	4 Hours	8 Hours	24 Hours
INCLINE LEG PRESS	48 reps	36 reps	44 reps	49 reps

Subjects also performed the bench press post-aerobic assessments with no difference in repetitions. This confirms previous studies that showed the negative effects of concurrent training are limited to muscles used for both endurance and strength applications.

Interestingly, there was no difference in the effect on strength-training repetitions between the submaximal and high-intensity interval groups.

Conclusion

Many of the studies that found concurrent training detrimental for strength gains assessed subjects with a high-weight/low-repetition range. Alternately, studies such as Sale et al.(184), which did not find strength to be compromised during concurrent training, evaluated subjects using a higher repetition/set range (15–22 reps, six-eight sets).

Therefore, it could be deduced that higher-repetitions (>15) protocols would have greater strength increases than low repetition when training concurrently.

Additionally, **none of the aforementioned studies on concurrent training implemented strength-training progression.** It could be theorized that decreases in strength, such as noted in Hickson's 1980 study, could be lessened or eliminated by establishing some form of progressive structure.

While this area of sports science warrants more research, there are some things to consider when designing a training program:

- Concurrent training may decrease strength gains as compared with strength training alone.
- Concurrent training does not appear to negatively affect endurance (aerobic training).
- Overtraining may be a factor as to why concurrent training results in decreased strength gains.
- Strength training within eight–10 hours after an aerobic-training session (utilizing the same muscles) will likely result in diminished effectiveness.
- Because of the duration of most individuals' workdays, a strength-training session could be implemented on the same day as another workout, assuming one is done in the morning

and the other in the evening with at least eight hours in between.
- Muscles used minimally during aerobic training can be strength trained the same day with minimal or no reduction in effectiveness.
- A progressive, structured strength-training program should be implemented using a concurrent training model to realize the highest strength gains.
- Strength training one or two times/week will maintain lean muscle mass and increase power output.

Summary

- Resistance training is an integral part of a running program.
- Performing resistance training only during the off-season violates the *use it or lose it* philosophy.
- There is no such thing as "almost correct form." Resistance training form is either correct or incorrect.
- Resistance training of large muscle groups stimulates the neuroendocrine response. This response increases testosterone and hormones (i.e., growth hormone) that stimulate muscle growth/strength.
    - Exercises that elicit the highest neuroendocrine response are those that involve large muscles groups (i.e., squat, dead lift).
- The genesis of any strength-training program is the ***overload principle***
- A muscle that is a *prime mover* is the primary muscle responsible for movement. A *secondary mover* is a muscle that assists the primary mover.
- Many endurance athletes do not strength train for fear of developing substantial muscle mass and, thus, lowering their power-to-weight ratio. This fear is largely unwarranted and inaccurate.
- Resistance training can functionally strengthen the LPHC.
- The ***strength-endurance continuum*** pertains to the number of repetitions in relation to the desired effect (i.e., gaining muscular endurance).
- ***Plyometrics*** are explosive movements characterized by a fast stretch followed immediately by a fast contraction.
- Resistance training programs should have a progressive structure.

- It is advised to have at least eight–10 hours between aerobic- and strength-training sessions to maximize strength gains.
- Muscles used minimally during aerobic training can be strength trained the same day with minimal or no reduction in effectiveness.
- Resistance-training programs can be used for many things other than increasing muscle mass. Following are some of these things:
  - Increase power
  - Reduce muscle imbalances
  - Replicate sport-specific movements

Module 11: Running Recovery

In this module, we will address common recovery methods such as stretching, icing, myofascial release, anti-inflammatories, and the value of quality sleep.

Running Recovery and Treatment

Don’t Fix an Injury, Manage It

The approach most runners take once injured is quite interesting. With respect to their training program, runners appreciate that there are multiple factors that lead to improvement, or lack thereof. For example, if a run goes poorly, a runner will often assess why that might have been the case. Perhaps it’s that they didn’t get enough sleep, trained too hard the day before, were overstressed from work, etc…

However, in respect to injury, runners often abandon this type of multi-factorial thinking and skip right to a treatment method, versus truly thinking about why the injury might have occurred in the first place and the various factors possibly involved. The biggest issue with this approach is that 'treating' an injury in isolation is a very myopic approach and one that will likely not have a positive and sustainable result. This is compounded by the fact that many treatments that runners utilize when injured are benign at best and at worst, may have the potential to exacerbate the injury. It seems that every few days, there is a fancy new device or 'method' that comes along promising to heal athletes of all their aches and pains. We're not saying that some of these things won't prove helpful, but they must be utilized in conjunction with the big picture and moreover, not make an injury worse.

Management

When a runner gets injured, the immediate reaction is to fix it so that they can get back to running. This is of course normal and expected. First, no one wants to be in pain and second, no one wants to have their training derailed due to an injury. For most endurance athletes, this initial response can best be described as panic. As noted above, this is likely why most runners don't take a minute to assess all of the possible reasons for injury and instead, go directly to Amazon.com to order the newest percussion therapy gun!

The unfortunate reality is that pain, for lack of a better word, is likely not just going to go away with the use of a singular 'treatment method.' **The desire of athletes to quickly fix any pain or injury to the point that it is eliminated almost immediately after an intervention (i.e., surgery, therapy device, medication) is not realistic.**

Therefore, the advised approach to responding to an injury is to pause and assess all of the possible reasons why an athlete might have become injured (ex: not enough sleep, increased training load substantially, work stress, too much time sitting at work, etc...) and using that information, devise a plan to manage the injury and more specifically, the issues that might have contributed to it.

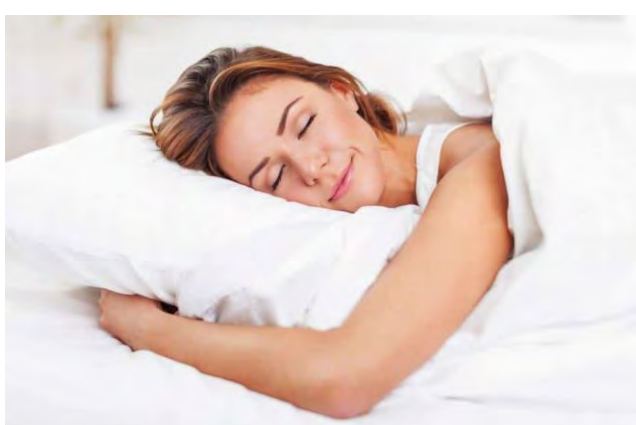
As you can see, the issue is not that a runner seeks out medical advice, or uses a product to help a specific area on the body to feel better. The issue is that these things are often utilized in isolation, without much thought, and immediately after an injury occurs. It is advised that if these and other areas are utilized, they should be done so as part of an overall injury management plan, and only after an athlete's self-assessment of the possible reasons for an injury or pain.

Summary

A 2018 paper by Lewis and O'Sullivan, published in the British Journal of Sports Medicine (816) summed up all of the above quite nicely. The paper states that due to an individual's desire to be pain-free, they often bypass the thinking stage regarding possible contributing factors to an injury and jump right to a quick fix intervention. In short, the paper states that individuals need to reframe how they think about non-traumatic musculoskeletal injury, specifically in regard to their expectations for a quick fix or cure.

If your athlete has not been running prior to the inception of the training program, you will need to be sensitive to the recovery aspect of running because of the physical stress placed on the body. As such, during a recovery period, running should not be done on back-to-back days, and if your athlete has a strain or sprain, there must be complete recovery before resuming running. Therefore, if there is any lingering pain, your athlete should hold off on resuming running. If your athlete's muscles are tight, sometimes a low-intensity run is fine. But you need to understand what your athletes mean if they say their legs are "tight." If you are unsure, err on the side of caution and pass on the run.

Sleep and Recovery



Contributor: Corrine Malcolm

Getting the right amount and right kind of sleep is critical not only for optimal sports performance but also for healthy cognitive functioning. Based on two sleep-consensus panels, the American Academy of Sleep Medicine and Sleep Research, it was found that adults need a minimum of seven hours of sleep per night (966, 967, 968). Let's dig a bit deeper...

Quality sleep is biologically critical for many of our body's cognitive, physical, and metabolic processes, including the immune system, testosterone, human growth hormone (HgH), and glucose metabolism. Quality sleep is also important for reducing the chance for injury and illness and recovering from injury or illness.

Sleep Type

There are two main types of sleep:

1. **Non-Rapid Eye Movement sleep (NREM)**
2. **Rapid Eye Movement sleep (REM)**

NREM sleep has four sleep stages and is the first type of sleep that occurs when falling asleep (969). During the third stage, HgH is released at maximum amounts, which is important for muscle recovery.

REM sleep occurs in 30-to-40 minute intervals after NREM sleep and is primarily the time when cognitive and mental recovery occurs.

Most of one's sleep at night is NREM sleep (75 – 80%), while REM sleep constitutes approximately 20 to 25%.

What Affects Sleep Patterns?

Several variables affect one's sleep patterns. Some of these are the body's natural internal clock (Circadian Rhythm), genetics, and the environment (ex: hot, cold, light, eating).

One's Circadian Rhythm is influenced by genetics and the environment (6) and is controlled by the hypothalamus in the brain. Regarding body temperature, the core temperature rises gradually throughout the day until approximately 9 pm and then starts to decrease (969). As the body's core temperature lowers, the metabolic processes also slow. External temperatures such as a hot room can negatively affect sleep patterns (969). This is often why people complain that they can't sleep in hot conditions such as summer nights without air conditioning.

Lastly, your body's natural inclination to sleep at a particular time is a genetic component referred to as one's chronotype (970). Most often associated with terms such as 'night owl or 'early riser,' from an athlete's point of view, this can either work for or against an individual. For example, if an athlete is a night owl, but because of work and family demands, they are forced to get in early morning training sessions, this goes against their chronotype and is likely not very enjoyable.

Health Issues

Many think the only downside to not getting enough sleep is that they will be tired the next day. While this is a downside, it is just one of many. Below are some of the other key downsides:

- **Decrease in immune function.** Specifically, there is a decrease in natural killer cell activity and cytokine interleukin-6 (IL-6) (968). Natural killer cells are white blood cells that fight viral infections, and IL-6 fights infections and inflammation.
- **Rise in cortisol levels**. Cortisol often gets a bad rap for being a stress hormone, but in moderation, cortisol is an important hormone for many critical physiological functions such as the breakdown of macronutrients and proper immune and nervous system function. However, when cortisol levels become too high, it can block serotonin receptors, decreasing melatonin secretion, preventing HgH release, and inhibiting adequate glycogen storage replenishment (971).
- **Decrease in cognitive function.** Based on a 2010 study by Cohen et al., 19 hours being awake resulted in the same performance decline/deficit at a blood alcohol concentration (BAC) of .05%, and 24 hours of being awake resulted in a BAC of .10%. (972)

Athlete Considerations

Aside from general health, there are several considerations to consider when working with athletes.

1. **Increased risk for injury.** Chronically getting less than eight hours of sleep per night has been associated with a 1.7 times greater chance of injury than individuals who chronically got eight or more hours of sleep per night (973). This is likely related to reduced psychomotor functioning (ex: timing, coordination) and reduced tissue and bone repair, primarily due to reduced HgH production (973).
2. **Decreased performance.** Sleep deprivation is associated with changes in insulin levels which is important for glucose metabolism. These changes can inhibit glucose synthesis (production) from glycogen stores, which can drastically affect endurance athletes as they can run out of energy (974).
3. **Take naps.** If an athlete is short on sleep, taking naps is a great way to extend one’s sleep time which correlates with increased physical performance (975).
4. **Sleep extension.** This refers to sleeping more via naps and more nighttime sleeping in the days/weeks leading up to a race. Like naps, this has been shown to increase performance on race day. (976)

Ice?

It has long been advised to ice not only injured areas, but also to ice areas on the body to prevent injury. The genesis behind this was primarily to reduce inflammation. However, it is now believed that inflammation is a critical aspect of the healing and muscle-repair process.

A 2015 study by Leeder et al. found that an ice bath did not promote recovery in subjects with DOMS (678). In fact, icing may slow the healing process. Dr. Jonathan Leeder stated that there is solid evidence to show that athletes feel better after an ice bath and this is likely why ice baths are popular and are believed to correlate with muscle healing and recovery. Despite this, there is no evidence to show that cold therapy results in increased performance and enhanced recovery (678).

While not injury-specific per se, a 2020 paper by Fuchs et al., found that cooling vis cold water immersion post-exercise during recovery from resistance-type exercise lowers the capacity of the muscle to take up dietary protein (1014). This may be one of the reasons why icing delays the healing process, as noted above.

For these reasons, runners should reconsider using ice as a means of recovery post-workout.

Peace And Love

As alluded to above, ice has long been the preferred treatment for injuries. As such, the acronym **R** (rest) **I** (ice) **C** (compression) **E** (elevation) was the recommended course of action for an injury. However, as time has progressed, we have a better understanding of what works and what doesn’t in respect to treating and managing an injury. The biggest issue with R.I.C.E is the ‘I’ – as, icing has been found to inhibit the healing process.

An April 2019 post in the British Journal of Sports Medicine by Dubois and Esculier (848) offers up two new acronyms that better suit the needs of athletes today – **Peace** and **Love**. Below is an explanation of these acronyms.

Protection: Avoid activities and movements that increase pain during the first few days after injury
Elevation: Elevate the injured limb higher than the heart as often as possible
Avoid Anti-Inflammatories: Avoid taking anti-inflammatory medication as they reduce tissue healing and avoid icing.
Compression: Use elastic bandages or taping to reduce swelling.
Education: Your body knows best. Avoid unnecessary passive treatments and medical interventions and let nature play its role.

Load: Let pain guide your gradual return to normal activities. Your body will tell you when it's safe to increase the load.
Optimism: Condition your brain for optimal recovery by being confident and positive.
Vascularization: Choose pain-free cardiovascular activities to increase blood flow to repair tissues
Exercise: Restore mobility, strength, and proprioception by adopting an active approach to recovery.

Anti-Inflammatory Drugs

Due to the pain-reducing and anti-inflammatory properties of NSAIDs, they are commonly used by those who experience DOMS and musculoskeletal injuries. However, as previously noted, non-steroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen and aspirin have been found to slow muscle recovery and muscle growth (133, 134). Acetaminophen (Tylenol®) is also commonly used to reduce pain and fever. However, it is not classified as an NSAID because of its limited anti-inflammatory response.

A 2015 study by Da Silva et al. looked at the effect of taking ibuprofen before running on male runners who had preexisting exercise-induced muscle damage (676). The study found that taking ibuprofen did not help prolong the subject's run time.

Additionally, a 2013 study by Kuster et al. found that ingesting over-the-counter pain relievers prior to a marathon resulted in increased gastrointestinal issues and muscle cramps (677). The higher the dose of painkillers, the greater the incidence of the aforementioned issues.

Based on the aforementioned studies, NSAIDs are not recommended for use as a method to reduce exercise-related soreness, regardless of the sport discipline.

Research has shown that NSAIDs actually slow healing of not just muscles, but on bones and tendons as well (133).

Myosatellite cells, or satellite cells, are found in muscles, and their job is to attach to existing muscle fibers to form new fibers. This process occurs during normal muscle growth as well as during recovery from injury. It is theorized that NSAIDs greatly reduce the spread of these cells, thus hampering muscle recovery and growth (134).

Another issue in taking any sort of pain-reducing medication is that it allows an individual to potentially train past what the person's body can physically handle because of the medication-induced reduction of pain. This can lead to greater injury down the road. However, other research has deemed that short-term use of NSAIDs would have minimal impact on the healing process (135).

In conclusion, while you cannot advise or prescribe any type of medication to athletes, you can inform them of the research in this area. While there is no definitive, agreed-upon effect of NSAIDs on muscle recovery, it would be wise to share this information with athletes as the abundance of research to support the negative effect of NSAIDs on muscle recovery is too great to ignore.

Starting Back Up

If your athlete has access to a pool, water jogging is a common cross-training practice that stimulates the same muscles in more or less the same range of motion as running (outdoors), although with substantially reduced impact on the body. Additionally, a specialized treadmill (Alter G®) allows individuals to run at varying percentages of their body weight. Cycling out of the saddle also positions the body similarly to running.

If your athlete is new to running, incorporating run/walk days into the program is often beneficial. When returning from an injury, slow and proper progression is of the utmost importance. For example, your athlete might run for one minute and then walk for two minutes, then repeat this cycle four more times. The primary purpose of run/walk programs is not for cardiovascular conditioning, but rather adapting the body slowly to the musculoskeletal demands of running.

Stretching

The primary reason for stretching is to increase range of motion at the joints. The purported benefits of stretching – or lack thereof – have recently become hot topics for debate within the realms of fitness and sports performance. The focus of these debates includes:

- Does pre-stretching (stretching before exercise) increase, decrease, or have no effect on athletic performance?
- Does stretching prevent injury?
- Do individuals need to warm up before stretching?
- How long should a stretch be held, and how often should stretching be done?

The generally accepted principle behind stretching is that when a muscle is stretched, muscle neuron activity decreases, thereby relaxing the muscle, which allows for greater ROM because of reduced resistance. For this reason, **it is theorized that pre-stretching a muscle decreases the force production of a muscle while running.**

In contrast, a 2013 study found that utilizing **a foam roller before exercise can increase range of motion without decreasing force production** (516).

Physiology

To understand how the principle of stretching works, it is important to define two proprioceptive parts of a muscle, the **G*olgi tendon organ,*** and the ***muscle spindle***.

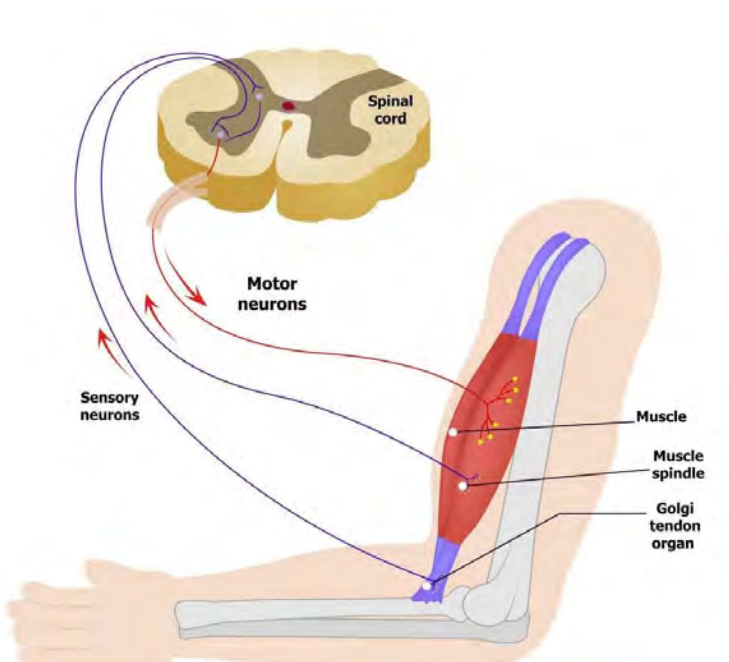
Golgi Tendon Organ (GTO)

The GTO is a nerve receptor located at the junction of a muscle and tendon. During a stretch, the GTO senses tension in the muscle/tendon and sends a signal via afferent neurons to the spinal cord. The spinal cord then sends a signal back to the muscle to relax via efferent neurons. This is called the ***Golgi tendon reflex*.** The GTO exists to regulate muscle tension so that injury does not occur to a muscle and its associated connective tissue (365).

Muscle Spindle

Muscle spindles are located deep within a muscle. Muscle spindles sense change in muscle length and the range of change in length and send a signal to the spine. This signal triggers the stretch reflex (also called *myotatic reflex*). The stretch reflex resists the increase in muscle length by causing a muscle to contract (365). The more rapidly muscle length increases, the stronger the stretch reflex is. Therefore, the primary function of muscle spindles is to protect the body from injury due to overstretching.

The goal of stretching is to minimize the stretch reflex so a muscle can be lengthened.

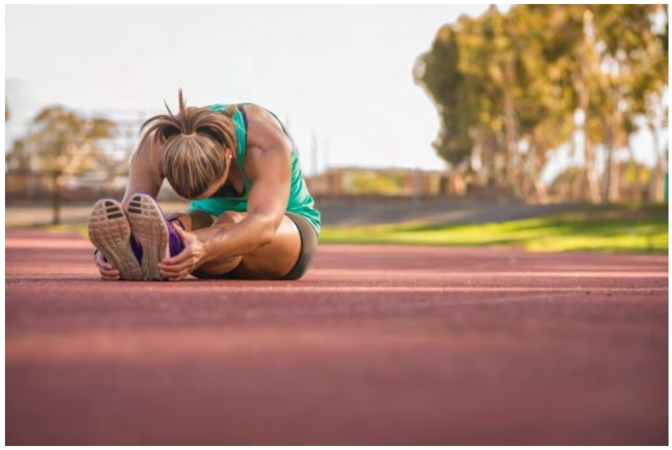


Elastic Response

As noted in the *Running Mechanics* module, **the stiffer the *muscle-tendon unit* is, the greater the elastic response**. In other words, the greater the stiffness of the *muscle-tendon unit*, the greater the potential energy source is. However, if an individual’s flexibility and range of motion do not match the degree of flexibility required for a particular movement, performance will decrease, and the potential for injury will increase. Therefore, the amount of flexibility and range of motion in a particular area should match the amount of flexibility required for the activity being participated in. **Substantially more or less flexibility than is needed will negatively impact an individual.** In particular, too much range of motion often equates to a lack of neuromuscular control for a given area and a decrease in force production (608).

Much of the research about stretching and performance enhancement has been based on different parameters, and thus the results of some of the studies contradict the results of others. For example, some studies show that pre-stretching a muscle prior to exercise does not hamper performance, while others show that it does due to decreased force production (111, 241, 242). It should be noted that many of the studies citing a decrease in force production are examining maximal force production, which is not typical for distance runners. Therefore, it could be theorized that pre-stretching is not as much of an issue for endurance sports as it would be for sports that predominantly require short, maximal efforts (ex: sprinting).

Types of Stretches



There are four primary types of stretches:

Static

An individual performs and holds a stretch at the end range of motion (ROM).

- **Active:** individuals perform the stretch themselves
- **Passive:** an external force (i.e., another person) acts upon an individual's body part being stretched

When holding a stretch for a relatively long period (> 40 seconds), the muscle spindles become accustomed to the new increase in length and reduce the signals sent to the spinal cord to tell the muscles to contract.

Dynamic

An individual moves a body part at a controlled speed through a range of motion that mimics the motions performed while running.

Ballistic

Similar to the dynamic stretch, but the movement is done with momentum to stretch past the current ROM limit. This is often considered a dangerous stretch and is not recommended for use with clients. Because of the speed with which a proper ballistic stretch is executed, the stretch reflex is triggered and therefore significantly limits the ability of a muscle to stretch.

Proprioceptive Neuromuscular Facilitation (PNF)

A PNF stretch consists of a passive stretch followed by an isometric muscle contraction. A partner passively moves a body part to its ROM limit, and then the client isometrically contracts the muscle being stretched for 10–15 seconds before relaxing the muscle briefly. The partner then passively moves the body part to a new ROM. This cycle is repeated two or three times in total.

PNF stretching works because of its effect on the *Golgi tendon organ*. During PNF stretching, a muscle is repeatedly contracted and relaxed, which triggers the Golgi tendon reflex numerous times. This expands the ROM of the targeted body part beyond what non-PNF stretches can accomplish (114).

Area of Caution

Overstretching a muscle and/or connective tissue can cause injury. **No matter how skilled you are or how long you have been stretching athletes, it is impossible to be sure of the integrity of a muscle and its connective tissue.** Many coaches and fitness professionals "*go by feel*" when stretching, which contributes to the injury rate associated with assisted stretching. Just because you feel substantial resistance during a stretch does not necessarily mean it is the end ROM; conversely, you could have passed the end ROM and caused an injury. When performing passive stretching or PNF stretching, you must consistently solicit verbal feedback about the muscle tension your athlete is feeling throughout the stretch.

Let your athletes dictate what the end ROM is, not you. By pushing past what your athletes communicate is the end ROM or by trying to guess the end ROM yourself, you risk injuring them. Before starting the stretch, let your athletes know what stretch you will be doing and why. Also, tell them they will need to tell you when they feel the muscle has been stretched to the end ROM. Whether or not at the actual end ROM, you must respect your athletes' feedback and stop when they tell you to.

If an athlete has an injury or a suspected injury such as a strain, the individual should not stretch or perform myofascial release on the injured muscle, as this may worsen the injury. With any injury or suspected injury, the first course of action should always be to seek out the advice of a specialist.

The safest type of stretch for your athletes to perform is static, active stretching (self-stretch). This is beneficial because they can do this on their own. While athletes could injure themselves while stretching during a static, active stretch, the likelihood is minimal.

If your athlete is experiencing muscle pain greater than normal or the result of ***delayed onset muscle soreness*** **(DOMS)**, do not perform or prescribe any stretches or myofascial release. Refer them to a specialist such as a physical therapist. Additionally, if an athlete has a postural issue that you feel could impact performance or safety, it is strongly advised to refer the person to a specialist.

Primary Stretches

The stretches that follow are all self-stretches. Instruct your athlete to hold each stretch for a minimum of 30–40 seconds.

Upper Body

Pectoralis Major

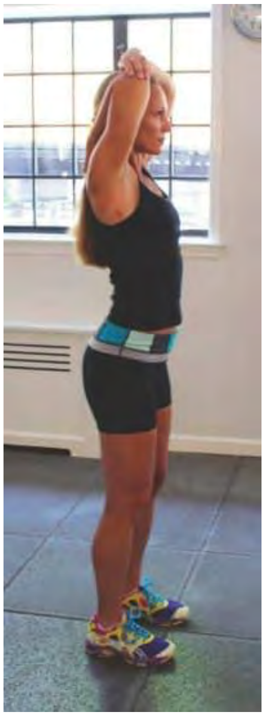
- Keep a slight bend in the elbow

Latissimus Dorsi

- Make sure hands are on top of each other

Rhomboids / Lower Trapezius

- Allow the shoulder blades to rotate forward (the upper back will flex)
- Do not overstretch; rather, hold the stretch as soon as it is felt.



Triceps

- Gently pull the elbow toward the midline of the body.

Levator Scapulae

- Tilt head to the side and slightly forward

Biceps

- Keep palms facing upward
- Keep torso as erect as possible while elevating arms behind the body

Rotator Cuff Muscles

Supraspinatus

Pull arm behind body. Keep scapula depressed.

Infraspinatus / Teres Minor

Slowly rotate the arms forward so the elbows are pointed as forward as possible in the sagittal plane.

Subscapularis

Slowly lower the forearm (as if you’re losing at arm wrestling) laterally in the frontal plane. With the elbow on a stable surface, you can also use the opposite arm to facilitate the stretch.

Lower the arm until slight tension is felt. Repeat with the other arm.

Lower Body

Hamstrings

This can be modified by propping one leg up on an elevated object such as a chair and bending forward at the hips. Additionally, this stretch can be done seated.

Gluteus Maximus

Pull the leg toward the midline of the body.

Piriformis / Gluteus Medius

Keep the tibia of the leg crossed as close to horizontal as possible.

Bend forward at the hips.

Gastrocnemius

Stretch slowly to avoid overstretching the Achilles tendon.

Soleus

Maintain a 90-degree knee bend during the stretch.

Stretch slowly to avoid over-stretching the Achilles tendon.

Hip Adductors

Keep pelvis horizontal during the stretch.

Hip Flexors (Iliopsoas)

Keep the upper body as vertical as possible with the arms overhead.

By performing this stretch with the arms overhead, the pectoralis major and latissimus dorsi are also stretched.

Obliques / Quadratus Lumborum

Keep knees locked out. Shift hips to the opposite side of the upper-body lean.

Rectus Abdominis

This stretch places the lumbar spine in extension.

DO NOT perform this stretch if the lumbar extension is contraindicated

Thoracic Spine Musculature

Rotate through the thoracic spine, not the lumbar spine.

Start

Finish

Myofascial Release

Myofascial release (MFR) is a term often used in sports and fitness. It is commonly used to describe a process in which tight muscles have compression applied to them to “release” tightness, thus restoring proper function (115). To understand exactly what MFR is, let us examine the name. Myo is a prefix that means muscle (333), and fascial relates directly to the fascia of the body, as described in an earlier module. Therefore, **the term myofascial release pertains to the reduction of tightness and restrictiveness of muscles and fascia.**

Fascia and its role in supporting and mobilizing the body have become known only in the last five or six years (330). Consequently, many coaches and health professionals do not fully understand fascia’s role in the body. Therefore, when MFR is noted and/or performed, it is usually done from the standpoint of reducing muscle tightness while overlooking the fascia element.

Fascia is a web-like connective tissue that surrounds all organs, muscles, nerves, blood vessels, and bones in the body. **Muscles do not function in isolation of the fascia** (330). If a muscle is too tight, the fascia surrounding that muscle and likely fascia in other areas will also be restricted. It is important to reiterate that fascia can also function independently of the muscles (270).

When athletes use the term **myofascial release,** they typically refer to it in the context of either massage (self or performed by a specialist) or utilizing a tool such as a foam roller or massage stick.

Maintain Properly Functioning Fascia

The following information by Julia Lucas provides valuable insight on how to properly care for the body and, more specifically, the fascia (330).

- **Let the body heal if coming back from injury – don't rush back into training:** If one's body mechanics are altered due to an injury, the fascia will adapt to new movement (incorrect) patterns.
- **Stretch gently:** Hold stretches for three to five minutes and do not force a stretch.
- **Stay hydrated:** Proper hydration levels are necessary for the fascia to move unrestricted.
- **Move around:** Adhesions can form between facial surfaces that are stationary for long periods. These adhesions can alter correct body mechanics.
- **Foam Roller:** Described below

Fascia Specialists

If you or your athletes feel they are not seeing the desired results with their stretching/MFR program, they should seek a fascia specialist. Specialists use various methods (i.e., stretching, hands-on manual bodywork) to work on fascia. Professionals such as massage therapists, chiropractors, physical therapists, and osteopaths are often trained in myofascial therapy.

Foam Roller

The foam roller is the most common and popular tool used to facilitate MFR. A foam roller is essentially a semi-hard, round tube of Styrofoam™. Foam rollers come in varying degrees of hardness as well as lengths. The concept behind a foam roller is that it “rolls” over a tight muscle and/or connective tissue to reduce muscle/connective tissue tightness. While foam rollers can be beneficial, they can make a tight muscle or fascia tighter.

The widely accepted theory surrounding foam rollers is that the more it hurts, the better it works. This could not be further from the truth. Too **much compression overstimulates the muscle spindles** (i.e., the nervous system) and causes the muscles to shorten in response as a protective measure against injury (274). Additionally, too much compression can damage muscles and connective tissue on a cellular level (fibroblasts), increasing inflammation rather than reducing it. Individuals typically apply much more pressure than is necessary and, frequently, in the wrong locations (274). Just because pressure is applied to a tender area via a foam roller or by other means, the pain and tightness will not necessarily go away. In some cases, and as noted above, it can actually make it worse, as excessive pressure from the roller can create trauma.

When fascia is irritated, it becomes inflamed. **Inflammation leads to the fascia shortening and becoming extremely tight in a localized area** (115). Reducing fascial tightness requires much lighter pressure in a specific area and for more extended periods than is typically utilized with standard foam-rolling protocol.

It is suggested that if your athletes utilize a foam roller or handheld massage stick, they apply light to medium amounts of pressure. There should be no pain due to too much pressure being applied. Therefore, in relation to working with fascia, the phrase pain, no gain is applicable.

Due to the size of foam rollers, they often do not compress just the desired muscle/fascia but ancillary muscles and connective tissue as well. When dealing with restricted fascia, it is advised to use a well-worn roller that is softer than a new one.

As noted previously, if you believe your athlete’s discomfort is the result of a sprain or strain, it is advised not to perform stretching or use a foam roller. In this situation, it is recommended to seek the attention of a specialist such as a physical therapist.

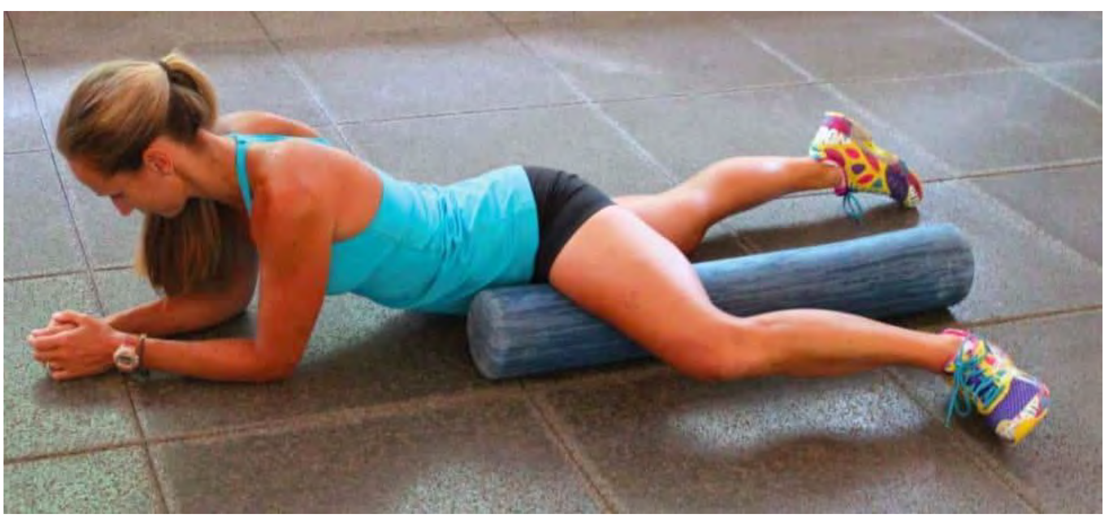
Foam Roller Applications
Lower Body

Hamstrings

Gluteals

Quadriceps

Tibialis Anterior (Shin)



Hip Adductors

Iliotibial Band (It Band)

Calves

Upper Body

Latissimus Dorsi

Low Back (Erector Spinae / Quadratus Lumborum)

Upper Back

Percussion Tools

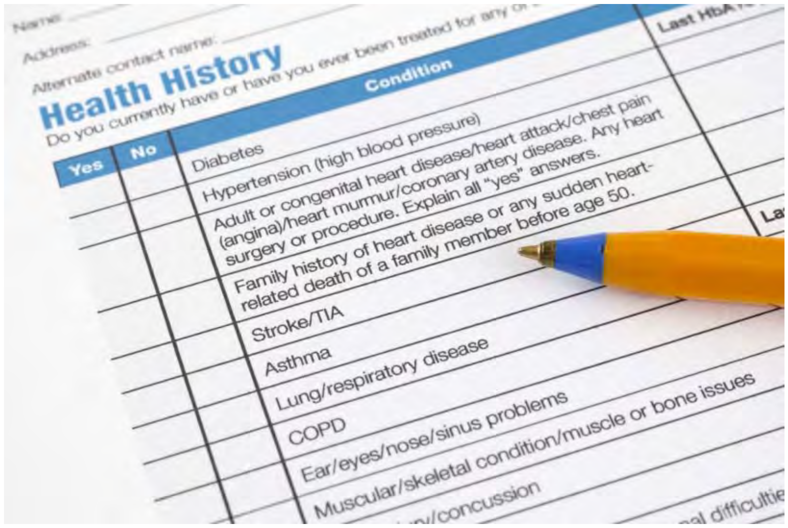
In the last few years, percussion tools, often termed, ‘massage guns’ have become quite popular amongst endurance athletes and fitness enthusiasts. The original massage gun was developed in 2007 by Dr. Werseland DC (2025). Since that time, many other massage guns have come into existence. Massage guns are essentially handheld power tools that generate extremely fast pulses (2,000 – 26,000 pulses per minute) to provide a vibration/massage effect to muscles. There are multiple reasons why athletes use massage guns such as reduction in pain, increased range of motion, recovery, etc.

Two research studies noted that the primary benefits of percussion therapy are increased range of motion and improved muscle recovery (2023, 2024). With respect to both studies, percussion therapy and traditional massage therapy demonstrated similar results. Additionally, another study by Imtiyaz et al., found that both massage therapy and percussion therapy helped in the prevention of delayed onset muscle soreness (2026).

Summary

- The *Golgi tendon organ* (GTO) and *muscle spindles* work together to influence the amount of stretch placed on a muscle.
- There are four types of stretches:
    - Static
    - Dynamic
    - Ballistic
    - Proprioceptive neuromuscular facilitation (PNF)
- Be careful not to overstretch a muscle.
- The safest way for athletes to stretch is for them to stretch themselves.
- When stretching statically, a stretch should be held for at least 40 seconds to allow the muscle spindles to reduce their signaling, thus relaxing the muscle and allowing for greater range of motion.
- A muscle-tendon unit that has too much flexibility may result in decreased force production.
- *Myofascial release* is a term that typically refers to compression of muscles for the purposes of reducing tightness.
- When fascia is irritated, it becomes inflamed and thus shortens, causing muscle tightness and pain.
- Fascia can be damaged when too much pressure is applied to it.
- Foam rolling prior to exercise has been shown to increase range of motion without a decrease in force production.
- Percussion tools such as massage guns have been shown to improve range of motion and recovery.

Module 12: Athlete Intake



The athlete intake is the most critical element of the training process, as it determines if an athlete is healthy enough to participate in a running program. Additionally, it establishes a starting benchmark regarding various training elements.

Upon completion of this module, you should have an understanding of the following areas:

- Elements of the athlete screening
- Anthropometric assessments
    - Girth measurements
    - Body composition
- Why BMI and BIA are not advised for assessments
- Athlete profile
- Athlete variables to consider
- Working with special populations

Athlete Screening

All athletes must be considered apparently healthy before receiving coaching. This is accomplished by using the HHQ and PARQ, as detailed below. The A*thlete Profile* will help you better understand your athlete to ensure the most efficient coach/athlete relationship.

Athlete Health Screening

Before you coach any athlete, two forms will need to be filled out:

– **Physical Activity Readiness Questionnaire (PARQ)**

– **Health History Questionnaire (HHQ)**

The gold standard for the PARQ is the *American College of Sports Medicine (ACSM),* which is reprinted from the *Canadian Society for Exercise Physiology* (497). As laws vary from state to state, **it is advised to seek legal advice before coaching individuals utilizing the PARQ and HHQ.**

Physical Activity Readiness Questionnaire (PARQ)

Based on ACSM protocol, if an athlete answers **yes** to any PARQ questions, the person must obtain written consent from a physician providing clearance to work out before coaching can begin. However, based on UESCA protocol, **any athlete, regardless of PARQ or HHQ results, *must* have been cleared by a physician in the last 12 months before beginning a running program.** The questions on the PARQ are (497):

- Has your doctor ever said you have a heart condition and recommended medically supervised physical activity?
- Do you have chest pain brought on by physical activity?
- Have you developed chest pain within the last month?
- Do you lose consciousness or fall over as a result of dizziness?
- Do you have a bone or joint problem that could be aggravated by physical activity?
- Has a doctor ever recommended medication for high blood pressure or a heart condition?
- Are you aware of any reason your doctor would advise you against physical activity without medical supervision?

Even if cleared, a physician may suggest specific guidelines such as keeping an athlete's heart rate below a certain limit or avoiding particular exercises. Regardless of your opinion of what a physician says, **the doctor's "prescription" for your athlete supersedes everything and must be followed precisely as intended**.

When administering the PARQ, it is advised to ask athletes the questions versus having them fill in the answers themselves. If you work remotely with an athlete, schedule a call to go over the PARQ instead of emailing the form.

Health History Questionnaire (HHQ)

The HHQ is important from both a liability and performance aspect. For example, past or present injuries of an athlete can affect biomechanical efficiency and overall performance. Medications that your athlete takes can also affect performance. This is especially true regarding the cardiovascular system. Common types of medications prescribed that affect heart rate are:

- Antidepressants
- Beta-Blockers
- Vasodilators
- Bronchodilators
- Calcium channel blockers
- Nitrates

If your athlete does not know what type or classification a drug is, it is the athlete's responsibility to contact a physician to understand its effects on the body. **Additionally, it is your responsibility to be aware of a drug's effects on an athlete and modify the program accordingly**. For example, beta-blockers are used for chest pain and high blood pressure and can lower one's heart rate. After a physician has cleared an athlete to exercise, you must modify the program to consider any effect medication has on the body (i.e., lowered heart rate). For athletes with asthma, it is strongly recommended that they always have a fast-acting inhaler on their person in the event of an acute asthma attack.

While there are various HHQ forms and questions, below are common questions found on HHQ forms:

1 – Current Height:
2 – Current Weight:
3 – Age:
4 – Sex:
5 – What is your resting heart rate upon waking:
6 – Do you take any drugs or medications?
7 – Date of your last medical physical:

8 – Do you have a family history of heart disease?
9 – Do you have emphysema?
10 – Are you, or have you been, recently pregnant?
11 – Do you have high cholesterol? If so, are you under the care of your physician for it?
12 – Have you had surgery in the past year?
13 – Have you ever had an injury that caused you to stop exercising for more than a week?
14 – Are you, or have you ever been, anorexic or bulimic?
15 – Are there any other physical or emotional problems affecting your training?
16 – Do you drink alcohol? If so, how much per week?
17 – Do you have any metabolic diseases such as diabetes, hyperthyroidism, etc.?
18 – Do you, or have you ever, smoked regularly?
19 – Do you currently or have you had any of the following conditions?

- Heart Attack
- Heart Disease
- Heart Surgery
- Heart Murmur
- Hypertension
- Thyroid Problems
- Asthma
- Epilepsy
- Anemia
- Stress Fracture

It is UESCA protocol for your athletes to be cleared by their physician within the last 12 months to train and participate in a coached running program. However, suppose your athlete answered yes to any of the questions 8, 9, 10, 11, 12, 13, 14, 15, 18, and 19 in the preceding list. In that case, it is advised to have them revisit a doctor to ensure that participation in a running program is medically approved.

Cardiac Screening

It is advised that your athlete receive a cardiac assessment (EKG stress test) within one year before beginning the running program. A successful stress test does not rule out the chance of having a cardiac event, nor does it identify all pre-existing cardiac issues.

However, this step ensures that the athlete has done everything to ensure that they are ‘apparently healthy’ before beginning a training program.

Keeping It Legal

As noted previously, it is strongly advised that the PARQ and HHQ questions noted in this section are not to be used *“as is”* without consulting legal counsel.

Anthropometric Assessments

Anthropometry relates to the measurements and proportions of the human body. Next, the anthropometric measurements represent the most common and relevant measurements relative to fitness and sports performance. The true value of anthropometric measurements is gauged by correlating initial assessments and reassessments of tests.

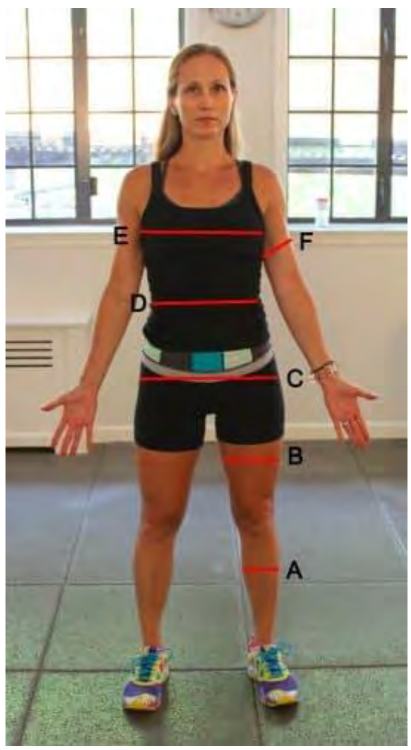
Anthropometric assessments are commonly performed at the beginning, end, and throughout a training program. These assessments are most commonly done for athletes who focus on weight loss.

The opportunity for error is significantly reduced by having the same person perform the initial assessments and reassessments.

Girth Measurements

Girth measurements are circumference measurements taken at various locations on the body. **The two most important factors in obtaining accurate measurements are the location and tension of the tape measure.** When retesting, the exact location and tape tension must be replicated. While it is always best to perform reassessment by the same individual who took the initial assessment, the testing apparatus plays a role in accuracy.
The *Gulick tape measure* is made for the purpose of girth measurements. It has a small spring at the end of the tape that expands after the tape becomes tight. Upon the initial stretch of the spring, the measurement is taken. This ensures that the correct amount of tension is on the tape, as it does not allow for overtightening.

While the locations differ based on what areas you are looking to assess, the most common areas are:



A: Calves: Measure at the widest part of the calves.

B: Thigh: Measure the middle section of the thigh. To get an accurate site measurement, have an athlete lift their leg until their femur is parallel to the ground. Then, place a finger at the hip angle formed by the *inguinal crease.* The midpoint between the inguinal crease and the top of the patella (kneecap) is the location of the testing site.

C: Hips: Measure at the widest part of the hips and take the measurement from the side of your athlete.

D: Abdomen: Measure level with the *navel* (belly button). As with the chest measurement, if you are a male assessing a female, have the athlete adjust the tape in front and take the measurement on the side or behind them.

E: Chest: Measure level with the nipples while ensuring the tape is horizontal to the ground. If you are a male assessing a female, have them adjust the tape in front and take the measurement from behind.

F: Arm: Measure at the widest part of the biceps (unflexed).

Body Composition

The two most prevalent methods for assessing body composition are:

– Skinfold Calipers
– Bioelectrical Impedance

Skinfold Calipers

As the name suggests, skinfold calipers take measurements of folds of skin. There are many different locations on the body to measure based on the particular formula or equation being used. The measurements are assessed in millimeter increments. Skinfold calipers are typically spring-loaded with a preset amount of tension. The spring tension is important because if the tension is too high or too low, the compression on the skinfolds will be inaccurate, and the percent of body fat will be wrong.

Excessive body fat and tester error can adversely affect the accuracy of the test. The more body fat an individual has, the higher the percent error will typically be.

If the tester does not administer the test correctly and places the calipers in the wrong location or pinches the skin with an incorrect amount of pressure, the test result will be inaccurate. UESCA references the *Jackson-Pollock 3-Site Test* (498).

Administering The Assessment

You will need to know your athlete's age and body weight for this test. After you get these pieces of information, you will want to inform your athlete that the measurements will pinch slightly and that you will assess each site three times and take the average. Perform this test in a private setting to make the individual feel more comfortable.

Pinch the fat and skin in the designated area and place the calipers right next to your pinch. Once you have placed the calipers in the right place, allow the calipers to "settle in," which takes about two or three seconds. Note the measurement, then open the calipers and remove them from the site that was tested. Pulling the calipers off the site without opening them will pinch the individual being tested.

Before measuring a site, instruct your athlete not to tighten the muscle at the site being tested. When performing a pinch, you want to make sure you pinch only the skin and fat, not muscle. There are two ways to do this:

1. After making the pinch and holding it, gently pull away from the surface of the body part being assessed. If a muscle is pinched, it will likely “slip” out of the pinch, leaving only the skin and fat.
2. After you have pinched the site and before you apply the calipers, ask the individual to tighten the muscle in the area being tested. If you feel the area you pinched tighten or slide out from the pinch, you pinched muscle.

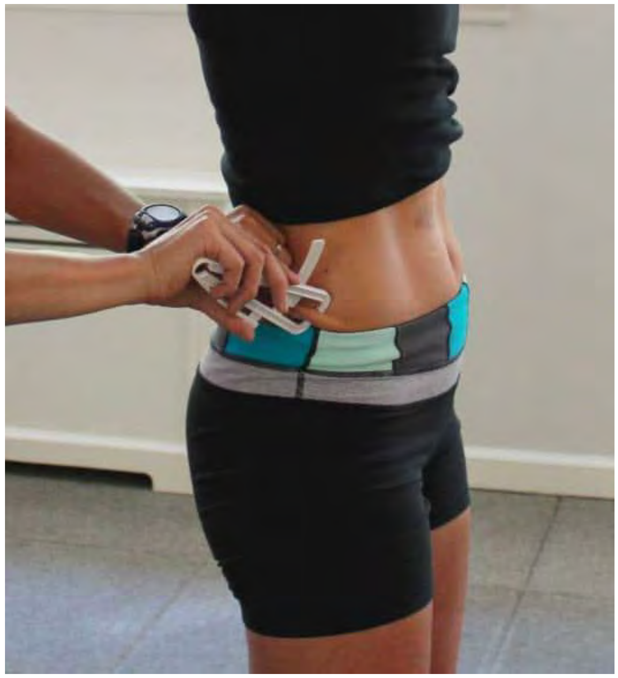
You should perform the pinch using your thumb and side of the forefinger (pointer finger). If you have long fingernails, use the sides of both fingers, so you do not scratch the subject.

Because of the tautness of the skin, the thigh measurement is often the most difficult one to perform accurately. To make this site easier to measure, have your athlete shift their weight onto the leg not being measured. The leg being measured should have a slight bend in the knee with all the body weight on the opposite leg.

All sites are on the right side of the body.

Female Measurement Sites

Tricep: (vertical pinch) In the middle of the upper arm, midway between the elbow and the top of the shoulder (*acromion process*).



Suprailiac Crest: (diagonal pinch) Have the individual find the top of the iliac crest by moving the thumb down the right side (mid-axillary *line*) of the body until hitting the top of the pelvic bone. This area is called the *iliac crest*. Once they have located the iliac crest, diagonally pinch the skin/fat directly in front of it.

Thigh: (vertical pinch) Same location as the thigh girth measurement noted previously, midway between the *inguinal crease* and the top of the *patella*.

Male Measurement Sites

Chest: (diagonal pinch) Halfway between the nipple and armpit (*axillary fold*).

Thigh: (vertical pinch) Same location as the female caliper site measurement noted above.

Abdomen: (vertical pinch) Approximately one inch laterally to the right of the belly button (*umbilicus*).

Body Fat Equation

Body density must first be determined to get one's body fat percentage. The Jackson-Pollock formula determines body density (85, 86). Once body density is determined, using a formula developed by W.E. Siri, one's body fat percentage can be ascertained (84). While the body-density equation differs between male and female, the body-fat equation (Siri) is the same for both genders.

Because of the labor-intensive nature of the body-density equations, use the following link (body-composition calculator):

http://www.linear-software.com/online.html

Following are categorizations of body composition, based on ACSM protocol, related to one's fitness level.

Source: *ACSM's Health-Related Physical Fitness Assessment Manual, 2nd Ed. (2008): pg. 59*

MALE

Percentile	Fitness Category	20-29	30-39	40-49	50-59	60 +
90	Well Above Average	7.1 - 11.7	11.3 - 15.8	13.6 - 18	15.3 - 19.7	15.3 - 20.7
70	Above Average	11.8 - 15.8	15.9 - 18.9	18.1 - 21	19.8 - 22.6	20.8 - 23.4
50	Average	15.9 - 19.4	10 - 22.2	21.1 - 24	22.7 - 25.6	23.5 - 26.6
30	Below Average	19.5 - 25.8	22.3 - 27.2	24.1 - 28.8	25.7 - 30.2	26.7 - 31.1
10	Well Below Average	25.9 +	27.3 +	28.9 +	30.3 +	31.2 +

FEMALE

Percentile	Fitness Category	20-29	30-39	40-49	50-59	60 +
90	Well Above Average	14.5 - 18.9	15.5 - 19.9	18.5 - 23.4	21.6 - 26.5	21.1 - 27.4
70	Above Average	19 - 22	20 - 23	23.5 - 26.3	26.6 - 30	27.5 - 30.8
50	Average	22.1 - 25.3	23.1 - 26.9	26.4 - 30	30.1 - 33.4	30.9 - 34.2
30	Below Average	25.4 - 32	27 - 32.7	30.1 - 34.9	33.5 - 37.8	34.3 - 39.2
10	Well Below Average	32.1 +	32.8 +	35 +	37.9 +	39.3 +

Bioelectrical Impedance Assessment (BIA)

The BIA is typically assessed using a handheld unit or scale. During the BIA, a very low electrical current is sent through the body. This current is facilitated by fat-free tissue (i.e., muscle and intercellular water). Fat impedes the electrical current, as there is little water in fat cells. The degree to which the electrical current is impeded is determined and converted to determine body density. That is then converted to percent body fat. These calculations are all done by the BIA unit.

The percent error with BIA is typically associated with one's hydration level. If an individual is dehydrated, the result will likely be inaccurate and show higher body fat levels than the person (87). Since hydration levels skew the results, it is advisable not to perform the BIA test in the morning or after exercise sessions, as your athlete will likely be dehydrated. Reassessments using the BIA should be done at the same time of day, with the athlete ingesting approximately the same amount of fluids before the assessment. However, this testing method is not advised because of the possibility of percent error.

Weight

Determining your athlete's weight as an independent variable is limited in practicality. By correlating weight and body fat, you will be able to determine what is occurring with body composition. For example, if an athlete's weight stays the same but body fat decreases, you can deduce that the person gained lean muscle. Many people associate weight as a function of health and fitness. However, it is your responsibility as a coach to educate your athletes that it is only one factor and that it needs to be correlated with percent body fat to have real value.

Body Mass Index (BMI)

BMI is calculated using a formula that determines if an individual is underweight, average weight, or overweight. This is determined by comparing an individual's height and weight. **While BMI might be a good assessment tool for the general population, it is not an accurate tool for an athlete, as it does not consider body fat or muscle** (88). Therefore, someone with a lot of muscle and low body fat would likely be considered "overweight" based solely on

BMI. While **UESCA does not use BMI as an assessment tool for these reasons,** it is important to be aware of what it constitutes and its shortcomings when working with athletes.

Athlete Profile

The ***Athlete Profile*** can be completed verbally or via a form sent to the athlete to fill out if they are being coached remotely. The purpose of the athlete profile is to obtain a wide variety of information to assess/coach an individual better and more accurately. This profile does have a degree of percent error based on the individual, as people might answer questions based on what they think they should say rather than reality.

Some of the areas that this profile addresses are:

- Athletic Background
- Goals
- Training Availability
- Pain Tolerance
- Mental Readiness to Train
- Best Type of Training Structure
- Personality Type

- Strengths/Weaknesses

UESCA Athlete Profile Questions

1. What type of physical activities do you currently participate in?
2. What competitive sports (if any) have you participated in?
3. Why, when, and how did you get started in running?
4. Will you be training by yourself or with others?
5. What are your short and long-term running goals?
6. When do you have the most energy: morning, midday, evening?
7. Does your schedule vary, or is it fairly consistent?
8. What days/times do you have to train?
9. How important is structure in your life?
10. What aspect of running are you most intimidated by (if any)?
11. What are your top three running-related strengths and weaknesses?
12. Do you respond better to tough love, handholding, or somewhere in between?
13. Do you crack, stay the same, or thrive under pressure?
14. What are you better at? Short, hard sprints or long endurance efforts?
15. How competitive of a person are you? Rate yourself 1-10 (10 = the most)
16. How much do you enjoy the intensity of pushing hard while training/racing (1-10)?
17. How confident are you in your abilities as a runner (1-10)?
18. Do you like to be challenged? 1-10 (1=untrue, 10 = very true)
19. Are you susceptible to mental burnout as a result of training? 1-10 (1=untrue, 10 very true)
20. What is your current weekly training volume (distance and/or time) and how long have you been running consistently?

Athlete Variables to Consider

When developing a program for an athlete, there are several variables to consider. While there are likely near-endless variables that a coach could consider when developing a running program, UESCA focuses on the following areas:

- Fitness Level of Athlete
- Experience of Athlete
- Training Duration
- Work Ethic of Athlete
- Trainability
- Special Populations

An athlete’s goals are a critical aspect of program development. Due to the importance, a full module is devoted to goal setting, and a full module is devoted to periodization and program design.

Fitness Level of Athlete

This is a crucial part of developing an effective program. You acquire an athlete who wants to compete in a 100-mile gravel race in two and a half months but has not started training and has poor aerobic conditioning. What do you do? Do you put the athlete on a fast-track program that might get the person ready for the event but would risk injury, or do you advise that it would be advantageous to choose another event given the time duration and the current fitness level?

It is important to understand the areas that comprise the fitness component of a training program. Too often, endurance athletes think of “fitness” solely in terms of the cardiopulmonary aspect. The four areas that make up fitness as defined by UESCA are:

1. **Cardiopulmonary**
2. **Strength/Power**
3. **Stamina/Endurance**
4. **Flexibility and Joint Range of Motion**

All four areas of fitness need to be incorporated into a training program. You likely have heard debates about which sport has the best athletes. Obviously, this is extremely subjective. For example, if people believe that power is the most important trait, they would be hard-pressed to find a sport more targeted than powerlifting. However, if one considers that cardiovascular endurance is the most important aspect, running would likely be high on the list. This is noted because this certification is not trying to determine which sport or which aspect of fitness is most important, but rather because it is attempting to identify all areas of fitness that must be included in a program for it to be comprehensive. **Each aspect of fitness needs to be appropriately scaled according to an athlete's individual needs.** For example, if your athlete has a substantially large percentage of body fat, it is advised to focus on fat/weight loss.

Experience of an Athlete

Generally speaking, experienced runners present more of a challenge to a coach than beginners, as the areas for improvement aren't as great. The principles and fundamentals of coaching a beginner are the same as an elite athlete. While the focus and specificity are often narrowed for an elite runner versus a beginner, the core coaching practices do not change. It is important to know how to scale your programming appropriately to accommodate each athlete as a coach.

Experienced Runner

Just because a runner is competing at a high level does not necessarily mean that the person is not deficient in one or more areas.

For many high-level athletes, the role of a coach is not so much to push them but to provide structure or hold them back when needed if they have the propensity to overtrain.

Experienced runners often challenge coaches because they may have established training methods and may be resistant to change

– even if their current training practices are incorrect and inefficient. **Being a quality coach means having the ability to communicate effectively with your athlete as to the *how's* and *why's* of the training process.**

If your athlete is both experienced and elite, the room for improvement is typically small. But while the total range for improvement is small, the effort required to coach this type of athlete is immense. The small, seemingly trivial insights often make all the difference at this level. Therefore, your attention to detail must be extremely high when working with these athletes. It is also suggested to bring in other professionals (i.e., biomechanists, sports psychologists, etc.) to assist. **If you do not feel you can assist certain athletes because they are above your level of expertise, you should not work with them and refer them to a more experienced coach.**

Novice Runner

When working with a beginner runner, you must make no assumptions.

When working with a novice runner, and especially one with no history of running, you must be sure to make no assumptions. Assuming a novice athlete understands and can execute seemingly basic things is a critical mistake.

It is advised that if you are working with novice runners with little to no base fitness, they should be limited to participating in a 10K event during their first year. Additionally, **the focus of a first-year runner should be on learning proper mechanics, increasing muscular endurance, gaining cardiovascular fitness, and learning about the sport – not speed.** While it is fine to incorporate high levels of intensity into a program, it should be done sparingly and only if the athlete is physically ready.

Training Duration

Training duration is the amount of time between the acquisition of an athlete and the race for which the person is training, assuming the athlete hired you to help train for a singular event. Before taking on an athlete, you must take entire stock of the following areas to determine if the duration of time to train for the race is enough.

- Current training volume
- Experience
- Time availability
- Biomechanical efficiency
- Fitness level
- Distance of event being trained for

Training Volume

The time commitment per week varies considerably based on where an athlete is in the training process and the race distance being trained for. For example, at the onset of a training program and during the taper, the weekly time commitment will likely be substantially less than during weeks at peak training volume.

Genetic Predisposition

No matter how hard most people work at it, they probably won't be able to sprint like Usain Bolt or run a marathon as fast as Emily Sisson. This does not mean, however, that they cannot improve substantially. You might know of runners who can without much training, run 10 miles without a problem whereas most others would cramp after just a few miles. **Whether it is muscular endurance, VO2 Max, ability to recover, or pain tolerance, your athlete will likely find certain areas of training and development easier than others.**

If your athlete is an experienced runner, they will likely have a good idea of where their inherent strengths and weaknesses lie. However, if your athlete is new to the sport or physical activity in general, it will take some time for you to be able to determine strengths. Once you ascertain this, it will help with program development, as you will be able to target areas that likely need more focus than others.

The Sport Gene by David Epstein examines the role of nature (genetics) versus nurture (training) regarding sports performance (466). While the book is full of research and interesting anecdotes regarding this topic, the gist of the book is that both genetics and training impact the success rate of an individual at a particular sport. Genetics influence many things, including how well an individual adapts to aerobic exercise, limb lengths, height, etc. Based on the sport, a genetic attribute such as extreme height might be an asset (basketball), whereas, in another sport, it is largely a liability (distance runner) (466).

As noted in the book, in a study, 1,900 Canadian firefighter applicants were given VO2 max tests to see if there was such a thing as a naturally gifted aerobic individual. As it turns out, **the study identified six individuals who had no history of aerobic training but had VO2 max levels associated with collegiate distance runners (**479).

Genetics may also play a role in how motivated an individual is to exercise and train. A study of 37,051 twins found that 48 to 71 percent of the variation in the amount of exercise performed by the subjects could be attributed to genetics (495).

Work Ethic of an Athlete

Some individuals will have a stronger work ethic than others. Some athletes will hang on to your every word and do anything and everything you say. For others, it will be like pulling teeth to get them to come close to adhering to the training program. The majority of athletes will fall somewhere between these two extremes.

The stronger an athlete’s work ethic is, the faster they will typically progress and see results. While one aspect of being a coach is the ability to motivate, it does not always ensure that an athlete will give 100 percent. If an athlete’s work ethic is poor, it is your responsibility to find a way to motivate the individual. Suppose the training program is modified to accommodate an athlete’s lack of training. In that case, you must reschedule the goal event if the training will not adequately prepare the athlete for the event.

You must be honest in your assessments of your athletes. This might call for tough conversations when addressing a lack of work ethic and adherence to the program. While tactful in your discussions, you must be direct and truthful concerning current training performance.

Trainability

This relates to how well an individual responds to the training process. Also, in *The Sport Gene*, Epstein notes a study called the HERITAGE Family Study (466). The general thesis of this study is that individuals respond differently to training, based mainly on genetic factors (467, 468). Specifically, in 2011 HERITAGE researchers identified specific gene variants associated with high aerobic capacity. Individuals with many of these gene variants improved their VO2 max three times more than those with a low number. Other gene variants correlated with the magnitude of the drop in heart rate due to aerobic training (466, 467, 468).

Trainability relates to training methodology. In other words, **if an athlete is not responding well to a particular training method or application, change the training stimulus** (466). A training method or application must be implemented for a long enough period to ensure that a training adaptation has enough time to occur.

Special Populations

This certification denotes three specific categories of runners that require additional awareness:

- **Youth Athletes (16 years or younger)**
- **Older Athletes (65 years or older)**
- **Athletes With a Physical Disability**

Youth Athletes

Below are six overriding themes when coaching any youth sport:

1. **Fun**
2. **Sportsmanship**
3. **Skill and Motor Development**
4. **Social and Emotional Development**
5. **Communication**
6. **Safety**

Fun

Participating in a running race should be fun for a child. If competing in running races is something a parent wants for a child, but the child hates it, this is a time when a conversation between you and the parent is appropriate. How you decide to make training fun is up to you, but you should try your hardest to balance fun and some form of structured training.

Youth under the age of 14 should have minimal specialization in the program.

Sportsmanship

Sportsmanship is winning with humility and losing with respect for each other and the opponent. Sportsmanship is more than just being nice to others. Someone displaying good sportsmanship is supportive (of teammates and opponents), participate with a positive attitude, are respectful, are willing to learn, and practice self-control. When these qualities are present, it makes for a more enjoyable and healthy sport environment. **Showing good sportsmanship is not just for the athletes; it is important for parents, coaches, volunteers, etc.**... to portray these traits so that youth can learn by example. And when learned at a young age, there is a much higher likelihood of youth to portray these traits later in life in their families, workplaces, and social circles.

Skill and Motor Development

The physical development of children largely relates to when they hit puberty. Kids hit puberty at different times. For this reason, **assessing a child's future potential is very difficult, as one's success may have more to do with hitting puberty early than it does with being a running prodigy.**

While a physically mature child might be able to handle a slightly higher workload than a child who has not yet hit puberty, this should be assessed individually. It is never a wrong decision to progress a child conservatively.

Social and Emotional Development

Research has shown that youth who participate in sports have a wide range of outcomes, including physical and social health, lower health-related issues, higher academic achievement, and a greater likelihood of attending and graduating college. **Sport has been shown to enhance the social and emotional competencies essential to success in relationships, school, and work.**

In a sports environment, youth are exposed to a structured need to engage in a social environment where decision-making, memory, teamwork, and problem-solving are necessary. All of which have been associated with greater social and emotional intelligence later in life. With mental health at the forefront of social priorities, it is essential to provide youth with a safe and healthy environment to provide youth with a supportive means for social and emotional skill development. For this to happen, **the sports environment needs to be safe, supportive, a place to succeed, and a community where youth can learn age-appropriate cognitively, emotionally, and socially skills.**

Communication

How you communicate with a child is different from how you should speak with an adult. A youth participant is less likely to be interested in the nuances of the training process than an adult would be. **Keeping the science out of the verbiage when coaching a child is highly advised.**

It is probable that a youth athlete won't have a long attention span; the younger the child, the more likely this is. Therefore, save your 20-minute dissertation on oxidative phosphorylation for your adult athletes and try to keep talks with youth clientele to a few minutes and on topics they understand.

The emphasis on communication should be lighthearted, fun, and supportive. Keep criticism to a minimum, and if criticism is used, it must always be framed in a supportive manner. Foul language should never be used when communicating with athletes, regardless of age.

Involving Parents

It is a parent's child, not yours. Therefore, you need to include the parents in the training process – as long as their participation is constructive and supportive. The exact verbiage of how you speak to a child's parents is up to you. The primary things to remember when involving parents are:

– Keep them involved in the training process
– Keep lines of communication open
– Be supportive of their goals for their child as long as they are healthy and attainable
– Manage their expectations
– Continue to reiterate your training philosophy throughout the coaching process
– Be respectful and professional at all times

It is important to understand that you are coaching a child who, more than likely, is not just participating in running events. **The child is most likely doing sports and non-sport-related activities.** Therefore, you should look at the training process as an opportunity to enhance their motor development. This means you should work in all planes of movement and engage the child in movements and activities that may not necessarily be running-specific.

Safety

W**hen training youth (and adult) athletes, the most important aspect is safety.** The musculoskeletal structure of a youth athlete is still in a developmental phase and cannot and should not have the same stressors placed upon it as an adult. The younger the athlete is, the more this is true. Areas, where this applies most are race distance, strength training, and race/training frequency and intensity.

Adolescents mature at very different rates, and as such, it is difficult to determine if an adolescent is physically mature enough to handle a set distance. Therefore, **when working with youth runners, a slow and steady progression is always the best**

course of action as you want to ensure the program is appropriately prescribed and scaled.

The focus on strength training should be on bodyweight exercises in all planes of movement, with the primary focus on overall strengthening and fun.

Education and awareness are essential to creating a network of coaches, trainers, doctors, volunteers, mentors, and parents that create a safe environment for youth athletes. All participants must be advocates of respect and an abuse-free environment. Emotional abuse, physical abuse, bullying, and harassment are primary examples that, if present, can lead to severe mental and physical health outcomes and potential cessation of sports participation altogether. Resources are available through the US Center for Safe Sport (if US-based) to educate and report concerns confidentially.

Interactions Between Coaches and Athletes

The most crucial thing in a coaching relationship is that it is always professional. This encapsulates many things ranging from physical touching, language used, physical actions, and mannerisms. As an example, a coach might say or do something that, in their mind, is professional; however, from the athlete's point of view, it is unprofessional and scary.

Older Athletes

It is a fact that as one ages, many physiological changes occur. Following are some examples of these changes (92):

- Cardiac output decreases
- The overall decrease in elasticity of muscles and connective tissue
- Decrease in nerve impulses (e.g., reaction time)
- Sarcopenia: a reduction in muscle strength
- Decrease in lung capacity
- Bones become weaker and more brittle (i.e., osteoporosis)
- Increase in recovery time

Does this mean that if you coach an athlete who is 65 or older, there is no hope? Not, and in fact, it is quite the opposite. **As a coach of an older adult, you have the opportunity to have a substantial impact on slowing the aging process.** As an example (running-specific) of what fitness can do to slow the aging process, at the 2021 Fifth Avenue Mile in NYC, the winner of the 70 to 74-year-old age group crossed the line in 6:18, and the winner of the 80–99 age group did it in 8:17 (93)!

A comprehensive training program can dramatically slow physiological changes associated with aging, especially muscle strength (639). Some important areas to identify in a training program are:

1. Strength training
2. Adequate recovery periods
3. Maintaining intensity in the training program

The overall structure of a running program for athletes 65 years and older should be based on the individual, not the age. This is because of the wide range of physical capacity.

Muscular Endurance

While muscle strength and power have been shown to decrease over 60 years of age (640), studies seem to indicate that muscular endurance is not affected (641). The neuromuscular system regarding running is primarily correlated to muscle fatigability. In other words, as someone ages, the fatigability does not change much, if at all.

Areas of Focus

While muscular endurance has not been shown to decrease with age, as noted previously, the same cannot be said for muscular power/strength, cardiovascular output, lung capacity, and reaction time (642).

Therefore, when creating a running program for an athlete over 65 years old, focusing on these areas safely is advised. Of course, age is a relative number, and while it is likely that you will come across a runner in their 40s who requires much more attention than a runner in their 60s, the general trend is that older athletes should have different areas of focus than younger athletes.

Athletes With a Physical Disability

Training methods differ based on the physical limitations of an athlete. Regardless of the disability, it would help to focus on the same areas of performance enhancement and components of fitness that you would with a non-disabled individual. How you customize the program and workouts to accommodate an athlete will differ.

Important areas on which to focus when working with a physically challenged athlete are:

- Safety
- Feedback
- Equipment

Depending on the area of disability, safety and equipment might be more of an issue than with an able-bodied athlete.

According to Catherine Sellers, director of *USA Paralympic Track and Field High Performance*, a coach goes through a growth process when first working with an athlete with a physical disability (342). The stages are as follows:

1 – Fear of the unknown
2 – Fear of saying something wrong
3 – Shock
4 – Acceptance
5 – Advocacy

When working with any athlete, the learning process is a two-way street. This is especially true when working with a physically disabled athlete. While the athlete is learning about training methodologies, physiology, and race tactics from you, you are learning about the person’s particular disability and its challenges.

As a coach, one of the most important and necessary traits is the ability to adapt. As stated above, the goal and basic structure of the program should not change; it is just a function of what modifications need to be implemented to meet the needs of an athlete.

Suppose your athlete has a video of him or herself racing or videos of others with the same disability. In that case, it can be beneficial in understanding the nature of the disability and its impact on performance.

Summary

- An *HHQ* and *PARQ* must be completed before training an athlete can commence.
  - If an athlete answers *yes* to any of the PARQ questions, the person *must* get a physician’s clearance prior to working with you.
- Based on UESCA protocol, any athlete, regardless of PARQ or HHQ results, *must* have been cleared by a physician in the last 12 months prior to beginning a running program.
- It is advised that your athlete receive a cardiac assessment within one year prior to beginning a training/racing program.

- A *Gulick tape measure* is used to take accurate girth measurements.
- It is advised not to use BMI or BIA methods to determine body composition because of a substantial chance for inaccuracies.
- Many variables come into play when creating a training program. Below are six variables:
    - Fitness level
    - Experience level
    - Training duration (how long before goal event)
    - Genetic predisposition
    - Work ethic of athlete
    - Special populations
- The athlete profile will help you to assess athletes in the following areas:
    - Athletic background
    - Goals
    - Training availability
    - Pain threshold
    - Mental readiness to train
    - Best type of training structure
    - Personality type
    - Strengths/weaknesses
- Fun, safety, and age-appropriate communication are focal points when coaching youth athletes.
- When working with youth athletes, parents must be involved in the training process.
- While physiological capacity decreases as people age, physical training can dramatically slow the aging process.
- Athletes with a physical disability present a unique set of challenges to a coach, primarily regarding the use of adaptive equipment. In these cases, a coach has an opportunity to learn as much as an athlete.

Module 13: Goal Setting

This lesson will discuss all aspects involved with understanding different goal types and how to effectively work with your athletes to establish them.

- Different types of goals and how to identify them
- Process, Performance, and outcome goals
- S.M.A.R.T goals
- Open and Closed goals
- Assessing potential
- Logging training and its value in goal setting
- Goal identification and identifying a goal event
- Health vs. Competitive goals
- Long and Short-Term goals

Goal Setting

Goals are a necessary component in effective program design. If your athlete has not identified a goal for which to train, you should work with them to come up with one. Without a specific goal, there is no focal point, and therefore intermediary goals cannot be established.

The goal of an athlete needs to be realistic and attainable. It is the responsibility of coaches to inform athletes if they feel their goal is too far-reaching and, thus, not realistic or potentially unsafe. Let's repeat this for those in the back… a goal MUST be realistic and attainable! Setting unrealistic and unachievable goals sets athletes up for perceived failure, sadness, negativity, and even depression.

If an athlete's goal race is months down the road, it is a good idea to choose preparatory races before the goal race. This is especially true for beginners as doing a "practice" race or races will help familiarize them with the process of preparing for, and racing in a running race. For more seasoned runners, practice races should be used for high-level training, maintaining motivation, and overall fitness/performance assessment.

For many individuals, the big picture goal might be something other than a running race, such as weight loss, increasing cardiovascular fitness, or lowering cholesterol. For this type of athlete, they typically use running as a motivating factor for their big picture goal.

Process, Performance, And Outcome Goals

In the following video, Jim Lehman discusses these different goal types. This topic is also covered later in the module by Ben Rosario. Note – the below video makes mention of cycling-specific scenarios, but the overreaching message is applicable to all athletes, including runners.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

The most foundational mental performance skills for athletes center on **Motivation, Goals, and Commitment**. It's tough to build a coaching plan for someone without knowing what they are pursuing, why the chosen goals are personally relevant to them, and what potential barriers are expected in their training. This trio of factors sets up the entire training cycle. It's important to understand and help guide the athlete through each of these areas to ensure alignment in the coaching-athlete relationship and minimize any potential points of conflict or disagreement. These three factors also wax and wane throughout an athlete's life. It's important to consider the changing nature of each domain throughout a season and a lifetime for those pursuing athletic-related goals.

Goals might be the most researched and discussed area related to performance. At its core, goal setting appears to be a relatively well-discussed concept embedded in all endurance performance. Still, there are important concepts and frameworks to consider when setting goals. The first of these factors center on Process, Performance, and Outcome goals.

Outcome Goals

Let's start with outcome goals, as most athletes have a specific, desired outcome they are trying to achieve. An outcome goal is a specific result that in and of itself is largely out of an athlete's control, such as winning a race, making it to the podium, or hitting a very specific time. Outcome goals are rooted in precision and often hold deep value and meaning, but often can contribute to anxiety and frustration, given the lack of controllability. Much of coaching is centered on agreeing to realistic Outcome goals and working to structure the process of physiological development to put the athlete in the best possible position to achieve these goals. But there can be a lot of outside influencers that stand in the way of Outcome goals becoming reality – weather can be non-conducive to someone's goal race for example, or a perfectly executed day may be eclipsed by another runner accidentally tripping them. In short, outcome goals are specific results with limited control.

Performance Goals

Performance goals lie just underneath Outcome goals and focus on a set of standards to engage that gives the athlete more control over effort and intensity while reducing an emphasis on an exact outcome. Performance goals are standards that are being pursued that are not necessarily tied to specific, exact outcomes, yet still hold high value. Performance goals will still likely be specific but will have fluidity, and flexibility behind them. They connect an athlete to more controllable, yet less specific outcomes – such as maintaining a hard effort when tired or fatigued or engaging in mental toughness (more on that topic later). **Performance goals can be modified and scaled given the unique needs of an athlete and/or changing circumstances of a specific event.**

Whereas an Outcome goal may be something along the lines of "Win the local 5K," a performance goal may be 'average a 6:30 per mile pace' in the race and continue to run hard when tired during the final stretch of a race." Focusing on those key standards of performance gives the athlete a specific, tangible area of control that in turn is likely to lead to the best possible outcome given the circumstances they find themselves in. Importantly, Performance goals are the defacto default for anyone who finds themself in a position in which their specific Outcome goal may be in jeopardy. Rather than reducing effort completely, continuing to push as hard as possible when a race outcome goal is unlikely gives the athlete a sense of agency and focus in a specific way, rather than becoming overwhelmed with the possible frustration of falling short of their intended Outcome goal.

Process Goals

Process goals are the most controllable set of goals in the life of an athlete and influence the daily and weekly training plan as they focus on actionable behaviors on a regular, consistent and trackable basis. Completion of Process goals puts the athlete in the best possible place to achieve a specific outcome goal. Examples of process goals could include how many days per week an athlete trains, what time of volume they work to complete, and what intensity ranges they perform their training.

Taken together, an example of these goals working together could realistically look something like this:

Outcome Goals (Desired Result): Get top 3 in the regional road race championship

Performance Goals (Standards): Execute a smart race day strategy. Focus on staying in the top 1/3 of the pack at all times. Follow any attack in the last 5 miles . In training – develop mental toughness skills of running hard when tired.

Process Goals (Behavior): weekly mileage, intervals, long runs, strength training, nutrition, and mental skills development over a 16-week training cycle. Percentage of training in specific zones that align with goal objectives.

S.M.A.R.T Goals

The acronym SMART has gained much attention in recent years due to goal setting and is worth considering when working with athletes. The acronym stands for:

Specific: goals need to be clear and specific. Some questions to consider include: What are you trying to accomplish? Why now? When and where is the event taking place?

Measurable: there are many ways to measure success. Completing a race is one level. Focusing on an outcome time goal is another. It’s also important to focus on measuring progress along the way. Whatever the outcome in focus, it needs to be clear and have some ability to be measured.

Attainable: there are two ways to think about the level of attainability. For many, making sure a goal is connected to a realistic level of achievement is important. A new runner who expresses a goal of winning the Bolder Boulder 10K in their first-ever race will likely be setting themselves up for huge levels of disappointment if they embark down this path. Yet, others may want to take on what we sometimes call “a big hairy goal,” which is of high challenge and a difficult level of attainability. It’s essential to consider the balance between challenge and risk when considering the level of attainability in goal creation. The conversation between athlete and coach regarding the expressed goal of risky attainability is critical here.

Relevant: Ensuring a goal is personally relevant and meaningful is arguably the most critical aspect of SMART. Essential questions to ask include: What makes this goal important to you? What makes the pursuit of this goal worthwhile? Is this the right time in your life to pursue this goal? Athletes often get caught in pursuing relatively meaningless personal goals because they see others pursuing them. For example, I've worked with plenty of runners who believe they can't call themselves a "runner" until they complete some arbitrary feat, such as completing a marathon or running a specific time in a particular event. Yet, they have no personal interest in pursuing such goals. Working to understand the personal significance of a goal is important.

Time-Based: Every goal needs a target date. When is the event? Is there adequate time to prepare? How will pursuing this goal impact my schedule for the next week, month, six months, year, etc.?

Open and Closed Goals

Open and closed goals are a further important consideration in goal setting, particularly in the connection to the underlying psychological frameworks that can be influenced by either, most notably that of flow state or clutch state (which will be discussed later). Flow has been described as an optimal experience of total immersion in an activity, which differs from a clutch state, a feeling of urgency in needing to perform in a particular way in a specific moment. Further, the concept of flow is often tied to Peak Performance attainment, defined as a state of accomplishment resulting from sustained effort and concentration.

Open Goals

Open goals are non-specific and exploratory. They do not have a specific target, focus, or outcome. Rather, they are tied to an ongoing process. Open goals are fluid, spontaneous, and responsive to the changing demands of any given situation. They are most often connected to performance standards over performance outcomes and permit adaptability and exploration. In the paper cited below, this description from a climber working on summiting Mt Everest is that of open goals: "I was just thinking, "Oh, I'll just see how it goes and take it as it comes." I climbed higher and higher, and the climb had got more and more engrossing and difficult and all-encompassing really ... until I discovered that I'd climbed like 40 meters without consciously knowing what I was doing." As a concept, the ability to tap into open goals based on performance standards is something that every athlete can work to adopt in their training and racing strategy/

Closed Goals

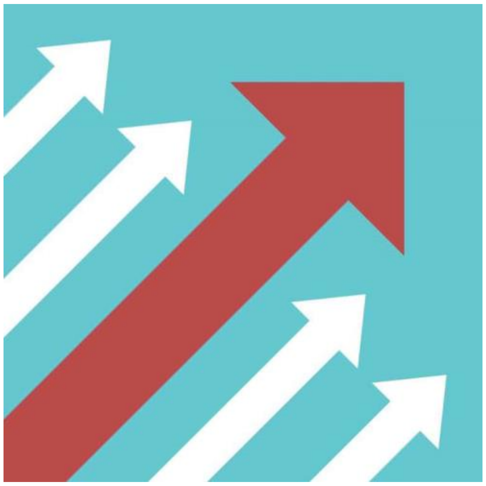
Closed goals are specific, rigid, and tied to the outcome. Due to their connection to an objective, measurable results such as winning or achieving a qualifying standard, for example, closed goals, are sometimes referred to as fixed goals. They can influence a sense of urgency during competition and lead to deliberate decision-making regarding increasing intensity, effort, and/or concentration in a given situation. An example of this type of goal is quoted by a tennis player, “To be honest, I just wanted to win the match. I didn’t care how; I just wanted to win.” A Clutch state is also an important psychological state for an athlete to work out. This state requires a decision to continue pushing despite the discomfort of pursuing a specific outcome. Clutch state is often tied to a deep sense of personal meaning with outcome goals (965).

Assessing Potential

For athletes who have a performance-based goal, you must assess their potential to determine if their goal is feasible. Athletes often ask about their potential in three main areas:

1. **Personal Potential:** What is my physiological potential?
2. **Event Potential:** What is my potential for placing in a particular event?
3. **Event Preparedness:** Will I be ready to participate in a specific event? Assessing potential can play a significant role in goal selection. For example, if you have an athlete who wants to win their local criterium the following year, you will need to assess their current fitness level and how it relates to their stated goal.

Assessing potential is very subjective, and while one can use data to assist in this determination, there is a high degree of bias involved at the end of the day. Two coaches could assess an individual and come up with two completely different views of their perceived potential. The longer one has coached an athlete; the more accurate the individual will be in assessing an athlete's potential.



Following are five primary things that affect the accuracy of "athlete potential" assessments:

- Past-performance history of running
- Physiological and performance testing
- Rate of progression in training
- Duration of working with an athlete
- Capacity to adapt to the training process

While these are five important areas to look at, there are unlimited areas that can be used to assess potential. As a coach, it is up to you to think outside the box and look for areas that you feel will most accurately assess your athlete's potential.

All the areas above affect how and what goals are set. The goals should be challenging, realistic, attainable, well-rounded, and most important, agreed upon by you and your athlete.

Following is a breakdown of each area regarding potential assessment:

Running Performance History

Perhaps your athlete was a top former division one track runner. While the time gap between their collegiate running career and present time affects potential to some degree, it does give a good picture of the individual's perceived potential concerning capacity to be a talented runner.

Physiological Testing

A test such as FTHR can provide you with good information regarding an athlete's current and potential physiological capacity.

Speed of Progression in Training

As there is always room for improvement, there is no such thing as one's "potential" or "capacity," as this is a moving target. It is just a function of how close an athlete is to this moving target that you should attempt to assess.

When working with a deconditioned athlete or an average-conditioned athlete, the fastest rate of progress occurs in the beginning. This is natural. The better conditioned an athlete becomes the slower the person's rate of quantitative progression.

As the name suggests, quantitative progression looks solely at the amount of improvement from a numerical standpoint, not the significance of the improvement. For example, let us say your athlete reduces his or her 54-minute 10K time by 10 seconds. While this is an improvement, it is not very significant given the relatively low starting benchmark and small-time improvement. However, if a runner's previous best 10K time was 29 minutes and 10 seconds and they improved by 10 seconds (29:00), this is very significant as improvements made on times close to one's current physical potential are difficult to achieve and usually measured in very small increments.

Therefore, rate of progress is a good way to assess where athletes are in relation to their current potential. Typically, the faster their progression, the further they are from their potential, whereas the slower they progress, typically the closer they are to their potential. While there are many reasons why an athlete's progress might speed up or slow down (i.e., skipping training sessions, overtraining, injury, etc.), a coach must take these factors into consideration to effectively and accurately assess an athlete's progress.

Duration of Athlete Relationship

As with any relationship, the longer two people know each other, the better they come to understand one another. The longer you work with athletes and track their performance, the more historical data you will have to help predict their potential.

Capacity to Adapt to the Training Process

Individuals adapt to training at different rates – remember *trainability*? If you were to take five athletes at equal fitness levels and give them the same training program, you would most likely see five different progression rates and results.

Training Log

In addition to aiding with documenting workouts, using a training log also holds substantial value for goal setting. Having athletes record what they did well and what they feel they need improvement on during a training session or race can significantly enhance the quality of the overall training program. Areas identified as having room for improvement can translate to goals throughout the training process. For example, perhaps an athlete noticed that their hips tilted more in the frontal plane at the end of their run when they got fatigued. While this is not a race-specific or fitness goal, a biomechanical improvement goal is just as important.

Sacrifice

Reaching a goal typically takes some form of sacrifice, whether getting up a half-hour earlier to get a training session in or cutting alcohol out of one's diet to lose weight.

Therefore, if athletes come to you with goals that you feel would require some sacrifice on their part, the first thing you should do is ask them if they are seriously willing to make the sacrifice(s) to attain them. It is your job to paint a realistic picture of what you believe will be required to reach a goal versus taking on an athlete without explaining what you envision will be needed on their part.

Goal Identification

Before identifying a goal race, it is advised to assess your athlete's fitness level, experience, and current training volume. Once these benchmarks have been established, based on the proper training volume progression, you will be able to project by what date the athlete will be ready to compete in a race of a certain distance.

A goal race is different from a non-goal race. Many runners identify one or two races per year to focus on (i.e., goal event) while participating in other races during the year strictly, focusing on using them as part of their training.

As is discussed later in this module, the primary goal of your athlete might not be race-based. For example, your athlete's primary goal might be to lose 10 pounds, and training for a marathon is the primary means to exercise and lose weight. However, even in these cases, a goal race should be identified as a focal point of the training process.

The most important aspect of a goal race is that it has meaning for an athlete. It could be the Boston Marathon, or a local running race that an athlete's family and friends will be able to attend. This is why it is critical that an athlete determines what their goal race(s) is versus the coach.

In the following two videos, Ben Rosario and Jim Lehman discuss the process for setting goals and goal races, respectively.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Identify Goal Event

The date of the goal event and any other preparation or training events are the genesis for assessing the distance and training time facets of the program.

Suppose your athlete already has a goal event in mind or has signed up for an event. In that case, you will need to assess if the person will be adequately prepared for the event, given the current level of conditioning and training volume.

The number of goal events per year is based on how many times an athlete can properly peak for an event. As noted previously, all goal events should be challenging but attainable.

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Determining Goal Events

Due to the various lengths of running races, race-specific goal(s), varied conditions of athletes, and individual physiology, it is impossible to determine how many goal events a runner can target in a season.

However, the longer the goal races are, the fewer races can be targeted per season as goals. Additionally, the more inexperienced a runner is and/or deconditioned, the fewer races should be targeted as goals.

Often, a runner's goal races are a mixed bag of distances and disciplines. For example, a runner might target a 5K road race, a 10K trail run, and a 40-mile ultramarathon. As you can see, these are all very different distances and disciplines – all of which require different areas of training focus.

In summary, **the number of goal races that a runner can target is highly individualistic based on many variables**, including but not limited to the following:

- Experience
- Fitness level
- Length of race(s)
- Ability to recover
- Time availability to train
- Race-specific goal(s)

Identify Preparation (Tune-Up) Events

Once you have established that an athlete's goal event(s) is feasible based on their current level of fitness and training volume, you will want to plot a preparation race or races. To do this, select races that are the appropriate distance concerning where the athlete is in the training program. There should always be at least one preparation race. As noted earlier, preparation races focus on gaining experience for new runners, whereas, for experienced races, the focus shifts to high-intensity training, assessing current fitness and performance levels.

Some coaches and athletes denote events by letters:

- **"A" event =** goal race
- **"B" event =** important preparatory race
- **"C" event =** training or minor race

Some runners choose to race a lot – like every weekend! For these runners, it is important for them to understand that most of their races are 'workouts' versus goal races. Regarding the event identification letter noted above, these would be 'C' events.

Quantitative Goals

Aside from providing experience, goals should be quantitative and measurable. Common goals are:

- Race or training times
- Performance testing (e.g.,heart rate, lactate)
- Race placing
- Weight or body fat loss/gain
- Weight or repetitions for strength-training assessments
- Range of motion for flexibility-based assessments
- Reduction or elimination of illness/disease/pain (e.g., high blood pressure, diabetes, high cholesterol, lower back pain)
- RPE

While not as quantitative, if assessed using a 10-point scale (0 being low, ten being high) initially and for future assessments, the following areas can be analyzed quantitatively.

- Stress
- Energy level
- Productivity
- Sleep

From a goal race standpoint, establishing quantitative goals throughout the season, such as performance testing and training/preparation race times, assists significantly in determining if an athlete is on track. These mid-season goals help athletes stay motivated and keep their heads in the game.

Goals need to be realistic. For example, if your athlete is currently averaging a nine-minute-mile pace for a 10K, it is highly unlikely the person will be able to average a seven-minute mile in one month's time. This is an example of a poorly established goal. Properly established, quantitative goals can be derived from some or all of the following:

1. Past and current testing data (if applicable)
2. Initial and current assessments
3. Athlete's rate of progression

By establishing goals with your athletes, you will ascertain what motivates them. Understanding this is critical to developing an engaging and fulfilling program for an individual.

Health vs. Competitive Goals

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Long and Short-Term Goals

What constitutes long and short-term goals is subjective. However, you must be aware that your athlete might be targeting a goal further out than one year, which may or may not be event specific. For example, your athlete may have the goal to qualify for the Boston Marathon in two years, or perhaps you are working with a very talented athlete whose goal is to participate in the Olympic Trials in three years. These are all examples of long-term goals that are more than 12 months away.

Conversely, whether short-term goals are weekly, biweekly, or monthly, they are important to maintain motivation and structure.

Summary

- Goals should be SMART:
    - Specific
    - Measurable

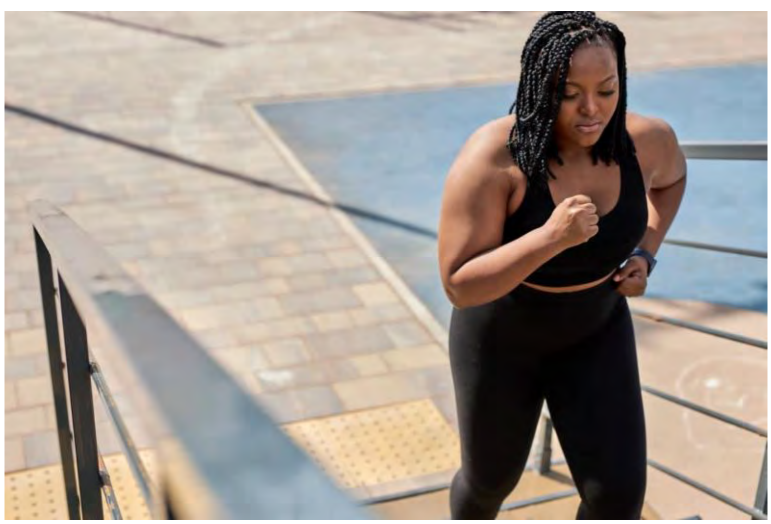
    - Achievable
    - Relevant
    - Time-Based
- The three primary types of goals are outcome, performance and process.
- There are five primary ways to address potential:
    - Past-performance history of specific sport (e.g., running)
    - Physiological and performance testing
    - Rate of progression in training
    - Duration of working with an athlete
    - Capacity to adapt to training process
- The most foundational mental performance skills for athletes are **Motivation, Goals, and Commitment.**
- Goals should be quantitative in nature, with the exception of training youth athletes, as the focus should be having fun.
- Identify goal and preparation events.
- Goals should be short and long term.
- Training logs assist with goal identification and setting.
- New runners, regardless of their physical condition, should not sign up for a marathon in their first year of training.

Module 14: Female Physiology and Programming

Contributor: Dr. Leah Roberts

This section is included in this module due to its importance with respect to female-specific programming.

The Menstrual Cycle and Female Athlete Performance



One of the main differences between males and females is the difference in dominant sex hormones: estrogen and progesterone for females and testosterone for males. In females, these hormones are responsible for maintaining a healthy menstrual cycle and contributing to bone health. They ultimately influence performance.

More than half of elite female athletes report hormonal fluctuations during their menstrual cycle that negatively affect their exercise training and performance capacity. (992) It has also been reported that many Olympic gold medals have been won during all phases of the menstrual cycle. (993)

What is the Menstrual Cycle?

The menstrual cycle starts with the first day of bleeding (menses), where women are shedding the uterine lining. Menses is the start of the follicular phase, or “low hormone” phase, characterized by low luteinizing hormone (LH), follicle stimulating hormone (FSH), progesterone, and slowly increasing levels of estrogens. Estrogen starts to increase on day 5-6 and surges around day 14-16, along with LH. The follicular phase is during the first 14 days of the menstrual cycle (assuming a 28-day cycle). On day 14, an egg is released, and estrogen then drops for several days. This is known as ovulation.

The luteal phase starts on days 15 thru 28, when the uterus prepares for a fertilized egg. Progesterone (and estrogen) begin to rise, peaking five days prior to the onset of bleeding. PMS symptoms start to occur when progesterone levels rise. If a fertilized egg isn’t implanted, progesterone and estrogen levels fall, causing the uterine lining to shed. This brings you back to day one of the menstrual cycle (1006).

The Female Sex Hormones

Hormones are chemical messengers produced by the endocrine glands and released into the bloodstream. As mentioned above, the women’s menstrual cycle has fluctuations in various hormones.

Estrogen: primarily produced by the ovaries. It plays an important role in reproductive and sexual development once a female reaches puberty

Progesterone: produced by the ovaries, adrenal glands, and placenta. It plays an important role in preparing the uterus for pregnancy (the luteal phase of the menstrual cycle) and maintaining pregnancy.

FSH: produced by the pituitary gland. Initiates the formation of follicles in the ovary and follicle cells to produce estrogen.

LH: produced by the pituitary gland. Triggers ovulation at the end of the follicular phase.

Testosterone: women produce testosterone too. Normal male testosterone: 250-900 ng/dL (Age 19-49); Normal female women testosterone: 15-70 ng/dL

Managing Periods with Contraceptives

Hormonal contraceptives (HCs) suppress the natural endogenous production of estrogen and progestin to prevent ovulation. Thus, females using HCs are always in a "low-hormone phase" regarding natural ovarian hormones. A recent study showed that 50% of female athletes use some kind of HC that affects their menstrual cycle (994). It was shown that the oral contraceptive pill is the most popular type of HC, used by almost 80% of the sample. With use, females reported adverse side effects that may affect performance, including weight gain, irregular periods, and poor skin being the most common.

In contrast to the adverse side effects reported, 13% of contraceptive users said that they liked the regularity of the pill and knew when they would experience their withdrawal bleed, which happens during the four or seven pill-free days of an oral contraceptive pill cycle. There is some evidence that the HC does not affect training-induced changes in performance; however, more research in this area is needed (1000).

Menstrual Cycle in Relation to Health

A natural menstrual cycle includes phases with high estrogen concentrations and is associated with good bone health and better fertility outcomes. The menstrual cycle is part of a much more significant health issue for female athletes. Low energy availability: meaning that energy intake and expenditure are not in balance, can lead to irregular periods or loss of menses entirely, as well as problems with bone health (or loss of bone mineral density), a condition termed the Female Athlete Triad (1003). Other disturbances to normal physiological function may also be observed, including, but not limited to, reductions in metabolic rate, immunity, protein synthesis, and cardiovascular health. Another concept, known as Red-S (Relative Energy Deficiency in Sport) (which can also be observed in male athletes), expands the definition of the Female Athlete Triad by considering other aspects of health and performance. (1004)

This new terminology Red-S for “Female Athlete Triad” includes amenorrhea, osteopenia, and disordered eating. The International Olympic Committee adopted it in 2014. The red-S underlying issue is that calorie expenditure is greater than caloric intake, creating a deficit in overall energy. Females and males of all ages and abilities can suffer from Red-S. It is estimated that 18-20% of all athletes suffer from disordered eating. It is also estimated that 62% of female and 33% of male athletes in aesthetic or weight-class sports (bodybuilding, wrestling, rowing, figure skating) suffer from disordered eating (1013)**.**

There are substantial health consequences of disordered eating. The body depends on energy for organ systems to function correctly; if energy is lacking, body systems start to fail. Other consequences include:

- Psychological: emotional lability/moodiness, increased anxiety and depression
- GI issues: constipation, irritable bowel syndrome (IBS)
- Bone health is related to estrogen and the endocrine system; men have estrogen too! So if energy is not available, men won’t be making testosterone (or estrogen). This can lead to osteopenia and stress.
- Endocrine: decreased melatonin causes impaired sleep which is associated with decreased cellular repair.
- Hematology: start seeing ferritin/iron drop. Decreased iron leads to anemia.

- Immunity: T cell function decreases, and the body can't fight off viral/bacterial infections. This is why an athlete tends to get a cold or sick when exhausted.
- Cardiovascular: resting HR increases, and HR becomes high for easy efforts
- Menstrual dysfunction: Seen in female athletes not on oral contraceptive pills (OCP), polycystic ovarian syndrome (PCOS), or peri-menopausal. When an athlete is starting to have irregular periods (spotting, irregular frequency, or missing), this is a red flag that the body's energy balance is off. Decreased energy availability means less estrogen produced by the ovaries. Decreased estrogen causes loss of ovulation and menstruation and decreased bone remodeling (bone strength). There is a strong correlation between menstrual dysfunction and the risk of stress fractures. Once energy balance is corrected, it can take up to 3-6 months for periods to return, and takes approximately one year for bone health to recover.

Warning Signs of Red-S

1. HR elevated for easy efforts.
2. The mindset of training is based in calories burned rather than effort
3. Ignoring rest days and recovery weeks
4. Lack of motivation and enjoyment for training
5. Menstrual dysfunction

Treatment For Red-S

- Reverse the lack of energy availability: To differentiate between overtraining and under-fueling, decrease training volume by 50-70% for 7-10 days (while keeping all other variables such as calorie intake the same). If the athlete starts feeling improved mentally and physically, the underlying issue is overtraining.
- Interdisciplinary approach:
    1. Physician (blood work, medication if needed)

2. Coach (balancing training volume and intensity appropriately)
3. Sports nutritionist (target caloric intake, education on quality foods)
4. Psychologist (address underlying pathology for disordered eating or overtraining)

Potential Effects of the Menstrual Cycle on Training and Performance

Approximately 75% of athletes experience adverse side effects due to menses with side effects including but not limited to: cramps, back pain, headaches, and bloating. (994) Fluctuations in strength, metabolism, inflammation, body temperature, and fluid balance changes coincide with hormonal fluctuations throughout the cycle. (995)

The effects of a female's menstrual cycle are still highly individual. Coaches can use an athlete's cycle to help determine where to fit in specific training. The potential effects of hormonal fluctuations during a female's menstrual cycle are as follows:

Follicular Phase

- Due to higher pain tolerance and experience of higher perceived energy levels, especially during the early follicular phase, this is the time when the female body is primed for high intensity.
- Carbohydrate loading the day before and during exercise might be necessary for female endurance athletes to be able to exercise at high intensities due to the rise in estrogen in the late follicular phase may hamper pre-exercise carbohydrate storage (996)
- Strength training may be more effective when estrogen levels are higher during the late follicular phase. (997)

Ovulation

- This could be the time to achieve strength gains. Compared to the follicular and luteal phases, a significant increase in quadriceps strength during ovulation has been documented (998).

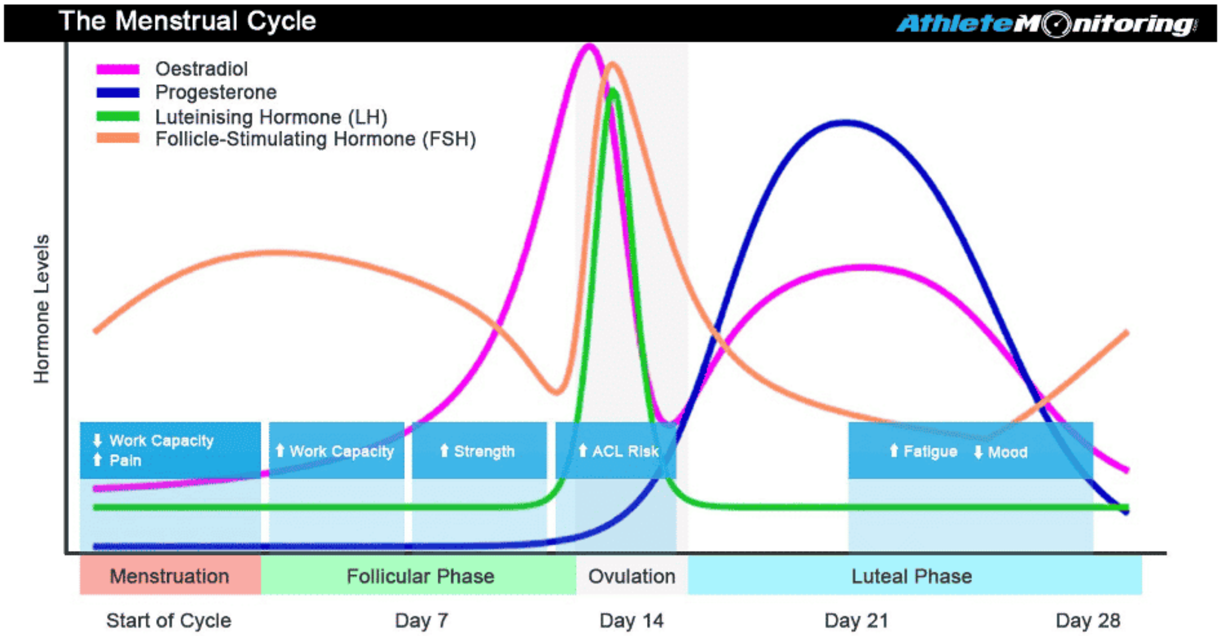
Luteal Phase

- The body is not primed for high intensity.
- PMS might interfere with training and performance (7-10 days before menses (999)
- Body mass might be higher due to fluid.
- It has been suggested that breathing and body temperature increases could make it harder to run in the heat. The onset of sweating is delayed so that sweating occurs only after higher body temperatures have been reached. Recent studies, however, indicate that this does not affect performance (1001).

Training with One's Natural Cycle

The Follicular Phase is considered the "low" hormone phase. When menstruation occurs, a female's hormone profile is most similar to a male. Estrogen rises a few days after menstruation and is associated with increased pain tolerance, higher testosterone, and improved recovery (1006)**. This is the best time for Intensity:** VO2 work, one rep max and heavy lifting, tempo intervals, and double training sessions. A weight training study showed increased strength gains when focused on training during the follicular phase (1008)**.**

The Luteal Phase is considered the "high" hormone phase. Progesterone is the dominant hormone associated with: increased core temperature, decreased plasma volume, increased HR (rest and with activity), and PMS symptoms (1006)**. This is the best time** for effort-based training sessions (rather than based strictly on HR or pace). Progesterone is catabolic (breaks down muscle); thus there is a need for increased BCAAs and proteins during recovery. Progesterone also impairs glycogen utilization, and athletes require an increased carbohydrate intake during training.

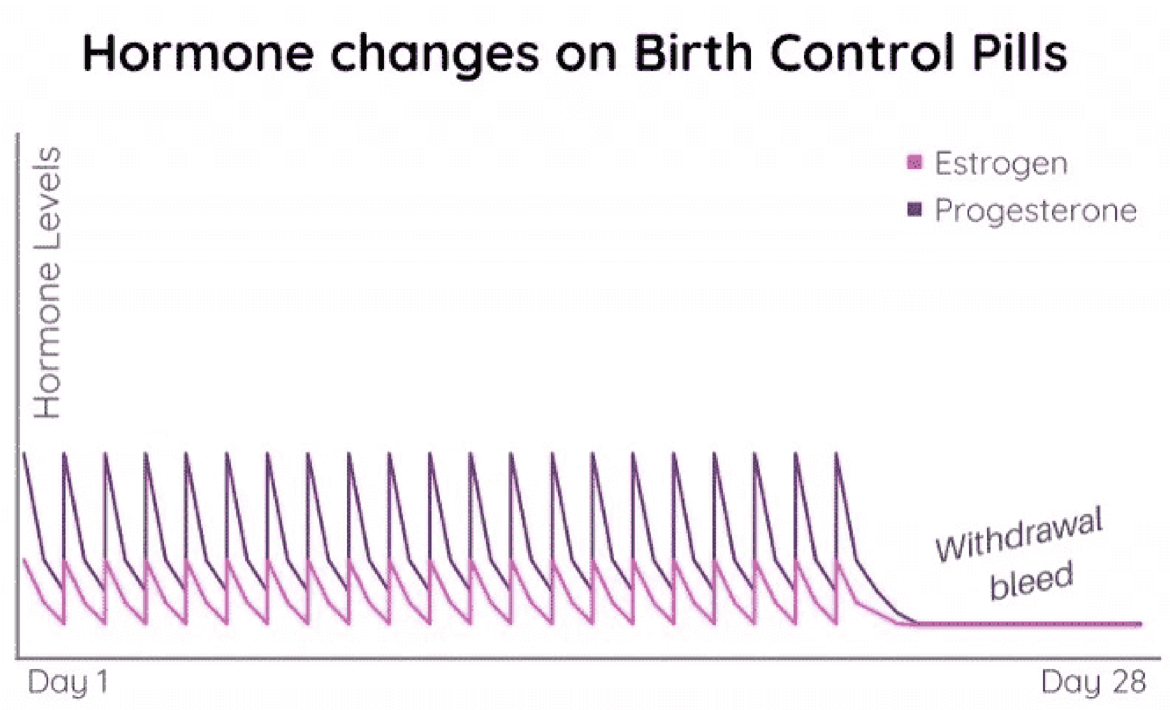


It is not normal to miss periods. A missed period is a red flag indicating an imbalance in the hormonal axis and likely underlying energy deficiency (RED-S).

Training with Hormonal Contraception

Oral Contraceptive Pills (OCPs) are typically a combination of estrogen and progestin for three weeks, then placebo pills for one week. Bleeding occurs during placebo week. Some OCPs are a continued cycle of estrogen and progesterone – this is called a minimal cyclical hormonal pattern; no ovulation, no follicular or luteal phase.

Bleeding while on OCP is withdrawal bleeding, not a true period. It takes a pill 8-12 hrs before a workout to minimize the effects of high progesterone. For example, if you are cycling in the morning, it is recommended to take OCPs at night.



Another contraceptive device is the Intrauterine Device (IUD) – either copper or hormonal IUD. Copper IUDs do not affect ovulation. Hormone IUDs exert hormones locally on the uterus, and some women still ovulate. Many women do not have monthly periods, although they still have cyclical hormone changes.

How to Mitigate PMS

Ways to mitigate PMS symptoms for athletes include:

1. Cramping: Dr. Stacy Sims Protocol – start 5-7 days BEFORE onset of bleeding
    1. Magnesium glycinate 250-400mg once a day
    2. Aspirin 81mg once a day
    3. Zinc 30-45mg once a day
    4. Fish oil 1000mg once a day
2. Bloating: simethicone (Gas X 1-2 tablets every 8 hours)
3. Headache and cramping: acetaminophen 650mg every 4-6 hours. Acetaminophen is preferred over NSAIDs (ibuprofen,

naproxen) during training and racing because of the risk of rhabdomyolysis.

Should an athlete want to mitigate PMS symptoms using any of the above-noted methods, they should reach out to their physician regarding the use of any medicine or supplement.

Menopause and Perimenopause

Strictly defined, menopause is the last menstrual period. It represents the end of a woman's reproductive years as her ovaries run out of eggs. Now the cells in the ovary are producing less and less hormones, and menstruation eventually stops.

Perimenopause is the period preceding, and just after menopause and, on average can last one to four years. In industrialized countries, the median age of onset of perimenopause is 47.5 years. However, this is highly variable. It is important to note that menopause itself occurs on average at age 51 and can occur between ages 45 to 55. The time to one's last menstrual period is defined as the perimenopausal transition. Often the transition can even last longer, five to seven years.

Symptoms of Perimenopause

The changes in hormone levels can lead to a varied set of physical and emotional symptoms. As with menstruation in younger life, all women will experience perimenopause differently. While symptoms are common in perimenopause, women will experience different combinations of the symptoms below, and to varying degrees:

Progesterone Deficiency Symptoms Bleeding

- Short cycles
- Heavier flow
- Premenstrual spotting

Premenstrual Syndrome (PMS) – often seen during pre-menopause:

- Moodiness
- Hot flushes

- Depression
- Poor concentration
- Irritability
- Anxiety
- Headaches (menstrual migraine)

Estrogen Excess Symptoms

- Bleeding (too much, too long, too soon)
- Erratic Breast tenderness
- Bloating
- Headaches
- Weight gain
- Vaginal discharge
- Estrogen Deficiency Symptoms
- Hot flushes, night sweats, and heat intolerance
- Insomnia
- Fatigue
- Headaches

Although the symptoms listed above are common during perimenopause, not everyone will have all of these. Women may have some symptoms, and they may have few or no symptoms.

Symptoms of Menopause

Estrogen plays a role in multiple organ systems. As estrogen drops during menopause, a woman may have:

- Changes in mood and irritability
- Decreased melatonin production → , sleep disruption
- Hot flashes
- Loss of elasticity and thinning of the skin
- Loss of menstruation and ovulation
- Decreased bone density
- Body fat redistribution: Estrogen plays a role in fat storage; decreased estrogen leads to increased visceral adipose Higher visceral adipose tissue is directly related to developing cardiovascular disease and diabetes (1009)
- Decreased muscle mass (sarcopenia): Estrogen sustains the number of satellite cells (muscle stem cells). As estrogen decreases, there is less stimulus for muscle synthesis.

Starting at menopause, a female will lose 50% of muscle fibers by age 80 (1010).

- Impaired glucose metabolism: Estrogen plays a role in insulin Less estrogen means blood glucose increases from insulin resistance. As estrogen decreases, the body has an impaired ability to process fructose, causing increased fatty acids in the bloodstream. (1011). This provides a higher risk of developing diabetes.

Hormone Replacement Therapy (HRT)

Hormone replacement therapy is used both during perimenopause and menopause to help alleviate the symptoms of decreased estrogen, such as hot flashes and vaginal dryness. There are two types:

1. Estrogen only (topical cream/gel, patch, pill, vaginal),
2. Estrogen plus progestin (pill). It is recommended to take HRT for the shortest time possible (1007).

HRT can provide several benefits for perimenopausal women beyond preventing pregnancy. These benefits are listed below. In addition to potential benefits, the use of hormonal contraceptives by perimenopausal women has been associated with an increased risk of blood clots. It may be associated with some increase in the risk of heart attack, stroke, and breast cancer.

Perimenopausal women should not use estrogen-containing contraceptives if they smoke or have a history of estrogen-dependent cancer, heart disease, high blood pressure, diabetes, or blood clots. Otherwise, hormonal contraceptives appear safe for healthy women over age 35, provided they do not smoke. Taking hormonal contraceptives can mask signs of approaching menopause, including menstrual irregularities. This can make it challenging to know when perimenopause is occurring.

The use of any combination estrogen-progestogen contraceptive will result in withdrawal bleeding (i.e., bleeding like a menstrual period) even after menopause. Some providers tell women to stop hormonal contraceptives at age 51 (the average age when menopause occurs). Still, this strategy is not always appropriate since not all women will have reached menopause by that age and will still need birth control.

Benefits of oral hormonal contraceptives beyond birth control:

- More regular menstrual cycles
- Reduced menstrual bleeding (and lower rates of iron deficiency anemia as a result)
- Decreased uterine pain during menstruation
- Reduced risk of ovarian and uterine cancer
- Reduced hot flashes
- Maintenance of bone strength
- Improvements in acne (which can flare up around menopause)

Training in Perimenopause and Menopause

Perimenopause training depends on perimenopausal symptoms: sleep disruption, and hot flashes. An athlete may need to decrease training intensity if recovery is impaired from menopausal symptoms (especially sleep disruption).

During menopause, it is recommended to lift HEAVY. Heavy load stimulates satellite cells to preserve muscle mass and stimulate Type II muscle fibers. It also decreases intramuscular fat deposits.

High-Intensity Interval Training (HIIT) is another way to train in menopause is HIIT increases insulin sensitivity. Multiple studies have shown that HIIT improves fasting glucose and insulin sensitivity in pre-diabetic patients. It decreases central obesity more than moderate-intensity continuous exercise (1012).

Adding balance work during menopause also preserves neuromuscular connections.

Summary

- The three phases of the menstrual cycle are follicular, ovulation and luteal.
- Estrogen and progesterone are the dominant female sex hormones
- Hormonal contraceptives (HCs) suppress the natural endogenous production of estrogen and progestin to prevent ovulation
- Low energy availability: meaning that energy intake and expenditure are not in balance, can lead to irregular periods

or loss of menses entirely, as well as problems with bone health

- This new terminology Red-S for “Female Athlete Triad” includes amenorrhea, osteopenia, and disordered eating
- Approximately 75% of female athletes experience adverse side effects due to menses
- Menopause is the last menstrual period
- Perimenopause is the period preceding, and just after menopause and, on average can last one to four years
- Hormone replacement therapy is used both during perimenopause and menopause to help alleviate the symptoms of decreased estrogen

Module 15: Periodization and Program Development

When one thinks of an endurance sports coach, regardless of the sport discipline, creating training programs is likely near the top of the list of responsibilities. While creating and updating training programs is just one aspect and responsibility of a coach, it is an important one. Moreover, often when an athlete seeks out a running coach, the primary thing they are looking for is a training program. Because of the fundamental importance to the training process, understanding the science behind periodization and how it relates to creating a safe, effective and progressive training program is critical for a running coach.

Periodization

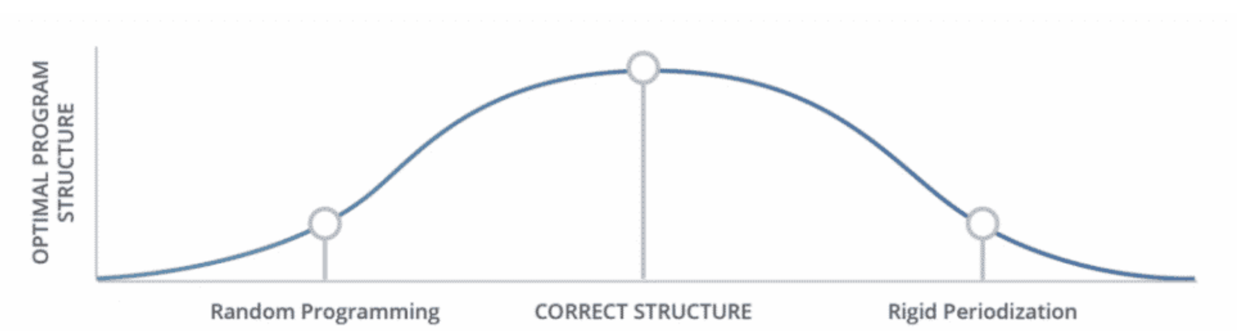
At the core of any effective training program is structure. A popular and well-known training structure is **periodization**. Periodization involves a progression of several cycles of training periods to get an individual to peak performance level by a particular date or time.

Each training period has a specific focus (e.g., increasing aerobic capacity). **The key element of periodization is that each training period builds on the prior training period.** Periodization also focuses on alternating stress and recovery phases to allow for the desired physiological training adaptations. While periodization is most often applied to training programs at least one year in duration, it is also applied to event-specific training programs. Periodization has garnered its fair share of criticism as being rigid, outdated, and flawed.

The opposite of periodization is “winging it” and randomly assigning workouts without much, if any, reasoning behind them. While this sounds crazy, this is often the training structure chosen by coaches who lack the knowledge of how the body functions. Often, these coaches apply the same training standards, structure, and protocol to all of their athletes. The advice and programs given to these athletes are often based on what the coach personally does or what they recently read in a magazine. This, of course, is not the advised way to structure a training program.

What Is the Right Approach?

This is the million-dollar question that has not and likely cannot be answered because the “correct” program structure is based on an athlete’s individual needs. That said, as with most things related to human performance, the optimal training structure can likely be represented as a bell curve – as noted in the figure below.



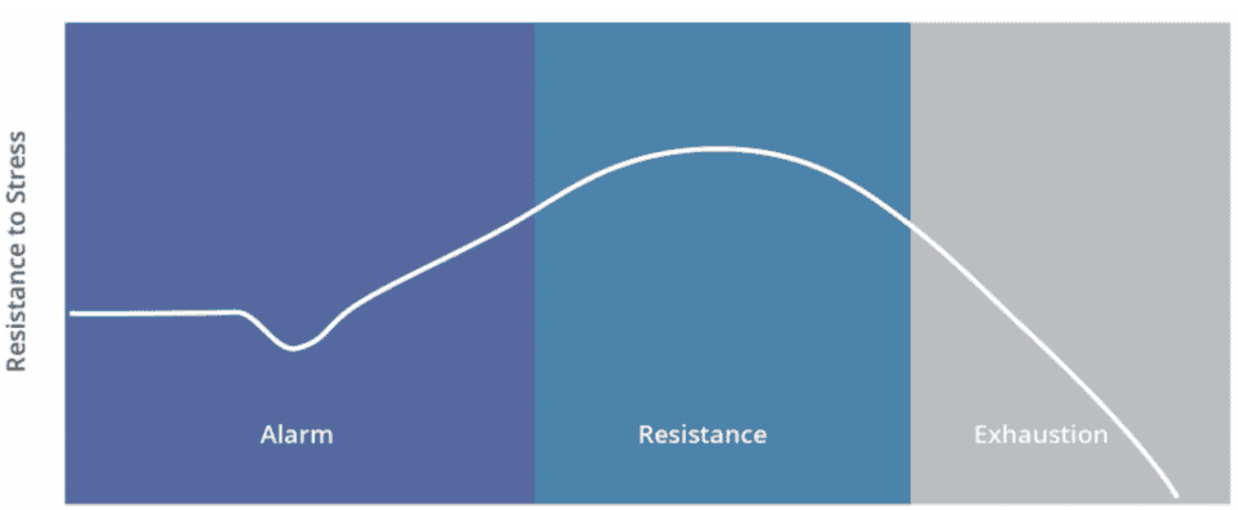
As you can see from the above image, the correct structure is not a finite point on the bell curve but rather a general area – meaning there is quite a bit of room for interpretation and customization.

The correct training structure must be based on an individual, not a predefined set of parameters. While it is fine to have phases within a training program that emphasize a particular type of physiological adaptation, these phases should not be rigid. They should exist more to provide general direction to a training plan than strictly dictate its content.

Periodization is a polarizing topic. As such, this module will discuss the different models of periodization and the pros and cons.

General Adaptation Syndrome

To understand how the body becomes stronger, it is important to understand how it adapts to stress. This can be traced back to findings from Hans Selye, an endocrinologist. He found that when stressors were placed on mice in a laboratory setting, all the mice demonstrated similar response levels. This adaptation is called **general adaptation syndrome (GAS)** (96).



GAS comprises three stages (96):

1. **Alarm:** When alarmed, the body immediately responds to fight or flight, shifting its resources to muscular and emotional needs.
2. **Resistance (also called *adaptation*):** The body adapts to stressful situations and becomes stronger.
3. **Exhaustion:** If high-stress levels persist for too long, the body will begin to break down.

When a runner trains or races, the body recognizes the effort as a stressor and adapts to deal with the stress. The goal of an athlete is to avoid the third category of GAS: exhaustion (i.e., overtraining). **Overtraining occurs when an individual continues to stress the body without adequate rest.** As a result, the body will eventually break down, decreasing performance. This is a crucial aspect of effective endurance-sports training, as athletes respond differently to the stressors put on them. For example, some athletes respond best to multiple back-to-back hard training days, while others respond best to at least one recovery day between hard training days.

It is important to note that the resistance stage of GAS differs in time and intensity depending on the athlete. For example, a professional runner would likely have a longer duration in the resistance phase than a recreational runner, and the professional would certainly have a higher effort level.

Having at least one day off between exercising the same muscle group via strength training is standard. What is not as well-known is how many days an endurance athlete should do cardiovascular training before taking a day off or adding an easy day. To answer this question, two primary factors need to be considered.

1. Intensity and/or distance of the training session
2. Recovery rate of the athlete

The harder the intensity or the longer the distance of a training session, the more required rest. **The level of fitness largely influences the recovery rate of an athlete.** The better conditioned an athlete is, the faster the recovery rate will typically be.

The optimal amount of rest required between training sessions is best ascertained through trial and error. The most important thing to be aware of is that athletes will respond differently to stress; therefore, training programs must reflect this.

The diagram below illustrates the basic concept of stress and recovery in relation to the construction of a training program.

Rest periods (decrease in volume/intensity) are integrated into a program to allow the body to recover and profit from periods of stress (i.e., intensity/volume). This, in turn, enables an athlete to attain a higher fitness level. This results in further increases in volume and intensity.

Getting Started: Easy Before Hard

While training programs differ in construction, a general theme is that training programs should progress from easy to hard regarding intensity and distance. As noted previously, a quote by Jay Johnson sums up the rationale behind this training practice:

"Metabolic fitness precedes structural readiness."

This means that individuals' cardiovascular adaptation occurs at a faster rate than their musculoskeletal adaptation. For example, from a cardiovascular standpoint, a runner might have no problem running for an hour at an intense level. However, their muscles and connective tissue might not be able to support or sustain this effort.

The overall trend when creating a training program should focus on proper progression. This is not to say that hard efforts cannot or should not be fused into the initial stages of a training program. However, they should be integrated intelligently.

Mount Everest Analogy

Perhaps the best analogy regarding the theory behind the creation of a training program is the method used to climb Mount Everest.

Because of the elevation, the mountain is climbed in stages. This allows climbers to acclimate properly to the altitude. Like altitude acclimation, training program progression is done in stages to properly progress athletes to their goals. If athletes were to progress too fast, they would risk injury and lack the required fitness level to perform on race day.

In mountaineering, there is an elevation termed the “death zone” (higher than 26K feet). Humans can stay at this elevation only for short periods before being forced to descend to a lower altitude … or die (303). The death zone is analogous to maintaining too high a volume/intensity based on an athlete’s fitness level.

While death is not the result (let’s hope not!), expanding the time spent doing high mileage and/or intensity beyond what the body can handle at a given point may result in long-lasting effects, including injury and extreme fatigue (i.e. overtraining).

Periodization Models

The concept of periodization was developed during the 1950s in the Soviet Union as a way to train for the Olympics. After the Russians did well at the 1952 Helsinki Olympics, it put periodization on the map as a novel and groundbreaking way to train. Because of this attention, periodization was established as a formal training practice in 1964 by Leo Matveyev. The establishment of periodization as a training practice primarily resulted from Matveyev's book, *Fundamentals of Sports Training* (312). Matveyev's periodization training principles were later adapted for use in Western countries in the mid-1980s. Tudor Bompa is often credited with being the founding father of modern periodization and, more specifically, responsible for bringing the concepts of periodization to Western countries (94, 95).

Specific to running, Arthur Lydiard created a periodization model (linear) that many runners have followed throughout the years.

Following are discussions of the three primary periodization models: classic (traditional), block, and intermediate (hybrid).

The goal of this section is not to recommend a specific periodization model but rather to educate you on the various models so that you can determine what model would be best for a particular athlete.

Classic Periodization

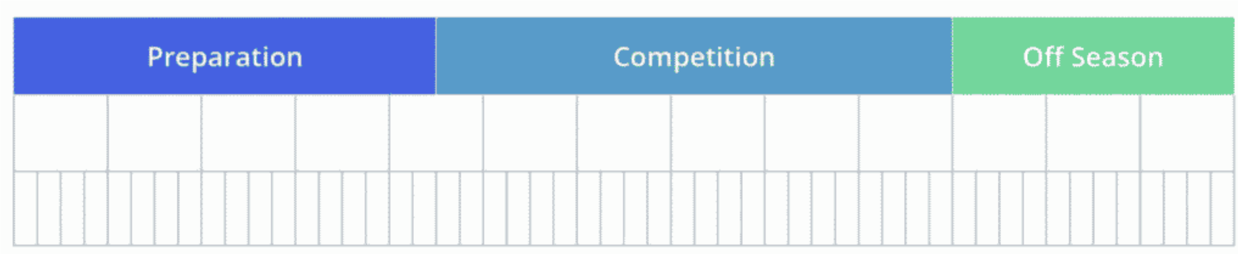
The classic periodization model (often termed Linear Periodization) is based on a one-year schedule with typically one peak. The year is broken down into cycles (94).

1. **Microcycle:** Typically represented by a week
2. **Mesocycle:** Consists of two to six micro cycles
3. **Macrocycle:** Refers to the annual training program and is broken into three phases:

    1. **Preparation**: represented by training done to prepare for the competition phase
    2. **Competition**: typically designed around one or two competitions and also includes the taper periods
    3. **Transition (off-season)**: used to recover physically and mentally

The image below illustrates the classic periodization model.

- **Bottom Line:** microcycles
- **Middle Line:** mesocycles
- **Top Line:** phases of macrocycles



The areas of classic periodization are as follows:

1 – **General Conditioning:** varied low-level fitness routines
2 – **Base Fitness:** sport-specific low intensity, increasing volume
3 – **Build Fitness:** integration of higher intensity, volume increases
4 – **Peak Fitness:** intensity continues to increase. While the overall volume may stay the same or decrease, this phase might contain the highest volume days (ex: long run).
5 – **Taper:** volume substantially decreases, periods of short intensity
6 – **GOAL EVENT**
7 – **Recover:** typically characterized by the transition season; similar to general conditioning

The overall trend of classic/linear periodization is that overall volume steadily decreases throughout the program while intensity steadily increases.

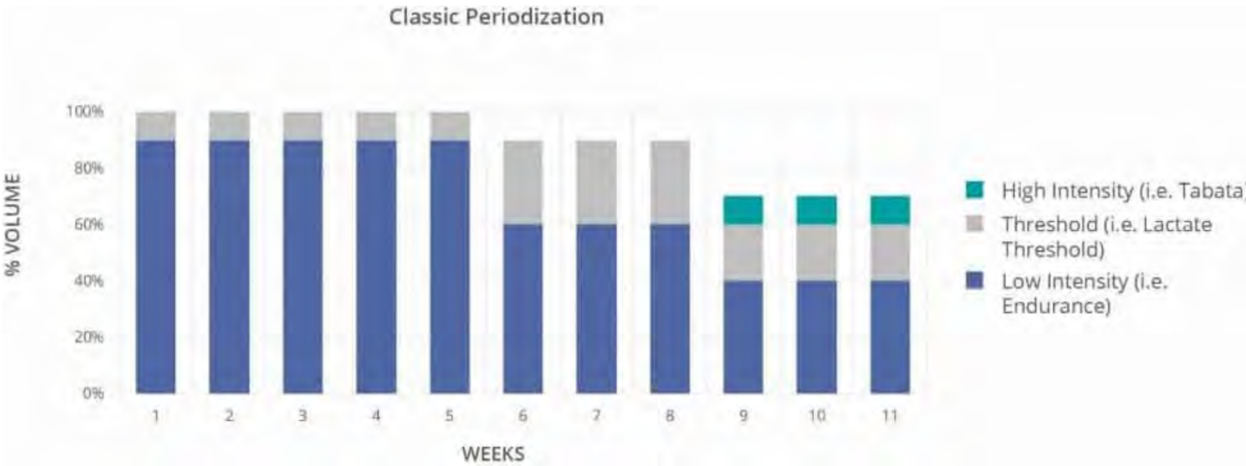
Below represents how these training cycles fit into the classic periodization model.

Preparation	Competition	Transition
General Conditioning	Peak Fitness	Recovery
Base Fitness	Taper	
Build Fitness	Goal Event	

In the video below, Ben Rosario discusses the importance of the transition season (i.e., off-season).

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

One of the hallmarks of classic periodization is that several fitness workloads are developed at the same time. The below chart visually denotes the simultaneous development of multiple fitness workloads. This chart does not accurately represent a training season as it is used solely to represent concurrent training workloads.

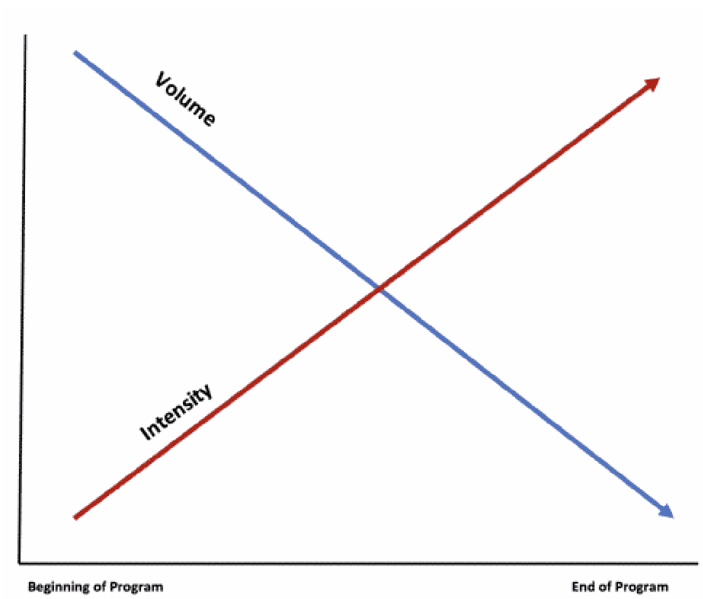


Lydiard Periodization

As noted above, Arthur Lydiard was a New Zealand running coach largely known for his linear periodization model. More specifically, Lydiard created a training (periodized) pyramid that many runners use to this day. This pyramid is illustrated below:

Regarding the above image, while most of the titles are selfexplanatory, hill training could be best described as the 'strength' training phase, and the integration phase could be best described as race-specific training.

As you can see from the above illustration, the Lydiard pyramid is based around a single season peak. In many respects, it reflects a 'block' periodization model due to its focus on a singular physiological adaptation at a time approach. In this sense, the Lydiard periodization is not a 'true' classic periodization model. However, as you will read below, a block periodization model allows for multiple peaks throughout the year and thus shorter, concentrated areas of focus than the Lydiard model.



Traditional (Linear) Periodization

The chart above illustrates the general trend of classic (linear) periodization. As you can see, from the beginning of the program to the end, there is a trend of decreasing volume and increasing intensity.

Criticisms of Classic Periodization

Initial criticism of classic periodization came from Dr. Yuri Verkhoshansky, a sports scientist deemed the "father of plyometrics" (233). Dr. Verkhoshansky's criticism of Matveyev's classic periodization model resulted in the creation of his own periodization model, *block periodization*, which is discussed in detail in the next section.

Dr. Vladimir Issurin is another detractor of the classic periodization model. Dr. Issurin is a sports scientist working with the Israeli Olympic Committee's Elite Sports Department (231, 232).

While Dr. Issurin and Dr. Verkhoshansky had many points of contention with the classic periodization model, the issues most applicable to endurance sports training are (233):

1. Originally developed for the 1952 Helsinki Olympic Games and constructed around one peak versus multiple peaks
2. Inflexible
3. Too many areas of simultaneous focus that result in insufficient training stimuli and less-than-desired outcomes
4. Incompatible workloads (i.e., high, low) result in conflicting and diminished physiological responses.

Of these points of contention, #3 and #4 represent the primary focus areas. When these two points are summarized, what is being proposed is that classic periodization is equivalent to multitasking. More specifically, too many areas of different intensity levels are being trained simultaneously, effectively negating the desired physiological responses.

While Dr. Issurin is largely critical of classic (linear) periodization, he notes in his book *Block Periodization: Breakthrough in Sport Training* that classic periodization is appropriate for beginner to intermediate athletes but not elite/advanced athletes (314).

Block Periodization

Dr. Verkhoshansky developed the theory of block periodization to correct what he viewed as flaws in the classic periodization model. **The genesis of this model is that multiple areas of focus are not addressed simultaneously but rather individually over a set period.** Block periodization is characterized by a series of connected groups or "blocks" (essentially *mesocycles*). Each block has a primary focus.

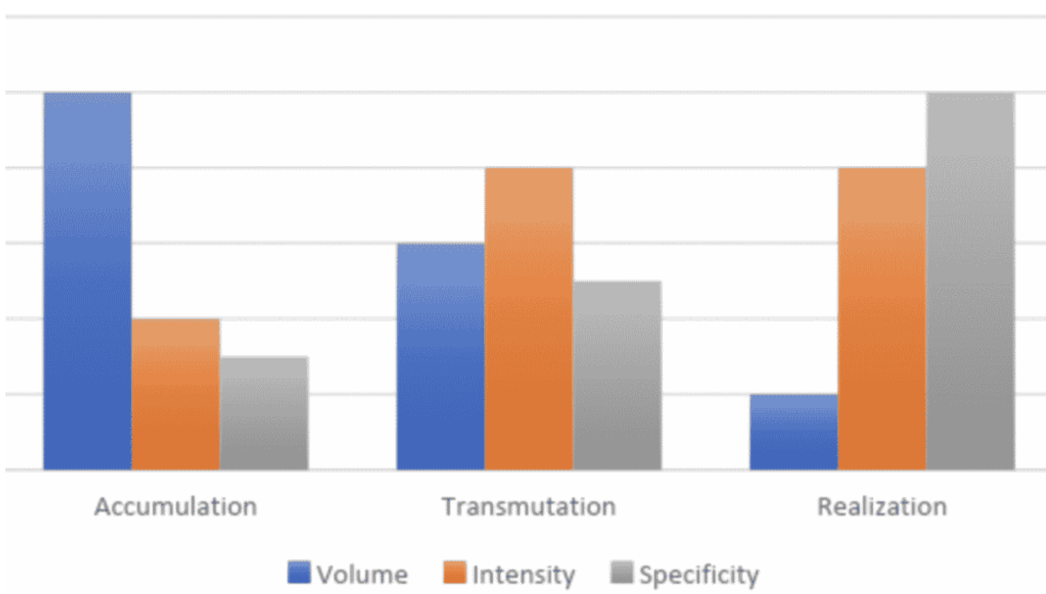
According to Plisk and Stone, 2003 (230), there are two types of blocks: ***accumulation*** and ***restitution***. During an accumulation block, one area of fitness is emphasized to the extent of intentional overreaching. The point is to stress the body with one aspect of fitness (i.e., high volume) over a set time duration – usually four weeks. Toward the end of the training block, fatigue typically sets in because of the extreme workload.

Following the accumulation block is the restitution block. This block looks to benefit from all of the hard work performed during the accumulation block. During this block, the fitness aspect focused on during the accumulation block is significantly reduced, while another fitness aspect is introduced at a moderate level (i.e., speed/intensity) (230).

This model by Plisk and Stone has been further developed to include three phases: Accumulation, Transmutation, and Realization. In this model, the accumulation phase (2-6 weeks) is akin to the base training phase, albeit shorter than a linear periodized base training phase. Therefore, the accumulation phase (2-4 weeks) focuses on low intensity and high volume. During the transmutation phase, the focus is on high intensity, typically focused on developing a singular energy system. The realization phase (7-14 days) or more commonly referred to as the 'taper' phase, is where the primary focus is on low volume and rest, with small doses of high-intensity workouts.

All three phases comprise a singular training block.

The image below denotes the areas of focus (i.e., volume, intensity, specificity) for each phase.



Hybrid (Intermediate) Periodization

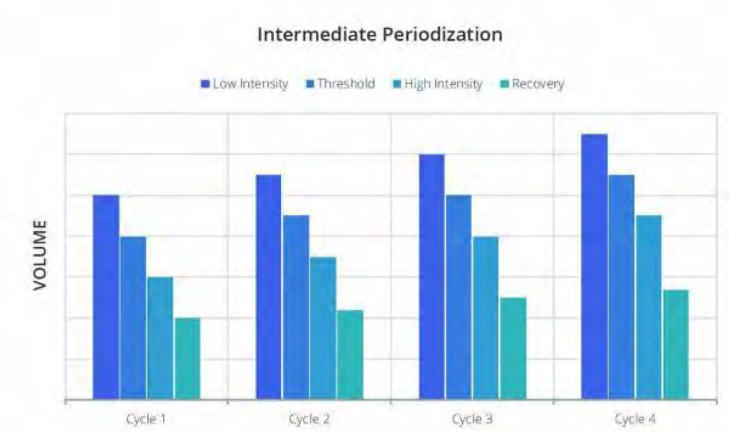
Based on the information regarding classic and block periodization, there are two constants:

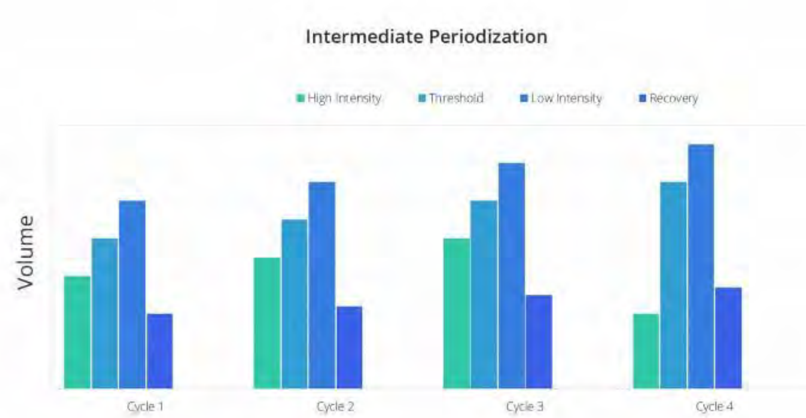
1. Classic periodization is structured for beginner and intermediate athletes
2. Block periodization is structured for advanced/elite athletes

As these two periodization models (classic and block) are at opposite ends of the spectrum, the question arises: Is there a model incorporating aspects of both models? Yes, a hybrid (intermediate) model.

Plisk and Stone reference an intermediate periodization model that uses a ***summated approach*** (230). This approach focuses on fitness versus fatigue and views each as having opposing effects on the body. Therefore, fitness = good, fatigue = bad. A summated approach looks at the sum of fitness and fatigue to maximize fitness and minimize fatigue. **A successful program based on this approach relies heavily on fatigue management practices**. A summation periodization strategy is believed to offer dual benefits of reducing the chance of overtraining and increasing fitness as the athlete trains in multiple aspects of fitness in parallel throughout the mesocycle (235, 236, 237).

The intermediate model theorized by Plisk et al. uses four-week mesocycles. The volume or intensity increases each week during the first three weeks (microcycles) of each mesocycle. The fourth and final week of the mesocycle is an “unloading” or recovery week characterized by a decrease in volume and intensity. This allows the body a chance to recover for the following week, which is the start of the next mesocycle. The following 16-week sample charts illustrate the basic concept of the Plisk et al. intermediate periodization model. Note each cycle represents four sequential weeks.





Criticism of Periodization

The concept of periodization has been widely criticized for being antiquated, rigid, and lacking a scientific and practical foundation. **In other words, detractors of periodization believe that it fails to meet the reality of training and competition.**

Additional criticisms of periodization include (782):

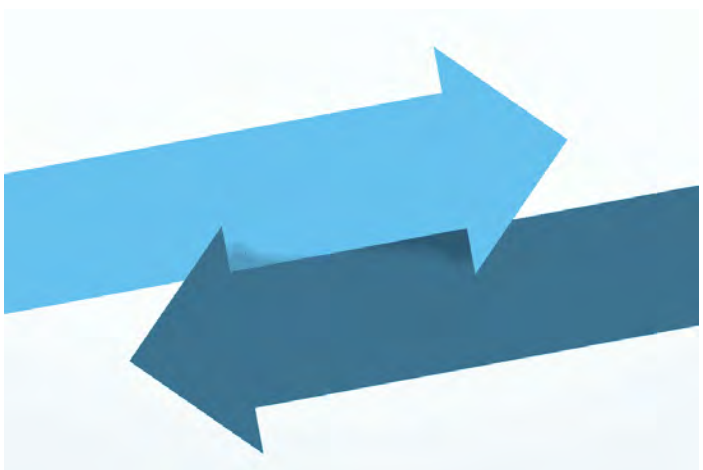
- Not based in science/physiology
- Does not consider the improvement of athletes and therefore slows the rate of improvement
- Rigid structure that is broken up into predefined periods and cycles

- Complex and confusing principles and guidelines
- Not practical in real-world training applications
- Arbitrary organization

It may be true that if just one type of training were focused on for a set period (i.e., endurance), an individual would see the most significant gains in that area compared to training multiple areas of fitness simultaneously. However, fitness elements and, more to the point, training aspects do not exist in isolation. **UESCA does not subscribe to the belief that multiple fitness areas cannot be trained within the same period because of diminished training adaptations.**

UESCA believes that multiple areas of focus should be trained simultaneously to ensure that a runner is well-rounded and not overtrained in any one area. This is not to say that there cannot be a primary focus, but only performing one type of training stimuli for a particular duration of time may not be advantageous for a runner. **The most important aspect of a properly constructed training program is that an athlete has individually based stressors placed upon them, followed by some recovery duration.** This process is repeated over and over to achieve the desired result.

Which Way?

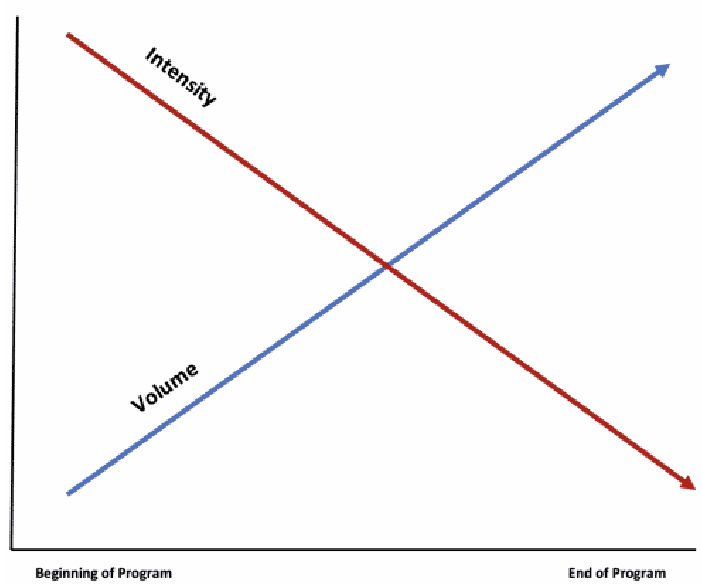


One of the key principles of training program construction is 'Least to Most Specific.' This means that you focus on the aspects least specific to the goal farthest away from the goal race and the most important aspects the closest to the event. As you can see, what is focused on at what point(s) in the training program will vary based on the race being trained for. It also varies based on the level of the runner. For example, a short race such as a 5K would likely focus on endurance toward the beginning of the program, and intensity closest to the event, due to the short distance. On the other end of the spectrum would be an ultramarathon which would place intensity toward the beginning of the program and a focus on distance/volume closer to the goal event.

Using the 5K and ultramarathon examples above as being on opposite ends of the spectrum, a half and full marathon would be somewhere in the middle. For most people, both of these events (half and full marathon) would be considered long-distance events and thus, would have a focus on intensity early in the program and volume later in the program. However, for many seasoned or elite runners, these distances do not represent 'long' distances as it relates to their ability to run those distances. In other words, they are not training to be able to run that far, they're training to race those distances as fast as they can. For these runners, their programs might be structured more like that of a regular runner's 5K program – meaning a focus on volume toward the beginning and a focus on intensity toward the end.

Reverse Periodization

As traditional (linear) periodization places a focus on volume at the beginning of a program and intensity toward the end. Flipping these focus areas in terms of where they are in the program is often termed 'Reverse Periodization.' The chart below illustrates the trend from beginning to end, where intensity decreases while volume increases over time. Following the 'Least to Most Specific' programming rule, the below model of reverse periodization would be most applicable to long events such as marathons or ultramarathons where the most specific aspect of training is volume versus intensity.



Reverse (Linear) Periodization

Periodization Summary

Conceptually, periodization is on the right track – to progress an athlete from a relatively low level of conditioning/intensity/volume to a high level in a systematic, progressive manner. This is the genesis and purpose of all training programs. So, what's the issue?

The problem is not the goal of periodization but rather how an athlete progresses and the time frame in which the progression occurs.

Perhaps the biggest issue is that periodization forces athletes into the same restrictive training parameters. What if athlete A attains the best results when performing high-intensity training, whereas athlete B realizes peak performances when training at or right below lactate threshold? Concerning periodization, if these two athletes were in a mesocycle that focused on below-lactate-threshold training, athlete A would likely progress slower than athlete B.

The correct way to progress an athlete is by taking an individual approach. **While it's fine to have a general theme for a training duration (i.e., general fitness), it should not be overly restrictive**, meaning it's fine to allocate the month of March to general fitness. However, the program should not state that every training session must be below XYZ heart rate. It's fine to have an overriding theme for a particular duration. However, it's not realistic, practical, or advantageous to have only one type of training intensity for an extended period. Training guidelines should primarily be directed by assessment checkpoints, an athlete's training history, and the individual's physiological response to training.

Therefore, **the proper way to progress an athlete through a training plan is to schedule a baseline assessment and periodic assessment checks** (i.e., training volume, physiological assessments, timed distances, etc.) throughout the plan to ensure that an athlete is on target. This ensures that it will be quickly identified and rectified if an athlete is behind schedule with preparedness.

Recovery Considerations

Travel

Many athletes travel to other time zones in their country or internationally to compete and for work and leisure. The farther away from the race or location is from their time zone, the more it will negatively affect an individual's Circadian Rhythm. In a 2016 study by Simpson et al., it is noted that traveling east more than two time zones wreaks the most havoc with one's Circadian Rhythm due to their wake-up and bedtime being shifted later in the day (976). This is termed, *'Circadian Misalignment.'* The same study noted that it takes approximately one day to adjust to each time zone crossed. The below strategy is recommended to combat jet lag due to Circadian Misalignment.

- Upon arriving at the destination, synchronize one’s watch and meals to the new time zone. Concerning meal timing, this might be the most effective as gastrointestinal tract signaling is an important factor in resetting one’s Circadian Rhythm.

Program Considerations

Determining the frequency of rest days in a training program can be challenging when considering all other training program variables. This section discusses how to best approach implementing rest and recovery days in a program.

An athlete’s ability to recover must be assessed on an individual basis. Unless an individual responds best to back-to-back hard days (discussed below), a rest day typically follows a hard day. A recovery day may constitute an easy run or a day off. “Hard days” traditionally include any form of training that puts a lot of stress on the body (ex: long runs, intervals).

“Hard days” are subjective. For a professional runner, running 10 miles (16.1 km) in an hour may not constitute a “hard day,” while for most everyone else who does not run for a living, it most certainly is!

Because of the individualized recovery rate, it is difficult – if not impossible – to decipher the optimal amount of recovery (in days) after a hard workout. There are so many variables that the best way to determine this is based on the history and progression of an athlete.

One day’s rest after a hard day might not be sufficient for an individual to be recovered from a physiological standpoint. Think about it. A marathon taper is typically two to three weeks in duration. Why? The theory is that it takes this long for the body to recover from all the training and build proper glycogen stores. It is highly probable that physiologically, the body is not completely recovered one or even several days after a hard workout. Also, keep in mind that fatigue is cumulative. **As a result, it is not just a singular hard workout that the body is recovering from but rather the total workload of training over a period of time.**

Therefore, unless an athlete is tapering for a race, the goal is not to be fully recovered physiologically but rather recovered enough to perform a prescribed workout at the desired pace/distance without risking injury or illness.

Trial and Error

The amount and type of rest will vary by the individual. **Some athletes might respond best to a whole day off, while others might respond best to active rest** (e.g., an easy run). Additionally, some individuals may respond best to two consecutive days off followed by two hard days, whereas others may respond best to just one day off followed by one hard day.

Unless an athlete already knows this and passes this information along to you, the only way to find out is through trial and error. Tracking all this information, including sleep hours, will help you understand how an athlete most efficiently recovers.

While this certification discusses rest regarding rest days or active rest days, rest should be viewed as a function of energy expenditure and muscle activation. Just because an athlete is not training on a rest day does not necessarily mean the person is resting. For example, spending the day at a concert standing in one place for hours or taking a four-hour walking tour of NYC are not rest days! If an athlete is serious about training, they must always consider energy conservation.

The more stress placed on the body – whether from racing, training, lack of sleep, or extended work hours – the more rest is required. Additionally, recovery time will vary based on many factors, such as injury, fitness level, muscle soreness, etc.

Hours, Not Days

It is advantageous to think of training days in terms of hours, not days. Let us say an individual gets off work at 9 p.m. and trains from 10 p.m. to midnight. The next day they have an after-work dinner, so they train in the morning from 5:30–7. While the training sessions are technically on consecutive days, as their training program states, only 5.5 hours passed between the end of one training session and the start of the next. This is not enough time between training sessions and, in this case, certainly not enough time for adequate sleep. It is ill-advised to perform evening-to-morning workouts on consecutive days or same-day workouts of high intensity and similar nature (strength to strength or cardiovascular to cardiovascular).

Back-to-Back Days

As athletes adapt to the same training stimulus differently, some runners may respond best to back-to-back hard days followed by one or more easy days. The exact number of recovery days that follow are based on the individual's recovery rate. However, if three or more recovery days are required, a program should not implement back-to-back hard days.

This type of training should be attempted only by a runner who recovers well daily.

Overtraining Syndrome

Contributor: Alexandra Coates

While everyone has heard of the terms "Overtraining" and "Overtraining Syndrome," there is much confusion about what overtraining really is. In this section, we will define the components of the overtraining spectrum and touch on how relative-energy deficiency in sport may drive underperformance. To begin, "overtraining" is defined as the *verb* to describe the process of intensified training that could lead to "acute fatigue," "overreaching," or "overtraining syndrome". Therefore, overtraining can be used interchangeably with "overload" training.

While this section could be appropriately included in many of the lessons, it is included here due to the correlation between programming and training load and more specifically, programming appropriate training loads to minimize the chance of overtraining your athletes.

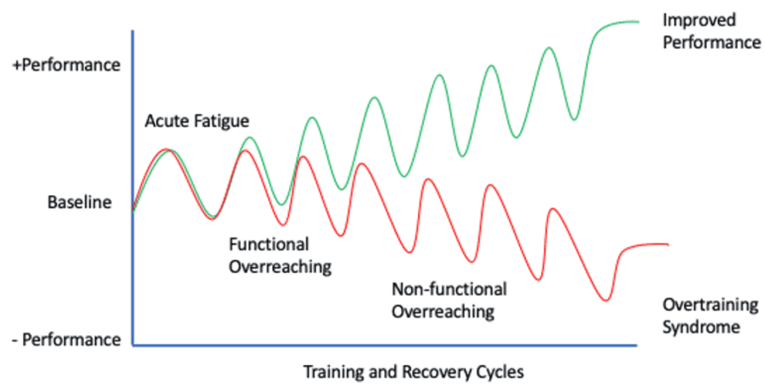
Acute Fatigue

It is well understood that a certain amount of overload training is required for training adaptation to occur. As the training load increases, the athlete should experience normal training fatigue, called *acute fatigue*. Acute fatigue should last a day or two upon recovery, and then there should be a micro-adaptation or *supercompensation.*

Functional Overreaching

Functional overreaching is different from acute fatigue primarily because there is a **decrease in performance**. While acutely fatigued athletes may feel tired, they can still perform if required, whereas functionally overreached athletes cannot perform to their normal abilities. This typically occurs following a two-to-three-week training camp or a particularly hard training block.

Functional overreaching takes about 1-2 weeks to recover from, but symptoms should resolve completely following appropriate rest. If adequate recovery is taken, there may be a performance super-compensation, or there may just be a return to the athlete's baseline level of performance (978-982). Eliciting acute fatigue with training has been shown to be a more reliable method of achieving super-compensation than functional overreaching (978), suggesting that overreaching is not required or even recommended for performance gains.



Overtraining Progression as described by Irene Margaritis, 2019.[7]

Non-Functional Overreaching

Non-functional overreaching occurs when athletes are not given sufficient recovery time following functional overreaching. Non-functional overreaching is different from functional overreaching simply because it takes weeks to months to recover from this state (once the athlete takes time off). The athlete will lose fitness due to the lengthy recovery period. Athletes may have difficulty determining whether they are having symptoms of overreaching or are simply unfit, which can be extremely frustrating. Non-functional overreaching should be avoided, and if an athlete is suspected of having non-functional overreaching, months of recovery may be required.

Overtraining Syndrome

Lastly, there is *Overtraining Syndrome* (OTS). The literature suggests that overtraining syndrome may not be as simple as training through a state of non-functional overreaching, as it likely requires some additional factors or circumstances in addition to training-stress for the syndrome to be initiated (977). It is believed that OTS may require additional factors simply because most athletes have likely experienced some form of overreaching. Still, only a very small percentage of athletes develop true OTS (977). The triggers of OTS are incompletely understood, but there may be a triggering virus, gastrointestinal issue, or excessive altitude/heat training stress, which may predispose the athlete towards developing OTS. OTS is defined as the “prolonged maladaptation of not only the athlete but several biological, neurochemical and hormonal regulation mechanisms” (977). It requires the exclusion of any other disease, nutrient deficiency, or other major disorders (977). The literature suggests that it takes months to years to recover from OTS and often spells the end of the athlete’s career (977). As these athletes seem particularly sensitive to training and life stress for many years afterward, most athletes do not return to high-performance sport after diagnosing OTS.

RED-S

Relative Energy Deficiency in Sport (RED-S) is not on the overtraining spectrum but shares several common overreaching symptoms. RED-S (984) is a progression of the Female Athlete Triad (985), which is the understanding that insufficient energy availability (i.e., not enough calories eaten compared to those expended) leads to a decrease in reproductive hormones, which disrupts the menstrual cycle, and which causes low bone mineral density leading to stress fractures (985). RED-S builds upon this model by suggesting that low energy availability can cause a host of other physiological symptoms, can drive underperformance and is not restricted to female athletes. RED-S and overreaching may overlap as it is very difficult to accurately assess calories in vs. calories out. Athletes may be training so hard that they are invertedly in a state of low-energy availability. Further, research hasn’t conclusively shown that overreached athletes aren’t underperforming simply because of an energy deficiency (986). However, while RED-S can be detected with standard blood tests, overreaching and overtraining syndrome rarely are identified with blood markers (987). As such, while there is without a doubt some overlap between RED-S and overtraining, there are also some differences which we will see in the symptoms section below.

Symptoms

To begin, mood states follow an inverse linear trend with training load so that the athlete will become more irritable and emotionally unstable as training load increases. A state of acute fatigue should feel like normal training fatigue, whereas athletes who are overreached will show more symptoms of depression and anxiety. Overreaching (functional and non-functional) is further characterized by a decrease in heart rate at exercise intensities over ~70% of max (not just at max) and a decrease in lactic acid production (978, 988) – resulting in the athlete not being able to sprint or push hard. Functionally overreached athletes also have increased sympathetic nerve activity, reduced maximal cardiac outputs, and increased arterial stiffness (980, 981). While coaches and athletes will not be able to detect all of these symptoms in training, they should be able to recognize underperformance in training and that training feels much harder than it should. If you use heart rate measures, heart rate recovery will be *faster* following a given effort (take HRR as the difference in HR at the end of exercise to 60sec post-exercise while the athlete is standing/sitting still). Waking heart rate variability may be higher or lower with overreaching but will likely be different from baseline (982). There isn't a resting blood marker that accurately predicts overreaching or OTS.

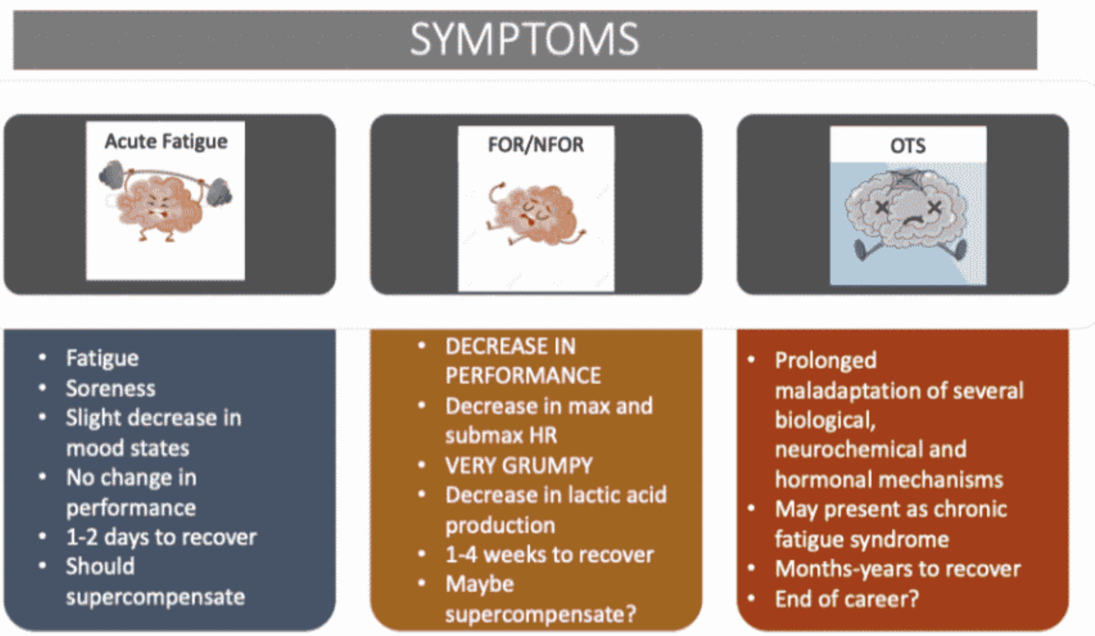
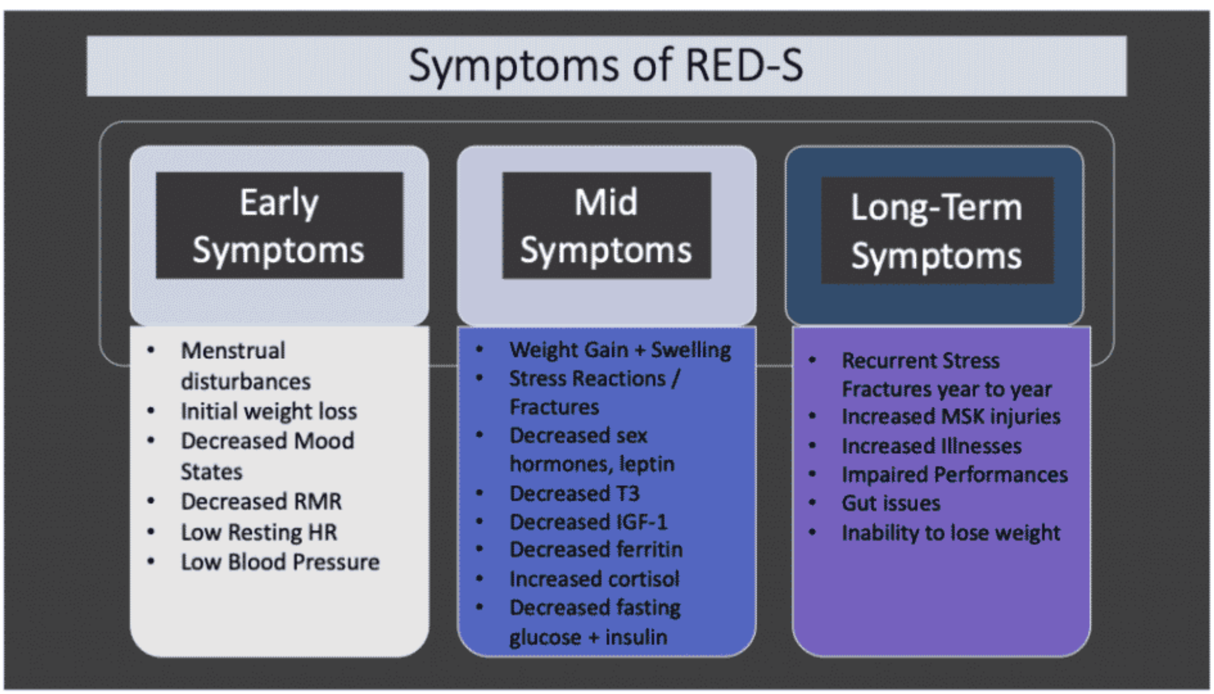


Figure 2. Symptoms of OR and OTS

Overtraining Syndrome symptoms are harder to pinpoint because due to the maladaptation of several physiological axes, there are a variety of symptoms individual to the athlete (977). Most athletes with OTS report extreme exhaustion, deep muscle and joint pain following even easy exercise, and strange disturbances to heart rate. Often these heart rate disturbances present as highly elevated heart rates during easy exercise and potential suppression of heart rate during strenuous exercise. Sleeping disturbances seem to be common with racing heart rates or anxiety that does not allow for sleep, or in extreme cases, athletes cannot stop sleeping. Athletes also report extreme brain fog, difficulty concentrating, anxiety, and depression. Many more symptoms likely represent the individual imbalances; however, standard blood test results are usually normal (987). Altogether, these symptoms are very similar to that of chronic fatigue syndrome.

Finally, RED-S is the easiest to diagnose as we have some clear symptoms and blood markers. The earliest marker of RED-S is suppression to resting metabolic rate, which means athletes with RED-S often gain weight rather than lose it, despite taking in fewer calories than they expend. There will be reductions in blood pressure and resting heart rate. In a standard blood test, there will be a combination of red flags such as low estrogen, testosterone, luteinizing hormone, follicle-stimulating hormone, T3, IGF-1, serum ferritin, leptin, fasting blood glucose and insulin, and increased LDL cholesterol and cortisol (985, 989-991). Finally, after a long period of RED-S, athletes will have reduced bone mineral density, likely resulting in stress reactions or stress fractures (984).



RED-S Symptoms

The below video by physiologist Alexandra Coates discusses Overtraining Syndrome in detail.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Training Program Development

Overview

A training program must be flexible and dynamic. Without a constant eye on an athlete's training process, good communication between athlete and coach, and modifications made to a program that reflect the athlete's condition, a training program quickly can be deemed ineffective. The quality of a coach is not so much judged by the ability to create a training program but rather by the ability to modify and manage it to get the most out of an athlete. This requires accurate and constant feedback from athletes to ensure programs are relevant and on track in relation to the goals.

Athletes will come to you with differing abilities, experience levels, goals, and starting points. Perhaps your athlete was working solo or with other coaches but not getting the desired results and is now three months away from the goal race. Conversely, your athlete may be new to running, unable to run more than a few miles and completely deconditioned. The problem with many training programs is that they make assumptions. Some of the more common assumptions are:

- Form and biomechanics are correct

- Base level of proficiency regarding skill and technique
- Training program starts at the beginning of a calendar year
- Training at a particular volume before the start of the program
- Prior running experience
- In good physical condition
- Training for a particular event

Most training programs focus on training volume. This is primarily because if an individual cannot complete distances in training near those of the race being trained for, the likelihood of successful completion on race day is slim.

This emphasis on volume versus intensity is typically greater with new and/or deconditioned runners.

Training programs range from generic to extremely involved and detailed. The training programs you create for your athletes are primarily based on their goals and should also consider their strengths and weaknesses.

An important thing to remember when developing a training program is that your athlete is not you! Many coaches design programs around what they like and what personally works for them. While this approach may work in some cases, it is important not to construct training programs this way. Individuals will have varying responses to the same training stimuli. This is a critical piece that separates a qualified coach from an unqualified coach. **A qualified coach will construct a training program grounded in science and customized for an individual athlete.**

Lack of Research

Despite the many articles in magazines and online regarding training programs for runners, there are a surprisingly low number of peer-reviewed research articles on this subject.

Therefore, most articles regarding pre-formatted programs and training program construction are either based on the foundations of exercise science and/or on the experience of coaches regarding their personal experiences and those they have coached.

This is not necessarily a bad thing. The training practices that coaches implement are often done because they know they work based on past experience, even if they don't know the physiological rationale behind the training methodologies. **In other words, the role of sports science is often to explain why a particular training methodology works versus developing a new and groundbreaking training theory.**

Because of this, most of the content in this module is formulated via applied exercise science and evidence-based findings from our team of advisers versus solely running-specific programming studies.

Preformatted Training Programs

Pick up any running magazine, and you will undoubtedly find that it contains training programs. They give specific plans to follow to achieve one's desired results. While one of these plans might work well for one person, it probably does not work so well for others. In other words, pre-formatted training plans are akin to a "one size fits all" model.

The primary trait that most pre-formatted programs have is a pre-determined starting training volume. For example, a marathon program might assume that an athlete has been running at least 20 miles (32 km) per week for the previous four weeks. This may or may not be the case. Regarding predetermined starting training volumes, this approach works only with someone not being professionally coached. Using the preceding example, an individual without a coach would train until reaching 20 miles (32 km)/week and then would begin following the program. However, in a coach/athlete relationship, **this buildup period of 20 miles (32 km)/week is part of the training program.** Therefore, the actual training program would be longer than noted in the pre-formatted plan.

This certification does not have pre-formatted training plans as part of the content. Why? **Pre-formatted programs simply do not adjust for an athlete's specific needs.** Following are several variables that cannot be accounted for with pre-formatted training plans:

- Recovery rate
- Accurate training volume
- Strengths/weaknesses
- Time availability
- Physiological adaptation to training
- Injury
- Scheduling conflicts

It is incorrect to force potentially hundreds of different training scenarios into just a few pre-formatted training programs. This leads to programs that are not individualized and, therefore, nowhere near as effective as they could be. **Training programs need to be individually designed for each athlete – period.** Pre-formatted training programs are a great place to get ideas but should not be used "as is" for athletes unless it happens to fit exactly with what a particular athlete needs.

Program Design Principles

Following are general trends concerning running program design:

1 – Deconditioned athletes generally start at a low volume and intensity level and spend the bulk of the beginning of the training program focused on building fitness and endurance.

- Athletes who begin a program at a moderate to high fitness level typically focus less on fitness and aerobic development than deconditioned athletes.
- Athletes with a relatively high fitness level at the inception of a training program typically integrate high-intensity training sooner than those with lower beginning-fitness levels.

2 – High-volume/intensity weeks are typically followed by a recovery week, especially with novice runners.

3 – While an athlete's ability to recover must be assessed individually, it is safe to say that after any hard training day, at least one recovery day should follow.

4 – As athletes progress through the training process, they should be able to handle a higher workload with faster recovery.

5 – Generally speaking, there should not be more than three consecutive weeks of increasing volume unless the training volume is low.

6 – Focus on developing physiology that is most race-specific closest to the race and conversely develop the least specific physiology furthest away.

7 – Focus on strengths closest to a race and weaknesses further away.

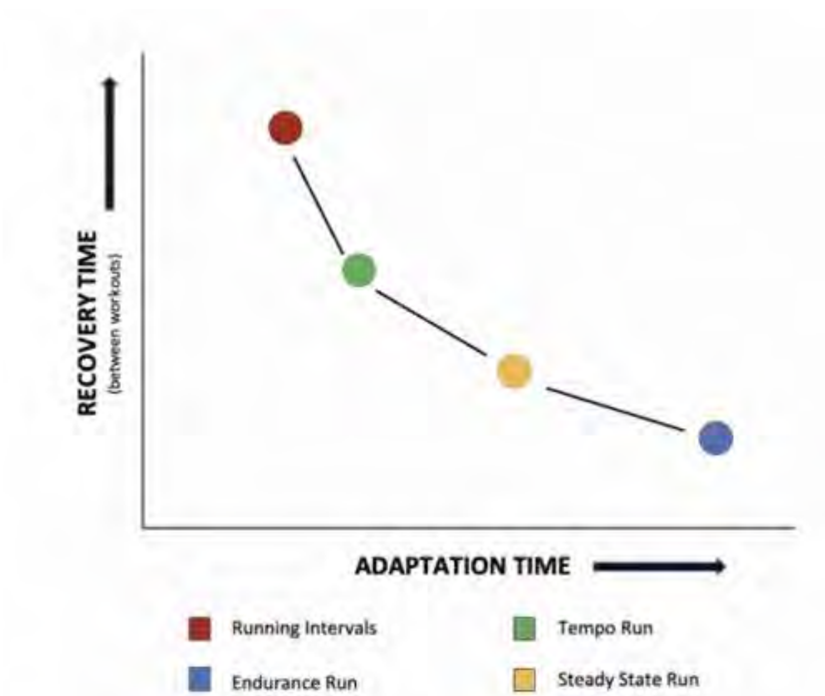
Trends six and seven are depicted in the visual below.

Common sense dictates that the most specific physiology to a particular race will not always align with a runner's strengths. So, what should you do as the coach if the most specific physiology to a race also happens to be a weakness of an athlete? Double down and start focusing on the athlete's area of weakness further from the race.

8 – During a training block, start with workouts with the highest training load and progress to workouts with the lowest load.

9 – The biggest bang for the buck workouts in respect to physiological adaptations are steady state runs, tempo runs and intervals. As such, these should be integrated into athletes' programs at some point(s).

The chart below denotes the four key workout types in respect to their associated recovery and adaptation times. For example, intervals have a short adaptation time, but due to the intensity, they require more recovery time than the other three workout types.



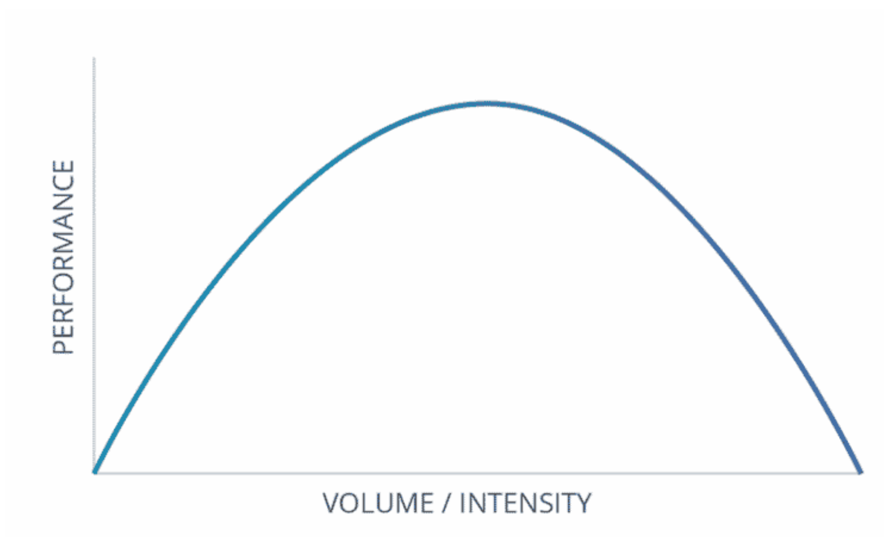
Do We Really Know What Training Volume Is Optimal?

To some degree, there is a trickle-down effect from professional runners to amateurs regarding training theory and methodology. As previously mentioned, this is largely due to magazine and online articles about how pros train. While most amateurs do not have the time to train at the same volume as the pros, and even if they did, most could not handle the volume without getting injured or overtrained. However, the message is clear: If you want to be the best, you must run a lot … period.

Is more always better?

Likely not. The following diagram illustrates that the ideal training volume and intensity exists on a bell curve, **meaning too much or too little volume and intensity will likely result in decreased performance.** The correct volume and intensity must be assessed and determined individually.

The main takeaway is that more is not (necessarily) better regarding training volume. It also needs to be stated that many runners run a lot of volume for one simple reason – they like to run! So long as they don't get injured or exhibit symptoms of overtraining, there is nothing inherently wrong with this. Still, it also might not be in the athlete's best interest from the standpoint of optimizing their training.



Not All Mileage Has the Same Value

According to running coach Jack Daniels, increases in training volume past a certain point follow the law of diminishing returns (720). Increasing volume to a certain point will substantially benefit a runner. However, volume increases past this point will have a reduced benefit. In other words, you don't get the same bang for your buck. This is a slippery slope as the benefits are reduced, and the chance for injury increases. This relates to the "sweet spot," as noted in the diagram above.

Conflicting Information

Historically, elite marathon runners have always trained at a high volume; therefore, it has been established that this is a prerequisite to running a fast marathon. At the elite level, the phrase *If it ain't broke, don't fix it* is very applicable. Changing a training program or strategy has few repercussions for an age-grouper. However, **for professionals, there is likely too much risk in drastically altering an annual training regimen.** This is likely why there is not a lot of research in the area of varied training volume among elite marathoners. As a result, high weekly mileage remains a constant among elite marathoners.

Several studies suggest that only high-intensity training increases fitness among well-conditioned aerobic athletes and that low-intensity training does not lead to increased aerobic capacity. As the majority of time spent running by elite marathon runners is at subthreshold levels, their training methodology seems to contradict the science behind improving one's cardiovascular capacity.

Conversely, there is evidence that an increase in running volume correlates with an increase in running economy (efficiency). Running long miles also increases one's muscular endurance and mental stamina. So is the mileage run by elite marathoners necessary or primarily based on historical training patterns, and how does this information relate to the average runner? At this point, no one really knows the answer.

Marathon Long Run(s)

At the most simplistic level, more time spent running equates to more stress on the body. This is especially the case with long runs.

Individuals who introduce high-mileage days before their body is adapted put excess stress on their musculoskeletal system and are at an increased risk for injury. This is especially the case for runners who have biomechanical inefficiencies and imbalances. These issues likely increase the chance of injury over those who are biomechanically sound.

A "fast" runner may be able to clip off 20 miles (32 km) in training at a seven-minute per mile pace (4:20 min/km) (2:20 elapsed time), while another runner may take 3:20 (10 minute per mile pace – 6:12 min/km) to cover the same distance. **The stress on the body for the slower runner is much greater than that of the faster runner, primarily because of the duration of time spent running (one hour more).**

Regarding marathon training, the average prescription for the maximum distance long run is ~20 miles (32 km). Therefore, for a slower runner, it may be advised for the long run to be 18 miles (29 km). Conversely, for faster runners and those with a low incidence of injury, 20-22 miles (32-35 km) may be the optimal long run distance.

The genesis of the longest run is that in order to finish a marathon, you must be able to run a particular distance in a single run.

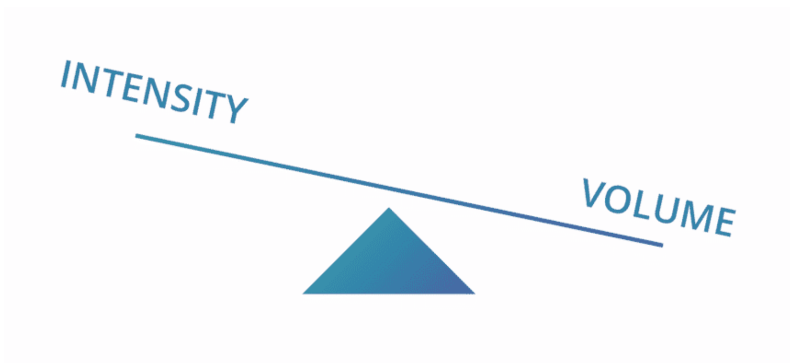
While long runs are part of a running program and more specifically, marathon programs – from the standpoint of making a singular long run the pinnacle of a training program is incorrect.

Why this is the case?

1. From an endurance standpoint, successful completion of a marathon is the result of the cumulative volume, not a singular run. Therefore, in the scheme of the overall program, a runner will log a lot of miles, making a singular long run fairly insignificant with respect to the whole training program.
2. Long runs put a lot of stress on the body, requiring more rest which from an overall program perspective, reduces the overall training volume.
3. The exact distance of long runs should be based on a runner's specific physiology and training adaptations.

Conclusions

A well-constructed training program must sufficiently stress the body and allow proper recovery. Regarding intensity versus volume, the ratio between the two must be configured individually. While long training runs are an important aspect of a marathon program, an athlete in good to great aerobic condition could likely benefit from increased intensity as much or more from increased volume.



As a general trend, volume and intensity have an inverse relationship, as noted in the image to the right. Therefore, during a week of volume increases, it is advised not to schedule large amounts of speedwork.

Programming Factors and Considerations

Noted below are three areas to be aware of when creating a program.

– Time vs. Distance (Long Run Specific)
– Factors That Influence Volume Increases
– The 10% Rule
– Least to Most Specific

Time vs. Distance (Long Run Specific)

Many factors influence one's ability to tolerate training stress on the musculoskeletal system. You probably know a runner who can barely even look at a pair of running shoes without getting injured, while another can seemingly run endless miles without ever getting so much as a blister! These two examples should be considered outliers. The rest of us fall somewhere in between.

Regarding long runs, some advise that the duration be prescribed by time versus distance. Famed running coach Dr. Jack Daniels suggests that for recreational runners (non-elite), the long run should be no longer than 2:30. His theory is that because of the lack of adaptation in the body from a musculoskeletal point of view, running over 2:30 would increase the chance for injury to the point where it's not worth the gamble (757). Moreover, Dr. Daniels states that even if an individual's goal time is 4:30 and the person's longest run is 2:30, this volume disparity should not be cause for alarm, as it is not necessary to train at or close to the race distance (757). In other words, Dr. Daniels looks at the long run from the perspective of stress on the body. **For example, if a top-level marathoner runs the marathon in 2:20 whereas a recreational runner does it in 4:40, from purely a time perspective, the recreational runner places twice the amount of stress on the body as the top-level marathoner.**

If prescribing long-run volume by time, the exact maximum should be determined by many factors, including training history, comparison between prescribed long-run time, and estimated marathon finish time and propensity for injury.

Factors That Influence Volume Increases

There is little argument that substantial increases in training volume (both long run and weekly volume) from week to week will likely increase the chance for injury. What qualifies as a *substantial increase* is subjective, as it is based on the individual. Many factors influence volume increases within a training program.

The date or distance of an event is not a factor in increasing volume. Volume should never be increased to accommodate an event. If an athlete is not ready for an event based on current training volume and proper volume progression, a new event must be selected.

Following are the most influential factors:

New Runners: Musculoskeletal system has not adapted to the impact of running.

Low Training Volume: The body has not adapted to high volume. Individuals with high training volumes typically tolerate large increases better than those with low training volumes.

Physiological Adaptation: The speed at which your athlete adapts to cardiovascular and musculoskeletal training will affect how fast the individual can progress. Of the two, musculoskeletal adaptation is more significant with respect to volume progression.

Biomechanics: If your athlete has substantial postural or biomechanical deviations that would likely increase the chance for injury, volume increases should be made sparingly or not at all and only after consultation with a specialist.

History: If your athlete is an experienced runner, you should note the individual's long-run volume increases during previous years and the results (e.g., performance, injury, etc.) to help estimate the appropriate volume increases. Additionally, any reasons for delays in the athlete's program progression, such as injury or burnout, should also be noted.

Injury: If your athlete is recovering from an injury or sickness, initial volume increases, if any, should be minimal.

Current Training Volume: Volume increases largely depend on the athlete's current training volume.

Genetics: Some athletes are naturally gifted in endurance and musculoskeletal adaptation. However, until you work with an athlete for a substantial amount of time, you will not be able to determine this.

Ability to Recover: Many factors affect one's ability to recover, including proper rest, nutrition, manageable volume, and intensity increases.

The 10% Rule – Fact or Fiction?

A 10 percent maximum increase in volume is commonly recommended for running programs. Most coaches reference and use the 10 percent rule to steer their athletes away from overtraining and potential injury. So where did this popular guideline originate from? No one knows. According to Carl Foster, director of the Human Performance Laboratory at the University of Wisconsin–La Crosse, the origin is "lost in history," and he notes that "whether it (the 10 percent rule) is right is undocumented" (130).

Dr. Ron Diercks from the University of Groningen in the Netherlands, who was a member of the team that conducted the study mentioned above, noted that regarding the 10 percent rule, "Nobody found out if it works or what is the basis of it." Dr. Diercks goes on to say, "People hear something, they read something, and then it's like a religion" (130).

Given the lack of an origin or scientific explanation of the 10 percent rule, it appears to be primarily based in running and training lore.

When using a set percentage (10 percent) for volume increases, the initial increases will be quite small (assuming the athlete is starting at a low training volume). As the volume increases, the increases get larger and larger. There are two primary issues with this approach:

1. Increases in volume are based on a preset, arbitrary percentage and therefore do not consider an individual's adaptation response to training.
2. If an athlete were starting at a very low training volume, the initial increases would be so small that they would be deemed inconsequential.
    1. For example, if your athlete starts at running one mile, a 10 percent increase is .1 mile. Based on the 10 percent increase logic, they would run 1.1 miles the following week.

While this section debates the validity of the 10 percent rule, it must be noted that weekly increases of less than 10 percent (or no weekly increases) are standard in training programs. Additionally, athletes respond differently to the same volume increases. In the below video, Ben Rosario discusses volume increases and the 10% rule.

At the end of the day, a 10% volume increase may or may not be correct for your athlete, but from a macro perspective, a ten percent volume increase would most likely be most applicable for mid-range volumes – meaning, for very low volumes or very high volumes, the 10% rule would likely be inaccurate due to the volume increases being too low and too high, respectively.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Challenge to the 10% Volume Guideline

The genesis of the 10 percent volume increase rule is to reduce the risk of injury. A 2008 study out of the University of Groningen in the Netherlands challenges this notion. There were 532 subjects in this study, split into two groups to train for a four-mile (6.4 km) running race (130). One group of runners was put on a 13-week program with 10 percent weekly increases in volume. The other group was put on an eight-week program with 50 percent weekly increases in volume. The study found that the 10 percent increase group had a 20.8 percent injury rate, whereas the 50 percent increase group had a 20.3 percent injury rate. In other words, the two groups essentially had the same injury rate, approximately one in four.

While this study seems to indicate that the rate of volume increase is not a factor for injury, it is important to note that many factors contribute to injury. Thus, volume increases must be based on the individual.

A 2012 study by Nielsen et al. also turned up similar findings (316). In this study, 60 novice runners tracked their distance over 10 weeks using GPS technology. Of the 60 runners, 13 sustained an injury. Those injured increased their weekly training volume 31.6 percent compared to those who were not injured (22.1 percent). While this study seems to demonstrate an increased chance for injury with significant increases in weekly training volume, it needs to be noted that the uninjured individuals increased their weekly training volume 22.1 percent, which is much greater than the customary 10 percent.

Weekly Volume Increases

Weekly volume increases are distributed unevenly across a week's worth of training. For example, if an athlete is running 40 miles (64 km) per week, a 10 percent increase equates to a four-mile (6.4 km) weekly increase for a total of 44 miles (71 km). Below is how the four miles might be distributed:

Long run: 1.5 miles (2.4 km)

Tempo run: 1 mile (1.6 km)

Easy run: 1 mile (1.6 km)

800-meter repeats: .5 mile (.8 km)

As the four additional miles (6.4 km) are spread unevenly across four days (workouts), the daily increases are likely not substantial when you distribute the increase across X number of days. While significant weekly increases in volume may or may not increase the chance for injury, when discussing percentage volume increases, it would be advantageous to focus on daily volume increases, as this likely has a stronger correlation to injury rate than weekly volume increases.

Weekly volume increases can also be accommodated via an additional day(s) of running.)

Programming Intensity

Introducing and/or increasing intensity in a training program requires a slow and disciplined progression. Training modalities such as sprints, hill repeats (for speed), and intervals require a runner to produce more power than normal running conditions. This increase in power typically occurs in conjunction with an increase in leg turnover, and therefore the running stride becomes more ballistic. Additionally, these training modalities often change the mechanics of a runner – specifically increased hip extension and transverse hip rotation.

Biomechanical Adaptations

Given the likely increase in hip extension and transverse hip rotation, injury is a likely result if one's body is not adapted to an increased range of motion. While a "tight" muscle(s) may assist a runner regarding the stretch-shortening cycle (SSC), the muscle(s) must have full range of motion. In addition to gradually increasing the speed and quantity of the aforementioned training modalities, performing stretches that target the hip flexors, hip adductors (groin muscles), and abdominal muscles are advantageous.

You must inform your athletes to take stock of their bodies during and after speedwork sessions. If they feel pain, excessive muscle tightness, or limited range of motion, they must stop to minimize the chance of injury.

Build Up Speed

Introducing speedwork and intensity into a program must be done in small amounts and with small speed increases. The theory of *walk before you run* applies to speedwork as there is a natural hierarchy to the type of workouts introduced in a training program. For example, before introducing speedwork, an athlete should already be performing threshold training (tempo runs, progression runs) and fartleks. While not pure speedwork, these training modalities help the body adapt to running at a faster pace.

The first interval-training session might look something like this:

- Two-mile warm-up at eight-min/mile pace
- 3 x 200-meter efforts at seven-min/mile pace with a 400-meter jog in between
- One-mile cool-down jog

As the runner progresses, the interval-training session might look like this:

- Two-mile warm-up at eight-min/mile pace
- 4 x 400-meter efforts at six-min/mile pace with a 200-meter jog in between
- One-mile easy jog
- 4 x 200-meter efforts at 5:30-min/mile pace with a 200-meter jog in between
- One-mile cool-down

Common Programming Mistakes

Creating a proper training program is a comprehensive and fluid process. Due to the many variables and moving parts, it’s a near certainty that no program is ‘perfect.’ However, there are four ultrarunning programming mistakes that are more common than others.

Your Athletes Are Not You

This is perhaps the most common coaching mistake – regardless of the sport. A lot of coaches are either current or prior athletes themselves in the sport that they are coaching. This is generally a good thing, as it gives a coach experience to draw from and appreciate the context of the sport and their athlete. However, often times, a coach uses their personal experience as the sole driving force to create their athletes' programs. In other words, this falls into the, *"it worked for me, so it will work for you"* training methodology. As you can see, this erroneously assumes that all ultrarunners (and humans in general) respond the exact same way to the same training stimuli.

While it doesn't take a Ph.D in exercise physiology to understand that this is incorrect, it is still the way that many coaches work with their athletes.

Too Much Volume

No doubt about it, training for a marathon takes time. However, a lot of runners and coaches overly focus on the volume aspect and therefore deduce that in order to be successful at a marathon, one must pile on huge amounts of volume.

There is nothing wrong with integrating long runs into a marathon program, as this is an important aspect of the training process. However, there is often a misplaced emphasis on volume as compared to other areas, such as aerobic conditioning.

If volume were the sole determinant of marathon success, there would be no need for workout types other than endurance runs. Thus, intervals, tempo and steady state runs would largely be deemed inconsequential. This of course is not the case.

Not Enough Intensity

In comparison to shorter distance events such as the 5K, marathons are run mostly at low intensities. Due to this, many marathoners and coaches don't integrate much, if any intensity into marathon programs. **This is a mistake.**

Programming intensity purely from the standpoint of specificity to an event is looking at it through the wrong lens. The benefits of intensity are multifaceted, with the primary benefit being increasing one's aerobic capabilities.

No Half Marathon Prerequisite

Among some athletes and coaches, there is a thought that in order to run a marathon, you must have done a half marathon previously. While on the surface this may appear to make sense. It is flawed logic. As an example, using this logic, if an athlete wanted to run a 5K, they must first race a mile or 3000-meter race. To be clear, in regard to being prepared to run a marathon – the most important thing is that the training that a runner does supports the distance and goals of the race they are targeting.

While having run a race before running a marathon is a good idea to gain an understanding of how the body and mind respond to race scenarios, it is not a prerequisite to have run a half marathon before attempting a marathon. So, is doing a half marathon before a marathon a bad idea? Not at all. It's just not a prerequisite.

Not Specific

All too often, marathon programs place an over-emphasis on two aspects of training: low intensity and high volume. While these aspects are included in a marathon program, they are often done at the exclusion of most everything else.

In order to be as best prepared for a marathon, or any type of race for that matter, dialing in the training to be as specific as possible to the race being trained for is critical. Some of these areas of specificity include:

- Environmental conditions: altitude, heat, humidity
- Running surface
- Length and percent grade of climbs

Mismatched Time Availability

While it's true that a lot of runners overestimate how much training time is required to train for a marathon, the reality is that training for a marathon does take a substantial amount of time.

Therefore, if an athlete does not have enough time to train for a marathon, they should look to select a shorter distance race that aligns with their training availability. 'Cramming' for a running race is never a good idea.

Training Environments

Course specificity is an overlooked aspect of program design. Race course terrain and characteristics must be incorporated as much as possible in training. Ideally, a race website will post the race profile detailing the elevation. This will help determine what the training environment for your athlete should consist of. If the course profile is not online, it would be prudent to email the race director to get this information. By knowing the course profile, you can tailor specific workouts and race strategies.

When running outside, the athlete must be aware of the surroundings and know the route and potential hazards.

Following are four primary types of training environments, each with its own advantages and disadvantages:

Track

A track is a great training environment for speedwork and, more specifically, interval-type training sessions. The inside of the inner lane (lane 1) is 400 meters long. Therefore, the distances of outer lanes are greater than 400 meters. Most tracks are made of cinder or a rubber-type compound.

Disadvantages: Boring; can be difficult to get track time if already utilized by a team or school.

Road

Often has varied terrain. Easiest for most people to access and most applicable for those training for road races.

Disadvantage: Often puts the runner far from home.

Trail

Running off-road offers many benefits over running on the road. Advantages are greater shock absorption, varied terrain, and an uneven surface that challenges a runner's balance and ankle/leg stability. Trail runs and cross-country races are run almost exclusively off-road. Additionally, many ultra-marathons integrate off-road sections for some or the entire course.

If your athlete participates in a race in which some or the entire course is off-road, the individual will need to do specific off-road training.

Running off-road requires much greater ankle stability and proprioception/kinesthetic awareness than running on the road. Without specific training, an individual greatly increases the chance of injury. As with implementing any new training element, a slow and progressive adaptation to trail running must occur.

In addition to incorporating trail runs into a training program, performing ankle-stability exercises and ballistic movements in the frontal plane can help an individual adapt to running off-road.

Disadvantages: Unstable surface; if in a hunting area, do not run during hunting season; unpredictable conditions.

Treadmill

A treadmill offers many advantages over running outdoors. Primary advantages are increased shock absorption, not weather dependent, controlled environment (control over speed and incline), and often a more time-efficient workout.

An ancillary benefit to treadmill running is the auditory feedback the treadmill provides. Unlike the ground, a treadmill running deck is suspended and allows users to hear the impact of their foot strike. The louder the foot strike is, the greater the impact on the feet and legs. This auditory feedback allows a runner to modify the gait to become smoother and more efficient.

Disadvantages: Boring, treadmill belt artificially enhances gait, thereby changing the leg muscular activity.

Treadmill vs. Outdoors

In many respects, running on a treadmill is the polar opposite of running outside. When running outside, an individual must push off the ground with the feet, whereas when running on a treadmill, a runner primarily moves the legs to keep up with the treadmill belt. Running on a treadmill requires a runner to push off the “ground” (i.e., belt) a little, but nowhere near the extent required when running outside.

Therefore, the muscles and how they are utilized differ between running on a treadmill and outside. While running on a treadmill exclusively is not advisable, it has benefits. Running inside is the best way for a coach to assess a runner’s gait. As treadmills tend to have better shock-absorption properties than outdoor running surfaces, for athletes who are injury prone because of the impact nature of running, treadmill workouts can be quite beneficial.

A 2014 study by Kaplan et al. found that when running outside, the foot strikes the ground with 201 percent of one’s body weight, whereas running on a treadmill results in 175 percent of one’s body weight (698). As a side note, exercising on an elliptical machine resulted in only 73 percent of one’s body weight. Therefore cross-training using closed-chain cardiovascular exercises such as ellipticals and bikes will likely reduce the chance of injury and increase recovery time over running. Nonimpact open-chain exercises, such as swimming and pool jogging, likely decrease the chance of injury as well.

Why Does Running on A Treadmill Suck?

While not everyone thinks treadmill running sucks, a large percentage of runners find running on one harder than running outside – especially mentally. After all, it doesn't get the nickname 'Dreadmill' for nothing!

As a treadmill has good shock-absorption properties and helps propel a runner's legs rearward, in theory it should feel easier than running outside. Additionally, the mechanics of treadmill running and running outside are the same (699) – so what gives?

A 2012 study by Kong et al. examined this exact phenomenon (700). In this experiment, runners ran on an outdoor track for three minutes and then immediately ran on a treadmill for three minutes, followed by another three-minute run outdoors. When running on a treadmill, subjects were instructed to run at the same perceived intensity and speed as on the track. The results showed that both outdoor runs occurred at the same pace while the treadmill run was performed at a substantially slower pace.

The study found that the likely culprit of this mismatch of perception of pace and speed is due to a distortion of visual inputs. **In other words, the perception of increased difficulty is purely psychological.**

This finding correlates with another study that found the perception of distance relates significantly to the exercise outcome (701). The closer subjects are to a perceived target or finish, the faster they go and the easier the task is perceived to be. In this study, participants walked a course with no defined finish line and then walked the same distance with a set finish line. Participants noted feeling more fatigued when no finish line was present.

Because there is no physical finish line when running on a treadmill, it is likely more mentally exhausting and thus is perceived to be more difficult than running outdoors.

This information is important to consider when working with athletes who integrate treadmill running into their program and/or prescribing treadmill workouts.

Workout Types

Repeats, fartleks, HIIT, base ... what do these things have in common? They are all specific workout/training types. While there are near-limitless names for workout types, training sessions fall under six primary categories:

- **Active Rest/Recovery**
- **Endurance**
- **Threshold**
- **Intervals**
- **High-Intensity Intermittent Training (HIIT)**
- **Multiple-Workout Training Days**

It is important to know that many athletes and coaches overemphasize one or more of the aforementioned areas. Any emphasis on a particular training area (i.e., HIIT) should correlate with a specific point in the training process. For example, when focusing on building fitness and a mileage base, endurance training should be the primary focus, not HIIT.

Active Rest / Recovery

While recovery can be a total rest day, active rest is often prescribed to keep the muscles “loose.” These workouts are characterized by extremely low intensity for relatively short amounts of time.

Run/Walk: Run/walk workouts are great for beginner runners and seasoned runners coming back from injury or doing a recovery workout. This type of workout combines running and walking, and it is up to you to determine the total time and ratio between running and walking.

Shakeout Run: These runs are done either the day before or the day of (a few hours before) a race. They are characterized by an easy pace (i.e., jog) and are relatively short in distance. They are done to loosen up the legs. If done directly before a race, it is considered some or all of the warm-up.

Cross-Training: Activities such as cycling and swimming can also be suitable for recovery, so long as the intensity is relatively low and the individual is used to the mechanics of the activity.

Endurance Training

The purpose of endurance workouts is to increase muscular endurance and aerobic capacity. They are characterized by exercising at a low intensity (sub LT) and for relatively long periods. What constitutes a 'long time' is subjective and based on an individual. The pace during endurance sessions is typically steady, with few fluctuations in intensity or speed.

Terminology: long steady distance (LSD), base, long run

Long Run

LSD running is generally structured around a slower pace and longer distance. The distance is based on where an athlete is in the training program.

A high training volume is often associated with an increased risk for injury. This is most correlated with total weekly training volume and dramatically increasing the volume during a training session. A 2012 study by Quinn et al. found that while muscular damage did occur during long runs, so long as an individual was adapted to this distance regarding a progressive training program, there were no adverse effects on a runner's running economy at submaximal speeds (688). Therefore, a runner well adapted to a training program should be able to adhere to the program in the days following a long run without negative consequences.

Base Run

These are used to help an athlete adapt to running from a musculoskeletal standpoint and gain cardiovascular fitness. These runs are typically done at a comfortable pace. The base run distances increase as an athlete adapts to running and gains cardiovascular fitness.

Intensity vs. Volume

Many runners try to get away with training as little as possible for their goal race. There are likely many factors for this, such as other time commitments, fear of injury due to overuse, boredom, etc.

While some of the aforementioned factors may be valid, the compensatory action of reducing training volume is typically to increase intensity. This is a slippery slope. Remember, the aerobic system adapts faster than the musculoskeletal system. Therefore, although an individual may be able to “handle” an intense cardiovascular workload, it is likely that the person’s musculoskeletal system cannot.

While there is some physiological carryover between aerobic and anaerobic-type exercises, they largely do not focus on the same thing. As such, swapping out endurance for intensity is not a shortcut that is advised. While there is nothing wrong with decreasing volume to a point, it must be done intelligently. You should have historical data on the athlete that would suggest that the proposed decrease in training volume would be sufficient for the event being trained for. However, even in this case, a drastic reduction in volume or a swap for high-intensity workloads is not recommended.

Laursen et al. note that while there is overlap between adaptations that occur at low and high intensities, the metabolic pathways to elicit these adaptations are likely different. Additionally, and more importantly, he notes that low-intensity training (base training) is largely a prerequisite to high-intensity training having an impactful and lasting effect on performance (738). In other words, low-intensity/high-volume training should precede low-volume/high-intensity training for the best results.

In summary, while high-intensity/low-volume training may initially elicit faster physiological adaptations than low-intensity/high-volume training, it is advised to perform low-intensity training at medium to high volumes before implementing high-intensity workloads.

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Threshold Training

The purpose of threshold training is to increase one's lactate threshold (LT). This training occurs in a range slightly below to slightly above one's LT. While threshold training can be steady efforts across a relatively long period, it can also be shorter or have intensity variations.

Terminology: Negative splits, tempo, time trial, progression runs

Tempo

Tempo workouts consist of five to twenty minutes efforts (excluding warm-up/cool-down), depending on the length of the race being trained for. The intensity is just below race pace. Tempo efforts are typically a minimum of five minutes and are steady-state efforts at or slightly above/below LT (317). If above LT, the pace is around one's MLSS.

These workouts are typically performed just below race pace and are done for five to 20 minutes (excluding warm-up/cooldown), depending on the length of the event being trained for. A warm-up is recommended when performing tempo efforts.

In the video below, Ben Rosario discusses why he believes that tempo workouts provide a runner the biggest bang for the buck.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Time Trial

This is essentially a tempo effort but is typically done over a set distance and often with a set time goal.

Progression Run

This run starts at a particular pace and ends at a faster pace and/or effort. For example, a six-mile run might begin with the first four miles at an eight-minute-mile pace, the fifth mile at 7:30/mile pace, and the last mile at 6:45/mile pace. The exact structure is up to you. However, the constant is an escalating pace throughout the workout. This type of workout or race strategy is also called a negative split. This refers to running the second half of a run faster than the first.

Intervals

Intervals are characterized by predetermined durations of intense cardiovascular efforts at or above threshold level and timed recovery periods. Recovery periods typically do not allow for full recovery. However, there may be full recovery between intervals for sprinters and middle-distance runners (ex: 800 meters).

The purpose of intervals is to increase aerobic capacity, increase LT, and improve recovery time. Interval intensity can be measured by pace, time, distance, and heart rate. While an interval effort can be short or long, it is advisable not to have an effort be shorter than 30 seconds and longer than five minutes. Efforts longer than five minutes would likely need to be done at a reduced intensity that would diminish the physiological benefits of the interval set.

Intervals have two primary components: intensity and time. Some intervals might increase in time and intensity, whereas other interval sets might increase in time but decrease in intensity. The structure of an interval set is dependent on what you deem most important for the progression of your athlete.

Intervals done at a very high intensity and for short durations could be considered a high-intensity interval (HIIT) workout due to the high intensity and short rest periods.

While the structures of intervals vary, five primary types are most applicable to running.

1. **Straight**

Efforts and recovery aspects remain the same throughout the session.

Ex: Three minutes at 150 beats per minute (bpm) followed by one minute of rest. Repeat this three times, then rest for five minutes. Repeat cycle two times.

Repeats: These are characterized by running at a relatively fast pace for a set amount of time (or distance), recovery, and then running the same distance and/or pace again (e.g., run a mile at seven-min/mile pace, easy jog quarter mile, run mile at seven-min/mile pace; repeat cycle two times).

Hill Repeats: A common interval-training workout is hill repeats. As the name suggests, this workout consists of an individual running up a hill at a particular pace, resting (usually by running back down the hill), and then running back up the hill. Hill repeats are often performed to become more efficient at running uphill and work on one's cardiovascular conditioning.

Float**:** Float runs are essentially repeats, but the pace is still relatively high instead of an easy jog in between efforts. Float workouts train the body to respond to mid-race surges while not dipping below race pace. An example of a float run would be a track workout with alternating 400-meter efforts. The alternating efforts would be at race pace and 30 seconds over race pace. Typically a float workout is assigned a set number of efforts. The workout ends if an individual can no longer maintain the assigned pace.

1. **Ascending**

Efforts increase in duration throughout the session.

Ex: One mile at 8:00 pace, one-minute recovery jog > two miles at 8:15 pace, two-minute recovery jog > three miles at 8:30 pace, five-minute recovery jog; then repeat.

1. **Descending**

Efforts decrease in duration throughout the session.

Ex: Three minutes at 7:00 pace, two-minute jog > two minutes at 6:00 pace, one-minute jog > one minute at 5:30 pace, five-minute jog; then repeat.

1. **Pyramid**

Effort durations increase then decrease to the starting level.

Ex: 400 meters at 75 seconds, 200-meter recovery jog > 800 meters at 2:30, 400-meter recovery jog > 1600 meters at 5:00, 400-meter recovery jog > 800 meters at 2:30, 400-meter recovery jog > 400 meters at 75 seconds.

1. **Fartlek**

Hard efforts and recovery periods characterize this type of interval, but the durations, intensities, and recovery times vary.

Ex: This is a Swedish word that means *speed play*. Fartleks are unstructured intervals. This means that an athlete randomly picks up the pace at random times throughout a workout. The duration of each “pickup” is also random. The only structure generally associated with fartleks is that your athlete might predetermine how many efforts to do during a session. An example of a fartlek would be upping the pace from one telephone pole to the next, but this is typically determined while running, not in advance. Fartleks are also termed *pick-ups* or *surges*.

High-Intensity Intermittent Training (HIIT)

There has been a recent trend of training at relatively low volume and high intensity for endurance events. The diminished training volume is typically replaced by increased intensity (often termed **high-intensity intermittent training** or **HIIT**) (427).

HIIT is characterized by an intensity level that is equal to or greater than that of intervals. While the time frame for bouts of intensity/recovery might be close to that of intervals, HIIT often incorporates varying training modalities (e.g., rowing, strength training, plyometrics, etc.)

The most cited and documented study done on the purported benefits of HIIT was performed by Tabata et al. The 1996 study has gained attention because of its findings related to the benefits of short, intense exercise on all areas of the body (170). Many people use the term Tabata training interchangeably with HIIT.

HIIT is not as well researched as many other areas of conventional endurance training. Therefore, it can be assumed that there are still many unknowns regarding the effectiveness of HIIT on distance running training, especially with respect to training for half and full marathons.

It is strongly advised to begin a training program with a solid base training phase that focuses on low heart rate training. This does not mean HIIT cannot be implemented during this phase. However, it should be done in moderation.

HIIT can place increased stress on the muscular system depending on the training modality. **Stress on muscles and connective tissue must be gradually progressed to ensure they can handle the increased workload.**

Assuming that HIIT could fulfill a runner’s cardiovascular, musculoskeletal, and endurance needs, the one variable that it cannot take into account is the psychological aspect. The time it takes to complete a marathon typically ranges from three to six hours, with the sub-three-hour range typically reserved for professional and top age-group competitors. Therefore a distance-running program made up almost exclusively of HIIT-type programming has a high probability of leaving a runner ill-prepared for the psychological demands of long-distance running, such as a marathon. This aspect cannot be ignored or underestimated.

Effect of HIIT on Well-Trained Endurance Athletes

One study of particular interest examined the effect of HIIT on well-trained endurance athletes versus sedentary individuals and recreational athletes. A 2002 study by Laursen and Jenkins found that submaximal training on well-trained athletes did not seem to increase their endurance capacity or VO2 max (171). Laursen and Jenkins theorize that the only way to elicit additional endurance and physiological gains in well-trained endurance athletes is through HIIT. Interestingly, studies examined by Laursen and Jenkins found that HIIT done by well-trained athletes did not increase their oxidative or glycolytic enzyme activity. **Therefore, they theorized that an increase in the muscle's ability to buffer hydrogen ions (delay muscle burn) was a possible reason for an increase in endurance capacity in well-trained endurance athletes.**

Several studies have demonstrated the physiological benefits of HIIT on endurance athletes who had not previously integrated HIIT into their training program (225, 226, 227). For these athletes, the benefits of HIIT were realized quickly. Conversely, increased intensity of HIIT sessions in athletes already performing HIIT did not show significant increases in cardiovascular adaptations (228, 229).

Multiple Workout Training Days

Whether it be because of a limited amount of time to train during a single session or an athlete is looking for performance gains by training twice in one day, this section discusses some things to consider.

Two-a-Days (TAD)

A TAD incorporates two workouts within a single day. There are some basic guidelines for TADs.

- No back-to-back TADs
- Must have at least eight hours between workouts

- TADs should be performed only by runners who have a solid mileage/fitness base

Common TAD Structures

- One of the TAD workouts is a running workout, while the other is some form of cross-training (e.g., swimming, weight training, cycling).
- One of the TAD workouts is cardiovascular-focused, while the other is focused on strength, recovery, or skill/form.
- Because of the mileage requirement, TADs are often performed by runners training for ultra-marathons. In these cases, typically, both workout sessions are mid-long distance runs at sub-maximal efforts.

Two recent studies suggest that training twice a day compared with training the same total volume (sum of the two sessions/day) every other day induces greater physiological adaptations (242, 243). Therefore, TADS should not be viewed solely from a perspective of convenience but also from a perspective of performance enhancement.

Program Creation Process

The below video by Ben Rosario discusses the creation process of building a training program.

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Before you can build out a program for an athlete, you need to make sure that the event selected to focus on is feasible based on the following factors. Therefore, these feasibility steps also represent the first four steps of the program design. The four steps to determine feasibility are:

1 – Goal Event Determination and Program Duration
2 – Determine Training Availability
3 – Initial Training Volume/Starting Fitness Level
4 – Long Range Strategy
5 – Integrate Training Blocks
6 – Integrate Weekly Volume and Structure

If an event is deemed feasible based on these four steps, continue building out the program with step 5.

Step 1: Goal Event Determination and Program Duration

A runner often comes to a coach with a goal event already in mind. If an athlete does not have a goal race in mind, they might leave the goal race selection up to the coach. However, regardless of who chooses the goal race, once it has been noted, it is time to determine how much time an athlete has to prepare for the race (i.e., program duration).

If it is determined that the goal race is not feasible based on an athlete's starting volume/fitness and the amount of time to train for the race, an athlete must select a goal race that is more aligned with the training load.

An athlete must be passionate about the goal event(s) they choose. This passion will help to keep them engaged and motivated throughout the training process. The passion might come from the desire to have a shiny belt buckle or perhaps an athlete wants to target an ultramarathon in their hometown. Whatever, the reason, race selection should start with the question, ***"What race am I most passionate about?"***

If your athlete is targeting an event that has a lottery system to get in, their participation in the event is solely in the hands of the lottery gods and as such, a plan 'B' non-lottery goal event needs to be selected.

Step 1B: Program Duration

The duration of an athlete's training program is multifactorial. The three primary factors are:

- **Intensity**
  - The more intense the training focus (ex: VO2 Max), generally the shorter the training block
- **Significance**
  - This relates directly to what training adaptation is most important for success come race day. The more important, the longer the duration.
- **Rate of athlete adaptation**
  - Of the three factors, this is the one that may change in duration based on how fast, or slow an athlete is adapting from a physiological perspective.

When creating a training program, a good starting place is to allocate 8 weeks per training block (VO2 Max/Lactate Threshold/Endurance). Consider this the blank slate approach. From this point and taking the aforementioned factors into consideration, you reduce or lengthen the duration of each block.

For example, the VO2 Max block may be as short as 2 or 3 weeks whereas the endurance block may be as long as 12 weeks.

The below illustration denotes two program trends with respect to duration and make-up. As you can see, the block durations can change, as well as the overall duration of a training program. The top trend would be more applicable to an experienced ultrarunner with a solid fitness/running base who is targeting a 100K or 100M race whereas the second trend would fit an athlete who is in good condition and targeting a fast 50K or 50-mile ultramarathon. As you can see, while the same training block types are the same for both athletes, the degree to which they are integrated varies substantially based on the athlete's needs, and the race being trained for.

Step 2: Determine Training Availability

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
NONE	30 Min	1 Hour	NONE	30 Min	1 Hour	1 Hour

Regarding misaligned goals, if the athlete whose time availability listed above has a goal to participate in a marathon, you must explain why the training availability is insufficient for the goal and recommend that a 10K distance or less would be a better and more feasible option.

It is better to err on the side of caution when selecting an event based on one's training availability. For example, if your athlete thinks their schedule will open up down the road and enable the person to target a half marathon, it is better to go for the sure bet and target a 10K.

Realistic Programming

Athletes have many obligations other than training, some of which come up unexpectedly. This is the norm, not the exception! It is your responsibility as the coach to develop a program that considers this while keeping the athlete on track to reach their goal. For example, if your athlete has business meetings that come up often at the end of the day, it is advisable not to schedule critical workouts for that time. You must work with them as much as possible to accommodate their schedule and personal obligations. However, athletes can aim to participate in an event only if they have the time to train for it.

Step 3: Initial Training Volume – Long Run

Once you know that an athlete's training availability is appropriate for the type/distance event they are training for, you need to know if they are currently training and, if so, what their longest run is. For this long run to have value in the creation of the training program, it must be recent, completed without injury, and have been progressively built up to.

Currently Not Running

If an athlete is deconditioned and/or not currently running, you must be very careful to start them at a low level and progress the individual slowly to allow the body to adapt from a musculoskeletal standpoint. If an athlete has never run before and has poor running form, a significant focus must be on proper running form/gait. The initial training volume for an athlete such as this would likely be:

Run/Walk–20 minutes

A run/walk program has an individual alternate between running and walking at set intervals (i.e., walk five minutes, run five minutes). **As an athlete progresses through the run/walk program, the run portion becomes longer, and the walk portion becomes shorter.**

Keep in mind that even if your athlete is in good aerobic shape unless the person has been running recently, their muscles and connective tissue need time to adapt. This type of athlete should adhere closely to the preceding volume guideline.

This is also relevant to athletes who trained and competed in running events the prior year and stayed in shape during the off-season but have not run recently. Assuming it has been at least one or two months since an athlete ran, they should start at or close to above the initial training guideline. These athletes typically adapt faster to running than someone new to the sport, yielding a high rate of progression.

Once an athlete is up to an hour run/walk program, the individual can advance to a mileage-based (versus time-based) program. The starting run mileage should start at two miles.

Step 4: Long Range Strategy

The long-range strategy is, for lack of a better word, a roadmap or outline for both the coach and athlete. It provides a bird's eye view of the whole training program and therefore is a great starting point as it serves as a framework to be built upon with more specific data such as training blocks, workouts, etc.

A long-term strategy also serves as a reference point to refer to throughout the training process to check if an athlete is on track for their goal race.

Step 5: Integrate Training Blocks

If the long-range strategy provides a roadmap, training blocks provide the directions. More specifically, training blocks (or themes, as noted in the periodization section) provide structure to the program based on the physiological adaptation that a coach is looking to develop at a given time.

Preparation Races

It is common for athletes to plot their preparation races at the same time as their goal race. However, it is more advantageous to plot them after determining the set training blocks/themes. This is because by plotting preparation races in specific blocks/themes, a coach can match a type of race that typically targets a specific intensity (i.e., energy system) to the corresponding block. For example, if the coach is looking to develop an athlete's lactate threshold, using a 20km time trial as a preparation event would likely be a good match.

In addition, these races are important for many reasons: check fitness level, practice fueling, get used to race-day jitters, strategy/pacing, and a lot more. Runners typically do not taper or do not taper to the same extent for tune-up races as they would for a goal race. As a result, most runners are not completely recovered for tune-up races. Many runners call this lack of tapering **training through a race**.

While a taper is typically not implemented for tune-up races, an athlete should still decrease intensity before one. The last hard workout before a tune-up should be two to five days in advance (720). The longer the tune-up race, the longer the duration between a hard workout and the tune-up race should be.

Trialing The Fueling Strategy

Preparation races are a great time to practice a fueling strategy. While this can and should be done during training runs, everything from pre-race, during the race, and post-race can be practiced with all of the dynamics and intricacies of race day! For example, a "nervous" stomach may react differently to the same fuel used in training and on race day.

Course Simulation

Most races denote the elevation profile on their websites. As such, you can get a pretty good idea of how difficult the course is and, specifically, where the most challenging sections of the race are. If an event has a course map but no elevation profile, you can use the website MapMyRun.com to create the race map. Clicking on the elevation feature will generate the elevation map. This information should not only help shape the overall training process but the selection of the tune-up race(s) as well.

Step 6: Integrating Weekly Volume and Structure

As noted in the video above, steps one through five represent the core steps of creating a training program. From this point on, the specifics of the program can be developed. For this section, we'll consider these four specifics the last step of the training program construction process.

These specifics consist of:

- Determining weekly volume increases
- Integrating recovery weeks
- Programming training weeks
- Integrating specific daily workouts

Weekly Volume Increases

This step is where you estimate the weekly training volume for each week until the goal race. Note, if you get to this step, you have already deemed that the goal race is feasible – therefore, this is solely to determine weekly training volume, not a feasibility work-up.

Like other areas of the program, there really is no standard or template to go off. For example, some weeks may increase in volume, while other weeks may stay the same or decrease. There are many factors, such as where in the training program an athlete is (ex: base phase, peak phase) and what the focus of the week is (ex: recovery, volume, intensity). This topic was discussed previously in this lesson regarding the 10% volume increase 'rule.'

Integrate Recovery Weeks

No more than three weeks should pass without being followed by a recovery week. The newer the runner, recovery weeks should occur more frequently than with an experienced runner.

– Characterized by a 30–50 percent reduction from the previous week's volume/distance.
– Intensity during recovery periods should be no higher than an RPE of 6.

Programming Training Weeks

Once the structure (duration, cycles, weekly volume, themes) of a program is in place, the weekly training program can be developed. The weekly volume must match that of the previous step. Weekly programs should be designed and sent weekly or biweekly to an athlete. It is not advised to give an athlete more than two weeks' worth of programming at a time to allow for feedback and changes in rate of progression, if necessary.

Following is a weekly template that denotes the following areas:

- **Weekly date range**
- **Training theme (if applicable)**
- **Morning and evening workouts (if applicable)**
- **Distance or time duration of workouts**
- **Intensity prescription (i.e., watts, speed, heart rate, RPE, etc.)**
- **Total weekly volume**

Integrating Daily Workouts

Workout types within a training program are repetitious in nature. Therefore, you should create a database of workouts for your athletes. This way, each workout type can be thoroughly explained, and the only variables that change are the intensity and volume. How you decide to define them is up to you. However, something simple and intuitive typically works best.

When creating a workout database, categorize exercises by activity. There can be as many workouts under each discipline as needed. For example, if you have six running workouts, they could be R1–R6. By having a workout database, all your athletes need to do is look at the referenced workout number to get the workout information and then refer to the weekly schedule for the volume and pace, if applicable.

Creating a workout database is initially time-consuming, but it is a time-saver throughout the training process. A database helps athletes, as more specific workout information can be provided for each exercise/workout. If you have an exercise or workout that does not fit into the database, note "custom" on the weekly workout sheet and specify the exercise/workout in an email or over the phone. If you think a workout/exercise might be repeated, add it to the database.

Workout Database Information

Information to address for workouts are:

- Workout type (ex: Interval)
- Workload (ex: heart rate, RPE, pace, etc.)
- Description/Notes
- Goal of workout

Programming Variables

Taper

[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Longest Run

The longest run is used to benchmark an athlete's preparedness with respect to a goal event. The reasoning behind this is simple – having the ability to run at or near the full distance of the goal event is a prerequisite to being able to compete and finish a race/event feeling good and with the least chance for injury.

For marathons, most programs have several long runs (typically between 3-9 long runs) throughout the program's duration, with the longest run typically done a week or two before the final taper.

The typical mileage range for the longest run is:

Half Marathon: 12–13 miles (19 – 21 km)

Marathon: 18–22 miles (29 – 35 km)

10K: 6–10 miles (9.7-16 km)

There are a few things to keep in mind concerning the long run. The long run is often associated as the primary benchmark to determine if a runner is prepared for race day from a volume standpoint. However, it's important to keep the long run volume in perspective with the total volume of the program. In reality, the volume of a singular long run is quite insignificant concerning the total volume of the whole program. Every time an athlete runs, they build fitness, strengthen their muscles and connective tissue, and add volume to their cumulative total. The point is that increasing endurance and fitness is not solely the domain of long runs. Endurance and fitness are being developed with each run. Therefore, while feeling good after some long runs is a good thing to benchmark, it is more important to view race preparedness from the perspective of the whole program, not just long runs.

Over-Distance Training

This is characterized by training at a longer distance than the race distance. **This type of training is most prevalent with 5K and 10K events.** Over-distance training should not be part of a marathon program. If the goal event is a half marathon, there is no reason for over-distance training. However, if a half marathon is being used as a tune-up race for a marathon, there will be training days at a greater distance than 13.1 miles (21 km). Following are several reasons why over-distance training should be left out of marathon programs:

Time: As it already takes a lot of time to train for a marathon, over-distance training will take time away from other aspects of training, and more importantly, it will take time away from other aspects of an athlete's life (e.g., family time).

Injury: Generally speaking, the longer an individual runs, the greater the risk of overuse injury. Therefore, even if there is a slight fitness advantage, which is debatable, this advantage would likely be offset by an increased chance of injury.

Catch Up: Training for a marathon is not like a school exam: You cannot cram for it. If the race date is near and an athlete is unprepared from a fitness or endurance perspective, it is better to pass on the event and choose a shorter event or select a marathon further out.

No Need: Above all else, there is no need to train at a further distance than a marathon. One exception is if your athlete is training for an ultra-marathon. This particular race distance(s) is not largely focused on in this certification.

Weekly Volume Percent Increases

The chart below provides rough guidelines for determining weekly volume percent increases based on the starting volume. These increases represent the maximum percent increases – therefore, there often will be just a small increase (or no increase) in training volume from week to week.

Weekly volume percent increases are not noted to determine if a goal event is feasible, as this should have been done previously. Rather, this step is a tool in assisting with the determination of overall weekly volume increases and, more specifically, what the maximum overall weekly volume increases should be. This step is most effectively utilized by determining the maximum overall weekly volume and subtracting the weekly long-run distance, if applicable. The remainder of the distance represents the rest of the weekly run distances.

CURRENT WEEKLY TRAINING VOLUME (miles)	MAXIMUM % INCREASE
< 5 Miles	50%
6 - 10 miles	35%
11 - 20 miles	25%
21 - 30 miles	20%
31 - 40 miles	15%
41 - 50 miles	10%
> 51 miles	7%

CURRENT WEEKLY TRAINING VOLUME (km)	MAXIMUM % INCREASE
< 8 km	50%
10 - 16 km	35%
17 - 32 km	25%
33 - 48 km	20%
49 - 64 km	15%
65 - 80 km	10%
> 81 km	7%

NOTE: If your athlete is running less than five miles per week, the focus should be on learning proper form and musculoskeletal adaptation.

Sample Training Scenarios

Athletes will come to you with unique situations and goals. You will need to be able to assess each athlete individually to devise an appropriate and effective training program.

Following are examples of types of athletes whom you might come across – all of whom require drastically different programs:

Scenario One

A deconditioned and overweight athlete who wants to lose weight and is motivated to participate in a 5K at the end of the year.

The first step would be to connect the athlete with a registered dietitian for assistance in creating a nutritional strategy that will enable the person to lose weight while still having enough energy to train properly.

Depending on height/weight, you might need to be sensitive to the impact of running on the body. This program would more closely resemble an overall fitness/weight-loss routine with a slant toward running-specific training.

Scenario Two

An experienced runner who is trying to qualify for the Boston Marathon.

You need to determine the individual's strengths and weaknesses to get the athlete to the next level. This would involve biomechanical assessments, weekly training program adjustments, nutritional strategies (in conjunction with an RD), and perhaps much more. Working with this type of athlete will require all of your expertise and, most likely, referrals to other specialists.

If you do not feel comfortable coaching an experienced athlete, it is best to refer the person to someone else. Just because you are a great coach to one particular level of runner does not necessarily mean you are a great coach to other levels. This is not a weakness but a function of reality; you cannot be everything to everyone. Good coaches know their strengths and weaknesses, and their clientele reflects this.

Scenario Three

An athlete who works 80 hours a week and has family obligations, signed up for a marathon, and now needs to find a way to train properly while not taking time or energy away from work or family.

This type of athlete is always a challenge to work with because of time constraints, and more than likely, you will not be able to prepare the individual perfectly for the event. It is your job to work with what you have and prepare the athlete as best as possible given the circumstances. In this case, it might mean advising the athlete against doing the goal event and shifting focus to a shorter distance. Important things to ascertain initially are starting fitness level, when the race is, and how much time the person has per week to commit to training. If the date of the event does not align with the amount of training time available, you should recommend a shorter event to focus on.

Athletes must be accountable. Just because you prescribe a workout does not necessarily mean an athlete will follow it. This is not always your athlete’s fault. For example, you might have prescribed an evening training session, but maybe a child got sick, or a business dinner popped up, and the person could not do the training. This is life, and you must adjust the training program accordingly. However, if these things become habitual or an athlete just doesn’t follow the training program, you need to make it clear that the lack of preparation is not your fault but rather due to the person’s lack of adherence to the program.

Scenario Four

A relatively fit person wants to do a 10K to stay motivated and have a goal.

This type of athlete is quite common. The emphasis of working with this type of athlete is overall fitness with a slant toward running.

Remember, this is not someone whose goal in life is to become an elite runner. The longer the distance of the running race being targeted, the more specialized the focus on running must be.

Athlete Case Study

Athlete Information

Athlete: Mike (40 years old)

Running Experience: Ran in college for soccer. Has been running for the last 3 years.

Goal: Finish a marathon in under 4 hours

Health History: No past running injuries. Had a physical and everything came back fine

Training Availability: Currently do most runs after work (1-1.5 hours availability) but due to a work schedule shift, will be doing most runs in the morning (have 1.5 hours availability).

Current Training Volume: 15 miles (24 km) per week

Assessment Results:

- Height/Weight: 5’ 11”, 175 lbs (79.4 kg)
- Body Fat: 21%
- Resting Heart Rate: 61 bpm
- FTHR: 155 bpm

Athlete Profile

1. **What type of physical activities do you currently participate in?**

Running, strength training, cycling, golf

1. **What competitive sports (if any) have you participated in?**

Running, cycling, soccer

1. **Why, when and how did you get started in running?**

In college (20 years ago) for conditioning for soccer

1. **Will you be training by yourself or with others?**

Group runs on weekends, during the week by myself

1. **What are your short and long-term running goals?**

Short-term goal – be able to run 10 miles non-stop

Long-term goal – Finish goal marathon in under 4 hours

1. **When do you have the most energy: morning, midday, evening? And when to you do your runs?**

Most energy in the morning, do runs in the evening after work.

1. **Does your schedule vary or is it fairly consistent?**

During the week it's fairly consistent. On the weekends it's less consistent due to my kid's schedule

1. **What days/times do you have to train?**

Monday through Friday: 6am-8am (some evening availability)

Saturday: 2-6pm

Sunday: 2-4pm

1. **How important is structure in your life?**

Quite important

1. **What aspect of running are you most intimidated by (if any)?**

Bonking during long runs or in the marathon

1. **What are your top three running-related strengths and weaknesses?**

Strengths: good aerobic fitness, speed, don't quit

Weaknesses: muscle cramps, run too fast at the start of races, endurance

1. **Do you respond better to tough love, handholding or somewhere in between?**

   Somewhere in between

2. **Do you crack, stay the same or thrive under pressure?**

Stay the same

1. **What are you better at: Short, hard sprints or long endurance efforts?**

   Short, hard sprints

2. **How competitive of a person are you? Rate yourself 1-10 (10 = the most)**

8

1. **How much do you enjoy the intensity of pushing hard while training/racing (1-10)?**

   9

2. **How confident are you in your abilities as a runner (1-10)?**

   5
3. **You like to be challenged? 1-10 (1=untrue, 10 = very true)**

   9

4. **You are susceptible to mental burnout as a result of training? 1-10 (1=untrue, 10 very true)**

   3

5. **What is your current weekly training volume (distance and/or time) and how long have you been running consistently?**

15 miles – have been running consistently at or above 10 miles per week for the last three months.

1. **Current and/or past injuries**

None

1. **What are your current best paces/times for races and/or distances, if applicable?**

1 mile – 5:03

5 miles – 30:28

Training Program Notes

Based on Mike's Athlete Profile, there are a few things that stick out.

- Mike notes that he gets exercise-related cramps. We know that most cramps are related to overloading the muscles versus hydration issues. Therefore, we can likely deduce that since Mike also noted that his weakness is endurance, he is likely progressing too fast in respect to training volume.
- As Mike noted that he likes to be pushed hard and challenged, as well as he's competitive by nature, this might also translate to him pushing too hard/far/fast in respect to his training progression.
- Given that Mike noted that he is a 5/10 with respect to his confidence, increasing his confidence will need to be integrated into the plan. It is likely that his lack of confidence is due to him pushing too hard in training/races and failing at his self-established goals. Therefore, realistic goal setting is likely needed.
- His self-reported time when he has the most energy is in the morning but his runs are in the evening. This mismatch must be discussed to see if he can run in the morning.
- As Mike noted that he participates in several sports and activities, it will need to be determined how he is structuring his weeks to accommodate these sports/activities.
- Due to the muscle cramping issue and that he is most intimidated by bonking during long runs/marathons, a focus on endurance is likely.
- Mike noted that he starts races too fast. This is also likely during training runs. Therefore, implementing some sort of pacing parameters at the start of runs/races such as heart rate, RPE or pace should be implemented.

- The date of Mike's goal marathon is October 18th, which is 20 weeks from today (May 30th).

Training Structure

- Mike exercises all year round. Therefore, he is always in pretty good shape. Due to this and the fact that he is targeting a marathon, using the 'least to most specific' rule of programming, Mike's program will start off with high-intensity work and gradually trend toward lower intensity and higher volume as the program progresses.
- Throughout the duration of the training program, Mike will reduce his cycling and golf and focus more on running and recovery.

Process Goals

Based on Mike's Athlete Intake, there are several important process goals to work on throughout the training process that was developed in conjunction with Mike.

1. Make time to run during the morning, when possible
2. Adhere to the programmed intensity and volume, especially at the start of runs to avoid running too fast/hard.
3. Work on mental training aspects to increase confidence

Periodization Phases

Mike is currently averaging ~ 15 miles (24 km) per week of running and has been doing this for three months. As such, Mike is likely in pretty good aerobic shape, as has a solid volume base. Due to this and the fact that he is training for a marathon, there is no need to implement a conditioning phase that focuses on building aerobic fitness. Moreover, implementing the 'least to most specific' rule, the periodization program will focus on high intensity first and focus on high volume closest to the event.

Assessment Results

Mike's FTHR based on a 20-minute assessment was 155 bpm. His body fat was 12%

Zone Creation

Based on Mike's FTHR of 155 bpm, below are his corresponding training zones:

Recovery (< 68% of FTHR): 105 bpm and below

Endurance (69-83% of FTHR): 106 – 128 bpm

Steady State (84-94% of FTHR): 129 – 145 bpm

Lactate Threshold (95-105% of FTHR): 146 – 162 bpm

VO2 Max (> 106% of FTHR): 163 bpm and above

- Note – as Mike becomes more aerobically fit, his LT will increase, thus likely changing his FTHR and thus, his heart rate-based training ranges. The mock-up weeks noted below primarily note rate of perceived exertion (RPE) as intensity benchmarks, however, you can also use training zones, as noted above.

Sample Weekly Training Programs for Each Phase

Below is a mock-up of the phases and dates of Mike's program:

Preparation Phase			Competition Phase			Transition Phase
General Conditioning	Base Fitness	Build Fitness	Peak Fitness	Taper	Goal Event	Recovery
Jun 1 - Jul 5th	Jul 6th - Aug 9th	Aug 10th - Sept 13th	Sept 14th - Oct 4th	Oct 5th - 17th	Oct 18th	Oct 19th - ???

Below are mock-up sample weeks from each sub-section from the phases of the macrocycle. With the exception of the transition week, please note that the below sample weeks do not denote cross-training or strength training.

The mock-up program is solely included in this section to illustrate what various weeks of a program could look like – including Mike's hypothetical program. Note – the below mock-up

Preparation

General Conditioning

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
OFF	2, 4 x 400M with 400 meter jog in between. 8/10 RPE	Easy Run (2 miles)	4 mile run with 10 fartleks	3, 4 x 200M with 400 meter jog in between. 9/10 RPE	Easy Run (2 miles)	Easy 6 mile run

Base Fitness

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
OFF	3, 4 x 400M with 400 meter jog in between. 8/10 RPE	Easy Run (2 miles)	2, 3 x 800M with 400 meter jog in between. 8/10 RPE	8 mile run with middle 4 miles at LT pace	Easy Run (3 miles)	Easy 8 mile run

Build Fitness

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
OFF	2, 2 x 1600M with 800 meter jog in between. 7/10 RPE	Easy Run (4 miles)	8 mile progression run - start at 8 min/mile, finish at 6:30 min/mile	8 mile run at 6/10 RPE	Easy Run (3 miles)	Easy 10 mile run

Competition

Peak Fitness

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
OFF	10 mile tempo run with middle six miles at LT pace	Easy Run (4 miles)	10 mile progression run - start at 8 min/mile, finish at 6:30 min/mile	3, 3 x 800M with 400 meter jog in between. 8/10 RPE	Easy Run (3 miles)	Easy 15 mile run

Taper

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
OFF	5 mile tempo run	Easy Run (2 miles)	5 mile run with 7, fartleks of ~ 20 seconds each	OFF	Easy Run (3 miles)	Easy 6 mile run

Transition

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
OFF	30 minutes of cycling	2 mile Easy Run	1 hour of cycling	Strength Training	Easy Run (3 miles)	OFF

Weekly Program

The weekly program below is a mock-up to illustrate how a weekly program could look. How you deliver programs to your athletes is up to you and will be based on your preference, what your athletes are looking for and the medium by which you are providing the information (ex: online coaching platform, spreadsheet, email, etc…)

(The Sample Weekly Program must be viewed in the online course as it can't be displayed in this pdf)

Whole Program

The below program highlights workouts, and total weekly volume and is noted for the purpose of giving an athlete a macro look at the whole program so that they can see how the workouts, training blocks and phases fit together.

(The Sample Whole program must be viewed in the online course as it can't be displayed in this pdf)

Summary

- *Periodization* is based on the principles of the *general adaptation syndrome* (GAS).
    - The three phases of GAS are:
        - Alarm
        - Resistance
        - Exhaustion
- *Classic periodization* consists of three phases: preparation, competitive, and transition.
- *Classic periodization* is broken down into cycles:
    - Macrocycle
    - Mesocycle
    - Microcycle
- *Classic periodization* is criticized by some for being too general and trying to focus on too many training variables simultaneously, thereby reducing a program's effectiveness.
- A *hybrid periodization* structure combines elements of both classic and block periodization.
- While a training program must have structure, it also must be flexible.

- Your athlete will miss training days. You may need to modify the program to accommodate this.
- Tapers are characterized by gradually reducing volume while still integrating some intensity.
- It is ideal to include assessment checks throughout a program to ensure an athlete is progressing properly.
- Peaking for an event gives your athlete the best chance to perform at physiological maximum on the day of the event.
    - For experienced athletes, historical data can help to estimate the peak with greater accuracy.
- Finding the right balance between exercise, rest, and other obligations (e.g., job, family,) is critical for a successful program. Individuals who do not successfully balance the demands of training can become mentally burned out and suffer symptoms of overtraining.
- Think of training days in terms of hours, not exact days.
- Overtraining occurs when the training volume and/or intensity exceeds the ability to recover.
- The off-season offers an opportunity for a well-deserved mental break.
- Reverse periodization focuses on intensity at the beginning of the program and volume toward the end.
- A training program should progress from least specific to most specific
- Not all mileage has the same value
- The 10% volume increase rule is largely erroneous
- The steps to create a training program are:

1 – Goal Event Determination and Program Duration
2 – Determine Training Availability
3 – Initial Training Volume/Starting Fitness Level
4 – Long Range Strategy
5 – Integrate Training Blocks
6 – Integrate Weekly Volume and Structure

Module 16: Pacing

Pacing is a crucial part of running and is perhaps one of the most important things to master, as it can be the difference between finishing strong and dropping out.

The pacing strategy for a short race (5K) versus a longer race (marathon) is quite different. This is often a difficult concept to grasp for new runners. Push too hard, too early, and pay for it later in the event. The best way to appreciate pacing is to consider the cost/benefit ratio. Too high an energy expenditure (cost), and the benefit (result) will decrease overall. Too low of energy expenditure, and the result will decrease as well. To perform optimally, the energy expenditure must be not too high or too low. In other words, running a well-paced race is a function of balance.

Race pace is determined during training and based on assessments of an athlete's fitness level. Correlating an athlete's effort level, such as LT, to assessment methods (i.e., heart rate, power, RPE, etc.) will enable the person to track pace throughout an event.

For example, while an athlete might want to break three hours for a marathon, this is not a feasible goal if the individual can average only a 10-min/mile. Assuming an athlete has a time goal, it must be realistic.

When participating in a race, unplanned events can occur, such as bathroom breaks, cramping, and falls, just to name a few. It is imperative that your athlete not exceed the planned pace after being delayed. Deviating from the planned pace almost always ends in a slower overall time.

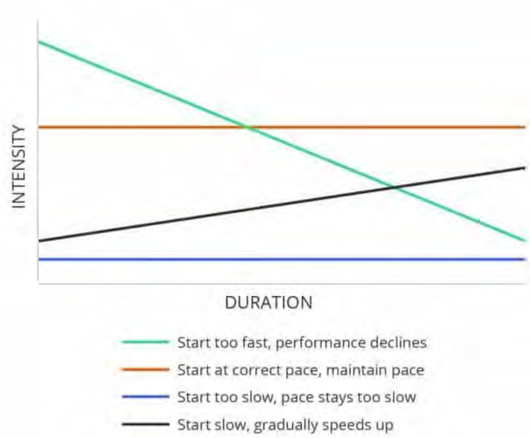
Pacing Overview

Smooth and steady is always the better pacing option than erratic and punchy. The more erratically a runner paces themselves, the less efficient they will be. For example, while it might seem faster to sprint up the hills and recover on the way down during a hilly race, this will likely result in a slower time than a competitor who keeps a constant effort. This does not mean that an athlete should not push harder in some sections of a course, but the efforts need to be calculated to ensure the overall pace is sustainable for the duration of the race.

While specific to cycling, a study that examined power output during Olympic distance triathlons found that large power variations equated to a higher overall workload versus cycling at a constant power output (470). Additionally, a study by Rapoport (705) found that running at a constant pace is the most efficient pacing strategy. **However, Rapoport noted that for running at a constant pace to be the most efficient strategy, the terrain must not vary.** Most outside courses have some sort of terrain variation. **In these cases, the most efficient way to run is not at a set pace but rather at a set level of exertion or effort.**

It is essential that you know the course profile of a race so that you can plan a detailed pacing strategy for your athlete.

The image below shows various pacing strategies. Of the four strategies, only the orange and black are recommended.



Impact on Glycogen in Relation to Varied Running Speed

To run a predetermined distance in a set time, a runner must maintain a specific average speed for the entire course. If a runner dips below this average pace, the individual must run faster than the average at some point to make up for going slower than the average pace. You may be thinking, *"Big deal, a little slower here, a little faster there … so long as I average the same overall pace, what does it matter?"*

From the standpoint of glycogen utilization, even if the finishing time is the same (i.e., same average pace) between a runner who maintains the same pace and one who varies pace, **there is a greater metabolic cost because of increased glycogen consumption with a varied pace.** This puts a runner at risk of a slower overall time or, in the worst-case scenario, hitting the wall.

Car Analogy

Two cars drive a distance of 65 miles. One drives at a steady speed of 65 mph while the other is constantly braking and accelerating between 40 to 80 mph. While both cars arrive at the finish in one hour, guess which car was more inefficient and thus used more gasoline? The car driving at varying speeds.

This example illustrates the importance of maintaining a steady pace and/or intensity level over the distance of an event.

Self-Regulation: Mind or Body?

Intensity and fatigue are invariably linked to endurance sports training and, thus, pacing. The longer and/or more intensely an individual continues to exercise past the initial sensation of muscle tightness, the more fatigued the person will become. If exercising to an extreme level, eventually, the level of performance will decrease, and if continued long enough, exercise cessation will likely occur.

The human body regulates itself to ensure survival and to operate efficiently. What is up for debate is the exact means by which the body regulates itself and to what extent.

Fatigue

Fatigue is a catchall term that is used in many different contexts. For example, fatigue can be used to denote mental or physical stress. Additionally, the exact definition of fatigue is also up for debate. Some view fatigue as a progressive process that begins at the onset of exercise, while others view it as the involuntary endpoint of an exercise bout (542, 554). Causes of physical fatigue likely differ based on the mode of exercise and its duration and intensity.

Physiological changes that occur within a motor unit regarding muscular fatigue are termed **peripheral fatigue (PF)**. This is considered the traditional model of fatigue. For example, when running, the quadriceps begin to “burn” and fatigue; this indicates a buildup of acidity within the muscle, which slows the runner and possibly causes the runner to stop. In other words, the physiological changes that result in fatigue are isolated to specific, local muscle motor units (545). To review, a motor unit comprises a motor neuron and the muscle fibers that the neuron influences (innervates). Below is an example of peripheral fatigue.

Cause: Running uphill

Response: Buildup of intramuscular acidity – causing muscle burn

Effect: Runner slows or stops to recover

The opposing model of fatigue is termed **central fatigue (CF).** Central fatigue results from events within the brain and spinal cord (541). Central fatigue and, more specifically, the **central governor model (CGM)** are discussed in the next section.

Performance Limiter Models

The central governor model (CGM) is the opposite of the peripheral fatigue model. **Peripheral fatigue states that the muscle itself is the cause of the fatigue, whereas the CGM states the brain, not the muscle, is the reason for a decrease in muscle performance** (534, 538). This means that the brain constantly assesses the body to ensure we don’t hurt ourselves. Regarding the functionality of the body, the baseline is called **homeostasis**. Homeostasis is defined as (535):

Any self-regulating process by which biological systems tend to maintain stability while adjusting to conditions that are optimal for survival.

The brain monitors the body to ensure that no aspect of the body's functioning deviates too far from homeostasis. In essence, the brain is the body's control room and thus its safety mechanism. If the brain senses that something is getting to a potentially dangerous level, it acts to decrease the intensity level of that specific area.

In the example used previously regarding a buildup of intramuscular acidity causing a muscle's function to decrease – under the CGM, this is the result of the brain reducing muscle fiber recruitment of the quadriceps (536).

Physiologist A.V. Hill originally proposed the CGM in 1924 (537, 539). Hill suggested that the heart was protected by some sort of a governor, likely either from the heart itself or the nervous system (539). More recently, the existence of a central governor was theorized by physiologist Tim Noakes (540, 544). Most of the information discussed today regarding a central governor is based on the research by Noakes et al.

Specifically, Noakes et al. summarize the CGM as follows:

"Exercise performance is regulated by the central nervous system specifically to ensure that catastrophic physiological failure does not occur during normal exercise in humans." **(542)**

"The subconscious brain sets the exercise intensity by determining the number of motor units that are activated and hence the mass of skeletal muscle that is recruited throughout the exercise bout." (543)

CGM regarding pacing:

"The brain determines the number of active motor units based on a pacing strategy that will allow completion of a task in the

most efficient way while maintaining internal homeostasis and a metabolic and physiological reserve capacity.” (543)

The existence of a central governor in respect to the CGM likely has ramifications regarding the definition of many physiological events and terms. VO2 max represents an individual’s maximal capacity to transport and use oxygen during exercise. However, under the CGM, an individual’s VO2 max does not represent the maximal capacity to transport and use oxygen; rather, the central governor determines this (602). While some may view this difference in definition as a nuance, physiologically speaking, the distinction is substantial because of its implications.

A study by Karlsson and Saltin noted that performance decreased before complete glycogen depletion (716). There is minimal evidence to support physiologically why this occurs. As such, a decrease in performance prior to total glycogen depletion may be associated with CGM.

The below excerpt from ‘Run Like a Pro,’ by Ben Rosario and Matt Fitzgerald explains the CGM in clear layman’s terms:

The brain’s number one job is to protect us at all times. When we engage in an activity that, if continued for a long enough period of time, would seriously harm us, certain areas of the brain begin to send signals to our conscious minds that we should stop said activity. When running a 400-meter repeat at our 5K race pace, even though that repeat only takes somewhere between one and two minutes for the typical high school cross country runner, the brain goes into Nostradamus mode and predicts that were we to keep running at this pace. Eventually, we would get dangerously tired. We might cramp up. We could pass out. So the brain gives us the high sign of physical discomfort and loss of motivation, which encourage us to quit long before we are in any real danger.

Realizing this fact, and acknowledging it, is a wonderful thing. It gives us the power to override these signals, first in practice and eventually, hopefully, on race day. And one of the best ways to achieve this effect is by doing a lot of very structured workouts where you’re forced to lock into a very specific pace. In locking into a pace, you’re essentially pretending you can’t slow down, forcing yourself to stick to it regardless of those danger signals. To succeed in this effort, you must learn to relax while still running *fast. Come race day, you can fall back on all of this practice and do essentially the same thing, running relaxed and fast so you can complete the distance at a pace that represents your true limit rather than what your brain tries to tell you is your limit.*

Key Aspects of the CGM

1. The CGM can be thought of as a modified central fatigue model, as the CGM does not identify the spinal cord or motor unit as having a regulatory role (538).
2. The CGM does not identify that changes within a motor unit (i.e., decrease in force production) can occur as the primary means of fatigue. In other words, the CGM views peripheral fatigue as a byproduct of central fatigue (538).
3. The CGM definition of fatigue is not quantitative but rather based on one's sensations (543). This is akin to using rate of perceived exertion (RPE) to assess fatigue.
4. The CGM theorizes that the subconscious brain determines the metabolic cost required to perform a physical task (543).
5. Homeostasis is maintained under all conditions of exercise (542).
6. CNS regulates motor-unit recruitment to ensure that ATP production and utilization are matched (538).

Challenges to the CGM

While the CGM raises many interesting propositions and theories, according to Henriette van Praag, Ph.D., a researcher at the National Institutes of Health, the CGM *"lacks a clear structural/physiological basis within the central nervous system. Thus, empirical evidence for the existence of a 'governor' remains to be established."* (546)

While there is no direct quantitative evidence to prove the existence of a central governor, does it mean that it does not exist? As noted earlier, the CGM is a modified version of the central fatigue model, and thus, it is likely that aspects of the CGM exist to regulate human performance.

Perhaps the most extensive critical review of the CGM is by Weir et al. (538). While the paper identifies many areas of the CGM that

Weir and his team feel are not valid regarding proving the existence of the CGM, Weir et al. conclude that **it is likely that both the central and peripheral fatigue models work together to influence and regulate human performance.**

Below are some aspects of the CGM that Weir et al. challenged regarding its validity.

1. While the CGM assesses fatigue based on one's perceived exertion levels, proponents of the CGM assess fatigue by quantitative means (e.g., VO2 max, blood lactate). Therefore the results are assessing exercise performance, not fatigue (538).
2. The CGM addresses a singular aspect of fatigue, whereas, in the real world, there are many limiting factors of performance such as motivation and environmental issues (538)
3. According to the CGM, homeostasis is maintained under all conditions of exercise. By definition, homeostasis is not maintained during exercise as varied exercise intensities exist and are likely necessary to stimulate the required physiological adaptations to induce advancements in fitness levels (547).
4. Changes in motor unit recruitment and firing rate can be influenced by reflex effects of the spinal cord (548, 549). This is at odds with the CGM.
5. During voluntary exercise, fatigue can occur that is not the result of CNS limiting motor-unit recruitment (550, 551, 552, 553).

CGM Discussion

When discussing the peripheral and central governor models regarding the origin of fatigue, it is essentially the classic argument of which came first, the chicken or the egg. Regarding fatigue, it is likely that both the chicken and the egg occur simultaneously. In other words, based on the studies reviewed in this section, it is likely that both the central and peripheral fatigue models coexist to monitor fatigue and, thus, regulate exercise intensity.

Specifically, regarding the CGM, while there are studies that seem to conflict with the existence of the model in its entirety, **it is likely that many aspects of the CGM are valid as no fatigue model or theory explains all of the aspects of fatigue** (538). More research is required regarding the existence of a “governor” and, more specifically, what exactly the term “governor” implies and encompasses. However, as Weir et al. propose, searching for a unifying theory of fatigue is “futile” (538).

Based on the likely contribution of both the peripheral and central fatigue models, the following definition of fatigue is suggested:

Any exercise-induced reduction in the ability to exert muscle force or power, regardless of whether or not the task can be sustained (554, 555).

The CGM and an updated model termed the ‘Psychobiological Model’ are discussed in the next lesson, ‘Mental Training.’

Stick to the Script

It is important not to get sucked into racing at another person’s pace but to stick to the planned pace. It is easy to get caught up in the excitement of a race and go too fast. Preparation events will allow your athlete to gain exposure to the excitement of race day while learning how not to translate that excitement into throwing off the pacing game plan.

Utilizing some or all of the assessment tools noted here during training will ensure that an athlete is racing at the correct pace on race day.

Although it is important to stick to the script, this applies only when your athlete is healthy, well-rested, and performing at their expected fitness level. If your athlete has an off day and feels terrible, the individual should reduce intensity below what was initially targeted. Your athlete must listen to their body.

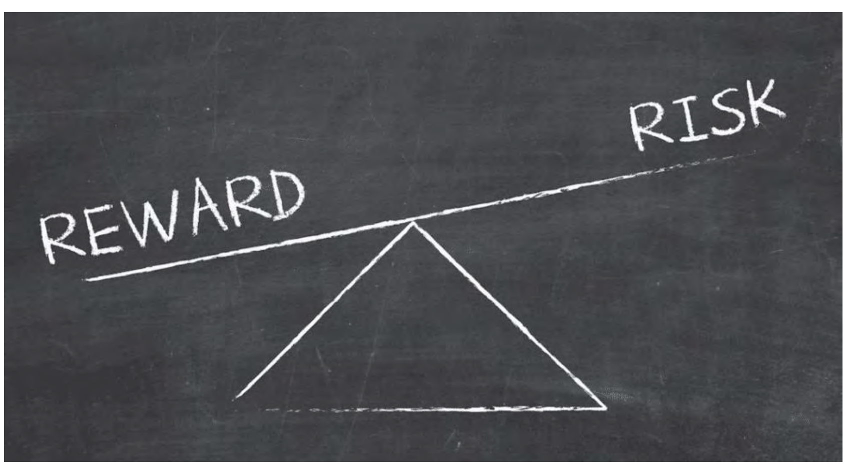
Many runners enjoy training with others for motivation and social interactions. While this is generally a good idea, a common mistake is to train with groups and do workouts that don't follow a runner's specific training program. Your athlete must be aware of the group workout plan and pass on it if it does not adhere to the plan.

While a warm-up is advised before starting a race, it is also advised to start off on the conservative side regarding pacing. Typically, only elite/professional athletes have the training to start at a high level of intensity and maintain that level throughout the event (318). The longer the event, the more accurate this is.

Risk Management

As with most things related to sports, proper pacing comes down to risk management. Go too easy, and an athlete will likely miss the time goal. Go too fast, and an athlete will likely blow up and miss the time goal. However, with proper pacing, an athlete can set themself up with the best chance to succeed by managing risk appropriately.

Risk vs. Reward



It is common knowledge that the pace for a marathon is slower than that of a 5K. **While there are physiological reasons for this,**

a runner's mentality also plays a large part. Specifically, distance will dictate how fast an individual will run. As an example, let's assume that Jessica can run a 5K at an eight-minute-mile pace and has never run a marathon. On a whim, Jessica decides to enter a marathon without any specific training (obviously not a smart idea). Jessica paces herself at a 10-minute-mile pace during the marathon. But why? She has no previous experience telling her how fast to run, so why did she slow down so much in relation to her 5K pace? The answer: While Jessica knows she can run at least 5K at an eight-minute pace, she believes that given the distance of a marathon, she would not be able to hold that pace throughout without blowing up. This is an example of risk management.

Let's assume that Jessica miraculously (in lieu of her lack of training) makes it to 24 miles at a 10-minute pace without blowing up. She decides that since there are only 2.2 miles left, she will pick up the pace. Jessica lifts her pace to 8:30 min/mile for the remaining distance. Why didn't Jessica pick up the pace before this? For the same reason that she initially determined to run a 10-minute pace at the start – there was too much risk of blowing up. With 2.2 miles left to run, she determined that the potential reward of a faster time was worth the risk. In other words, she estimated that the reward (i.e., picking up the pace and finishing faster) outweighed the risk (i.e., blowing up and not finishing or finishing slower).

This represents that the intensity one runs at correlates to the distance left to run. **The farther from the finish, the higher the risk – and the closer to the finish, the lower the risk**.

In 2010, a group of researchers (de Koning et al.) sought a way to quantify this phenomenon. In a paper titled, *Regulation of pacing strategy during athletic competition*, de Koning et al. created a formula called the **hazard score** (567).

The hazard score correlates the intensity at which an individual is performing to the distance or time remaining in the exercise bout and thus determines the likelihood that an individual will change pace. The formula is as follows (534):

Hazard = Momentary RPE * Fraction of the distance remaining

While this certification won't go into the specifics of the hazard score paper by de Koning et al., the primary thing to understand is that the RPE of an individual at a set point in time and the distance remaining of an event influence the pace of a runner (567).

Pace Assessment Tools

While there are likely countless pace assessment tools, UESCA has identified four that are likely the most popular.

- **Heart rate monitor**
- **GPS that gives real-time pace**
- **Time**
- **Rate of perceived exertion scale (RPE)**

Regarding heart rate and as noted earlier in the certification, *cardiac drift* is a phenomenon that elevates one's heart rate without increasing other performance metrics. It is natural for cardiac drift to occur while running, especially in the latter parts of a training run or race. Cardiac drift can artificially increase one's heart rate by as much as 10–15 bpm. Therefore, it is important to pace not only by heart rate but also by RPE.

Sample Workout Structures Using Pace Tools

You can create structured workout sessions for your athlete using some or all pace assessment tools. Following are examples of workout structures using the pace assessment tools:

Heart Rate Monitor: Easy run, keep heart rate between 110 and 130 bpm for 30 minutes.

GPS: Five-mile run, run the first mile at nine-minute pace, then run the next three miles at 6:30 pace, and cool down at nine-minute pace for the last mile.

Time: Run 400 meters in two minutes, jog 100m. Repeat the cycle two more times.

Based on the preceding pace tools, it is very easy to allow numbers and quantitative metrics to rule your coaching practice. It is also easy for runners to obsess about their quantitative stats. While pace assessment tools are extremely valuable and necessary in the training and racing process, it is also important to unplug from this data stream now and then. Have your athlete go for a run purely for fun and check out the scenery versus staring at a GPS watch the whole time! Unplugging is easiest to accomplish and most practical on easy workout days.

Pacing By Feel

Many experienced runners can accurately determine their pace by feel. This comes from lots of experience and correlating their pace to pacing tools and RPE. So how does one become accustomed to pacing by feel? Following is a way to test one's "feel" for pace.

Using a stopwatch, run a set distance at the desired pace (ex: run three miles in 24 minutes – eight-min/mile pace), but look at the time only once the distance has been run. If your athlete has a real-time GPS watch, they can guess the pace and then look at the watch to see how close they were.

While this is just one example of how to test one's ability to feel the pace, the tests must be done frequently to increase one's proficiency at correctly pacing by feel.

If individuals use a pacing tool(s) (i.e., heart rate monitor) to determine effort level, they need to correlate the pacing tool pace with how they feel. For example, if an athlete's pacing strategy is to maintain an 8:30-min/mile pace during a race, but the person cannot maintain this pace because of exhaustion or injury, the runner *must* slow down. Failure to do so will likely result in a slower overall time at best and a "Did Not Finish (DNF)" at worst. Learning how to pace by feel is important to develop and master.

Pacing Variables

Long vs. Short

Pacing for long- and short-distance events varies. Short-distance events (e.g., 5K) are typically raced at a higher intensity than long-distance events (e.g., marathon), assuming an athlete's fitness level is very good, and the individual has trained at high intensities. Short events are often considered 10K or less, and long-distance events are typically considered 10K or longer.

Experience / Fitness Variables

Individuals with race-proven fitness levels can typically perform at a higher intensity than those who are new to the sport or not at a high fitness level. We're not talking about a jump from a Zone 2 effort to a Zone 4 effort – we're talking about a relatively small difference. For example, a new runner might run a half marathon in the low to middle part of Zone 2, whereas an experienced participant who has a high fitness level might race it in the middle to high Zone 2 or even low Zone 3.

Many of the pacing guidelines described in this section are identified in ranges based on LT. Less-fit athletes should race toward the bottom of the range, and more-fit athletes should target a pace in the upper range.

In a study of runners at the IAAF Half Marathon World Championships, the fastest runners more or less maintained their pace throughout the race, whereas slower runners saw their pace decrease from 5K onward (657). While this is likely an example of experience versus fitness, regardless of fitness or experience, a runner can run an even-paced race.

Pacing By Time

More than any other metric, most runners shoot for a particular overall time. During training and racing, the total event distance is often broken into sections, each with a set time goal. From a racing perspective, this is done to check one’s progress. For example, during a marathon, your athlete might have set time goals at the 10K, 13.1 miles (halfway), and 30K (18.6 miles) points.

If your athlete is not at the targeted time at certain checkpoints, it is important not to lift the pace to regain time. This will typically result in an overall slower time and perhaps a DNF.

The best way for your athlete to reduce overall time without putting in more effort is to focus on running the tangents of the race course.

Pack Pacing

In the study of the IAAF Half Marathon runners previously noted (657), runners who ran in packs had smaller pace reductions than those who did not run in packs. Therefore running with others in race scenarios is advised to run at or near a specific pace.

Age, Gender, and Speed Variables

A 2011 study (March et al.) looked at determinants of pacing pertaining to marathons (658). The findings:

- Older runners maintained a more consistent pace than younger runners
- Women maintained a more consistent pace than men
- Faster runners maintained a more consistent pace than slower runners
    - This finding was also consistent in other studies (659, 660).

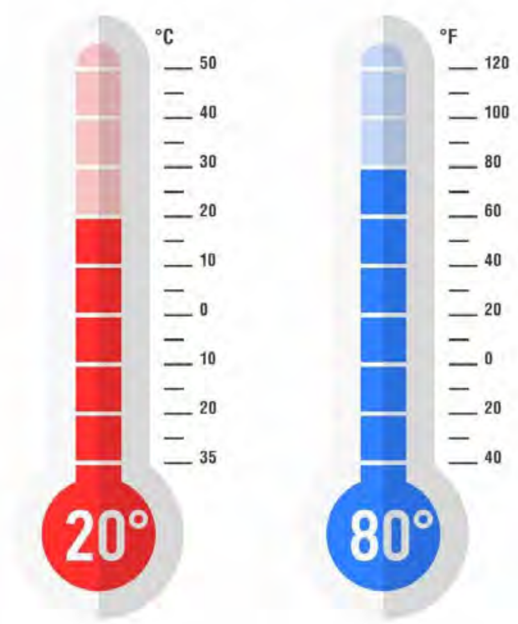
These results were consistent with a 2014 study by Trubee et al. that showed women were better at pacing than men, especially in hot weather. The study also found that elite runners were better at pacing than non-elite runners. Unsurprisingly, there was no difference in pacing aptitude between elite men and women (659).

Effect of Temperature on Pace

The ideal temperature for running (especially long distances) is between 50 and 53.6 degrees F (729). Within this range, the temperature does not have a negative effect on a runner's pace. **Temperatures above or below this range will likely negatively affect pacing.** Temperatures outside this range likely have a more significant adverse effect the longer the distance is.

A table by Magill et al. demonstrates the effect temperature can have on pace (723). Like the aforementioned study, this table assumes that 53 degrees F is the ideal temperature for running. At 90 and 10 degrees, the recommended adjusted pace is 27 seconds per mile slower. For example, if a runner's pace at 53 degrees F is 8:00 min/mile, the pace at 90 and 10 degrees is 8:27 min/mile (723). Slight temperature variations from 53 degrees F to 60 degrees have little to no change in recommended pace changes. However, the more extreme the temperature change, the more effect it has on the recommended pace.

Be aware that any temperature/pace guideline chart is just that, a guideline. Individuals' tolerance for temperature changes varies substantially, and adaptations such as heat-acclimation training will lessen the negative effect of heat on an individual.



Effect of Heat on the Central Governor (CG)

Concerning the central governor model, research shows that an increase in core temperature stimulates the CG regarding anticipatory regulation – meaning, as one's core temperature rises to the point where fatigue would be likely, the body self-regulates to slow a runner, thereby reducing the person's core temperature (733).

Physiology

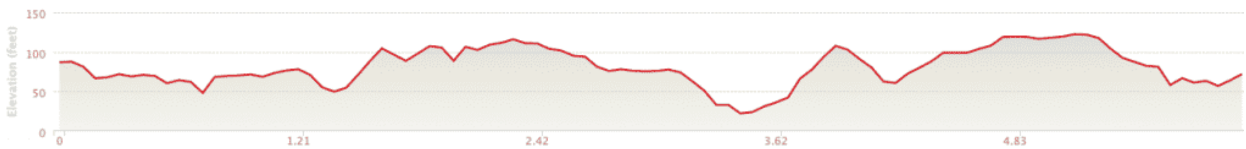
As an individual's core temperature rises, an increase in blood is directed to the skin to aid in cooling. This results in a decrease in blood flow to the muscles, reducing the amount of oxygen to the muscles and thus likely decreasing performance (520). Additionally, there is a decrease in ATP production at elevated core temperatures (732).

Mitigating Effects of High Temperatures

In addition to heat-acclimatization training, pre-cooling the body before running (i.e., use of an ice vest) and ingesting cold fluids have been shown to decrease core body temperature, which may result in increased performance (534).

Course Profile (Elevation)

As noted in the *Training Program* module, utilizing an elevation course profile can be very helpful in determining a pacing strategy for your athlete. Even if the advised strategy is to maintain a consistent pace, your athlete needs to know where any hills are. Following is a sample profile of a 10K course.



Determining Percent Gradient

Most course profiles equate to little more than a squiggly line and distance/elevation markers. How do you figure out the steepness of the hills? For this, you need to think back to middle school and channel your inner 13-year-old – remember “rise over run”?

The equation for this is: 100 x rise/run. For example, let’s say that a road rises 100 feet in one mile (i.e., run). Therefore, the equation would be:

100 x 100/5,280 (feet in a mile) = **1.9 percent grade**

Sample Race Pace Guidelines

Regarding the sample elevation profile above, a sample pacing strategy might look something like this:

Use miles 0–1.5 as a chance to warm up, maintain an RPE of no more than 6/10, and target a pace of 8:00 min/mile. Once you hit the hill at 1.5 miles, increase your RPE to 7/10 (note, your pace will drop slightly).

Continue running at this 7/10 pace until you hit mile 3.5 when the big hill begins. Your RPE will likely increase, but try not to go above a 7.5/10 RPE on this hill. Maintain this RPE until mile five. At this point, if you feel good, increase your RPE to 8/10 and target a 7:45-min/mile pace if possible throughout the end.

As this example is based on a six-mile course, the pace and RPE increases are likely more aggressive than would be considered appropriate for a longer event, such as a half or full marathon. Longer events would likely have a more consistent pace (RPE-based) throughout the race.

Pacing Guidelines

The Start

Most runners have some level of anxiety before a race. Combine this with adrenaline, and it's easy to see how your athlete can get lulled into running a pace faster than initially planned.

The run start is a chaotic and stressful environment. Most individuals will have elevated heart rates because of nervousness and stress – and this is before the gun goes off! This is termed ***'anticipatory anxiety.'*** In other words, an athlete is anxious about the start of a race and experiences many physiological changes typically associated with physical exercise (e.g., increased respiration and heart rate). **Once the race starts, the stress and anxiety typically decrease.**

Your athlete should focus on pacing by feel and settling into the planned pace. Athletes who find themselves running too fast must slow down immediately and get back on track.

A 2010 study by Hausswirth et al. analyzed how the pace of the first kilometer of a run affected overall run performance (367). **The results found that running 5 percent slower than one's 10K race pace resulted in the best overall run time.** Conversely, running at 5 percent faster or 10 percent slower than one's 10K race pace resulted in an overall slower time. In other words, too slow or too fast a pace in the first kilometer/mile is likely to negatively affect the overall run time (specifically for distances of 10K and below).

Body Awareness

The ability to pace properly is dependent on body awareness, specifically concerning one's form, breathing rate, and overall fatigue/exertion level.

Running Form

Being aware of one's form is very important while running. If an athlete notices form is breaking down, they are advised to slow down and work to regain proper form. Common causes of form breakdown are exhaustion and overexertion. Runners often find that doing a body scan helps check their form. For example, the athlete might assess the following areas:

– **Head:** Is my neck vertical or substantially flexed or extended?

– **Back:** Is my back hunched (flexed) or extended?

– **Foot strike:** Is my foot strike at the correct size and rate? Am I shuffling?

– **Hips:** Are my hips static or rotating?

– **Shoulders:** Am I elevating my shoulders, or are they depressed and retracted?

Breathing / Stride Rate

Breathing rate can be correlated to one's stride rate. By taking a breath after a set number of strides, your athletes will be able to regulate the stride rate and the breathing rate. Being aware of the speed and frequency of the stride/breathing rate will allow your athletes to stay on pace.

Once an athlete has established a pace that they feel is appropriate, they need to get a sense of their stride and breathing rate and how it relates to RPE.

Overall Fatigue Level

If an athlete becomes overly fatigued at any time, they must slow down. While some degree of fatigue is normal and expected with running, if your athlete experiences fatigue to the extent that it could impact their safety or their ability to complete a workout or race, they must slow down or stop.

Break It Up

Breaking a large project into smaller, more manageable pieces helps with motivation and maintaining focus and discipline to accomplish a task. This strategy is often used when running and, more specifically, during longer events such as half and full marathons.

A sample strategy for a marathon might be to break the race up into three parts:

- **Start–13.1:** Run at an RPE of 6/10
- **Miles 13.2–22:** Run at an RPE of 7/10
- **Mile 22–Finish:** Assuming you feel good, run at an RPE of 8/10.

This strategy can also be used in shorter events.

Common Pacing Mistakes

Starting Off Too Fast

This is the most common pacing mistake, as even seasoned runners fall victim to it. It is very easy to start off too fast. This is because of many factors:

- Overly excited
- When starting off, a runner doesn’t feel fatigued at a faster pace

- Desire to keep up with surrounding runners

While this can occur at any distance event, it is most common in short races (10K or less).

Running Someone Else's Pace

It is common to run at another person's pace. This is especially true if an individual runs near another runner for long periods and that runner either speeds up or slows down. Regarding slowing down, this speed change is often not a conscious effort. Lastly, some runners get competitive with other runners during a race and change their speed based on beating those around them.

Running at Substantially Varied Intensities

The most common issue in this regard involves hills and descents. Many runners try to maintain the same speed (pace) up hills as on the flats. Conversely, many runners relax while going downhill, causing their form to break down. Running at varying intensities is less efficient than running at a constant intensity. Therefore, typically results in a slower time and potentially a "Did Not Finish (DNF)" because of excessive energy expenditure.

Running Positive Splits?

If a runner runs a race with positive splits, it means that if the race was split up into halves, the second half is run slower than the first. Running a race with positive splits is likely viewed as negative due to the assumption that the runner started too fast, which is the reason for the slower pace later in the race.

However, 'positive split' does not denote how much slower the second half of a race was compared to the first. For example, was a runner one minute slower or 45 minutes slower during the second half? Both are examples of positive splitting, but as you can see, both are very different from a time perspective.

Therefore, when addressing whether or not a positive split on the part of your athlete was a poor pacing strategy or something went wrong, you must first look at the time difference between the two halves to see if the time difference is substantial.

In contrast, a ***'negative split'*** denotes a race in which a runner runs the second half of the race faster than the first, and an ***'even split'*** denotes a race in which a runner runs both the first and second half at the same pace. Lastly, as common-sense dictates, a race course's terrain significantly affects the runner's speed and, thus, the pacing.

Distance Specific Pacing Guidelines

The following guidelines are rough recommendations, as the exact pacing prescriptions are individually based. Typically, the more in shape and/or advanced a runner is, the longer the person can run at MLSS. Additionally, an in-shape runner can typically run at paces higher than MLSS for longer periods than someone who is not as aerobically conditioned.

One thing to keep in mind – the information in this section is for you, the coach... not for your athletes. Why? Because all of this information can be overwhelming for an athlete. It is your job as a coach to understand the below information and then convey a strategy to your athlete in an easily consumable manner.

Race-Specific Advice

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Functional Threshold Heart Rate-Based Pacing

The following guidelines are rough recommendations, as the exact pacing prescriptions are individually-based on many factors, including environmental factors such as heat and humidity – as they affect heart rate.

TRAINING ZONE	PRIMARY TRAINING APPLICATION(S)	AVERAGE HEART RATE (% OF FTHR)
ZONE 1: RECOVERY (Active)	Recovery	< 68%
ZONE 2: ENDURANCE	Improve fat metabolism, increase endurance, Increase cardiovascular/base fitness	69 - 83%
ZONE 3: STEADY STATE	Increase aerobic fitness	84 - 94%
ZONE 4: LACTATE THRESHOLD	Increase lactate threshold, improve sustainable pace	95 - 105%
ZONE 5: VO2 MAX	Increase ceiling for aerobic fitness	> 106%
ZONE 6: ANAEROBIC POWER	Anything above VO2 Max (ex: sprinting)	N/A

Below are sample training zones based on an individual's (Brian) functional threshold heart rate (FTHR) 0f 165 bpm.

TRAINING ZONE	PRIMARY TRAINING APPLICATION(S)	% OF FTHR	BRIAN'S FTHR ZONES
ZONE 1: RECOVERY (Active)	Recovery	< 68%	**< 112 bpm**
ZONE 2: ENDURANCE	Improve fat metabolism, increase endurance, Increase cardiovascular fitness	69 - 83%	**113 - 137 bpm**
ZONE 3: STEADY STATE	Increase aerobic fitness	84 - 94%	**139 - 155 bpm**
ZONE 4: LACTATE THRESHOLD	Increase lactate threshold	95 - 105%	**157 - 173 bpm**
ZONE 5: VO2 MAX	Increase ceiling for aerobic fitness	> 106%	**> 175 bpm**
ZONE 6: ANAEROBIC POWER	Sprinting	N/A	

Short Distance

Assuming your athlete is in good physical condition, race distances of 5K and less can be run at a relatively high level of intensity compared with long-distance events. Depending on one's fitness level, the advised intensity range is between **95 and 110 percent of one's FTHR**.

Events around 10K in length are done at a more conservative pace. This typically equates to racing at **90-100 percent of one's FTHR.**

Long Distance

For half marathons, the pace should be between **75 and 85 percent of one's FTHR.** From an RPE perspective, the intensity should feel like a long-tempo run (368).

Pacing for marathons should be between **70 and 80 percent of one's FTHR**. The pace should be five to 10 bpm lower than a long tempo run from a heart rate perspective. The RPE should feel like a long endurance-based run.

Summary

- The general pacing theme is to start conservative and finish strong.
- Race pace is determined during training.
- Intensity benchmarks such as heart rate, GPS pace, and rate of perceived exertion (RPE) are ways to assess one's pace.
- It is important to stick to the predetermined pace plan and not deviate from it and, more specifically, not go over the prescribed pace.
    - The only time it is OK to go above the prescribed pace is toward the very end of a run, and only if an athlete feels able to do so without "hitting the wall."
- It is common for runners to be nervous before a race. This is especially true for new runners and/or before important events.
- Peripheral fatigue relates to fatigue starting within the skeletal muscle whereas central fatigue corresponds to fatigue controlled by the mind. These issues relate to the *central governor model*.
- Even though short-distance events like 5Ks are short compared with marathons, they are still endurance events and therefore proper pacing is important for a successful race.
- Pacing is typically based either off pacing metrics (GPS), heart rate, RPE, or a combination of them.
- Generally speaking, the shorter the event, the faster the pace.
- Overreaching too early in a distance-running race will reduce the energy and performance later in the race.
- Pacing must be practiced in training to be effective on race day.

- Default to body awareness: Regardless of what a heart rate monitor says, if your athlete is feeling fatigued the individual should slow down.

Module 17: Mental Training

- History of performance psychology
- Psychology 101
- Myths of sports psychology
- Motivation
    - Theories
- High-performance commitment and barriers
- Historical beliefs and models of endurance limits
- Self-talk
- Self-concept and self-efficacy
- Mindfulness
- Emotional regulation
- Executing when it counts
    - Mental imagery
- Flow and clutch state
- Yerkes-Dodson Law
- Performance anxiety
- Mental toughness
- Pain threshold, tolerance, and sensitivity
- Psychology and Injury
- Attentional control

Introduction

Much like physical training, mental skills or sports performance psychology can be trained in a specific, deliberate way that optimizes skill development and gives the athlete the best possible chance of success.

Although goal attainment alone is not the key defining marker of success for many recreational athletes and is only part of their journey, endurance sports offer many athletes a unique opportunity to pursue their self-discovery. Many come to endurance sports to redefine themselves or out of a deep desire to change their life direction. Understanding what brings an athlete into the endurance world is critical in any coach-athlete relationship.

Periodization of skills helps enhance this process, with the athlete learning how to engage in specific sports psychology practices at particular times. Mental skills can unfold across three broad domains, including The Foundation (much like base training, skills that are best utilized throughout the year and set up the next order of trainings); High-Performance Psychology Skills (the meat of many programs, especially when an athlete has identified and enters a training block targeting a set of goals or races for the year or season); and Race Day Specific Skills (as an event draws near, working on specific aspects of the mental game that align with the specific aspects of the event at hand). All of these skills work together in concert and build upon one another. Much like a physical race, the training of the mind is done deliberately throughout the athlete’s life. It is specific to the details of that individual, their personal history, their time commitment, dedication level, their specific goals, and the race day specifics of the event they are partaking in.

Sport psychology training need not occur in a vacuum. Some skills can be developed outside of training, such as mindfulness, goal setting, and self-talk, but a number of these skills are best developed alongside the physical training itself.

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The Foundation

Brief History of Performance Psychology

Performance psychology has grown in leaps and bounds over the past 100+ years, with a recent acceleration of interest over the past few decades. Norman Triplett is widely believed to have been the first to study how psychological factors impact performance, citing social facilitation factors of cyclists performing better when riding in groups than riding solo in 1898. He was perhaps the first to show interest in understanding social and psychological impacts on performance. In 1925 Coleman Griffith (widely labeled the father of sports psychology) founded the first sports psychology research lab and facility at the University of Illinois. His early quotes on the impact of the mind on performance were groundbreaking at that time: “The more mind is made use of in athletic competition, the greater will be the skill of our athletes, the finer will be the contest, the higher will be the ideals of sportsmanship displayed, the longer will our games persist in our national life, and the more truly will they lead to those rich personal and social products which we ought to expect of them. Because of these facts, the psychologist may hope to break into athletic competition, just as he has already broken into the realms of industry, commerce, medicine, education, and art.” Coleman was likely the first psychologist to work alongside a professional sports team, helping the Chicago Cubs in the 1930s. He also authored two books, Psychology of Coaching and Psychology of Athletics.

Performance psychology development mainly focused on motor skill acquisition and was conducted in lab settings over the following few decades, with renewed enthusiasm in the 1960s and 1970s. The International Society of Sport Psychology (ISSP) was created in 1973. In 1979, a paper by Rainer Martens titled “About Smocks to Jocks” was published, calling for more research to examine the real-life applicability of psychological factors in the field of play rather than just within laboratory settings. This paper was one of the cornerstones in increasing recognition, awareness, and growth of psychology in the sport over subsequent years.

The Association for Applied Sport Psychology was founded in 1985, and one year later, Division 47, The Society for Sport, Exercise and Performance Psychology, was created in the American Psychological Association.

Today, there is significant interest in performance psychology practices across nearly all spectrums of sport, spanning all age ranges and performance levels. Sport and performance psychology today typically involves three areas of focus, including a Clinical Focus (general mental health, athlete mental health, and aspects of injury and retirement from sport, etc.), a Performance Enhancement Focus (managing performance pressure, developing focus, and concentration skills, self-talk, and visualization, etc.), and a Coaching Collaboration Focus (improving communication and team cohesion, etc.). Most major sports franchises, including the United States Olympic Committee (USOC), now employ a team of mental health professionals, clinical psychologists, and Certified Mental Performance Coaches (CMPC), a credentialing program for those with a master's or doctorate through the AASP.

Myths of Sports Psychology

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Why Train the Mind?

"Mind is everything — muscle pieces of rubber. All that I am, I am because of my mind." Paavo Nurmi, aka The Flying Finn (9 time Olympic Gold Medalist, Running)

"Your mind is what makes everything else work." Kareem Abdul-Jabbar (Basketball)

"Of the top 100 players, physically, there is not much difference. It's a mental ability to handle the pressure, to play well at the right moments." Novak Djokovic (Tennis)

"I believe a champion wins in his mind first; then he plays the game, not the other way around. It's powerful stuff." Alex Rodriguez (Baseball)

Motivation

Perhaps the most common definition of motivation in the sporting context is "The direction and intensity of one's efforts. (Sage, 1977)"

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Self-Determination Theory (Intrinsic and Extrinsic)

Psychologists Edward Deci and Richard Ryan developed the self-determination theory, which describes motivation as “what moves us to act.” Their theory states that humans have an inherent tendency to move toward growth and suggests that humans have three core needs that help move towards growth and mastery: Autonomy, Competence, and Relatedness (939).

Autonomy is defined as the ability to engage in voluntary and self-endorsed behavior and feel in control.

Competence is defined as experiencing our behaviors as effective, feeling as though we’ve done, are doing, and will continue to do a good job.

Relatedness is defined as the need to interact and be connected with others.

Self-determination theory describes two forms of motivation: intrinsic and extrinsic. **Intrinsic motivation** is inherently self-satisfying and engaging and is often described as “behavior for its own sake.” Autonomy, competence, and relatedness are the three primary core need drivers connected to intrinsic motivation.

On the other hand, extrinsic motivation is related to being rewarded in some tangible fashion. For athletes, this can come in completing an event, earning a finisher’s medal, or hitting a qualifying standard for entry into a specific race. This can also come in social media and a need to be recognized by others.

Self-determination theory has been studied about a number of factors in sport, including drop-out rates from participation in adolescents.

Goal Orientation Theory

Goal orientation theory is another popular theoretical framework for understanding drivers for sports participation and performance, with the theory proposing two underlying orientations: a **task or**

ego orientation. This theory centers on the idea that goals are created with differing underlying reasons or purposes besides goal content (i.e., that which the person is attempting to accomplish). This helps explain the different approaches and responses to achievement-based situations and tasks. **Task orientation** is often called mastery-goal orientation and focuses on three primary factors: engaging in challenging activities, exerting effort within an activity, and persisting during the challenge. Task orientation may differ from one activity to another, given the individual's interest level (for example – a triathlete may be more interested in cycling than swimming). Mastery goals focus on learning and mastering new skills, with a desire for increased understanding, awareness, or competence. This is most often seen with the drive for learning and improvement.

Ego orientation is often referred to as performance-goal orientation and tends to focus primarily on evaluation, typically in comparison to others. Competence is comparative under this framework, with an individual's feelings of worth or self-efficacy being based on completing a task or event compared to a similar reference point, person, or self-imposed standard. Performance goal orientation tends to focus on an attempt to be better than others, win, or make social comparisons that are judgmental in nature, emphasizing seeking favorable judgment from others.

What Can You Learn From Ten Million Marathon Finishers?

A seminal paper, "Reference-Depended Preferences: Evidence from Marathon Runners" (940), published in 2016, reviewed nearly 10 million marathon finishing times (9,789,093 to be exact) from 1970-to 2013 with the hypothesis that athletes are driven by the motivation to finish ahead of a given reference point (i.e.a particular time goal, typically either ending with 0 or 5 in connection to the marathon). This was seen most clearly with the 4 hour marathon finishing time, with a total of 300,324 finishers in the 3 minutes before 4 hours (i.e., 3:57, 3:58, 3:59), compared to 212,477 finishing in the 3 minutes just after 4 hours (i.e., 4:00, 4:01, and 4:02). The difference of 87,847 finishers was the most distinct with time stamps on a round number at either the half-hour (i.e., 3:30, 4:30, 5:30, etc.) or hour mark (3, 4, 5, etc.), although this trend of "excess mass" before a clear time stamp was seen when plotted across the 10 minute and 5-minute marks as well (i.e., 3:35, 3:40, 3:45, 3:50, 3:55, etc.).

This paper provides evidence for the internal and extrinsic factors associated with specific race times, particularly the marathon, but that likely translates for other endurance activities.

Outcome goals for recreational endurance athletes most often fall into one of the following categories: 1. Just Finish Goals, 2. Break a time goal (most often that of a clear number reference point such as a 4-hour marathon). 3. Set a PR, 4. Qualify for an event (again, most often targeting a set reference point). This paper supports reference points as a key motivating factor in goal orientation and the decision to speed up or increase intensity in a racing environment.

Commitment And Barriers

Barriers are inevitable in any pursuit of a goal. Some barriers are known in advance to encompass the reality of daily life. This includes work obligations, family responsibilities, outside commitments, and overall time to train daily and weekly. Barriers need to be systematically identified, emphasizing a structured plan that begins with awareness and recognition by coach and athlete alike, with a mutually agreed-upon plan focused on how barriers will be integrated as an expected part of the training plan.

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High-Performance Psychological Barriers

Psychology 101: Understanding Your Athlete

An individual's overall psychological framework is essential in understanding how to work with them. At any given moment in our lives, our basic individual psychology can be broken down into three main categories: Thoughts, Feelings, and Actions. Collectively, over time and with repetition, this constellation of experiences makes up what we commonly refer to as personality. Thousands of research articles, books, and countless models dive into a theoretical understanding of personality. Personality is a word and idea deeply woven into our everyday lexicon and intuitive experience. And for purposes here, we certainly do not have time to review all the potential models.

The core of our psychological framework involves those three basic categories. Thoughts include beliefs, biases, and values. Feelings encapsulate our mood and the frequency, intensity, and range of our emotional experiences. Actions describe our behaviors. You can consider each of these areas to be impacted in typical ways (how we may usually feel, think and act regardless of the situation) along with situation-dependent ways (how you might approach an easy versus a difficult task, for example).

We must be cognizant of the environmental factors in which our athletes and we live – the cultural, religious, financial, and community aspects that occupy a significant portion of the athlete's life. Combining the internal framework outlined above and the external environment, we thus have a Bio-Psycho-Social understanding of the psychological factors that comprise the athlete's life from a whole-person perspective.

One of the most widely used models for considering personality is **The 5 Factor Model**, sometimes more commonly referred to as The Big 5. These traits include extroversion, agreeableness, conscientiousness, neuroticism, and openness to experience. A 2006 meta-analysis reviewed 33 peer-reviewed studies that correlated extroversion and conscientiousness with high physical activity levels, citing that the former "may be the most important factor associated with physical activity." Neuroticism was correlated with low physical activity (941).

In my practice, I often view many athletes pursuing endurance sports as a journey of self-discovery. Many who seek endurance-based challenges are doing so to re-define themselves, push their limits of what they believe to be possible, or seek curiosity in the connection to challenge. **As a coach, you have a fantastic opportunity to partner with your athlete in their process of self-discovery by being able to ask pertinent questions about their background, motivation level, and pursuit of goals.**

The below video discusses this section and the next section – the three various models of one's limiting factors.

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Historical Beliefs on Endurance Limits

Peripheral and Central Fatigue

Historical models of exhaustion, fatigue, and limiting factors in endurance sports focused on a culmination of muscle fatigue on a central (central nervous system) and peripheral (muscular system) basis. Thus, traditional models have focused on improving muscular and cardiovascular strength, stamina, power, and endurance while largely ignoring psychological variables.

Central Governor

The central governor theory focuses on the central nervous system monitoring physiological signals during exercise, with the premise that the brain is in a state of constant monitoring and will prevent overexertion to the point of physical detriment or death (942). "It is hypothesized that a central governor controls physical activity in the brain and that the human body functions as a complex system during exercise." This theory claims that the brain is the ultimate regulator, i.e., the central governor, of all physical and physiological systems and regulates physical exertion and exercise through monitoring and subsequent regulation controls to prevent serious bodily damage and/or death. The central governor theory largely ignores conscious psychological processes such as motivation of self-determination and labels events such as fatigue through subconscious awareness, "Fatigue is a sensation that results from the conscious perception and interpretation of subconscious regulatory processes in the brain. It is therefore not the expression of a physical event" with the idea that the brain will override any other system to limit physically pushing past one's physiological limits.

Timothy Noakes is largely credited with the central governor theory in 2005 (943). However, the original idea was proposed by Archibald Hill as early as 1924.

Critics of the central governor largely posit the role of mediating factors of several psychological variables (including motivation and conscious control), alongside citing several examples of athletes who have experienced "physiological catastrophes" during sport, thus providing evidence of the ability to override the purported monitoring mechanism of which the central governor gets its name.

The Psychobiological Model (The New Formulation for Understanding Limits in Endurance)

The Psychobiological Model of Endurance Performance is the newest development in the theoretical understanding of limiters to performance. It gives increased attention to psychological factors, including cognitive perception, motivation, and willingness. This model considers the decisions to slow, stop, continue or accelerate in endurance-based tasks as being a highly conscious process of decision making, intensity regulation, and behavioral regulation, even going so far as to make their declarative statement, "The psychobiological model of endurance performance proposes that perception of effort is the ultimate determinant of endurance performance. Therefore, any physiological or psychological factor affecting the perception of effort will affect endurance performance." The basic tenets of this model are being examined by researchers in several different applications and even helped spawn the book, Endure by Alex Hutchinson.

This model is primarily based largely on Brehm's Motivational Intensity Theory (944), which consists of two main aspects of motivation: potential motivation and motivational intensity. Potential motivation is the maximum effort a person is willing to expend to reach a goal. In contrast, motivational intensity is the amount of effort that a person will exert. Of critical importance, this theory suggests that people will continue to engage in a task with the same level of intensity as long as they view their goal as remaining possible.

The psychobiological model suggests that an athlete will stop not necessarily when they reach a limit of physiological capacity but rather "when the effort required by the tasks exceeds the greatest amount of effort that the individual is willing to exert during the task (motivation) or when the maximal effort is considered to have occurred, and continuation of the task is perceived as impossible." Therefore, motivation alongside the perception of effort becomes the ultimate determinant guiding an athlete's decision-making in endurance sports in this theory.

Perception of effort is primarily mediated by cognition, cognitive appraisals (i.e., how one is thinking about the physical experiences they are having), and self-talk, the latter of which can function through either instructional or motivational channels.

Instructional self-talk is the specific language around desired movement and mechanics. This type of self-talk is best done when needing to guide through a particular task or remembering the basics of a new skill. For an endurance athlete, I'll often recommend instructional self-talk when guiding through specifically challenging aspects of a race course, such as charging up a hill, or descending a technical trail, with specific guidance around the mechanics of particular aspects of movement and posture or body position. Instructional self-talk is best used to remind yourself about a specific mechanical aspect of movement, which can be a helpful strategy in many instances, including open-water swimming for triathletes, especially those experiencing anxiety over this portion of the race. Clinically, I've found guiding instructional self-talk language to be a critically helpful factor in overcoming open water swim anxiety by giving the athlete a direct focal point of focus in bodily movement over anxious thoughts.

Motivational self-talk often focuses on psyching ourselves up for a challenge, boosting confidence, or maintaining effort. In short, motivational self-talk is often the framework that many think of when discussing self-talk and can sometimes be best summarized by thinking of it being your positive coach or cheerleader. The three areas of focus most motivational self-talk tend to center on include arousal and emotional control (i.e., you got this, stay present), mastery (i.e.learning and improving), and drive (i.e., you're doing great, keep pushing).

Studies have found that motivational self-talk reduces rates of perceived effort and increases time to exhaustion, especially in time-to-exhaustion laboratory tests (947, 948). There is also prevailing evidence suggesting that instructional self-talk is best used for tasks requiring fine motor control in specific movements. In contrast, motivational and instructional self-talk show effectiveness in both strength and endurance.

Further, how you direct your self-talk also matters in terms of the voice being used, specifically whether that includes first person (i.e., "I") or second person (i.e., "You") pronouns. Research indicates that using second-person pronouns in motivational self-talk, such as "You got this," produces faster time-trial performances than first-person self-talk (945, 946).

Strategies

Self Talk – Cognitive Beliefs and Biases

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Organic vs. Strategies

Additional strategies for engaging in self-talk center on the differences between organic and strategic. Organic self-talk is a natural, responsive, and instinctual process that all of us develop over the course of our lives. It's the predominant voice that arises in our mind without having to direct control or effort to how we're speaking. A good starting point for developing sports psychology skills with the athletes you are working with begins by having them pay attention to the organic self-talk voices that naturally arise during the course of their training. This will give both you and them a glimpse into the types of self-talk that unfold naturally and in differing circumstances (easy days vs. hard days etc.).

Strategic self-talk is planned, rehearsed, and practiced pre-determined statements and used in specific situations for a particular purpose. Mantras are the most widely used type of strategic self-talk. They are typically short, specific, positive, instructional, or goal-oriented statements that an athlete can use for a specific reason (i.e., instructional or motivational) and a certain time in either training or racing scenarios (949).

One of my favorite sayings is, **"you can't change what you're unaware of."** From this coaching perspective, helping an athlete become aware of their organic self-talk through monitoring thus allows for the specific self-talk voices to be developed over the course of a training plan.

Cognitive Appraisals

The last distinction in self-talk is cognitive appraisals, which are subjective interpretations and/or automatic thoughts about perceived events. For an athlete, this includes how they are interpreting and responding to stimuli they come across, which could consist of internal and/or external sources. How a runner interprets the various data being displayed on their watches or other technological areas providing information is important. With so many potential measurements (i.e., pace, distance, time, heart rate, power, zones, etc.), it's important to understand how an athlete is making sense of, potentially judging, and/or potentially making decisions based on this information. For example, an athlete may become quickly consumed by monitoring pace or speed and have biased beliefs, judgments, or negative self-talk about their paces. One of the most common elements a coach will run into is an athlete having predetermined beliefs (often tying back to individual self-worth, ego, identity, and/or comparison to others on platforms such as Strava) around these numbers, which can often lead to anxiety, stress, or refusal about running slower than what they've deemed comfortable, particularly on easier days. For example, an athlete may have predetermined that 8:30 per mile is their recovery running pace, perhaps not based on any physiological metric but rather their perception of what slow/recovery means (more mentally than physically). But in reality, as a coach, you've determined their recovery pace based on physiological markers is closer to 9:30 per mile. This is often an area of consternation between coach and athlete alike and ties back to understanding the underlying core beliefs and identity structure that an athlete may have when working together.

Other cognitive appraisals for endurance athletes include that of stress, threat, or challenge as it relates to their performance. The biopsychosocial approach embeds a framework for understanding the view of challenge vs. threat related to potential stressors or moments of pressure (950). The premise is that situations can be appraised through a perspective of either challenge or threat, which then differ in physiological and psychological reactions. A challenge appraisal occurs when a problem is self-determined to be relevant and important. The individual perceives to have sufficient (or nearly sufficient) personal resources to meet or exceed the task's demands.

On the other hand, a threat appraisal involves a situation that is also determined to be important and self-relevant but with an underlying perception of lacking personal resources or abilities to meet the demands of the task. For an athlete, this determining level of thinking will likely be noticed in the form of self-talk as they approach demanding workouts or races. Not surprisingly, a challenging state is typically associated with improved performance.

Self-Concept And Self-Efficacy

Psychologist Albert Bandura defines self-efficacy as “the belief in your capacity to execute behaviors necessary to produce specific performance attainments.” This includes an individual’s ability to control own’s one actions, intentions, motivations, and behaviors related to goal pursuit. At its core, self-efficacy is connected to the underlying beliefs and ideas an athlete has regarding their ability in any number of areas, including but not limited to: following a training plan, executing workouts, and ultimately reaching their goals. Self-efficacy can be broken down into granular areas and relates to someone’s belief about success in a particular situation. An athlete may have differing levels of self-efficacy related to differing tasks – for example; someone may believe strongly that they can accomplish their weekly assigned volume but have skepticism regarding their ability to maintain tempo paces in training. A triathlete may have high efficacy in running but low efficacy in swimming.

According to Albert Bandura’s theory, self-efficacy beliefs are developed by four primary sources of influence:

Mastery Experiences/Performance Outcomes

The most influential source of efficacy is our previous performance, specifically the interpreted results of past success. “Mastery experiences are the most influential source of efficacy information because they provide the most authentic evidence of whether one can muster whatever it takes to succeed. Success builds a robust belief in one’s personal efficacy. Failures undermine it, especially if failures occur before a sense of efficacy is firmly established” (Bandura, 1997).

Vicarious Experiences/Social Role Models

Watching, hearing, or learning about others’ successes, especially those coming from personally relevant others, influences our beliefs around what we deem possible. Bandura (1977) posits that “Seeing people similar to oneself succeed by sustained effort raises observers’ beliefs that they too possess the capabilities to master comparable activities to succeed.” In today’s current climate, social media can play a significant role in comparing to other athletes and helping (or harming) self-efficacy beliefs.

Social Persuasion

Positive verbal feedback following tasks reinforces the belief that the person has the ability, skills, or capabilities to succeed. Coaches play a vital role in the verbal feedback following workouts, training sessions, and peak performance attempts. Therefore, the coaching-athlete relationship must have a well-understood pathway for providing this type of feedback that helps build efficacy.

Emotional States

We're all human and have changing emotional, mental, and psychological states. Those specific states can influence how a person interprets their abilities in any given situation. Therefore, it is much easier to bolster self-efficacy when feeling healthy, strong, well, and/or successful.

There has been a 5th addition to this model by James Maddux (2013) regarding "imaginal experiences," which is "the art of visualizing yourself behaving effectively or successfully in a given situation."

Mindfulness

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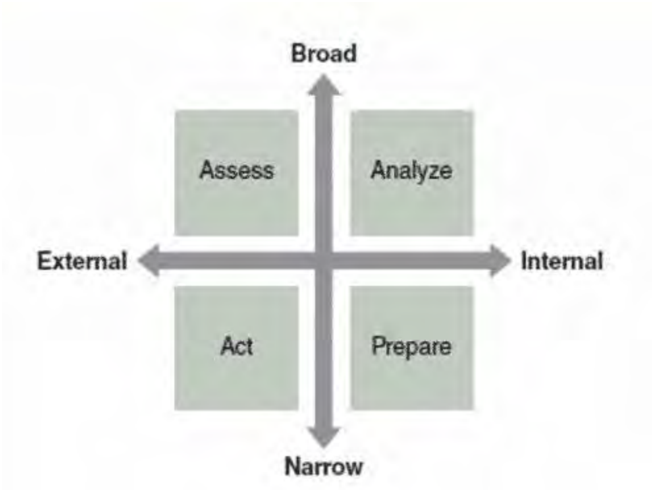
In recent years, mindfulness has been given much attention for a broad range of concerns and is being utilized by athletes to improve concentration and focus and help regulate emotions. In short, mindfulness carries the definition, "Paying attention in a particular way; on purpose, in the present moment, and non-judgmentally." (Jon Kabat-Zinn) There are typically five facets of Mindfulness involved in practice: Observing, Describing, Acting with Awareness, Non-Judging, and Non-Reacting. These skills are critical for an athlete in overall well-being and goal pursuit.

Mindful Sport Performance Enhancement is one of several models aimed at implementation with an athlete population (951). In general, research findings are mixed with specific outcome goals, in part because connecting mindfulness practice directly to the achievement of goals proves difficult, given the many other factors at play in an athlete's life (i.e., training cycle, overall volume/intensity, etc., length of time in sport, weather demands of the day, the overall skill of an athlete, emotional regulation skills, etc.). In short, available research points to the potential for increased flow state and decreased performance anxiety. More specific studies have demonstrated increased flow and running economy (952), correlated with higher performance levels in track athletes (953), and improved bodily awareness, self-regulation, and trust (954).

There has never been an easier time to practice mindfulness, given the availability of online resources and apps. Insight Timer is my recommendation, in part because I am a teacher on that platform and have created a course, Unlock Your Athletic Potential (https://insighttimer.com/meditation-courses/course_dr-justin-ross-final) but more so because the general platform is free.

Attentional Control

Attention control is the ability to choose what we pay attention to and ignore. This theory suggests there are two dimensions involved in sports: direction (internal-external) and Width (Broad – Narrow). Demand refers to where we drive attention, moving from the internal experiences (ranging from bodily sensations to psychological experiences) to the external environment (landscape, the world around us, etc.). The width refers to the degree of expansion in shifting focus, moving from a Broad focus where we take in many things at once to the Narrow, where we focus on a single reference point. There is thus an overlapping continuum in these dimensions, most often presented with four quadrants of potential focus.



Four Quadrants of Attentional Focus – Adapted from Nideffer, 1976

Broad – External

A general view of the overall landscape and environment. Assessment and evaluation of the various elements in the environment occur here, including the ability to scan a wide array of changing stimuli. For an endurance athlete in competition, this category includes scanning the external features of the race course, including the general landscape, weather, and surrounding participants/competitors.

Broad – Internal

A general view of the internal landscape includes physical sensations and psychological experiences. This is sometimes referred to as the Analyzing quadrant. Reviewing one's general thoughts, emotional experience, and bodily sensations occurs here.

Narrow – Internal

Maintaining a specific focus on some aspect of the mind or body. This range could include a negative hyper-focus on an area of discomfort or pain in the body to a rhythmic repetition of a mental mantra. This quadrant is known as Preparing – readying oneself to execute in the next moment or maintain a series of calm conversations in the mind.

Narrow – External

Finding a specific external cue in the environment to drive attention and focus. This quadrant is sometimes called Acting, in which there is a deliberate outward focus of execution in the environment.

Of importance in this model is the skill development in shifting focus given changing demands. For example, I often recommend athletes learn how to shift attention through each of the four quadrants during their training cycle to recall this ability in key events. In the Leadville 100 MTB race, for example, I often recommend developing the ability to execute narrowly focused internal and external skills on key sections of the course (St Kevvins, Powerline, and Columbine, for example) while working with a broad focus of attention on other areas requiring less cognitive demands.

Emotional Regulation

In short, our ability to regulate emotions is at the heart of performing well in just about any area in our lives, both in and out of the sport. Emotional regulation refers to monitoring and then managing your own psychological experiences effectively, in mind, body, emotion, and spirit. This first requires the ability to generate self-awareness. There are hundreds if not thousands of different ways to practice this set of skills, and you, along with your athletes, are already employing a number of them in your daily life. There are several general categories to consider when thinking about emotion regulation, including (955):

Situation Selection: we can modify when and where we place ourselves, the races we tackle, and the training environments.

Situation Modification: once committed to a specific situation (i.e., training environment or race), we are no longer necessarily able to escape – we are, however, able to modify some aspect of the environment to align with peak performance.

Attentional Deployment: cognitive perspective is a choice, and learning where, what, and how we focus is critical in determining our emotional experiences. Athletes need to learn how to focus and channel their concentration, knowing what each focal point may trigger emotional experiences.

Psychological Change: we have a tremendous impact on choosing our self-talk. Maintaining consistent, non-judgmental self-talk focused on forward momentum and aligned with goals is critical in managing emotions. Cognitive appraisals are also necessary.

Response Modulation: our best possible mechanism for changing emotional experiences is to focus on addressing our thoughts (above), actions, and physiology. We can choose how we respond in our posture and breathing, working to maintain a confident body position and regulate our breathing (some sports provide modifying breathing opportunities more readily than endurance sports).

How To Practice or Recommend Practicing These Skills

Implementing sport psychology training into ongoing physical training requires a creative, collaborative process. There is no “gold standard” or “best practices” recommendation for how to teach and guide an athlete through these steps. I believe these skills need to coincide with the physical training and are best learned while engaging in regular physical training. I also recommend that most workouts (say 75% of an athlete’s workouts) focus on developing at least one of these mental skills. An athlete need not spend the entire 60 minutes of an hour-long session focused entirely on these skills. Instead, a dedicated, deliberate 5-10 minute block is often enough at a described time in the session. The key point is to make mental skills an ongoing process to coincide with the physical aspects of training.

Executing When It Counts

This set of psychological theories and applications is best tailored for the specific event that an athlete is preparing for. Whereas the previously mentioned skills are more broadly applied throughout an athlete’s life, the following set of skills are event/race-specific and will be guided/tailored for the specific nuances of the race details and the current fitness level of your athlete.

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Mental Imagery

Visualization, better known in the performance psychology world as Mental Imagery, may be one of the most misunderstood and misused skills. And as referenced above, visualization can play a vital role in our development of self-efficacy and preparation for key events. Over time, researchers have tightened the framework to a specific set of steps to take when engaging in mental imagery, with the most well-known and thoughtful method following the acronym **PETTLEP**. Mental Imagery has shown to help impact psychological states, decrease anxiety, increase confidence, self-efficacy, and concentration.

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Physical

Imaging all the relevant physical characteristics associated with the performance. In some instances, athletes may choose to wear their race day attire when practicing mental imagery to connect to the physical sensations of the specific kit they will be wearing at their key event.

Environment

The location of the key event, imagining in as much detail as possible the sights, sounds, smells, and tactile experiences involved. This can include aspects of the course itself, the spectators, and surrounding environmental cues.

Task

Focusing on the specific actions, movements, and details about the upcoming experience, including form and flow.

Timing

Picturing in real-time how the movement will proceed at race pace or how certain aspects of a course will unfold, although the timing aspect of PETTLEP can be utilized to slow down, rewind, or fast forward specific components as deemed necessary.

Learning

The ability to use visualization to adapt, review and change to meet changing demands and tasks throughout the season. For example, the speed at which an athlete performs may likely change as the season progresses, or as new skills are developed through a training cycle, so must the visualization sequence adapt with newly acquired skills.

Emotion

Focuses on the same emotional experiences as will be felt during the key event. For performance anxiety/pressure, this can be a helpful technique for an athlete to practice developing a pre-race routine to calm nerves and stay focused.

Perspective

Perspective can change in PETTLEP from first person (i.e., as if the athlete is performing) to third person (i.e., watching from a distance). Perspective shifts can be beneficial for working on timing and flow in the first person to working on form or mechanics in the third person.

Flow State and Clutch State

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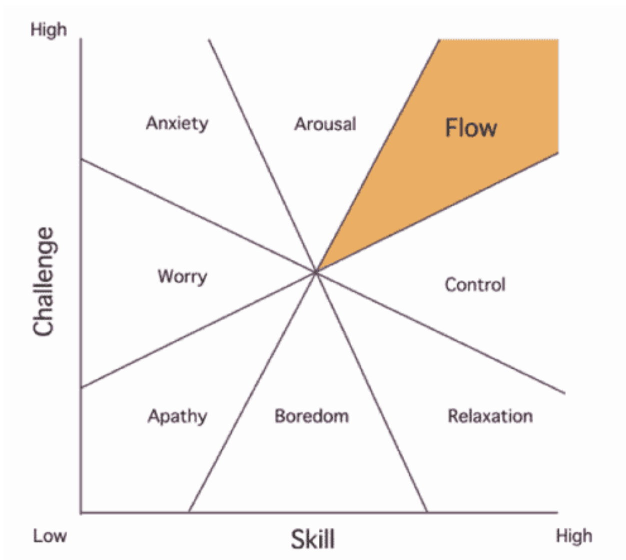
Endurance athletes love flow state, that marvelous experience coined by psychologist Mihaly Csikszentmihalyi that embodies perceived ease and effortless absorption of movement connecting us to the pure enjoyment of the sport. That beautiful feeling when we let go of caring about the metrics displayed on our watch, removing obsession about pace defined by mile splits, and shifting our focus primarily to the feeling of moving smoothly and rhythmically through open space. Time can become distorted, and our experience guides us into a natural trance of feeling alive. We may be chasing big goals or running/riding/swimming/moving to achieve a personal best when experiencing flow. However, the spotlight of our focus shines most brightly on the experience of movement, and we absorb into the moment. Flow state can be an elusive beast for many, one we aspire to tap into, yet challenging to conjure up at our fancy at any particular time.

There are noted to be several dimensions of flow, with three being of primary importance in achieving this state.

1. Clear goals.
2. Unambiguous feedback.
3. Merging and balance of challenge and skill (see below). This step is crucial, as too high of a challenge with too little skill is met with anxiety, whereas too little challenge with too much skill is met with boredom. There needs to be a goldilocks match between high challenge and high skill to tap into flow.

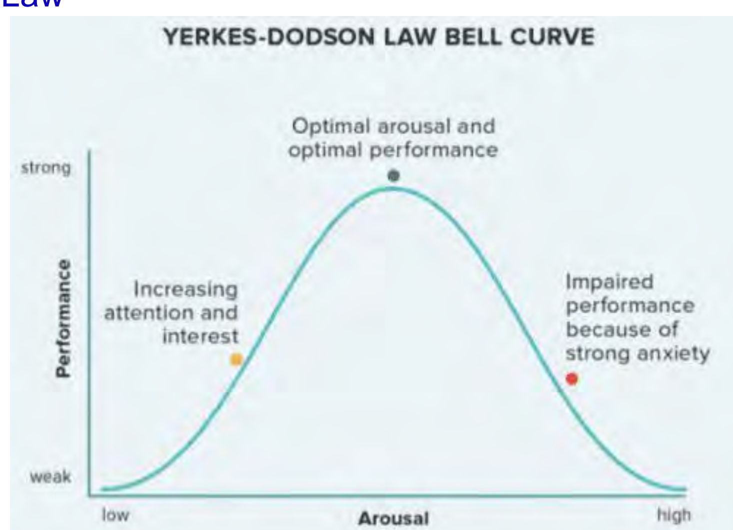


Csikszentmihalyi went on to describe certain personality traits as being better able to achieve a flow state, described as an autotelic personality (in Greek translation, Auto means self and Telos means goals) in which a person is curious and persistent, with low self-centeredness (ego) and pursues activities for intrinsically motivating reasons.

Clutch state is flow states counter and not nearly reference as frequently. Whereas flow state connects us to how it feels to be tapped into movement, clutch state resides in the sense of pressure or urgency to execute in a specific manner at a specific time. A study published in 2017 looked at 26 peak performances through the perspective of each of these psychological states and had some interesting findings and takeaways. Notably, there appeared to be three key differentiators when operating from a clutch state perspective, including the purposeful mental “switching on” in response to situational demands that kicked this process into motion. From a clutch state perspective, fixed goals, challenge appraisals, and a deliberate decision to increase effort and/or intensity guided the psychological framework.

Fixed goals become a necessary starting point when wanting to work on clutch state. Fixed goals are specific and outcome-focused, such as wanting to run a personal best in the 5k or maintaining a certain pace during workout repeats (compare this to open goals that focus on process more than outcome). Having a clear number in mind with a targeted objective greatly influences the second factor in the algorithm, challenge appraisals. A challenge appraisal is a cognitive appraisal that references how we internally think about or narrate our experiences in mind and body. When we are purposefully pushing the pace, hunting down fast splits, how we talk to ourselves has to change otherwise, we may talk ourselves back into a state of comfort. The opportunity to work on task-specific self-talk in challenge appraisal moments is an important factor in tapping into a clutch state. In these moments, our beliefs about our capabilities and our limits come to light and have a dramatic impact on our choices to either keep pushing or back down, which hits squarely on that third differentiator – the deliberate decision to increase effort or intensity. That moment when the choice to continue is met with knowledge that it’s going to hit head-on with mounting discomfort. And you or your athletes do it anyway because the goal matters.

Yerkes-Dodson Law



The Yerkes-Dodson Law is a model of the relationship between stress and performance that is sometimes referred to as the Inverted-U model, given its graphed shape. This theory was first developed in 1908 by psychologists Robert Yerkes and John Dodson, who described “arousal” when they first began experimenting on mice. They noted a relationship between “arousal” (i.e., stress) and performance, noting that performance peaked with medium levels of arousal/stress. If too low on the stress continuum, performance diminished. Likewise, too much arousal or stress caused performance to suffer as well. Therefore, they concluded that optimal arousal leads to optimal performance. Too little arousal can be described as boring and unchallenging. Too much can lead to heightened anxiety and situations in which demands exceed capabilities.

Performance Anxiety

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Performance anxiety is a common experience. In my experience working with athletes, I will often begin the discussion surrounding performance anxiety as a marker showing that the athlete cares. This often sets and validates the importance of sport and performance in life and normalizes the experience as common. I’ve even taught my seven-year-old daughter that pre-performance anxiety simply means that you care, and she now shares this statement to help her gymnastics teammates prepare for their routines.

A biopsychosocial model of anxiety performance centers on the role of challenge and threat appraisal as key factors in the experience and subsequent ability to manage performance anxiety states. This theory suggests that the cognitive evaluations (i.e., what is being thought) precede any physiological responses to an event. It also suggests that a challenge appraisal is associated with a better cardiovascular pattern in the body and improved performance.

A threat state comes from a situation being appraised as both important and with insufficient “resources to meet the demands of the task.” For an athlete, the threat appraisal is often centered on not being fully prepared or not capable of rising to the challenge to meet race day goals or expectations. In short, the athlete focuses on doubt about their ability to succeed in a meaningful event. Further, when that event is tied deeply to identity, not only can the event feel as though it is being threatened, but so too can the very fabric of how a person view's themself given their performance. I firsthand experienced this after running the Boston marathon in 2017 on a sweltering day. Not many performed well, given the conditions, and I was struck by how many athletes were devastated by their performance after the fact. Whereas I was elated, it was my first Boston, and I thoroughly enjoyed the experience despite one of my slowest marathon times, others were distraught. The more I talked to athletes that day, the more I found that they had tied a core part of their identity to their race times, and many were concerned that they had not re-qualified for the 2018 marathon, further disrupting and threatening their sense of identity.

On the other hand, a challenge state occurs when a situation is perceived as both important along with sufficient (or close to enough) abilities to meet the demands of the task. In short, a challenge state occurs when the athlete deems the event as meaningful and believes they have what it takes to meet the specific demands of race day to reach their goals.

This model is extended by considering how self-efficacy, perceived control, and goals influence someone's perceived resources to cope with situations. Higher levels of perceived control paired with self-efficacy and mastery-oriented goals are the most likely to elicit challenge appraisal. Lower perceived control and self-efficacy, and ego-oriented goals are the most likely to elicit threat appraisal.

The Stress Response

Our bodies have a physiological pathway that has evolved to screen for and prepare for the threat. This system includes the sympathetic nervous system, in which the Fight or Flight response is activated upon threat detection. A cascade of physiological changes unfolds from the Hypothalamic-Pituitary-Adrenal Axis (HPA), including cortisol and adrenaline.

The HPA system can be activated from 1 of 3 general categories, with names that are often used interchangeably but are quite different.

Fear: a real-time threat to your physical safety or well-being in this moment. If you've ever been in a car accident or come across a mountain line on a hiking trail, the experience in mind and body comes from a place of fear.

Stress: when demands exceed the ability to handle them effectively, for example, having 12 hours of work to complete in an 8-hour day. Importantly, in the life of an athlete, stress can accumulate in mental form (mental demand from work, school, household, and family responsibilities) and physical form (daily and weekly training impacts on the body).

Anxiety: thoughts, beliefs, or forecasting about the future, specifically when perceiving the future as a threat state. Of the three listed conditions, this is the one that will be predominant in the life of an endurance athlete. It will often come from the questioning of "what if…" and be attached to any number of factors ranging from the uncontrollable (such as the weather before a race) to the outcome (setting a PR, qualifying for a subsequent event, etc.), to the identity impact most often tied to outcome(not succeeding is perceived as a negative impact on identity).

As discussed earlier, challenge versus threat appraisal is a significant perception that will dictate this experience for your athlete's experience, particularly that of performance anxiety. In a race day context, it's also important to get a better sense of what the athlete may be identifying as a potential threat. The deeper the threat (i.e., the more the event's outcome is perceived as meaningful, tied to identity, and socially judged), the higher the likelihood of anxiety (956).

Mental Toughness

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"Mental toughness is probably one of the most used but least understood terms used in applied sport psychology. It appears, therefore, that virtually any desirable positive psychological characteristic associated with sporting success has been labeled as mental toughness at one time or another." This quote comes from a 2002 paper aptly titled, "What Is This Thing Called Mental Toughness? An investigation of Elite Sport Performers" highlights the lack of unification regarding labeling, identifying, and understanding the concept of Mental Toughness. This concept appears to have widespread usage and a general sense of intuitive understanding despite the lack of research clarity. A few models will be presented here (957).

Peter Clough (psychologist) describes four essential traits of mental toughness that he labels the 4 C's: Confidence, Challenge, Control, and Commitment. In addition, he describes Resilience (coping with life's difficulties) and Positivity (seeing and seizing opportunity) as key elements in his model. He is often criticized for taking the concept of Hardiness and curtailing it into his model of mental toughness. Other conceptual frameworks include the role of coping skills in the context of performance, optimism, and resiliency (958).

Further ideas include the bioecological model of mental toughness, defining this state as "A personal capacity to deliver high performance regularly despite varying degrees of situational demands."

A 2017 review of Mental Toughness and Success in Sport reviewed the quantitative literature available and found 19 studies that met inclusion criteria. Results indicated that mentally tougher athletes "participated at higher levels of competition and tended to achieve more or perform better." (959)

A subsequent meta-analysis in 2020, "Developing and training mental toughness in sport; a systematic review and meta-analysis of observational studies and pre-test and post-test experiments" (960), examined the various types of mental toughness training practices to understand best practices in developing this approach with athletes. The study suggests a high range of variability in the approach and training methodology used in sport.

The Mental Toughness Questionnaire is the most used objective report.

Obsessive vs. Harmonious Passion

The dualistic model of passion defines passion as a "strong inclination for a self-defining activity that we love, value and spend a considerable amount of time on." This model proposes two types of passion, harmonious (having autonomous control in choosing when to and when to not engage in the activity that results in adaptive outcomes) and obsessive (a feeling of being unable to help oneself and surrendering to the desire to engage in that activity despite potential harm). (961)

In short, harmonious passion involves an activity living in harmony with all other areas of a person's life and not overpowering decision making or identity. Sport is valued and of importance, but a person can modify the urge, tendency, or desire to participate, or perhaps more importantly, not participate when needed. Engagement is completed in large part for engagement's sake. This is shown in athletes who can take rest days and listen to changing needs of their bodies for recovery practices. In general, harmonious passion fuels greater subjective well-being and sustains higher performance.

Obsessive passion can feel as though it is outside a person's control. Often this can come from an over-identification of participation in sport as a person's singular identity. This can fuel a sense of urgency about training and performing well, which is often tied tightly to self-worth. Behaviors can manifest in not resting or recovering properly and fuel injury cycles. Obsessive passion is associated with poorer outcomes and a lower sense of subjective well-being (962).

Pain Threshold, Tolerance, And Sensitivity

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Pain threshold is the minimum amount of stimulus or intensity that someone will describe as shifting from a place of comfort to discomfort to painful. Pain tolerance is based on willingness and is defined as the amount of time someone is willing to tolerate pain or the maximum amount of intensity they are willing to tolerate. Pain sensitivity is the subjective experience of discomfort, an effort to quantify how intensely someone rates discomfort (963).

Reaction to pain, particularly in psychological responses through cognitive appraisal and emotional regulation, is vital in an endurance athlete's life.

Research has attempted to examine these concepts with athletes and found, in general, that elite and high-level athletes tend to have higher pain tolerance and threshold levels along with lower pain sensitivity compared to controls. Further, endurance athletes may have better tolerance than other types of sports.

Importantly, endurance sports offer an opportunity for athletes to work on psychological skills for managing these experiences through motivation, goals, and self-talk.

Psychology and Injury

Injuries are one of the most frustrating aspects of being an endurance athlete. As a coach, it's essential to understand the general psychological and emotional impact and timeline most athletes face when working through an injury. The Three Phases of Injury Recovery is the model most commonly used and includes:

Injury And Illness Phase

This is the time frame in which the athlete is first injured. Accompanying this phase tend to be negative emotions – sadness, anger, and frustration. Notably, one of the most challenging aspects of this phase is the uncertainty that the athlete experiences related to the diagnosis, prognosis, and potential negative impact on pursuing their goals. Uncertainty is one of the biggest drivers of anxiety we have as humans. Therefore, anxiety is a common concern at this stage as an athlete seeks answers and a treatment plan.

Rehabilitation And Recovery Phase

The time frame in which an athlete has a working diagnosis and begins a treatment plan. This could include taking a few days off to, having reconstructive surgery, or starting physical therapy. This phase often includes mixed emotions, but more control as the athlete works to regain strength, mobility, and general healing. Increased control often leads to reduced anxiety.

Return To Sport Phase

The time in which an athlete is permitted to get back to training. This phase is often connected to increased anxiety related to potential concerns around re-injury and the subsequent concern this has on perceived goal pursuit or a tendency to try to take on more than is advised by returning to the pre-injury workload without proper ramping. In the case of the former, it can often be helpful to have the athlete complete an anxiety hierarchy rating their concerns on a scale of 0 – 10, then working systematically from the bottom to the top to rebuild confidence and work through any emotional regulation needed. For example, a runner overcoming a stress fracture may rate walking without a boot as a 1, alternating a 20-minute session with jog-walking every 2 minutes as a 5, and running a set of 800 repeats on a track as a 10. It's the coach's work to help that athlete progress through each level (first by adhering to PT guidelines) and to help manage the emotional experiences involved with each increasing level of concern. It's advised not to move on to the next step in the hierarchy until the athlete has demonstrated anxiety reduction and improved trust (964, 965).

Summary

- **Ego orientation** is often referred to as performance-goal orientation and tends to focus primarily on evaluation, typically in comparison to others
- **Task orientation** is often called mastery-goal orientation and focuses on three primary factors: engaging in challenging activities, exerting effort within an activity, and persisting during the challenge.
- Outcome goals for recreational endurance athletes most often fall into one of the following categories: 1. Just Finish Goals, 2. Break a time goal (most often that of a clear number

reference point such as a 4-hour marathon). 3. Set a PR, 4. Qualify for an event (again, most often targeting a set reference point).

- The core of our psychological framework involves those three basic categories: Beliefs, biases, and values.
- The Psychobiological Model of Endurance Performance is the newest development in the theoretical understanding of limiters to performance and it gives increased attention to psychological factors, including cognitive perception, motivation, and willingness.
- Strategic self-talk is planned, rehearsed, and practiced pre-determined statements and used in specific situations for a particular purpose.
- Mantras are the most widely used type of strategic self-talk.
- Attention control is the ability to choose what we pay attention to and ignore.
- Mental Imagery has shown to help impact psychological states, decrease anxiety, increase confidence, self-efficacy, and concentration.
- Performance anxiety is a common experience.
- Our bodies have a physiological pathway that has evolved to screen for and prepare for the threat.
- The three phases of injury are the illness and injury phase, the rehabilitation and recovery phase and the return to sport phase.

Module 18: Sports Nutrition

Content Provided by Bob Seebohar, RD

Proper nutrition plays a critical role in the health and performance of a runner. Underestimating or misunderstanding proper nutrition and elements of fueling can have disastrous consequences on one's physical performance and health. It is important to understand that while nutritional/fueling guidelines are important to be aware of, individuals will respond differently to the same nutritional/fueling strategy.

Scope Of Practice

It is essential first to discuss scope of practice when it comes to nutrition. Unless you are a Registered Dietitian (RD) or Registered Dietitian Nutritionist (RDN), many states limit the amount of nutritional information you can legally provide to an athlete. Therefore, in many states, you cannot prescribe an eating plan or supplements, but you can educate athletes about the concepts discussed in this module. You **can educate and inform an athlete on nutritional information. Still, you cannot prescribe a nutrition program, diet, or supplement protocol, especially if your athlete has any pre-existing medical conditions.** It is recommended that all endurance coaches have a team of professionals incorporated into their business. One of these should be a reputable Registered Dietitian, preferably one who is a Board

Certified Specialist in Sports Dietetics (CSSD) and specializes in endurance athlete nutrition.

This module will not discuss specific amounts of macro/micronutrients or caloric intake, as this varies widely due to several variables. An athlete should seek an RD or RDN for advice and guidance specifically tailored to them.

Upon completion of this module, you should understand the following areas:

- Macro and Micronutrients
- Foundation nutrition
    - Energy systems
    - Physical periodization
    - Nutrition periodization
    - Meal/nutrient timing
- Training and racing strategies
    - Carbohydrate loading theories
    - Role of insulin
    - Pre-race fueling
    - Fueling during and after a race
- Supplements

Terminology

Carbohydrate: Organic compound made up of oxygen, hydrogen, and carbon. Categorized as simple and complex, carbohydrates provide energy to the body.

Hypoglycemia: Low blood sugar

Hyperglycemia: High blood sugar

Iron: A mineral needed to help hemoglobin carry oxygen in red blood cells to the muscles.

Electrolytes: Regulate the body’s fluids, help to maintain a healthy blood pH balance, and create electrical impulses essential to all aspects of physical activity – from basic cell function to complex neuromuscular interactions needed for athletic performance.

Calcium: Element that is important for growth, maintenance, and repair of bone tissue, maintenance of blood calcium levels, regulation of muscle contraction, nerve conduction, and normal blood clotting.

Dehydration: Occurs when the body loses excessive amounts of fluids.

Hyponatremia: A metabolic condition that occurs when there is not enough sodium (salt) in the body fluids outside the cells. It is often associated with overhydration.

Carbohydrate loading: Characterized by eating additional carbohydrates in the days leading up to an event with an associated decrease in training volume. Carb loading aims to maximize endurance performance by increasing muscle glycogen levels.

Insulin: A hormone produced in the pancreas plays a crucial role in regulating blood glucose (sugar) levels.

Macronutrients

Carbohydrates

Carbohydrate intake is a vital part of a runner's diet, as it provides energy for training and racing. Carbohydrates maintain blood glucose levels during exercise and replace muscle glycogen. The carbohydrate required depends on an athlete's total daily energy expenditure based on ***resting metabolic rate*** **(RMR)**, training intensity and duration, race distance, gender, and environmental conditions (1 kg = 2.2 lbs).

- **Kg = Kilogram**
- **BW = Body Weight**

The type of carbohydrate ingested is just as important as the amount. During carbohydrate-loading periods and when eating meals, it is advised to focus on complex carbohydrates found in whole foods and avoid junk carbohydrates such as highly processed foods as they provide little nutrition. Simple carbohydrates are recommended before running races and during training sessions as they are easily broken down for immediate energy.

Several variables influence carbohydrate and caloric requirements:

– **Intensity:** The higher the intensity, the greater the carbohydrate requirement.

– **Heat/Humidity:** Heat and humidity typically increase the carbohydrate requirement.

– **Altitude:** Metabolic rate increases at altitude and will necessitate a higher carbohydrate intake.

Adaptation

Some individuals find that dramatically increasing one's carbohydrate intake causes gastrointestinal issues. If this is the case, gradually introduce higher carbohydrate levels as a program progresses.

Rebound Hypoglycemia

Rebound hypoglycemia refers to a process by which carbohydrates are consumed close to the start of an exercise session. This ingestion causes one's insulin levels to elevate, which causes blood sugar levels to decrease. The net effect of this process is that an athlete feels like they have run out of energy or, in endurance athletics terms, "hit the wall."

A 2010 study out of the University of Birmingham analyzed studies pertaining to rebound hypoglycemia and found that while there was a substantial metabolic change in the body (e.g., rebound hypoglycemia), there was no consistent data to point to a decrease in performance (447). This was corroborated by a similar study out of Deakin University (448). However, the University of Birmingham study did find that some subjects experienced effects similar to hypoglycemia, even if they did not have low glucose levels.

It is recommended to experiment with ingesting carbohydrates in varying amounts and at various times prior to exercise bouts to assess the effect on an athlete (447).

Relationship Between Carbohydrates (Sports Drinks) and Dehydration

Most sports drinks have a high concentration of carbohydrates. This is often marketed to maintain energy and avoid hitting the dreaded “wall.” While carbohydrates help to slow the body’s rate of glycogen depletion, is it possible to have too much of a good thing?

According to physiologist and nutrition expert Dr. Stacy Sims, the answer is yes. Dr. Sims notes that when too many carbohydrates are taken in by way of a sports drink, the intestines are forced to take in water from other parts of the body to dilute the ingested carbohydrates (760). **Drawing away water from other parts of the body can cause dehydration.** Therefore, Dr. Sims recommends a lower carbohydrate concentration when drinking fluids that contain carbohydrates – with the sole purpose of the carbohydrates being for fluid absorption – not fuel. As a result, Dr. Sims recommends keeping fluid intake for hydration purposes and food for fueling purposes (671).

Insulin

The Energy Systems module notes that insulin is a hormone produced in the pancreas and plays a crucial role in regulating blood glucose (sugar) levels. Individuals who have type 1 diabetes are unable to produce insulin and require insulin shots for the body to uptake glucose from food. The roles of insulin are noted below (449):

- Facilitates muscles, the liver, and fat to uptake glucose from the blood. Glucose stored in the muscles and liver is converted to and stored as glycogen, and glucose stored in fat is stored as triglycerides.
- Prevents the use of fat as an energy source. Therefore, when insulin levels are low or nonexistent, the body utilizes fat as fuel.
- When glucose levels rise, insulin is secreted by the pancreas. When glucose levels fall, the secretion of insulin slows or stops.
- Blood glucose levels rise after ingesting carbohydrates. Simple carbohydrates such as candy bars elicit a more significant insulin response (greater insulin secretion) than complex carbohydrates such as beans. When blood glucose levels rise, this sets off an alarm that signals insulin secretion. This insulin secretion can cause blood sugar levels to plummet because of the uptake of blood glucose for storage by the muscles, liver, and fat.
- Insulin regulates the metabolism of carbohydrates and fat.

Gels, Drinks, Jelly Beans

A 2008 study by Campbell et al. examined the effectiveness of ingesting various forms of carbohydrates mid-race (gels, drink, jelly beans) in relation to each other and water (710). The study found that while all carbohydrate sources increased blood glucose over solely water, there was no significant difference in blood glucose levels between the three carbohydrate sources. This study did not discuss individuals' gastrointestinal tolerance to the different carbohydrate sources. This factor and one's tolerance to the quantity of carbohydrate intake are equally important and must be determined during training and on an individual basis.

Protein

Protein from animal and plant sources provides the amino acids needed for muscle growth, repair, and maintenance.

Protein can be a reserve fuel source during an endurance event or training session once glycogen stores have been exhausted. Protein is found in beans, poultry, eggs, dairy, lean meats, and fish. Some energy/sports bars and shakes can supply the appropriate amounts of protein and carbohydrates and are easily digested. While protein is important for building and repairing muscle, it is not a necessary food source right before a race.

Fat

Fat, a source of energy, is important in athletes' diets. High-fat diets are not recommended for athletes. Heart-healthy fats like those found in fatty fish, avocados, nuts, and olive oil can help with recovery and the inflammation resulting from exercise.

Micronutrients

Iron

Iron is needed to help hemoglobin carry oxygen in red blood cells to the muscles. Therefore, iron is a critical dietary element for endurance athletes. While there is typically little difference in vitamin and mineral levels of athletes and non-athletes, athletes are often iron deficient. An increased iron requirement for women is often recommended due to menstruation.

Electrolytes

Electrolyte is a term used to describe ions. The major electrolytes in the body are:

- Chloride
- Phosphate
- Calcium
- Sulfate
- Sodium
- Magnesium
- Potassium
- Bicarbonate

Electrolytes are minerals that break down into small, electrically charged particles called ions. Present wherever there's water in the body, electrolytes regulate the body's fluids, helping maintain a healthy blood pH balance and creating the electrical impulses essential to all aspects of physical activity – from basic cell function to complex neuromuscular interactions. The kidneys work to keep the electrolyte concentrations in the blood constant despite changes in the body.

Foundation Nutrition

Foundation nutrition principles include educating athletes about the importance of optimizing blood sugar within a yearly periodized nutrition model.

A runner-focused study (2013 by Hansen et al.) found that **marathon runners who followed a scientifically-based nutritional strategy were approximately 5 percent faster than runners who self-selected their nutritional approach** (702).

Energy Systems

This section is largely a review of Lesson 4 (Energy Systems). The duration and intensity of exercise dictate which energy system will be utilized the most, making it easier to deliver nutrition education strategies to athletes. Therefore, understanding energy systems is critical for understanding sports nutrition.

The body's three energy systems include the phosphagen system, glycolytic system, and the aerobic pathway. **These systems are engaged at different times and amounts based on the intensity and duration of training.** Warm-up and lower intensity training will exert a much different metabolic response and energy system demand on the body than moderate to high-intensity exercise. Strength and power training will be somewhat different from long endurance training. These alterations in training load (volume and intensity) dictate what, when, and how much of the energy systems contribute to fuel workouts. Carbohydrates, protein, and fat consumed daily follow different metabolic paths, and their utilization depends mainly on the intensity and duration of training.

The phosphagen system, also known as the phosphocreatine or creatine phosphate system, is an anaerobic (without oxygen) pathway that supplies immediate energy to the working muscles. A limited amount of phosphocreatine is stored in the body, and this system only provides enough energy for approximately 10 seconds of high-intensity exercise. After the initial 10 seconds of this type of training, athletes will typically require about 2-4 minutes of rest to allow regeneration of the phosphocreatine used. **It is very important for athletes participating in this type of training to allow this rest interval between sets to account for the energy system to recover during this maximal energy use.**

The glycolytic system, also known as glycolysis, is another anaerobic metabolic pathway that breaks down glucose or glycogen for energy. As with the first energy system, the glycolytic system also has limited stores and will provide only enough fuel for about 1-2 minutes of high-intensity exercise. This system also yields lactate molecules, which can be thought of more as friends than foes. Lactate can be used as an energy source to fuel muscular work at certain intensity levels.

The last energy system, aerobic, uses oxygen and can thus produce a larger amount of energy. Pyruvate, a product of glycolysis, enters the mitochondria and generates a constant energy supply to fuel working muscles for hours and hours and lower to moderate intensities.

Generally, high-intensity training will rely more on anaerobic metabolism, while lower intensity and longer duration training will rely more on aerobic metabolism. It is important to remember that there is very rarely one energy system doing all of the work at any given point throughout exercise.

Physical Periodization

Physical periodization is a training strategy that promotes an improvement in performance by providing varied training parameters such as specificity, intensity, and volume, in an annual training plan. Physical periodization is important for nutrition planning because each training cycle has specific physiological goals. These goals are necessary to consider when educating an athlete about daily nutrition, nutrient timing strategies, and the effective use of supplements.

There are three primary periodization cycles: 1) macro, 2) meso, and 3) microcycle. The macrocycle is generally viewed as the big picture and is defined as the entire year for most recreational endurance athletes. The mesocycle usually encompasses 2-3 month blocks separated into three specific sub-categories: 1) base training, 2) competition and 3) transition. Each mesocycle sub-category has its own very specific physical goals based on an athlete’s goals and competitive calendar. Lastly, the microcycle is a block of 7-10 days and is focused mainly on daily training and recovery adaptations.

Understanding the body's energy system usage coupled with energy expenditure information during different training cycles is the first step in knowing how to approach educating endurance athletes about their daily nutrition and nutrient timing plans. Using the "eat to train, don't train to eat" mantra is important. It will assist an athlete with eating in preparation for high-quality training sessions rather than using food as a training-based reward system.

While not a formal lesson in periodization, it is important to note that coaches use many different types of periodization with athletes. The above information depicts a traditional physical periodization model where the athlete progresses throughout the year from aerobic foundation to speed/intensity, then competition followed by the transition cycle or off-season.

Nutrition Periodization

I created the concept of nutrition periodization in the early 2000s to assist coaches and athletes in aligning their physical periodization planning with nutritional needs. As different training cycles are followed throughout the year, energy demand, energy expenditure, and body stressors differ. Proper nutrition planning and periodization will allow maximal physiological adaptation at the right times of the year for athletes. It is typical for many athletes to only concern themselves with their nutrition plan a few days before competitions. This "old school" method of sports nutrition will not assist with supporting health or performance improvements. Nutrition must be aligned with physical training as training load ebbs and flows from day to day, week to week, and month to month throughout an athlete's annual training plan.

Nutrition has been a complex set of numbers for both athletes and coaches, cumulating into meal plans that have traditionally been calorie-based. Unfortunately, even though this is what athletes think they need to be successful, it has not proven to be efficacious nor long-standing in their athletic careers. The only reason athletes think in this manner is because nutrition professionals present this process to them year after year. The coach's role is to provide more successful nutrition periodization planning techniques that can be easily adapted to an athlete's lifestyle, including more qualitative strategies and engaging the athlete in their nutrition planning and monitoring.

Compared to calorie counting and measuring and weighing food, the concept of nutrition periodization is not meant to be complex. The **simpler the messaging, the more successful athletes will be at adopting a successful nutrition periodization plan.** The following are the five components of the nutrition periodization concept:

1. Manipulate body weight and body composition safely and effectively
2. Improve the hormonal influence of appetite
3. Support the immune system and the microbiome
4. Periodize supplement usage
5. Support a healthy relationship with food

Training load changes alter the level of physical stress on the body, which leads to different nutrient needs. Athletes who are not nutritionally prepared before training sessions will not receive the same positive physiological adaptations as athletes who are prepared and place their nutrition on the top of their priority list. As mentioned previously, nutrition periodization does not need to be complicated. The basic premise is to educate an athlete on which nutritional strategies should be modified to best support training cycles' energy demands and stressors.

The following section highlights each of the main physical periodization cycles and specifically emphasizes nutrition education that can be promoted during each of the specific mesocycles.

Macrocycle Nutrition Guidelines

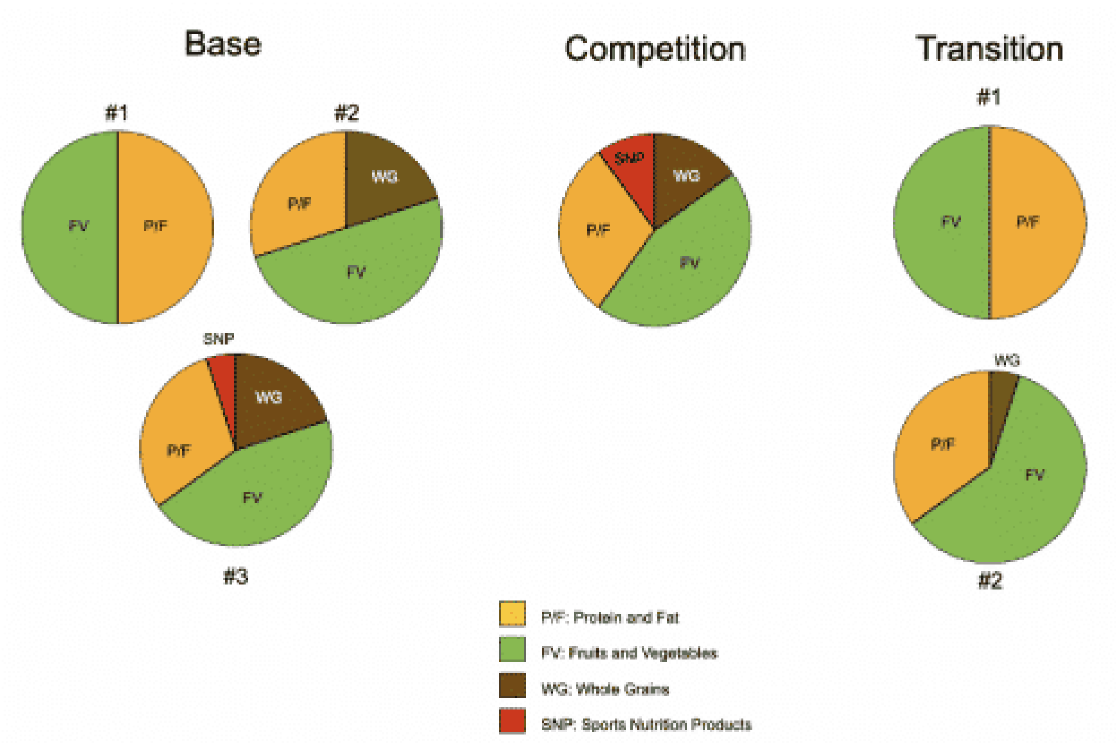
The main nutrition concept that should be emphasized year-round to athletes is how to optimize blood sugar best. This one simple nutrition emphasis creates a more successful approach to daily eating and provides physical benefits, including steady energy, reduced cravings, better recovery, and a better likelihood of achieving body weight and body composition goals. There are also cognitive benefits such as better focus, concentration, reaction time, and overall brain health.

Blood Sugar Optimization

The "how" of optimizing blood sugar is quite simple. The ebb and flow of blood sugar is facilitated by food consumption and is regulated by hormones, particularly glucagon and insulin. To optimally control blood sugar, a proper ratio of carbohydrates and protein should be consumed. Because most protein sources contain fat, there is not a strong emphasis on discussing a quantifiable amount of fat to consume per day. Instead, the main focus should be on educating athletes about seeking a balance of saturated versus unsaturated, emphasizing mono-unsaturated and polyunsaturated fats, especially Omega-3 fats.

Consuming many carbohydrates without protein or fat will raise blood sugar quickly, causing an increase in insulin secretion from the pancreas. When this happens, an athlete will feel like they have more energy, but it is usually short-lived. One of the roles of insulin is to lower circulating blood sugar, and when this happens, an athlete's energy level may drop significantly soon thereafter. There is a difference in the type of carbohydrate consumed and the response on blood sugar. Carbohydrates with more fiber will have a slower effect on the rise and fall of blood sugar, but it is not until adding a protein-rich food that blood sugar will be more stable. Therefore, it is ideal for the coach to educate athletes on consuming carbohydrate and protein-rich foods to elicit a positive blood sugar response.

The easiest strategy to teach athletes how to proportion and periodize macronutrients properly is via the Periodization Plates™. I designed these as a qualitative only food pairing model, so it is best not to assign quantitative percentages when using this model. Adopting this qualitative daily nutrition model will allow you to educate without prescribing (remember the scope of practice).



Based on the training cycle and associated health goals, daily nutrition goals and macronutrients will need to be shifted to account for the training load, energy expenditure, and recovery needs.

The Periodization Plates™ model comprises four nutrient categories: 1) protein/fat, 2) fruits and vegetables, 3) whole grains, and 4) sports nutrition products. Each nutrient category can be shifted higher or lower based on the training cycle and training load changes throughout the year.

It is important to remember that the Periodization Plates™ model guides athletes in learning how to structure and periodize their daily nutrition needs. It is not a meal planning tool in the traditional sense but rather a qualitative model that athletes can adapt to their lifestyle without depending on calorie counting for their daily nutritional needs. The various plates represent general shifts in nutrients, and there is not necessarily only one plate that will be correct for an athlete. Daily nutrition may likely change based on health and performance goals, and there should be flexibility in moving between plates to account for individual athlete goals and needs.

The following are guidelines on educating an athlete to use the Periodization Plates™ model based on the specific training cycle.

Base Training Cycle

Plate #1

This plate yields most carbohydrates from fruits and vegetables and is ideal for an athlete who is just beginning a training program or has weight loss and/or body composition goals. It is important to note that carbohydrates may be lower using this plate since it is comprised of fruits and vegetables and does not include whole grains. Accordingly, it is important to ensure the athlete understands this and does not have higher quality (high intensity and/or longer duration) training while using this plate.

Plate #2

This plate contains a higher amount of carbohydrates and is more beneficial to support an increased training load and energy expenditure, which often happens around the middle to end of base training. While still necessary to emphasize protein-rich foods, whole grains are introduced to increase total carbohydrate needs, which will help performance as the athlete progresses to a higher training load.

Plate #3

This plate is very similar to Plate #2, with the exception of introducing sports nutrition products. Suppose an athlete plans to use sports nutrition products during competition. In that case, it is important to introduce these various products into the training plan around the middle to end of base training so the athlete can test the product’s response on their digestive system and begin to develop successful nutrient timing strategies. These products should only be used within a nutrient timing window (before, during, or after training) and not throughout the day when the athlete has access to real food.

Competition Training Cycle

During most of the competition training cycle, there is really only one plate that athletes will need to use. While plates from the other training cycles can be used, it is ideal to stay in the confines of the competition training cycle plate. Because this training cycle typically includes higher intensity and longer duration training sessions, ample carbohydrates are necessary for energy management and support performance goals and faster recovery. There is still adequate protein and fat to support blood sugar control. Sports nutrition products should be used during most quality training sessions to determine if they will positively or negatively affect an athlete's digestive system during higher intensity training. The digestive system shunts more blood away from the gut to the working muscles during higher intensity training. It is more difficult for an athlete to digest calories when this happens. Because of this, an athlete needs to simulate race pace efforts during high-quality training sessions to determine how well certain sports nutrition products will be tolerated.

Transition Training Cycle

Because the transition cycle is classically known as the off-season, and this is synonymous with more rest and little to no structured training, it is beneficial to teach the athlete the importance of scaling back their daily nutrition plan to support lower energy expenditure training. While it is still paramount that an athlete consumes enough food throughout the day, blood sugar control and implementing a more balanced macronutrient profile become the most important nutrition goals for this cycle.

Plate #1

This is the same plate offered during the first part of base training, and for good reason, as it is optimal for controlling blood sugar. This plate is ideal for an athlete who is truly taking a break from structured training and may only be incorporating fun exercise during this cycle.

Plate #2

While the transition cycle is known for rest and recovery, a few athletes still train at lower intensities during this cycle. This plate introduces more carbohydrates with whole grains and will help an athlete manage a moderate energy expenditure to support their exercise training.

It is important to remember that it is tough for athletes to discontinue the use of sports nutrition products following their last competition. This is due to behavior change, and the coach should emphasize energy intake from whole foods and slowly encourage the athlete to wean off sports nutrition products. Be patient as this could take upwards of 3-4 weeks.

Additionally, it may be prudent to recommend plate #2 in the first part of the transition cycle and encourage the athlete to move to plate #1 after a few weeks. This would support a more positive behavior change process and would assist the athlete in their preparation for base training nutrition once again.

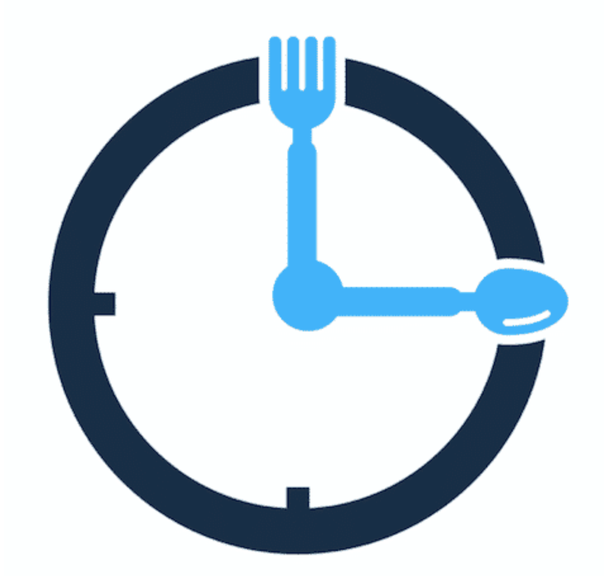
Fluid / Hydration

There are no exact daily hydration guidelines that are 100% supported by research. The most effective qualitative strategies to teach athletes about proper hydration strategies include urine color and urine frequency. While neither is an exact science, these two methods are easy for athletes to learn and utilize daily.

The normal color of urine throughout the day should ideally be pale yellow. Many factors will disrupt this color, such as the first void of the morning and taking supplements, specifically B-vitamins. A good rule of thumb is to teach athletes to begin assessing the color of their urine after the first void in the morning.

The second strategy is urine frequency. Ideally, an athlete will urinate approximately every 2 hours throughout the day. Urinating too often or not often enough can indicate improper daily hydration techniques. Remember, fruits and vegetables contain water, as do other beverages such as tea, milk, and coffee. There is not an exact amount of plain water an athlete should drink throughout the day. It is much more effective to teach athletes about urine color and frequency since water, other beverages, fruits, and vegetables can all be used to improve daily hydration status.

Nutrient Timing



Nutrient timing involves specific strategies to improve hydration status, carbohydrate delivery, and electrolyte usage before, during, and after training sessions. Specifically, pre-training involves consuming calories and fluid 1-3 hours before training, and post-training involves consuming calories and fluid immediately and up to 1 hour after training.

Like daily nutrition principles, it is ideal for periodizing nutrient timing strategies based on the athlete's training cycle. It is important to note that nutrient timing strategies introduce more quantitative-based information supporting current sports nutrition research.

In nutrient timing terms, five major nutrition areas are discussed:

1. Fluid
2. Carbohydrate
3. Protein
4. Fat
5. Electrolytes, specifically sodium

The following are specific nutrient timing strategies you can teach an athlete based on their training cycle. Energy expenditure will fluctuate throughout the year, and nutrient timing strategies should support these changes.

Base Training Cycle – Before Training (1-3 Hours)

Fluid

Current fluid recommendations are very quantitative-based with fluid intake before training. However, it is good to begin the new training year by teaching athletes about thirst instead of jumping right into the numbers game. I will present quantitative fluid strategies for the Competition Training Cycle. Still, to begin this new training cycle and year of training, athletes will benefit more from coaches continuing to teach them about urine color and frequency of urination throughout the day.

Of course, this method of non-quantitative fluid timing does have a higher risk to reward ratio in the beginning. Still, it usually only takes a few weeks for athletes to learn more about their bodies and fluid needs based on their training load, environmental stress, and individual sweat responses.

I refer to this qualitative hydration education as the “goldilocks” method. Teaching an athlete how much fluid is too little, too much, and just right. Successful daily fluid balance includes consuming plain water and other beverages and higher water-containing foods such as fruits and vegetables. Because daily consumption of these will likely vary, it is even more important for coaches to begin the training year with subjective measures that athletes can utilize throughout the entire year to assess their bodies and hydration needs better.

For early morning training sessions, the athlete will be dehydrated. There is no way around this due to water loss during sleep. Encourage the athlete to drink according to thirst when they first wake up. This will be a bit of trial and error, but most athletes will usually consume around 8-20 ounces of fluid automatically in the first hour of waking. Again, **emphasize the process of teaching an athlete about their thirst response and that they should start listening to their body’s hydration needs more closely before a training session.**

For later day training sessions, it is much easier to help athletes implement a proper nutrient timing hydration strategy because they will have more time in the day to consume fluids and higher water-containing foods. Encourage athletes to pay closer attention to their thirst at least two hours before a later day workout. This will establish more awareness of their hydration needs and will assist them in making better hydration choices throughout the day.

Carbohydrates, Protein and Fat

Following along the same pattern as educating on a more qualitative basis, nutrient timing of carbohydrates is highly individual. Because this training cycle should be lower in intensity and duration, the messaging is geared more toward the time of the day an athlete trains and the overall goal of the training session.

If the workout is less than 60 minutes, there is not an immediate need to eat to fuel the session unless a runner is hungry. If that is the case, encourage the consumption of 1 serving of carbohydrates and a smaller amount of protein and fat (about 1/4-1/2 serving).

If the workout is longer than 60 minutes, it is recommended to consume 2-3 servings of carbohydrates and one serving of protein and fat. This combination will stabilize blood sugar before the training session and ensure steady energy during training.

Electrolytes

The five main electrolytes in sports nutrition for endurance athletes are sodium, chloride, potassium, magnesium, and calcium. Therefore, sodium is the primary electrolyte lost in sweat and is more frequently discussed. Sodium intake is highly individual because of sweat rate, sweat sodium concentration, and family history of certain disease states such as hypertension. Because of this, general guidelines should only be provided to athletes. The range of sodium loss in sweat is typically reported to be 1000 milligrams per liter. However, sweat sodium concentration testing has indicated that athletes can lose between 200-2300 milligrams of sodium per liter of sweat. This makes it nearly impossible to provide an accurate sodium consumption plan for athletes unless they have had a sweat sodium concentration test. Therefore, most sports nutrition professionals recommend a range of 500-1000 milligrams of sodium per liter of sweat lost per hour to be on the conservative side. Athletes who require a more aggressive sodium supplementation plan should consult a qualified Registered Dietitian specializing in sports nutrition for endurance athletes.

Base Training – During Training

Calorie needs will be minimal during training due to this training cycle's lower energy expenditure nature. In general, athletes usually do not need to consume any calories for workouts under 60 minutes. For workouts between 1-3 hours, current sports nutrition research recommends consuming 30-90 grams of carbohydrate per hour. However, because of the lower training stress of this cycle, athletes can easily consume a lower range of carbohydrate intake per hour. Almost all endurance athletes will not need to consume any protein or fat during training during this cycle.

Fluid needs vary greatly depending on the sport. For example, cyclists are able to consume fluids on the higher end of the recommendation and runners on the lower end. Current recommendations based on sports nutrition research suggests athletes should consume between 12-32 ounces of fluid per hour during training.

Base Training – After Training (0-60 Minutes)

Post-workout nutrition is highly individualized based on the goals of the athlete. There are two main goals for post-workout nutrition: 1) speed glycogen replenishment and 2) optimize blood sugar to improve fat oxidation. A key learning message is that it takes the body 16-24 hours to completely replenish glycogen stores when an athlete follows their normal, periodized daily nutrition plan. In other words, if an athlete does not have quality (glycogen depleting) workouts within 16-24 hours of each other, there is no need to speed glycogen replenishment. This is often true during this training cycle. The general recommendation of eating a more balanced post-workout meal or snack consisting of carbohydrates, protein, and fat will be more supportive of an athlete's goal during this training cycle. Remember, nutrient timing must be periodized just as daily nutrition is periodized throughout the year.

While somewhat dependent on how well the athlete was pre-hydrated and stayed hydrated during the workout, fluid recommendations post-workout ranges from 16-24 ounces of fluid.

The addition of electrolytes is very important post-workout. While a specific recommendation does not exist, consuming 200-500 milligrams of sodium in a fluid or post-workout meal/snack will usually meet an athlete's needs.

Competition Training Cycle – Before Training (1-4 Hours)

This training cycle usually includes either longer duration, higher intensity, or a combination of both. Because of this, pre-training nutrition becomes much more important to provide nutrients for training and set up a more successful post-training recovery process. Because of the higher stress of intensity-effort-based training in this training cycle, more quantitative recommendations are adopted from the published research, which will be discussed.

Fluid / Hydration

Current quantitative recommendations suggest drinking 0.07-0.10 ounces of fluid per pound of body weight 4 hours before training and an additional 0.04-0.10 ounces per pound of bodyweight 2 hours before training. For example, a 140-pound (63.5 kg) female would consume 10-14 ounces of fluid 4 hours before training and an additional 6-14 ounces of fluid 2 hours before training. A 185-pound (84 kg) male would consume 13-19 ounces of fluid 4 hours before training and an additional 7-19 ounces of fluid 2 hours before training.

Early morning training sessions will be challenging for athletes to be fully hydrated beforehand. In these cases, it is recommended that the athlete begin the morning rehydration process by drinking small amounts of fluids that contain sodium and carbohydrates. Water alone is not the best rehydrating option; thus, athletes should not over-consume water without eating food or consuming electrolytes. Low-fat milk or chocolate milk, or a smoothie can also be used in small quantities in the early morning to begin the rehydration process.

Carbohydrates, Protein and Fat

Carbohydrates are essential during this training cycle due to the higher training load. Because of this, the periodization of carbohydrates should somewhat follow the athlete's training to best support energy needs and recovery. However, carbohydrate intake does not need to be the same daily but rather periodized to match training and adaptation needs. For example, suppose an athlete has a higher quality training session (defined as a threshold or higher effort or longer than 2-3 hours). In that case, it is important to shift daily carbohydrates to support this. Alternatively, if an athlete has more aerobic or tempo-based training, this could suggest a lower carbohydrate intake before training.

While more complex, once an athlete understands this, it is a straightforward concept with an easy implementation strategy. Suppose an athlete wakes up on a day when higher intensity intervals are to be performed. In that case, the first inclination is to eat a higher carbohydrate meal to help supply the body with the energy needed for the workout. As stated previously, the ability to improve glycogen stores usually takes 16-24 hours, so the athlete may not have had enough time to fully replenish their glycogen from the previous day. This can complicate matters because the athlete must not only know when their higher quality training sessions are, but they must also plan ahead by increasing carbohydrate consumption in the day prior to a higher effort workout.

Protein and fat are still important macronutrients for athletes to consume but must be periodized more carefully in this training cycle. As the intensity of exercise increases, this will shunt blood from the digestive system to the working muscles. This reduces the body's ability to digest foods, especially protein and fat. The quantities of protein and fat before higher intensity training will be much lower than in the previous training cycle.

Because of the higher calorie expenditure and calorie needs during this training cycle, the nutrient timing window is expanded to 1-4 hours before training. An athlete should be educated to eat more calories when there is a longer period before training and eat fewer calories when there is a shorter period before training.

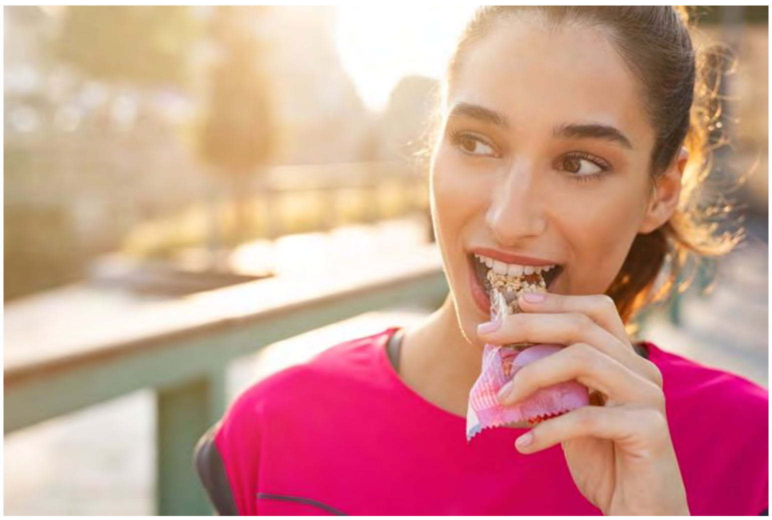
Electrolytes

Electrolyte consumption can mostly mimic the base training cycle nutrition guidelines. The main difference will be the environmental strain since this training cycle usually falls in hotter and sometimes more humid weather conditions. The athlete should be encouraged to begin their electrolyte consumption based on the previous training cycle and adjust accordingly due to their sweat rate, sweat sodium concentration (if they have this data), and performance and recovery markers.

Fiber

Research by Burke et al., found that a reduction in pre-race intestinal fiber content lowered the risk of GI issues (2027) of distance runners. However, the optimal time period to reduce fiber pre-race is dependent on an individual's gut transit times (2028).

Competition Training Cycle – During Training



Fluid / Hydration

Research supports drinking 3-8 ounces of fluid every 15-20 minutes during training. While many theories either support or dispute this range, the athlete needs to have a range to start with and then alter based on their individual needs. It is advisable not to consume more than approximately 33-36 ounces of fluid per hour to avoid gastric emptying challenges.

Carbohydrates, Protein and Fat

Calorie needs during training will be increased due to this training cycle's higher energy expenditure nature. Similar to the last training cycle, most athletes will not need to consume any calories during workouts under 60 minutes, as long as they are fed and hydrated before these workouts. For workouts over 60 minutes, current sports nutrition research recommends consuming 30-90 grams of carbohydrate per hour. This is an expansive range, and it is recommended that athletes begin on the lower end at first and make necessary adjustments throughout this training cycle to customize their needs.

Most endurance athletes will not need to consume any protein or fat during training, except ultrarunners. Many of these endurance athletes will consume whole food during more extended training. While specific recommendations per hour do not exist for protein and fat, it is advisable to encourage these athletes to consume more carbohydrates with a smaller ratio of energy coming from protein and fat.

Competition Training Cycle – After Training (0-60 Minutes)

Carbohydrates, Protein and Fat

Most of the post-workout nutrition messaging in this training cycle is to quickly replenish glycogen stores due to a higher training load and more frequent training. Current research recommendations suggest athletes consume 1-1.2 grams of carbohydrates per kilogram of body weight. For a 120-pound (54.4 kg) female athlete, this would equal 55-65 grams of carbohydrates. For a 180-pound (81.6 kg) male athlete, this would be 82-99 grams of carbohydrate. A typical serving of fruit or grain typically has around 20 grams of carbohydrate.

Fluid and electrolyte consumption post-training are the same as in the previous training cycle: fluid recommendations range from 16-24 ounces of fluid, and electrolytes, specifically sodium, range from 200-500 milligrams.

Transition Training Cycle

The most important nutrition message during the transition cycle is that a specific nutrient timing system will usually not be necessary since the physiological goals of this cycle center on recovery and pre-habilitation/rehabilitation without a high degree of volume or intensity training.

The transition cycle should be thought of more as a lifecycle phase, and, as such, athletes should be educated about the differences in energy expenditure. This alone will help guide them in periodizing their nutrition more successfully and shift their paradigm from “eating to compete” to “eating for a lifestyle.” During this cycle, the main nutrition education points are to avoid the daily nutrition plan of sports nutrition products used in the previous training cycle and to re-focus on blood sugar optimization using the Periodization Plates.

Exceptions

If an athlete engages in light training during this cycle, it will likely not be extremely high in energy expenditure. It is usually unnecessary to have a specific nutrient timing plan during the transition cycle.

If an athlete engages in strength training with hypertrophy goals, a nutrient timing plan consisting of carbohydrates and protein immediately before and after can be implemented. In this case, consuming two servings of protein and 2-4 servings of carbohydrate is recommended to maximize muscular adaptations.

Carbohydrate Loading

Carbohydrate loading has been a prevalent practice among endurance athletes for many decades, usually in the form of the somewhat casual pasta loading dinner the night before a competition. However, some athletes have also followed a more structured carbohydrate loading protocol, of which many versions have existed over the decades.

When carbohydrate loading protocols were first created, there were quite a few negative side effects athletes experienced, specifically gastrointestinal (GI) distress.

In the 1960s, researchers began using the biopsy needle to collect small muscle tissue samples to measure glycogen (stored form of carbohydrates). Based on the ability to measure glycogen, researchers were able to conclude the following:

- There was a direct correlation between glycogen concentration and daily carbohydrate intake
- Glycogen concentration declined during exercise, especially during high intensity exercise
- Higher glycogen concentrations resulted in less fatigue and better performance

Of course, if an athlete follows a high carbohydrate daily nutrition plan, glycogen stores should be high in theory. However, with so many different dietary strategies being followed by athletes currently, it cannot be assumed that athletes have fully loaded glycogen stores.

Carbohydrates are linked to performance, which is why carbohydrate loading strategies have become so popular. Unfortunately, early carbohydrate loading protocols were difficult to implement and caused a few side effects that had negative performance consequences.

Glycogen Super Compensation

Glycogen super-compensation (above normal levels) was studied in the early stages of carbohydrate loading protocols by manipulating both training and dietary intake of carbohydrates. In brief, researchers would deplete muscle glycogen in athletes first through high-intensity workouts one week before a competition. This was followed by three days of dietary carbohydrate restriction with very light training if any at all. The idea was to deplete glycogen stores as much as possible.

Following the depletion state, athletes would follow an extremely high carbohydrate daily nutrition plan for three days before competition with little to no training. While this improved to super-compensate glycogen stores, athletes experienced GI distress symptoms from the low carbohydrate/high fat intake and did not appreciate the lack of training the few days before competition.

For these reasons, this glycogen super-compensation strategy was short-lived.

Updated and Current Strategies

The second carbohydrate loading strategy, proposed in the 1980s, deleted the glycogen-depleting exercise part of the protocol and focused on simply increasing daily carbohydrates over a 6-7 day period before competition. This protocol has withstood the test of time and has, by far, been the most popular carbohydrate loading protocol that many athletes continue to follow.

This updated strategy seemed to work well, especially since athletes were usually instructed to eat as many carbohydrates as possible the week leading up to the competition. Unfortunately, this strategy had some challenges, including weight gain (mainly from water), an overall feeling of heaviness, and more frequent daily bowel movements. Also important to note is the psychological impact of overeating which typically accompanied this approach. It is often difficult for an athlete to approach eating almost unlimited carbohydrates for a short period and then return to normal daily carbohydrate consumption levels after a competition. Nevertheless, this became the mainstay carbohydrate loading protocol for endurance athletes.

The current recommendations for the most up-to-date carbohydrate loading protocol have shifted and have been reduced to a shorter loading protocol (933, 934). This is advantageous for various reasons that were mentioned previously.

The two most current carbohydrate loading strategies are outlined below:

International Society of Sports Nutrition (ISSN)

8-10 grams of carbohydrate per kilogram of body weight for 1-3 days before an endurance event.

International Olympic Committee (IOC)

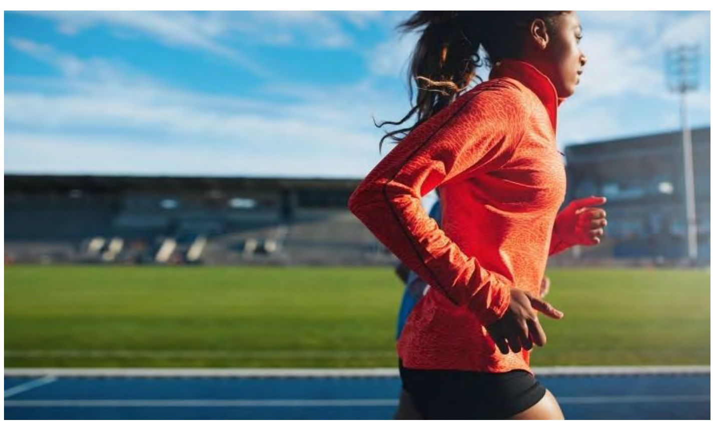
7-12 grams of carbohydrate per kilogram of body weight for 24 hours before an endurance event.

OR

10-12 grams of carbohydrate per kilogram of body weight for 36-48 hours before an endurance event.

The IOC carbohydrate loading recommendations are founded on more recent evidence that super-compensation of glycogen stores can be done in 24-36 hours with increased daily carbohydrate intake along with a properly implemented physical training taper and rest strategy leading up to competition (935). This may lead to lower rates of GI distress and, overall, more gut comfort and psychological well-being as athletes prepare for competition.

Female-Specific Considerations



Most research on carbohydrate loading strategies has been done on males, and while there have been a handful of studies on females, the evidence is far from conclusive. Much of the early research on males versus females centered on increasing the total percentage of daily carbohydrates. This research found that women did not respond as well as men (lower glycogen stores and smaller performance improvement). The conclusions focused on moving away from recommending percentages of carbohydrates and shifted to implementing a grams of carbohydrate per kilogram of bodyweight strategy. Additionally, increasing total energy intake (calories) and increasing total carbohydrates were also beneficial for females who wanted to reap the benefits of a carbohydrate-loading protocol.

Factors that determine the success of carbohydrate loading in female athletes include total daily energy intake, amount of daily carbohydrate consumed, and menstrual cycle phase. Women may have a greater capacity for storing glycogen during the luteal phase of the menstrual cycle. The natural fluctuations of hormones, specifically estradiol, may be the critical factor determining carbohydrate sensitivity and the ability to super-compensate glycogen stores throughout the normal menstrual cycle.

There are differences in carbohydrate loading in male and female endurance athletes. Females can super-compensate glycogen stores but require a higher daily carbohydrate intake and a higher daily calorie intake during the loading phase. Current recommendations suggest that female athletes need 8-12 grams of carbohydrate per kilogram of body weight per day and a higher daily energy intake during the carbohydrate loading phase (937, 938). Additionally or alternatively, it may be beneficial to add more carbohydrates during exercise depending on menstrual cycle patterns. However, there is no conclusive evidence to support specific recommendations.

Supplements

Discussing supplements with athletes is somewhat of a slippery slope. Be sure to keep in mind the scope of practice. As a coach, you want to provide education surrounding supplements rather than providing a prescription.

It is easy to educate athletes about supplements once you understand the three main categories of supplements:

1. Micronutrient
2. Sport
3. Ergogenic aid

Micronutrient

These are usually daily supplements, primarily vitamins and minerals, taken either as an "insurance policy" or for specific deficiencies that have been identified from biomarker testing and a qualified professional's interpretation.

Common micronutrients endurance athletes will take include vitamin B12, iron, vitamin D, and magnesium. Many whole foods provide ample micronutrients, which should be the first line of education for athletes. It is only recommended that athletes take micronutrient supplements only after having specific biomarker testing done and reviewed by a Registered Dietitian or Sports Physician. Aligning micronutrient supplement use with biomarker testing will provide the athlete with a more individualized approach to taking these supplements and will support taking them due to micronutrient deficiencies.

Sport

This category includes energy bars, sports drinks, gels, gummies/chews, electrolytes, and anything else that will be used immediately before, during, or after training sessions. These supplements are usually necessary for most endurance athletes at some point throughout their competitive year, but coaches should educate athletes on the correct timing of using these. Whenever possible, athletes should be encouraged to use real food to meet most of their daily nutritional needs and sports supplements to meet training nutritional needs when digestive challenges are present.

Ergogenic Aids

Ergogenic means performance-enhancing. This is the more complicated category related to “pills, powders, and potions” and usually supplements that do not have much scientific support of safety or efficacy. Examples of ergogenic aids for endurance athletes include caffeine, creatine, nitrates, beta-alanine, sodium bicarbonate, and branched-chain amino acids. Athletes should be advised when considering ergogenic aids as specific dosing, and implementation strategies are required to elicit physiological adaptations. This category of supplements will be the lowest on the priority list for endurance athlete recommendations. Coaches should always refer to a qualified Registered Dietitian who is a Board Certified Specialist in Sports Dietetics (CSSD) and has a specialty in working with endurance athletes.

Summary

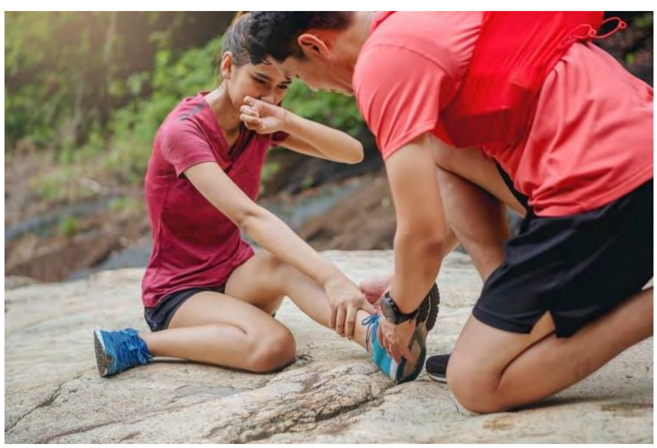
- Unless you are a Registered Dietitian (RD) or Registered Dietitian Nutritionist (RDN), many states limit the amount of nutritional information you can legally provide to an athlete.
- You can educate and inform an athlete on nutritional information but you cannot prescribe a nutrition program, diet, or supplement protocol, especially if your athlete has any pre-existing medical conditions.
- Macronutrients are carbohydrates, fat and protein
- Two of the more important micronutrients for endurance athletes are iron and electrolytes
- Carbohydrate intake is a vital part of a runner’s diet, as it provides energy for training and racing. Carbohydrates maintain blood glucose levels during exercise and replace muscle glycogen.

- Protein from animal and plant sources provides the amino acids needed for muscle growth, repair, and maintenance.
- Marathon runners who followed a scientifically based nutritional strategy were approximately 5 percent faster than runners who self-selected their nutritional approach
- As different training cycles are followed throughout the year, energy demand, energy expenditure, and body stressors differ.
- The following are the five components of the nutrition periodization concept:
    - Manipulate body weight and body composition safely and effectively
    - Improve the hormonal influence of appetite
    - Support the immune system and the microbiome
    - Periodize supplement usage
    - Support a healthy relationship with food
- Nutrient timing is a critical aspect of effective fueling for training and racing
- Carbohydrate loading strategies have lessened in duration over time.
- It is recommended for your athlete to consult a registered dietitian if they are interested in taking any form of supplement program.

Module 19: Safety

In many respects, this is the most important section of the whole running coach certification course because safety is paramount to everything else. While this lesson will discuss various issues that encompass safety-related areas, the safety of your athlete and your programming methodologies must first and foremost take safety into consideration.

Scope of Practice and Knowledge



Successfully passing this certification will qualify you to advise and educate an athlete in all areas about training a runner. It is important to remember that there is a big difference between advising/educating and prescribing/diagnosing. As a running coach, you cannot do the following:

- Diagnose or treat an injury or illness
- Prescribe a nutritional or supplementation program in most states unless you are a registered dietician or certified nutritionist
    - Check with local or state laws.
- Hands-on muscular (i.e., massage) or skeletal manipulation
- While not illegal, any substantial biomechanical deviation should be referred to a specialist

If you do not feel qualified to advise or educate on a specific area and decide to proceed anyway, you are practicing outside your scope of knowledge.

This is an important distinction. **You can work within your scope of practice but outside your scope of knowledge. Conversely, you can work within your scope of expertise but outside your scope of practice.** An example of someone working within the scope of practice but outside the area of knowledge would be a running coach whose athletes want advice on the difference between two separate shoe brands. While this question is within a running coach's scope of practice, an individual who does not know the answer and proceeds to fabricate one is practicing outside the scope of knowledge.

An example of someone practicing within the scope of knowledge but outside the scope of practice would be a seasoned running coach who is 99.9 percent sure an athlete has a patellofemoral syndrome and manually massages the athlete's leg to relieve the symptoms. While the running coach may be correct in the "diagnosis," the individual cannot legally diagnose a condition and cannot massage the athlete's leg to relieve the symptoms – even if sure that this will work.

The distinction between what is legal and illegal when working with athletes can be unclear. It is always best to avoid caution when working with athletes and refer them to specialists.

Team Approach

The legal and professional obligation of a coach to refer athletes may also present learning opportunities. Often, a specialist will advise a coach on what protocol to follow or why an athlete is experiencing a particular issue.

Knowledge gained in this fashion is invaluable and will help further the development and education of a coach.

It is suggested that a coach coordinate a team of specialists to whom they feel comfortable referring athletes. **This team approach generally elicits the best and fastest results for athletes and releases a coach from potential liability by overstepping professional boundaries.**

A coach can't have all the resources an athlete requires. A coach not implementing a team approach is not providing athletes with the best possible training. Unless a coach is a doctor, physical therapist, registered dietician, podiatrist, sports psychologist, etc., the individual needs other members on the team. **Just because a coach has knowledge or experience in a particular area does not necessarily mean that there is not someone better equipped to assist an athlete in that area.** For example, a registered dietician focused on endurance sports is likely better suited to advise an athlete on nutritional strategy than a running coach. This is not a negative reflection on a running coach but demonstrates a dietician's specialized expertise. Referring athletes to others is not a sign of weakness but indicates a coach's intelligence and dedication to their athletes.

Weather

Temperature extremes not only negatively affect one's performance but can be dangerous as well.

If an athlete knows that competition will occur in extreme temperatures, the individual needs to simulate these temperatures, if possible, during training safely. Accurately predicting how the body will react and perform in adverse conditions is critical for an athlete's safety. Preparing for extreme temperatures is covered in detail in the Environmental Physiology lesson.

Additionally, wet weather can also pose a risk in respect to slippery surface conditions. This is exacerbated on smooth surfaces (ex: painted lines, manhole covers, smooth rocks, etc...), uneven terrain and leaves. Unlike the picture above, it is not a good idea to run through water (if possible), as a runner does not know the depth or surface condition beneath the water.

Another issue in cold weather is ice. While all ice can be dangerous, the most dangerous type of ice is often referred to as 'black ice,' this is ice that blends in with the asphalt and a runner does not often notice it until it is too late.

Regarding heat, it is advised if possible, to run during the coolest times of the day and ideally, on a route that provides ample shade. Regardless of the route, a runner must take enough fluids with them for the distance that they are running. While it's never a bad idea to run on a route that passes by a water fountain(s) or a store where a runner can purchase drinks, a runner cannot depend on this. For example, perhaps the water fountain is out of order and the store is closed.

Safety Issues

Runners must always stay alert on roads and always run on the left-hand side unless there is a wider shoulder or running lane on the right.

Runners need to be careful not to put themselves in harm's way. Below are several examples of this:

- If running in low light or at night, runners must wear reflective clothing and have a light source.
- Runners *must* be aware of their surroundings at all times!
- It is not a good idea to run alone in areas that could be considered dangerous. Athletes should run only on roads they are familiar with and, ideally, roads with a wide shoulder or running lane.
- If your athlete is trail running, instruct the individual to always run with a partner and know the route. If your athlete runs in areas that allow hunting, trail running should not be done during hunting season.
- If wearing headphones, make sure to wear only one earpiece or headphones that allow outside noise to be heard.
- When running near parked cars, a runner must be far enough to the side so that if someone opens a door suddenly, the runner will not be hit.
- An asthmatic athlete should always carry a rescue (fast-acting) inhaler.

Hydration is an important area to focus on when discussing safety. Many runners often run long distances without fluids. This can lead to dehydration which can cause serious medical complications. Two ways to prevent this are to carry fluids during a run or make sure there are stops along a route to get fluids and food. Specific hydration systems are made for runners, such as hydration backpacks, belts that carry small fluid containers, and handheld bottles with a strap for easy carrying.

An athlete must always have emergency information on hand. Products allow individuals to put their name and emergency information on wrist or ankle bracelets. Additionally, athletes should tell someone of their proposed route.

If running on trails in unfamiliar or remote areas, having a watch that has navigation capability such as Global Navigation Satellite System (GLONASS) is important.

When doing a long run, if possible, an athlete should bring a cell phone and run in an area with stops for food and water. This is especially important when running in hot weather.

Summary

- As a running coach, you cannot do the following:
    - Diagnose or treat an injury
    - Prescribe a nutritional or supplementation program in most states unless you are a registered dietician or certified nutritionist
        - Check with local or state laws
    - Any hands-on muscular or skeletal manipulation
    - While not illegal, any substantial biomechanical deviation should be referred to a specialist
- As a coach, you must always practice within your scope of practice and knowledge.
- It is never the wrong approach to refer your athlete to a specialist.
- It is advised to have a team of specialists to whom you can refer your athletes to.
- When running in low light conditions, reflective clothing and a light source are musts.
- Always run in familiar locations and ideally with a partner.
- Always carry emergency contact information when training and racing.

Module 20: Running Shoes and Apparel

Running shoe and apparel selection has a significant influence on a runner. The primary areas of impact are:

1. **Safety**
2. **Performance**
3. **Comfort**

This module will cover all of these areas and more!

Running Shoes

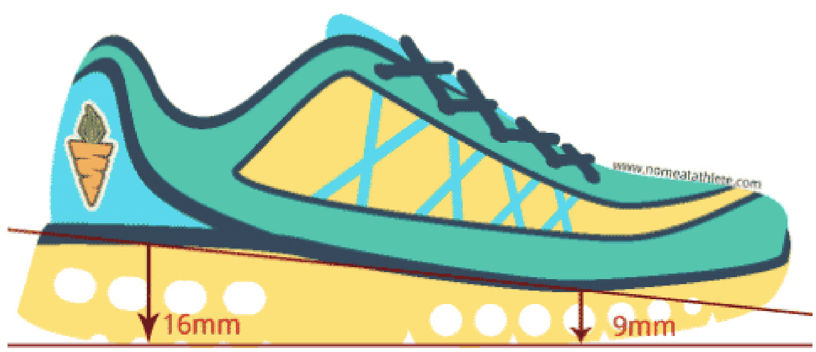
There are three primary classifications of shoes: ***conventional, minimalist*** and **maximal/over cushioned.**

Conventional shoes are designed to provide support and stability. Minimalist shoes are designed to allow the feet to use their inherent strength to support a runner instead of relying on the structure of a shoe. Minimalist shoes are typically used in conjunction with a midfoot strike. Maximal or over cushioned shoes have substantially more cushioning than conventional shoes. While most shoe companies now make some version of a maximal shoe, *Hoka One One* was the originator and is largely still the most popular.

Heel-To-Toe Drop

This relates to the difference in the height of the sole at the heel and forefoot (in millimeters). Because of its padding and support, a conventional running shoe typically has a drop of 10 or more millimeters, whereas a minimalist shoe typically has a drop of less than 10mm (364). The heel-to-toe fall has become a focal point with the increasing popularity of minimalist shoes.

Individuals converting to or currently running in minimalist shoes tend to look for shoes with small to zero heel-to-toe drop. While this level of the heel-to-toe drop might be fine for some, it is not advised for those converting to a midfoot strike or for individuals with excessively tight calves and/or Achilles tendons. Successful progression to a minimalist shoe often takes time.



As noted previously, during a midfoot stride, the ball of the foot impacts the ground first, but almost immediately after that, the midfoot and heel lightly impact the ground to provide support. Assuming your athlete runs with correct form, a small- to the zero-drop shoe will place extra eccentric stress on the calves and Achilles tendon. Without an elevated heel, the ankle is forced to dorsiflex more. Therefore, you need to factor in ankle flexibility (flexion) concerning your athlete’s shoe selection.

In the image above, the heel drop corresponds to 16 minus 9 in millimeters (7 mm drop). Note, that a *zero-drop* minimalist shoe has the same sole height at the heel and forefoot.

Conventional Shoes

Conventional running shoes are geared toward runners who heel strike. More specifically, traditional running shoes primarily focus on the following three areas:

- Cushioning: primarily through midsole EVA foam
- Control rear foot motion
- Arch support

The foot arch type often (but not always) correlates to the amount of rear foot motion. Below is a chart of these correlations:

ARCH TYPE	REAR FOOT MOTION	SHOE TYPE OFTEN RECOMMENDED
Low	Over Pronation	Motion Control
Normal	Mild Pronation	Stability/Neutral
High	No Pronation or Slight Supination	Neutral

Conventional running shoes typically have some or all of the following characteristics:

- Substantial heel padding
- Large heel-to-toe drop
- Relatively rigid shoe

The results of most studies regarding the impact of shoe type (e.g., stability, motion control, neutral) with respect to controlling foot motion are conflicting. Some studies show that the kind of running shoe does not influence rear foot motion (197, 198) or minimally, whereas other studies show that rear foot motion can be substantially affected by the type of shoe (199).

It is well-known that the body can adapt to change. Studies have demonstrated that runners can adapt to changes in the type of shoe and the type of running surface (200). This tends to support the argument that the specific shoe type minimally affects rear foot motion because of the body's ability to adapt to varying shoe types. Adaptability is one of the main reasons behind the barefoot/minimalist shoe movement. Those who run barefoot or in minimalist shoes often believe that conventional shoes take over the stabilization and support aspects the body should naturally provide. This, in turn, makes for a weaker foot and ankle complex.

Many shoe/foot studies are done with the feet in a static position or with runners who run for short bouts and therefore for whom fatigue is not a factor. It has been proposed that most injuries occur when a runner is fatigued and have reduced muscle strength levels to control the ankle joint (199). Therefore, it could be deduced that a decrease in the body's ability to stabilize the ankles and feet would place greater reliance upon a shoe's structure for support.

While it is standard to recommend a specific type of running shoe based on foot type, according to a 2009 study by Richards et al., only minimal evidence exists to support that this practice reduces injury rates (201). Additionally, **Richards et al. found no randomized, controlled trials correlated the prescription of the shoe type to injury rates.** Richards et al. concluded that prescribing shoe type based on foot type is not an evidence-based practice (201).

Perhaps the most interesting study to support the findings of Richards et al. is a 2009 study by Knapik et al. This study examined the correlation of shoe prescription and injury risk among new military trainees (195). **Knapik et al. found that regardless of a trainee's static foot type (high/low arch), individual shoe prescription based on foot type did not lower injury risk compared with when all the trainees ran in the same type of shoe (stability).**

The one caveat to these studies is that the participants ran low mileage relative to distance runners, who typically run long miles. For example, participants in Knapik's study ran between .5 to three miles at a time. Therefore, **more research is needed on high-mileage runners to assess the relationship between shoe type and injury risk** (202).

Minimalist Shoes

Characteristics

- Minimal heel padding
- Small heel-to-toe drop (typically 10mm or less)
- Minimal arch support
- Flexible sole and shoe fabric ("uppers")
- Light
- Wide toe box to take into account the expansion of the forefoot during impact

Unless your athletes specifically ask for a minimal shoe at a running store, the staff will watch them run and put them in a conventional shoe based on their visual assessment of rear foot motion (e.g., neutral, motion control, stability).

Most track and cross-country-specific racing shoes are minimalist in design. Some of these shoes have small spikes on the sole for traction on the track and off-road.

Hybrid shoes fall in between minimal and conventional shoes. They are characterized by a medium level of heel-toe drop, heel padding, and weight and arch support.

Barefoot

If your athlete wants to run barefoot, the most important thing to realize is that the body needs to adapt biomechanically, and the skin also needs to become more calloused. Walking indoors in bare feet is the starting point to toughen up the feet, then walking outside in an area void of any debris that could puncture the skin. Before running outside, running on a treadmill in bare feet (if allowed) or on a running track is also helpful in the adaptation process.

Following are several reasons why people believe that running barefoot is more advantageous than running in shoes:

- Direct power transfer
- It is the way the body was biomechanically meant to run
- Less mass (i.e., shoe) to accelerate/decelerate to increased efficiency
- Strengthens the foot arch

When beginning to run barefoot outside, it is strongly advised to use shoes such as *Vibram Five Fingers®.* These are essentially a foot glove with a rubber tread on the bottom. The standpoint of this certification is that if your athlete wants to run barefoot outside, it should be done in shoes such as *Vibram's.* This is because the chance for injury, and thus infection, due to glass or other debris penetrating the skin on the sole is high when running barefoot. However, **an athlete who insists on running barefoot must know the risks and follow a proper adaptation process.**

A 2009 study by Kerrigan et al. found that while conventional running shoes provide good support to the foot, they place more significant stress on the ankles, knees, and hips than running barefoot (323). The study consisted of 68 young, uninjured, active runners who averaged 15 miles/week. The study theorizes that the reason for increased stress on the three joints above is because conventional shoes lessen the amount of pronation of the foot/ankle. This reduces the body's natural shock absorption properties via foot/ankle pronation, and the shock is then transferred to the three joints (hip, knee, ankle).

A 2012 study by Franz et al. found that running barefoot offered no metabolic benefit over running in lightweight, cushioned running shoes (787). As noted in the below section, a study by Kram et al. found that running barefoot increased one's energy requirement by 2% compared to running in shoes with 10mm of EVA foam (327).

Over Cushioned (Maximal) Shoes

In contrast to running barefoot or in minimalist footwear, over-cushioned shoes (aka maximal cushioned) have up to twice the cushioning of standard running shoes.

The genesis for creating over-cushioned shoes was to serve the ultramarathon market (673). However, since then, the key demographic has expanded to encompass runners of all ability levels and apply to events of varying distances.

While there are not many studies regarding maximal shoes at the point of this certification, some aspects of the research are contradictory. For example, some research (1019, 1020) found that maximal shoes increased the loading rate and impact forces. Conversely, another study that looked at minimal, traditional, and maximal shoes found that maximal shoes did not increase the loading rate (1021). There are likely many variables that influence the outcome of these studies, such as the foam type, thickness, overall shoe construction, etc... As such, more research is needed in this area.

That said, if an athlete likes to run in maximal shoes and does not see an increase in injuries. As a result, running in this type of shoe should not be discouraged.

Other Shoe Types

TRAIL

TRACK

Efficiency

A 2012 study by Kram et al. found that running in cushioned shoes was more efficient regarding VO2 Max and metabolic power than running barefoot (326). This is because there is an increased metabolic cost with running barefoot. **While running in shoes does increase the metabolic cost by adding mass to a runner, it is hypothesized that the cushioning cancels out the metabolic cost** (327).

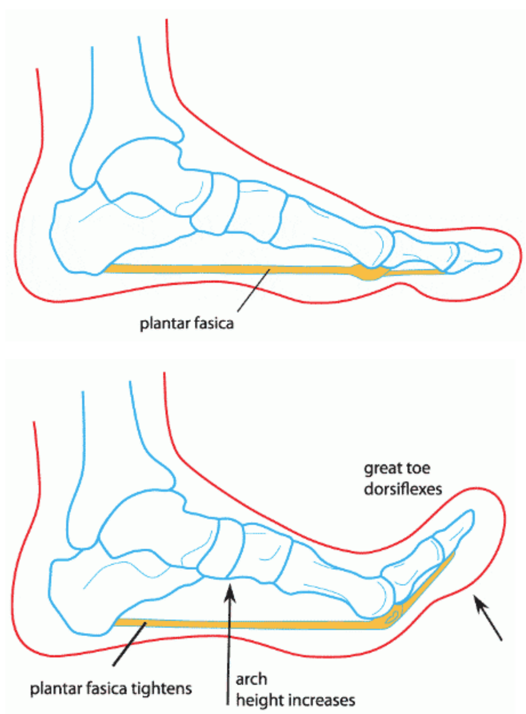
A study by Logan et al. researched the difference in ground reaction forces (GRF) between three types of running shoes: cushioned, racing flats, and racing spikes (609). **Ground reaction force** pertains to the force the body exerts on the ground (e.g., impact). The primary finding was that racing flats and spikes have a substantially higher GRF and loading rate than cushioned shoes.

The **loading rate** relates to the speed at which GRF occurs – in other words, the speed at which loads are applied to the body. While minimally cushioned shoes such as racing flats and spikes likely enhance a runner's efficiency from a biomechanical standpoint because of a direct power transfer (e.g., minimal energy lost through the absorption of shoe cushioning), it is likely that the metabolic cost is much higher than with a more cushioned shoe and possibly puts a runner at an increased risk of injury (609). Because of the increased GRF and loading rates of racing flats and spikes, it could be theorized that this type of footwear would be most applicable to shorter-distance events.

Carbon-Soled Shoes

It seems that most all shoe companies nowadays are offering carbon-soled shoes. One can only guess that this is largely due to the hype and the results of pro runners wearing the most well-known carbon-soled shoe, the Nike Vaporfly. The Vaporfly 4% is named due to the purported energy savings (4%) over the best existing, non-carbon plated marathon shoe. So, the million-dollar question… do carbon shoes make a difference, and to what degree?

While there is not much research on carbon-soled shoes at the point this section was written (2022), like many other research areas, it appears to be a mixed bag. However, before we go any further, it's important to discuss the reasoning behind a carbon fiber sole. As you've learned throughout this certification, the foot has a built-in tension system called the 'Windlass Mechanism.' As the image below illustrates, this mechanism is an energy source by tightening and releasing energy upon foot push-off. The carbon sole plate is essentially an external 'Windlass Mechanism' that looks to enhance one's energy return and thus, reduce the metabolic cost of running, hopefully translating to faster running times.



Credit: https://www.docpods.com/foot-pain-info/the-windlass-mechanism-in-the-foot-and-foot-pain/

Research

As noted above, the carbon-plated sole research is a mixed bag. Specifically, regarding the Vaporfly 4%, one 2019 study by Hoogkamer et al. found that the prototype shoe (assumed to be the Vaperfly) did reduce the metabolic cost of a runner by 4% (1016).

Conversely, research by Beck et al. (1017) and Healey et al. (1018) did not find a correlation between carbon-plate soles and improved running the economy or metabolic efficiency.

Of the studies noted, the 2022 study by Healey was the most notable. In this study, two shoes were examined. One was the Vaporfly 4%, and the other was the Vaperfly 4%, but the carbon sole had six lateral cuts into the middle of the sole, thus significantly reducing its effectiveness. The finding was that between both shoes, there was not a significant difference between the two shoes regarding running economy (energy savings) or mechanics. Healey et al. suggest that the shoe's effectiveness likely results from a combination and interaction of the carbon sole, foam, and the shoe's geometry.

In summary, based on the current research, it appears that the carbon plate in isolation is not a significant factor in the Vaporfly 4%'s success in real-world running scenarios. Still, the shoe's construction in totality is likely the most crucial variable.

Note: Like the studies in this section focused on the Nike Vaporfly 4%, we cannot speak to other shoe brands that offer carbon plate sole shoes concerning their effectiveness or purported benefits.

Arch Support Debate

Following are common theories about why individuals are for and against arch support.

For Arch Support

- Favors orthotics and shoes that offer arch support. Shoe types that provide arch support are *stability* and *motion control.*
- An individual's arch is not strong enough to support the foot when running. Therefore, orthotics and arch-supporting shoes are needed to compensate for the lack of inherent arch strength/support.
- Overpronation can lead to injury and therefore needs to be lessened or eliminated via support shoes or orthotics.

Against Arch Support

- The running style most associated with a need for arch support is a heel strike, which is biomechanically incorrect. The correct form is a midfoot strike.
- Pronation is normal and does not need to be corrected (stabilized).
- Arch support weakens the arch. Conversely, shoes without arch support strengthen the arch.
- As the longitudinal foot arch produces elastic energy when running, arch support reduces this energy source, reducing the runner's efficiency.
- Arch support reduces pronation, acting as the body's natural shock absorber. This, in turn, increases stress in lower-extremity joints (323).

Running Shoe Fit and Replacement Guidelines

Regarding shoe fit, the most important thing is to try on shoes with the same socks and orthotics (if used) for training and racing.

Correct Fit

Front of Shoe: There should be at least half a thumb's width but not more than an entire thumb's width between the longest toe and the front of the shoe. Running for extended periods and in hot weather causes the foot to expand. Selecting a shoe that is too small will likely cause the foot to become cramped when running for long periods and in hot weather.

A shoe should fit snugly but not tight in any one spot. The heel should not feel like it will come out of the shoe at any point. As noted in the video in the biomechanics model with Dr. Studholme, a shoe should have enough room in the toe box, have good heel support, and feel good.

Breaking In Shoes

While there is almost no break-in period with running shoes today because of their simple structure (203), this does not mean that the specific design and nuances of a shoe might not take getting used to. **Shoes should feel comfortable when trying them on.** If your athlete experiences pain or the shoe just doesn't feel right in the store, they should not purchase them with the hope that they will get more comfortable once broken in. It is advised that an athlete wear new shoes around the house or while running errands before running in them. When running in new shoes initially, it is ideal to start with shorter distances to see how one's feet respond to the shoes. For example, perhaps the shoes fit great, but the tongue moves to the side while running and puts pressure on the side of the feet. This is not something your athlete wants to discover on a long training run.

Seeing how shoes feel when running in them is essential. Many running stores have treadmills for this purpose, and some stores allow customers to run around the block. Running outside is ideal, but a treadmill or running in the store is fine if this is not an option. Some running stores also allow customers a long return window (i.e., ten days) to take them home and run in them to know if they fit correctly.

Replacement Guidelines

Depending on the source, running shoes should be replaced between 400 and 600 miles (203). While this range is sufficient, three key variables ultimately should be the determining factor in shoe replacement:

1. Comfort
2. Fit
3. Shoe Condition

If any of these factors are substantially compromised, the shoes should be replaced – regardless of age or mileage. Continued use of shoes that are deficient in any of the areas above would at best diminish performance and, at worst, cause injury.

Effect Of Shoe Breakdown on Biomechanics

While it is well-known that running shoes lose some of their cushioning over time, what is less known is how the loss of cushioning affects a runner's form.

In a 2009 study by Kong et al., 24 runners (14 men, ten women) were divided into three groups, each wearing a different type of cushioned shoe. Over the course of the study, the subjects ran 200 miles (200). Runners were assessed biomechanically before and after the 200 miles. The three types of shoes used in regard to cushioning were air, gel, and spring-cushioned.

After 200 miles of running, there were only minimal changes in subjects' running mechanics, and, interestingly, there were no reported changes in impact forces. So, what does this mean? **This information seems to support the notion that the body adapts to changes in foot cushioning and support.** As shoe cushion and support decrease, the body adapts to maintain proper running mechanics. Additionally, there was no difference in running mechanics from one type of shoe cushioning to the next. This finding relates to the effectiveness of different kinds of shoe cushioning. Results from the Kong et al. study demonstrate that regardless of shoe brand and shoe cushioning type (i.e., air, gel, spring), the primary source of cushioning is midsole EVA foam. Concerning EVA midsole foam, there is no discernible difference between shoe brands (200).

Shoe Compression

The exact percentage of cushioning lost in relation to miles run is unknown as there are many variables (e.g., running style, runner's weight, etc.). However, it has been theorized that the most significant cushion loss occurs within the first 200 miles (204). It is important to stress to athletes that while they do not want to run in a shoe that is uncomfortable, over time, they should not expect shoes to feel the same as when they were brand new.

A common practice among distance runners is to swap out identical pairs of running shoes during training. Generally speaking, this is a good idea, as it will provide more cushioning for an athlete during runs. Another common reason for doing this is to "rest" or allow shoes to "decompress" before the next run. The legal advice in this area is to let shoes rest between 24 and 48 hours before running in them again, the theory being that it takes this long for the EVA midsole foam to return to its decompressed state. **A study by Cook et al. found no difference in the retention of cushioning between shoes that were rested and those that were not when used within 24- to 48 hours** (204).

Shoe Cushioning: Effect on Performance

According to a study by Kram et al., the degree of foot/shoe cushioning concerning efficiency resembles a bell curve (327). Too much or too little cushioning decreases efficiency (metabolic power). This study had ten barefoot runners run with a midfoot strike on a rigid treadmill (no shock absorption) using the following scenarios:

1. Barefoot
2. Barefoot on a treadmill that was covered in 10mm EVA foam
3. Barefoot on a treadmill that was covered in 20mm EVA foam

Note: 10mm is roughly the same height of cushioning in the forefoot of lightweight running shoes

The conclusion: Running with 10mm of EVA foam required approximately 2 percent less energy than running barefoot while running with 20mm of EVA foam required 1.7 percent less energy than running barefoot.

Apparel Selection

Proper selection of clothing is not a function of only comfort but also safety and performance. Fabric materials and clothing design are constantly advancing to reflect the increasing demands of endurance athletes.

There are several factors to consider when selecting clothing:

- Fabric composition (wicking fabrics)
- Proper fit
- Clothing for all types of weather
- Function
- Comfort

Wicking Fabrics

Wicking refers to the ability of a fabric to “breathe.” Unlike fabrics such as cotton that absorb sweat, wicking fabrics allow body heat and sweat to pass through, so the body stays cool and dry.

It is important to note that the quality and effectiveness of wicking apparel vary greatly. While you don’t have to break the bank to get good-wicking running clothing, be aware that inexpensive, non-running, brand-specific clothing might not be as performance-oriented as clothes from a running company.

Proper Fit

Proper fit of clothes is essential for performance clothing. Depending on the preference of a runner, clothing might fit snugly or be more relaxed. An important factor to be aware of is that loose-fitting clothes tend to have a higher chance of chafing a runner than tighter-fitting clothes.

Weather-Based Clothing Options

An athlete must be ready for all types of weather. This means it is imperative to have a wide selection of clothing. Not having a proper choice of clothing compromises not only an athlete’s comfort but also performance and safety.

Cold Weather

A base layer (layer next to the skin) that provides warmth and wicks sweat away from the body is necessary when exercising during cold weather. The colder the weather, the more layers are needed. It is advised to have some sort of windproof outer layer for cold and windy days to minimize wind chill. Following is a chart of cold-weather running clothing concerning applicable temperatures. A few things to keep in mind:

- Individuals respond differently to hot and cold weather; therefore, the below chart is only rough guidelines.
- A clothing product will likely have varying insulation properties depending on the purpose. For example, tights come in varying thicknesses based on the temperature for which they are designed.
- Many cold-weather clothing items indicate on the tag what range of temperatures they are recommended for.
- It is better to overdress and shed layers than underdress
- Dark-colored clothing will absorb more heat than light-colored clothing.
- Normally, less clothing is worn during a race than in training, as the effort level is usually more significant, and thus more body heat is generated.
- It is essential to wear sunscreen on any exposed skin, regardless of the time of year.

ITEM	TEMPERATURE
Arm Warmers	< 45 (7.2 C)
Thin Tights	< 45 (7.2 C)
Thermal Tights	< 35 (1.7 C)
Vest	< 45 (7.2 C)
Long Sleeve Base Layer	< 35 (1.7 C)
Long Sleeve Shirt	< 55 (12.8 C)
Wind Jacket	< 50 (10 C)
Insulated / Thermal Jacket	< 30 (-1.1 C)
Thin Finger Gloves	< 45 (7.2 C)
Insulated / Thermal Gloves	< 35 (1.7 C)
Full Face Mask	< 25 (-3.9 C)
Hat	< 40 (4.4 C)
Winter Running Socks	< 40 (4.4 C)

This chart takes into account the wind chill

Guidelines

While a wide selection of clothing options makes training and racing in most conditions possible, it can be challenging to figure out which pieces of clothing to wear. A few rough guidelines can help with making these decisions:

1. Always dress in layers.
2. When running in cold weather, an athlete should be somewhat hard for the first five to 10 minutes.
3. The more intense the training or racing, the fewer layers should be worn in cold weather.
4. Always try out different clothing options during training before racing in them.

Apparel Inventory

The following are common types of running apparel for training and racing:

- Running top (sleeveless/short sleeve/long sleeve)
- Jacket
- Sports bra
- Hat/Visor
- Headband
- Vest
- Shorts
- Tights
- Arm warmers
- Gloves
- Socks/Compression socks
- Sunglasses

Some areas to consider when selecting apparel are:

- **Seams:** Make sure that the seams will not rub the skin.
- **Pockets:** What and how much do you need to carry?
- **Reflective clothing:** Never a bad idea and mandatory for low light conditions.
- **Fit:** Make sure clothes don’t chafe during training before using them in races.

- **Brand:** Don’t assume that just because a piece of apparel from a brand fits well, all other pieces in the same size from the same brand will also work well.
- **Keys:** Will you be carrying your keys when running? Critical pockets that attach to shoelaces may be a good option.

Compression Socks / Calf Sleeves

Many runners wear compression socks and compression calf sleeves. Compression socks cover the foot and extend to the top of the calf. Compression sleeves slide on over the calf and just protect the calf.

The science behind compression socks comes from clinical settings where they are used to treat edema (buildup of fluid beneath the skin) and other medical issues. The theory behind compression socks is to increase the blood flow rate back to the heart (venous return) and reduce blood pooling in the legs. Another reason some people use them is to reduce the vibration of the calf muscles, thereby decreasing fatigue.

A 2011 study out of Stellenbosch University in South Africa sought to determine the effectiveness of compression socks in post-exercise recovery for male long-distance runners (268). The study assessed 83 runners competing in a 56 km (35-mile) ultra-endurance run. Following is a summary of the findings regarding the effectiveness of compression socks:

- Runners wearing compression socks had more minor muscle damage post-race than those without compression socks
- Those wearing compression socks exhibited more minor calf swelling
- Some compression socks may not provide enough compression to elicit the desired results
- Theorized that compression socks provide dynamic stability to the calf (i.e., flexible cast)
    - Calf stability reduces muscle vibration, which is theorized to increase muscular endurance and reduce soreness and muscle damage.
- Benefits are limited to calf muscles (gastrocnemius and soleus) via reduced structural damage.
- Increased cellular repair
- Increased venous return is inconclusive

These findings are substantiated by several other studies (265, 266, 267).

While the primary data show that compression socks may decrease muscle damage (calf) while running and improve recovery, another study found that compression socks did not increase running performance. A 2015 study by Stickford et al. found that lower-leg compression sleeves did not improve an individual’s running mechanics (ground contact time, step frequency, and length) or VO2 max. In other words, the running economy of test subjects was not altered using calf compression sleeves (675).

Lastly, a 2014 study by Hill et al. examined the effect of calf-compression apparel concerning accelerating recovery post-marathon (689). There was no physiological recovery improvement in the group that wore compression socks over the group that did not. Interestingly, the group that wore compression socks noted a significantly lower level of perceived muscle soreness (after 24 hours) than the group that did not wear compression socks. **As compression socks improved the sense of recovery without actually enhancing recovery from a physiological standpoint, the benefit of compression wear may be more psychological than physiological.**

Conclusion

The physiological benefit of wearing compression socks/sleeves during and after a run is largely inconclusive, with the most recent data leaning toward the apparel showing no benefit. However, because of potential psychological benefit and the fact that there is no physiological downside to wearing compression socks, it should not be discouraged if an athlete wants to wear them while running and for recovery.

While not a performance factor per se, due to the skin coverage of compression socks and calf sleeves, they also likely keep a runner slightly warmer in cold weather.

Other Running Gear

- Smartphone/*iPod* arm holder
- Hydration belt/vest
- Handheld water bottle
- Anti-chafe balm/spray
- Backpack
- Nipple covers
- Jogging stroller
- Foam roller
- Headlamp (if running at night)
- Sweat-proof sunscreen
- GPS watch
- Reflective clothing/straps
- Waist pack/*SPIbelt*™

Summary

- The major factors in apparel selection are:
    - Fabric composition (wicking fabrics)
    - Proper fit
    - Clothing for all types of weather
    - Function
    - Comfort
- In cold weather, clothing should be layered.
- Compression socks can assist in recovery and may improve running performance.

- It is important to have a wide range of clothing options so that an athlete is
prepared for any weather condition.
- The primary types of shoes are: conventional, minimalist and over-cushioned (maximal)
- While EVA foam in running shoes compresses over time, allowing shoes
24–48 hours to "decompress" before running in them again is a myth.
- Conventional running shoes are characterized by substantial heel
padding, cushioning, and arch support.
- Minimalist shoes are characterized by minimal heel padding, cushioning,
and arch support. Minimalist shoes have a low heel-to-toe drop.
- Shoe cushioning may assist in forward propulsion but likely diminishes the
effect of the stretch-shortening cycle (SSC). A large heel-to-toe drop also
reduces the SSC of the Achilles tendon.
- There is little to no break-in time required with today's running shoes.
- Carbon soled shoes look to use the energy return of the carbon fiber sole to reduce the energy cost of a runner.

Module 21: Race Preparation and Execution

Race day represents the culmination of all of your athlete's hard work. As such, there should be no surprises on race day, and a strategy should be in place to give your athlete the best chance of hitting their goal. This module discusses the areas and aspects necessary to ensure that your athlete has a positive experience on race day!

Upon completion of this module, you should have an understanding of the following areas:

- General preparations
    - Course knowledge
    - Weather
    - Prepare for the unexpected
- Rules
- Areas of focus the week before a race
- Areas of focus on the day of a race
- Race-day schedule and logistics

General Preparations and Considerations for an Athlete

It should go without saying that the more prepared your athlete is for their race, the smoother it will go and the more enjoyable it will be. Below are some areas your athlete should likely focus on to be as prepared as possible for the big day.

Course Knowledge

The course substantially influences how tough a running race is and, more specifically, how a race is trained for. Prior knowledge of a course will tell athletes where challenging parts of the course are, the locations of water stops, and where they can conserve energy. The Pacing module notes that races often provide course and elevation maps. This is very helpful in the design of a training program as well as for pacing and race strategies. Another way to gather information about a race course is to speak to those who have done the race before. Online blogs and message boards are also good sources of event information, assuming that it is not the first year it's being held or that the course hasn't changed. Some of the things to focus on are:

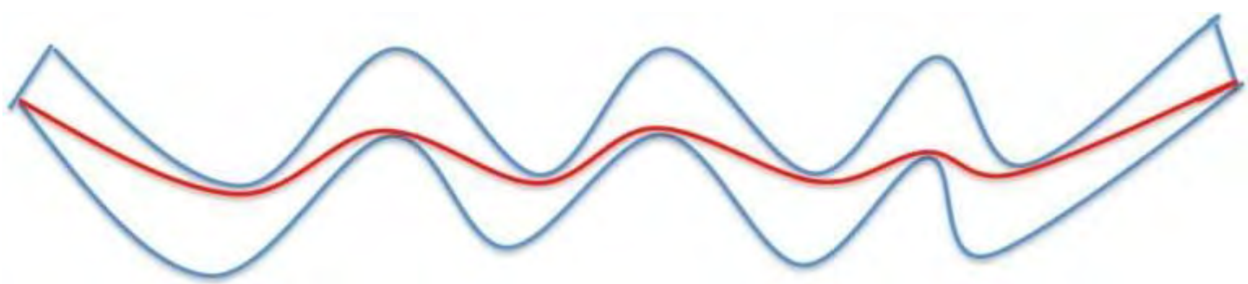
- Notable climbs and descents
- Road conditions
- Final few miles and especially the last mile
- Location of aid stations/bathrooms

Tangent Lines

A tangent line is a straight line that touches a curve at a point. By following "tangent lines," a runner can run a course in the most efficient way possible. While it's likely impossible to run exactly the distance of a race (ex: 26.2 miles), the closer a runner can get to the actual race distance, the better job they did running the tangents.

This is done by making the course as “small” and “fast” as possible. This is accomplished by running the shortest distance between points A and B. This tactic is called *running the tangents*. **Course distances are typically measured at the tangents of turns, so by not racing the tangents, a runner effectively races farther than the course actually is.** *Racing the tangents* is the best way to decrease one’s race time without expending extra physical effort. By some estimates, a runner could run an additional half-mile by not running the tangents over the course of a marathon!

In the image below, the blue lines represent the road, and the red line represents a runner’s path. This illustrates how racing the tangents can substantially “shorten” a course. This is possible only on a race course that is closed to traffic.



As noted above, running the tangents throughout an entire course is difficult, if not impossible – especially when there are a lot of participants.

Therefore, in large, crowded races, your athlete should never cut someone or a group of people off to run a tangent line, as this is dangerous and impolite.

As such, faster runners at the head of the race generally have fewer people to contend with and, therefore, can run tangent lines much easier than slower runners that run in large packs.

Google Maps

If a race provides a course map, you can utilize the street view function of *Google Maps* to view the course. This is an excellent tool as you can virtually "drive" the entire course. Some of the aspects of a course that can be ascertained are:

- Hills and descents (steepness)
- Landmarks (ex: one mile to go, there is a red barn on the left of the road)
- Turns

Be aware that if you are using the map for landmarks, depending on when the mapping was done, the landmarks might have changed (ex: the red barn was torn down).

Weather

The weather influences race strategy as well as race preparation. The temperature affects many things, including apparel, hydration strategy, and effort level. While training in the expected weather conditions of an event is essential, your athlete must be prepared for unforeseen temperature fluctuations.

Rain and inclement weather do not impact just performance but also the overall safety of your athlete. Strong winds also affect an athlete's overall time.

To prepare for bad weather, both physically and mentally, one must train in it. Any training done in bad weather must also be done safely.

Heat acclimatization is necessary for athletes competing or training in extreme heat. Information on this is provided in the *Illness and Injury* module.

Adjusting Race Plans for Prior Illness or Injury

If an athlete has missed training due to injury or illness over some part of the training process and you think they've missed enough that they aren't as fit as they need to be to hit their original time goal, you have to be the one to convince them to change their plan. While often challenging to come to terms with, it takes a lot for an athlete to admit where they are in terms of fitness and try to execute the best race possible, given that fitness. This is in contrast to an athlete not admitting where their fitness is and trying to pace themselves like they were in perfect condition.

Prepare for the Unexpected

While there will always be unexpected issues that come up on race day, it is your responsibility to identify and adapt an athlete to as many potential problems that can occur on race day during training. This serves three primary purposes:

1. Reduce the anxiety of an athlete by building their confidence
2. Safety
3. Increase performance

Potential Issues

- Bad weather: rain, wind, extremely hot/cold temperatures
- New course
- Adjusted start times
- Sickness
- Body pains (e.g., knee, shin, cramps, hamstring)
- Injury
- Food/drink offerings at rest stops are not the same as noted in the race information.
- Extreme fatigue
- Tripping/falling
- Blisters/chafing
- Getting lost
- GPS watch malfunction

You must identify potential issues and possible solutions – or at least ways to mitigate their severity. In addition to the solutions discussed in this module, it is advantageous to speak to other coaches to get additional resources to minimize the impact of unexpected issues that can pop up on race day.

Common Issues

Of the issues above, some of the more common ones that a runner may encounter and experience on race day are listed below. As a coach, you must explain the issue and how to minimize the impact.

Weather

Your athletes should keep an eye on the forecast. Regardless of the forecast, athletes are advised to pack a wide assortment of clothes to be ready for any weather condition.

Sickness

Athletes who are sick should not race; if they are in the race already, they should not continue.

Exercise-Related Cramps

If a cramp is localized and can be treated via stretching and self-massage, it is OK to continue. However, once cramps become so severe that rest, self-massage, and stretching do not eliminate the cramping, it is advisable to stop. Cramping is typically associated with the following:

- Running at a high-intensity level
- Running for longer than having previously trained for
- Low glycogen/electrolyte stores*
- Dehydration*

* Most current research has debunked the myth that low glycogen and mild to moderate dehydration levels cause muscle cramps (763-765).

Cramping, and more specifically severe cramping, is typically preventable.

Injury

Through proper training, many injuries can be prevented. However, if an injury does occur during an event, athletes need to be able to assess accurately if they should or should not continue. Reasons not to continue are:

- The injury will get worse if they continue
- Head or back injury
- Excessive pain
- Suspected ligament or tendon injury
- Potential for infection
- Broken bone

Athletes who are unsure if they should continue in an event should seek medical advice at an aid station, if possible.

Rules

Part of having a successful strategy is having a clear understanding of the rules. You and your athlete must fully understand and comply with the rules of an event and that of the race’s sanctioning body, if applicable. The rules of an event should be posted on the event’s website.

Coach Checklist and Considerations

As is noted in this module, having an athlete checklist is fairly commonplace. However, what is not so commonplace is having a coach checklist on race day. The below list by Ben Rosario highlights key aspects that a coach must be aware of on race day.

Weather

Athletes spend months with a goal in mind, and it can be tough to explain that because of extreme heat, crazy wind, or something entirely unforeseen, they will need to adjust their time goal. The strategy is to explain to an athlete that a time goal is chosen because that time goal represents the best an athlete can perform on a particular day. Adjusting that time goal doesn't change that. An athlete is still trying to achieve the best they can on a specific day, but now, because of the weather, it's no longer X; it's X + Y. Additionally, coaches have to do their homework. They need to look at past results from a particular race when the weather's been bad. Consult heat calculators because the athlete will look to you to give them a new pace goal. You've got to get it right.

Injury or Illness

If an athlete has missed training due to injury or illness over some part of the training process and you think they've missed enough that they aren't as fit as they need to be to hit their original time goal, you have to be the one to convince them to change their plan. While often challenging to come to terms with, it takes a lot for an athlete to admit where they are in terms of fitness and try to execute the best race possible, given that fitness. This is in contrast to an athlete not admitting where their fitness is and trying to pace themselves like they were in perfect condition.

Uncertainty

Be prepared, as a coach, to handle uncertainty on race weekend. Face it- if your athlete has travel delays on race weekend, forgets to pack their racing flats, or can't get a reservation at the restaurant they wanted to eat at, etc., you will be the one they call. You have to stay calm, confident, and pragmatic. Athletes feed off of you.

Importance of the Day

Coaches need to remember how vital race weekend is to the athlete. Even if you have a hugely successful coaching business and are working with 50 athletes, with multiple races on any one weekend, you can never let an athlete think you don't care. The point is, as a coach–even an online coach–you work on the weekends, and you have to be prepared just like you expect your athletes to be. You have to have a system for communicating with athletes the day before the race, the morning of (if necessary), and of course, right afterward. They want to know you care.

Ben Rosario Quote: *"I made this mistake once a long time ago when I was coaching an adult training group. We had about 30 or so people running either a marathon or half marathon. It was a loop course so the marathoners had to run two loops. One of the athletes was coming in toward the end of the loop and I started cheering that he was almost done. The problem was that he was one of the marathoners. He was really hurt that I didn't remember which race he was running. That stuck with me big time."*

Local In-Person Race

For coaches who work in person with athletes in their town, it is highly recommend making a big deal out of whatever the "bragging rights" race is in their town. If you have multiple athletes running a certain race, you should have a meet-up spot for your athletes before and after. You should set up a post-race get-together. Perhaps you could create shirts specifically for your crew. It's a great chance to build camaraderie and also great advertising!

Week Before the Race

Following are areas to focus on the week before a race (most applicable to half and full marathons and ultra-marathons):

1. Rest and Recovery
2. Nutrition and Hydration
3. Supplies
4. Confirm Travel Plans (if any)

5. Athlete Tracking
6. Mental Preparation

Rest And Recovery

Conserving energy must be a primary focus before a race. This is an all-encompassing focus. Aside from any workout sessions, excess energy expenditure should be avoided or kept to a minimum. The longer the race, the more this applies.

Nutrition And Hydration

Eating healthy, familiar foods is always important during the training process but especially the week before an event. An athlete must also place a significant focus on staying well hydrated. Refer to the *Sports Nutrition* module for more information on nutrition and hydration.

Supplies

Anything and everything needed for an event must be sourced and organized before or during this week. Following is a list of items/supplies that athletes should reference to make sure they are prepared for their event:

- Nipple covers for men; if chafing is a concern
- Running shoes
- Socks
- Anti-chafing balm such as *Body Glide*
- Running hat/visor
- Running clothing (i.e., shorts, shirt, tights, sports bra)

- Sunglasses
- Compression socks, if used
- Throwaway clothes if it's a cold-weather event
- Pre-post race food
- On-the-run food/drinks
- Dry clothing and comfortable shoes for after the race
- Waterproof sunscreen
- Photo ID/passport
- Towels
- Toilet paper
- Medical needs (e.g., insulin, inhaler)
- Heart rate/GPS watch
- Watch/alarm clock
- Directions to race
- Camera
- Wallet/money
- Phone
- Reservation information (if any) for lodging, restaurants, transportation
- Race information, race number/timing chip/confirmation of race entry
- Rain gear
- Extra running clothing
- Hydration sources
- Emergency contact information

If your athlete is flying to an event, **running shoes *must* be in their carry-on.** Running shoes are specific to an individual and can spell disaster if lost in transit. This is not to say other equipment/apparel is not important. However, running shoes are critical.

Confirm Travel Plans

If traveling to an event, it is suggested that your athlete get confirmation of travel arrangements such as plane, hotel, restaurant, and car rental reservations. athlete

Athlete Tracking

Many large half and full marathons have athlete tracking services that follow the runners' mid-race progress and finish time. This is facilitated by participants running over timing strips that capture their time. If family and friends enter a participant's race number in the athlete tracker (typically a mobile phone app), they will get updates directly to their phone.

Note: As glitches with tracking apps and timing strips can occur, runners should inform their friends and family not to worry if they don't get mid-race splits.

Day Before the Race

There should be three areas of focus the day before a race.

1. Take Care of All Event Preparations

This includes all operational items such as race number pickup and pick up any last-minute items needed for race day, such as food, hydration, and throwaway clothes.

Also, layout your race clothes and any other things you'll need on race day (ex: race number, fuel, etc.

2. Proper Fueling

This should have been practiced during training. Aside from meals, it is a good idea for your athlete to bring along familiar and healthy snacks. Depending on the size of a race, the host town or city may be overrun with hungry runners looking for food. This means hundreds – if not thousands – of runners and their families/friends all vying for the same places to dine. Therefore, advanced planning such as reservations (months ahead) is advisable. Also, hydrate, hydrate, hydrate!

3. Rest

It's tempting to walk around the expo area or check out the sights, especially if a race is in a fun place and your athlete is with friends and family. However, you must remind athletes that they have trained very hard for their race and must conserve their energy for the next day. More specifically, your athletes should keep walking around to a minimum and get enough sleep.

If an athlete wants to go sightseeing and do tourist things, they are best done after the event and ideally one or two days after the event. The longer the event, the more recovery time is needed.

At Event: Day Before the Event

The race schedule will inform athletes where they need to be and when. Below are some things to consider:

Athlete Check-In

This is where all registered athletes go to confirm participation in the event. At this point, athletes usually get a race packet with all the information about the event. This is often the same location as the race expo for more significant events. It is a good idea for your athlete to have the following items on hand:

- Photo ID
- Confirmation of entry (i.e., printed copy of race entry confirmation)

Race Expo

Expositions are commonplace at races, big and small. Some are held on-site at the race course, while others are "off-site" (often in the host hotel or a convention hall). Expos are fabulous for checking out new gear and present an excellent opportunity to pick up fueling samples to use at a later time.

Remember, *Nothing* new on race day! Race day is when your athlete uses what is tried and tested during training to make everything familiar and safe.

Pasta Dinner

For participants and family members, many events have a pasta dinner the evening before the race. These are great to attend because they are often cheaper than dining at restaurants. They are typically held at the host hotel or somewhere nearby. It's also an excellent time for your athlete to meet fellow competitors.

The one potential downside is that your athlete typically has no control over what food is available, how it is prepared, or the exact content. For example, this might not be the best idea if your athlete has special dietary needs. In addition, the times when the pasta dinner is offered might not fit in with your athlete's fueling schedule.

Race Day

Your athlete will likely go through a wide range of emotions on race day, including being excited, nervous, or even scared. This is natural, and to some degree, it would be unnatural not to have anxiety the day of an event. You are responsible for reminding athletes that they have trained for this day and should feel confident in their preparation. Remind them to stick to the game plan and to have fun.

Four things need to be focused on in the weeks leading up to an event:

- Assure athletes that they have adequately prepared.
- Have athletes write down a schedule for race day.
  - Make sure they go over the race-day checklist to ensure they have everything they will need for the event.
- Ensure that they have read and understand the rules, directions, and schedule of events (e.g., race number pickup).
- Have them mentally review the strategy you have laid out for them the morning of the event.

Through stomach butterflies and nervous energy, it is easy for your athlete to get thrown off track and lose focus on what needs to be done on race day. For this reason, **they need to write down a race-day timeline.** Once athletes get the race schedule that notes where and when things need to be done (e.g., race start time, shuttle bus, deadline to be in start corrals), they should create a race-day timeline. This will give athletes confidence that they do not forget anything and are where they need to be at particular times.

A sample race-day event timeline might look like this:

4:45 a.m. Wake up

5:15 a.m. Breakfast

6:15 a.m. Get changed into running clothing and throw away clothing

7:30 a.m. Check out of the hotel

8:00 a.m. Eat pre-race food

8:15 a.m. Bathroom break

8:30 a.m. **Warm-up**

9:00 a.m. **Race Start**

Wake Up!

The months of training and sacrifices all culminate on race day. So, what is the most important thing to remember? **Get Up!**

Whether oversleeping an alarm or forgetting to set one, the result is the same: missing the event. Therefore, it is in your athlete's best interest to set multiple alarms and, if available, arrange for a wake-up call. There are many reasons for a DNF. However, not making the start line should not be one of them.

Eat

Regardless of the length of an event, your athlete needs to be fueled appropriately. The timeline of when and what to eat should be worked through in training to ensure ample time to eat and digest food without negative consequences. The longer the event, the more critical pre-race fueling is.

Warm-Up

A proper warm-up is an essential part of race day for shorter-distance events. Physiologically speaking, a warm-up has many positive effects on the body (261, 262).

- Warm blood delivers more oxygen to the muscles.

- Dilates blood vessels, which are constricted during rest
- Muscles contract faster and with greater force when warm versus “cold.”

For short-distance events, the minimum time spent warming up should be five minutes and the maximum 30 minutes.

Like all other aspects of racing, a warm-up should be practiced in training to determine the optimal duration and intensity. A warm-up is meant to enhance performance and decrease the chance of injury, not to pre-fatigue an athlete. The time from the end of the warm-up to the beginning of the race should be no longer than 15 minutes.

Research by Tillaar et al., found that when trialing two different warm-up strategies (10 minutes vs. 8 x 60M sprints), there was no significant difference in running performance using the two different strategies (2029). Therefore, it was concluded that athletes should choose what warm-up method they feel most comfortable with.

Regardless of the strategy, a warm-up will have a positive effect on performance in relation to no warm-up (2030).

In the following two videos, Ben Rosario discusses warm-up routines and suggested protocol.

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[There is a Video here. Videos cannot be viewed from this PDF. To View Video Content, please refer to the online Course]

Race Day Logistics

Timing Chip

This is typically included in the registration packet. The chip is typically attached to the race number or is embedded on a strap that gets fastened around shoelaces. As this chip records an individual's time, an athlete must know where it is on race day. No chip = no official race time.

Last Bathroom Stop

Increased anxiety often equates to increased bathroom stops. The port-a-potty area is often very hectic before the start of a race. As such, lines can be extended. Athletes should factor in time to use a port-a-potty before the race. Additionally, as there is a better-than-average chance the toilet paper will be gone, your athlete should have a personal supply!

The Start

Starting areas of significant events will likely be divided into corrals and waves based on participants' estimated paces – fastest up front, slowest in the back. The corrals are often denoted by the color of the race number bibs and the race numbers. The event schedule will identify when and where your athlete will start.

Before the start, have your athletes replay the race game plan in their mind – nothing should come as a surprise. Having a game plan will help to reduce pre-race anxiety and increase performance.

If an athlete must be at the start line long before the gun goes off and cold is expected, it is advised to wear warm clothes that can be discarded before the start. These are often termed *throwaway clothes*, and many events donate the garments left behind by runners. Thrift stores are great places to pick up cheap, warm clothes!

Adhering to the Race Strategy

The 'strategy' for a race should contain multiple strategies that encompass all areas of the race. Below are a few of these strategies:

- Nutrition
- Pacing/intensity
- Mental aspects
- If things go wrong (more on this in the next section)

It is important to stress to an athlete the importance of adhering to the race strategy. The training process leading up to a race is long and meticulous – especially in regard to nutrition and pacing/intensity for various sections of the course. Deviation, especially substantial deviation from the race day strategy in one more or areas can spell disaster for an athlete.

Things such as running at a higher intensity level, eating unfamiliar food or drinking unfamiliar drinks or forgetting about one's mental game plan can substantially affect the success of an athlete on race day.

What To Do When Things Go Wrong

"Everyone has a plan until they get punched in the mouth." This is a famous Mike Tyson quote that more or less sums up what happens when an athlete fails to plan for adversity. When asked to elaborate on his quote, Tyson stated, "People were asking me [before a fight], 'What's going to happen?,' " Tyson said. *"They were talking about his style. 'He's going to give you a lot of lateral movement. He's going to move, he's going to dance. He's going to do this, do that.' I said,* ***"Everybody has a plan until they get hit. Then, like a rat, they stop in fear and freeze."***

Unless an athlete has prepared in training for specific situations, they will likely under or overcompensate or as Tyson stated, they will freeze.

A.D.A.P.T.

Due to the number of variables, there is a better than average chance that things will go wrong at some point in a race. As noted previously, minimizing issues on race day is largely predicated on the preparation that a runner does in training.

Hopefully, the things that go wrong during a race will be minimal, both in terms of scope and frequency. However, regardless of the scope and severity of an issue on race day, an athlete needs to know how to deal with it. The **A.D.A.P.T** acronym provides a framework to deal with issues that arise on race day.

This strategy also seeks to reduce the chance of an athlete panicking and throwing their race plan out the window. **Most of the time, making small, incremental changes is a much better course of action than revising a whole race plan.**

Accept: During a race, it's easy to blow things out of proportion and lose a rational perspective on things. It is also easy for an athlete to project their current situation to the future, and often in a worst-case scenario sort of projection. An example of this would be getting a blister and instead of appreciating the situation for what it currently is and how to deal with it, an athlete solely focuses on how it will negatively affect their race.

Staying in the present is an important aspect of accepting one's current situation. Anticipatory anxiety is a term that describes having anxiety about an event in the future. In terms of ultrarunning, this would apply with respect to how one's current situation will affect them later in the race.

Diagnose: This is a pretty cut and dry step. **"What is the issue?"**

Whether it be a rolled ankle, it's pouring down rain, or a side stitch, an athlete needs to figure out what the issue is.

Analyze: This is the step that an athlete assesses what's going on. For example, an athlete might have an upset stomach. Thinking through the current situation will allow an athlete to plan a course of action.

An example of this step would be an athlete that rolled their ankle:

"My ankle hurts but not too bad, I can walk on it which is good and I'm only one mile from the finish.

Plan: In this step, an athlete makes an actionable plan to deal with the current situation. It goes without saying that the fewer issues an athlete has to deal with, the easier it is to create a plan. Conversely, the greater the number of issues, the more thinking is involved to figure out a solution. While perhaps oversimplified, when facing multiple issues at the same time, issues that have the most potential to be catastrophic (ex: overheating) should be dealt with first.

Take Action:

This is where the rubber hits the road. The prior step of establishing a plan now must be put into action.

With respect to the athlete that is overheated, below might be their action plan:

"I'm going to walk to minimize my effort and focus on drinking to cool my body and if I need to, I'll stop under a shaded area to get out of the sun. Once I feel better, I'll start a walk/run strategy."

Did Not Finish (DNF)

While a *'Did Not Finish'* (DNF) is not the goal of any athlete on race day, the reality is that if an athlete races for long enough, statistically speaking, a DNF is a high probability. Moreover, some races have a higher DNF percentage than others.

From a coaching standpoint, **the main thing to get across to an athlete is that a DNF should not have a negative connotation associated with it**. Regardless of how well an athlete prepares for a race, there may be various scenarios out of their control that makes continuing ill advised. Additionally, the reality is that things do not always go as planned and despite an athlete having a DNF, there are always positives to take away from the race as well as learning experiences.

There are many reasons for a DNF, including but not limited to being unprepared, under-trained, risky race strategy, injury/illness, etc... Let's analyze some of these below to gain a bit more insight.

Injury/Illness: As the saying goes, *"Sh#t Happens."* Perhaps an athlete falls on a descent and hits their knee so hard that they cannot continue. While unfortunate, there is nothing else that could be done and stopping is the correct decision.

Aggressive Strategy: The bigger the risk, the greater the chance for failure. An athlete has to understand this before adopting an aggressive race strategy. That being said, should an athlete set an ambitious goal that forces them to run at or near their maximum capacity and they DNF, there is nothing to be ashamed of.

Undertrained/Unprepared: These two areas are synonymous and unlike injury or an aggressive race strategy, these two issues should not be a reason for a DNF. Whether an athlete didn't put in enough training (mental/physical), didn't trial their nutrition program or didn't do enough research about the course... at the end of the day, issues such as these are training mistakes that should not occur.

Unsafe Conditions: Maybe the temperature plummeted unexpectedly and put an athlete at risk of hypothermia or perhaps a freak lightning storm rolled in making continuing unsafe. In cases such as these, the only correct choice is that of personal health and safety.

Lastly, regardless of the reason for a DNF, any race analysis with an athlete must not be done until a few days after the race... at a minimum. Emotions run high after a race and likely continue for a few days after it – especially if the race didn't go as planned.

During The Race

By the time the start gun goes off, your athlete should know the pacing and fueling strategy and how to deal with the most unexpected things that might come up. Other tactics, such as *running the tangents*, must be remembered and implemented on race day.

The Finish

If an athlete has family/friends there to cheer them on, they should mutually agree upon a location to meet at. Some events do not allow spectators in the finish area, so selecting a pre-determined meeting location is important. Large events often have a *family and friends area* near the finish line. Additionally, directly after a race, an athlete is advised to walk around to prevent blood pooling in their legs.

Post-Race Fueling

The event is over, and it's time to celebrate ... almost. Your athlete will need to refuel and hydrate at the end of the event, either using the food and drinks made available by the event or brought by the individual's family, friends, or coach… ideally within 30 minutes of finishing.

Completing a running race is a huge accomplishment and deserves celebrating, but not at the expense of one's well-being. Your athlete should wait a few hours after the race and preferably at dinner that evening to have alcohol – and only then in moderation. Alcohol has a much more significant effect on an athlete post-race than under normal conditions. The primary post-race drink should be water and an electrolyte replacement drink.

Runners sacrifice a lot to train for a race. Some of these things are their free time and their favorite (and maybe not so healthy) food and drinks. Because of these sacrifices, many runners go crazy at dinner after finishing a race to reward themselves for their sacrifices. While this is deserved, the body is probably not used to a triple-decker cheeseburger with a plate of gravy fries. Therefore, it is advisable to eat relatively healthy and in moderation.

Post-Race Care

Many events have complimentary post-race massages and stretches available for participants. This is a service that will help in the recovery process. While hopefully, your athlete is not a candidate for the medical unit at an event, they need to know that there is that level of support, if need be.

Depending on a runner's level of fatigue post-race, some walking around is helpful in the recovery process. However, it is ill-advised to spend the remainder of the day after a race constantly walking around. This will likely hinder the recovery process and, more to the point, your athlete might not be able to do so. Therefore, it is probably not a good idea for your athlete to make plans that require lots of energy after a race. The longer the race, the more applicable this is.

Post-Race Assessment

Upon finishing a race, your athlete will likely be exhausted and feeling a wide range of emotions from elation to disappointment. **Regardless of the emotional response, a self-assessment of one's performance directly after an event is not recommended.** Your athlete should wait until having a chance to clean up, refuel, and relax before assessing how the race went.

Self-assessment will provide both you and the athlete with valuable information. The following areas represent common assessments:

- Areas in which the athlete felt prepared and underprepared
- Strengths and weaknesses
- Accuracy of pacing
- Fueling strategy

Odds and Ends

- Most hotels have a checkout time in the late morning. Your athlete should try to get a late checkout or book the hotel for another day to have somewhere to get cleaned up and recover after the race.

- Please remember to tell your athlete to thank the event organizers and volunteers for all their hard work!
- Maintaining a proper diet and hydration levels during travel is essential. Do not rely on fast food at airports and gas stations for fuel while traveling.
- Allow ample time for pre-race activities. Your athlete does not want to be on such a tight schedule that a slight delay might negatively affect other aspects of race preparation.
- Big races often have a host hotel where the expo/race pick-up is also at. This is often preferable as it minimizes the running around that an athlete will need to do.

Summary

- Before leaving for an event, your athlete must go through a checklist to make sure nothing is forgotten.
- If flying to an event, running shoes must be taken as a carry-on.
- If traveling to a popular event, it is advised to make lunch/dinner reservations ahead of time.
- Knowledge of a race course is extremely important.
- Once athletes have their race program, they should create a time-based schedule to ensure they are where they need to be at certain times.
- In order to be comfortable racing in bad weather, one has to train in it.
- Your athlete must be aware of all the rules of the race.
- Nothing new on race day!
- The athlete should do some, but not too much, walking after the race.
- If a race has an athlete tracking system, your athlete should inform friends and family so they can track the runner during the race.
- Have athletes set multiple alarms on race day to ensure they do not oversleep.
- Post-race fueling and care are critical to recovery.
- A runner should have a plan/strategy for if and when things go wrong
- When things go wrong, use the A.D.A.P.T strategy
- If staying in a hotel, it is advised to book a room for an additional day so athletes have somewhere to clean up and rest after their event.

- Following are areas that must be focused on the week before a race:
    - Rest and recovery
    - Nutrition and hydration
    - Supplies
    - Confirm travel plans (if any)

Module 22: Keeping it Legal

When starting up a running coach practice, many individuals forget about the legal aspect. This is especially the case with individuals who start up a coaching practice as a side gig to their full-time job.

The reality is that no matter how large or small your coaching business is, you must appreciate the legal aspects that go along with starting a business.

This lesson primarily serves as a reminder to ensure that you have researched and checked all of the necessary boxes in terms of running (pun intended) your coaching business.

You’ve likely heard the acronym COA (cover your a$$). Well, it could not be truer when working with athletes since you are advising them on matters that pertain to their body and physical health.

UESCA has made a significant effort to cover all pertinent legal areas as it relates to starting and running a coaching practice in this lesson. However, you should not view the content in this lesson as an exhaustive list. You are strongly advised to speak with an attorney in your state/country that can advise you on matters pertaining to starting and running a coaching business.

Legal Considerations

As a running coach, you will be advising clients on matters that involve their physical well-being. Because of this and the fact that we live in a highly litigious society, protecting yourself from liability is of paramount importance.

The information in this module should not be construed as legal advice. Consult a licensed attorney in your jurisdiction regarding any topic in this module.

Upon completion of this lesson, you should have an understanding of the following areas:

- Insurance and certifications
- Athlete health screening
- Liability waiver
- Personal conduct
- Scope of practice
- Documenting workouts
- What constitutes a performance-enhancing aid
- Physiological testing regarding a physician’s presence
- Legal business structure

As noted above, this lesson will focus on the nine areas above. This lesson is critical as many full-time and part-time coaches do not have the correct structure and operations to protect themselves from lawsuits. Moreover, many coaches operate some aspects of their businesses that are not legal.

Insurance

Let’s start by discussing working with an athlete in a health club or like environment. Most health clubs are required to carry insurance that covers their staff, so if a health club employs you, more than likely, you do not need to purchase additional insurance.

Suppose you perform coaching sessions at a personal training studio or apartment gym. In that case, the facility will most likely require you to show proof of liability insurance and possibly a personal training or coaching certification. Most facilities require at least a $1 million policy and require inclusion on the insurance policy as “additional insured.” If this is the case, the apartment complex/studio will inform you of the exact name to put on the insurance policy. There is typically no extra cost to add an additional insured party to a policy. Depending on where you purchase your insurance, the price typically ranges between $100 and $400 annually for a $1 million policy.

If you are training a client via the internet or elsewhere, it is still advisable to get a $1 million policy to ensure that you are covered for almost anything that could happen.

Before purchasing insurance, make sure that the policy covers the type of training you will be providing, as in some cases, fitness-based insurances do not cover sport-specific training. For example, a policy might cover road running but not trail running.

CPR/AED Certification

You need to be CPR/AED certified if you coach in person. CPR stands for *cardiopulmonary resuscitation,* and AED stands for an *automated external defibrillator*. The two primary organizations that provide CPR/AED training in the United States are the *American Red Cross* (ARC) and the *American Heart Association* (AHA). The other basic certification offered by both ARC and AHA is *first aid,* which is suggested but required by UESCA.

CPR/AED is not a prerequisite to taking the UESCA Running Coach certification exam, as many coaches solely work with athletes in a virtual capacity. However, as noted above, if you are going to be working with athlete’s in-person, you need to be CPR/AED certified.

Athlete Health Screening

All athletes must complete the ***Health History Questionnaire (HHQ)*** and ***Physical Activity Readiness Questionnaire (PARQ).*** The gold standard for health screening forms within the US is the *American College of Sports Medicine* (ACSM). ACSM provides the guidelines for exercise/testing prescription in the fitness industry. Assessing an athlete using these two tools (HHQ and PARQ) will give you a good picture of their overall health and potential risk factors.

Sample PARQ and HHQ questions are noted in *Module 12: Athlete Intake*. It is strongly advised that the PARQ and HHQ questions noted in *Module 12* not be used “as is” with clients without consulting legal counsel.

Liability Waiver

It would help if you had your athletes sign liability waivers. The waiver should absolve you of all liability and responsibility for all potential situations, including death. An attorney should draw up this waiver. However, many coaches/trainers get their waivers off the internet. While a waiver does not prevent you from being sued, it helps protect you in the event of a lawsuit. **If you decide to use a liability waiver from the internet, it is advised to have it reviewed by an attorney in your local jurisdiction.**

Personal Conduct

You must conduct yourself professionally and understand that how others perceive your actions and verbiage is subjective. **Therefore, you must always err on the side of caution regarding your actions and language.** While this holds for athletes of all ages, it is especially true when working with minors. For example, any physical contact not necessary from a coaching perspective should not occur. Examples of normal physical contact within a coach/athlete relationship are physical cueing of exercises, taking anthropometric measurements, and assisted stretching. Note that even within these areas, proper judgment must be exercised. For example, when taking the chest girth measurement of a female athlete, the correct method is to instruct the athlete to hold the tape measure at chest level and to assess the girth measurement from behind them. If performing a hip girth measurement, it should be taken from the athlete’s side rather than from the front or rear.

Before performing any action that requires physical contact, you must inform an individual what it is you intend to do and the purpose of the action. You must obtain consent regarding this type of physical contact. Do not assume that this pertains only to coaching the opposite sex. Regardless of the age or gender of an athlete, this protocol must be followed. Following is an example of a proper way to inform an athlete of your proposed action and ask for consent:

“Sara, during the back-row exercise, I typically place my fingers on the midback muscles to ensure that the correct muscles are being activated. However, before I do this, I want to make sure you are comfortable with this.”

If an athlete’s answer is *“No,”* you cannot physically cue them during this exercise or likely during any exercise.

Anything that could constitute physical or sexual abuse (verbal or physical) is prohibited. What is considered abuse is highly subjective, so you must always be aware of your actions. If your athlete states that something you are doing makes them uncomfortable, you *must* discontinue this action immediately.

Scope Of Practice

Upon passing the running certification exam, you will be a certified coach – not a doctor, not a physical therapist, or a registered dietician. It is critical that this is understood and, more to the point, practiced and respected. As a professional, you must understand your professional limits and responsibilities. As mentioned previously in this certification, the following is a listing of things that you **cannot** do as a certified UESCA Running Coach:

1. Prescribe medications or sports supplements
2. Prescribe a nutritional program*
3. Diagnose and/or treat any injury or perceived medical condition
4. Manual therapy such as massage, trigger point, etc…….

* Check with the local and state laws

You can inform and educate but never prescribe any medication, sports supplement, or nutrition program. Nor can you diagnose or treat any injury or medical condition.

Regarding medications and supplements, most anything that can be purchased in a drugstore falls into things that cannot be prescribed or administered by you. Some of the things that fall into this category, even though they are not necessarily prescription medications, are:

- Pain relievers/anti-inflammatories (e.g., *Advil*®, *Tylenol*®)
- Decongestants (e.g., *Sudafed*®)
- Diet pills (e.g., water pills)

Scope of Practice Scenarios

Following are three scenarios along with the correct way to handle them:

Scenario One

Athlete: Your athlete comes to you with a sore ankle that has not been getting better over the last two training sessions and is seeking your advice on how to deal with it.

Coach: Assuming you have evaluated their biomechanics before working with the individual and have advised the person on proper form, at this point, the professional thing to do is to make a referral to a specialist.

Scenario Two

Athlete: Your athlete just read about a new diet in a recent magazine article and wants you to put together a similar diet for her.

Coach: You need to inform her that you cannot do so because this is the realm of a registered dietician (RD), not a running coach. It would be advisable to refer her to an RD to get her questions answered and a nutrition plan made for her.

Scenario Three

Athlete: Your athlete wants to incorporate an over-the-counter ergogenic sports supplement into his program and wants to know which is the best one to get and how much to take per day.

Coach: You can only inform your athlete about the purported effects of the supplement, assuming you are knowledgeable about it. Advising the athlete to take it and advising on how much to take is outside your scope of practice. Just because most sports supplements are over-the-counter does not mean they cannot harm an individual.

Document Workouts

Anything you do or advise your athletes on needs to be documented. Not doing so leaves you legally exposed if an athlete questions something you did or suggested that caused the person harm. Additionally, it is advised to be as specific as possible. For example, let us say your athlete accused you of assigning push-ups that resulted in a shoulder injury. If you did not document the exercise, it is their word against yours. However, if you document your sessions, you can state with certainty if you did or did not perform push-ups during the session. This also has to do with asking for and receiving feedback during sessions. You should be asking athletes if they are experiencing any pain during exercises, especially during strength and flexibility training. If pain is felt, you need to modify or cease the activity and document feedback.

Before working with athletes, you must inform them that if they feel any pain or discomfort past what is expected, they must let you know at that time (assuming you are working with them in person). If you're not working with an athlete in person, the individual must cease the activity and let you know that day via email or phone.

Detailed information you can provide about a session and informing your athlete what you expect regarding feedback will reduce your potential exposure to legal action against you.

Ergogenic Aids

It is essential to discuss what is legal and illegal regarding ergogenic aids. Many people automatically think of doping when they hear the words performance-enhancing or ergogenic aids. However, this could be anything from a coffee in the morning to the use of illegal drugs. What is important to note is that the rules and nothing else make a substance or product illegal. A sport's governing body could ban orange juice tomorrow, and what was a legal, healthy drink today is a banned substance tomorrow. For example, caffeine is considered a banned substance within some sports but only in high concentrations.

Some drugs get a bit trickier, particularly involving individuals with asthma. Most sports ban the drug albuterol (bronchodilator), a performance-enhancing aid. However, a therapeutic use exemption (TUE) that requires a doctor's approval allows usage for those who need it. A TUE states that an individual needs to use a banned substance to bring the person up to the level of "regular" participants. An athlete needs to be familiar with banned substances to know where the line between legal and illegal lies. Many athletes who test positive for controlled substances do not consciously do so. They may have unknowingly taken a vitamin or sports supplement containing trace amounts of a banned substance. **It is always an athlete's responsibility to know the contents of everything they ingest.**

If a US-based runner has a question regarding a TUE or whether a specific drug is on the banned list, the individual should contact the US Anti-Doping Agency (USADA) for more information. If residing outside the US, the country's national governing body, national anti-doping agency, and/or the World Anti-Doping Agency (WADA) should be contacted.

Physiological Testing

The legal aspect of physiological testing has a lot to do with the administrator's scope of practice and local laws. Let us take blood lactate testing, for example. It is the responsibility of a coach to research all local laws about the drawing and disposal of blood – before performing a blood lactate test.

Another physiological test worth mentioning is the VO2 max test. The VO2 max test takes an individual to the physical limit regarding cardiopulmonary stress. Many labs that test VO2 max require a doctor's presence. While having a doctor on hand during a VO2 max test is not required by law in most places, it is worth performing due diligence to confirm this. If based on an athlete's HHQ and PAR-Q results, you feel the individual should not do a VO2 max test, a sub-maximal test might be a better option for this individual, or perhaps no test would be best. Regardless, any sub-maximal or maximal test should be done with an AED present and with the athlete haven gotten physician clearance for the test. If a VO2 max test is administered, the administrator is assumed to be qualified to run the test and fully understands the protocol.

Unless you are qualified to administer a VO2 max test and have a physician and AED on-site, it is advised to outsource this test or perform a sub-maximal VO2 assessment.

Business Formation

When starting your coaching business, you will need to determine what type of legal business structure it will operate. The four primary types of business structures (US-based) are:

- Sole Proprietorship
- Partnership
- Limited Liability Corporation (LLC)
- Corporation

Of these four structures, only the LLC and corporate structures create a legal identity that is separate from you. This means that if you were on the losing end of a lawsuit, assets could be collected only from the company and not your personal assets, such as your house or personal finances. This is because the lawsuit would be against your company and not against you personally.

Therefore, it is highly recommended that you structure your coaching business as a legal entity that is entirely separate from you.

It is strongly advised to work through an accountant, attorney, or company/individual specializing in business formations. The following are three things that you will likely need to be established if setting up your company as an LLC or corporation.

1. Local Business License(s)
    - If doing business under a name other than your own, you might need to file a *doing business as* (DBA) form with your business name.
2. Federal Tax ID # (also called Employee Identification Number or EIN)
3. Business checking account

Summary

- If you are coaching independently, you need to purchase professional liability insurance.
- As a coach, being CPR/AED certified is an absolute requirement if working with athletes in person.
- Prior to working with athletes, they must fill out a HHQ, PARQ, and liability waiver.
    - Ideally, you should have an attorney draft and/or review these documents.
- You must always appreciate and work within your scope of knowledge and practice.
- Document workouts
- Be sure of all local laws prior to performing tests such as blood lactate and VO2 Max.
- If your athlete is performing a sub-maximal aerobic assessment, the individual must get physician clearance prior to performing the assessment.
- If coaching independently, it is advised to create a separate legal entity for your business.
- Your personal conduct must always be professional. This includes correspondence via phone and email.